

(19) **United States**
(12) **Patent Application Publication**
An et al.
(10) **Pub. No.: US 2025/0283105 A1**
(43) **Pub. Date: Sep. 11, 2025**

(54) **USE OF CCT-DOMAIN PROTEINS TO IMPROVE AGRONOMIC TRAITS OF PLANTS**
(71) Applicants: **Donald Danforth Plant Science Center, St. Louis, MO (US); The United States of America as Represented by the Secretary of Agriculture, Washington, DC (US)**

(72) Inventors: **Yong-Qiang An, Frontenac, MO (US); Wolfgang Goettel, St Louis, MO (US); Hengyou Zhang, Harbin (CH)**

(21) Appl. No.: **18/849,813**
(22) PCT Filed: **Mar. 23, 2023**
(86) PCT No.: **PCT/US2023/064890**
§ 371 (c)(1),
(2) Date: **Sep. 23, 2024**

Related U.S. Application Data

(60) Provisional application No. 63/323,026, filed on Mar. 23, 2022.

Publication Classification
(51) **Int. Cl.**
C12N 15/82 (2006.01)
C07K 14/415 (2006.01)
(52) **U.S. Cl.**
CPC **C12N 15/8261** (2013.01); **C07K 14/415** (2013.01)

(57) **ABSTRACT**

Genetically modified plants having improved agronomic traits are disclosed. The plants comprise nucleic acid modifications that modify CCT proteins thereby improving the agronomic trait of the plants, including seed quality, seed oil content and seed protein content. Nucleic acid modification systems and nucleic acid constructs encoding the engineered nucleic acid modification system are also disclosed. Further, methods of improving agronomic traits of plants using the nucleic acid modification systems and nucleic acid constructs.

Specification includes a Sequence Listing.

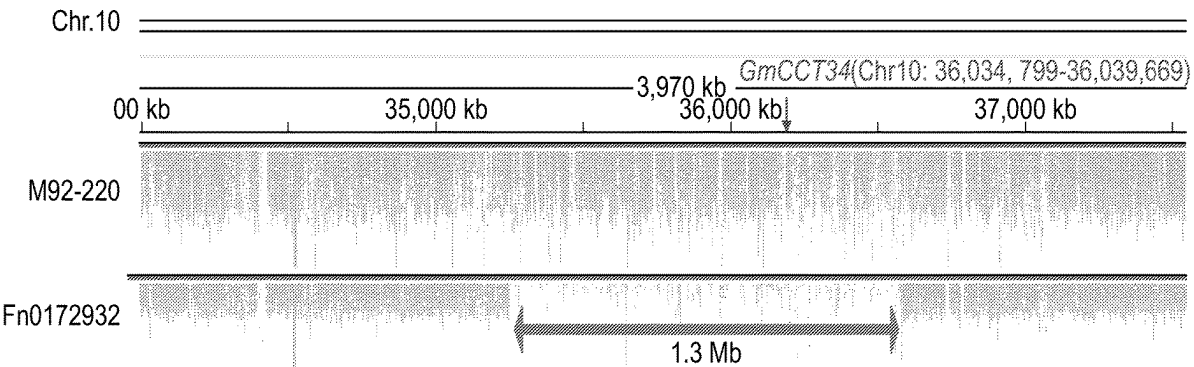




FIG. 1

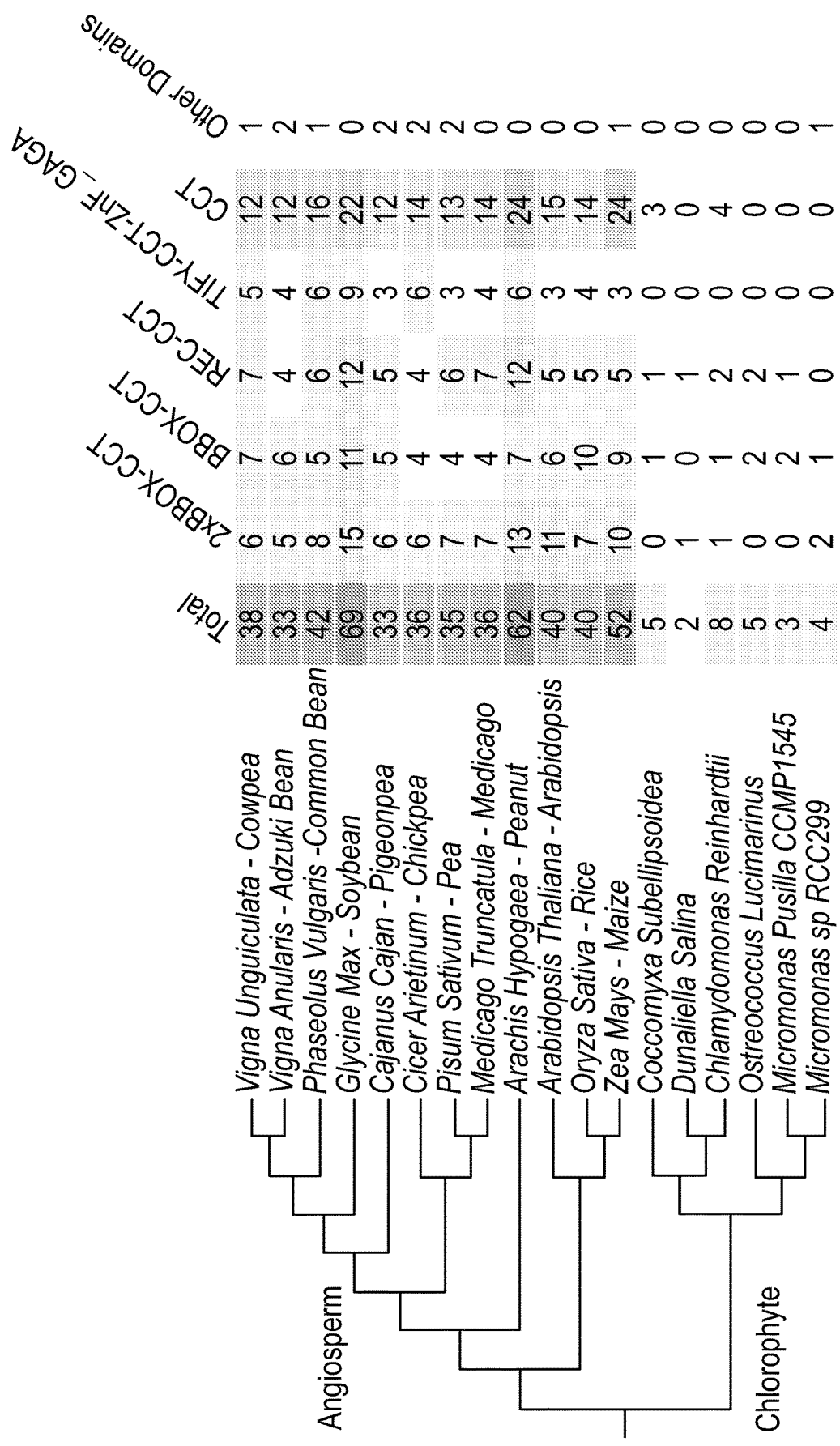


FIG. 2A

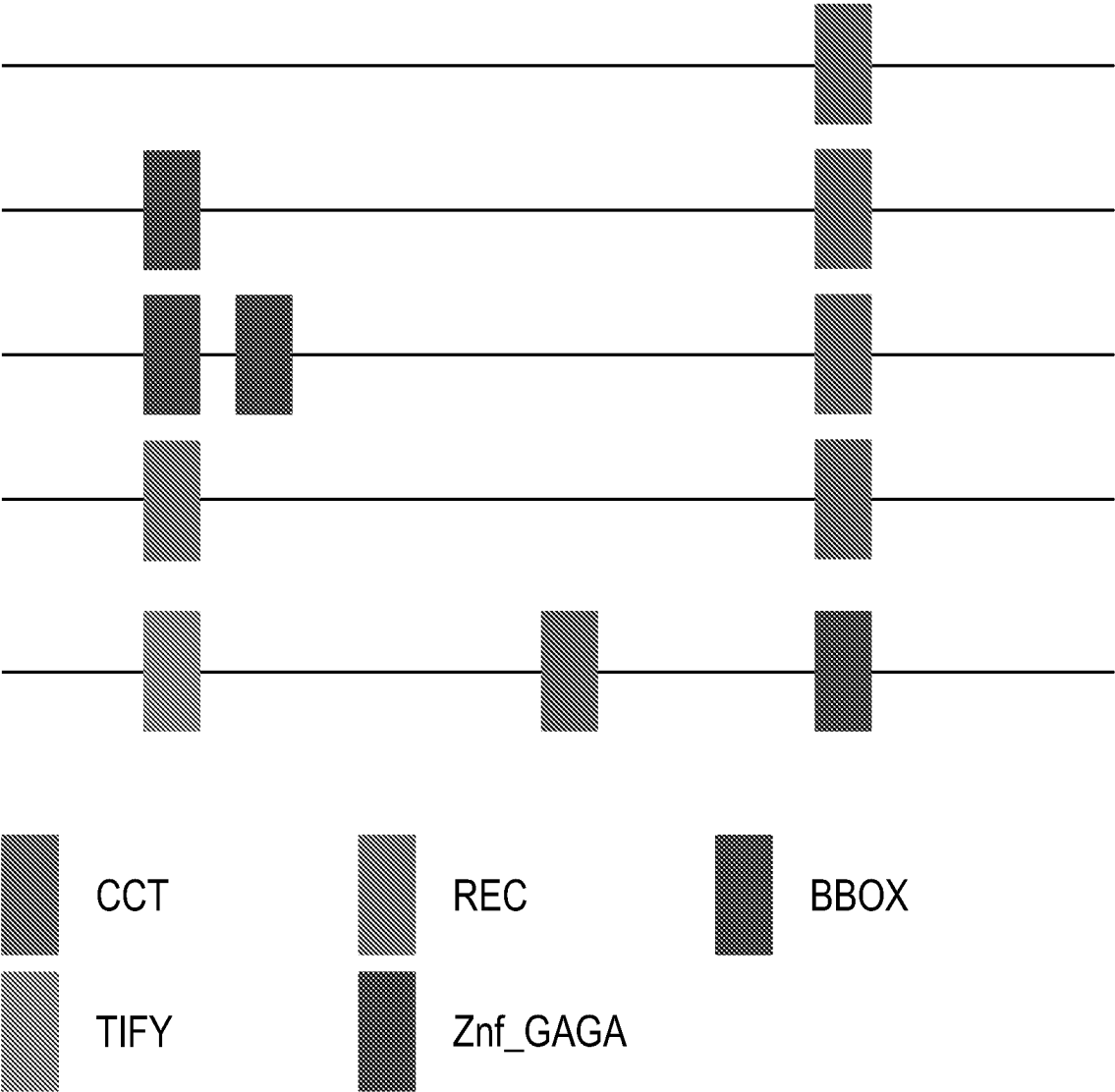


FIG. 2B

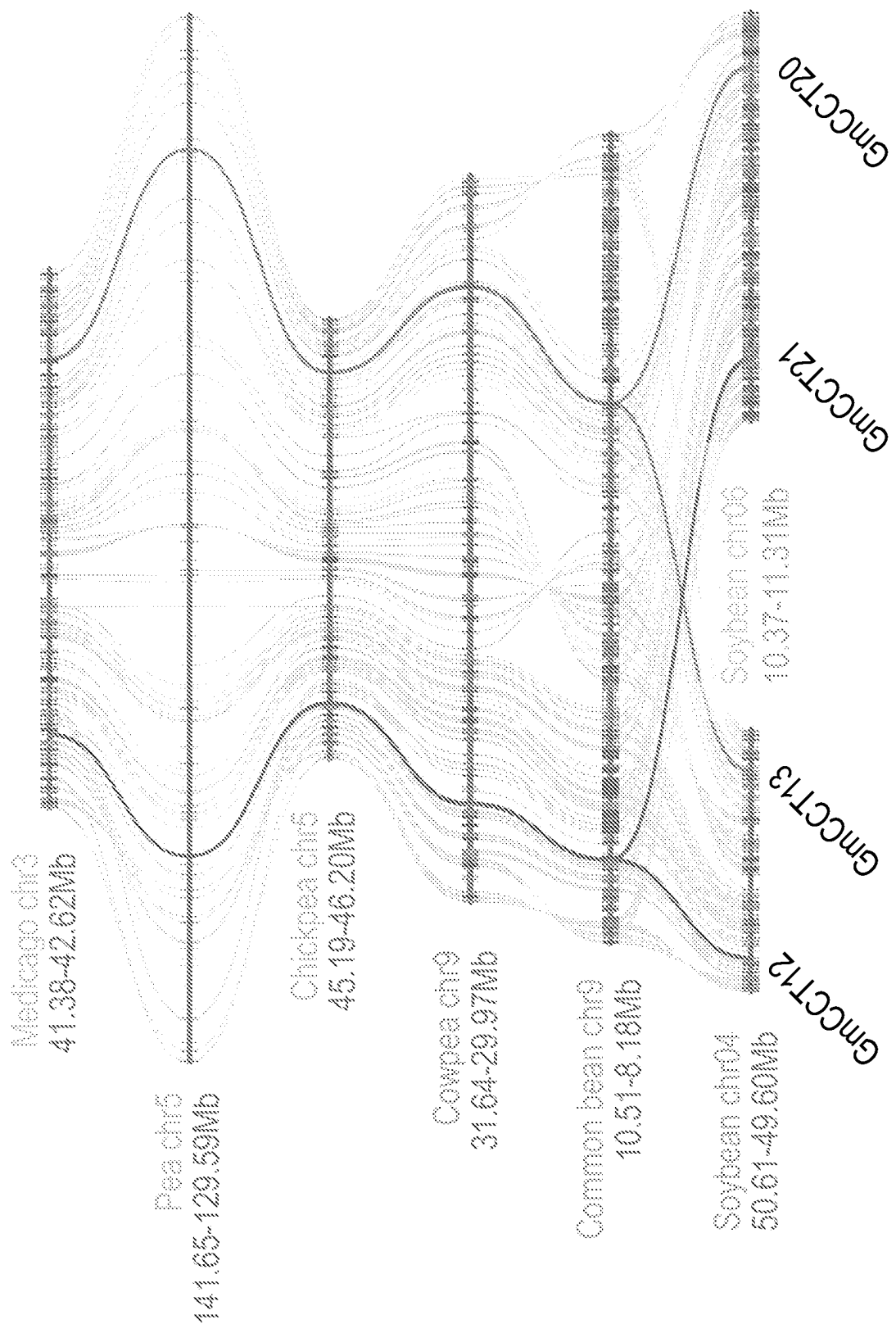


FIG. 3A

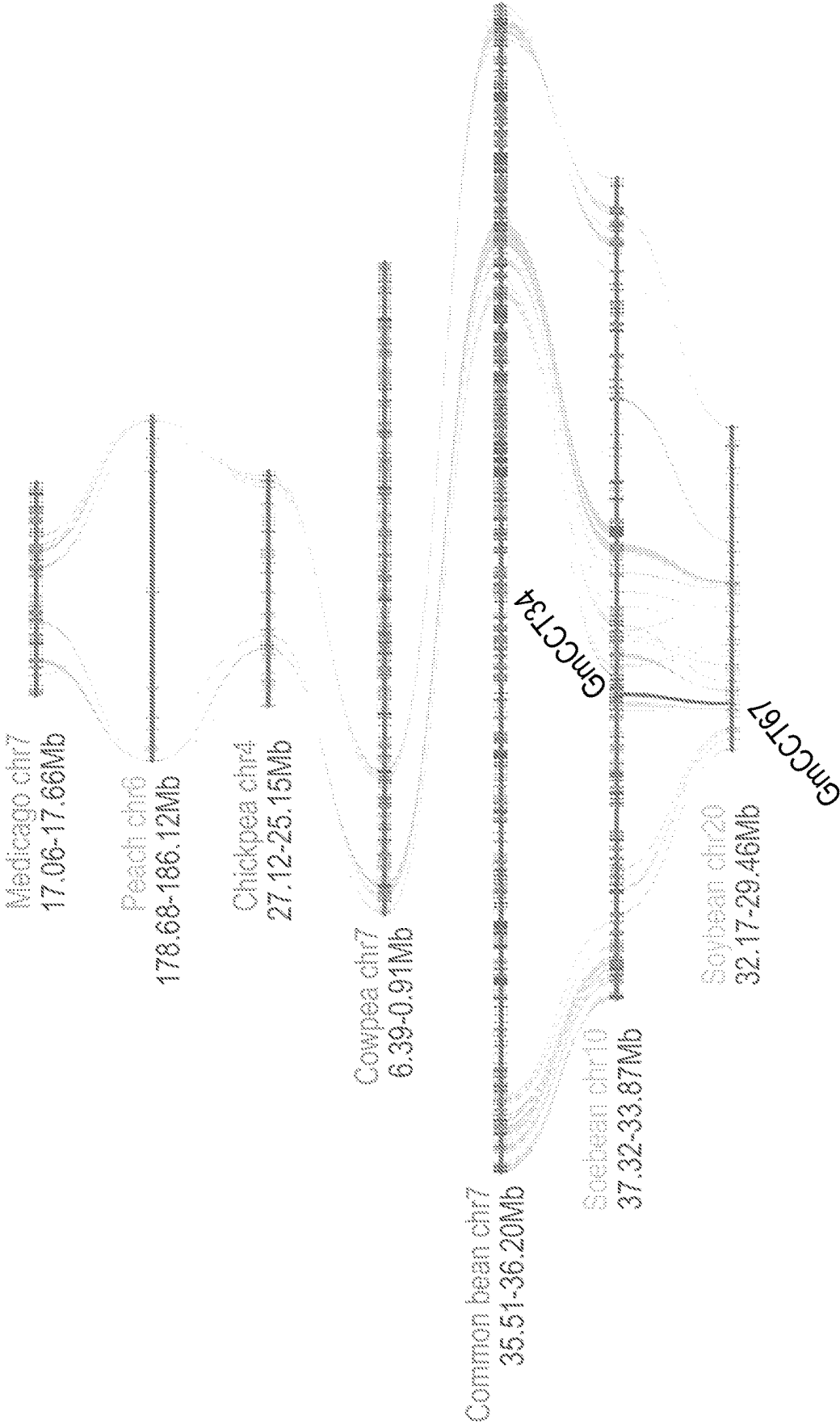


FIG. 3B

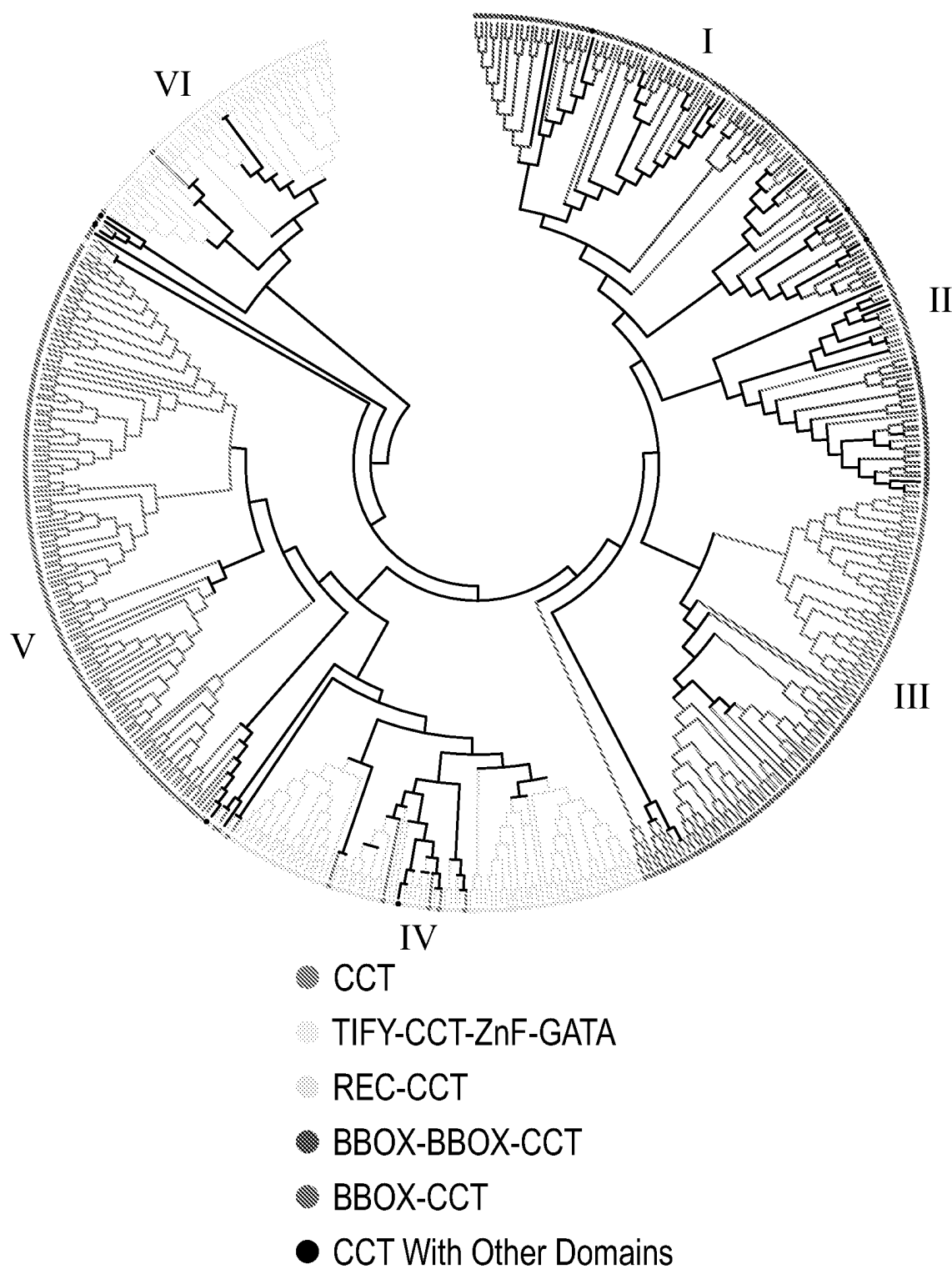
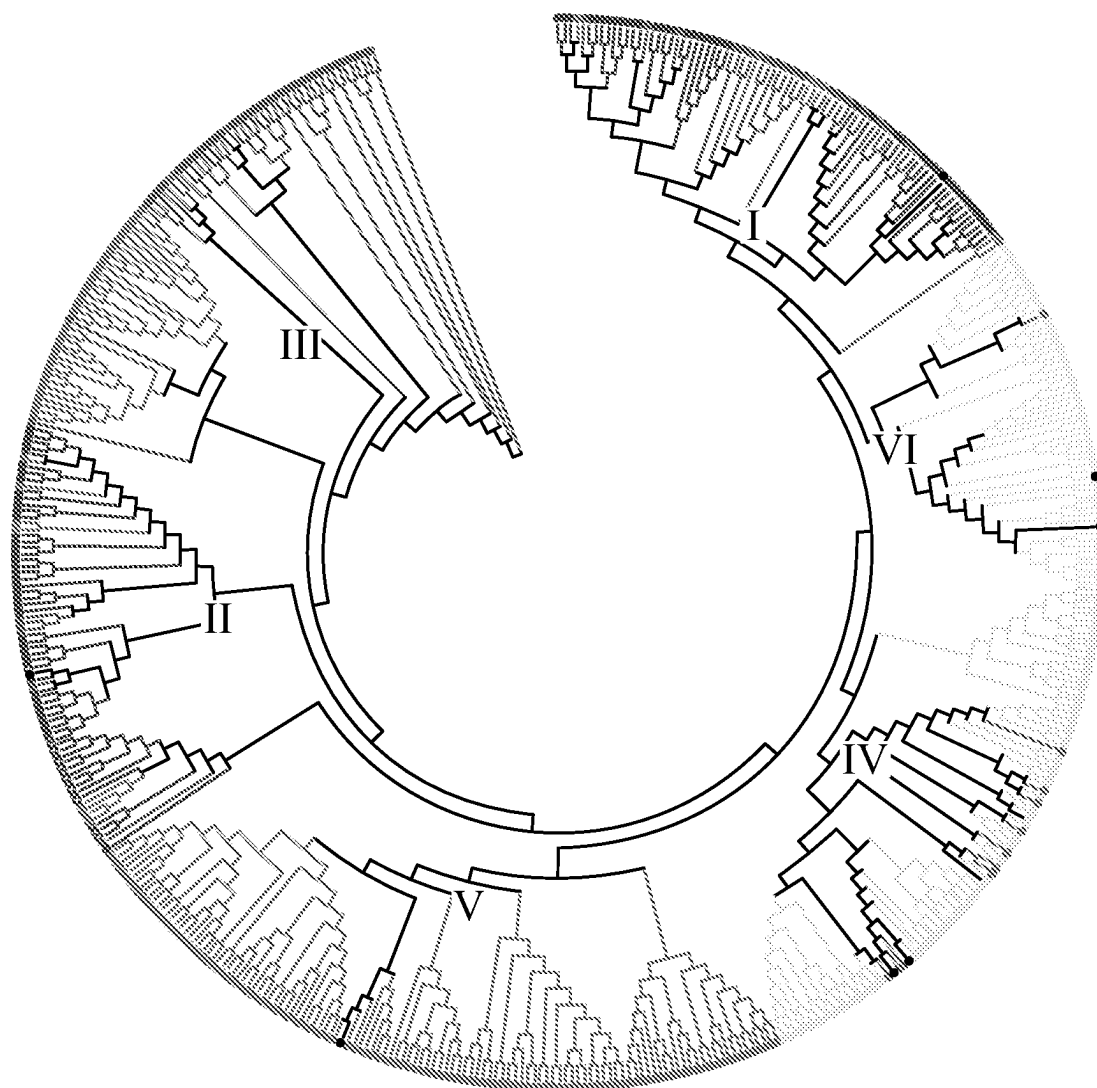


FIG. 4A



- CCT
- TIFY-CCT-ZnF-GATA
- REC-CCT
- BBOX-BBOX-CCT
- BBOX-CCT
- CCT With Other Domains

FIG. 4B

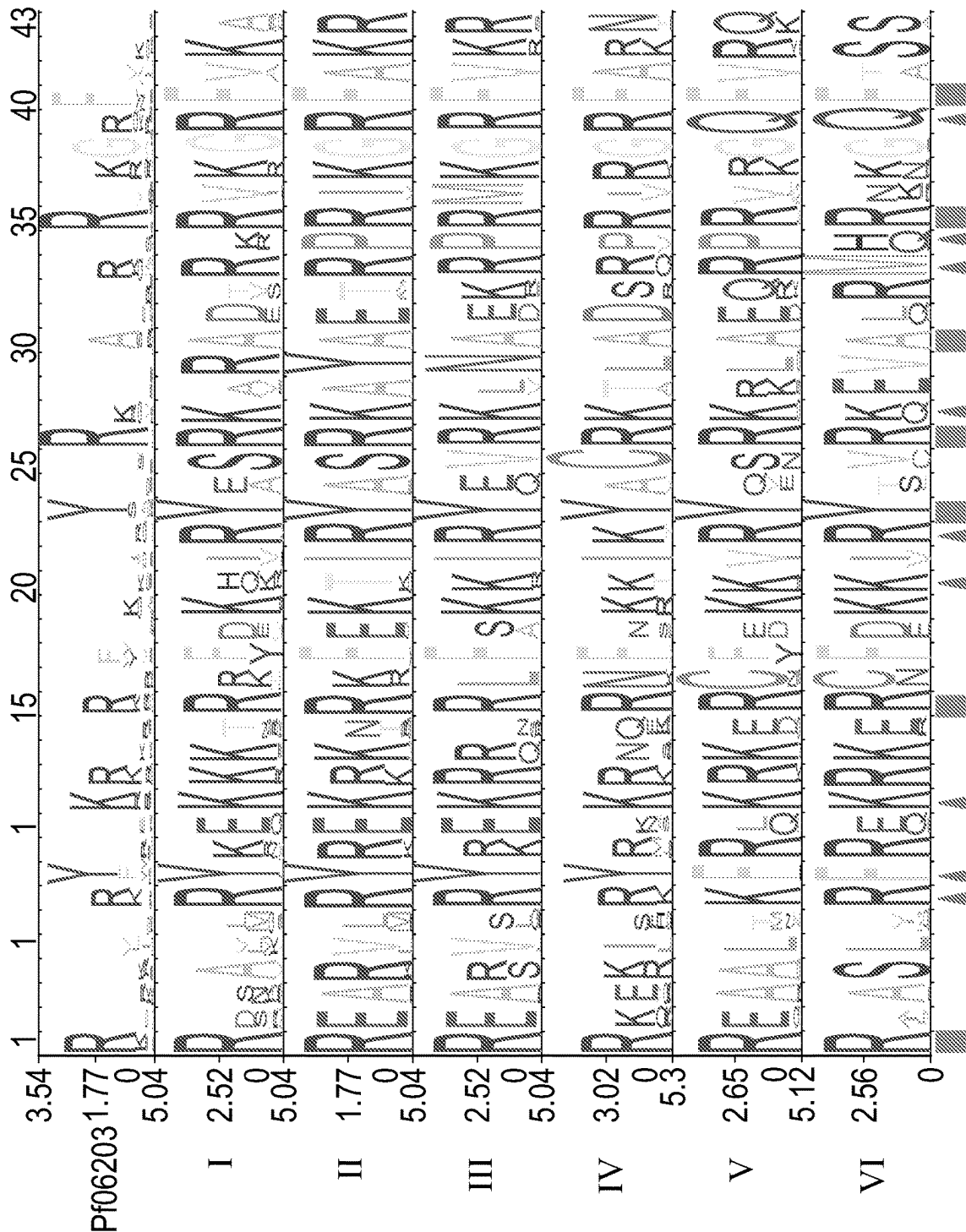


FIG. 4C

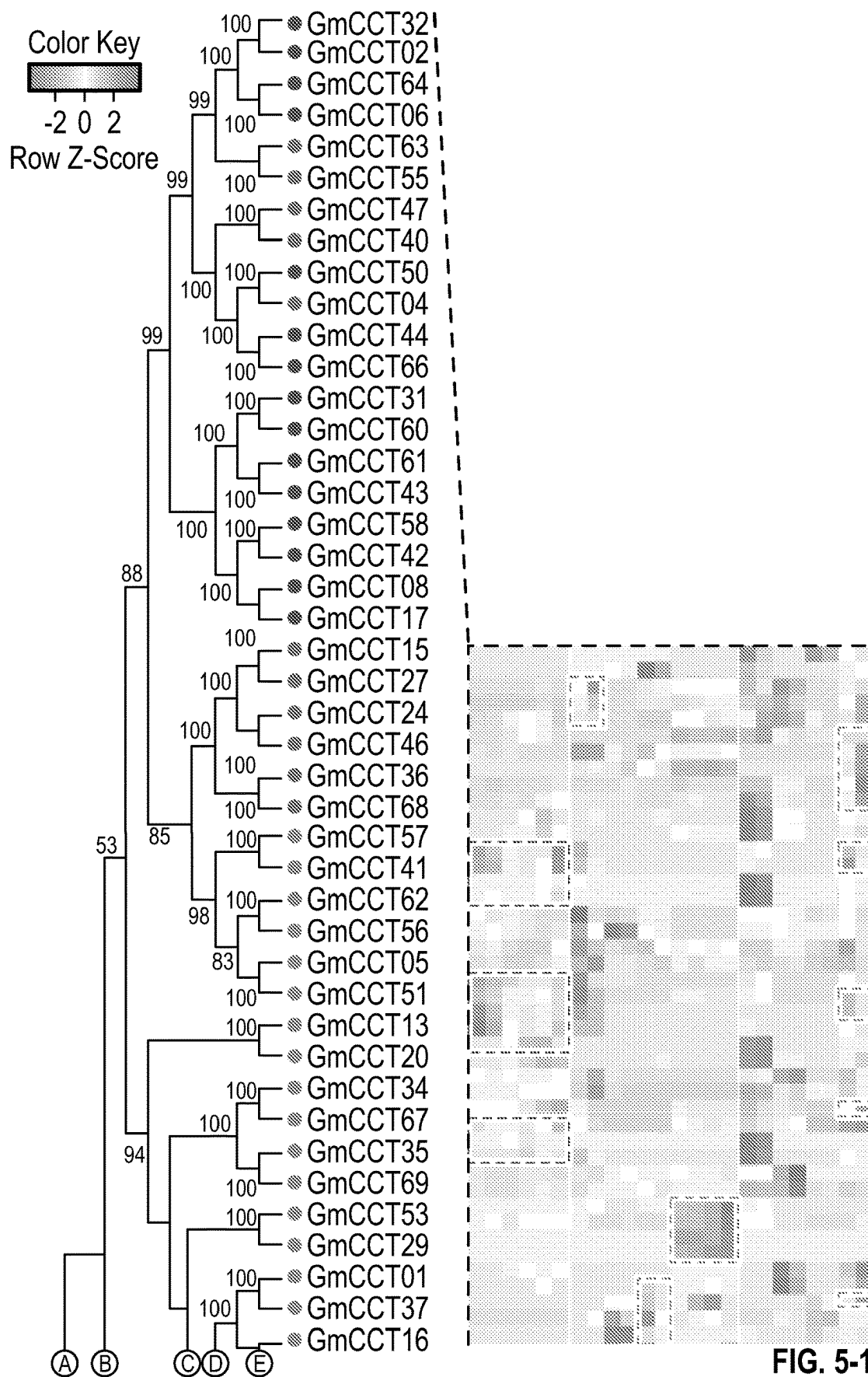


FIG. 5-1

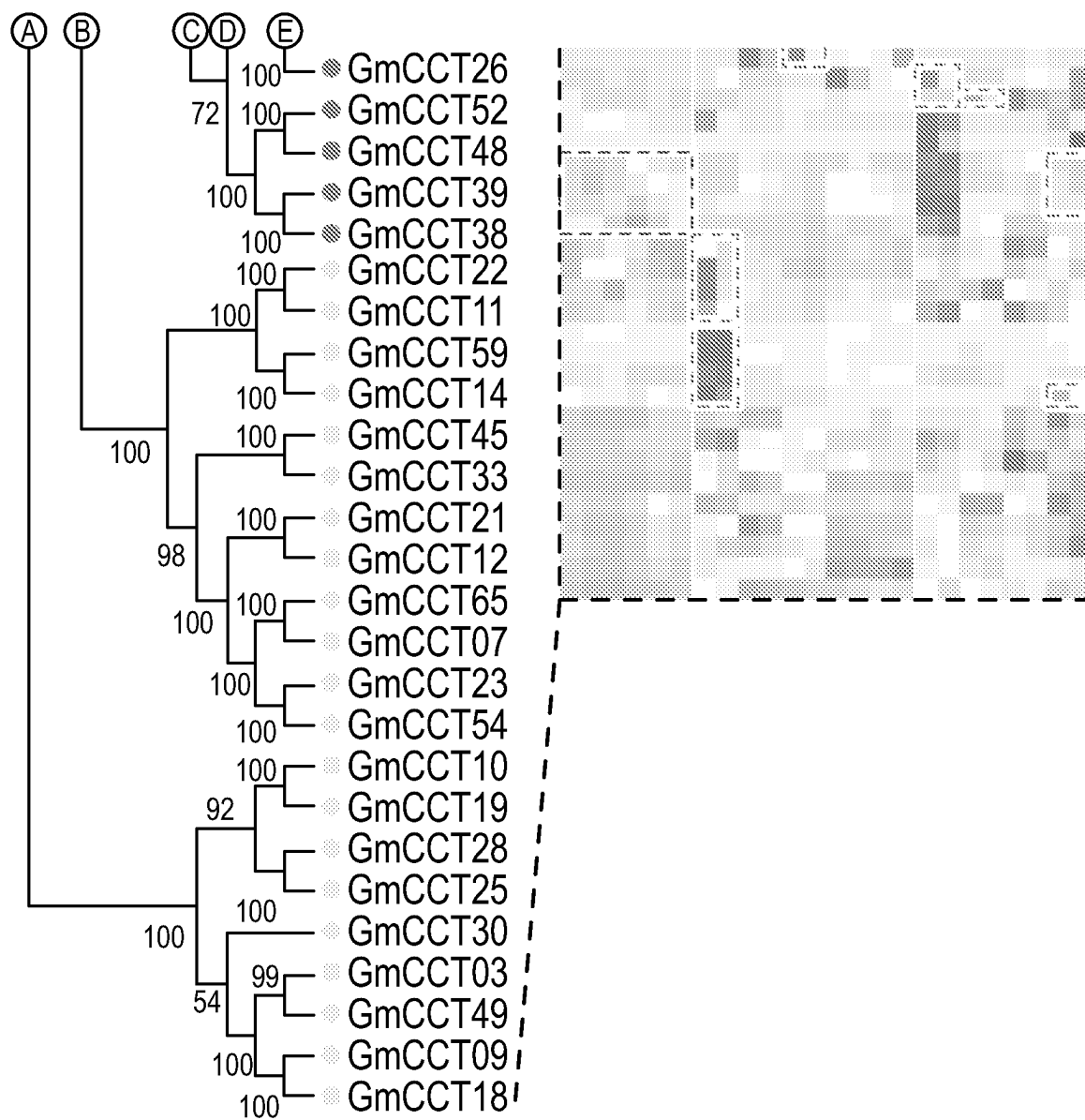


FIG. 5-2

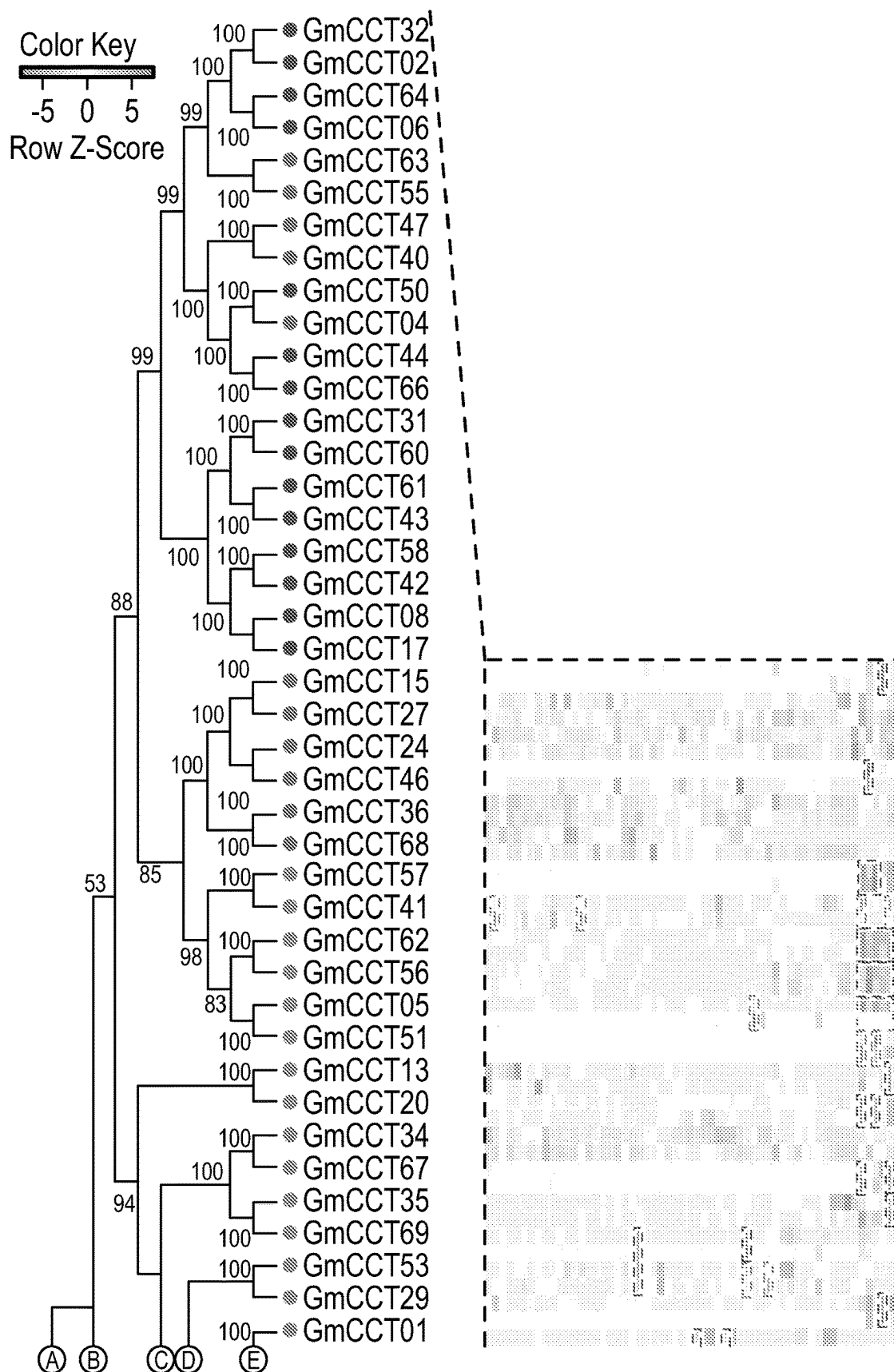


FIG. 6-1

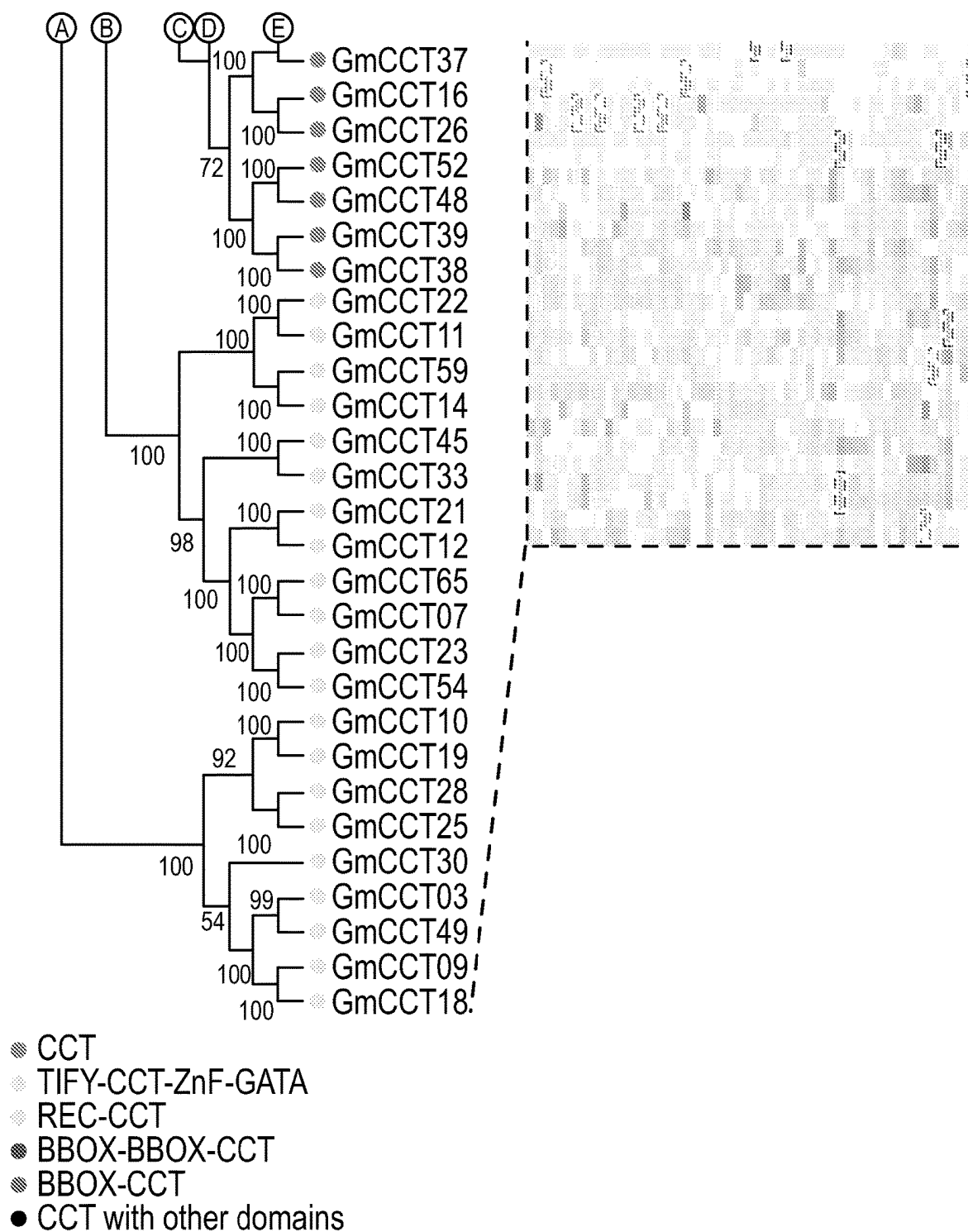


FIG. 6-2

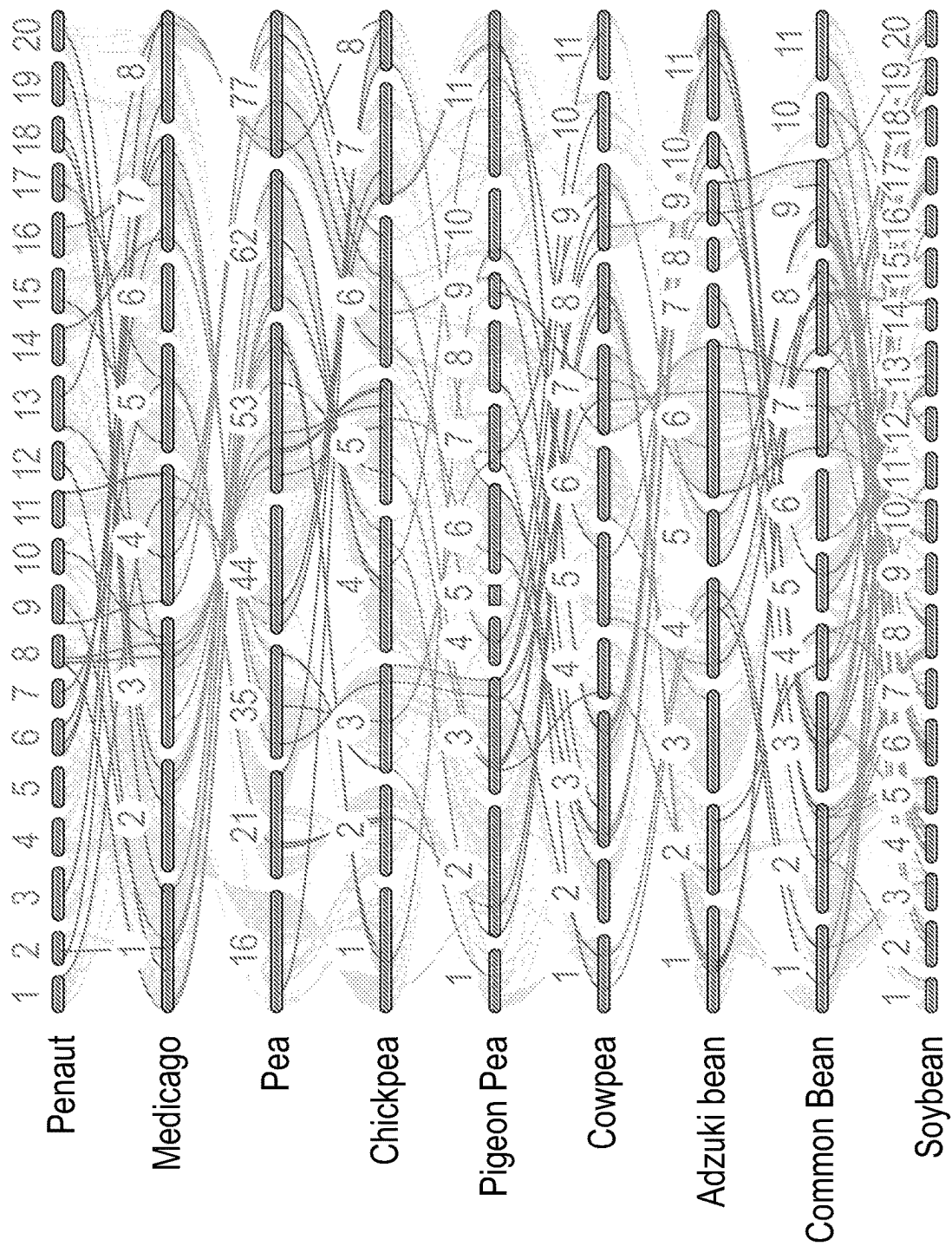


FIG. 7

Ref-AT1G51600/1-43	RLASLVREHEKPKGKNEFKKIRYTVRKEVAALRMQPNKGQFTSA	40
Medtr4g008D90/1-29	RLASLVREHEKPKGKNEFKKIRYTVRKEVAALRMQPNKGQFTSA	30
Vang0017s00340/1-32	RLASLVREHEKPKGKNEFKKIRYTVRKEVAALRMQPNKGQFTSA	20
Vigun09g276400/1-38	RLASLVREHEKPKGKNEFKKIRYTVRKEVAALRMQPNKGQFTSA	10
Ref-AT5G81380/1-43	REALVKEFRQKPKGKNEFKKIRYTVRKEVAALRMQPNKGQFTSA	
Vang09g02620/1-37	REALVKEFRQKPKGKNEFKKIRYTVRKEVAALRMQPNKGQFTSA	
Phvul.003G125700/1-30	RRQSLQRFLEKPKGKNEFKKIRYTVRKEVAALRMQPNKGQFTSA	
Vang11g15830/1-31	RRQSLQRFLEKPKGKNEFKKIRYTVRKEVAALRMQPNKGQFTSA	
Ca.16604/1-27	RRQSLQRFLEKPKGKNEFKKIRYTVRKEVAALRMQPNKGQFTSA	
C.cajan.03709/1-37	REASVLRKYKEKPKGKNEFKKIRYTVRKEVAALRMQPNKGQFTSA	
Arahy.YRC4PH/1-40	REASVLRKYKEKPKGKNEFKKIRYTVRKEVAALRMQPNKGQFTSA	
Ca.03219/1-39	REASVLRKYKEKPKGKNEFKKIRYTVRKEVAALRMQPNKGQFTSA	
Medtr4g127420/1-37	REASVLRKYKEKPKGKNEFKKIRYTVRKEVAALRMQPNKGQFTSA	
Psat4g013600/1-37	REASVLRKYKEKPKGKNEFKKIRYTVRKEVAALRMQPNKGQFTSA	
C.cajan.00921/1-38	REASVLRKYKEKPKGKNEFKKIRYTVRKEVAALRMQPNKGQFTSA	
GmCCT62/1-37	REASVLRKYKEKPKGKNEFKKIRYTVRKEVAALRMQPNKGQFTSA	
Ca.17243/1-38	REASVLRKYKEKPKGKNEFKKIRYTVRKEVAALRMQPNKGQFTSA	
GmCCT05/1-41	REASVLRKYKEKPKGKNEFKKIRYTVRKEVAALRMQPNKGQFTSA	
C.cajan.05381/1-37	REASVLRKYKEKPKGKNEFKKIRYTVRKEVAALRMQPNKGQFTSA	
Ref-AT1G73870/1-43	REASVLRKYKEKPKGKNEFKKIRYTVRKEVAALRMQPNKGQFTSA	
Vang06g09620/1-26	REASVLRKYKEKPKGKNEFKKIRYTVRKEVAALRMQPNKGQFTSA	
Arahy-FA5JII/1-23	REASVLRKYKEKPKGKNEFKKIRYTVRKEVAALRMQPNKGQFTSA	
Ca.20458/1-34	REASVLRKYKEKPKGKNEFKKIRYTVRKEVAALRMQPNKGQFTSA	
Vang04g15980/1-27	REASVLRKYKEKPKGKNEFKKIRYTVRKEVAALRMQPNKGQFTSA	
Ref-AT2G33350/1-43	REASVLRKYKEKPKGKNEFKKIRYTVRKEVAALRMQPNKGQFTSA	
Psat6g107360/1-39	REASVLRKYKEKPKGKNEFKKIRYTVRKEVAALRMQPNKGQFTSA	
GmCCT67/1-27	REASVLRKYKEKPKGKNEFKKIRYTVRKEVAALRMQPNKGQFTSA	

FIG. 8

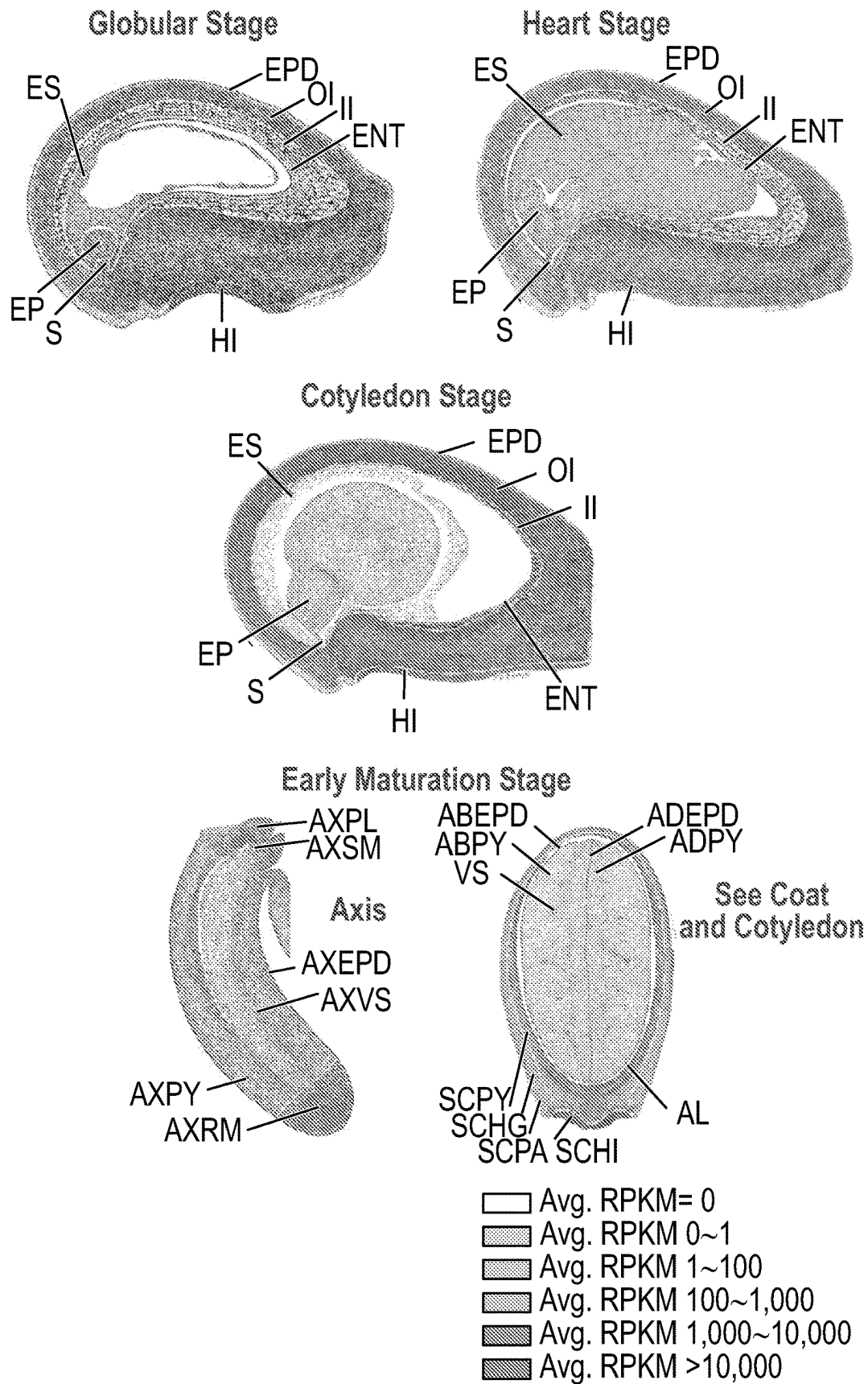


FIG. 9A

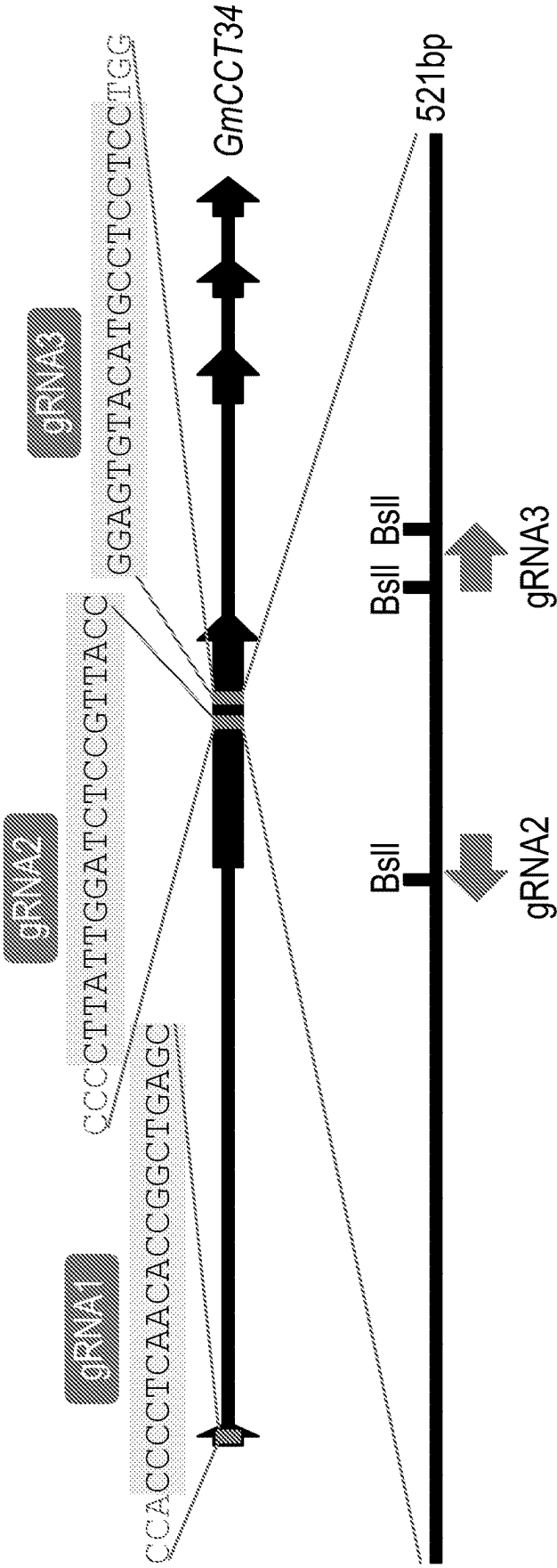


FIG. 9B

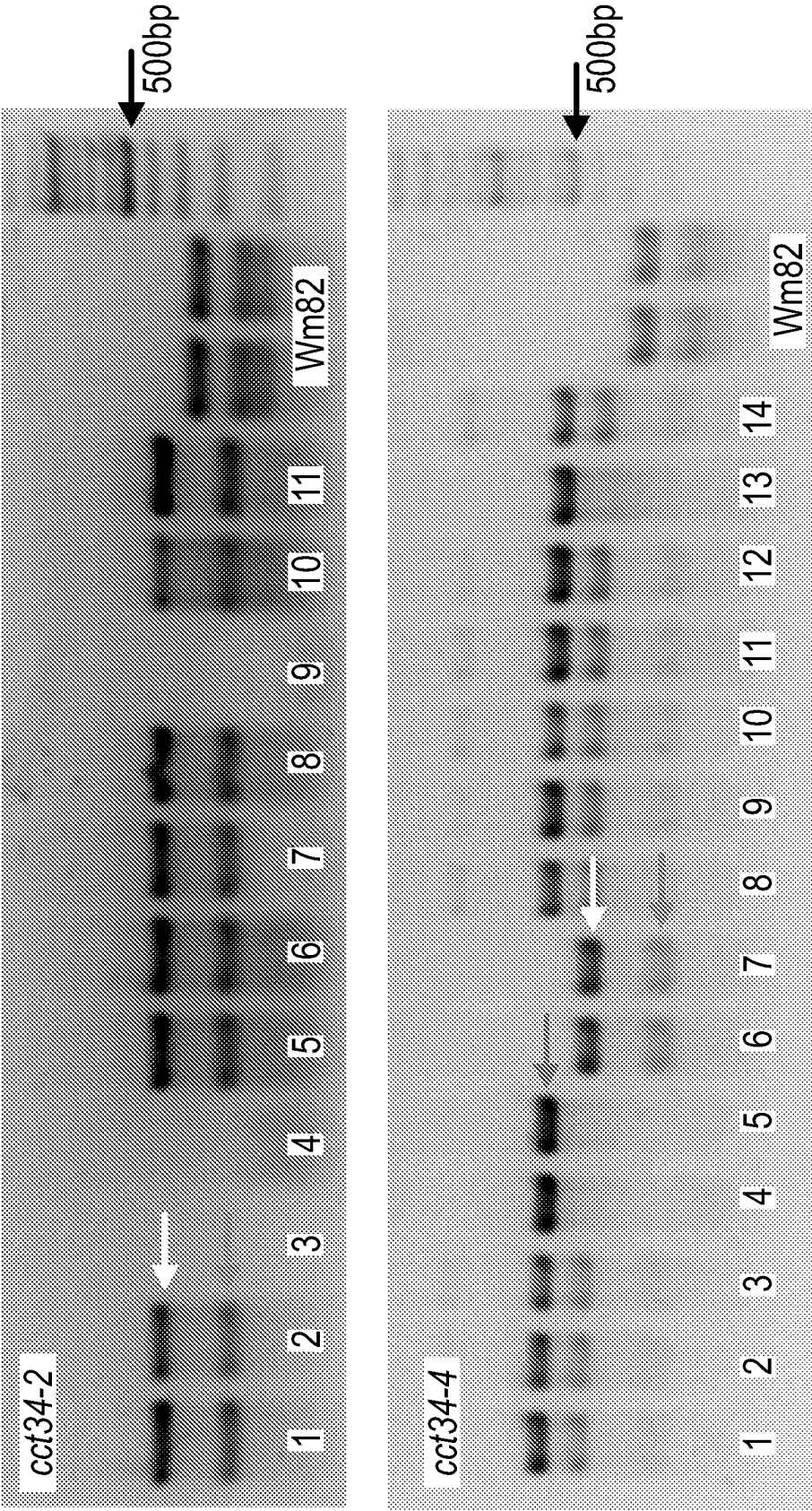


FIG. 9C

	gRNA1	gRNA2	gRNA3
Wm82	ATGTTGCAGGATGTTATCCACCCCTCAACACCCGGGTGAGC	ATGTTGTGATTGCAACCACTTATTGGATCTCCGTTACC	CTCAGTCCCTGGCATAGGAGTGTACATGCCCTCTCTCC
ccf34-2-2	ATGTTGCAGGATGTTATCCACCC-----ACACCGGTGAGC	ATGTTGTGATTGCAACCACTTA-----TCTCCGTTACC	CTCAGTCCCTGGCATAGGAGTGTACATGCCCT--TCC
ccf34-4-5	ATGTTGCAGGATGTTATCCACCC-----CACCGGTGAGC	ATGTTGTGATTGCAACCACTTA-----TCTCCGTTACC	CTCAG-----ATTCT--TCC
ccf34-4-7	ATGTTGCAGGATGTTATCCACCC-TCAACACCGGTGAGC	ATGTTGTGATTGCAACCACTTA-----TCTCCGTTACC	CTCAGTCCCTGGCATAGGAGTGTACATG-----TCC

FIG. 9D

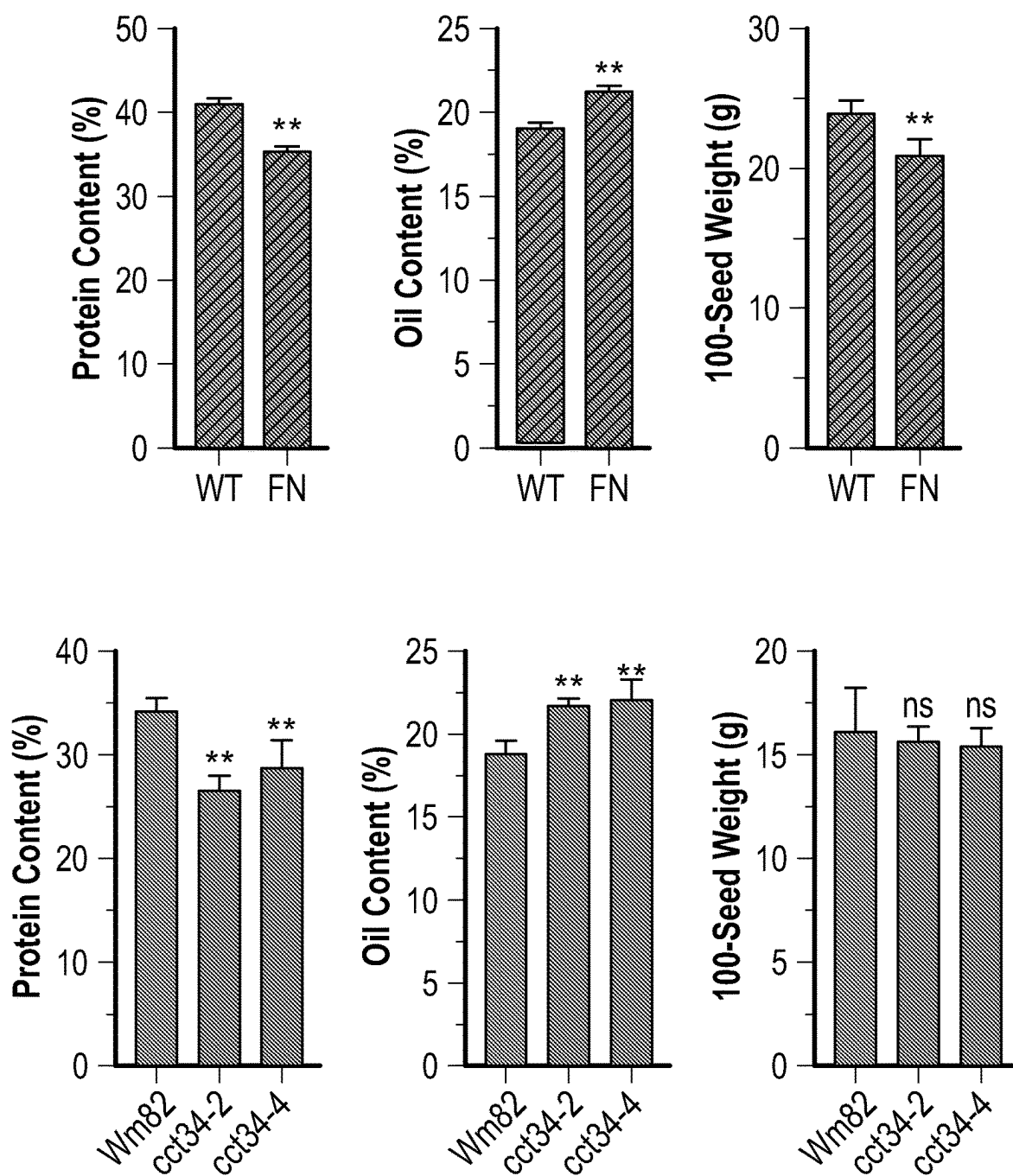


FIG. 9E

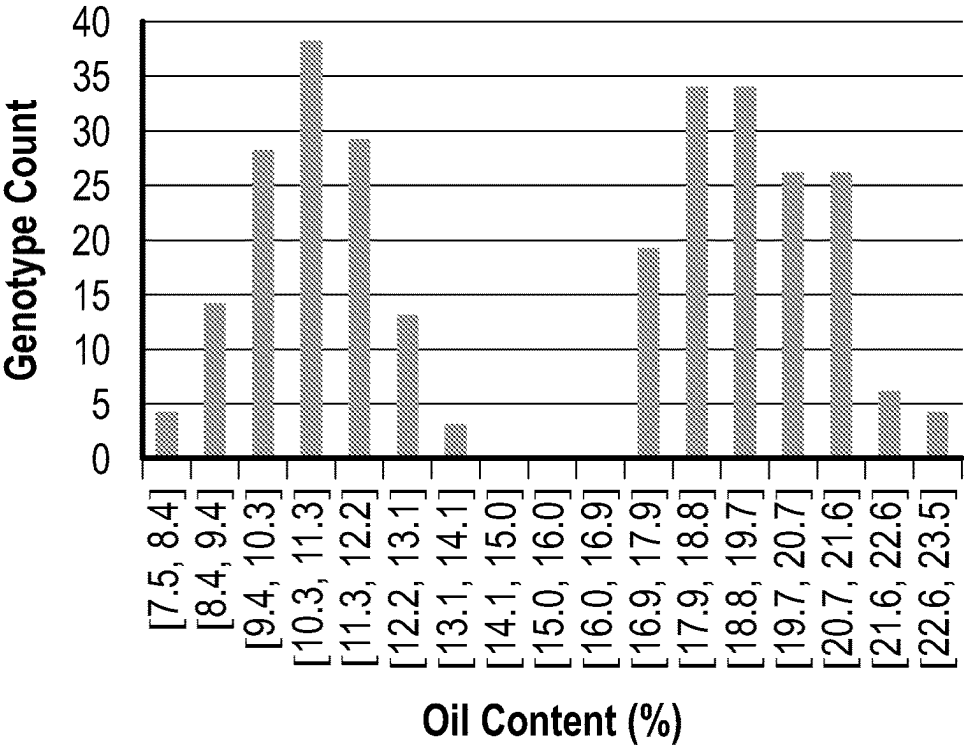


FIG. 10A

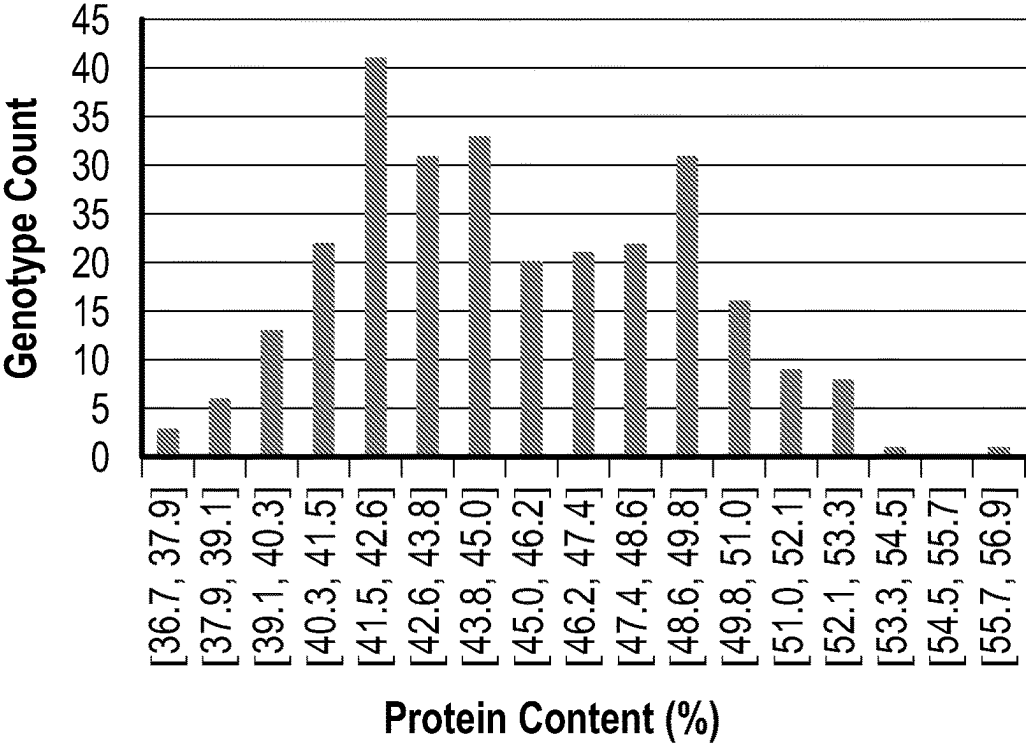


FIG. 10B

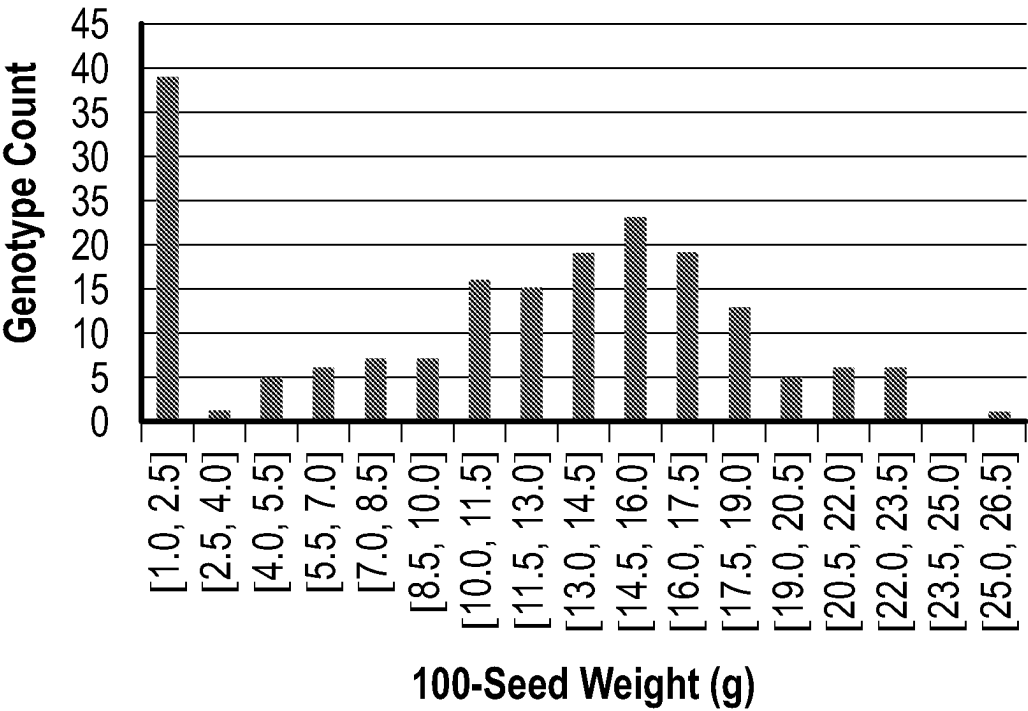
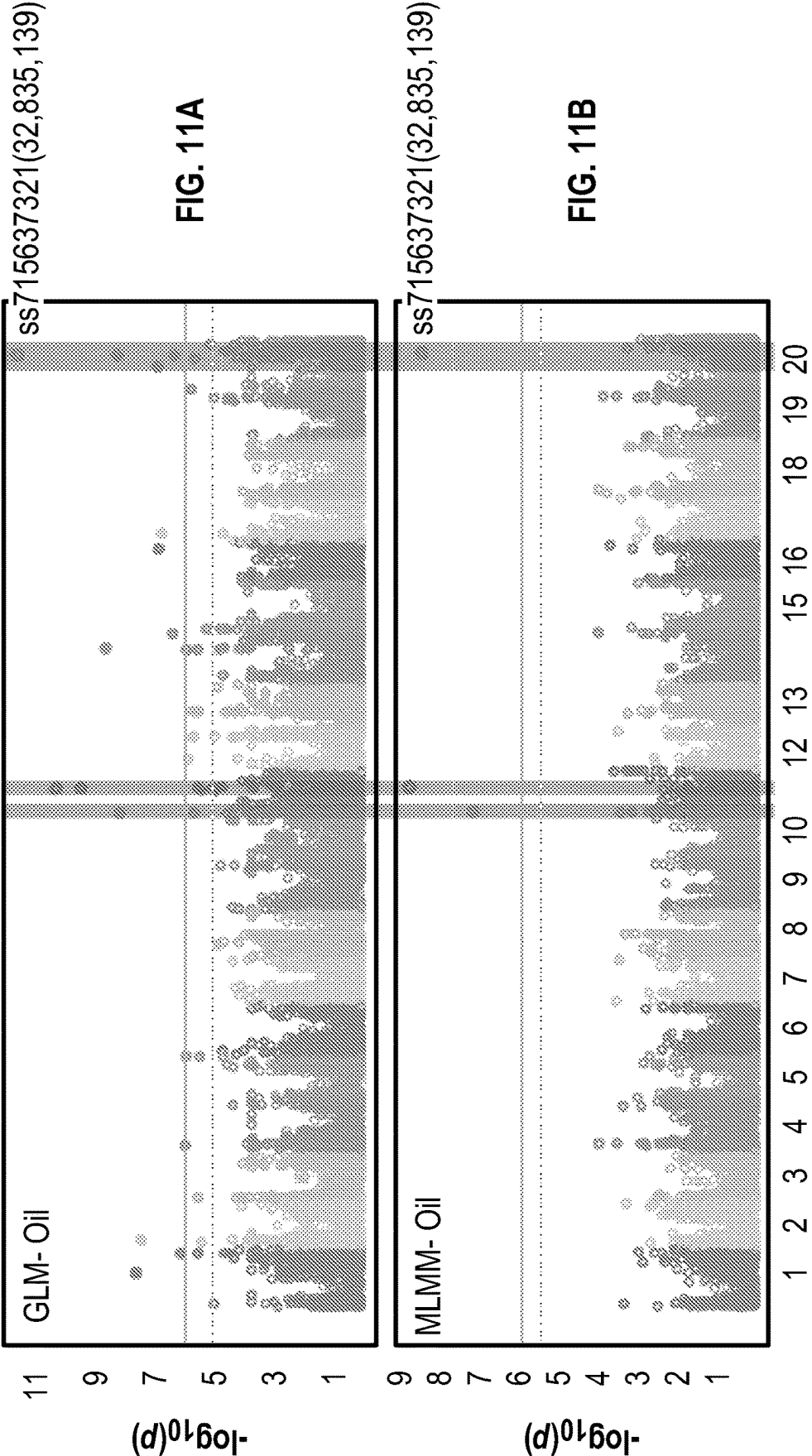
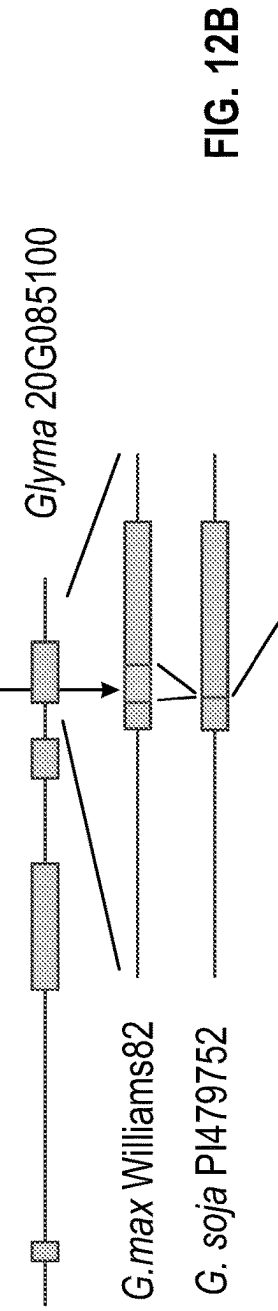
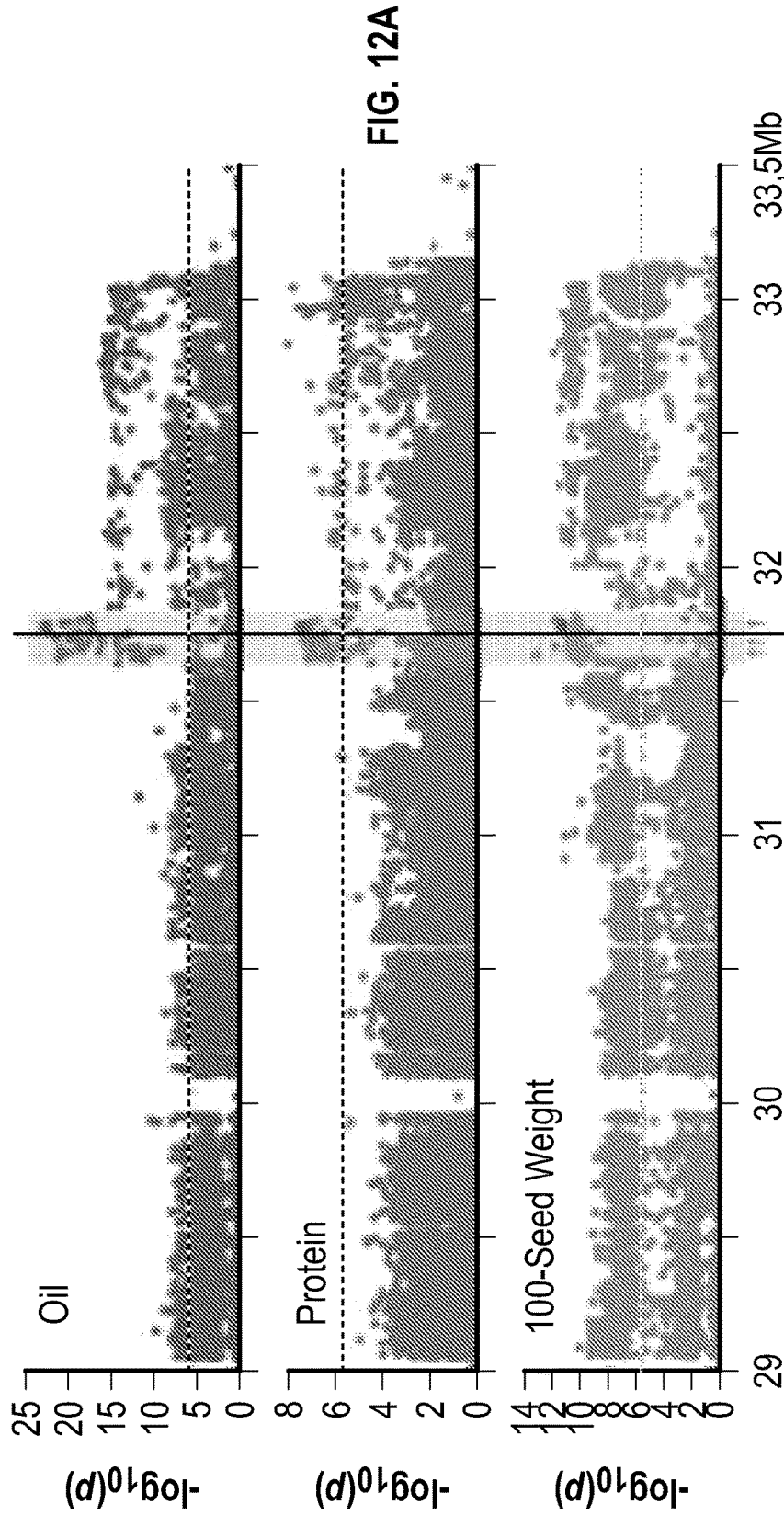


FIG. 10C





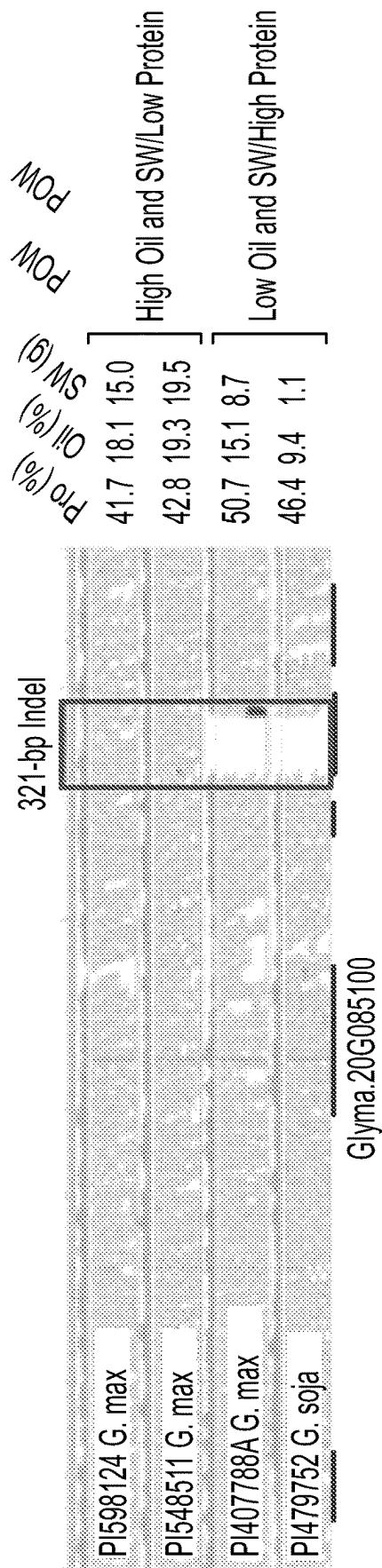


FIG. 12C

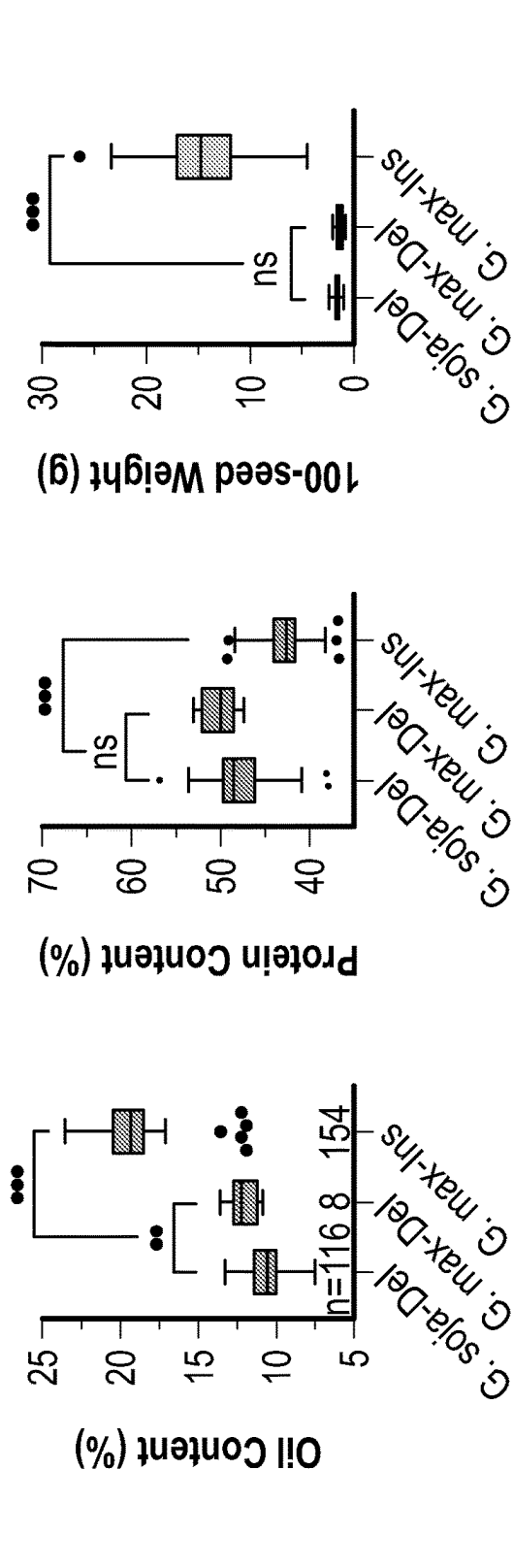


FIG. 12D

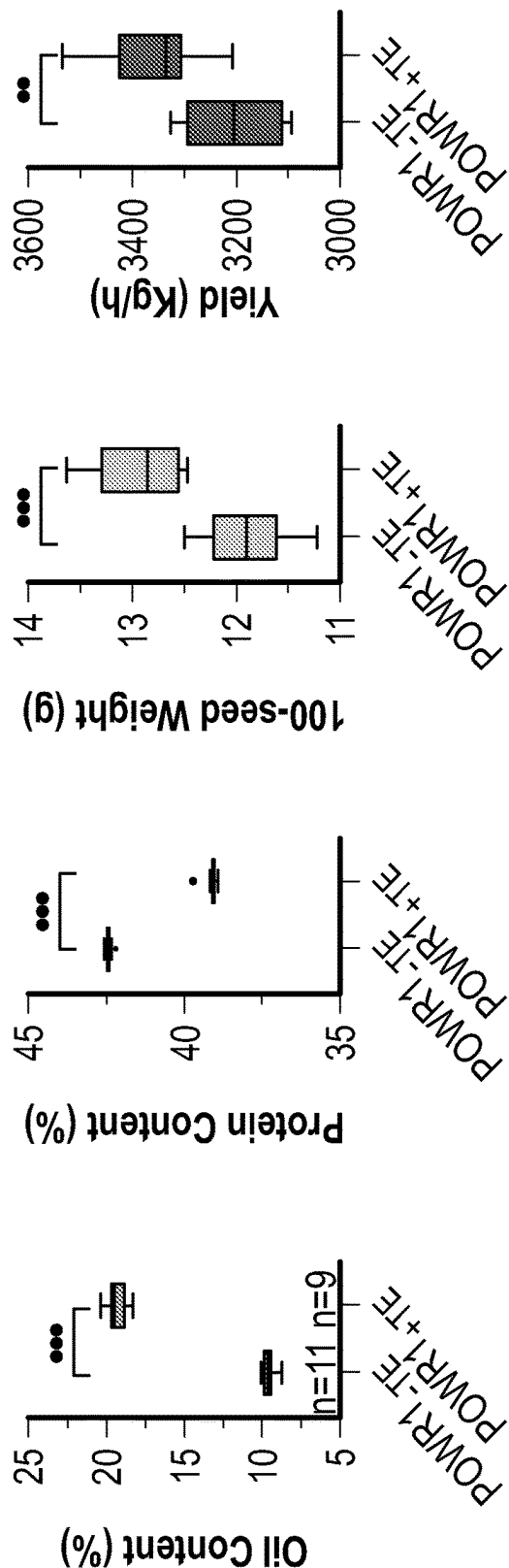


FIG. 12E

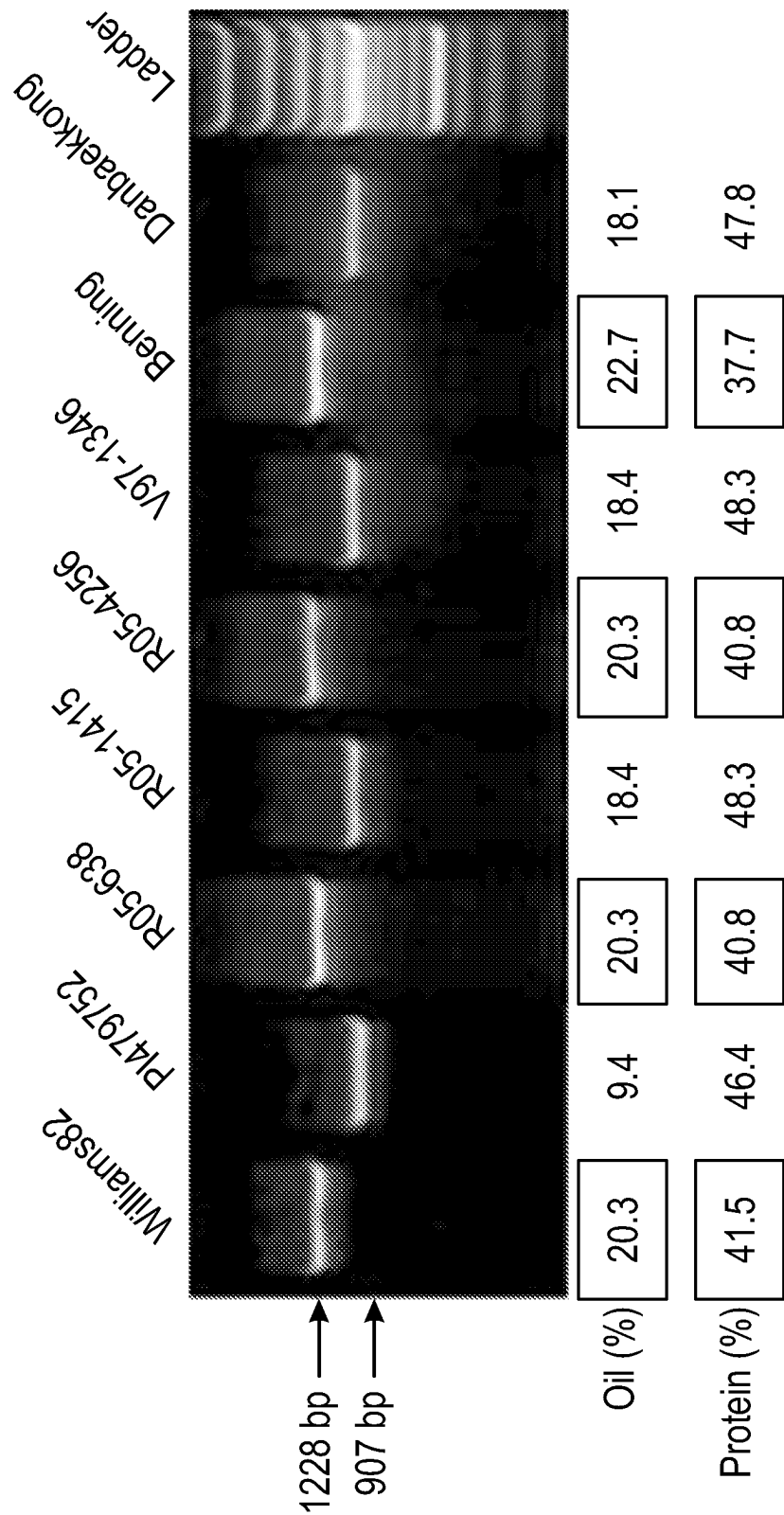


FIG. 12F

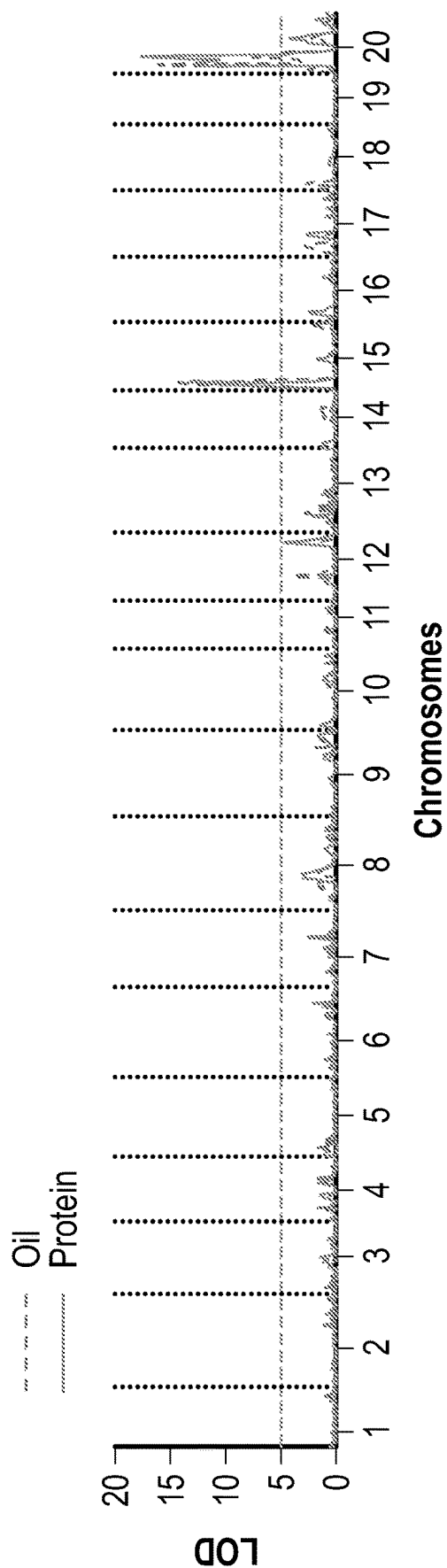


FIG. 13A

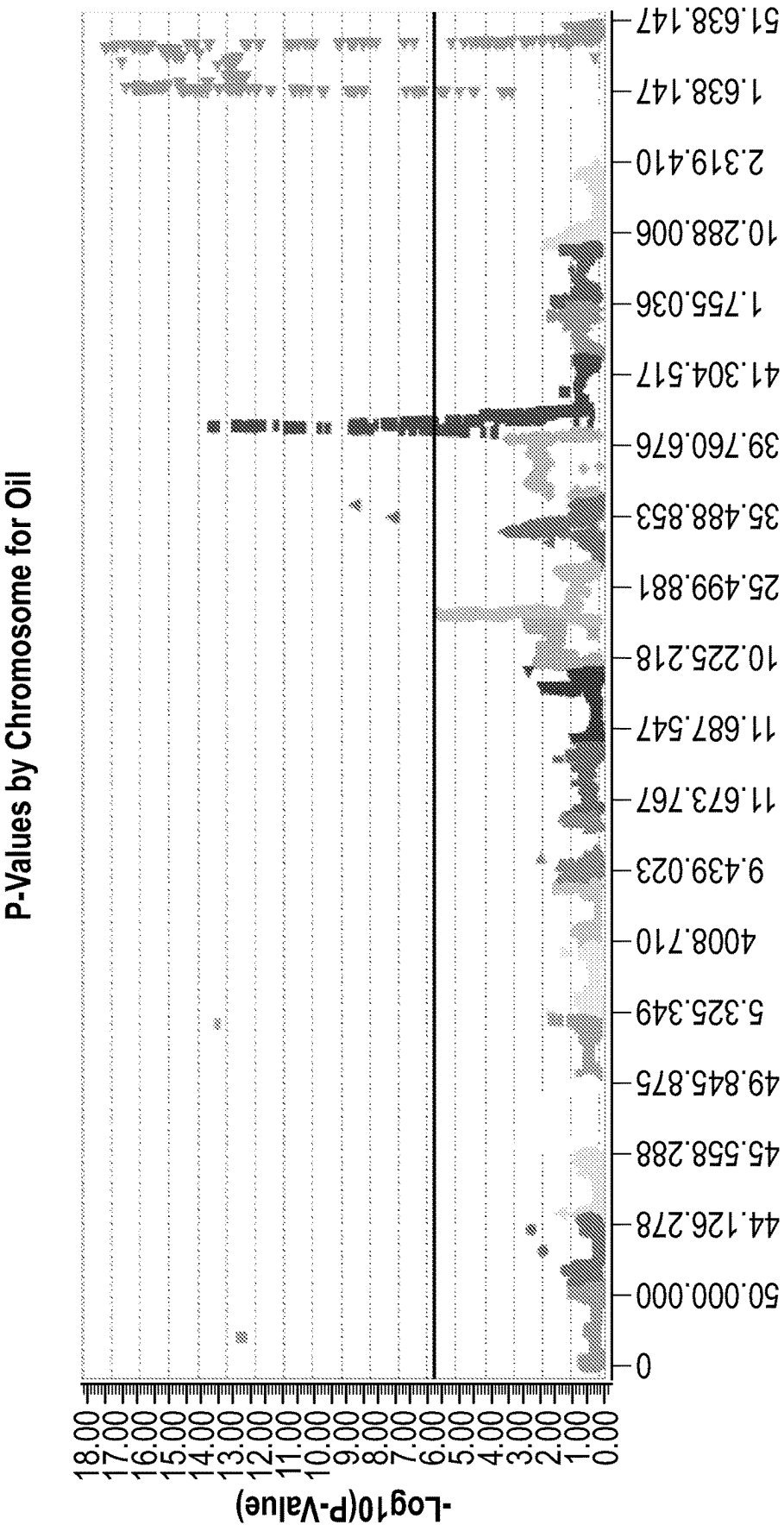


FIG. 13B-1

		⌘ GM01 ⌘ GM02 ⌘ GM03 GM04 ⌘ GM05 ⌘ GM06 ⌘ GM07 ⌘ GM08 ⌘ GM09 ⌘ GM10 ⌘ GM11 ⌘ GM12 ⌘ GM13 ⌘ GM14 ⌘ GM15 ⌘ GM16 ⌘ GM17 ⌘ GM18 GM19 ⌘ GM20				
Population	Trait	Soy SN P50K Marker	Chr	Position [bp]	p-value	perm p-value
RJLs	Oil	ss715637271	20	31,728,036	1.19E-17	0.001
		ss715637273	20	31,776,855	1.19E-17	0.001
		ss715637274	20	31,810,008	1.19E-17	0.001
		ss715637268	20	31,702,354	1.69E-17	0.001
		ss715637277	20	31,904,512	1.81E-17	0.001
		ss715637281	20	32,001,829	3.65E-17	0.001
		ss715637294	20	32,282,623	3.65E-17	0.001
		ss715637309	20	32,576,952	3.65E-17	0.001
		ss715637334	20	33,245,025	4.44E-17	0.001
		ss715637283	20	21,083,902	5.05E-17	0.001

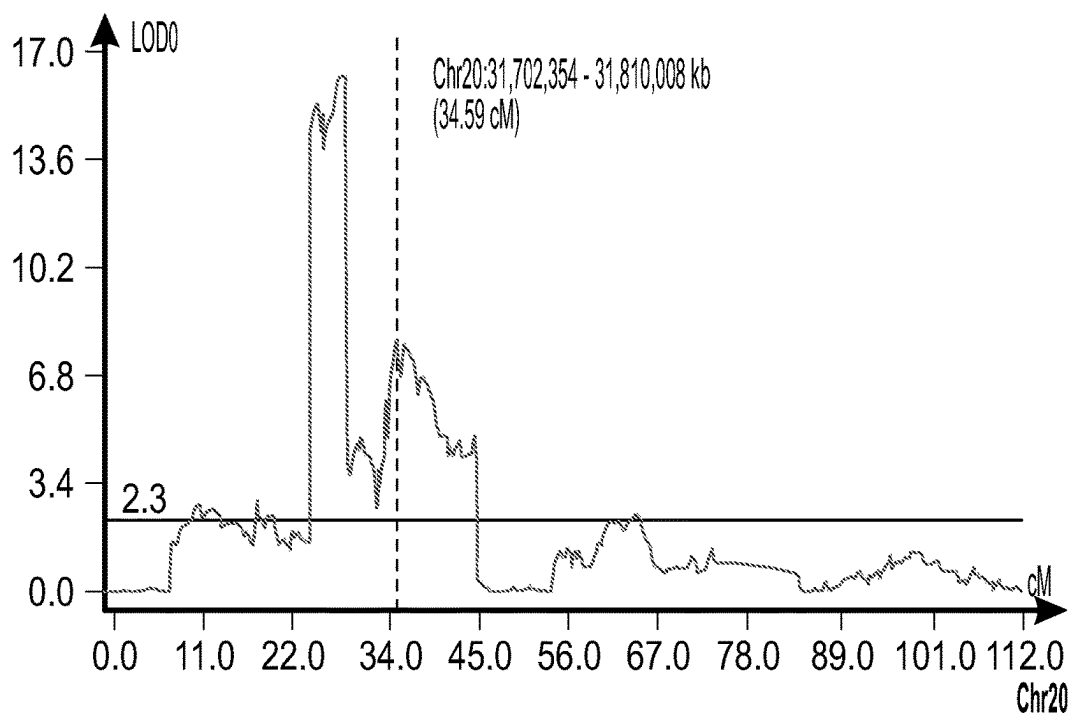


FIG. 13B-2

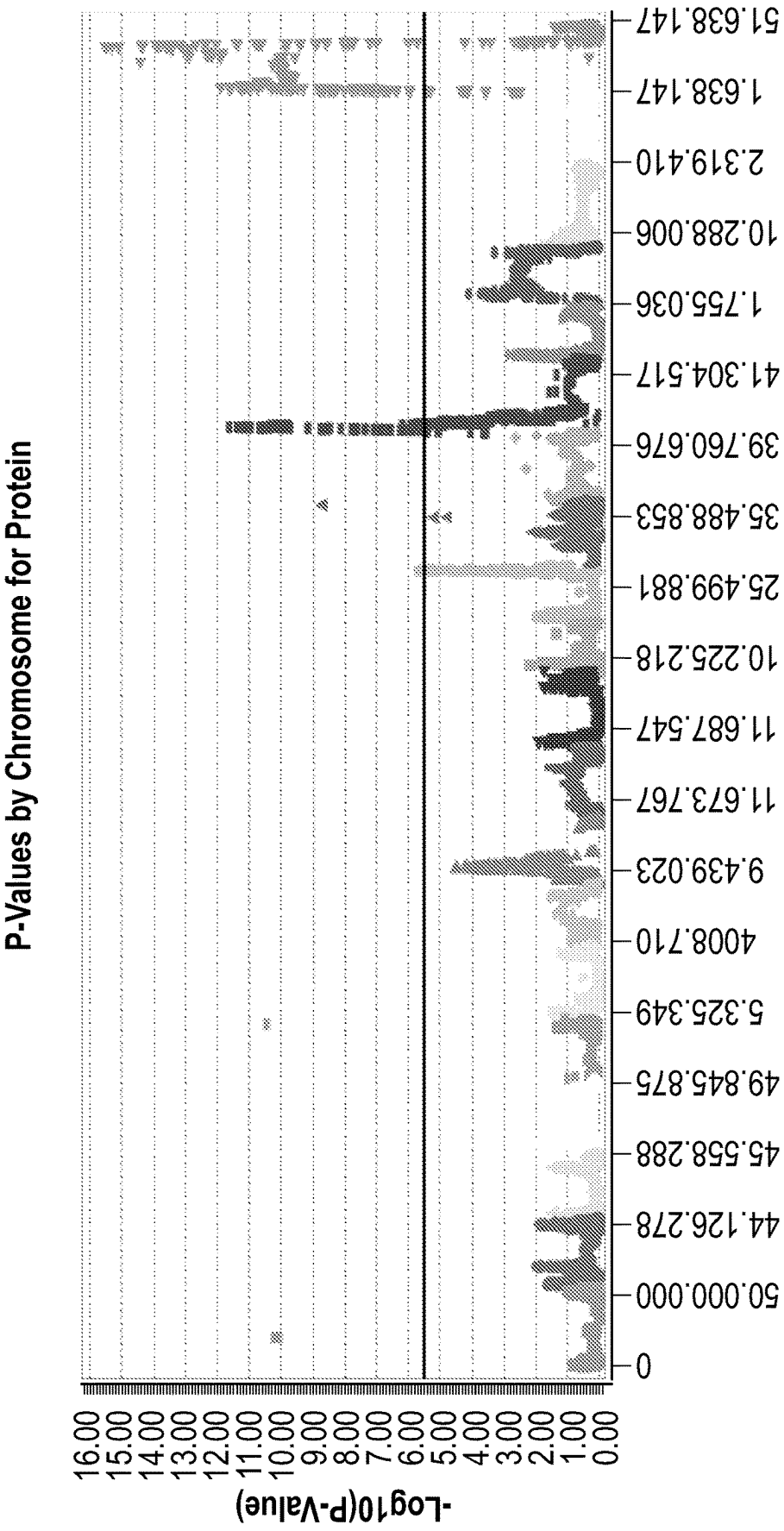


FIG. 13C-1

⌘ GM01 ⌘ GM02 ⌘ GM03 GM04 ⌘ GM05 ⌘ GM06 ⌘ GM07 ⌘ GM08 ⌘ GM09 ⌘ GM10 ⌘ GM11 ⌘ GM12 ⌘ GM13 ⌘ GM14 ⌘ GM15 ⌘ GM16 ⌘ GM17 ⌘ GM18 GM19 ⌘ GM20						
Population	Trait	Soy SN P50K Marker	Chr	Position [bp]	p-value	perm p-value
RJLs	Protein	ss715637271	20	31,728,036	8.15E-17	0.001
		ss715637273	20	31,776,855	8.15E-17	0.001
		ss715637274	20	31,810,008	8.15E-17	0.001
		ss715637268	20	31,702,354	1.08E-17	0.001
		ss715637277	20	31,904,512	2.0E-17	0.001
		ss715637279	20	31,974,972	7.57E-15	0.001
		ss715637283	20	32,282,623	1.05E-14	0.001
		ss715637281	20	32,576,952	1.15E-14	0.001
		ss715637294	20	33,245,025	1.15E-14	0.001
		ss715637309	20	21,083,902	1.15E-14	0.001

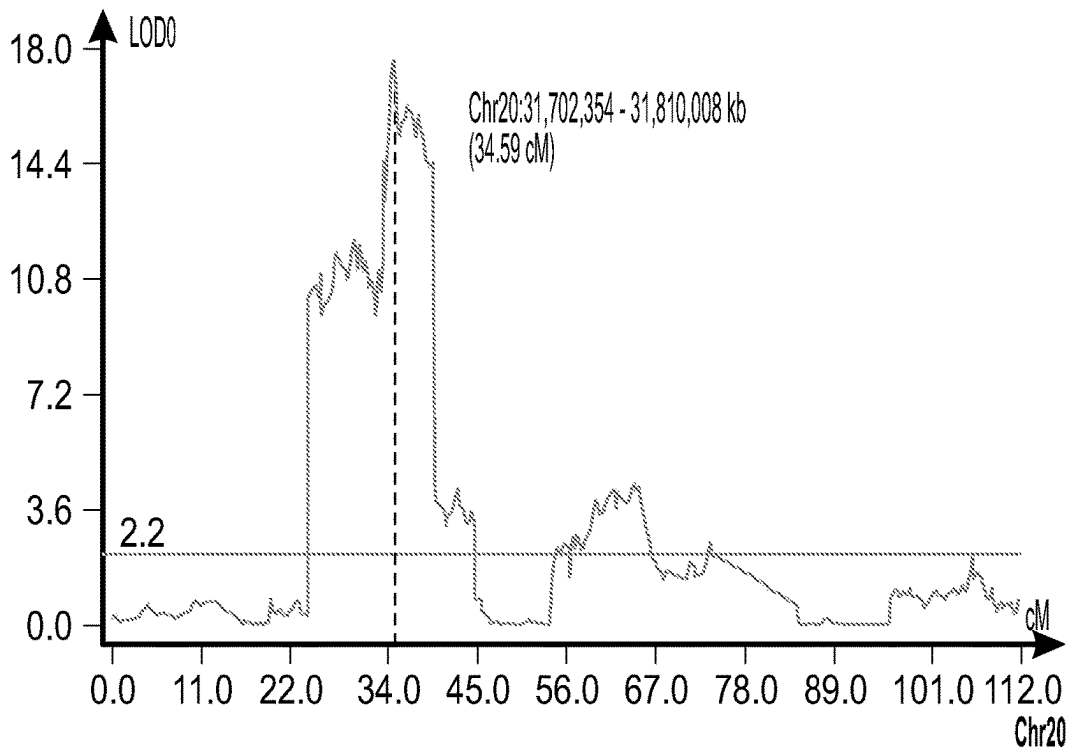


FIG. 13C-2

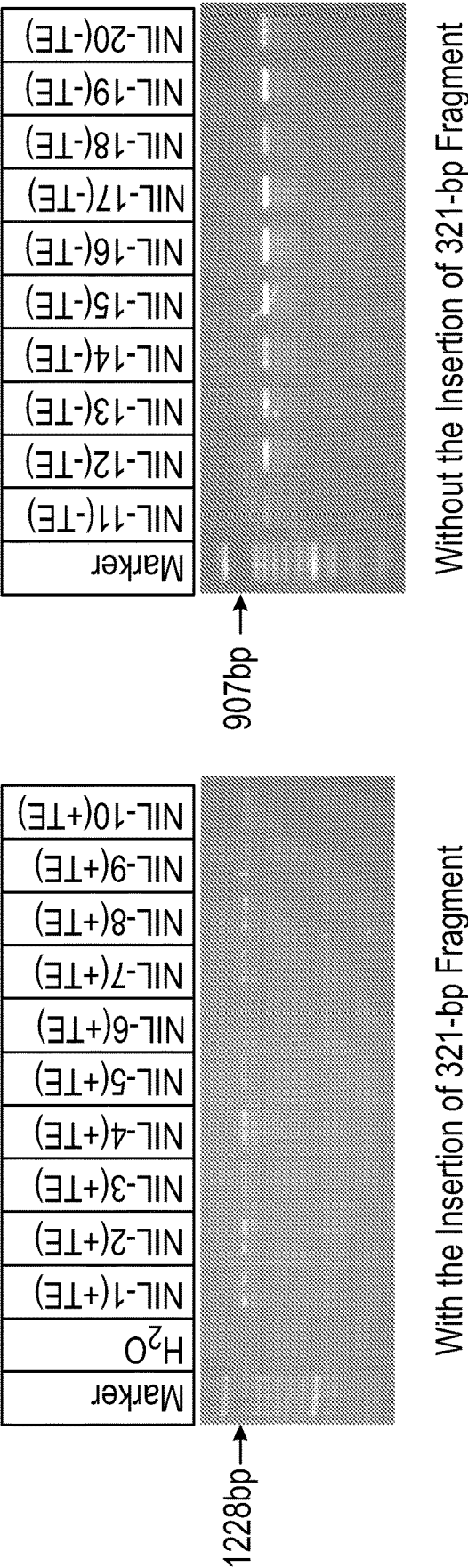


FIG. 14

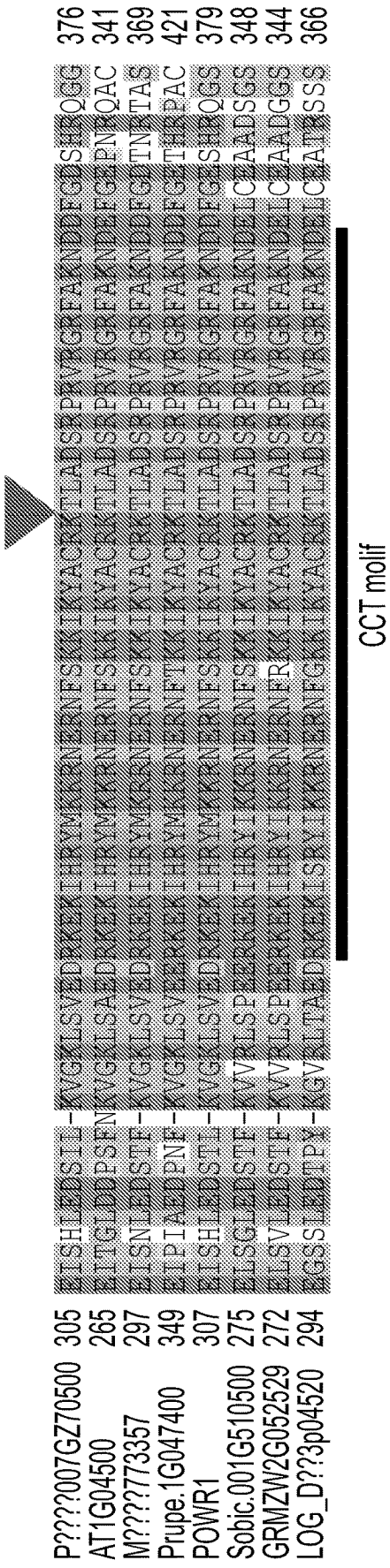


FIG. 15A

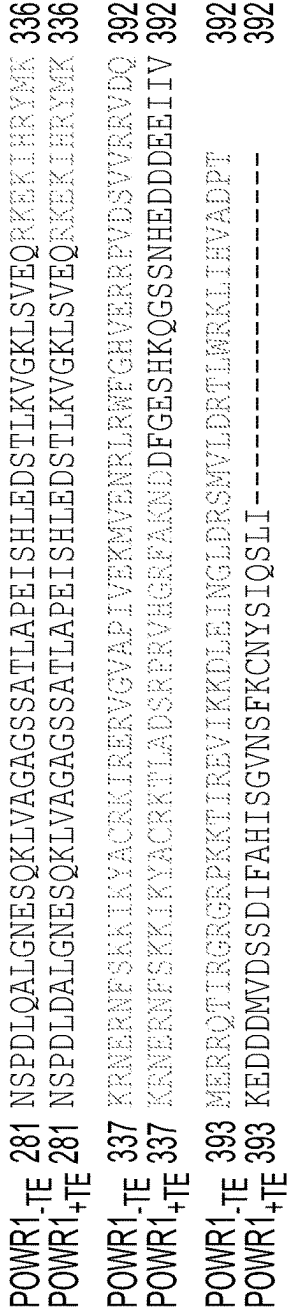


FIG. 15B

17-bp TSD: GTATGGTTGCCGCAAAA + 304-bp TE → 321-bp Insertion

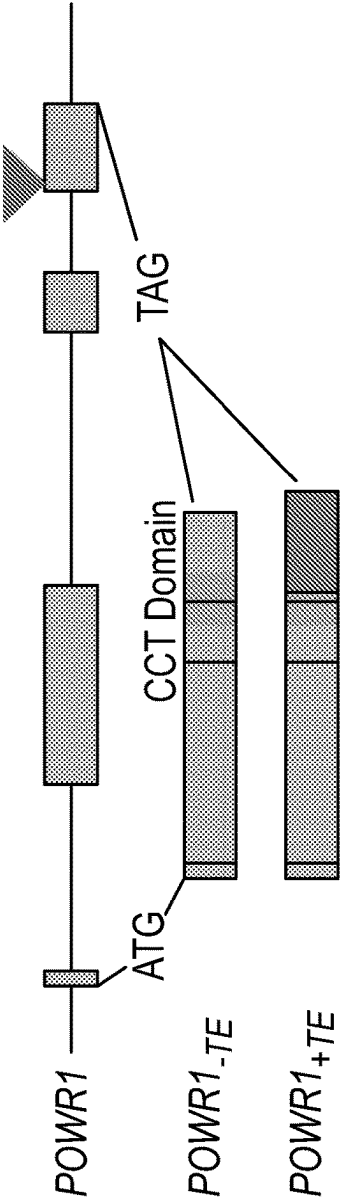


FIG. 15C

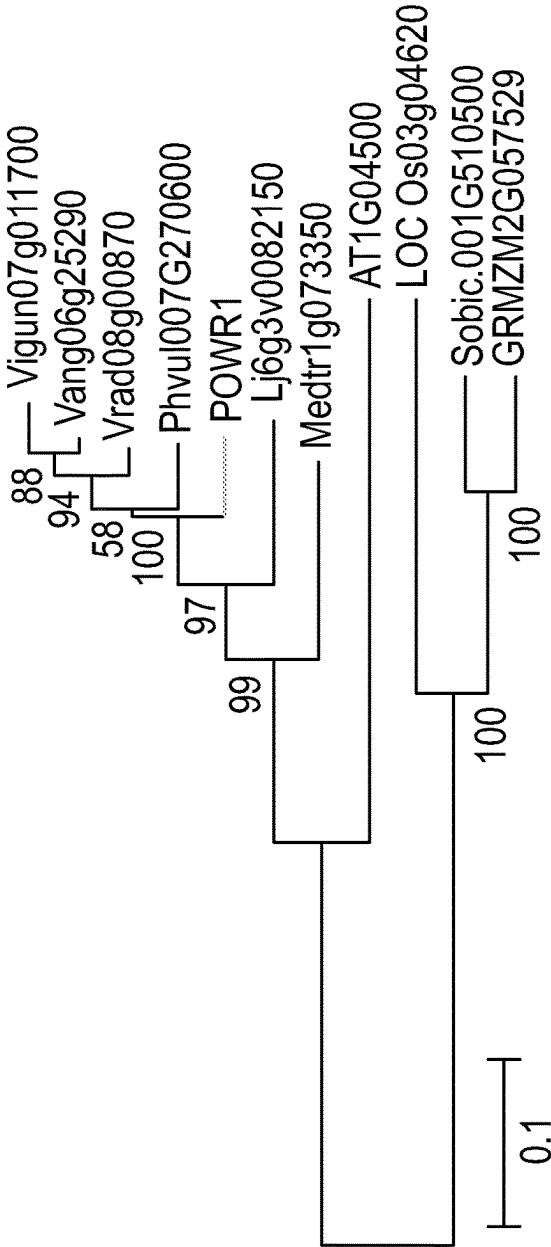


FIG. 15D

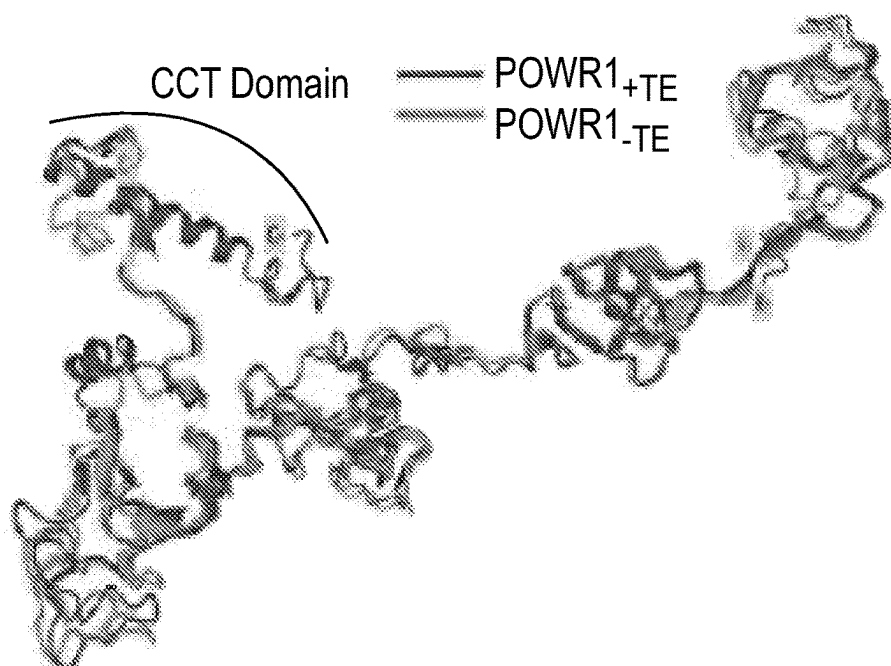


FIG. 15E

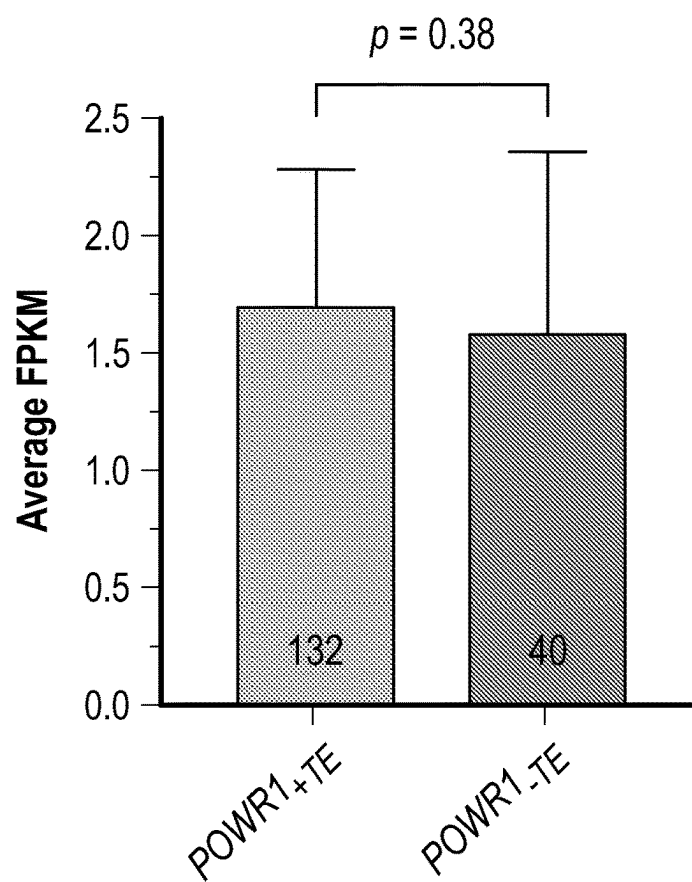


FIG. 15F

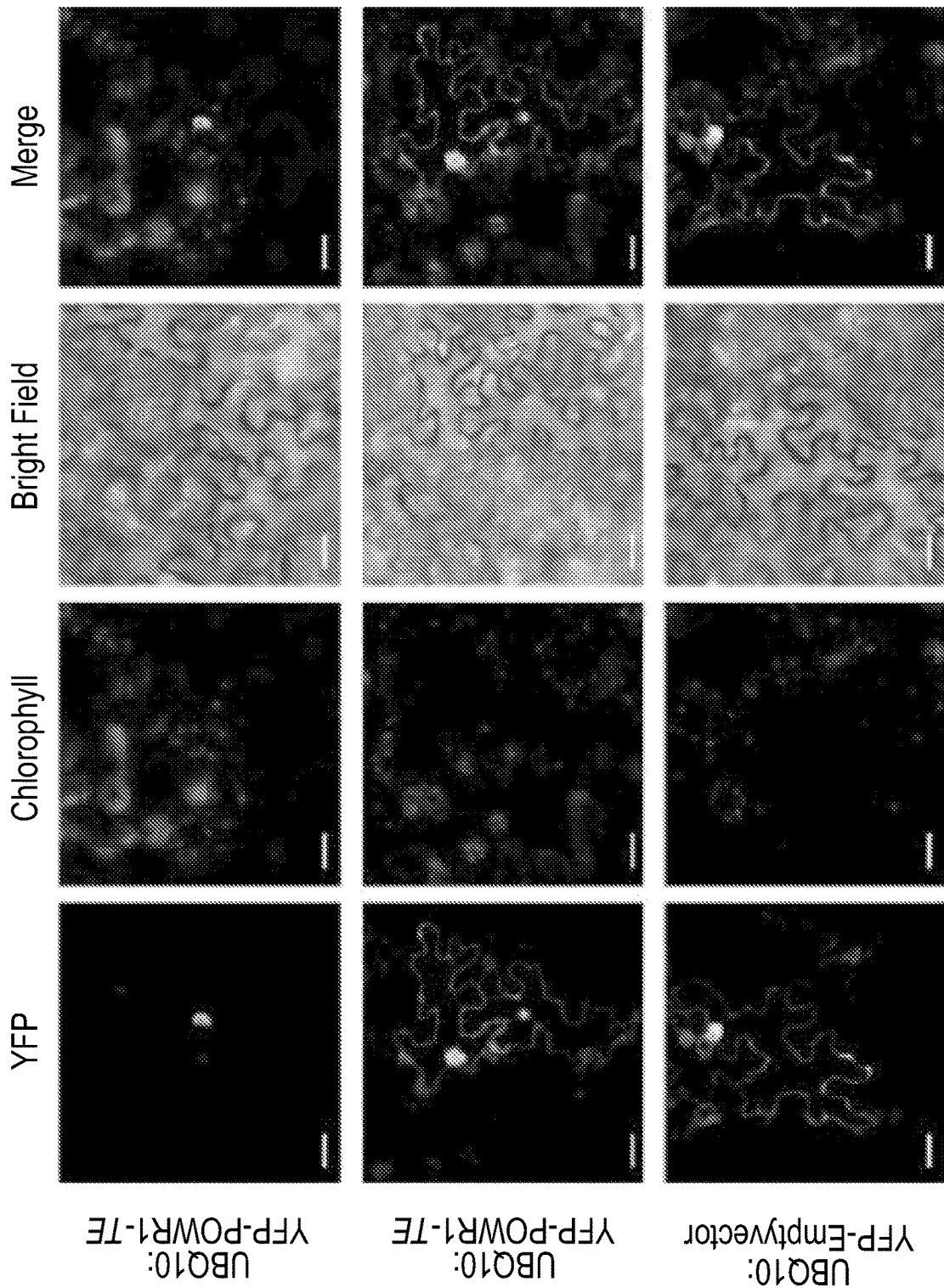


FIG. 15G

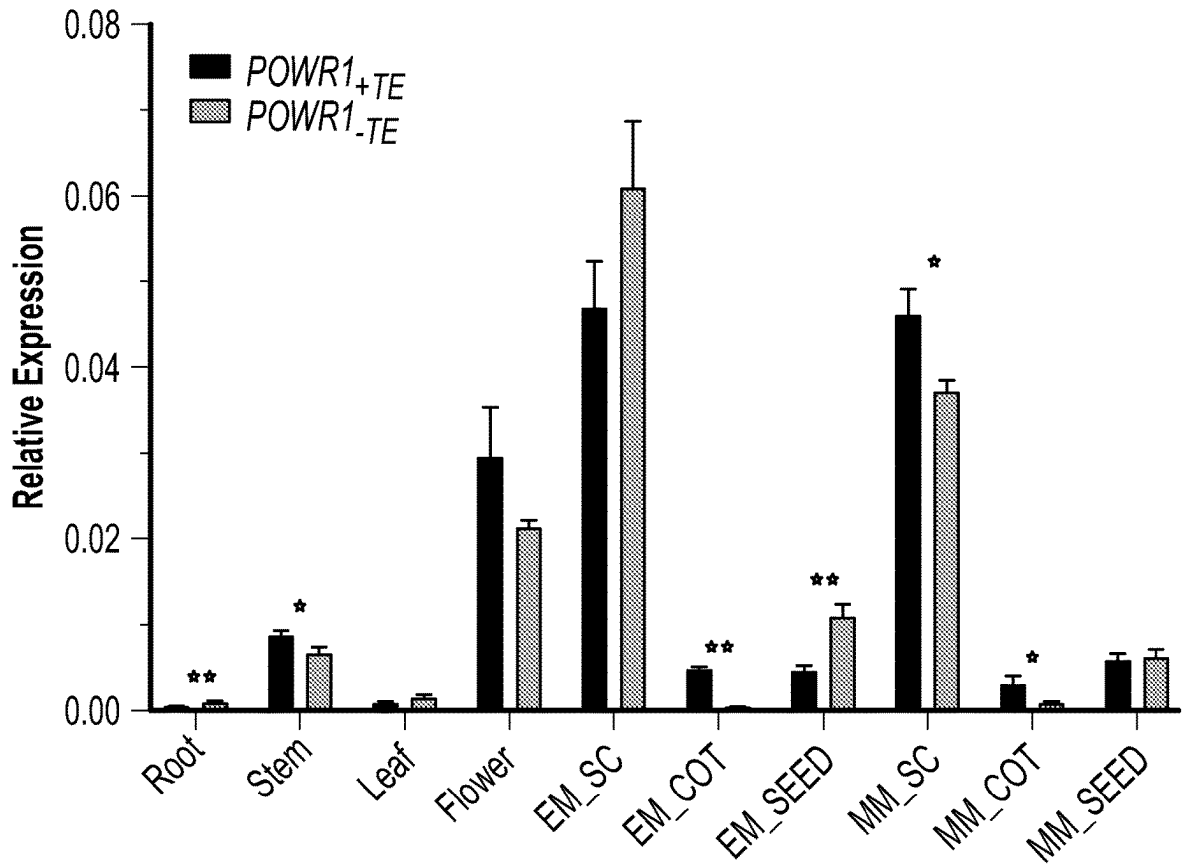


FIG. 15H

	Term	#	Corrected <i>P</i>
KEGG	Linoleic acid metabolism (gmx00591)	8	2.01E-04
	Fatty acid degradation (gmx00071)	10	4.46E-04
	Starch and sucrose metabolism (gmx00500)	22	1.41E-03
	Glycerolipid metabolism (gmx00561)	10	3.84E-03
BP	Transmembrane transport (GD;0055085)	75	7.67E-07
	Carbohydrate metabolic process (GD;0005975)	56	9.02E-05
	Regulation of transcription (GD;0006355)	81	2.87E-04
MF	DNA-binding transcription factor activity (GD;0003700)	44	1.13E-04
	transporter activity (GD;0005215)	52	2.53E-02
CC	apoplats (GD;0048046)	13	3.37E-02

FIG. 15I

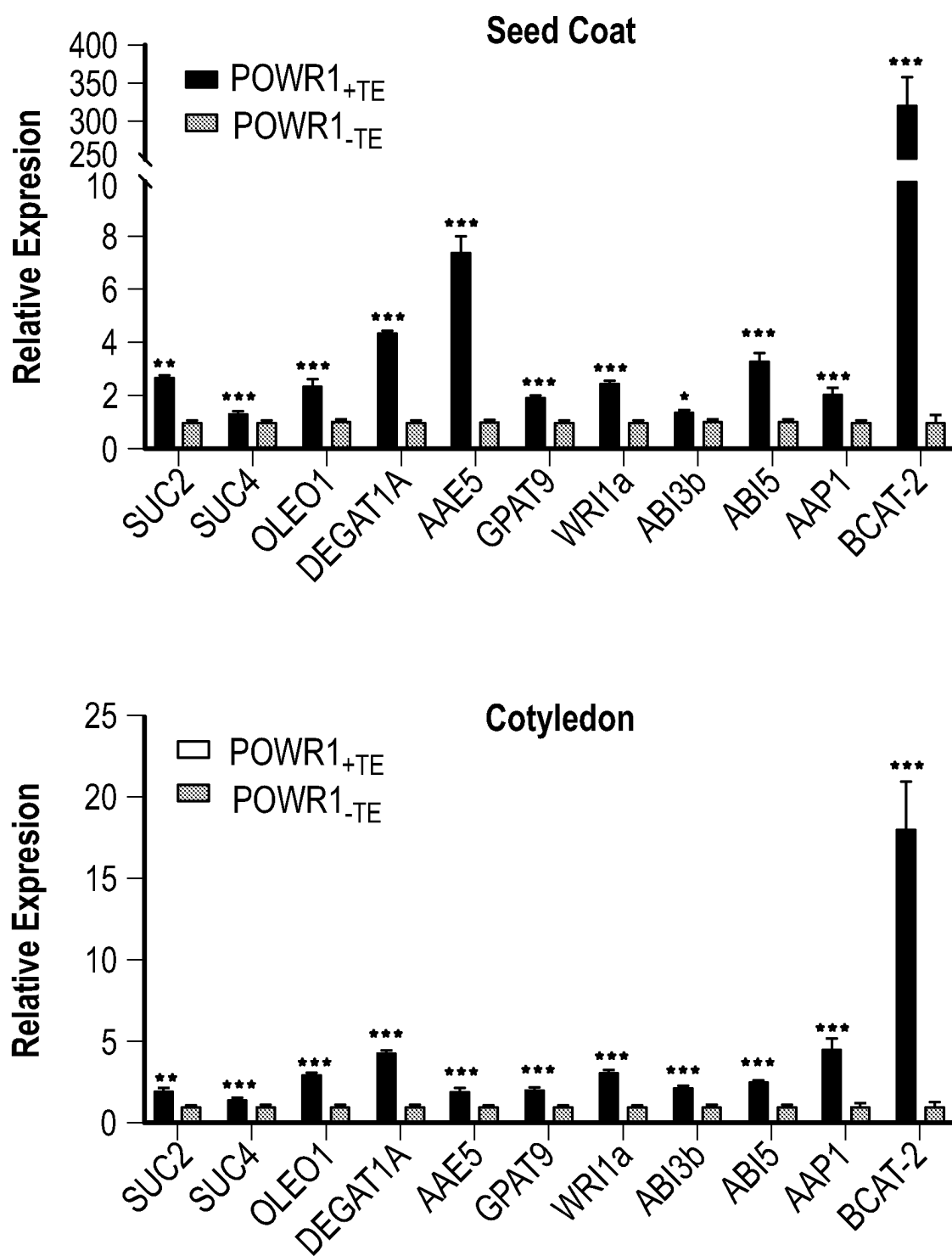
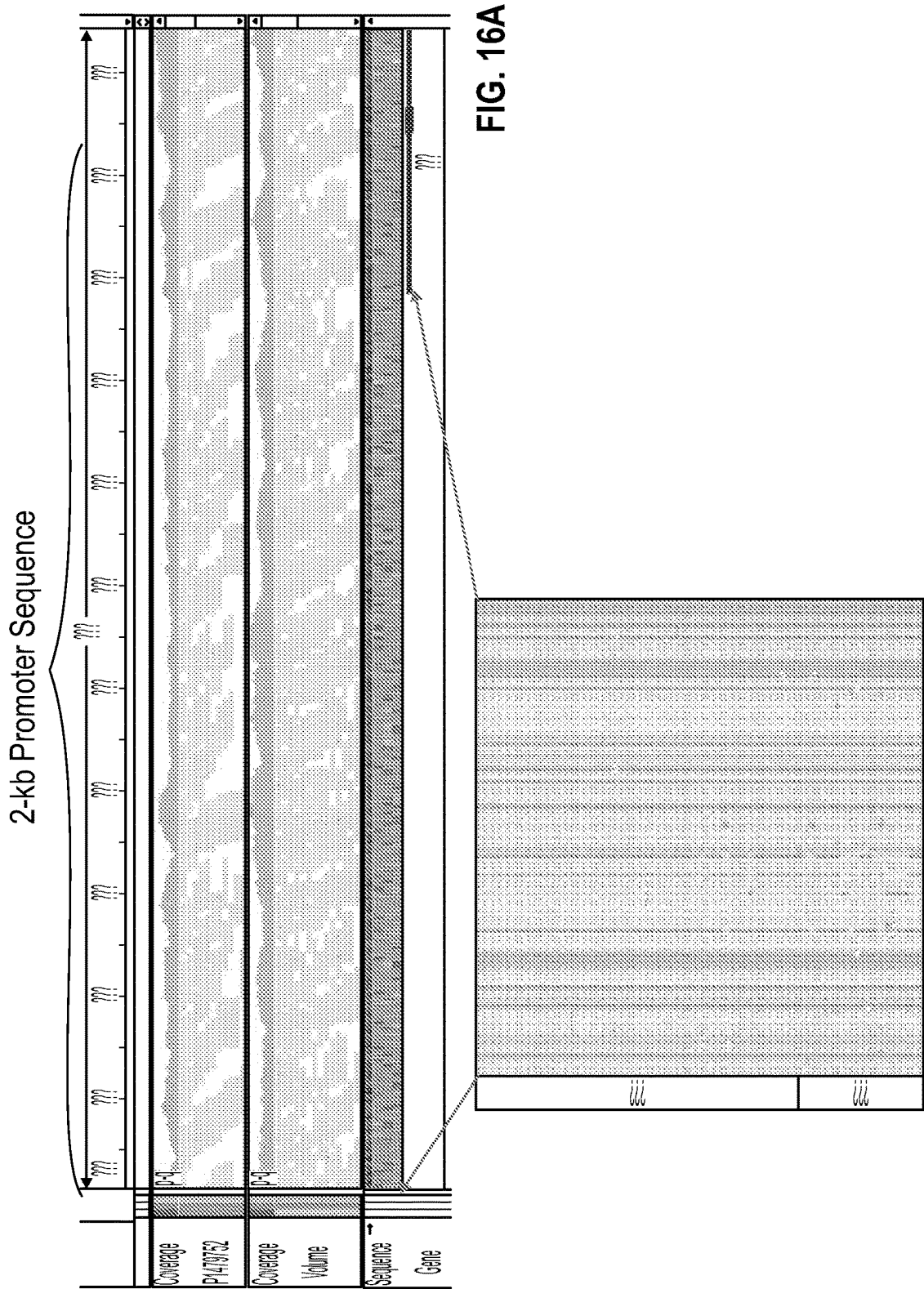


FIG. 15J



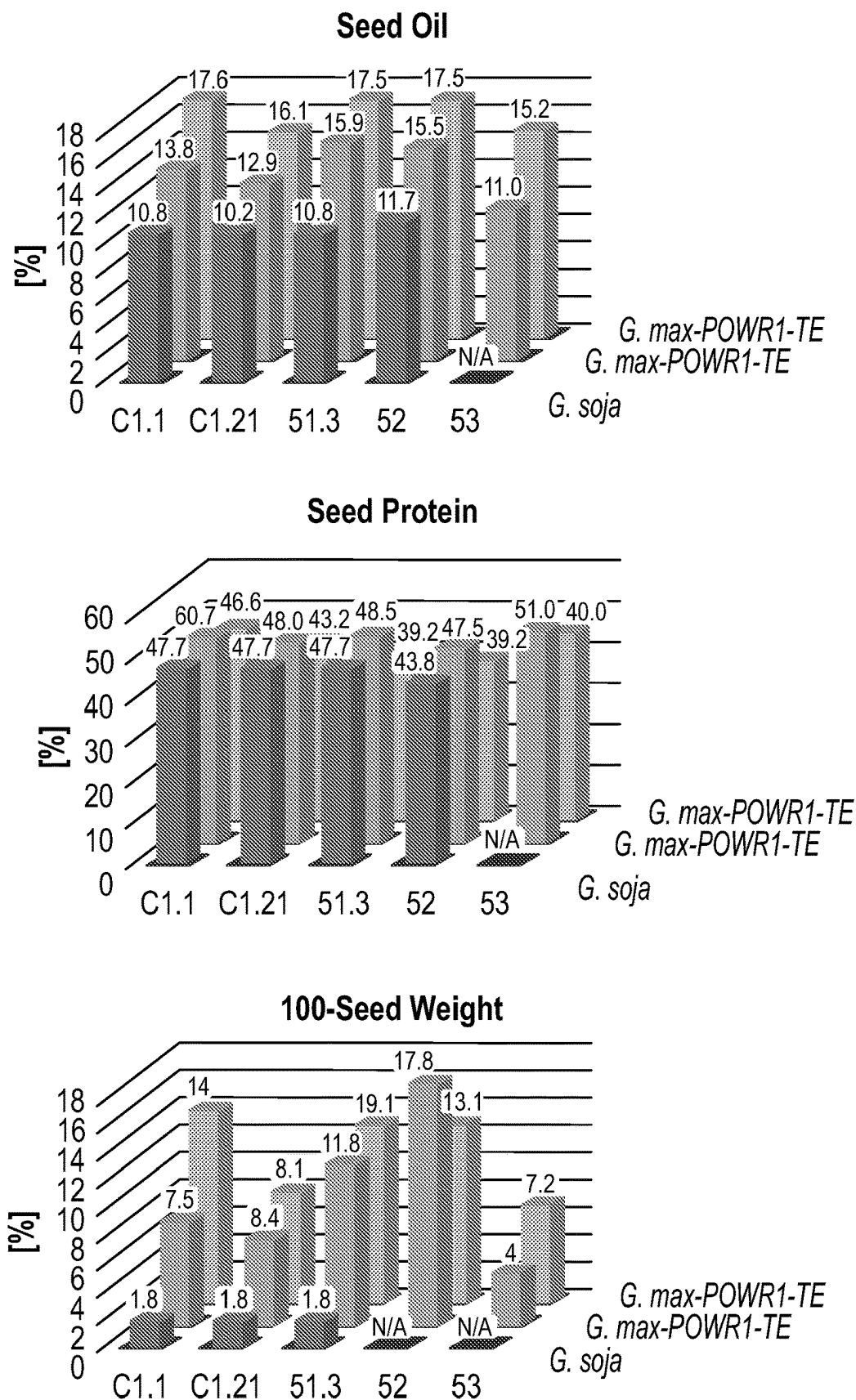


FIG. 17



FIG. 18A

Control

Transgenic plants

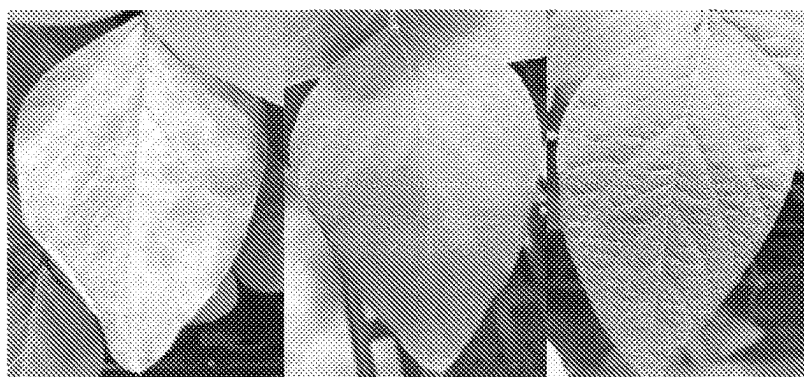


FIG. 18B

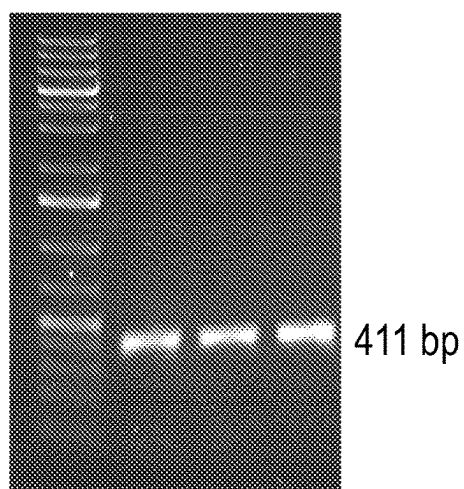


FIG. 18C

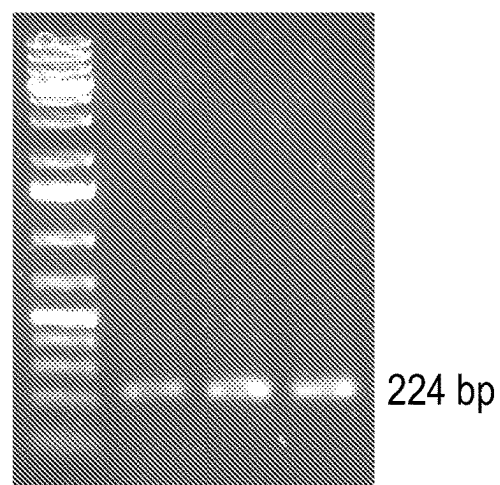


FIG. 18D

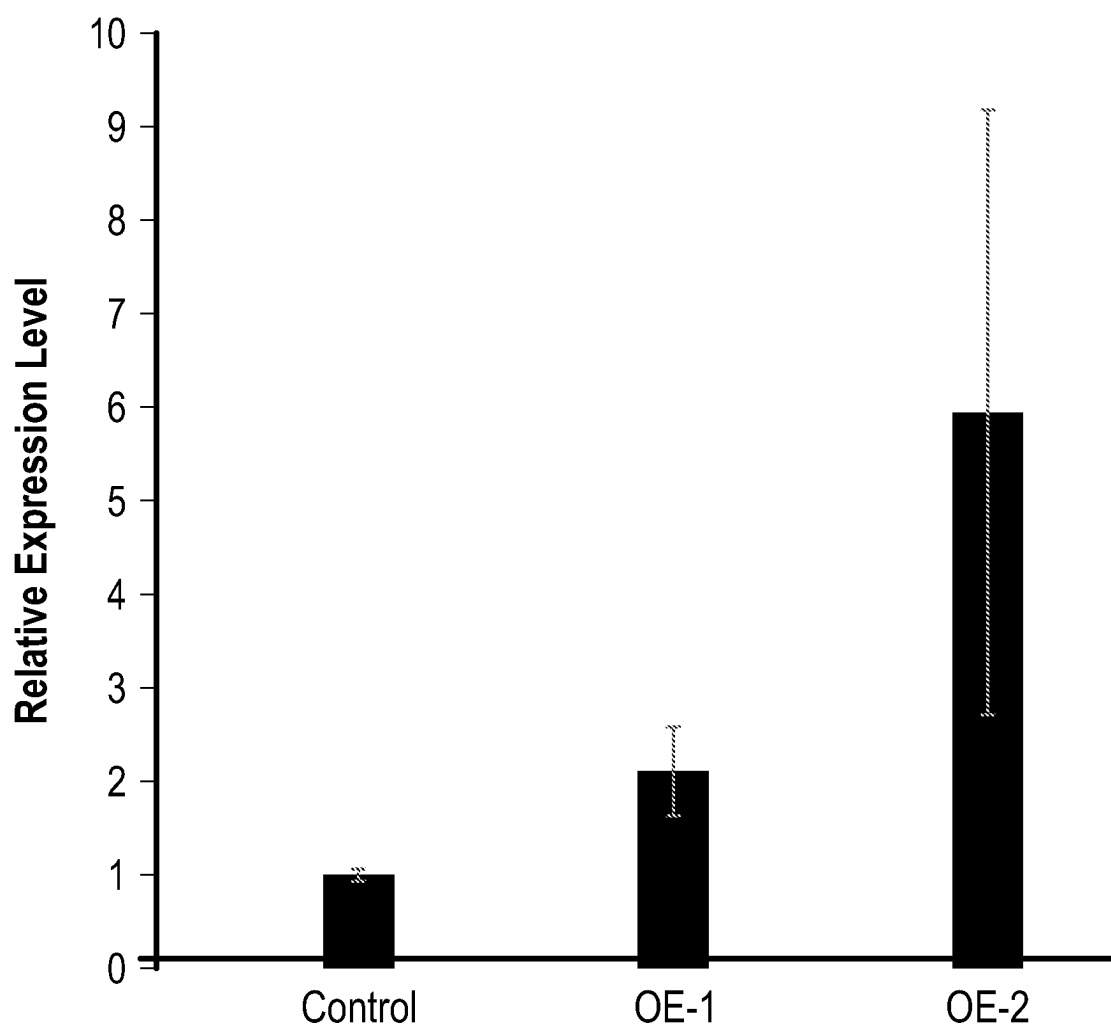


FIG. 18E

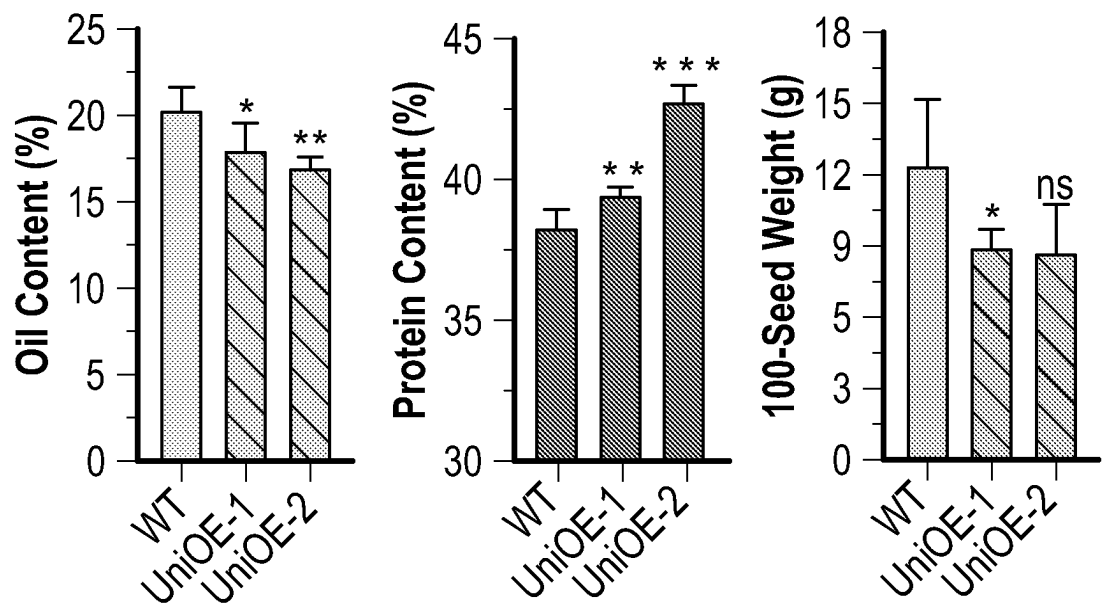


FIG. 19A

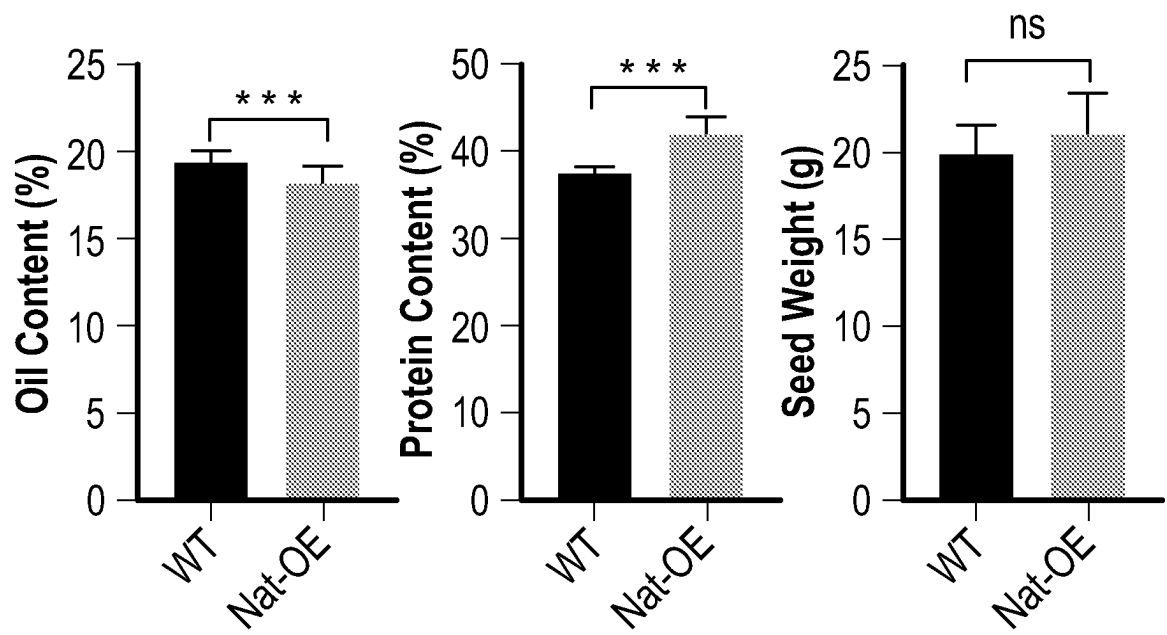


FIG. 19B

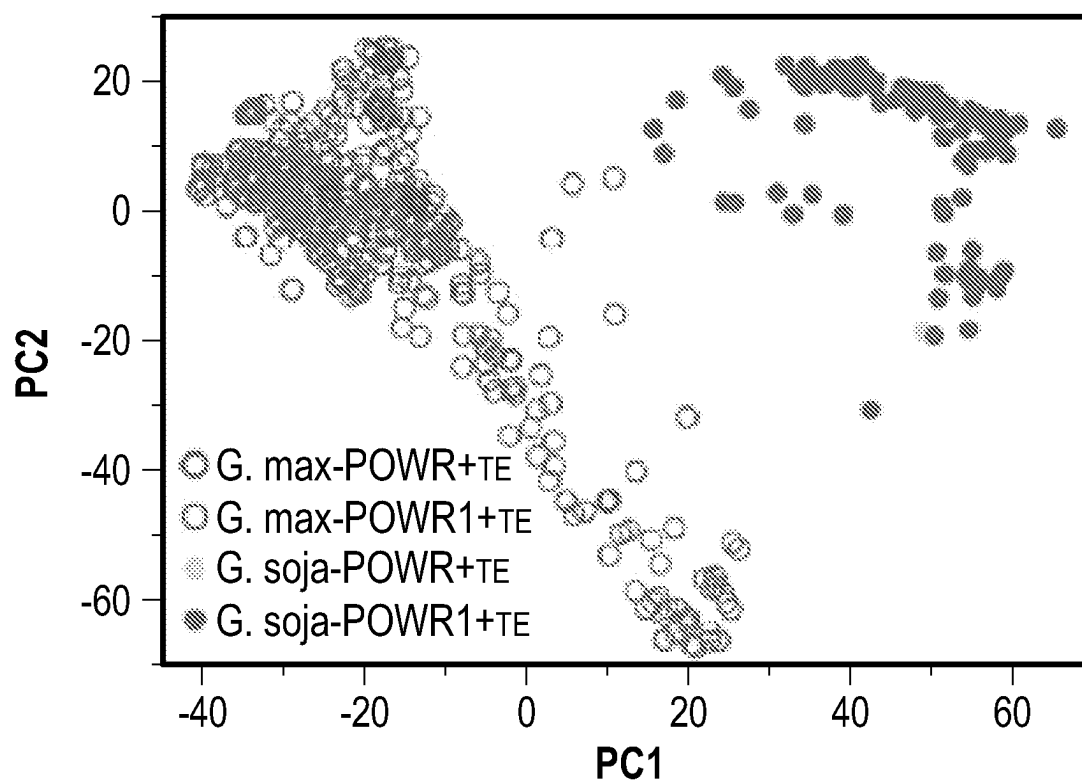


FIG. 20A

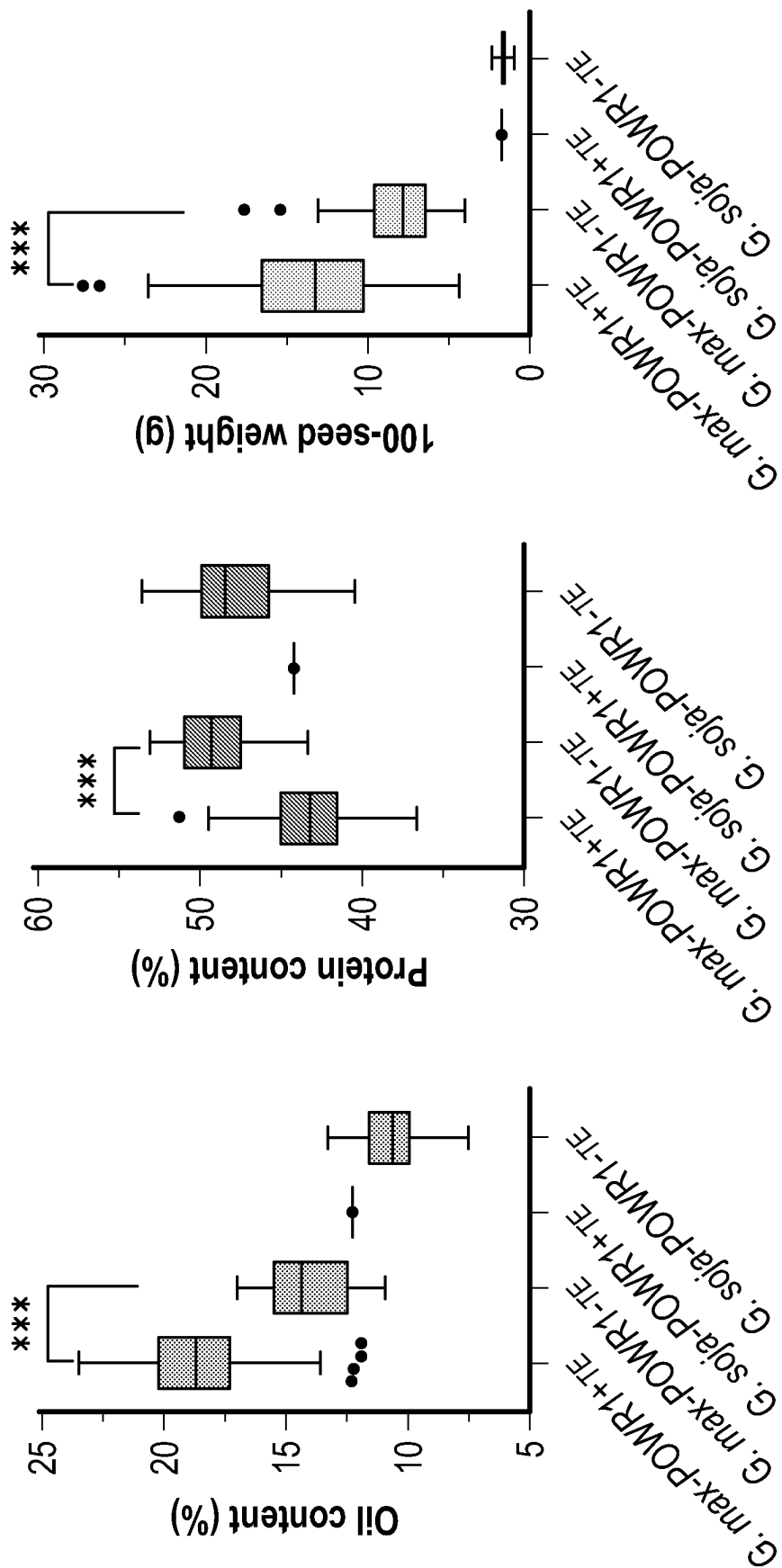
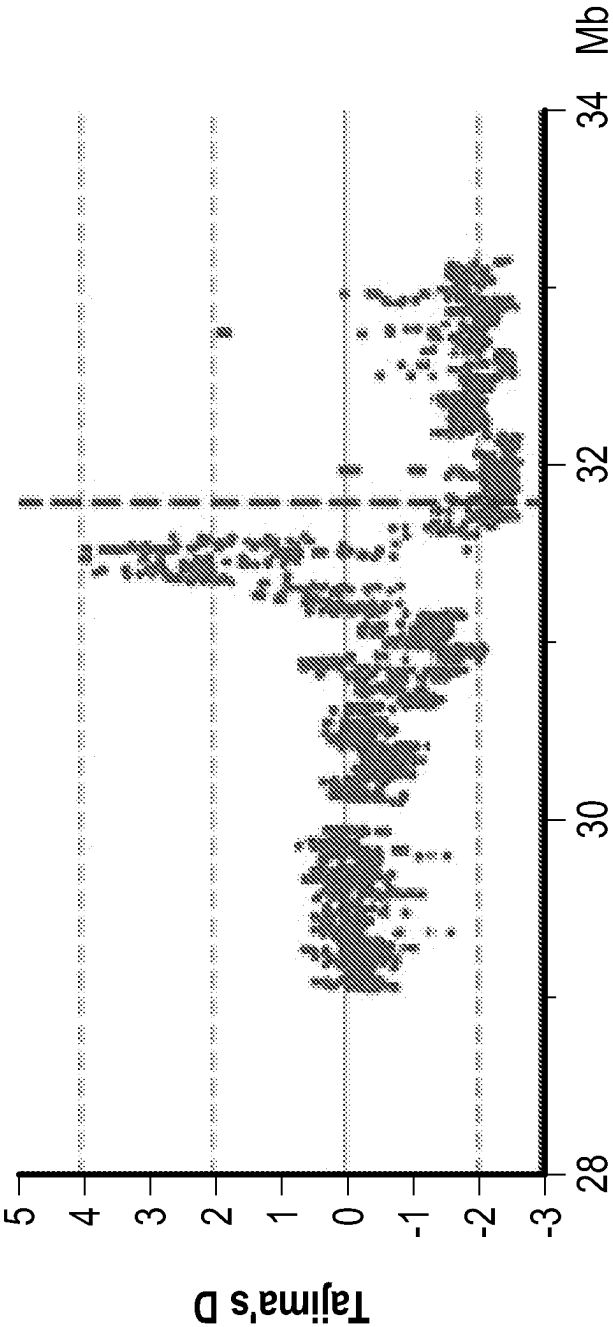


FIG. 20B



Position
FIG. 20C



Position
FIG. 20D

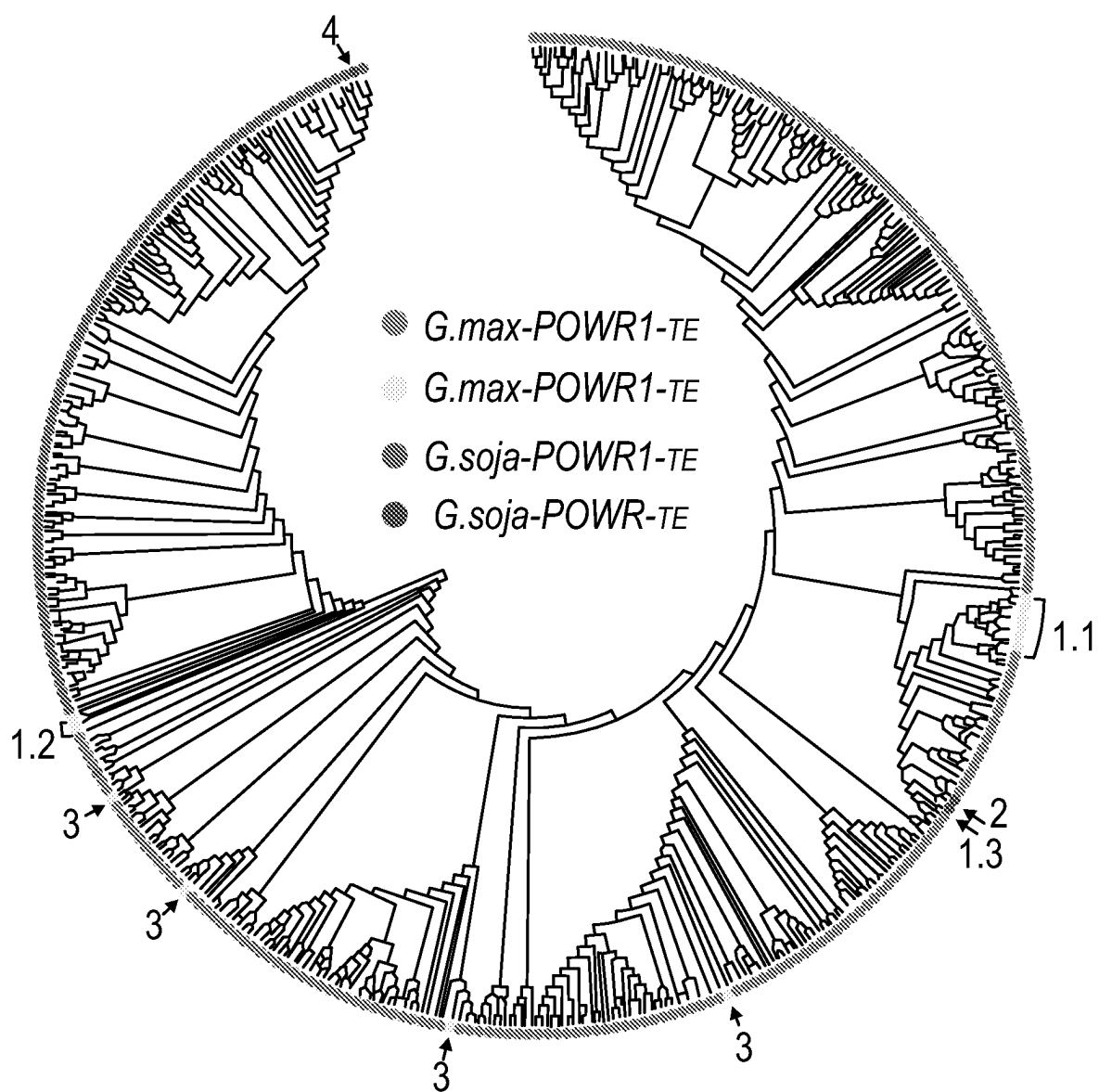


FIG. 21A

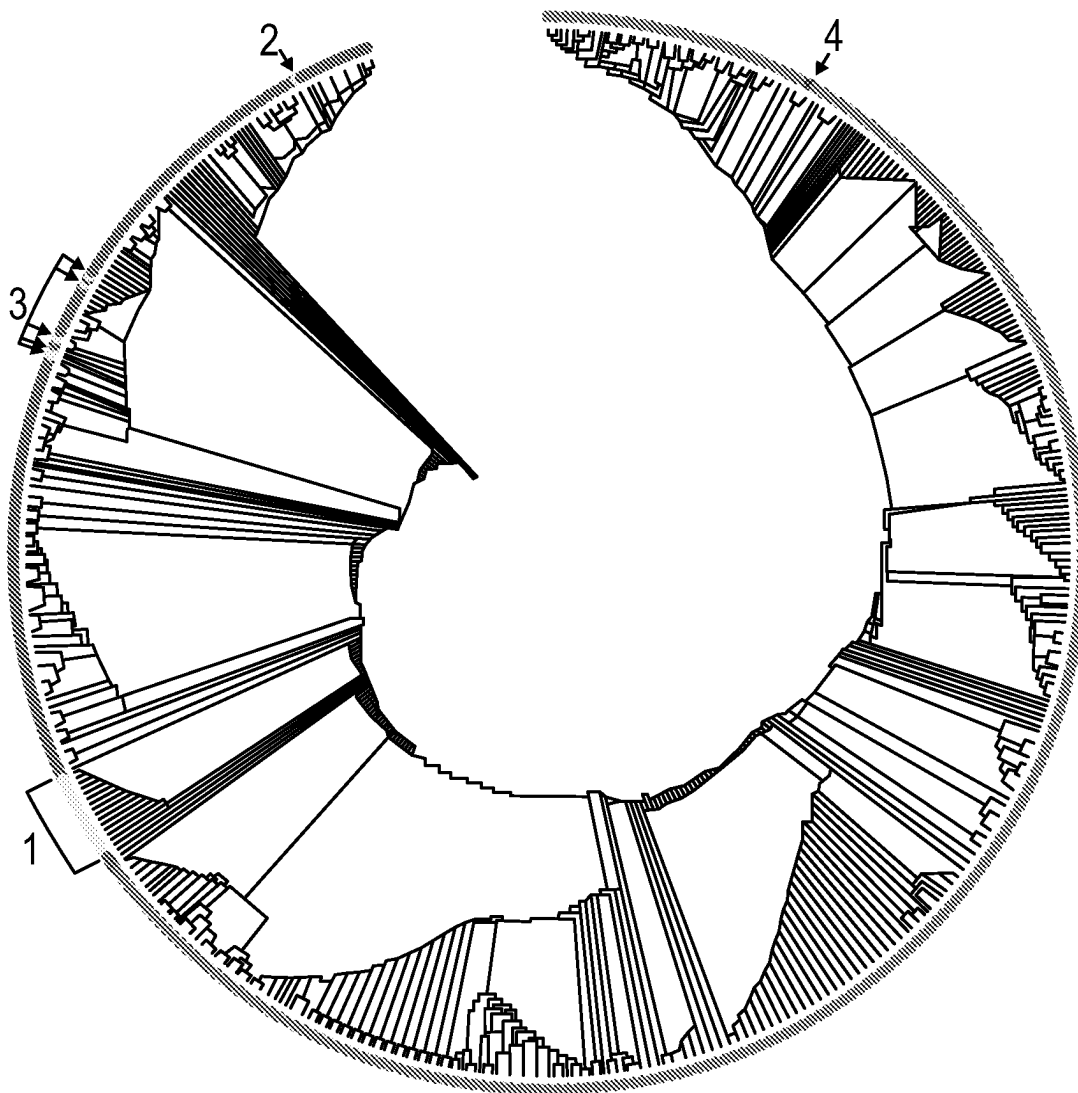


FIG. 21B

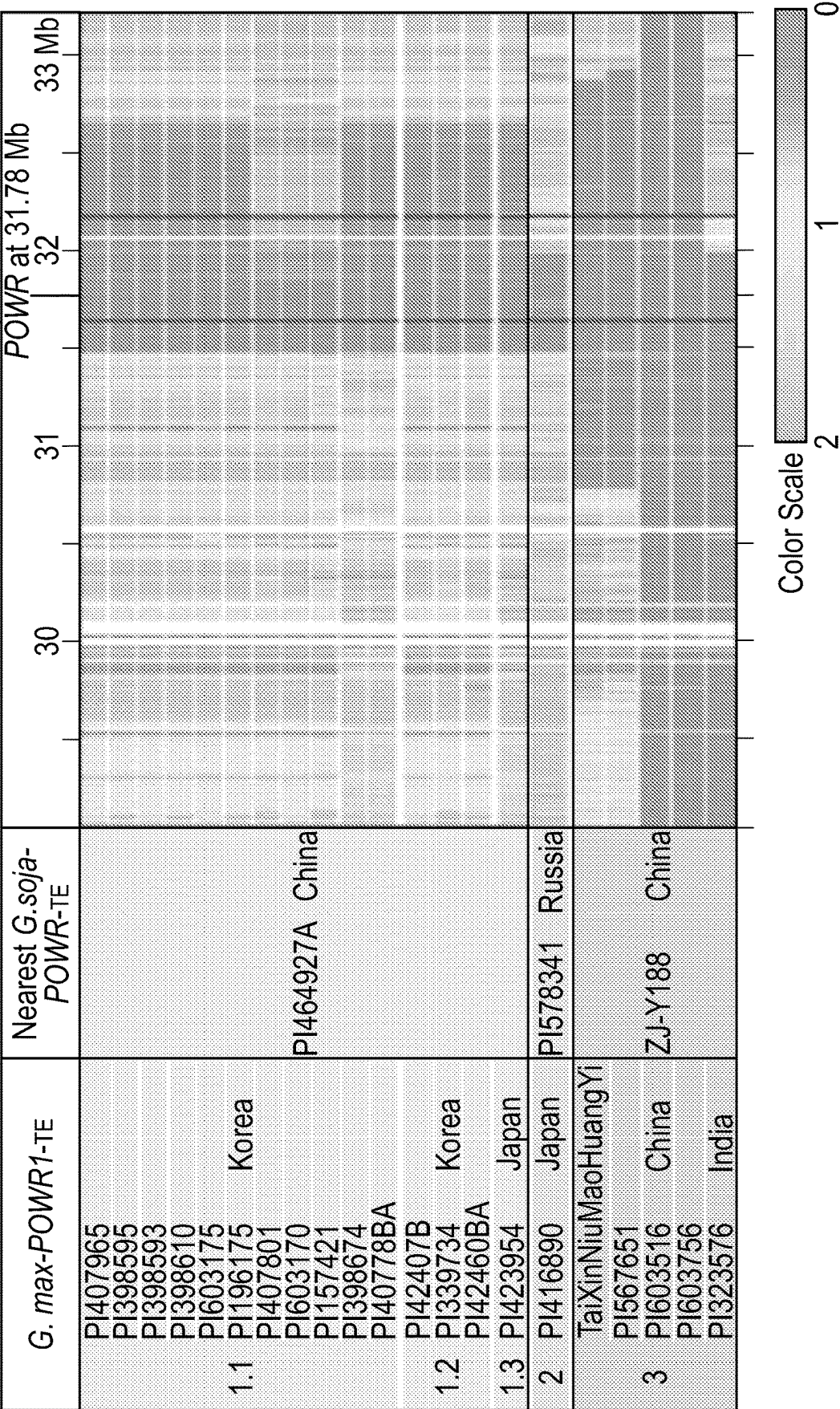


FIG. 21C

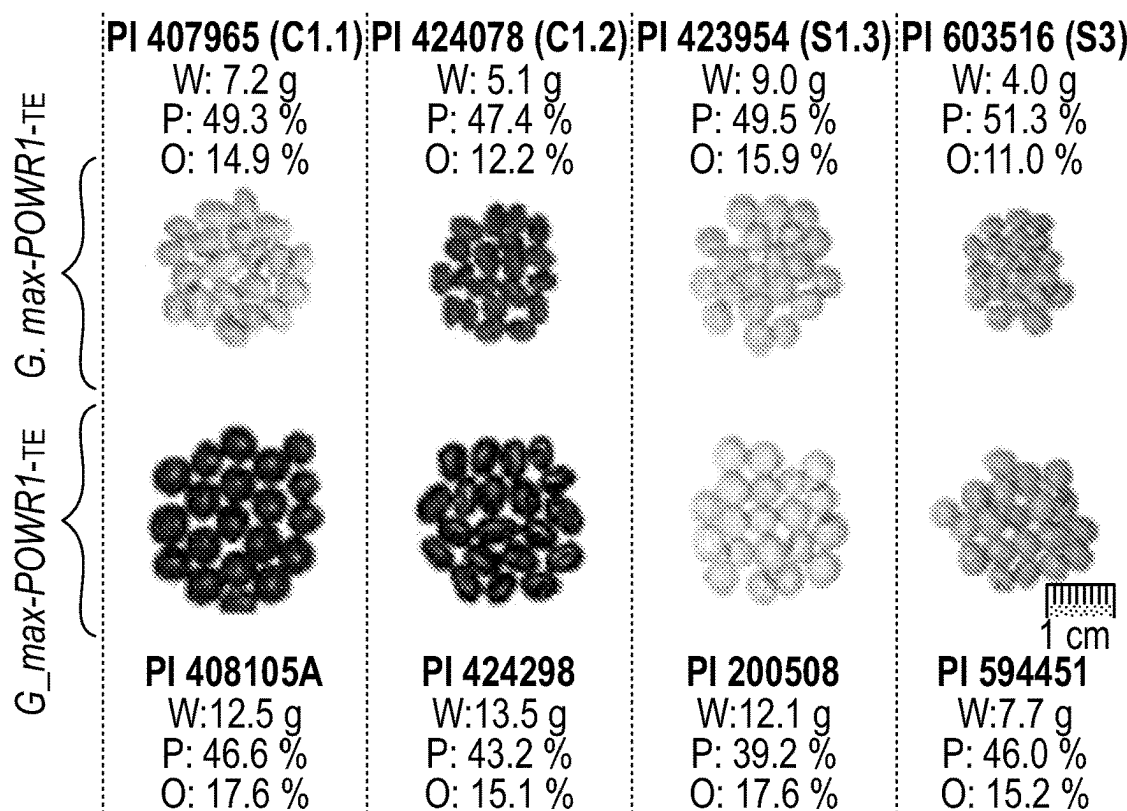


FIG. 21D

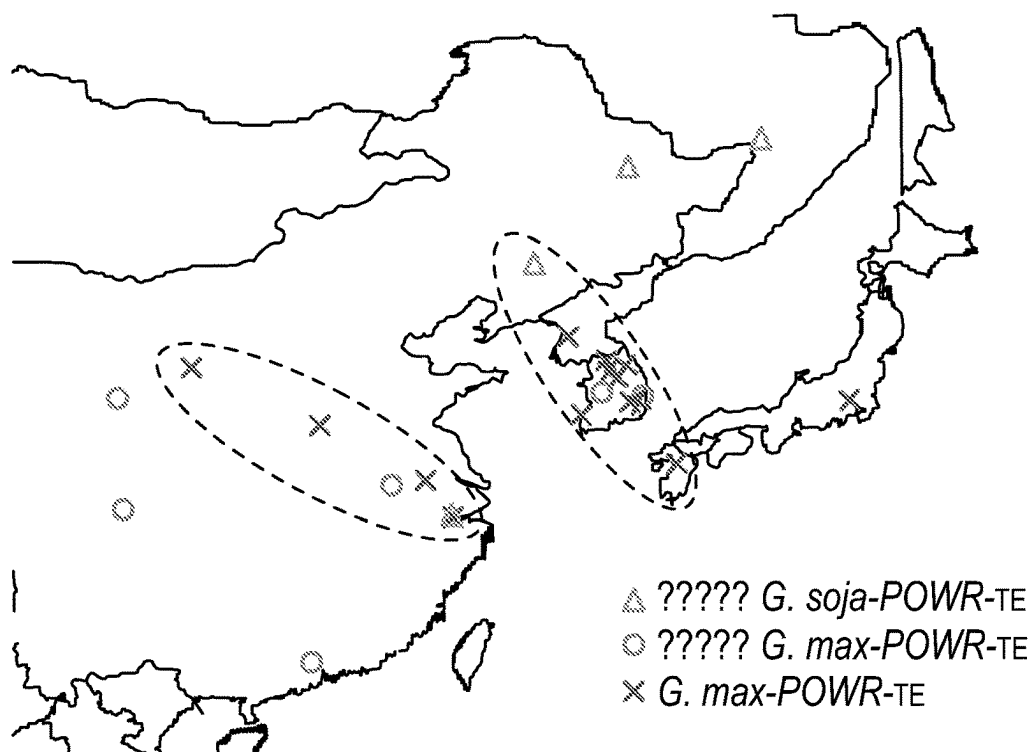


FIG. 21E

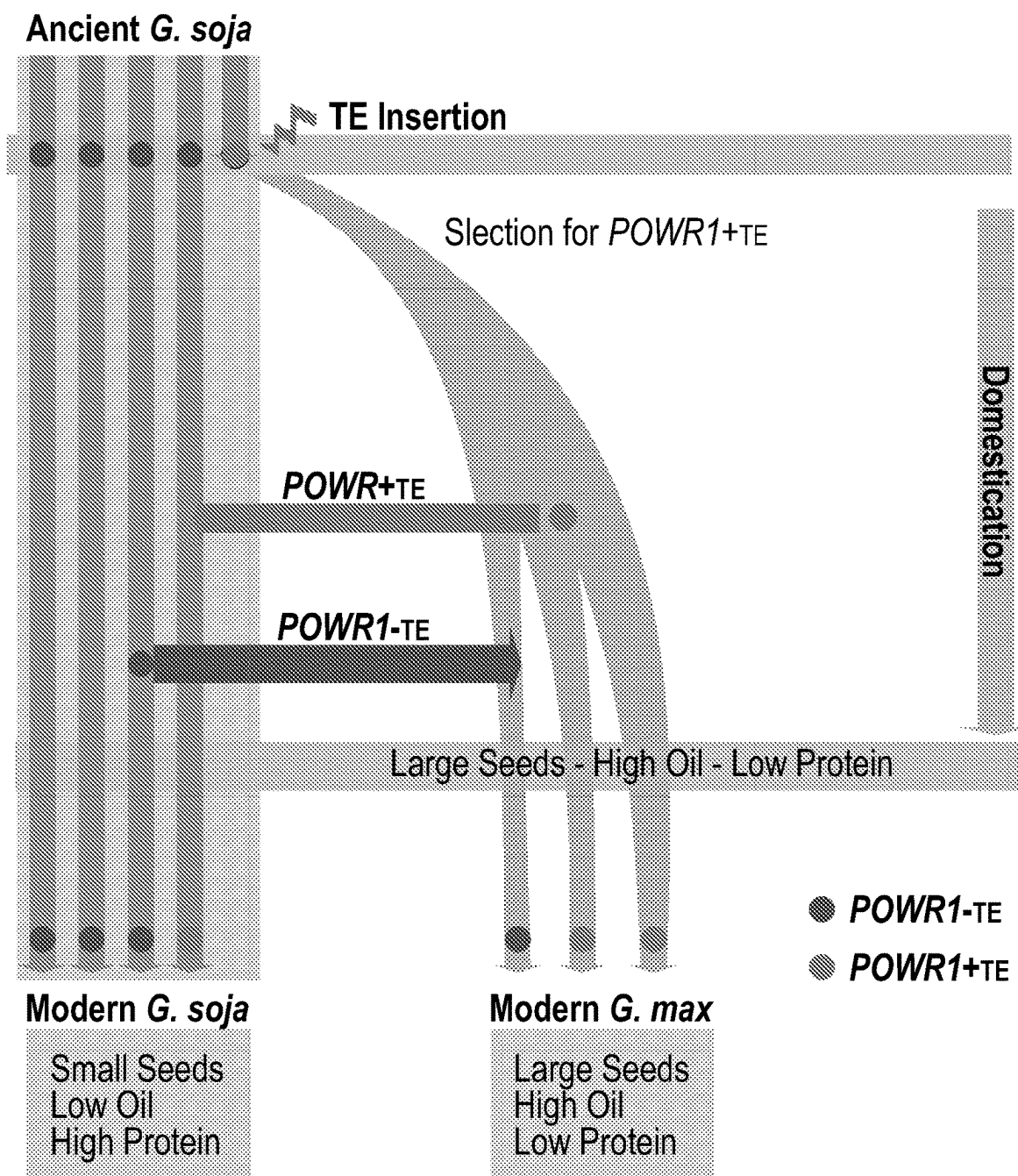


FIG. 22



FIG. 23A

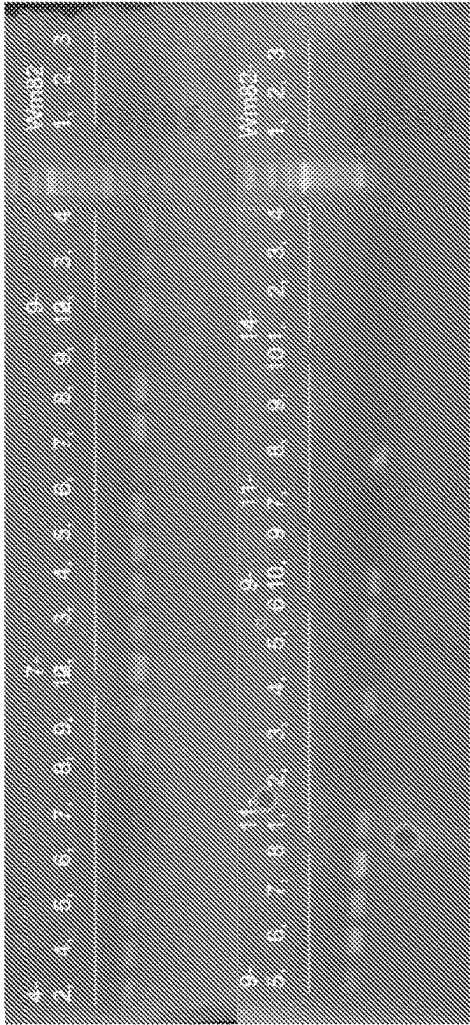


FIG. 23B

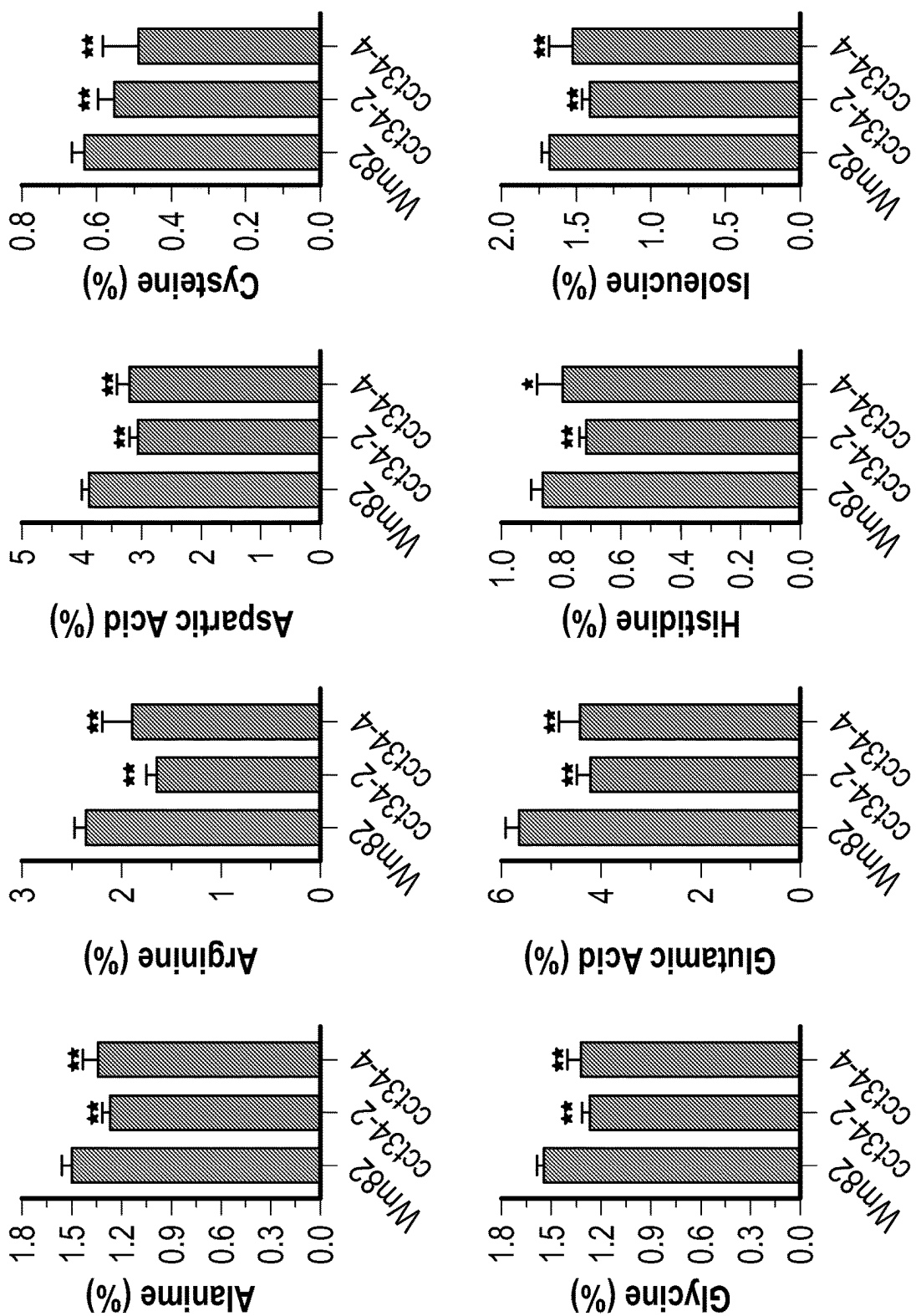


FIG. 24-1

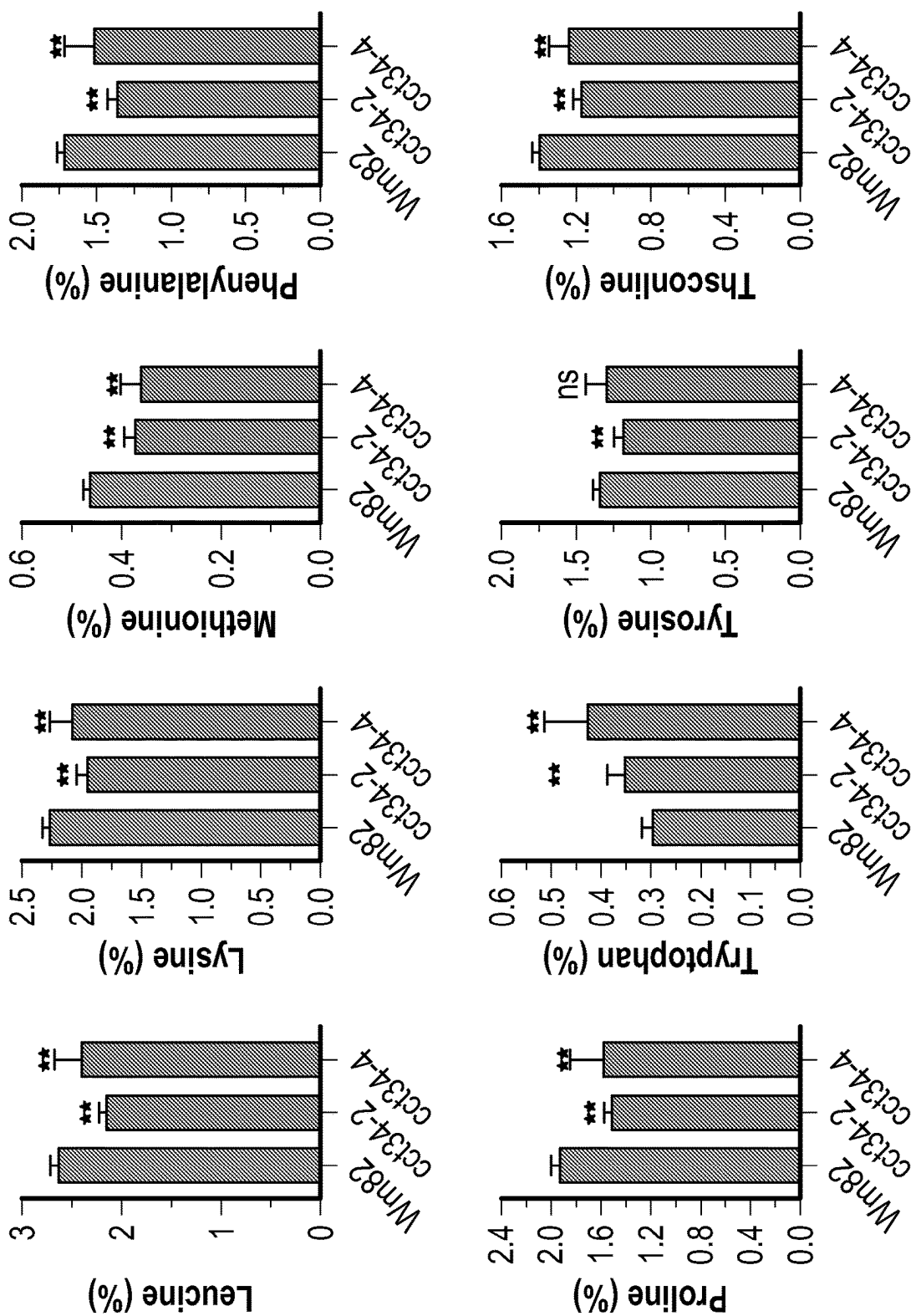


FIG. 24-2

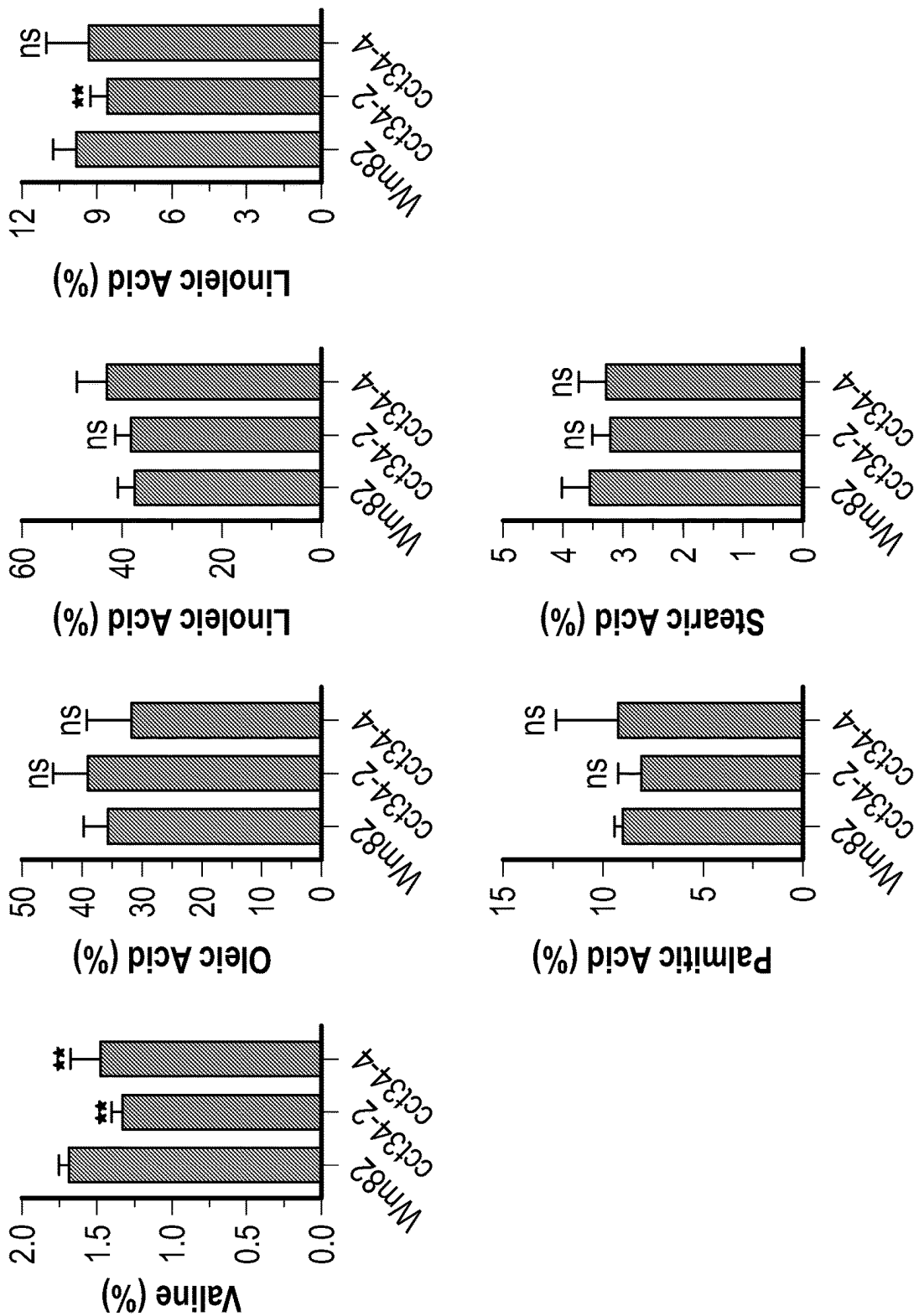


FIG. 24-3

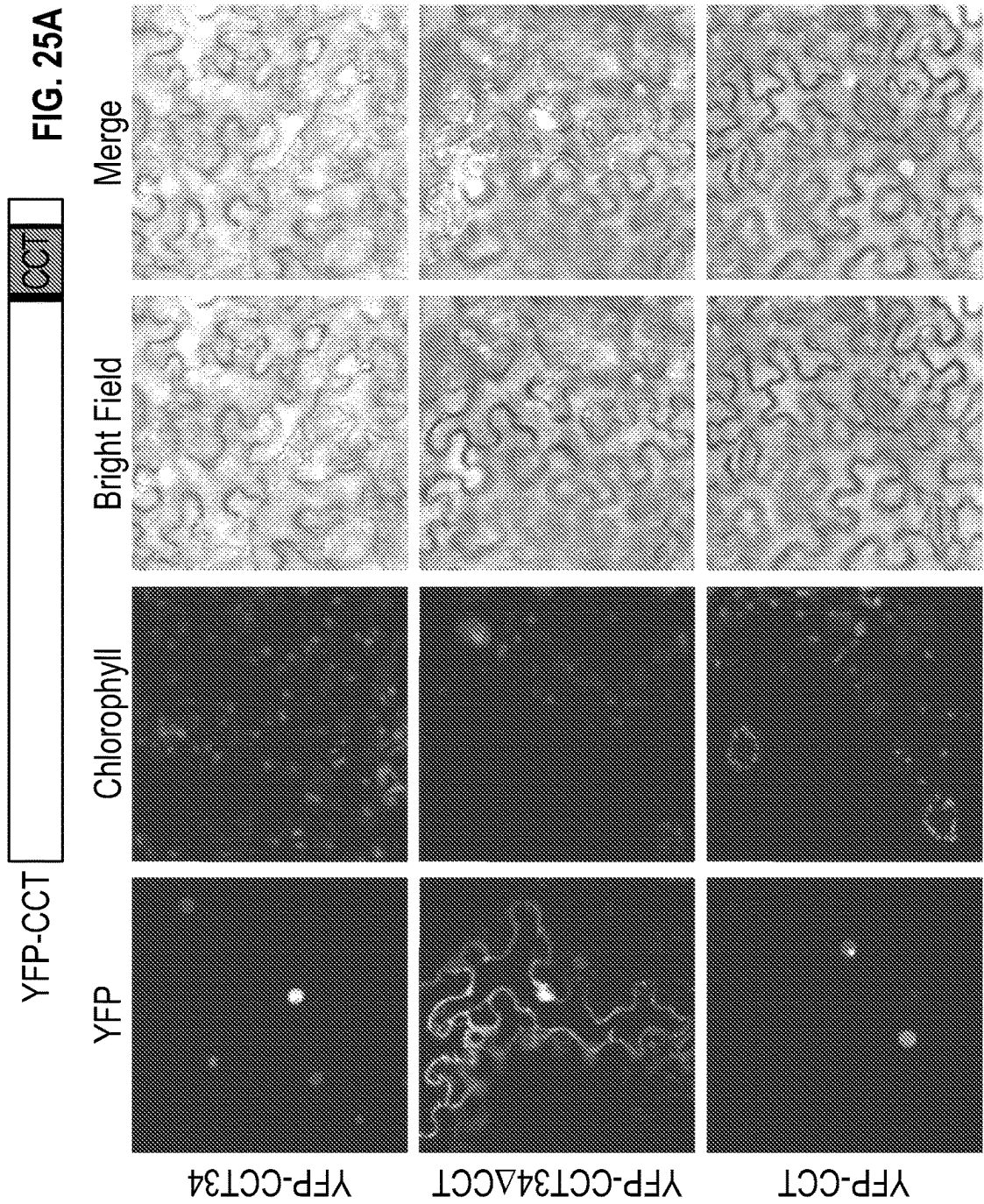


FIG. 25B

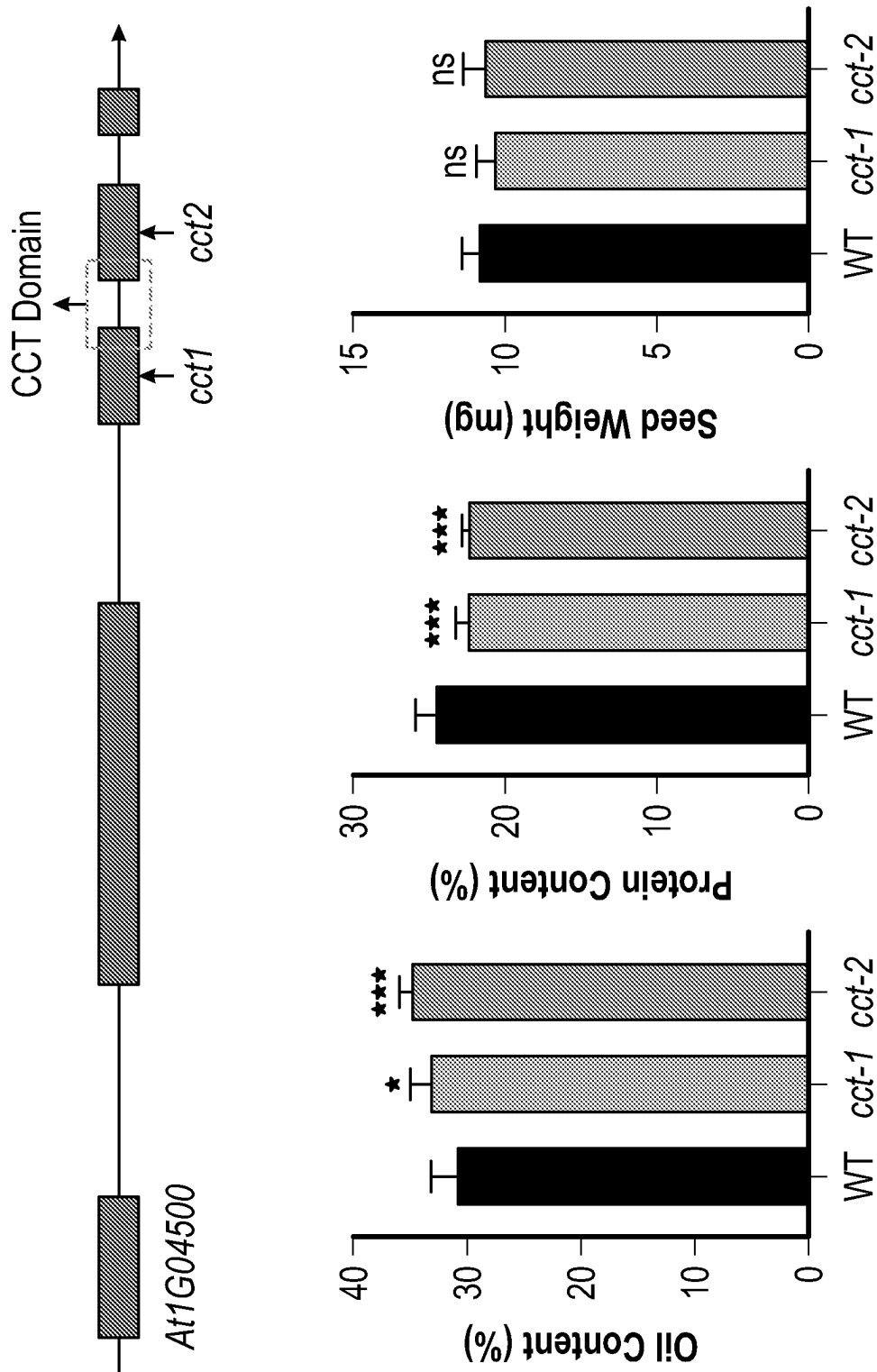


FIG. 26

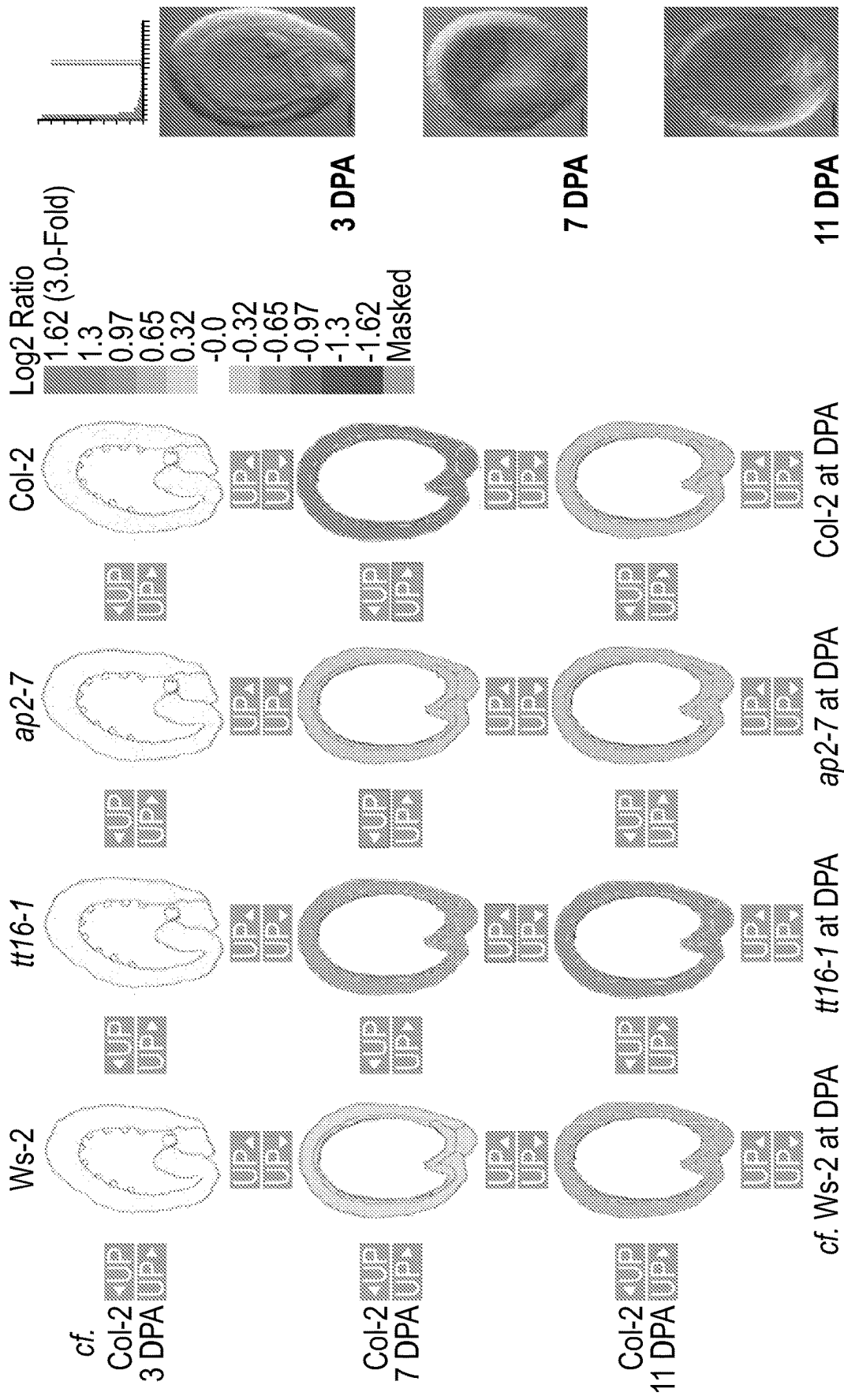


FIG. 27

USE OF CCT-DOMAIN PROTEINS TO IMPROVE AGRONOMIC TRAITS OF PLANTS

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority from Provisional Application No. 63/323,026, filed Mar. 23, 2022, the entire contents of which are hereby incorporated by reference.

GOVERNMENTAL RIGHTS

[0002] This invention was made with government support under USDA-ARS 5070-21000-043-000-D and 5070-21000-043-021-A awarded by the United States Department of Agriculture. The government has certain rights in the invention.

SEQUENCE LISTING

[0003] The present application contains a Sequence Listing which has been submitted in .XML format via Patent Center and is hereby incorporated herein by reference in its entirety. Said WIPO Sequence Listing was created on Mar. 23, 2023 is named 077875-751278 Seq. List and is 95.4 kilobytes in size.

FIELD OF THE INVENTION

[0004] The present disclosure provides genetically modified plants having improved agronomic traits.

BACKGROUND OF THE INVENTION

[0005] According the United Nations Food and Agricultural Organization (UN FAO), the world's population will exceed 9.6 billion people by the year 2050, which will require significant improvements in agricultural production to meet growing food demands. At the same time, conservation of resources (such as water, land), reduction of inputs (such as fertilizer, pesticides, herbicides), environmental sustainability, and climate change are increasingly important factors in how food is grown. Improvement of agronomic traits of cultivated plants such as seed quality and yield has proven challenging for conventional paradigms for crop improvement. This challenge is in part due to the complex genetic and environmental factors that can affect agronomic traits in plants. The complex correlation of important agronomic traits poses a great challenge in simultaneously improving more than one desirable agronomic trait to increase the overall economic value of a cultivated plant. For instance, in soybean, one of the most important seed crops grown worldwide, seed protein content and oil content appear to be negatively correlated, posing a great challenge in simultaneously improving both seed quality traits and yield to increase the overall economic value of soybean. Thus, there is a need for cultivated plants with improved agronomic traits.

SUMMARY OF THE INVENTION

[0006] One aspect of the instant disclosure encompasses a genetically modified plant having an improved agronomic trait. The plant comprises a nucleic acid sequence encoding a CCT motif-containing protein (CCT protein) wherein the CCT protein is a single-CCT domain polypeptide, wherein the nucleic acid sequence encoding the CCT protein com-

prises a nucleic acid modification and wherein the nucleic acid modification modifies the expression of the CCT protein in the plant thereby improving the agronomic trait of the plant.

[0007] The agronomic trait can be seed quality, seed protein content, seed protein composition, seed oil content, seed oil composition, yield, seed set, response to photoperiod, abiotic stress tolerance, biotic stress tolerance, flowering time and maturity, regulation of circadian clock light response-related flowering, high latitude adaptation, or any combination thereof. In some aspects, the improved agronomic trait is an agronomic trait of Table 14. In other aspects, the improved agronomic trait is an agronomic trait associated with a QTL of Table 15. In other aspects, the agronomic trait is: (a) seed quality and the CCT protein is encoded by a nucleic acid sequence comprising a gene of Table 5; (b) yield-related traits and the CCT protein is encoded by a nucleic acid sequence comprising a gene of Table 6; (c) response to abiotic/biotic stress tolerance and the CCT protein is encoded by a nucleic acid sequence comprising a gene of Table 7; (d) flowering time and maturity and the CCT protein is encoded by a nucleic acid sequence comprising a gene of Table 8; and (e) development-related traits and the CCT protein is encoded by a nucleic acid sequence comprising a gene of Table 9.

[0008] In some aspects, the plant is a legume (Fabaceae). The legume can be common bean, cowpea, soybean, chickpea, pea, or *Medicago*. In some aspects, the legume is a soybean species (*Glycine max*, *hispida*). When the legume is soybean, the agronomic trait can be seed protein, oil content, 100-seed weight, or any combination thereof, and the CCT protein is GmCCT34 (POWR2), GmCCT67 (POWR1), GmCCT35 (POWR3), GmCCT69 (POWR4), any variant thereof, or any combination thereof.

[0009] In some aspects, the CCT protein is GmCCT67 (POWR1). In one aspect, the nucleic acid modification reduces the expression of the GmCCT67 protein in the plant. In some aspects, oil content of seeds is increased by about 0.5% to about 5% wt/wt and wherein protein content of seeds is reduced by about 1% wt/wt to about 20% wt/wt. When the CCT protein is POWR1, the nucleic acid modification can increase the expression of the GmCCT67 protein in the plant. In some aspects, oil content of seeds is decreased by about 0.5% to about 5% wt/wt and wherein protein content of seeds is increased by about 1% wt/wt to about 20% wt/wt. In some aspects, the GmCCT67 protein comprises an amino acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the amino acid sequence of SEQ ID NO: 1.

[0010] In some aspects, the GmCCT67 protein is encoded by a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 2. In some aspects, the nucleic acid modification in the nucleic acid sequence encoding the GmCCT67 protein comprises a transposable element (TE) insertion. In some aspects, the nucleic acid sequence comprising the TE insertion comprises at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 3. In some aspects, the nucleic acid modification in the nucleic acid sequence encoding the GmCCT67 protein comprises an expression construct for

expression of the GmCCT67 protein, wherein the expression construct comprises a nucleic acid sequence encoding the GmCCT67 protein operably linked to a ubiquitin promoter or a native promoter.

[0011] In some aspects, the CCT protein is GmCCT34 (POWR2). When the CCT protein is GmCCT34, the nucleic acid modification reduces the expression of GmCCT34 (POWR2) in the plant such that the oil content of seeds can be increased by about 0.5% to about 5% wt/wt and protein content of seeds can be reduced by about 1% wt/wt to about 20% wt/wt.

[0012] In some aspects, the nucleic acid modification increases the expression of GmCCT34 (POWR2) in the plant. The oil content of seeds can be decreased by about 0.5% to about 5% wt/wt and wherein protein content of seeds is increased by about 1% wt/wt to about 20% wt/wt. In some aspects, the GmCCT34 protein comprises an amino acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the amino acid sequence of SEQ ID NO: 5. In some aspects, the GmCCT34 protein is encoded by a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 6.

[0013] The nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein can also comprise an expression construct for expression of the GmCCT34 protein, wherein the expression construct can comprise a nucleic acid sequence encoding the GmCCT34 protein operably linked to a promoter.

[0014] In some aspects, the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a genomic deletion of a 1.3-Mb region of the soybean genome (Chr10: 35253890-36584337) comprising a nucleic acid sequence encoding the GmCCT34 CCT protein. In one aspect, the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid modification generated using a CRISPR/Cas programmable nucleic acid modification system targeted to a nucleic acid sequence in a nucleotide sequence encoding the GmCCT34 protein. In some aspects, the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 8 to 10 or any combination thereof. In other aspects, the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 11 to 13 or any combination thereof. In additional aspects, the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 14 to 16 or any combination thereof. In some aspects, the plant is a soybean species (*Glycine max*, *hispida*), the CCT protein is GmCCT34 (POWR2), the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a genomic deletion of a 1.3-Mb region of the soybean genome (Chr10: 35253890-36584337) comprising a nucleic acid sequence encoding the GmCCT34 CCT protein or a nucleic acid sequence of SEQ ID NO: 8 to 16 or any combination thereof, and wherein the nucleic acid modification reduces the expression of the GmCCT34 protein in the plant.

[0015] The CCT protein can be GmCCT35 (POWR3). In some aspects, the GmCCT35 protein comprises an amino

acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the amino acid sequence of SEQ ID NO: 25. In some aspects, the GmCCT35 protein is encoded by a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 26. In some aspects, the plant is a soybean species (*Glycine max*, *hispida*), the CCT protein is GmCCT35 (POWR3), the nucleic acid modification in the nucleic acid sequence encoding the GmCCT35 protein comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 27.

[0016] In some aspects, the CCT protein is GmCCT69 (POWR4). The GmCCT69 protein can comprise an amino acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the amino acid sequence of SEQ ID NO: 28. The GmCCT69 protein can be encoded by a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 29. In some aspects, the plant is a soybean species (*Glycine max*, *hispida*), the CCT protein is GmCCT69 (POWR4), the nucleic acid modification in the nucleic acid sequence encoding the GmCCT69 protein comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 30.

[0017] In some aspects, the plant is a soybean species (*Glycine max*, *hispida*), wherein; the CCT protein is GmCCT67 (POWR1) and the nucleic acid modification in the nucleic acid sequence encoding the GmCCT67 protein comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 3; and the CCT protein is GmCCT34 (POWR2), wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a genomic deletion of a 1.3-Mb region of the soybean genome (Chr10: 35253890-36584337) comprising a nucleic acid sequence encoding the GmCCT34 CCT protein, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 8 to 10, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 11 to 13, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 14 to 16, or any combination thereof.

[0018] In some aspects, the plant is a soybean species (*Glycine max*, *hispida*), wherein the CCT protein is GmCCT67 (POWR1) and the nucleic acid modification in the nucleic acid sequence encoding the GmCCT67 protein comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 3; the CCT protein is GmCCT34 (POWR2), wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein com-

prises a genomic deletion of a 1.3-Mb region of the soybean genome (Chr10: 35253890-36584337) comprising a nucleic acid sequence encoding the GmCCT34 CCT protein, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 8 to 10, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 11 to 13, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 14 to 16, or any combination thereof; and the CCT protein is GmCCT69 (POWR4) and the nucleic acid modification in the nucleic acid sequence encoding the GmCCT69 protein comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 30.

[0019] In other aspects, the plant is a soybean species (*Glycine max*, *hispida*), wherein the CCT protein is GmCCT67 (POWR1) and the nucleic acid modification in the nucleic acid sequence encoding the GmCCT67 protein comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 3; and the CCT protein is GmCCT35 (POWR3) and the nucleic acid modification in the nucleic acid sequence encoding the GmCCT35 protein comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 27.

[0020] In yet other aspects, the plant is a soybean species (*Glycine max*, *hispida*), wherein the CCT protein is GmCCT34 (POWR2), wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a genomic deletion of a 1.3-Mb region of the soybean genome (Chr10: 35253890-36584337) comprising a nucleic acid sequence encoding the GmCCT34 CCT protein, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 8 to 10, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 11 to 13, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 14 to 16, or any combination thereof; and the CCT protein is GmCCT35 (POWR3) and the nucleic acid modification in the nucleic acid sequence encoding the GmCCT35 protein comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 27.

[0021] In some aspects, the plant is a soybean species (*Glycine max*, *hispida*), wherein the CCT protein is GmCCT67 (POWR1) and the nucleic acid modification in the nucleic acid sequence encoding the GmCCT67 protein comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 3; the CCT protein is GmCCT34 (POWR2), wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein com-

prises a genomic deletion of a 1.3-Mb region of the soybean genome (Chr10: 35253890-36584337) comprising a nucleic acid sequence encoding the GmCCT34 CCT protein, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 8 to 10, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 11 to 13, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 14 to 16, or any combination thereof; and the CCT protein is GmCCT35 (POWR3) and the nucleic acid modification in the nucleic acid sequence encoding the GmCCT35 protein comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 27.

[0022] In some aspects, the plant is a soybean species (*Glycine max*, *hispida*), wherein the CCT protein is GmCCT67 (POWR1) and the nucleic acid modification in the nucleic acid sequence encoding the GmCCT67 protein comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 3; the CCT protein is GmCCT34 (POWR2), wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a genomic deletion of a 1.3-Mb region of the soybean genome (Chr10: 35253890-36584337) comprising a nucleic acid sequence encoding the GmCCT34 CCT protein, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 8 to 10, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 11 to 13, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 14 to 16, or any combination thereof; the CCT protein is GmCCT35 (POWR3) and the nucleic acid modification in the nucleic acid sequence encoding the GmCCT69 protein comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 27; and the CCT protein is GmCCT69 (POWR4) and the nucleic acid modification in the nucleic acid sequence encoding the GmCCT69 protein comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 30.

[0023] In some aspects, the plant is *Arabidopsis thaliana*. When the plant is *Arabidopsis*, the CCT protein can be AtPOWR1, any variant thereof, or any combination thereof, and a nucleic acid modification can reduce the expression of the AtPOWR1 protein in the plant. In some aspects, the oil content of the seeds is increased and wherein the protein content of the seeds is reduced. In some aspects, the AtPOWR1 protein comprises an amino acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the amino acid sequence of SEQ ID NO: 33. In other aspects, the AtPOWR1 protein is encoded by a nucleic acid

sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 31. In some aspects, the *Arabidopsis* plant comprises a first T-DNA-insertion mutant of AtPOWR1 (WiscDsLox297300_13A.1, Atcct1), a second T-DNA-insertion mutant of AtPOWR1 (SALK_036731.1; Atcct-2).

[0024] Another aspect of the instant disclosure encompasses an engineered nucleic acid modification system for modifying the expression of a CCT protein in a plant. The system comprises a nucleic acid expression construct comprising: a promoter operably linked to a nucleic acid sequence encoding a programmable nucleic acid modification system targeted to a nucleic acid sequence in a nucleotide sequence encoding the CCT protein; or a nucleotide sequence encoding the CCT protein operably linked to a promoter. Expressing the programmable nucleic acid modification system or expressing the CCT protein introduces a nucleic acid modification of the nucleic acid sequence encoding the CCT protein, thereby modifying the expression of a CCT protein in a plant.

[0025] In some aspects, the CCT protein is GmCCT34 (POWR2), GmCCT67 (POWR1), GmCCT35 (POWR3), GmCCT69 (POWR4), or any combination thereof. The GmCCT67 (POWR1) protein can be encoded by a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 2. In some aspects, a nucleic acid modification can be an expression construct comprising a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 4.

[0026] In some aspects, the CCT protein is GmCCT34 encoded by a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 6. In some aspects, the expression construct for expression of GmCCT34 POWR2 comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 7. In another aspect, the nucleic acid expression construct comprises a promoter operably linked to a nucleic acid sequence encoding a programmable nucleic acid modification system targeted to a nucleic acid sequence in a nucleotide sequence encoding the GmCCT34 protein. In yet other aspects the programmable nucleic acid modification system is CRISPR/Cas system comprising a guide RNA (gRNA) having a sequence complementary to a target sequence within the nucleotide sequence encoding the GmCCT34 protein. The gRNA can comprise a nucleic acid sequence of SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, or any combination thereof.

[0027] In some aspects, the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises an expression construct for expression of the GmCCT34 protein, wherein the expression construct comprises a nucleic acid sequence encoding the GmCCT34 protein operably linked to a promoter. In some aspects, the nucleic acid expression construct comprises a nucleotide sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100%

sequence identity with the nucleotide sequence of SEQ ID NO: 4. The nucleic acid expression construct can comprise a nucleotide sequence encoding the GmCCT34 protein operably linked to a promoter. In some aspects, the nucleic acid expression construct comprises a nucleotide sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleotide sequence of SEQ ID NO: 7. In some aspects, the construct can further comprise a nucleic acid delivery vector comprising the nucleic acid expression construct for delivering the nucleic acid expression construct to the target cell.

[0028] Yet another aspect of the instant disclosure encompasses one or more nucleic acid constructs encoding an engineered nucleic acid modification system for modifying the expression of a CCT protein in a plant. The engineered nucleic acid modification system can be as described herein above.

[0029] An additional aspect of the instant disclosure encompasses a plant comprising one or more nucleic acid constructs encoding an engineered nucleic acid modification system for modifying the expression of a CCT protein in a plant. The nucleic acid constructs can be as described herein above.

[0030] One aspect of the instant disclosure encompasses a method of identifying a plant having an improved agronomic trait using marker-assisted selection (MAS). The method comprises identifying in a population of plants one or more plants comprising a molecular marker, wherein the molecular marker demonstrates linkage with a nucleic acid modification that modifies the expression of a CCT protein in the plant. The molecular marker can be a quantitative trait locus (QTL) selected from QTLs of Table 15. In some aspects, the population of plants comprises progeny of a cross between parent plants. In other aspects, a parent plant can be a plant described herein above.

[0031] Another aspect of the instant disclosure encompasses a method of generating a genetically modified plant having an improved agronomic trait. The method comprises: introducing one or more nucleic acid constructs encoding an engineered nucleic acid modification system of any one of claims 64-81 into a plant or plant cell; and growing the plant or plant cell for a time and under conditions sufficient for the nucleic acid expression construct to express the programmable nucleic acid modification system or the CCT protein in the plant or plant cell. Expressing the programmable nucleic acid modification system or expressing the CCT protein introduces a nucleic acid modification in the nucleic acid sequence encoding the CCT protein, thereby modifying the expression of a CCT protein in a plant and improving the agronomic trait of the plant.

[0032] One aspect of the instant disclosure encompasses a method of improving an agronomic trait of a plant. The method comprises introducing one or more nucleic acid constructs encoding an engineered nucleic acid modification system of any one of claims 64-81 into a plant or plant cell; and growing the plant or plant cell for a time and under conditions sufficient for the nucleic acid expression construct to express the programmable nucleic acid modification system or the CCT protein in the plant or plant cell. Expressing the programmable nucleic acid modification system or expressing the CCT protein introduces a nucleic acid modification in the nucleic acid sequence encoding the

CCT protein, thereby modifying the expression of a CCT protein in a plant and improving the agronomic trait of the plant.

[0033] Another aspect of the instant disclosure encompasses a kit for improving an agronomic trait of a plant. The kit comprises: one or more genetically modified plant having an improved agronomic trait; one or more nucleic acid constructs encoding an engineered nucleic acid modification system for modifying the expression of a CCT protein in a plant; a plant comprising one or more nucleic acid constructs encoding a programmable nucleic acid modification system for modifying the expression of a CCT protein in a plant; or any combination of (a)-(c). The plants constructs, and systems can be as described herein above

BRIEF DESCRIPTION OF THE FIGURES

[0034] The patent or application file contains at least one drawing executed in color. Copies of this patent or patent application publication with color drawing(s) will be provided by the Office upon request and payment of the necessary fee.

[0035] FIG. 1 depicts sequencing comparison between FN0172932 and the wild type M92-220.

[0036] FIG. 2A depicts species tree and the number of identified CCT domain-containing proteins in each species.

[0037] FIG. 2B depicts the number of identified CCT domain-containing proteins in each species and constituent domains and organization.

[0038] FIG. 3A depicts the chromosomal details of GmCCT genes in soybean genome and microsynteny relationship in representative legumes by showing the microsynteny comparison of 573 GmCCT12/21 and GmCCT13/20 paralogs among soybean, common bean, cowpea, chickpea, pea, and *Medicago*.

[0039] FIG. 3B depicts the chromosomal details of GmCCT genes in soybean genome and microsynteny relationship in representative legumes by showing microsynteny comparison of 573 GmCCT12/21 and GmCCT34/67 paralogs among soybean, common bean, cowpea, chickpea, pea, and *Medicago*.

[0040] FIG. 4A depicts the phylogeny analysis of CCT protein and domains, showing the global phylogenetic tree of all CCT domain-containing proteins.

[0041] FIG. 4B depicts the phylogeny analysis of CCT protein and domains, showing the phylogenetic tree constructed by 43-bp CCT domain.

[0042] FIG. 4C depicts the phylogeny analysis of CCT protein and domains, showing HMM logos representing amino acids of CCT domains as illustrated in different clusters in FIG. 4A and FIG. 4B. Conserved and cluster-specific amino acids are indicated in green rectangle and red triangles, respectively.

[0043] FIG. 5 depicts the phylogenetic tree of GmCCTs in soybean and the expression patterns in circadian clock response. C and T indicate control and treatment. Blue, green, and red dotted rectangles highlight the circadian clock-responsive GmCCTs, condition-specifically expressed GmCCTs, and condition-responsive GmCCTs, respectively.

[0044] FIG. 6 depicts the phylogenetic tree of GmCCTs in soybean and the expression patterns in the compartments of developing seeds at globular, heart, cotyledon, early maturation stages, and major vegetative tissues. Blue and green rectangles indicate the conserved expression and divergent expression of GmCCT paralogs. AB—Abaxial;

AD—Adaxial; AL—Aleurone; AX—Axis; COT—Cotyledon; EP—Embryo Proper; EPD—Epidermis; ENT—Endothelium; ES—Endosperm; FBUD—Floral Bud; HG—Hourglass; HI—Hilum; II—Inner Integument; OI—Outer Integument; PA—Palisade; PL—Plumule; PY—Parenchyma; RM—Root Meristem; S—Suspensor; SC—Seed Coat; SM—Shoot Meristem; VS—Vascular Bundle; WM—Whole Mount Seed; SDLG—seedling; FLUB—floral bud; STEM—stem; ROOT—root; LEAF—leaf.

[0045] FIG. 7 depicts macrosyntenic visualization of syntenic relationships among CCT proteins between legume genomes.

[0046] FIG. 8 depicts the CCT proteins with truncated domains.

[0047] FIG. 9A shows the generation of GmCCT34 knockout mutant cct34 using CRISPR/Cas9 editing technology and seed composition measurements by an illustration depicting the preferential expression of GmCCT34 in the seed coats of cotyledon and early maturation seeds of Williams 82. Abbreviation: AB—Abaxial; AD—Adaxial; AL—Aleurone; AX—Axis; EP—Embryo Proper; EPD—Epidermis; ENT—Endothelium; ES—Endosperm; HG—Hourglass; HI—Hilum; II—Inner Integument; OI—Outer Integument; PA—Palisade; PL—Plumule; PY—Parenchyma; S—Suspensor; VS—Vascular Bundle.

[0048] FIG. 9B depicts a schematic representation of GmCCT34 and the guide RNAs (gRNAs) sequences for gene knockout. PAM sites (NGG for gRNAs on the forward DNA strand and CCN for gRNAs on the reverse DNA strand) are indicated in blue. A 521-bp fragment containing both gRNA2 and gRNA3 targeting sites was used for BslI digestion to confirm the mutation.

[0049] FIG. 9C: Screening results for mutations on gRNA2 and gRNA3 targeting sites by BslI digestion. PCR amplicons carrying any mutations on either or both targeting sites showed different patterns of digested products from those (four bands: 248 bp, 144 bp, 108 bp, and 21 bp) of wild type Williams 82 (Wm82). White and red arrows indicate the resulting two-band pattern from the cct34-2 lines carrying two mutations in both gRNA2 and gRNA3; green arrows indicate band pattern of many cct34-4 lines.

[0050] FIG. 9D depicts the targeting sequence comparison of cct34-2-2, cct34-4-5, cct34-4-7 with the wild type Wm82 as indicated in FIG. 9C.

[0051] FIG. 9E indicates the comparisons of seed oil, protein, and 100-seed weight between FN0172932 (FN) and the wild type (WT), cct34 and the wild type Wm82, respectively.

[0052] FIG. 10A depicts phenotype distribution of the seed traits used for the association studies, illustrating the phenotypic distribution of seed oil content.

[0053] FIG. 10B depicts phenotype distribution of the seed traits used for the association studies, illustrating the phenotypic distribution of protein content.

[0054] FIG. 10C depicts phenotype distribution of the seed traits used for the association studies, illustrating the phenotypic distribution of 100-seed weight.

[0055] FIG. 11A depicts GWAS of oil content in the 278 diverse accessions using a GLM model.

[0056] FIG. 11B depicts GWAS of oil content in the 278 diverse accessions using a MLM model.

[0057] FIG. 12A depicts the identification of QTL on chr20 for oil, protein and seed weight and functional validation. Manhattan plots illustrating the regional association

results for oil, protein, and seed weight. Red solid dots highlight the 321-bp InDel significantly associated with the three traits. The Bonferroni-corrected genome-wide significance threshold is depicted in the horizontal dotted lines. The three most significantly associated SNPs (ss715637271, ss715637273, ss715637274, left to right) from SoySNP50K data set that were identified in the RILs using GWAS approach are indicated with red arrows below the bottom panel.

[0058] FIG. 12B depicts the identification of QTL on chr20 for oil, protein and seed weight and functional validation. Gene structure of Glyma.20G085100 harboring the most significant 321-bp InDel and indication of the InDel between two parental lines (Williams82 and PI479752) of RILs.

[0059] FIG. 12C depicts the identification of QTL on chr20 for oil, protein and seed weight and functional validation. Sequencing read alignments of Glyma.20G085100 gene model from two high oil/low protein and two low oil/high protein accessions to that of the soybean reference genome from Williams 82 shown. The 321-bp insertion is present in high-oil/low-protein genotypes but absent in two low-oil/high-protein genotypes. Seed oil (Oil), protein (Pro) and 100-seed weight (SW) of each genotype were provided beside the panel.

[0060] FIG. 12D depicts the identification of QTL on chr20 for oil, protein and seed weight and functional validation. Box plots showing the allelic effects of the InDel on oil, protein, and 100-seed weight in the association panel.

[0061] FIG. 12E depicts the identification of QTL on chr20 for oil, protein and seed weight and functional validation. Box plots showing the comparison of the three seed traits (oil, protein, seed weight) and field yield in NILs polymorphic for the TE, n=12.

[0062] FIG. 12F depicts the identification of QTL on chr20 for oil, protein and seed weight and functional validation. Genotypes of TE in four pairs of parental lines where POWR1 locus was successfully mapped in previous studies. PCR amplification using primers flanking the TE give rise to an amplicon of 1228 bp or an amplicon of 907 bp based on the presence or absence of the TE insertion in tested genotypes. Oil and protein levels from corresponding genotypes are given below the image and are highlighted with a gray background for the genotypes carrying the TE insertion. *, **, and *** are used to indicate significance at $p < 0.05$, $p < 0.01$, and $p < 0.001$, respectively, in all figures in the study.

[0063] FIG. 13A depicts GWAS and linkage mapping of oil content and protein content using 300 RILs

[0064] FIG. 13B depicts GWAS and linkage mapping of oil content and protein content using 300 RILs by showing association and linkage mapping results of protein and oil content. The most significant associations for both traits are provided in the corresponding Manhattan plot.

[0065] FIG. 13C depicts GWAS and linkage mapping of oil content and protein content using 300 RILs by showing association and linkage mapping results of protein and oil content. The most significant associations for both traits are provided in the corresponding Manhattan plot.

[0066] FIG. 14 depicts PCR-based genotyping of the 321-bp TE in NILs for POWR1. NILs show a 1228-bp PCR amplicon with the 321-bp TE insertion while NILs show a 907-bp fragment without the 321-bp TE insertion.

[0067] FIG. 15A depicts gene structure and expression of POWR1, by showing sequence alignment of CCT domain from different plant species.

[0068] FIG. 15B depicts sequence comparison between C terminus of POWR1_{+TE} and POWR1_{-TE}. The conserved CCT domain is colored in green. POWR1_{+TE} is 19 amino acids longer than POWR1_{-TE}. Amino acids in red indicate the distinct peptide sequence at the C terminus of POWR1_{+TE} as the result of the 321-bp TE insertion.

[0069] FIG. 15C depicts gene structure of POWR1 with and without the 321-bp TE insertion and the position of TE insertion (red arrow) in POWR1. The insertion caused a codon reading frameshift, which truncated the CCT domain (in orange) and generated a longer C terminus with distinct amino acid sequence (in blue).

[0070] FIG. 15D depicts a phylogenetic tree showing the evolutionary relationship among the POWR1_{-TE} homologous proteins from monocot and dicot plant species.

[0071] FIG. 15E depicts predicted structures of POWR1_{-TE} and POWR1_{+TE} had almost identical N-termini but distinct C-termini.

[0072] FIG. 15F depicts the comparable expression levels of POWR1_{-TE} in 40 soybean accessions and POWR1_{+TE} of 132 accessions in seeds at mid-maturation stages.

[0073] FIG. 15G depicts the comparison of subcellular localization of POWR1. TE and POWR1_{+TE} in tobacco cells. Scale bar=20 μ m.

[0074] FIG. 15H depicts the comparison of expression patterns of POWR1_{-TE} and POWR1_{+TE} in different soybean tissues. Y axis indicates the expression levels relative to GmCYP2.

[0075] FIG. 15I depicts enriched GO and KEGG terms for the differentially expressed genes between *G. max* accessions containing POWR1_{-TE} and POWR1_{+TE}.

[0076] FIG. 15J depicts relative expression levels of selected genes in seed coat and cotyledon of NILs containing POWR1_{-TE} or POWR1_{+TE}.

[0077] FIG. 16A depicts the comparison of promoter sequences between two POWR1 alleles, by showing IGV visualization of read alignment in the 2-kb region upstream of the start codon of POWR1 in the parental lines of the RIL population, PI479752 and Williams 82.

[0078] FIG. 16B depicts sequence comparison of promoter sequences between two POWR1 alleles, by revealing nearly identical promoter sequences between two groups carrying POWR1-TE (20 *G. soja* accessions) and POWR1+TE (51 *G. max* accessions). No correlation of seed traits with any DNA variants in their promoters.

[0079] FIG. 17 depicts the phenotypic changes associated with the transfer of a POWR1-TE from *G. soja* into *G. max*. Seed oil content, seed protein content and 100-seed weight of *G. max*-POWR1-TE accessions are compared to their closest *G. soja* accessions and *G. max*-POWR1+TE accessions based on local and global phylogenetic analyses. The average phenotype values for the Korean clusters 1.1 (C1.1) and 1.2 (C1.2) are given. Both S1.3 and S2 only contain one Japanese accession. A representative accession for S3 is shown. NA: data not available.

[0080] FIG. 18A depicts the identification of positive transgenic plants by Basta leaf painting assay by showing schematic illustration of the construct (Ubi917::POWR1) that was used for overexpression of POWR1-TE in soybean.

[0081] FIG. 18B depicts the basta leaf painting assay showed basta resistance in two transgenic lines and yellowish wilting leaves in control plants.

[0082] FIG. 18C depicts PCR verification of three positive transgenic plants using bar-specific and POWR1-cDNA-specific primers.

[0083] FIG. 18D depicts another PCR verification of three positive transgenic plants using bar-specific and POWR1-cDNA-specific primers.

[0084] FIG. 18E depicts relative seed expression of POWR1 in control and two transgenic plants.

[0085] FIG. 19A depicts the seed oil and protein content and weight in transgenic soybean overexpressing (OE) POWR1-TE, by showing seed protein, oil and weight of T2 plants in each of two transgenic events containing Ubi-promoter driven POWR1-TE CDNA.

[0086] FIG. 19B depicts seed protein, oil and weight of T1 plants from 18 independent transgenic events. *, **, and *** indicate significance at $p < 0.05$, $p < 0.01$, and $p < 0.001$, respectively.

[0087] FIG. 20A depicts the distribution of both POWR1 alleles in soybean population and diversity analyses, by showing PCA of the soybean accessions with assigned germplasm and allele type.

[0088] FIG. 20B depicts comparison of seed oil and protein content and 100-seed weight of *G. max* and *G. soja* accessions carrying POWR1-TE or POWR1-TE.

[0089] FIG. 20C depicts Tajima's D and $\text{Ln}(\pi-G. soja)-\text{Ln}(\pi-G. max)$ between *G. max* and *G. soja* population within the 4.1 Mb region. The vertical solid red line indicates the physical position of POWR1.

[0090] FIG. 20D depicts another Tajima's D and $\text{Ln}(\pi-G. soja)-\text{Ln}(\pi-G. max)$ between *G. max* and *G. soja* population within the 4.1 Mb region. The vertical solid red line indicates the physical position of POWR1.

[0091] FIG. 21A depicts the dynamic interspecific introgressions of POWR1, showing global phylogenetic tree consisting of 548 *G. soja* and *G. max* accessions using genome-wide SoySNP50K SNPs and 1,000 SNPs in the 154-kb region containing POWR1 respectively. Labels (1, 2, 3, 4) in the local tree indicate four clusters of accessions containing unusual genotypes (*G. max*-POWR1-TE (1,2,3), *G. soja*-POWR1-TE (4)) in the tree. The labels in the global tree are corresponding to the labels in the local tree. Notably, cluster 1 in the local tree is split into subclusters 1.1, 1.2, and 1.3 in the global tree.

[0092] FIG. 21B depicts the dynamic interspecific introgressions of POWR1, showing a local phylogenetic tree consisting of 548 *G. soja* and *G. max* accessions using genome-wide SoySNP50K SNPs and 1,000 SNPs in the 154-kb region containing POWR1 respectively. Labels (1, 2, 3, 4) in the local tree indicate four clusters of accessions containing unusual genotypes (*G. max*-POWR1-TE (1,2,3), *G. soja*-POWR1-TE (4)) in the tree. The labels in the global tree are corresponding to the labels in the local tree. Notably, cluster 1 in the local tree is split into subclusters 1.1, 1.2, and 1.3 in the global tree.

[0093] FIG. 21C depicts the pairwise nucleotide distance analyses across a 4.1-Mb region of each *G. max*-POWR1-TE accession with their closest *G. soja*-POWR1-TE accessions. Their clusters and origins are labeled. The pairwise distance is indicated by a color scale from red (close) and green (distant).

[0094] FIG. 21D depicts *G. max* accessions with POWR1-TE alleles transferred from *G. soja*. Top of the panel shows representative accessions from clusters 1.1 (C1.1) and 1.2 (C1.2), 1.3 (S1.3) and 3 (S3) that carry POWR1-TE originated from *G. soja*. The bottom row shows their closest related *G. max*-POWR1-TE accessions. Each accession PI number with corresponding 100-seed weight (W), seed protein content (P) and seed oil content (O) are provided.

[0095] FIG. 21E depicts geographic origins of *G. max*-POWR1-TE accessions and closest *G. soja*-POWR1-TE accessions from the local phylogenetic tree and the closest *G. max*-POWR1-TE accessions from the global tree. Dotted circles include the geographic regions where interspecific transfer might occur.

[0096] FIG. 22 depicts a proposed model of POWR1 in soybean domestication. The insertion of the LINE transposon represents an important event in transition from *G. soja* to *G. max* during soybean domestication. Following TE insertion event, the offspring or diversified populations from the plant containing POWR1+TE were expanded likely from the selection for bigger seeds by ancient farmers. Selection for the larger seed together with other human-favorite domestication traits such as seed shattering resistance and loss of seed dormancy resulted in complete fixation of POWR1+TE in all modern *G. max* accessions with increased oil but reduced protein content in seeds because of its pleiotropy on these traits. The interspecific transfers of POWR1-TE from *G. soja* to *G. max* during the post-domestication was likely driven by local needs for high-protein soybean in Asia. Fixation of POWR1+TE in *G. max* contributes to much larger seeds for modern *G. max* with higher oil and lower protein content than those of contemporary *G. soja*.

[0097] FIG. 23A depicts the vector and transgenic plant by showing diagram for the vector used for transformation.

[0098] FIG. 23B depicts the vector and transgenic plant by showing PCR examination for selected lines containing native promoter-driven POWR1-TE. PCR produced 266 bp in transgenic plants, but not in non-transformed soybean. Wm82 plants is used as a negative control.

[0099] FIG. 24 depicts the frequency of POWR1 alleles in a diverse population consisting of 3,956 accessions and the allele effects on protein, oil and seed weight from analyzing their whole genome resequencing data.

[0100] FIG. 25A depicts the subcellular localization of GmCCT34.

[0101] FIG. 25B depicts another subcellular localization of GmCCT34.

[0102] FIG. 26 depicts the seed oil-protein content phenotype of Arabidopsis thaliana T-DNA insertion mutants of the GmPOWR ortholog gene AT1G04500. The top panel shows the AtPOWR1 gene structure with exon regions highlights as a gray box, the arrowheads representing the T-DNA insertion locations for two T-DNA lines, WiscD-*resLox297300_13A.1* and *SALK_036731.1*, respectively. The red rectangle shows the CCT domain location spanning exons three and four. The bar graphs show the oil phenotypes. * denotes the statistical significance (p -value < 0.05).

[0103] FIG. 27 depicts AtPOWR1 expression in the seed coat tissues with red color indicating the AtPOWR1 expression in the seed coat.

DETAILED DESCRIPTION

[0104] The present disclosure is based in part on the identification and characterization of genes encoding CCT motif-containing proteins (CCT proteins) and their comprehensive roles in the regulation of a variety of development and physiological processes critical for multiple agronomically important traits in agricultural plants such as legumes. For instance, the inventors surprisingly discovered a role for a subfamily of CCT proteins in regulating seed protein, seed oil accumulation, and seed weight and field seed yield in economically important legumes such as soybean. The inventors further demonstrated the ability to genetically manipulate these agronomic traits by manipulating expression of the identified CCT proteins. Accordingly, the present disclosure encompasses plants with improved agronomic traits, and compositions and methods for modifying the expression of CCT proteins in a plant to improve an agronomic trait. The present disclosure also encompasses methods of marker-assisted selection (MAS) plant breeding to improve agronomic traits of a plant using molecular markers identified by the inventors through extensive experimentation.

I. Genetically Modified Plants

[0105] One aspect of the present disclosure encompasses a genetically modified plant having an improved agronomic trait. The plant comprises a nucleic acid sequence encoding a CCT motif-containing protein (CCT protein). The nucleic acid sequence encoding the CCT protein comprises a nucleic acid modification that modifies the expression of the CCT protein, thereby improving one or more agronomic traits of the plant. The present disclosure also encompasses agricultural products produced by any of the described genetically modified plants.

(a) Plants

[0106] The present disclosure provides a genetically modified plant having an improved agronomic trait. The plant comprises a nucleic acid sequence encoding a CCT protein. The nucleic acid sequence comprises a nucleic acid modification that modifies the expression of the CCT protein in the plant. As explained in Section I(b) below, CCT proteins are associated with many developmental functions which affect agronomic traits. Accordingly, modifying the expression of the CCT protein in the plant can be used to improve an agronomic trait of the plant. CCT proteins, nucleic acid sequences encoding CCT proteins, and nucleic acid modifications that modify the expression of the CCT protein in the plant can be as described in Section I(b) herein below.

[0107] As used herein, a “plant” refers to any of various photosynthetic, eukaryotic multi-cellular organisms of the kingdom Plantae, characteristically producing embryos, containing chloroplasts, having cellulose cell walls and lacking locomotion. As used herein, a “plant” includes any plant or part of a plant at any stage of development, including seeds, suspension cultures, plant cells, embryos, meristematic regions, callus tissue, leaves, roots, shoots, gametophytes, sporophytes, pollen, microspores, and progeny thereof. Also included are cuttings, and cell or tissue cultures. As used in conjunction with the present disclosure, plant tissue includes, without limitation, whole plants, plant cells, plant organs, e.g., leaves, stems, roots, meristems,

plant seeds, protoplasts, callus, cell cultures, and any groups of plant cells organized into structural and/or functional units.

[0108] Non-limiting examples of suitable plants may include, for example, species of the Family Gramineae, including *Sorghum bicolor* and *Zea mays*; species of the genera: *Cucurbita*, *Rosa*, *Vitis*, *Juglans*, *Fragaria*, *Lotus*, *Medicago*, *Onobrychis*, *Trifolium*, *Trigonella*, *Vigna*, *Citrus*, *Linum*, *Geranium*, *Manihot*, *Daucus*, *Arabidopsis*, *Brassica*, *Raphanus*, *Sinapis*, *Atropa*, *Capsicum*, *Datura*, *Hyoscyamus*, *Lycopersicon*, *Nicotiana*, *Solanum*, *Petunia*, *Digitalis*, *Majorana*, *Cichorium*, *Helianthus*, *Lactuca*, *Bromus*, *Asparagus*, *Antirrhinum*, *Heterocallis*, *Nemesis*, *Pelargonium*, *Panicum*, *Pennisetum*, *Ranunculus*, *Senecio*, *Salpiglossis*, *Cucumis*, *Browaalia*, *Glycine*, *Pisum*, *Phaseolus*, *Lolium*, *Oryza*, *Avena*, *Hordeum*, *Secale*, and *Triticum*.

[0109] In some embodiments, plants may include, for example, those from corn (*Zea mays*), canola (*Brassica napus*, *Brassica rapa* ssp.), *Brassica* species useful as sources of seed oil, alfalfa (*Medicago sativa*), rice (*Oryza sativa*), rye (*Secale cereale*), sorghum (*Sorghum bicolor*, *Sorghum vulgare*), millet (e.g., pearl millet (*Pennisetum glaucum*), proso millet (*Panicum miliaceum*), foxtail millet (*Setaria italica*), finger millet (*Eleusine coracana*), sunflower (*Helianthus annuus*), safflower (*Carthamus tinctorius*), wheat (*Triticum aestivum*), duckweed (*Lemna*), soybean (*Glycine max*), tobacco (*Nicotiana tabacum*), potato (*Solanum tuberosum*), peanuts (*Arachis hypogaea*), cotton (*Gossypium barbadense*, *Gossypium hirsutum*), sweet potato (*Ipomoea batatas*), cassava (*Manihot esculenta*), coffee (*Coffea* spp.), coconut (*Cocos nucifera*), pineapple (*Ananas comosus*), citrus trees (*Citrus* spp.), cocoa (*Theobroma cacao*), tea (*Camellia sinensis*), banana (*Musa* spp.), avocado (*Persea americana*), fig (*Ficus casica*), guava (*Psidium guajava*), mango (*Mangifera indica*), olive (*Olea europaea*), papaya (*Carica papaya*), cashew (*Anacardium occidentale*), macadamia (*Macadamia* spp.), almond (*Prunus amygdalus*), sugar beets (*Beta vulgaris*), sugarcane (*Saccharum* spp.), oats, barley, vegetables, ornamentals, and conifers.

[0110] Non-limiting examples of suitable vegetable plants may include, for example, tomatoes (*Lycopersicon esculentum*), lettuce (e.g., *Lactuca sativa*), green beans (*Phaseolus vulgaris*), lima beans (*Phaseolus limensis*), peas (*Lathyrus* spp.), and members of the genus *Cucumis* such as cucumber (*C. sativus*), cantaloupe (*C. cantalupensis*), and musk melon (*C. melo*).

[0111] Non-limiting examples of suitable ornamental plants may include, for example, azalea (*Rhododendron* spp.), hydrangea (*Macrophylla hydrangea*), hibiscus (*Hibiscus rosasanensis*), roses (*Rosa* spp.), tulips (*Tulipa* spp.), daffodils (*Narcissus* spp.), petunias (*Petunia hybrida*), carnation (*Dianthus caryophyllus*), poinsettia (*Euphorbia pulcherrima*), and chrysanthemum.

[0112] Non-limiting examples of suitable conifer plants may include, for example, loblolly pine (*Pinus taeda*), slash pine (*Pinus elliotii*), ponderosa pine (*Pinus ponderosa*), lodgepole pine (*Pinus contorta*), Monterey pine (*Pinus radiata*), Douglas-fir (*Pseudotsuga menziesii*), Western hemlock (*Isuga canadensis*), Sitka spruce (*Picea glauca*), redwood (*Sequoia sempervirens*), silver fir (*Abies amabilis*), balsam fir (*Abies balsamea*), Western red cedar (*Thuja plicata*), and Alaska yellow-cedar (*Chamaecyparis nootkatensis*).

[0113] Non-limiting examples of suitable forage and turf grass may include, for example, alfalfa (*Medicago s. sp.*), orchard grass, tall fescue, perennial ryegrass, creeping bent grass, and redtop.

[0114] Non-limiting examples of suitable crop plants and model plants may include, for example, *Arabidopsis*, corn, rice, alfalfa, sunflower, canola, soybean, cotton, peanut, sorghum, wheat, tobacco, and lemna.

[0115] In some aspects, the plant is a legume (fabacea). Non-limiting examples of suitable leguminous plants may include, for example, guar, locust bean, fenugreek, soybean (*Glycine*), garden beans, cowpea, mungbean, lima bean, fava bean, lentils, chickpea, peanuts (*Arachis sp.*), crown vetch (*Vicia sp.*), hairy vetch, adzuki bean, lupine (*Lupinus sp.*), trifolium, common bean (*Phaseolus sp.*), field bean (*Pisum sp.*), clover (*Melilotus sp.*), Lotus, trefoil, lens, and false indigo.

[0116] In some aspects, the plant is a soybean (*Glycine sp.*). Soybean is one of the most important seed crops grown worldwide. It was domesticated from wild soybean (*G. soja*) in East Asia about 6,000-9,000 years ago. Domestication and improvement have shaped soybean as the most important dual-function crop to provide both highly valuable seed protein and oil, which together account for almost all of soybean economic value. Non-limiting examples of *Glycine sp.* include *Glycine hispida*, *Glycine max*, and *Glycine soja*. In another aspect, the plant is *Glycine hispida*. In some aspects, the soybean plant is a domesticated soybean plant. In one aspect, the plant is *Glycine max*.

(b) Agronomic Traits

[0117] Any agronomic trait of a plant can be improved by regulating the expression of one or more CCT protein provided the trait depends on the expression of a CCT protein. Non-limiting examples of agronomic traits that can be improved using compositions and methods of the instant disclosure can be an agronomic trait of Table 14. In some aspects, the agronomic trait is seed quality, seed protein content, seed protein composition, seed oil content, seed oil composition, yield, seed set, response to photoperiod, abiotic stress tolerance, biotic stress tolerance, flowering time and maturity, regulation of circadian clock light response-related flowering, high latitude adaptation, or any combination thereof.

[0118] In some aspects, the plant is soybean. In some aspects, the agronomic trait is seed quality, seed protein content, seed protein composition, seed oil content, seed oil composition, yield, seed set, response to photoperiod, abiotic stress tolerance, biotic stress tolerance, flowering time and maturity, regulation of circadian clock light response-related flowering, high latitude adaptation, or any combination thereof.

[0119] Seed protein content, oil content, and yield are considered as three of the most important traits in soybean improvement. On average, commodity-type soybean varieties contain about 40% seed protein and 20% seed oil. However, the three traits vary greatly in wild soybean populations and often correlate with each other. Seed protein frequently shows a negative correlation with seed oil content and yield. However, its underlying genetic mechanism remains largely unknown. The complex correlation of the three important traits poses a great challenge in simultaneously improving both the soybean seed quality traits and yield to increase the overall economic value of soybean. In

addition, cultivated soybean also contains higher seed yield and oil content, but lower protein content than their ancestry, wild soybean. The identification of the CCT proteins in soybean that underlie these important traits by the inventors after extensive experimentation provides the genetic and molecular basis underlying the three traits and their trait correlation. CCT proteins that underlie seed oil content, seed protein content, seed weight, or any combination thereof can be as described in Section I(c).

(c) CCT Family of Proteins

[0120] A plant of the instant disclosure comprises a nucleic acid sequence encoding a CCT motif-containing protein (CCT protein), any variant thereof, or any combination thereof. A CCT protein variant can comprise a naturally occurring variant of a CCT protein, an ortholog of a CCT protein, a paralog of a CCT protein, a CCT protein comprising a loss-of-function mutation, a CCT protein comprising altered expression in the plant, a CCT protein comprising an introduced mutation, or any combination thereof. Non-limiting examples of CCT protein variants include a naturally occurring variant of the CCT protein, an ortholog of the CCT protein, a paralog of the CCT protein, a CCT protein comprising a loss-of-function mutation, a CCT protein comprising altered expression in the plant, a CCT protein comprising an introduced mutation, or any combination thereof.

[0121] Proteins comprising a CCT motif (CCT proteins) were initially identified in three proteins in *Arabidopsis thaliana*, namely CO (CONSTANS), COL (CO-LIKE) and TOC1 (TIMING OF CAB1). CCT proteins play comprehensive roles in the regulation of a variety of development and physiological processes. The CCT motif comprises about a 43-amino acid conserved sequence in the carboxy-terminus of the proteins. CCT proteins form a large family of proteins in plants with demonstrated roles in adaptation or agronomic traits. Proteins comprising CCT domains are generally classified into three subfamilies: (1) CMF (CCT motif family) containing a single CCT domain, (2) COL proteins carrying an additional one or two B-box (BBOX) domains, and (3) PRR (Pseudo Response Regulator) proteins also containing a response regulator (REC) domain. Accordingly, a CCT protein of the instant disclosure can be a CCT protein classified into the CMF sub-family of CCT proteins, a CCT protein classified into the COL sub-family of CCT proteins, a CCT protein classified into the PRR sub-family of CCT proteins, any variants thereof, or any combination thereof. In some aspects, the CCT protein is a protein classified in the CMF sub-family of CCT proteins. In some aspects, the CCT protein is a protein classified in the COL sub-family of CCT proteins. In some aspects, the CCT protein is a protein classified in the PRR sub-family of CCT proteins.

[0122] A CCT protein can be a single-CCT domain polypeptide, a 1 or 2×BBOX-CCT domain polypeptide, a REC-CCT domain polypeptide, a TIFY CCT-ZnF_GATA domain polypeptide, a CCT protein comprising non-canonical domains, any variants thereof, or any combination thereof. Non-limiting examples of CCT proteins comprising non-canonical domains include DUF740-DUF740-CCT in Vang06g17920 from adzuki bean, Adaptin_N-CCT in Psat0s3732g0120 from pea, S_TKc-CCT in Ca. 14621 from chickpea, any variant thereof, or any combination thereof.

[0123] CCT proteins of the instant disclosure can be selected from a CCT protein of Table 2 any variants thereof, or any combination thereof. Genes interacting with and genes in the biological pathways underlying the CCT genes can also be genetically modified to improve the traits.

[0124] As explained in Section I(a) herein above, CCT proteins are used to improve agronomic traits. In some aspects, the improved agronomic trait is an agronomic trait associated with a QTL of Table 15. In some aspects, the agronomic trait is seed quality, and the CCT protein is encoded by a nucleic acid sequence comprising a gene of Table 5. In some aspects, the agronomic trait is seed set and the CCT protein is encoded by a nucleic acid sequence comprising a gene of Table 6. In some aspects, the agronomic trait is abiotic/biotic stress tolerance and the CCT protein is encoded by a nucleic acid sequence comprising a gene of Table 7. In some aspects, the agronomic trait is flowering time and the CCT protein is encoded by a nucleic acid sequence comprising a gene of Table 8. In some aspects, the agronomic trait is development-related traits and the CCT protein is encoded by a nucleic acid sequence comprising a gene of Table 9.

[0125] In some aspects, the plant is a soybean plant. In some aspects, a CCT protein of the instant disclosure is a CCT protein of Table 1. In some aspects, the agronomic trait is seed oil content, seed protein content, seed weight, or any combination thereof. In some aspects, the CCT protein is a protein of Table 10. In some aspects, a CCT protein of the instant disclosure is GmCCT05 or any variant thereof. In some aspects, a CCT protein of the instant disclosure is GmCCT34 (POWR2), GmCCT67 (POWR1), GmCCT35 (POWR3), GmCCT69 (POWR4), any variant thereof, or any combination thereof.

[0126] In some aspects, the CCT protein is GmCCT67 (POWR1). When the CCT protein is GmCCT67 (POWR1), reducing the expression of the GmCCT67 protein can increase the level of oil in soybean seeds. In some aspects, reducing the expression of the GmCCT67 protein in a soybean plant increases the level of oil in soybean seeds by about 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 1% 7, 18%, 19%, or about 20% w/w or more when compared to the level of oil in seeds of the plant before the level of expression of the GmCCT67 protein is reduced in the plant. When the CCT protein is GmCCT67 (POWR1), reducing the expression of the GmCCT67 protein can also reduce the level of protein in soybean seeds. In some aspects, reducing the expression of the GmCCT67 protein in a soybean plant reduces the level of protein in soybean seeds by about 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, or about 20% w/w or more when compared to the level of protein in seeds of the plant before the level of expression of the GmCCT67 protein is reduced in the plant.

[0127] In some aspects, the GmCCT67 protein comprises an amino acid sequence comprising about 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity with the amino acid sequence of SEQ ID NO: 1. In some aspects, the GmCCT67 (POWR1) protein comprises an amino acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the amino acid sequence of SEQ ID NO: 1. In some aspects, GmCCT67 is encoded by a nucleic acid sequence

comprising about 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 2. In some aspects, the GmCCT67 (POWR1) protein is encoded by a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 2.

[0128] In some aspects, the nucleic acid modification in the nucleic acid sequence encoding the GmCCT67 protein comprises a transposable element (TE) insertion. In some aspects, the nucleic acid sequence comprising the TE insertion comprises at least about 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 3. In some aspects, the nucleic acid sequence encoding the GmCCT67 CCT protein comprises at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 3.

[0129] In some aspects, the nucleic acid modification in the nucleic acid sequence encoding the GmCCT67 protein comprises an expression construct for expression of the GmCCT67 protein, wherein the expression construct comprises a nucleic acid sequence encoding the GmCCT67 protein operably linked to a promoter. In some aspects, the promoter is a ubiquitin promoter or a native promoter.

[0130] In some aspects, the expression construct for expression of GmCCT67 comprises a nucleic acid sequence comprising about 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 4. In some aspects, the expression construct for expression of GmCCT67 POWR1 comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 4.

[0131] In some aspects, the CCT protein is GmCCT34 (POWR2). When the CCT protein is GmCCT34 (POWR2), reducing the expression of the GmCCT34 protein can reduce the level of oil in soybean seeds. In some aspects, reducing the expression of the GmCCT34 protein in a soybean plant increases the level of oil in soybean seeds by about 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, or about 20% w/w or more when compared to the level of oil in seeds of the plant before the level of expression of the GmCCT34 protein is reduced in the plant. When the CCT protein is GmCCT34 (POWR2), reducing the expression of the GmCCT34 protein can also reduce the level of protein in soybean seeds. In some aspects, reducing the expression of the GmCCT34 protein in a soybean plant reduces the level of protein in soybean seeds by about 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, or about 20% w/w or more when compared to the level of protein in seeds of the plant before the level of expression of the GmCCT34 protein is reduced in the plant.

[0132] In some aspects, the GmCCT34 (POWR2) protein comprises an amino acid sequence comprising about 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%,

86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity with the amino acid sequence of SEQ ID NO: 5. In some aspects, the GmCCT34 (POWR2) protein comprises an amino acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the amino acid sequence of SEQ ID NO: 5. In some aspects, GmCCT34 (POWR2) is encoded by a nucleic acid sequence comprising about 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 6. In some aspects, the GmCCT34 (POWR2) protein is encoded by a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 6.

[0133] In some aspects, the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises an expression construct for expression of the GmCCT34 protein, wherein the expression construct comprises a nucleic acid sequence encoding the GmCCT34 protein operably linked to a promoter. In some aspects, the promoter is a ubiquitin promoter or a native promoter. In some aspects, the expression construct comprises a nucleic acid sequence encoding the GmCCT34 protein a GmCCT34 variant selected from a wild soybean (*G. soja*, PI479752 accession).

[0134] In some aspects, the expression construct for expression of GmCCT34 POWR2 comprises a nucleic acid sequence comprising about 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 7. In some aspects, the expression construct for expression of GmCCT34 POWR2 comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 7.

[0135] In some aspects, the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid modification generated using a CRISPR/Cas programmable nucleic acid modification system targeted to a nucleic acid sequence in a nucleotide sequence encoding the GmCCT34 protein. In some aspects, the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 8 to 16 or any combination thereof. In some aspects, the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 8 to 10 or any combination thereof. In one aspect, the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 11 to 13. In another aspect, the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 14 to 16.

[0136] In some aspects, the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a genomic deletion of a 1.3-Mb region of the

soybean genome (Chr10: 35253890-36584337) comprising a nucleic acid sequence encoding the GmCCT34 CCT protein.

[0137] In some aspects, The genetically modified plant of claim 8, wherein the CCT protein is GmCCT35 (POWR3). In some aspects, the GmCCT35 protein comprises an amino acid sequence comprising at least about 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity with the amino acid sequence of SEQ ID NO: 25. In some aspects, the GmCCT35 protein comprises an amino acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the amino acid sequence of SEQ ID NO: 25. In additional aspects, the GmCCT35 protein is encoded by a nucleic acid sequence comprising at least about 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 26. In additional aspects, the GmCCT35 protein is encoded by a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 26. In yet other aspects, the plant is a soybean species (*Glycine max*, *hispida*), the CCT protein is GmCCT35 (POWR3), the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 27.

[0138] In some aspects, the CCT protein is GmCCT69 (POWR4). In some aspects, the GmCCT69 protein comprises an amino acid sequence comprising at least about 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity with the amino acid sequence of SEQ ID NO: 28. In some aspects, the GmCCT69 protein comprises an amino acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the amino acid sequence of SEQ ID NO: 28. In additional aspects, the GmCCT69 protein is encoded by a nucleic acid sequence comprising at least about 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 29. In additional aspects, the GmCCT69 protein is encoded by a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 29. In yet other aspects, the plant is a soybean species (*Glycine max*, *hispida*), the CCT protein is GmCCT69 (POWR4), the nucleic acid modification in the nucleic acid sequence encoding the GmCCT69 protein comprises a nucleic acid sequence comprising at least about 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 30. In yet other aspects, the plant is a soybean species (*Glycine max*, *his-*

pida), the CCT protein is GmCCT69 (POWR4), the nucleic acid modification in the nucleic acid sequence encoding the GmCCT69 protein comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 30.

[0139] The mutations in POWR3 (GmCCT35) and POWR4 (GmCCT69) genes were generated by using CRISPR-Cas9 mediated gene editing approach. For POWR3, the gRNAs were designed to target exon 2 and 3 regions. The CRISPR-Cas9 mediated 4 bp deletion (in exon 3 by using gRNA-ctggcagaacttcagccccc SEQ ID NO: 34), and 39 bp deletion (in exon 2 by using gRNA-ccaggactgagataaagtgc SEQ ID NO: 35) were generated. Similarly, for POWR4, exon 2 region was targeted by gRNA-ccaggactgagataaagtgc SEQ ID NO: 36, which generated a 39 bp deletion.

[0140] In some aspects, the CCT protein is AtPOWR1, any variant thereof, or any combination thereof. In some aspects, the nucleic acid modification reduces the expression of the AtPOWR1 protein in the plant. When the nucleic acid modification reduces the expression of the AtPOWR1 protein in the plant, the oil content of the seeds can be increased and the protein content of the seeds can be reduced. In some aspects, the AtPOWR1 protein comprises an amino acid sequence comprising at least about 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity with the amino acid sequence of SEQ ID NO: 33. In some aspects, the AtPOWR1 protein comprises an amino acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the amino acid sequence of SEQ ID NO: 33. In other aspects, the AtPOWR1 protein is encoded by a nucleic acid sequence comprising at least about 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 31. In other aspects, the AtPOWR1 protein is encoded by a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 31.

(d) Aspects of Plants

[0141] One aspect of the present disclosure encompasses a genetically modified plant having an improved agronomic trait. The plant comprises a nucleic acid sequence encoding a CCT motif-containing protein (CCT protein) selected from CCT proteins of Table 2, a variant of any thereof, or any combination thereof. The nucleic acid sequence encoding the CCT protein comprises a nucleic acid modification, wherein the nucleic acid modification modifies the expression of the CCT protein in the plant thereby improving the agronomic trait of the plant. The nucleic acid modification can be a nucleic acid sequence comprising a single nucleotide polymorphism of Table 4, Table 10, or any combination thereof.

[0142] A CCT protein variant can comprise a naturally occurring variant of the CCT protein, an ortholog of the CCT protein, a paralog of the CCT protein, a CCT protein comprising a loss-of-function mutation, a CCT protein having altered expression in the plant, a CCT protein comprising an introduced mutation, a functional fragment, or any

combination thereof. In some aspects, the CCT protein is a single-CCT domain polypeptide, a 1 or 2×BBOX-CCT domain polypeptide, a REC-CCT domain polypeptide, a TIFY CCT-ZnF_GATA domain polypeptide, a CCT protein comprising one or more non-canonical domains, any variants thereof, or any combination thereof. The CCT protein comprising non-canonical domains can be DUF740-DUF740-CCT in Vang06g17920 from adzuki bean, Adaplin_N-CCT in Psat0s3732g0120 from pea, S_TKc-CCT in Ca. 14621 from chickpea, any variants thereof, or any combination thereof. In some aspects, the CCT protein is a single-CCT domain polypeptide.

[0143] In some aspects, the CCT protein is a CCT protein of Table 1. In some aspects, the CCT protein is GmCCT05 and wherein the agronomic trait is drought tolerance. In some aspects, the agronomic trait is seed protein, oil content, 100-seed weight, or any combination thereof, and the CCT protein is GmCCT34 (POWR2), GmCCT67 (POWR1), GmCCT35 (POWR3), GmCCT69 (POWR4), any variant thereof, or any combination thereof. In one aspect, the CCT protein is GmCCT35 (POWR3). In another aspect, the CCT protein is GmCCT69 (POWR4).

[0144] The agronomic trait can be seed quality, seed protein content, seed protein composition, seed oil content, seed oil composition, yield, seed set, response to photoperiod, abiotic stress tolerance, biotic stress tolerance, flowering time and maturity, regulation of circadian clock light response-related flowering, high latitude adaptation, or any combination thereof. In some aspects, the improved agronomic trait is an agronomic trait of Table 14. In some aspects, the improved agronomic trait is an agronomic trait associated with a QTL of Table 15. In some aspects, the agronomic trait is (a) seed quality and the CCT protein is encoded by a nucleic acid sequence comprising a gene of Table 5; (b) yield-related traits and the CCT protein is encoded by a nucleic acid sequence comprising a gene of Table 6; (c) response to abiotic/biotic stress tolerance and the CCT protein is encoded by a nucleic acid sequence comprising a gene of Table 7; (d) flowering time and maturity and the CCT protein is encoded by a nucleic acid sequence comprising a gene of Table 8; and (e) development-related traits and the CCT protein is encoded by a nucleic acid sequence comprising a gene of Table 9.

[0145] In some aspects, the CCT protein is GmCCT67 (POWR1). A nucleic acid modification can reduce the expression of the GmCCT67 protein in the plant. When the nucleic acid modification reduces the expression of the GmCCT67 protein in the plant, the oil content of the seeds is increased by about 0.5% to about 5% wt/wt and wherein the protein content of the seeds is reduced by about 1% wt/wt to about 20% wt/wt. Alternatively, a nucleic acid modification can increase the expression of the GmCCT67 protein in the plant. When the nucleic acid modification increases the expression of the GmCCT67 protein in the plant, the oil content of the seeds can be decreased by about 0.5% to about 5% wt/wt and wherein the protein content of the seeds is increased by about 1% wt/wt to about 20% wt/wt.

[0146] The GmCCT67 protein can comprise an amino acid sequence comprising at least about 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity with the amino acid sequence of SEQ ID NO: 1, and can be encoded by a nucleic

acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 2. The GmCCT67 protein can also comprise an amino acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the amino acid sequence of SEQ ID NO: 1, and can be encoded by a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 2.

[0147] In some aspects, the nucleic acid modification in the nucleic acid sequence encoding the GmCCT67 protein comprises a transposable element (TE) insertion. In one aspect, the nucleic acid sequence comprising the TE insertion comprises at least about 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 3. In one aspect, the nucleic acid sequence comprising the TE insertion comprises at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 3.

[0148] In some aspects, the nucleic acid modification in the nucleic acid sequence encoding the GmCCT67 protein comprises an expression construct for expression of the GmCCT67 protein, wherein the expression construct comprises a nucleic acid sequence encoding the GmCCT67 protein operably linked to a ubiquitin promoter or a native promoter. In one aspect, the expression construct comprises a nucleic acid sequence comprising at least about 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 4. In one aspect, the expression construct comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 4.

[0149] In some aspects, the CCT protein is GmCCT34 (POWR2). When the CCT protein is GmCCT34, the nucleic acid modification reduces the expression of GmCCT34 (POWR2) in the plant, and the oil content of the seeds is increased by about 0.5% to about 5% wt/wt and wherein the protein content of the seeds is reduced by about 1% wt/wt to about 20% wt/wt.

[0150] The GmCCT34 protein can comprise an amino acid sequence comprising at least about 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity with the amino acid sequence of SEQ ID NO: 5, and can be encoded by a nucleic acid sequence comprising at least about 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 6. The GmCCT34 protein can comprise an amino acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the amino acid sequence of SEQ ID NO: 5, and can be encoded by a nucleic acid sequence comprising at least about 75% or more, at

least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 6.

[0151] The nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein can comprise an expression construct for expression of the GmCCT34 protein, wherein the expression construct comprises a nucleic acid sequence encoding the GmCCT34 protein operably linked to a promoter. In some aspects, the expression construct for expression of GmCCT34 POWR2 comprises a nucleic acid sequence comprising at least about 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 7. In some aspects, the expression construct for expression of GmCCT34 POWR2 comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 7.

[0152] In some aspects, the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a genomic deletion of a 1.3-Mb region of the soybean genome (Chr10: 35253890-36584337) comprising a nucleic acid sequence encoding the GmCCT34 CCT protein. In one aspect, the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid modification generated using a CRISPR/Cas programmable nucleic acid modification system targeted to a nucleic acid sequence in a nucleotide sequence encoding the GmCCT34 protein. The nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein can comprise a nucleic acid sequence of SEQ ID NO: 8 to 10 or any combination thereof, a nucleic acid sequence of SEQ ID NO: 11 to 13 or any combination thereof, or a nucleic acid sequence of SEQ ID NO: 14 to 16 or any combination thereof.

[0153] The plant can be a legume (Fabaceae) such as common bean, cowpea, soybean, chickpea, pea, or Medicago. In some aspects, the legume is a soybean species (*Glycine max*, *hispida*). In some aspects, the CCT protein is GmCCT67 (POWR1) and wherein the nucleic acid modification reduces the expression of the GmCCT67 protein in the plant. In other aspects, the CCT protein is GmCCT34 (POWR1) and wherein the nucleic acid modification reduces the expression of the GmCCT34 protein in the plant.

[0154] In some aspects, the plant is a soybean species (*Glycine max*, *hispida*), the CCT protein is GmCCT67 (POWR1), the nucleic acid modification in the nucleic acid sequence encoding the GmCCT67 protein comprises an expression construct for expression of the GmCCT67 protein, wherein the expression construct comprises a nucleic acid sequence encoding the GmCCT67 protein operably linked to a promoter, and wherein the nucleic acid modification increases the expression of the GmCCT67 protein in the plant. In some aspects, the oil content of the seeds is decreased by about 0.5% to about 5% wt/wt and wherein the protein content of the seeds is increased by about 1% wt/wt to about 20% wt/wt.

[0155] The plant can be a soybean species (*Glycine max*, *hispida*), the CCT protein is GmCCT67 (POWR1), the nucleic acid modification in the nucleic acid sequence encoding the GmCCT67 protein comprises a transposable element (TE) insertion, and wherein the nucleic acid modi-

fication reduces the expression of the GmCCT67 protein in the plant. In some aspects, the oil content of the seeds is increased by about 0.5% to about 5% wt/wt and wherein the protein content of the seeds is reduced by about 1% wt/wt to about 20% wt/wt.

[0156] In some aspects, the plant is a soybean species (*Glycine max*, *hispida*), the CCT protein is GmCCT34 (POWR2), the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises an expression construct for expression of the GmCCT34 protein, wherein the expression construct comprises a nucleic acid sequence encoding the GmCCT34 protein operably linked to a promoter, and wherein the nucleic acid modification increases the expression of the GmCCT34 protein in the plant. In one aspect, the oil content of the seeds is decreased by about 0.5% to about 5% wt/wt and wherein the protein content of the seeds is increased by about 1% wt/wt to about 20% wt/wt.

[0157] In some aspects, GmCCT34 (POWR2), the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a genomic deletion of a 1.3-Mb region of the soybean genome (Chr10: 35253890-36584337) comprising a nucleic acid sequence encoding the GmCCT34 CCT protein or a nucleic acid sequence of SEQ ID NO: 8 to 16 or any combination thereof, and wherein the nucleic acid modification reduces the expression of the GmCCT34 protein in the plant. In one aspect, the oil content of the seeds is increased by about 0.5% to about 5% wt/wt and wherein the protein content of the seeds is reduced by about 1% wt/wt to about 20% wt/wt.

[0158] In some aspects, the plant is a soybean species (*Glycine max*, *hispida*), the CCT protein is GmCCT35 (POWR3), the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 27.

[0159] In other aspects, the plant is a soybean species (*Glycine max*, *hispida*), the CCT protein is GmCCT69 (POWR4), the nucleic acid modification in the nucleic acid sequence encoding the GmCCT69 protein comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 30.

[0160] In yet other aspects, the plant is a soybean species (*Glycine max*, *hispida*), wherein the CCT protein is GmCCT67 (POWR1) and the nucleic acid modification in the nucleic acid sequence encoding the GmCCT67 protein comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 3; the CCT protein is GmCCT34 (POWR2), wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a genomic deletion of a 1.3-Mb region of the soybean genome (Chr10: 35253890-36584337) comprising a nucleic acid sequence encoding the GmCCT34 CCT protein, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 8 to 10, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid

sequence of SEQ ID NO: 11 to 13, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 14 to 16, or any combination thereof.

[0161] In additional aspects, the plant is a soybean species (*Glycine max*, *hispida*), wherein the CCT protein is GmCCT67 (POWR1) and the nucleic acid modification in the nucleic acid sequence encoding the GmCCT67 protein comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 3; the CCT protein is GmCCT34 (POWR2), wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a genomic deletion of a 1.3-Mb region of the soybean genome (Chr10: 35253890-36584337) comprising a nucleic acid sequence encoding the GmCCT34 CCT protein, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 8 to 10, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 11 to 13, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 14 to 16, or any combination thereof; and the CCT protein is GmCCT69 (POWR4) and the nucleic acid modification in the nucleic acid sequence encoding the GmCCT69 protein comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 30.

[0162] In some aspects, the plant is a soybean species (*Glycine max*, *hispida*), wherein the CCT protein is GmCCT67 (POWR1) and the nucleic acid modification in the nucleic acid sequence encoding the GmCCT67 protein comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 3; and the CCT protein is GmCCT35 (POWR3) and the nucleic acid modification in the nucleic acid sequence encoding the GmCCT69 protein comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 27.

[0163] In additional aspects, the plant is a soybean species (*Glycine max*, *hispida*), wherein the CCT protein is GmCCT34 (POWR2), wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a genomic deletion of a 1.3-Mb region of the soybean genome (Chr10: 35253890-36584337) comprising a nucleic acid sequence encoding the GmCCT34 CCT protein, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 8 to 10, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 11 to 13, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 14 to 16, or any combination thereof; and the CCT protein is GmCCT35 (POWR3) and the nucleic acid modification in the nucleic acid sequence encoding the

GmCCT69 protein comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 27.

[0164] In some aspects, the plant is a soybean species (*Glycine max*, *hispida*), wherein the CCT protein is GmCCT67 (POWR1) and the nucleic acid modification in the nucleic acid sequence encoding the GmCCT67 protein comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 3; the CCT protein is GmCCT34 (POWR2), wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a genomic deletion of a 1.3-Mb region of the soybean genome (Chr10: 35253890-36584337) comprising a nucleic acid sequence encoding the GmCCT34 CCT protein, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 8 to 10, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 11 to 13, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 14 to 16, or any combination thereof; and the CCT protein is GmCCT69 (POWR4) and the nucleic acid modification in the nucleic acid sequence encoding the GmCCT69 protein comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 30.

[0165] In additional aspects, the plant is a soybean species (*Glycine max*, *hispida*), wherein the CCT protein is GmCCT67 (POWR1) and the nucleic acid modification in the nucleic acid sequence encoding the GmCCT67 protein comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 3; and the CCT protein is GmCCT35 (POWR3) and the nucleic acid modification in the nucleic acid sequence encoding the GmCCT69 protein comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 27.

[0166] In yet other aspects, the plant is a soybean species (*Glycine max*, *hispida*), wherein the CCT protein is GmCCT67 (POWR1) and the nucleic acid modification in the nucleic acid sequence encoding the GmCCT67 protein comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 3; the CCT protein is GmCCT34 (POWR2), wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a genomic deletion of a 1.3-Mb region of the soybean genome (Chr10: 35253890-36584337) comprising a nucleic acid sequence encoding the GmCCT34 CCT protein, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 8 to 10, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid

sequence of SEQ ID NO: 11 to 13, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 14 to 16, or any combination thereof; and the CCT protein is GmCCT35 (POWR3) and the nucleic acid modification in the nucleic acid sequence encoding the GmCCT69 protein comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 27.

[0167] In some aspects, the plant is a soybean species (*Glycine max*, *hispida*), wherein the CCT protein is GmCCT67 (POWR1) and the nucleic acid modification in the nucleic acid sequence encoding the GmCCT67 protein comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 3; the CCT protein is GmCCT34 (POWR2), wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a genomic deletion of a 1.3-Mb region of the soybean genome (Chr10: 35253890-36584337) comprising a nucleic acid sequence encoding the GmCCT34 CCT protein, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 8 to 10, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 11 to 13, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 14 to 16, or any combination thereof; the CCT protein is GmCCT35 (POWR3) and the nucleic acid modification in the nucleic acid sequence encoding the GmCCT69 protein comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 27; and the CCT protein is GmCCT69 (POWR4) and the nucleic acid modification in the nucleic acid sequence encoding the GmCCT69 protein comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 30.

[0168] In some aspects, the plant is *Arabidopsis thaliana*. When the plant is *Arabidopsis thaliana*, the CCT protein can be AtPOWR1, any variant thereof, or any combination thereof. In some aspects, the nucleic acid modification reduces the oil content of the seeds is increased and wherein the protein content of the seeds is reduced. In some aspects, the AtPOWR1 protein comprises an amino acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the amino acid sequence of SEQ ID NO: 33. In additional aspects, the AtPOWR1 protein is encoded by a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 31. In yet other aspects, the *Arabidopsis* plant comprises a first T-DNA-insertion mutant of AtPOWR1 (WiscDsLox297300_13A.1, Atcct1), a second T-DNA-insertion mutant of AtPOWR1 (SALK_036731.1; Atcct-2).

II. Engineered Nucleic Acid Modification System

[0169] One aspect of the present disclosure encompasses an engineered nucleic acid modification system for modifying the expression of a CCT protein in a plant. Non-limiting examples of suitable protein expression modification systems include programmable nucleic acid modification systems, an expression construct encoding a protein or variants thereof, and any combination thereof.

[0170] In some aspects, the nucleic acid modification system is an expression construct comprising a nucleotide sequence encoding the CCT protein operably linked to a promoter. Expression constructs comprising a nucleotide sequence encoding the CCT protein operably linked to a promoter can be as described in Section I(c).

[0171] In some aspects, the nucleic acid modification system is a programmable nucleic acid modification system targeted to a sequence within a gene encoding the CCT protein. As used herein, a “programmable nucleic acid modification system” is a system capable of targeting and modifying the nucleic acid or modifying the expression or stability of a nucleic acid to alter a protein or the expression of a protein encoded by the nucleic acid. The programmable nucleic acid modification system can comprise an interfering nucleic acid molecule or a nucleic acid editing system. The programmable protein expression modification system is specifically targeted to a sequence within a gene encoding the CCT protein.

[0172] In some aspects, the programmable expression modification system comprises an interfering nucleic acid (RNAi) molecule having a nucleotide sequence complementary to a target sequence within a gene encoding the CCT protein used to inhibit expression of the CCT protein. RNAi molecules generally act by forming a heteroduplex with a target RNA molecule, which is selectively degraded or “knocked down,” hence inactivating the target RNA. Under some conditions, an interfering RNA molecule can also inactivate a target transcript by repressing transcript translation and/or inhibiting transcription. An interfering RNA is more generally said to be “targeted against” a biologically relevant target, such as a protein, when it is targeted against the nucleic acid encoding the target. For example, an interfering RNA molecule has a nucleotide (nt) sequence which is complementary to an endogenous mRNA of a target gene sequence. Thus, given a target gene sequence, an interfering RNA molecule can be prepared which has a nucleotide sequence at least a portion of which is complementary to a target gene sequence. When introduced into cells, the interfering RNA binds to the target mRNA, thereby functionally inactivating the target mRNA and/or leading to degradation of the target mRNA.

[0173] Interfering RNA molecules include, inter alia, small interfering RNA (siRNA), microRNA (miRNA), piwi-interacting RNA (piRNA), long non-coding RNAs (lncRNAs or lncRNAs), and small hairpin RNAs (shRNA). lncRNAs are widely expressed and have key roles in gene regulation. Depending on their localization and their specific interactions with DNA, RNA and proteins, lncRNAs can modulate chromatin function, regulate the assembly and function of membraneless nuclear bodies, alter the stability and translation of cytoplasmic mRNAs, and interfere with signaling pathways. Piwi-interacting RNA (piRNA) is the largest class of small non-coding RNA molecules expressed in animal cells. piRNAs regulate gene expression through interactions with piwi-subfamily Argonaute proteins.

siRNA are double-stranded RNA molecules, preferably about 19-25 nucleotides in length. When transfected into cells, siRNA inhibit the target mRNA transiently until they are also degraded within the cell. miRNA and siRNA are biochemically and functionally indistinguishable. Both are about the same in nucleotide length with 5'-phosphate and 3'-hydroxyl ends, and assemble into an RNA-induced silencing complex (RISC) to silence specific gene expression. siRNA and miRNA are distinguished based on origin. siRNA is obtained from long double-stranded RNA (dsRNA), while miRNA is derived from the double-stranded region of a 60-70nt RNA hairpin precursor. Small hairpin RNAs (shRNA) are sequences of RNA, typically about 50-80 base pairs, or about 50, 55, 60, 65, 70, 75, or about 80 base pairs in length, that include a region of internal hybridization forming a stem loop structure consisting of a base-pair region of about 19-29 base pairs of double-strand RNA (the stem) bridged by a region of single-strand RNA (the loop) and a short 3' overhang. shRNA molecules are processed within the cell to form siRNA which in turn knock down target gene expression. shRNA can be incorporated into plasmid vectors and integrated into genomic DNA for longer-term or stable expression, and thus longer knock-down of the target mRNA.

[0174] Interfering nucleic acid molecules can contain RNA bases, non-RNA bases, or a mixture of RNA bases and non-RNA bases. For example, interfering nucleic acid molecules provided herein can be primarily composed of RNA bases but also contain DNA bases or non-naturally occurring nucleotides. The interfering nucleic acids can employ a variety of oligonucleotide chemistries. Non-limiting examples of oligonucleotide chemistries include, without limitation, peptide nucleic acid (PNA), linked nucleic acid (LNA), phosphorothioate, 2'O-Me-modified oligonucleotides, and morpholino chemistries, including combinations of any of the foregoing. In general, PNA and LNA chemistries can utilize shorter targeting sequences because of their relatively high target binding strength relative to 2'O-Me oligonucleotides. Phosphorothioate and 2'O-Me-modified chemistries are often combined to generate 2'O-Me-modified oligonucleotides having a phosphorothioate backbone.

[0175] In some aspects, the programmable nucleic acid modification system is a nucleic acid editing system. Such modification system can be used to edit DNA or RNA sequences to repress transcription or translation of an mRNA encoded by the gene, and/or produce mutant proteins with reduced activity or stability. Non-limiting examples of programmable nucleic acid editing systems include, without limit, an RNA-guided clustered regularly interspersed short palindromic repeats (CRISPR)/CRISPR-associated (Cas) (CRISPR/Cas) nuclease system, a CRISPR/CpfI nuclease system, a zinc finger nuclease (ZFN), a transcription activator-like effector nuclease (TALEN), a meganuclease, a ribozyme, or a programmable DNA binding domain linked to a nuclease domain. Other suitable programmable nucleic acid modification systems will be recognized by individuals skilled in the art.

[0176] Such systems rely for specificity on the delivery of exogenous protein(s), and/or a guide RNA (gRNA) or single guide RNA (sgRNA) having a sequence which binds specifically to a gene sequence of interest. When the programmable nucleic acid modification system comprises more than one component, such as a protein and a guide nucleic

acid, the multi-component modification system can be modular, in that the different components may optionally be distributed among two or more nucleic acid constructs as described herein. The system components can be delivered by a plasmid or viral vector or as a synthetic oligonucleotide. More detailed descriptions of programmable nucleic acid editing systems can be as described further below.

[0177] In some aspects, the programmable nucleic acid modification system is a CRISPR/Cas tool modified for transcriptional regulation of a locus. In some aspects, the programmable nucleic acid modification system is a CRISPR/Cas transcriptional regulator driven by cell-specific promoters using a catalytically dead effector (dCAS9) to modulate transcription of a nucleic acid sequence encoding a CCT protein.

[0178] In some aspects, the programmable nucleic acid modification system is a CRISPR/Cas system comprising a guide RNA (gRNA) having a sequence complementary to a target sequence within the nucleotide sequence encoding the CCT protein. In some aspects, the CCT protein is a GmCCT34 protein. In some aspects, the GmCCT34 protein is encoded by a nucleic acid sequence comprising about 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 5. In some aspects, the GmCCT34 (POWR2) protein is encoded by a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 5. When the programmable nucleic acid modification system is a CRISPR/Cas system and the CCT protein is a GmCCT34 protein, the gRNA can comprise a nucleic acid sequence of SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, or any combination thereof.

[0179] In some aspects, the CCT protein is a GmCCT35 protein. When the programmable nucleic acid modification system is a CRISPR/Cas system and the CCT protein is a GmCCT35 protein, the gRNA can comprise a nucleic acid sequence of SEQ ID NO: 34, SEQ ID NO: 35, or a combination thereof.

[0180] In some aspects, the CCT protein is a GmCCT69 protein. When the programmable nucleic acid modification system is a CRISPR/Cas system and the CCT protein is a GmCCT69 protein, the gRNA can comprise a nucleic acid sequence of SEQ ID NO: 36.

[0181] Another aspect of the present disclosure encompasses an engineered nucleic acid modification system for modifying the expression of a CCT protein in a plant. The system comprises a nucleic acid expression construct comprising: a promoter operably linked to a nucleic acid sequence encoding a programmable nucleic acid modification system targeted to a nucleic acid sequence in a nucleotide sequence encoding the CCT protein; or a nucleotide sequence encoding the CCT protein operably linked to a promoter; and wherein expressing the programmable nucleic acid modification system or expressing the CCT protein introduces a nucleic acid modification of the nucleic acid sequence encoding the CCT protein, thereby modifying the expression of a CCT protein in a plant. In some aspects, the engineered nucleic acid modification system further comprises a nucleic acid delivery vector comprising the nucleic acid expression construct for delivering the nucleic acid expression construct to the target cell.

[0182] The CCT protein can be GmCCT34 (POWR2), GmCCT67 (POWR1), GmCCT35 (POWR3), GmCCT69 (POWR4), or any combination thereof. In some aspects, the CCT protein is GmCCT67 (POWR1) encoded by a nucleic acid sequence comprising at least about 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 2. In some aspects, the CCT protein is GmCCT67 (POWR1) encoded by a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 2. The GmCCT67 (POWR1) protein can comprise an amino acid sequence comprising at least about 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity with the amino acid sequence of SEQ ID NO: 1. The GmCCT67 (POWR1) protein can comprise an amino acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the amino acid sequence of SEQ ID NO: 1. The nucleic acid expression construct can comprise a nucleotide sequence encoding a GmCCT67 protein operably linked to a promoter. In some aspects, the expression construct comprises a nucleic acid sequence comprising at least about 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 4. In some aspects, the expression construct comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 4.

[0183] In some aspects, the CCT protein is GmCCT34 encoded by a nucleic acid sequence comprising at least about 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 6. In some aspects, the CCT protein is GmCCT34 encoded by a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 6. The GmCCT34 can comprise an amino acid sequence comprising at least about 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% or more, or 100% sequence identity with the amino acid sequence of SEQ ID NO: 5. The GmCCT34 can comprise an amino acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the amino acid sequence of SEQ ID NO: 5.

[0184] In some aspects, the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises an expression construct for expression of the GmCCT34 protein, wherein the expression construct comprises a nucleic acid sequence encoding the GmCCT34 protein operably linked to a promoter. The expression construct for expression of GmCCT34 POWR2 can comprise a nucleic acid sequence comprising at least about 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%,

87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 7. The expression construct for expression of GmCCT34 POWR2 can comprise a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 7.

[0185] The nucleic acid expression construct can also comprise a promoter operably linked to a nucleic acid sequence encoding a programmable nucleic acid modification system targeted to a nucleic acid sequence in a nucleotide sequence encoding the GmCCT34 protein. The programmable nucleic acid modification system can be CRISPR/Cas system comprising a guide RNA (gRNA) having a sequence complementary to a target sequence within the nucleotide sequence encoding the GmCCT34 protein. In some aspects, the gRNA comprises a nucleic acid sequence of SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, or any combination thereof.

[0186] In some aspects, the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises an expression construct for expression of the GmCCT34 protein, wherein the expression construct comprises a nucleic acid sequence encoding the GmCCT34 protein operably linked to a promoter. The nucleic acid expression construct can comprise a nucleotide sequence comprising at least about 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity with the nucleotide sequence of SEQ ID NO: 4. The nucleic acid expression construct can comprise a nucleotide sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleotide sequence of SEQ ID NO: 4. In some aspects, the nucleic acid expression construct comprises a nucleotide sequence comprising at least about 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity with the nucleotide sequence of SEQ ID NO: 7. In some aspects, the nucleic acid expression construct comprises a nucleotide sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleotide sequence of SEQ ID NO: 7.

i. CRISPR Nuclease Systems.

[0187] The programmable targeting nuclease can be an RNA-guided CRISPR endonuclease system. The CRISPR system comprises a guide RNA or sgRNA to a target sequence at which a protein of the system introduces a double-stranded break in a target nucleic acid sequence, and a CRISPR-associated endonuclease. The gRNA is a short synthetic RNA comprising a sequence necessary for endonuclease binding, and a preselected ~20 nucleotide spacer sequence targeting the sequence of interest in a genomic target. Non-limiting examples of endonucleases include Cas1, Cas1B, Cas2, Cas3, Cas4, Cas5, Cas6, Cas7, Cas8, Cas9 (also known as Csn1 and Csx12), Cas100, Csy1, Csy2, Csy3, Cse1, Cse2, Csc1, Csc2, Csa5, Csn2, Csm2, Csm3, Csm4, Csm5, Csm6, Cmr1, Cmr3, Cmr4, Cmr5, Cmr6, Csb1, Csb2, Csb3, Csx17, Csx14, Csx10, Csx16, CsaX, Csx3, Csx1, Csx15, Csf1, Csf2, Csf3, Csf4, or Cpf1 endonuclease, or a homolog thereof, a recombination of the

naturally occurring molecule thereof, a codon-optimized version thereof, or a modified version thereof, or any combination thereof.

[0188] The CRISPR nuclease system may be derived from any type of CRISPR system, including a type I (i.e., IA, IB, IC, ID, IE, or IF), type II (i.e., IIA, IIB, or IIC), type III (i.e., IIIA or IIIB), or type V CRISPR system. The CRISPR/Cas system may be from *Streptococcus* sp. (e.g., *Streptococcus pyogenes*), *Campylobacter* sp. (e.g., *Campylobacter jejuni*), *Francisella* sp. (e.g., *Francisella novicida*), *Acaryochloris* sp., *Acetohalobium* sp., *Acidaminococcus* sp., *Acidithiobacillus* sp., *Alicyclobacillus* sp., *Allochromatium* sp., *Ammonifex* sp., *Anabaena* sp., *Arthrosira* sp., *Bacillus* sp., *Burkholderiales* sp., *Caldicelulosiruptor* sp., *Candidatus* sp., *Clostridium* sp., *Crocospaera* sp., *Cyanotheca* sp., *Exiguobacterium* sp., *Finegoldia* sp., *Ktedonobacter* sp., *Lactobacillus* sp., *Lyngbya* sp., *Marinobacter* sp., *Methanohalobium* sp., *Microscilla* sp., *Microcoleus* sp., *Microcystis* sp., *Natranaerobius* sp., *Neisseria* sp., *Nitrosococcus* sp., *Nocardioopsis* sp., *Nodularia* sp., *Nostoc* sp., *Oscillatoria* sp., *Polaromonas* sp., *Pelotomaculum* sp., *Pseudoalteromonas* sp., *Petrotoga* sp., *Prevotella* sp., *Staphylococcus* sp., *Streptomyces* sp., *Streptosporangium* sp., *Synechococcus* sp., or *Thermosipho* sp.

[0189] Non-limiting examples of suitable CRISPR systems include CRISPR/Cas systems, CRISPR/Cpf systems, CRISPR/Cmr systems, CRISPR/Csa systems, CRISPR/Csb systems, CRISPR/Csc systems, CRISPR/Cse systems, CRISPR/Csf systems, CRISPR/Csm systems, CRISPR/Csn systems, CRISPR/Csx systems, CRISPR/Csy systems, CRISPR/Csz systems, and derivatives or variants thereof. Preferably, the CRISPR system may be a type II Cas9 protein, a type V Cpf1 protein, or a derivative thereof. In some aspects, the CRISPR/Cas nuclease is *Streptococcus pyogenes* Cas9 (SpCas9), *Streptococcus thermophilus* Cas9 (StCas9), *Campylobacter jejuni* Cas9 (CjCas9), *Francisella novicida* Cas9 (FnCas9), or *Francisella novicida* Cpf1 (FnCpf1).

[0190] In general, a protein of the CRISPR system comprises an RNA recognition and/or RNA binding domain, which interacts with the guide RNA. A protein of the CRISPR system also comprises at least one nuclease domain having endonuclease activity. For example, a Cas9 protein may comprise a RuvC-like nuclease domain and an HNH-like nuclease domain, and a Cpf1 protein may comprise a RuvC-like domain. A protein of the CRISPR system may also comprise DNA binding domains, helicase domains, RNase domains, protein-protein interaction domains, dimerization domains, as well as other domains.

[0191] A protein of the CRISPR system may be associated with guide RNAs (gRNA). The guide RNA may be a single guide RNA (i.e., sgRNA), or may comprise two RNA molecules (i.e., crRNA and tracrRNA). The guide RNA interacts with a protein of the CRISPR system to guide it to a target site in the DNA. The target site has no sequence limitation except that the sequence is bordered by a protospacer adjacent motif (PAM). For example, PAM sequences for Cas9 include 3'-NGG, 3'-NGGNG, 3'-NNAGAAW, and 3'-ACAY, and PAM sequences for Cpf1 include 5'-TTN (wherein N is defined as any nucleotide, W is defined as either A or T, and Y is defined as either C or T). Each gRNA comprises a sequence that is complementary to the target sequence (e.g., a Cas9 gRNA may comprise GN17-20GG). The gRNA may also comprise a scaffold sequence that forms

a stem loop structure and a single-stranded region. The scaffold region may be the same in every gRNA. In some aspects, the gRNA may be a single molecule (i.e., sgRNA). In other aspects, the gRNA may be two separate molecules. Those skilled in the art are familiar with gRNA design and construction, e.g., gRNA design tools are available on the internet or from commercial sources.

[0192] A CRISPR system may comprise one or more nucleic acid binding domains associated with one or more, or two or more selected guide RNAs used to direct the CRISPR system to one or more, or two or more selected target nucleic acid loci. For instance, a nucleic acid binding domain may be associated with one or more, or two or more selected guide RNAs, each selected guide RNA, when complexed with a nucleic acid binding domain, causing the CRISPR system to localize to the target of the guide RNA.

ii. CRISPR Nickase Systems.

[0193] The programmable targeting nuclease can also be a CRISPR nickase system. CRISPR nickase systems are similar to the CRISPR nuclease systems described above except that a CRISPR nuclease of the system is modified to cleave only one strand of a double-stranded nucleic acid sequence. Thus, a CRISPR nickase, in combination with a guide RNA of the system, may create a single-stranded break or nick in the target nucleic acid sequence. Alternatively, a CRISPR nickase in combination with a pair of offset gRNAs may create a double-stranded break in the nucleic acid sequence.

[0194] A CRISPR nuclease of the system may be converted to a nickase by one or more mutations and/or deletions. For example, a Cas9 nickase may comprise one or more mutations in one of the nuclease domains, wherein the one or more mutations may be D10A, E762A, and/or D986A in the RuvC-like domain, or the one or more mutations may be H840A (or H839A), N854A and/or N863A in the HNH-like domain.

iii. ssDNA-Guided Argonaute Systems.

[0195] Alternatively, the programmable targeting nuclease may comprise a single-stranded DNA-guided Argonaute endonuclease. Argonautes (Ago) are a family of endonucleases that use 5'-phosphorylated short single-stranded nucleic acids as guides to cleave nucleic acid targets. Some prokaryotic Ago use single-stranded guide DNAs and create double-stranded breaks in nucleic acid sequences. The ssDNA-guided Ago endonuclease may be associated with a single-stranded guide DNA.

[0196] The Ago endonuclease may be derived from *Alisipites* sp., *Aquifex* sp., *Archaeoglobus* sp., *Bacteroides* sp., *Bradyrhizobium* sp., *Burkholderia* sp., *Cellvibrio* sp., *Chlorobium* sp., *Geobacter* sp., *Mariprofundus* sp., *Natronobacterium* sp., *Parabacteriodes* sp., *Parvularcula* sp., *Planctomyces* sp., *Pseudomonas* sp., *Pyrococcus* sp., *Thermus* sp., or *Xanthomonas* sp. For instance, the Ago endonuclease may be *Natronobacterium gregoryi* Ago (NgAgo). Alternatively, the Ago endonuclease may be *Thermus thermophilus* Ago (TtAgo). The Ago endonuclease may also be *Pyrococcus furiosus* (PfAgo).

[0197] The single-stranded guide DNA (gDNA) of an ssDNA-guided Argonaute system is complementary to the target site in the nucleic acid sequence. The target site has no sequence limitations and does not require a PAM. The gDNA generally ranges in length from about 15-30 nucleotides. The gDNA may comprise a 5' phosphate group. Those skilled in the art are familiar with ssDNA oligonucleotide design and construction.

iv. Zinc Finger Nucleases.

[0198] The programmable targeting nuclease may be a zinc finger nuclease (ZFN). A ZFN comprises a DNA-binding zinc finger region and a nuclease domain. The zinc finger region may comprise from about two to seven zinc fingers, for example, about four to six zinc fingers, wherein each zinc finger binds three nucleotides. The zinc finger region may be engineered to recognize and bind to any DNA sequence. Zinc finger design tools or algorithms are available on the internet or from commercial sources. The zinc fingers may be linked together using suitable linker sequences.

[0199] A ZFN also comprises a nuclease domain, which may be obtained from any endonuclease or exonuclease. Non-limiting examples of endonucleases from which a nuclease domain may be derived include, but are not limited to, restriction endonucleases and homing endonucleases. The nuclease domain may be derived from a type II-S restriction endonuclease. Type II-S endonucleases cleave DNA at sites that are typically several base pairs away from the recognition/binding site and, as such, have separable binding and cleavage domains. These enzymes generally are monomers that transiently associate to form dimers to cleave each strand of DNA at staggered locations. Non-limiting examples of suitable type II-S endonucleases include BfiI, BpmI, BsaI, BsgI, BsmBI, BsmI, BspMI, FokI, MboII, and SapI. The type II-S nuclease domain may be modified to facilitate dimerization of two different nuclease domains. For example, the cleavage domain of FokI may be modified by mutating certain amino acid residues. By way of non-limiting example, amino acid residues at positions 446, 447, 479, 483, 484, 486, 487, 490, 491, 496, 498, 499, 500, 531, 534, 537, and 538 of FokI nuclease domains are targets for modification. For example, one modified FokI domain may comprise Q486E, 1499L, and/or N496D mutations, and the other modified FokI domain may comprise E490K, 1538K, and/or H537R mutations.

v. Transcription Activator-Like Effector Nuclease Systems.

[0200] The programmable targeting nuclease may also be a transcription activator-like effector nuclease (TALEN) or the like. TALENs comprise a DNA-binding domain composed of highly conserved repeats derived from transcription activator-like effectors (TALEs) that are linked to a nuclease domain. TALEs are proteins secreted by plant pathogen *Xanthomonas* to alter transcription of genes in host plant cells. TALE repeat arrays may be engineered via modular protein design to target any DNA sequence of interest. Other transcription activator-like effector nuclease systems may comprise, but are not limited to, the repetitive sequence, transcription activator like effector (RipTAL) system from the bacterial plant pathogenic *Ralstonia solanacearum* species complex (Rsse). The nuclease domain of TALEs may be any nuclease domain as described above in Section II (i).

vi. Meganucleases or Rare-Cutting Endonuclease Systems.

[0201] The programmable targeting nuclease may also be a meganuclease or derivative thereof. Meganucleases are endodeoxyribonucleases characterized by long recognition sequences, i.e., the recognition sequence generally ranges from about 12 base pairs to about 45 base pairs. As a consequence of this requirement, the recognition sequence generally occurs only once in any given genome. Among meganucleases, the family of homing endonucleases named LAGLIDADG has become a valuable tool for the study of genomes and genome engineering. Non-limiting examples

of meganucleases that may be suitable for the instant disclosure include I-SceI, I-CreI, I-DmI, or variants and combinations thereof. A meganuclease may be targeted to a specific nucleic acid sequence by modifying its recognition sequence using techniques well known to those skilled in the art.

[0202] The programmable targeting nuclease can be a rare-cutting endonuclease or derivative thereof. Rare-cutting endonucleases are site-specific endonucleases whose recognition sequence occurs rarely in a genome, such as only once in a genome. The rare-cutting endonuclease may recognize a 7-nucleotide sequence, an 8-nucleotide sequence, or longer recognition sequence. Non-limiting examples of rare-cutting endonucleases include NotI, AscI, PacI, AsiSI, SbfI, and FseI.

vii. Optional Additional Domains.

[0203] The programmable targeting nuclease may further comprise at least one nuclear localization signal (NLS), at least one cell-penetrating domain, at least one reporter domain, and/or at least one linker.

[0204] In general, an NLS comprises a stretch of basic amino acids. Nuclear localization signals are known in the art (see, e.g., Lange et al., J. Biol. Chem., 2007, 282:5101-5105). The NLS may be located at the N-terminus, the C-terminal, or in an internal location of the fusion protein.

[0205] A cell-penetrating domain may be a cell-penetrating peptide sequence derived from the HIV-1 TAT protein. The cell-penetrating domain may be located at the N-terminus, the C-terminal, or in an internal location of the fusion protein.

[0206] A programmable targeting nuclease may further comprise at least one linker. For example, the programmable targeting nuclease, the nuclease domain of the targeting nuclease, and other optional domains may be linked via one or more linkers. The linker may be flexible (e.g., comprising small, non-polar (e.g., Gly) or polar (e.g., Ser, Thr) amino acids). Non-limiting examples of suitable linkers are well known in the art, and programs to design linkers are readily available (Crasto et al., Protein Eng., 2000, 13 (5): 3096-312). In alternate aspects, the programmable targeting nuclease, the cell cycle regulated protein, and other optional domains may be linked directly.

[0207] A programmable targeting nuclease may further comprise an organelle localization or targeting signal that directs a molecule to a specific organelle. A signal may be a polynucleotide or polypeptide signal, or may be an organic or inorganic compound sufficient to direct an attached molecule to a desired organelle. Organelle localization signals can be as described in U.S. Patent Publication No. 20070196334, the disclosure of which is incorporated herein in its entirety.

III. Nucleic Acid Constructs

[0208] A further aspect of the present disclosure provides a system of one or more nucleic acid constructs encoding the components of the engineered nucleic acid modification system described above in Section II.

[0209] Any of the multi-component systems described herein are to be considered modular, in that the different components may optionally be distributed among two or more nucleic acid constructs as described herein. The nucleic acid constructs may be DNA or RNA, linear or circular, single-stranded or double-stranded, or any combination thereof. The nucleic acid constructs may be codon-

optimized for efficient translation into protein, and possibly for transcription into an RNA donor polynucleotide transcript in the cell of interest. Codon optimization programs are available as freeware or from commercial sources.

[0210] The nucleic acid constructs can be used to express one or more components of the system for later introduction into a cell to be genetically modified. Alternatively, the nucleic acid constructs can be introduced into the cell to be genetically modified for expression of the components of the system in the cell. In some aspects, the nucleic acid constructs transiently express the various components of the system. Transiently expressing the system in a plant overcomes the cumbersome regulatory hurdles required for traditionally genetically modified crops.

[0211] Expression constructs generally comprise DNA coding sequences operably linked to at least one promoter control sequence for expression in a cell of interest. Promoter control sequences may control expression of the transposase, the programmable targeting nuclease, the donor polynucleotide, or combinations thereof in bacterial (e.g., *E. coli*) cells or eukaryotic (e.g., yeast, insect, mammalian, or plant) cells. Suitable bacterial promoters include, without limit, T7 promoters, lac operon promoters, trp promoters, tac promoters (which are hybrids of trp and lac promoters), variations of any of the foregoing, and combinations of any of the foregoing. Non-limiting examples of suitable eukaryotic promoters include constitutive, regulated, or cell- or tissue-specific promoters. As explained above, methylation of the MeSWEET10a gene can be targeted in leaves by specifically expressing the system in leaves using a leaf-specific promoter, allowing for fine-tuning pathogen resistance and normal plant growth and development.

[0212] Suitable eukaryotic constitutive promoter control sequences include, but are not limited to, cytomegalovirus immediate early promoter (CMV), simian virus (SV40) promoter, adenovirus major late promoter, Rous sarcoma virus (RSV) promoter, mouse mammary tumor virus (MMTV) promoter, phosphoglycerate kinase (PGK) promoter, elongation factor (EF1)-alpha promoter, ubiquitin promoters, actin promoters, tubulin promoters, immunoglobulin promoters, fragments thereof, or combinations of any of the foregoing. Non-limiting examples of suitable eukaryotic regulated promoter control sequences include, without limit, those regulated by heat shock, metals, steroids, antibiotics, or alcohol. Non-limiting examples of tissue-specific promoters include B29 promoter, CD14 promoter, CD43 promoter, CD45 promoter, CD68 promoter, desmin promoter, elastase-1 promoter, endoglin promoter, fibronectin promoter, Flt-1 promoter, GFAP promoter, GPIIb promoter, ICAM-2 promoter, INF- β promoter, Mb promoter, NphsI promoter, OG-2 promoter, SP-B promoter, SYN1 promoter, and WASP promoter.

[0213] Promoters may also be plant-specific promoters, or promoters that may be used in plants. A wide variety of plant promoters are known to those of ordinary skill in the art, as are other regulatory elements that may be used alone or in combination with promoters. Preferably, promoter control sequences control expression in cassava, such as promoters disclosed in Wilson et al., 2017, The New Phytologist, 213 (4): 1632-1641, the disclosure of which is incorporated herein in its entirety.

[0214] Promoters may be divided into two types, namely, constitutive promoters and non-constitutive promoters. Constitutive promoters are classified as providing for a range of

constitutive expression. Thus, some are weak constitutive promoters, and others are strong constitutive promoters. Non-constitutive promoters include tissue-preferred promoters, tissue-specific promoters, cell-type specific promoters, and inducible promoters. Suitable plant-specific constitutive promoter control sequences include, but are not limited to, a CaMV35S promoter, CaMV 19S, GOS2, *Arabidopsis* At6669 promoter, Rice cyclophilin, Maize H3 histone, Synthetic Super MAS, an opine promoter, a plant ubiquitin (Ubi) promoter, an actin 1 (Act-1) promoter, pEMU, Cestrum yellow leaf curling virus promoter (CYMLV promoter), and an alcohol dehydrogenase 1 (Adh-1) promoter. Other constitutive promoters include those in U.S. Pat. Nos. 5,659,026; 5,608,149; 5,608,144; 5,604,121; 5,569,597; 5,466,785; 5,399,680; 5,268,463; and 5,608,142.

[0215] Regulated plant promoters respond to various forms of environmental stresses, or other stimuli, including, for example, mechanical shock, heat, cold, flooding, drought, salt, anoxia, pathogens such as bacteria, fungi, and viruses, and nutritional deprivation, including deprivation during times of flowering and/or fruiting, and other forms of plant stress. For example, the promoter may be a promoter which is induced by one or more, but not limited to one of the following: abiotic stresses such as wounding, cold, desiccation, ultraviolet-B, heat shock or other heat stress, drought stress or water stress. The promoter may further be one induced by biotic stresses including pathogen stress, such as stress induced by a virus or fungi, stresses induced as part of the plant defense pathway or by other environmental signals, such as light, carbon dioxide, hormones or other signaling molecules such as auxin, hydrogen peroxide and salicylic acid, sugars and gibberellin or abscisic acid and ethylene. Suitable regulated plant promoter control sequences include, but are not limited to, salt-inducible promoters such as RD29A; drought-inducible promoters such as maize rab17 gene promoter, maize rab28 gene promoter, and maize lvr2 gene promoter; heat-inducible promoters such as heat tomato hsp80-promoter from tomato.

[0216] Tissue-specific promoters may include, but are not limited to, fiber-specific, green tissue-specific, root-specific, stem-specific, flower-specific, callus-specific, pollen-specific, egg-specific, and seed coat-specific. Suitable tissue-specific plant promoter control sequences include, but are not limited to, leaf-specific promoters [such as described, for example, by Yamamoto et al., *Plant J.* 12:255-265, 1997; Kwon et al., *Plant Physiol.* 105:357-67, 1994; Yamamoto et al., *Plant Cell Physiol.* 35:773-778, 1994; Gotor et al., *Plant J.* 3:509-18, 1993; Orozco et al., *Plant Mol. Biol.* 23:1129-1138, 1993; and Matsuoka et al., *Proc. Natl. Acad. Sci. USA* 90:9586-9590, 1993], seed-preferred promoters [e.g., from seed-specific genes (Simon et al., *Plant Mol. Biol.* 5: 191, 1985; Scofield et al., *J. Biol. Chem.* 262:12202, 1987; Baszczynski et al., *Plant Mol. Biol.* 14:633, 1990), Brazil Nut albumin (Pearson et al., *Plant Mol. Biol.* 18:235-245, 1992), legumin (Ellis et al., *Plant Mol. Biol.* 10:203-214, 1988), Glutelin (rice) (Takaiwa et al., *Mol. Gen. Genet.* 208:15-22, 1986; Takaiwa et al., *FEBS Letts.* 221:43-47, 1987), Zein (Matzke et al., *Plant Mol Biol.* 143:323-32, 1990), napA (Stalberg et al., *Planta* 199:515-519, 1996), Wheat SPA (Albanietal, *Plant Cell*, 9:171-184, 1997), sunflower oleosin (Cummins et al., *Plant Mol. Biol.* 19:873-876, 1992)], endosperm specific promoters [e.g., wheat LMW and HMW, glutenin-1 (*Mol Gen Genet* 216:81-90, 1989; *NAR* 17:461-2), wheat a, b, and g gliadins (*EMBO* 3:

1409-15, 1984), Barley Itrl promoter, barley B1, C, D hordein (*Theor Appl Gen* 98:1253-62, 1999; *Plant J* 4:343-55, 1993; *Mol Gen Genet* 250:750-60, 1996), Barley DOF (Mena et al., *The Plant Journal*, 116 (1): 53-62, 1998), Biz2 (EP99106056.7), Synthetic promoter (Vicente-Carbajosa et al., *Plant J.* 13:629-640, 1998), rice prolamin NRP33, rice-globulin Glb-1 (Wu et al., *Plant Cell Physiology* 39 (8) 885-889, 1998), rice alpha-globulin REB/OHP-1 (Nakase et al., *Plant Mol. Biol.* 33:513-S22, 1997), rice ADP-glucose PP (*Trans Res* 6:157-68, 1997), maize ESR gene family (*Plant J* 12:235-46, 1997), sorghum gamma-kafirin (PMB 32:1029-35, 1996)], embryo-specific promoters [e.g., rice OSH1 (Sato et al., *Proc. Natl. Acad. Sci. USA*, 93:8117-8122), KNOX (Postma-Haarsma et al., *Plant Mol. Biol.* 39:257-71, 1999), rice oleosin (Wu et al., *J. Biochem.*, 123:386, 1998)], and flower-specific promoters [e.g., AtPRP4, chalcone synthase (chsA) (Van der Meer et al., *Plant Mol. Biol.* 15, 95-109, 1990), LAT52 (Twell et al., *Mol. Gen Genet.* 217:240-245; 1989), *apetala-3*].

[0217] Any of the promoter sequences may be wild type or may be modified for more efficient or efficacious expression. The DNA coding sequence also may be linked to a polyadenylation signal (e.g., SV40 polyA signal, bovine growth hormone (BGH) polyA signal, etc.) and/or at least one transcriptional termination sequence. In some situations, the complex or fusion protein may be purified from the bacterial or eukaryotic cells.

[0218] Nucleic acids encoding one or more components of an engineered DNA methylation system and/or transcription activation system may be present in a construct. Suitable constructs include plasmid constructs, viral constructs, and self-replicating RNA (Yoshioka et al., *Cell Stem Cell*, 2013, 13:246-254). For instance, the nucleic acid encoding one or more components of an engineered DNA methylation system and/or transcription activation system may be present in a plasmid construct.

[0219] Non-limiting examples of suitable plasmid constructs include pUC, pBR322, pET, pBluescript, and variants thereof. Alternatively, the nucleic acid encoding one or more components of an engineered DNA methylation system and/or transcription activation system may be part of a viral vector (e.g., lentiviral vectors, adeno-associated viral vectors, adenoviral vectors, and so forth).

[0220] The plasmid or viral vector may comprise additional expression control sequences (e.g., enhancer sequences, Kozak sequences, polyadenylation sequences, transcriptional termination sequences, etc.), selectable reporter sequences (e.g., antibiotic resistance genes), origins of replication, T-DNA border sequences, and the like. The plasmid or viral vector may further comprise RNA processing elements such as glycine tRNAs, or Csy4 recognition sites. Such RNA processing elements can, for instance, intersperse polynucleotide sequences encoding multiple gRNAs under the control of a single promoter to produce the multiple gRNAs from a transcript encoding the multiple gRNAs. When a cys4 recognition site is used, a vector may further comprise sequences for expression of Csy4 RNase to process the gRNA transcript. Additional information about vectors and use thereof may be found in "Current Protocols in Molecular Biology", Ausubel et al., John Wiley & Sons, New York, 2003, or "Molecular Cloning: A Laboratory Manual", Sambrook & Russell, Cold Spring Harbor Press, Cold Spring Harbor, NY, 3rd edition, 2001.

[0221] In some aspects, the nucleic acid modification comprises an expression construct for expression of POWR1, wherein the construct comprises a nucleotide sequence encoding the CCT protein operably linked to a promoter. In some aspects, the CCT protein is GmCCT67. In some aspects, the promoter is a ubiquitin promoter.

[0222] In some aspects, the expression construct for expression of GmCCT67 POWR1 comprises a nucleic acid sequence comprising about 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO:4. In some aspects, the expression construct for expression of GmCCT67 POWR1 comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 4.

[0223] In some aspects, the expression construct for expression of GmCCT34 POWR2 comprises a nucleic acid sequence comprising about 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 7. In some aspects, the expression construct for expression of GmCCT34 POWR2 comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 7.

IV. Methods

[0224] A further aspect of the present disclosure encompasses a method of generating a genetically modified plant having an improved agronomic trait. The method comprises introducing one or more nucleic acid constructs encoding an engineered nucleic acid modification system for modifying the expression of a CCT protein in a plant into a plant or plant cell. The plant or plant cell is then grown under conditions whereby the nucleic acid expression construct expresses the programmable nucleic acid modification system or the CCT protein in the plant or plant cell. Expressing the programmable nucleic acid modification system or expressing the CCT protein introduces a nucleic acid modification in the nucleic acid sequence encoding the CCT protein, thereby modifying the expression of a CCT protein in a plant and improving the agronomic trait of the plant. The CCT protein and the plant can be as described in Section I. The engineered nucleic acid modification system for modifying the expression of a CCT protein in a plant can be as described in Section II, and nucleic acid constructs expressing the engineered nucleic acid modification system can be as described in Section III.

[0225] Another aspect of the present disclosure encompasses a method of improving an agronomic trait of a plant. The method comprises introducing one or more nucleic acid constructs encoding an engineered nucleic acid modification system for modifying the expression of a CCT protein in a plant into a plant or plant cell, growing the plant or plant cell under conditions whereby the nucleic acid expression construct expresses the programmable nucleic acid modification system or the CCT protein in the plant or plant cell. Expressing the programmable nucleic acid modification system or expressing the CCT protein introduces a nucleic

acid modification in the nucleic acid sequence encoding the CCT protein, thereby modifying the expression of a CCT protein in a plant and improving the agronomic trait of the plant. The CCT protein and the plant can be as described in Section I. The engineered nucleic acid modification system for modifying the expression of a CCT protein in a plant can be as described in Section II, and nucleic acid constructs expressing the engineered nucleic acid modification system can be as described in Section III.

(a) Marker-Assisted Selection

[0226] Yet another aspect of the present disclosure encompasses a method of identifying a plant having an improved agronomic trait of a plant using marker-assisted selection (MAS). The method comprises identifying in a population of plants one or more plants comprising a molecular marker that demonstrates linkage with a nucleic acid modification that modifies the expression of a CCT protein in the plant. Through extensive experimentation, the inventors identified genetic markers that are linked to nucleic acid sequences encoding CCT proteins wherein the nucleic acid modification modifies the expression of the CCT protein in the plant thereby improving the agronomic trait of the plant.

[0227] Molecular markers suitable for a method of the instant disclosure are known in the art and include, without limitation, restriction fragment length polymorphisms (RFLPs), isozyme markers, allele specific hybridization (ASH), amplified variable sequences of plant genome, self-sustained sequence replication, simple sequence repeat (SSR), single base-pair change (single nucleotide polymorphism, SNP), random amplification of polymorphic DNA (RAPDs), SSCP (single stranded conformation polymorphisms); amplified fragment length polymorphisms (AFLPs), a quantitative trait locus (QTL), and microsatellites DNA. In some aspects, the molecular marker is a QTL selected from SNPs of Table 15. In some aspects, the population of plants is a progeny of a cross between parent plants. In some aspects, a parent plant is a plant described in Section I.

[0228] Molecular markers can be used in a variety of plant breeding applications. Molecular markers can be used to increase the efficiency of identifying progeny plants of a cross between parent plants using marker-assisted selection (MAS), wherein one or more of the progeny plants comprise a favorable nucleic acid modification. As used herein, the term “favorable nucleic acid modification” is a nucleic acid modification that modifies the expression of the CCT protein in the plant thereby improving the agronomic trait of the plant.

[0229] A molecular marker that demonstrates linkage with a locus affecting a desired phenotypic trait provides a useful tool for the selection of the trait in a plant population. This is particularly true with traits that are difficult to phenotype due to their dependence on environmental conditions. This category includes traits related to an improved agronomic trait. This category also includes traits that are very expensive to phenotype because of laborious artificial inoculation or maintenance of managed stress environments. Another category of traits includes those which are associated with destruction of plant per se. Destructive phenotyping has been a bottleneck to implement MAS for the seed quality traits. Because DNA marker assays are not environmentally dependent, are robust, reliable, less laborious, less costly and take up less physical space than field phenotyping, much

larger populations can be assayed, increasing the chances of finding a recombinant with the target segment from the donor line moved to the recipient line. The closer the linkage, the more useful the marker, as recombination is less likely to occur between the marker and the gene causing the trait, which can result in false positives. Having flanking markers decreases the chances that false positive selection will occur as a double recombination event would be needed. The ideal situation is to have a marker in the gene itself, so that recombination cannot occur between the marker and the gene. Such a marker is called a 'perfect marker'.

[0230] When a gene is introgressed by MAS, it is not only the gene that is introduced but also the flanking regions. This is referred to as "linkage drag." In the case where the donor plant is highly unrelated to the recipient plant, these flanking regions carry additional genes that may code for agronomically undesirable traits. This "linkage drag" may also result in negative agronomic characteristics even after multiple cycles of backcrossing into the elite plant line. The size of the flanking region can be decreased by additional backcrossing, although this is not always successful, as breeders do not have control over the size of the region or the recombination breakpoints. In classical breeding it is usually only by chance that recombinations are selected that contribute to a reduction in the size of the donor segment. Even after 20 backcrosses in backcrosses of this type, one may expect to find a sizeable piece of the donor chromosome still linked to the gene being selected. With markers however, it is possible to select those rare individuals that have experienced recombination near the gene of interest. In 150 backcross plants, there is a 95% chance that at least one plant will have experienced a crossover within 1 cM of the gene, based on a single meiosis map distance. Markers will avoid unequivocal identification of those individuals. With one additional backcross of 300 plants, there would be a 95% chance of a crossover within 1 cM single meiosis map distance of the other side of the gene, generating a segment around the target gene of less than 2 cM based on a single meiosis map distance. This can be accomplished in two generations with markers, while it would have required on average 100 generations without markers. When the exact location of a gene is known, flanking markers surrounding the gene can be utilized to select for recombinations in different population sizes. For example, in smaller population sizes, recombinations may be expected further away from the gene, so more distal flanking markers would be required to detect the recombination.

(b) Introduction into the Cell

[0231] The method comprises introducing a nucleic acid construct expressing an engineered protein into a cell of interest. As explained above, an engineered protein can be encoded on more than one nucleic acid sequence. Accordingly, a method of the instant disclosure comprises introducing more than one nucleic acid construct into the cell.

[0232] The one or more nucleic acid constructs described above may be introduced into the cell by a variety of means. Suitable delivery means include microinjection, electroporation, sonoporation, biolistics, calcium phosphate-mediated transfection, cationic transfection, liposomes and other lipids, dendrimer transfection, heat shock transfection, nucleofection transfection, gene gun delivery, dip transformation, supercharged proteins, cell-penetrating peptides, viral vectors, magnetofection, lipofection, impalefection, optical

transfection, *Agrobacterium tumefaciens* mediated foreign gene transformation, proprietary agent-enhanced uptake of nucleic acids, and delivery via liposomes, immunoliposomes, virosomes, or artificial virions. The choice of means of introducing the system into a cell can and will vary depending on the cell, or the system or nucleic acid nucleic acid constructs encoding the system, among other variables.

(c) Culturing a Cell

[0233] The method further comprises culturing a cell under conditions suitable for expressing the engineered protein. Methods of culturing cells are known in the art. In some aspects, the cell is from an animal, fungi, oomycete or prokaryote. In some aspects, the cell is a plant cell, plant, or plant part. When the cell is in tissue ex vivo, or in vivo within a plant or within a plant part, the plant part and/or plant may also be maintained under appropriate conditions for insertion of the donor polynucleotide. In general, the plant, plant part, or plant cell is maintained under conditions appropriate for cell growth and/or maintenance. Those of skill in the art appreciate that methods for culturing plant cells are known in the art and may and will vary depending on the cell type. Routine optimization may be used, in all cases, to determine the best techniques for a particular cell type.

V. Kits

[0234] A further aspect of the present disclosure provides kits comprising one or more genetically modified plant having an improved agronomic trait, an engineered nucleic acid modification system for modifying the expression of a CCT protein in a plant, one or more nucleic acid constructs encoding an engineered nucleic acid modification system for modifying the expression of a CCT protein in a plant, a plant comprising the one or more nucleic acid constructs encoding a programmable nucleic acid modification system, or any combination thereof.

[0235] The genetically modified plant having an improved agronomic trait can be as described in Section I. The engineered nucleic acid modification system for modifying the expression of a CCT protein in a plant can be as described in Section II. The one or more nucleic acid constructs encoding an engineered nucleic acid modification system for modifying the expression of a CCT protein in a plant can be as described in Section III. A plant comprising the one or more nucleic acid constructs encoding a programmable nucleic acid modification system can be as described in Section I herein above.

[0236] The kits may further comprise transfection reagents, cell growth media, selection media, in vitro transcription reagents, nucleic acid purification reagents, protein purification reagents, buffers, and the like. The kits provided herein generally include instructions for carrying out the methods detailed below. Instructions included in the kits may be affixed to packaging material or may be included as a package insert. While the instructions are typically written or printed materials, they are not limited to such. Any medium capable of storing such instructions and communicating them to an end user is contemplated by this disclosure. Such media include, but are not limited to, electronic storage media (e.g., magnetic discs, tapes, cartridges, chips), optical media (e.g., CD ROM), and the like. As used herein,

the term “instructions” may include the address of an internet site that provides the instructions.

Definitions

[0237] Unless defined otherwise, all technical and scientific terms used herein have the meaning commonly understood by a person skilled in the art to which this invention belongs. The following references provide one of skill with a general definition of many of the terms used in this invention: Singleton et al., *Dictionary of Microbiology and Molecular Biology* (2nd ed. 1994); *The Cambridge Dictionary of Science and Technology* (Walker ed., 1988); *The Glossary of Genetics*, 5th Ed., R. Rieger et al. (eds.), Springer Verlag (1991); and Hale & Marham, *The Harper Collins Dictionary of Biology* (1991). As used herein, the following terms have the meanings ascribed to them unless specified otherwise.

[0238] When introducing elements of the present disclosure or the preferred aspects(s) thereof, the articles “a”, “an”, “the” and “said” are intended to mean that there are one or more of the elements. The terms “comprising”, “including” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements.

[0239] A “genetically modified” plant refers to a plant in which the nuclear, organellar or extrachromosomal nucleic acid sequences of a cell has been modified, i.e., the cell contains at least one nucleic acid sequence that has been engineered to contain an insertion of at least one nucleotide, a deletion of at least one nucleotide, and/or a substitution of at least one nucleotide.

[0240] As used herein, the term “gene” refers to a DNA region (including exons and introns) encoding a gene product, as well as all DNA regions which regulate the production of the gene product, whether or not such regulatory sequences are adjacent to coding and/or transcribed sequences. Accordingly, a gene includes, but is not necessarily limited to, promoter sequences, terminators, translational regulatory sequences such as ribosome binding sites and internal ribosome entry sites, enhancers, silencers, insulators, boundary elements, replication origins, matrix attachment sites, and locus control regions.

[0241] As used herein, the term “engineered” when applied to a targeting protein refers to targeting proteins modified to specifically recognize and bind to a nucleic acid sequence at or near a target nucleic acid locus. A “genetically modified” plant refers to a cell in which the nuclear, organellar or extrachromosomal nucleic acid sequences of a cell have been modified, i.e., the cell contains at least one nucleic acid sequence that has been engineered to contain an insertion of at least one nucleotide, a deletion of at least one nucleotide, and/or a substitution of at least one nucleotide.

[0242] The term “nucleic acid modification” refers to processes by which a specific nucleic acid sequence in a polynucleotide is changed such that the nucleic acid sequence is modified. The nucleic acid sequence may be modified to comprise an insertion of at least one nucleotide, a deletion of at least one nucleotide, and/or a substitution of at least one nucleotide. The modified nucleic acid sequence is inactivated such that no product is made. Alternatively, the nucleic acid sequence may be modified such that an altered product is made.

[0243] As used herein, “protein expression” includes but is not limited to one or more of the following: transcription

of a gene into precursor mRNA; splicing and other processing of the precursor mRNA to produce mature mRNA; mRNA stability; translation of the mature mRNA into protein (including codon usage and tRNA availability); production of a mutant protein comprising a mutation that modifies the activity of the protein, including the calcium channel activity; and glycosylation and/or other modifications of the translation product, if required for proper expression and function. The term “heterologous” refers to an entity that is not native to the cell or species of interest.

[0244] The terms “nucleic acid” and “polynucleotide” refer to a deoxyribonucleotide or ribonucleotide polymer, in linear or circular conformation. For the purposes of the present disclosure, these terms are not to be construed as limiting with respect to the length of a polymer. The terms may encompass known analogs of natural nucleotides, as well as nucleotides that are modified in the base, sugar and/or phosphate moieties. In general, an analog of a particular nucleotide has the same base-pairing specificity, i.e., an analog of A will base-pair with T. The nucleotides of a nucleic acid or polynucleotide may be linked by phosphodiester, phosphothioate, phosphoramidite, phosphorodiamidate bonds, or combinations thereof.

[0245] The term “nucleotide” refers to deoxyribonucleotides or ribonucleotides. The nucleotides may be standard nucleotides (i.e., adenosine, guanosine, cytidine, thymidine, and uridine) or nucleotide analogs. A nucleotide analog refers to a nucleotide having a modified purine or pyrimidine base or a modified ribose moiety. A nucleotide analog may be a naturally occurring nucleotide (e.g., inosine) or a non-naturally occurring nucleotide. Non-limiting examples of modifications on the sugar or base moieties of a nucleotide include the addition (or removal) of acetyl groups, amino groups, carboxyl groups, carboxymethyl groups, hydroxyl groups, methyl groups, phosphoryl groups, and thiol groups, as well as the substitution of the carbon and nitrogen atoms of the bases with other atoms (e.g., 7-deaza purines). Nucleotide analogs also include dideoxy nucleotides, 2'-O-methyl nucleotides, locked nucleic acids (LNA), peptide nucleic acids (PNA), and morpholinos.

[0246] The terms “polypeptide” and “protein” are used interchangeably to refer to a polymer of amino acid residues.

[0247] As used herein, the terms “target site”, “target sequence”, or “nucleic acid locus” refer to a nucleic acid sequence that defines a portion of a nucleic acid sequence to be modified or edited and to which a homologous recombination composition is engineered to target.

[0248] The terms “upstream” and “downstream” refer to locations in a nucleic acid sequence relative to a fixed position. Upstream refers to the region that is 5' (i.e., near the 5' end of the strand) to the position, and downstream refers to the region that is 3' (i.e., near the 3' end of the strand) to the position.

[0249] The term “Molecular marker” shall refer to any type of nucleic acid based marker, including but not limited to, Restriction Fragment Length Polymorphism (RFLP), Simple Sequence Repeat (SSR), Random Amplified Polymorphic DNA (RAPD), Cleaved Amplified Polymorphic Sequences (CAPS), Amplified Fragment Length Polymorphism (AFLP), Single Nucleotide Polymorphism (SNP), Sequence Characterized Amplified Region (SCAR), Sequence Tagged Site (STS), Single Stranded Conformation Polymorphism (SSCP), Inter-Simple Sequence Repeat (ISR), Inter-Retrotransposon Amplified Polymorphism

(IRAP), Retrotransposon-Microsatellite Amplified Polymorphism (REMAP), an RNA cleavage product (such as a Lynx tag), and the like.

[0250] The term “allele” as used herein refers to one of two or more different nucleotide sequences that occur at a specific locus.

[0251] An allele, a nucleic acid modification, or a CCT protein is “associated with” an agronomic trait when it is linked to it and when the presence of the allele, nucleic acid modification, or CCT protein is an indicator that the desired trait will occur in a plant comprising the allele, nucleic acid modification, or CCT protein.

[0252] “Backcrossing” refers to the process whereby hybrid progeny are repeatedly crossed back to one of the parents. In a backcrossing scheme, the “donor” parent refers to the parental plant with the desired gene or locus to be introgressed. The “recipient” parent (used one or more times) or “recurrent” parent (used two or more times) refers to the parental plant into which the gene or locus is being introgressed. The initial cross gives rise to the F1 generation: the term “BC1” then refers to the second use of the recurrent parent; “BC2” refers to the third use of the recurrent parent, and so on.

[0253] The term “crossed” or “cross” means the fusion of gametes via pollination to produce progeny (e.g., cells, seeds or plants). The term encompasses both sexual crosses (the pollination of one plant by another) and selfing (self-pollination, e.g., when the pollen and ovule are from the same plant). The term “crossing” refers to the act of fusing gametes via pollination to produce progeny.

[0254] As used herein, an “elite line” is any line that has resulted from breeding and selection for superior agronomic performance.

[0255] A “favorable allele” is the allele at a particular locus that confers, or contributes to, a desirable phenotype, e.g., increased GS tolerance, or alternatively, is an allele that allows the identification of plants with decreased GS tolerance that can be removed from a breeding program or planting (“counterselection”). A favorable allele of a marker is a marker allele that segregates with the favorable phenotype, or alternatively, segregates with the unfavorable plant phenotype, therefore providing the benefit of identifying plants.

[0256] “Genome” refers to the total DNA, or the entire set of genes, carried by a chromosome or chromosome set.

[0257] The terms “phenotype”, or “phenotypic trait” or “trait” refer to one or more traits of an organism. The phenotype can be observable to the naked eye, or by any other means of evaluation known in the art, e.g., microscopy, biochemical analysis, or an electromechanical assay. In some cases, a phenotype is directly controlled by a single gene or genetic locus, i.e., a “single gene trait”. In other cases, a phenotype is the result of several genes.

[0258] The term “genotype” is the genetic constitution of an individual (or group of individuals) at one or more genetic loci, as contrasted with the observable trait (the phenotype). Genotype is defined by the allele(s) of one or more known loci that the individual has inherited from its parents. The term genotype can be used to refer to an individual’s genetic constitution at a single locus, at multiple loci, or, more generally, the term genotype can be used to refer to an individual’s genetic make-up for all the genes in its genome.

[0259] “Germplasm” refers to genetic material of or from an individual (e.g., a plant), a group of individuals (e.g., a plant line, variety or family), or a clone derived from a line, variety, species, or culture. The germplasm can be part of an organism or cell, or can be separate from the organism or cell. In general, germplasm provides genetic material with a specific molecular makeup that provides a physical foundation for some or all of the hereditary qualities of an organism or cell culture. As used herein, germplasm includes cells, seed or tissues from which new plants may be grown, or plant parts, such as leaves, stems, pollen, or cells, that can be cultured into a whole plant.

[0260] A “haplotype” is the genotype of an individual at a plurality of genetic loci, i.e. a combination of alleles. Typically, the genetic loci described by a haplotype are physically and genetically linked, i.e., on the same chromosome segment. The term “haplotype” can refer to sequence, polymorphisms at a particular locus, such as a single marker locus, or sequence polymorphisms at multiple loci along a chromosomal segment in a given genome. The former can also be referred to as “marker haplotypes” or “marker alleles”, while the latter can be referred to as “long-range haplotypes”.

[0261] A “heterotic group” comprises a set of genotypes that perform well when crossed with genotypes from a different heterotic group. Inbred lines are classified into heterotic groups, and are further subdivided into families within a heterotic group, based on several criteria such as pedigree, molecular marker-based associations, and performance in hybrid combinations. The two most widely used heterotic groups in the United States are referred to as “Iowa Stiff Stalk Synthetic” (BSSS) and “Lancaster” or “Lancaster Sure Crop” (sometimes referred to as NSS, or Iron-Stiff Stalk).

[0262] The term “heterozygous” means a genetic condition wherein different alleles reside at corresponding loci on homologous chromosomes.

[0263] The term “homozygous” means a genetic condition wherein identical alleles reside at corresponding loci on homologous chromosomes.

[0264] The term “hybrid” means a progeny of mating between at least two genetically dissimilar parents. Without limitation, examples of mating schemes include single crosses, modified single cross, double modified single cross, three-way cross, modified three-way cross, and double cross wherein at least one parent in a modified cross is the progeny of a cross between sister lines.

[0265] “Hybridization” or “nucleic acid hybridization” refers to the pairing of complementary RNA and DNA strands as well as the pairing of complementary DNA single strands.

[0266] The term “hybridize” means the formation of base pairs between complementary regions of nucleic acid strands.

[0267] The term “inbred” means a line that has been bred for genetic homogeneity.

[0268] The term “indel” refers to an insertion or deletion, wherein one line may be referred to as having an insertion relative to a second line, or the second line may be referred to as having a deletion relative to the first line.

[0269] The term “introgression” or “introgressing” refers to the transmission of a desired allele of a genetic locus from one genetic background to another. For example, introgression of a desired allele at a specified locus can be transmitted

to at least one progeny via a sexual cross between two parents of the same species, where at least one of the parents has the desired allele in its genome. Alternatively, for example, transmission of an allele can occur by recombination between two donor genomes, e.g., in a fused protoplast, where at least one of the donor protoplasts has the desired allele in its genome. The desired allele can be, e.g., a selected allele of a marker, a QTL, a transgene, or the like. In any case, offspring comprising the desired allele can be repeatedly backcrossed to a line having a desired genetic background and selected for the desired allele, to result in the allele becoming fixed in a selected genetic background. For example, the GS locus described herein may be introgressed into a recurrent parent that has increased GS tolerance. The recurrent parent line with the introgressed gene or locus then has increased GS tolerance.

[0270] As used herein, the term “linkage” is used to describe the degree with which one marker locus is associated with another marker locus or some other locus (for example, a GS locus). The linkage relationship between a molecular marker and a phenotype is given as a “probability” or “adjusted probability”. Linkage can be expressed as a desired limit or range. For example, in some embodiments, any marker is linked (genetically and physically) to any other marker when the markers are separated by less than 50, 40, 30, 25, 20, or 15 map units for cM). In some aspects, it is advantageous to define a bracketed range of linkage, for example, between 10 and 20 cM, between 10 and 30 cM, or between 10 and 40 cM. The more closely a marker is linked to a second locus, the better an indicator for the second locus that marker becomes. Thus, “closely linked loci” such as a marker locus and a second locus display an inter-locus recombination frequency of 10% or less, preferably about 9% or less, still more preferably about 8% or less, yet more preferably about 7% or less, still more preferably about 6% or less, yet more preferably about 5% or less, still more preferably about 4% or less, yet more preferably about 3% or less, and still more preferably about 2% or less. In highly preferred embodiments, the relevant loci display a recombination frequency of about 1% or less, e.g., about 0.75% or less, more preferably about 0.5% or less, or yet more preferably about 0.25% or less. Two loci that are localized to the same chromosome, and at such a distance that recombination between the two loci occurs at a frequency of less than 10 (e.g., about 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2%, 1%, 0.75%, 0.5%, 0.25%, or less) are also said to be “proximal to” each other. Since one cM is the distance between two markers that show a 1% recombination frequency, any marker is closely linked (genetically and physically) to any other marker that is in close proximity, e.g., at or less than 10 cM distant. Two closely linked markers on the same chromosome can be positioned 9, 8, 7, 6, 5, 4, 3, 2, 1, 0.75, 0.5 or 0.25 CM or less from each other.

[0271] The term “linkage disequilibrium” refers to a non-random segregation of genetic loci or traits for both). In either case, linkage disequilibrium implies that the relevant loci are within sufficient physical proximity along a length of a chromosome so that they segregate together with greater than random (i.e., non-random) frequency (in the case of co-segregating traits, the loci that underlie the traits are in sufficient proximity to each other). Markers that show linkage disequilibrium are considered linked. Linked loci co-segregate more than 50% of the time, e.g., from about 51% to about 100% of the time. In other words, two markers that

co-segregate have a recombination frequency of less than 50% (and by definition, are separated by less than 50 cM on the same chromosome.) As used herein, linkage can be between two markers, or alternatively between a marker and a phenotype. A marker locus can be “associated with” (linked to) a trait, e.g., decreased green snap. The degree of linkage of a molecular marker to a phenotypic trait is measured, e.g., as a statistical probability of co-segregation of that molecular marker with the phenotype.

[0272] Linkage disequilibrium is most commonly assessed using the measure r^2 . When $r^2=1$, complete LD exists between the two marker loci, meaning that the markers have not been separated by recombination and have the same allele frequency. Values for r^2 above $\frac{1}{3}$ indicate sufficiently strong LD to be useful for mapping. Hence, alleles are in linkage disequilibrium when r^2 values between pairwise marker loci are greater than or equal to 0.33, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, or 1.0.

[0273] As used herein, “linkage equilibrium” describes a situation where two markers independently segregate, i.e., sort among progeny randomly. Markers that show linkage equilibrium are considered unlinked (whether or not they lie on the same chromosome).

[0274] A “marker” is a nucleotide sequence or encoded product thereof (e.g., a protein) used as a point of reference. For markers to be useful at detecting recombinations, they need to detect differences, or polymorphisms, within the population being monitored. For molecular markers, this means differences at the DNA level due to polynucleotide sequence differences (e.g. SSRs, RFLPs, AFLPs, SNPs). The genomic variability can be of any origin, for example, insertions, deletions, duplications, repetitive elements, point mutations, recombination events, or the presence and sequence of transposable elements. Molecular markers can be derived from genomic or expressed nucleic acids (e.g., ESTs) and can also refer to nucleic acids used as probes or primer pairs capable of amplifying sequence fragments via the use of PCR-based methods.

[0275] Markers corresponding to genetic polymorphisms between members of a population can be detected by methods well established in the art. These include, e.g., DNA sequencing, PCR-based sequence specific amplification methods, detection of FLPs, detection of isozyme markers, detection of polynucleotide polymorphisms by allele specific hybridization (ASH), detection of amplified variable sequences of the plant genome, detection of self-sustained sequence replication, detection of SSRs, detection of SNPs, or detection of FLPs. Well established methods are also known for the detection of expressed sequence tags (ESTs) and SSR markers derived from EST sequences and RAPDs.

[0276] A “marker allele”, alternatively an “allele of a marker locus”, can refer to one of a plurality of polymorphic nucleotide sequences found at a marker locus in a population that is polymorphic for the marker locus.

[0277] “Marker assisted selection” (or MAS) is a process by which phenotypes are selected based on marker genotypes.

[0278] “Marker assisted counter-selection” is a process by which marker genotypes are used to identify plants that will not be selected, allowing them to be removed from a breeding program or planting.

[0279] A “marker locus” is a specific chromosome location in the genome of a species when a specific marker can

be found. A marker locus can be used to track the presence of a second linked locus, e.g., a linked locus that encodes or contributes to expression of a phenotypic trait. For example, a marker locus can be used to monitor segregation of alleles at a locus, such as a QTL or single gene, that are genetically or physically linked to the marker locus.

[0280] A “marker probe” is a nucleic acid sequence or molecule that can be used to identify the presence of a marker locus, e.g., a nucleic acid probe that is complementary to a marker locus sequence, through nucleic acid hybridization. Marker probes comprising 30 or more contiguous nucleotides of the marker locus (“all or a portion” of the marker locus sequence) may be used for nucleic acid hybridization. Alternatively, in some aspects, a marker probe refers to a probe of any type that is able to distinguish (i.e. genotype) the particular allele that is present at a marker locus.

[0281] The term “molecular marker” may be used to refer to a molecular marker, as defined above, or an encoded product thereof (e.g., a protein) used as a point of reference when identifying a linked locus. A marker can be derived from genomic nucleotide sequences or from expressed nucleotide sequences (e.g., from a spliced RNA, a cDNA, etc.), or from an encoded polypeptide. The term also refers to nucleic acid sequences complementary to or flanking the marker sequences, such as nucleic acids used as probes or primer pairs capable of amplifying the marker sequence. A “molecular marker probe” is a nucleic acid sequence or molecule that can be used to identify the presence of a marker locus, e.g., a nucleic acid probe that is complementary to a marker locus sequence. Alternatively, in some aspects, a marker probe refers to a probe of any type that is able to distinguish (i.e., genotype) the particular allele that is present at a marker locus. Nucleic acids are “complementary” when they specifically hybridize in solution, e.g., according to Watson-Crick base pairing rules. Some of the markers described herein are also referred to as hybridization markers when located on an indel region, such as the non-collinear region described herein. This is because the insertion region is, by definition, a polymorphism *vis a vis* a plant without the insertion. Thus, the marker need only indicate whether the indel region is present or absent. Any suitable marker detection technology may be used to identify such a hybridization marker, e.g., SNP technology is used in the examples provided herein.

[0282] A “physical map” of the genome is a map showing the linear order of identifiable landmarks (including genes, markers, etc.) on chromosome DNA. However, in contrast to genetic maps, the distances between landmarks are absolute (for example, measured in base pairs or isolated and overlapping contiguous genetic fragments) and not based on genetic recombination.

[0283] A “plant” can be a whole plant, any part thereof, or a cell or tissue culture derived from a plant. Thus, the term “plant” can refer to any of: whole plants, plant components or organs (e.g., leaves, stems, roots, etc.), plant tissues, seeds, plant cells, and/or progeny of the same. A plant cell is a cell of a plant, taken from a plant, or derived through culture from a cell taken from a plant.

[0284] A “polymorphism” is a variation in the DNA that is too common to be due merely to new mutation. A polymorphism must have a frequency of at least 1% in a population.

A polymorphism can be a single nucleotide polymorphism, or SNP, or an insertion/deletion polymorphism, also referred to herein as an “indel”.

[0285] The term “progeny” refers to the offspring generated from a cross.

[0286] A “progeny plant” is generated from a cross between two plants.

[0287] A “reference sequence” is a defined sequence used as a basis for sequence comparison. The reference sequence is obtained by genotyping a number of lines at the locus, aligning the nucleotide sequences in a sequence alignment program (e.g. Sequencher), and then obtaining the consensus sequence of the alignment.

[0288] A “single nucleotide polymorphism (SNP)” is an allelic single nucleotide-A, T, C or G-variation within a DNA sequence representing one locus of at least two individuals of the same species. For example, two sequenced DNA fragments representing the same locus from at least two individuals of the same species, contain a difference in a single nucleotide.

[0289] The term “quantitative trait locus (QTL)” means a locus that controls to some degree numerically representable traits that are usually continuously distributed.

[0290] Techniques for determining nucleic acid and amino acid sequence identity are known in the art. Typically, such techniques include determining the nucleotide sequence of the mRNA for a gene and/or determining the amino acid sequence encoded thereby, and comparing these sequences to a second nucleotide or amino acid sequence. Genomic sequences may also be determined and compared in this fashion. In general, identity refers to an exact nucleotide-to-nucleotide or amino acid-to-amino acid correspondence of two polynucleotides or polypeptide sequences, respectively. Two or more sequences (polynucleotide or amino acid) may be compared by determining their percent identity. The percent identity of two sequences, whether nucleic acid or amino acid sequences, is the number of exact matches between two aligned sequences divided by the length of the shorter sequences and multiplied by 100. An approximate alignment for nucleic acid sequences is provided by the local homology algorithm of Smith and Waterman, *Advances in Applied Mathematics* 2:482-489 (1981). This algorithm may be applied to amino acid sequences by using the scoring matrix developed by Dayhoff, *Atlas of Protein Sequences and Structure*, M. O. Dayhoff ed., 5 suppl. 3:353-358, National Biomedical Research Foundation, Washington, D.C., USA, and normalized by Gribskov, *Nucl. Acids Res.* 14 (6): 6745-6763 (1986). An exemplary implementation of this algorithm to determine percent identity of a sequence is provided by the Genetics Computer Group (Madison, Wis.) in the “BestFit” utility application. Other suitable programs for calculating the percent identity or similarity between sequences are generally known in the art, for example, another alignment program is BLAST, used with default parameters. For example, BLASTN and BLASTP may be used using the following default parameters: genetic code=standard; filter=none; strand=both; cutoff=60; expect=10; Matrix=BLOSUM62; Descriptions=50 sequences; sort by=HIGH SCORE; Databases=non-redundant, GenBank+EMBL+DDBJ+PDB+GenBank CDS translations+Swiss protein+Spupdate+PIR. Details of these programs may be found on the GenBank website. With respect to sequences described herein, the range of desired degrees of sequence identity is approximately 80% to 100% and any

integer value therebetween. Typically the percent identities between sequences are at least 70-75%, preferably 80-82%, more preferably 85-90%, even more preferably 92%, still more preferably 95%, and most preferably 98% sequence identity.

[0291] As various changes could be made in the above-described cells and methods without departing from the scope of the invention, it is intended that all matter contained in the above description and in the examples given below, shall be interpreted as illustrative and not in a limiting sense.

EXAMPLES

[0292] All patents and publications mentioned in the specification are indicative of the levels of those skilled in the art to which the present disclosure pertains. All patents and publications are herein incorporated by reference to the same extent as if each individual publication was specifically and individually indicated to be incorporated by reference.

[0293] The publications discussed throughout are provided solely for their disclosure before the filing date of the present application. Nothing herein is to be construed as an admission that the invention is not entitled to antedate such disclosure by virtue of prior invention.

[0294] The following examples are included to demonstrate the disclosure. It should be appreciated by those of skill in the art that the techniques disclosed in the following examples represent techniques discovered by the inventors to function well in the practice of the disclosure. Those of skill in the art should, however, in light of the present disclosure, appreciate that many changes could be made in the disclosure and still obtain a like or similar result without departing from the spirit and scope of the disclosure, therefore all matter set forth is to be interpreted as illustrative and not in a limiting sense.

Example 1. Comprehensive Genome-Wide Analysis of CCT Domain Family Genes GmCCT34 Identified as Involving in Seed Protein and Oil Accumulation in Soybean

[0295] CCT domain is included in a large family of proteins in plants with demonstrated roles in adaptation or agronomic traits, however, such an important family in economically important legumes has yet to be systematically investigated. In the current study, a combination of comparative genomics, transcriptomics, and population genomics was used to comparatively investigate CCTs in legumes with a prioritized analysis on GmCCTs in soybean and conducted gene functional validation with fast-neutron mutation and gene editing analyses.

[0296] Four subfamilies of CCT domain-containing proteins were identified with conserved domain constitution and arrangement across plant species. The soybean genome contained 69 CCT-domain proteins, approximately two times of those in other legumes. Whole-genome duplication was a major driven force of GmCCT family expansion. Further analysis has revealed domain sequence divergence, domain shuffling, and syntenic CCTs in legumes. GmCCTs were rich in natural variation and twelve have the signature of artificial selection. GmCCTs exhibited diversified expression patterns with some showing specificities to circadian clock or environment stressors, or in certain seed tissues.

[0297] The current studies demonstrated a newly discovered role of CCT regulating seed protein and oil accumulation and seed weight. The current results provided an overview of molecular evolution, phylogeny, conserved and novel functions of GmCCTs, shedding insight into the role of CCT domain proteins for legume improvement.

INTRODUCTION

[0298] Advance in DNA sequencing has been greatly promoting the accumulation of reference genome sequences in individual species. Gene function characterization in model plant species advanced the annotation and prediction of functions for orthologous genes in crop species of the lineage. Comparative genomics studies provided genomic evidence of indicating conservation in the evolutionary process of genomes and functionality of orthologous genes across species falling within the same genus or family, especially for those with insufficient experimental evidence in underexplored species. Information gathered from systematic studies of gene families, ortholog groups, and gene functions across species provided important insights into the evolutionary processes and functions of gene families of importance.

[0299] The CCT motif genes were initially identified in three proteins in *Arabidopsis thaliana*, namely CO (CONSTANS), COL (CO-LIKE) and TOC1 (TIMING OF CAB1) and they generally contained 43-amino acid conserved sequence in the carboxy-terminus of the proteins. CCT genes generally were classified CCT family into three subfamilies, CMF (CCT motif family) containing a single CCT domain, COL proteins carrying an additional one or two B-box (BBOX) domains, and PRR (Pseudo Response Regulator) proteins also containing a response regulator (REC) domain. Extensive studies revealed that CCT proteins played important roles in the regulation of flowering by controlling photoperiod response or circadian clock and abiotic stress responses or plant development. CCT domains played a role in DNA binding and it was also required for the interaction of CO with COP1 or NF-YB2 to affect flowering time. The results suggested comprehensive roles of CCT family genes involved in the regulation of a variety of development and physiological processes in the model plant *Arabidopsis*. The knowledge gained from the studies would be helpful to infer the roles of CCT orthologs in other species with the potential to facilitate crop improvement.

[0300] However, knowledge about the function of the CCT family genes and the agricultural significance in crop species was so far limited to cereal crops. For example, Ghd7 and Ghd7.1 (BBOX-CCT) from rice and ZmCCT and ZmCCT9 (a single CCT) from maize underly respective major QTLs for rice or maize adaptation from tropical cultivation to longer-day higher latitudes, some of which were subjected to artificial selection. These genes were also critical for multiple agriculturally important traits that can favor human needs such as higher grain production. Despite the importance, the evolution and biological/agricultural significance of the CCT family have yet been systematically explored.

[0301] Legumes (Fabaceae) comprised the most economically important bean species that can be used for both grain and forage and has a contributing role to the ecosystem by nitrogen fixation, whereas non-legume crops rarely do. Legumes' grains account for 33% of the protein needs of humans and have been a major plant-based protein provider

to meet a great demand for a legume-rich diet. However, legumes were less researched, lagging greatly behind the cereals in both yielding and planting acreages. In legumes, soybean was the most cultivated legume crop with dual uses for both vegetable oil and high-quality proteins, and was also deemed to be a model legume providing tremendous insights into legume research. Protein and oil content accumulation was investigated primarily in soybean in the last decade, mainly via genetic approaches, while rare gene underlying the mechanism has been identified. Therefore, the mechanism of protein and oil accumulation remains largely unclear, hindering the practical improvement of protein and oil. Thus far, the genetic or molecular link between CCT genes and seed proteins has yet to be reported. Currently, the rapid advances in genomic technologies have led to the development of the reference genome assemblies for several legumes (legumeinfo.org). The availability of the reference genomes provided an unprecedented opportunity to explore and compare CCT motif-containing genes to seek opportunities of using them for legume improvement.

[0302] In the present study, a detailed overview is provided of the quantities, evolution, phylogeny of CCT proteins in eight representative legume genomes with a focus on soybean genome, along with the comparisons with those in *Arabidopsis*, and two cereals, rice and maize. Further provided is a detailed characterization of expression patterns of soybean CCT family genes in different tissues or under different conditions. The natural variation of GmCCTs was explored and the candidacy for previously identified QTLs associated with a variety of agronomically important traits in soybean was predicted. Finally, the role of GmCCT34 involved in protein and oil accumulation was experimentally validated using both fast neutron mutation and gene editing approaches. The results presented here provide a fundamental frame for and new insight into the evolution and biological mechanism of CCT genes in soybean and legumes.

Materials and Methods

A. Identification of CCT Genes in Soybean, Legumes, and Other Species

[0303] The complete protein profiles for soybean (Glycine max, Wm82.a2.v1), eight legumes (adzuki bean (*Vigna angularis*, 3.0), chickpea (*Cicer arietinum*), common bean (*Phaseolus vulgaris*, 2.0), cowpea (*Vigna unguiculata*, v1.0), Medicago (*Medicago truncatula*, Mt4.0v1), pea (*Pisum sativum*, v1.0), peanut (*Arachis hypogaea*), pigeon pea (*Cajanus cajan*), and *Arabidopsis* (*Arabidopsis thaliana*, TAIR10), and two monocot species (maize (*Zea mays*, Ensembl-18), rice (*Oryza sativa*, v7_JGI)) were downloaded from the Phytozome v12 and Legume Information System. Protein profiles from chlorophyte species including *Chlamydomonas reinhardtii* v5.5, *Dunaliella salina* v1.0, *Coccomyxa subellipsoidea* C-169 v2.0, *Micromonas pusilla* CCMP1545 V3.0, *Micromonas* sp. RCC299 v3.0, *Ostreococcus lucimarinus* v2.0 were downloaded from the Phytozome for comparison. The HMM file of the CCT domain (PF06203) was downloaded from Pfam database. Search for the CCT domain proteins ($p < 10^{-10}$) in each protein dataset was carried out with the HMM file using HMMER software (v3.3.1). The search hits obtained from the HMMER were manually checked in the online database SMART (Simple Modular Architecture Research Tool) to ensure the presence of CCT domains ($p < 10^{-10}$). Sequence alignment of the CCT

complete protein sequences or extracted CCT domain sequences were performed using ClustalX2.1 followed with unrooted phylogenetic tree construction using MEGA 7.0 with the neighbor-joining method and 1000-bootstraping analysis.

B. Expression Analysis

[0304] The raw sequencing data for the transcriptomes of seed tissues and compartments in developing seeds of cv. Williams82 at different developmental stages, generated by Goldberg-Harada laboratories, and the sequencing data for circadian clock, abiotic/biotic stress analyses (PRJNA285677, PRJNA288296, PRJNA259941, PRJNA432861, PRJNA207354, PRJNA285880, PRJNA348534) were retrieved from NCBI SRA database and re-analyzed. The raw sequencing reads were aligned to the Williams 82 soybean reference genome (Wm82.a2.v1) with TopHat (v2.1.1). Transcript abundance for each gene was estimated using Cufflinks followed by normalization across samples using the quartile method in Cuffdiff. The heatmap was drawn in R with the function heatmap.2 from the gplots package.

C. Genotyping and Genetic Diversity Analyses

[0305] Genotyping and genetic diversity analysis for GmCCTs were carried out using the 32mSNPs identified in a panel consisting of 1,556 diverse soybean genomes. SNPs and indels with a minimum allele frequency of greater than 0.01 were reported. Genetic diversity (Pi) was calculated in the wild and landrace soybean subpopulations with 10-kb window 5-kb step window as previously described. Pi value for a CCT was calculated by the 5-kb window that harboring the gene, and the ratio of Pi-wild versus Pi-landrace greater than 4 was deemed as a putative selective sweep. The Phyton version of MCScan was used to identify gene blocks and syntenic genes in genomes across the species. OrthoFinder was used to identify single orthologs across the species and the single orthologs were used to construct species phylogenetic tree. Whole-genome duplication data were downloaded from the Plant Genome Duplication Database and duplicated segment pairs ≥ 2 Mb were illustrated as background events in Circos. Information for the previously-identified QTLs in the last decades (1992-2018) were retrieved from the SoyBase, including those associated with flowering time and maturity, seed composition traits (such as oil, protein content, fatty acids, and amino acids), development (such as plant height, lodging, pubescence density, root length, branching, canopy height, leaflet length, yield-related traits (seed set, seed weight, seed yield), as well as responses to abiotic and biotic stressors (such as *Phytophthora sojae*, *Spodoptera litura*, *Helicoverpa zea*, *Fusarium solani* f. sp. *glycines* infection, *Sclerotinia sclerotiorum*, *Heterodera glycines*; drought, flooding). The genomic intervals of the QTLs, traits, flanking markers, authors, and publication titles were included in Table 15. QTL intervals greater than 10 Mb were not included in the analysis.

D. Fast-Neutron Mutant Identification

[0306] Fast neutron (FN) mutant line FN0172932 was selected from the M2 generation of irradiated elite line M92-220 in 2007 and further planted for homozygous mutants (FIG. 1). FN-induced genomic deletion region in M4-generation FN0172932 was previously determined

using Comparative Genomic Hybridization (CGH) and further validated by whole genome sequencing (Illumine NovaSeq PE150, depth of coverage=16). The 1.3-Mb deletion region contains 52 gene models (Glyma. Wm82.a2) Plants were grown in the environment-controlled greenhouse in the Donald Danforth Plant Science Center with regular management (day 25° C./night 22° C., 40% humidity, 16 h/8 h day length for light/dark). Seed protein and oil content were measured on a pre-calibrated Perten DA 7250 analyzer (Perten Instruments, Inc., Springfield, IL, USA). Table 16 below provides details of POWR CCT-subfamily genes and their knockout and overexpression mutants. Table 17 provided field performance details about the POWR1 (CCT-subfamily gene) overexpression mutants.

TABLE 17

Details of POWR CCT-Subfamily Genes and Their Knockout and Overexpression Mutants				
Sr. No.	Gene ID	Gene Name	CRISPR-MEDIATED MUTATION COMBINATIONS	Oil Phenotype (% change)
1	Glyma.20G085100	POWR1	POWR1, 2, 3, 4 POWR1 OVER-EXPRESSION	~3.78% ↑ ~1.34% ↓
2	Glyma.10G134400	POWR2	POWR1, 2, 3 POWR2, 3 POWR2	~3.91% ↑ ~3.44% ↑ ~0.69% ↑
3	Glyma.10G190300	POWR3	POWR1, 3	~3.84% ↑
4	Glyma.20G200400	POWR4	POWR 1, 2, 4	~4.29% ↑

Note:
Arrows (↑ and ↓) indicate the increase or decrease in oil content in the seeds of the mutants grown in the greenhouse condition.

TABLE 17

Field performance details about the POWR1 (CCT-subfamily gene) overexpression mutants					
Sr. No.	Gene ID	Line Description	Oil Phenotype (% change)	field performance (preliminary data)	
1	Glyma.20G085100 (POWR1)	Ubi promoter driving POWR1 expression transgenic line	~2.0% ↓	high	Early (~25 days)
		Native promoter driving POWR1 expression transgenic line	~2.00% ↓	high	Early (~21 days)

E. Generation of Gene Edited Soybean Lines

[0307] Three guide-RNA (gRNA) sequences specific to the exons of GmCCT34, one for exon 1, two for exon 2, were designed using the web tool CRISPOR. The gRNA sequences were synthesized and annealed to the CRISPR/Cas9 expression vector and transformed into soybean cv. Williams 82 by the Wisconsin Crop Innovation Center using an *Agrobacterium*-mediated transformation protocol. A pair of primers specific to the vector was used to confirm positive transformants via PCR amplification (Forward:

CTGCTGTTGATGGAGGACTT SEQ ID NO: 22; Reverse: CTCCTGGAGAAGCAGAAAGTT SEQ ID NO: 23). T1 seeds from 10 independent TO plants were obtained and further grown in the environment-controlled greenhouse in the Donald Danforth Plant Science Center with the same condition as earlier mentioned. Unifoliate leaves were sampled from T1 plants to confirm the gene editing via PCR amplification followed by restriction enzyme digestion (Bs/I). Editing-generated deletion was further confirmed using Sanger sequencing. T2 seeds from the two homologous cct34 mutants were used to measure the seed composition traits as mentioned above.

F. Subcellular Localization Analyses

[0308] The assay was performed through transient expression in *Nicotiana benthamiana* following a known method. The full-length CCT34 coding sequence (CDS), CCT34 lacking the CCT domain, and the CCT domain only were subcloned into the expression vector to generate UBQ10: YFP-CCT34, UBQ10: YFP-CCT34ACCT, and UBQ10: YFP-CCT, respectively. UBQ10: YFP was used as the empty vector. The vectors were individually transformed into *Agrobacterium tumefaciens*, and cultures of each construct were infiltrated into young leaves of *N. benthamiana* plants (4~6 weeks) using a 3-ml syringe without the needle. Leaves were imaged 48 h after infiltration. Imaging was carried out a Leica TCS SP8 confocal microscope using the 63× water immersion lens. Samples were excited with a 514-nm laser line and 649-nm laser line to detect YFP and chlorophyll signals, respectively. Fluorescence emission was collected for best signals of indicated fluorescent probes. This experiment was repeated twice.

G. Arabidopsis Mutant Analysis

[0309] Two independent T-DNA insertions mutant lines (WiscDsLox297300_13A.1 (cct1) and SALK_036731.1 (cct2)) were obtained from ARBC (*Arabidopsis* Biological Resource Center). These two T-DNA insertion regions lie with different sites of the 3'end of the CDS of AT1G04500 (FIG. 29), the closest homolog of soybean GmCCT67 (POWR1) and GmCCT34 (POWR2). The homozygous mutants were identified by PCR with specific primer sets listed in Table 18.

TABLE 18

Primers used		
T-DNA mutant line ID	Left Primer (5'-3')	Right Primer (5'-3')
SALK_036731.1	ATTTATCCCGAATGGTAACGC SEQ ID NO: 37	TACCGGAGATGTGTTTGAAGG SEQ ID NO: 38
WiscDsLox297300_13A.1	AACGTGAAACGGATTGTCTAG SEQ ID NO: 39	AAATTGGTGAAACCAAGACC SEQ ID NO: 40
T-DNA specific Border Primer-LBb1.3	5'-ATTTGCCGATTTCGGAAC-3 SEQ ID NO: 41	

Results

A. Identification of CCT Proteins in Soybean and Other Species

[0310] The CCT domain is a highly conserved basic module with ~43 amino acids at the protein's C-terminus. The Hidden Markov Model (HMM) and the CCT domain (Pfam ID-PF06203) were used to search for the CCT proteins in selected plant species covering all members of the plant kingdom, including algae, mosses, ferns, conifers, and flowering plants. A set of 543 CCTs across the 24 plant species were identified (Table 2), including 69 soybean CCT domain-containing proteins (FIG. 2A, Table 1) and a range from 33 to 62 in other legumes, 40 and 52 CCT proteins, respectively, in the cereal crops rice and maize, and 13 to 29 in non-angiosperm land plants. (FIG. 2A). Traditionally, CCT proteins are classified into three subfamilies according to their constituent domains: single CCT (CCT Motif Family (CMF)), 1-2×BBOX-CCT (CONSTANTlike (COL) Family), and REC-CCT (Pseudo-Response Regulator Family). The present disclosure identified an additional protein group that carries the CCT domain, TIFY-CCT-ZnF_GATA. In these proteins, the CCT domain was located between two different domains, TIFY and ZnF_GATA. It is irrational to exclude the possibility that the CCT domain is involved in the function. Therefore, TIFY-CCT-ZnF_GATA was included in the analysis (FIG. 2B). The numbers of CCT protein genes in the tetraploids soybean and peanut were nearly doubled those in other diploid legumes. The CCT genes identified in Arabidopsis and the two cereal crops were generally more than those in legumes except for common bean and peanut. A small number of CCT genes (2-8) were present in chlorophyte species.

B. Conservation in Domain Composition and Organization

[0311] To better understand the characteristics of CCT domain-containing proteins in plant species, the inventors analyzed domain features in sequences. According to constituent domains, CCT proteins could be classified into four subfamilies including single CCT, 1-2×BBOX-CCT, REC-CCT, TIFY-CCT-ZnF_GATA (FIG. 2B). All four subfamilies can be identified in higher plant species, suggesting a highly conserved domain architecture of CCT proteins among higher species (FIG. 2B). In contrast, only one or two of the four subfamilies were identified in the chlorophyte species that contain significantly fewer CCTs. Other than

these canonical domains, CCT proteins carrying non-canonical domains were also identified, such as DUF740-DUF740-CCT in Vang06g17920 from adzuki bean, Adap-tin_N-CCT in Psat0s3732g0120 from pea, S_TKc-CCT in Ca. 14621 from chickpea. Non-typical CCT proteins were not identified in soybean and model plants *Arabidopsis*. All identified CCT genes in this study were summarized in Table 2.

[0312] The inventors observed that the total numbers of CCT genes in soybean and peanut are approximately 2-3 times of those in other legumes or higher species, and so do for the subfamily. For example, the soybean genome contains 22 single-CCT proteins, which is more than those in legumes (12-16), *Arabidopsis* (15), and rice (14) (FIG. 2A). Interestingly, four subfamily members per species are generally in proportion across the higher plant species, approximately 2:1:1:2 for 1-2×BBOX-CCT:REC-CCT:TIFY-CCT-Zn_GATA:single CCT (FIG. 2B). Differing from the three families containing the CCT domain in the C terminus, TIFY-CCT-ZnF_GATA subfamily contains the CCT in the middle of the sequences.

C. Evolution and Expansion of CCT Family in Legumes

[0313] To gain insight into the evolution of CCT proteins in soybean and legumes, individual phylogenetic trees using the CCT proteins from each species were constructed. It was observed that the majority of the CCT proteins in soybean (68 of 69, 98.6%) and peanut (60 of 62, 96.8%) tree were clustered in pairs, leaving 1-2 unpaired CCT proteins (1.45-3.23%). In contrast, legumes and two cereal species have unpaired CCTs ranging from 6-12, representing 15.0-33.3% of total CCT members. A phylogenetic tree encompassing all investigated CCT proteins in plants was next re-reconstructed. Phylogeny analysis indicated that nearly every CCT family member from legumes corresponds to or evolutionarily closes to a pair of CCT paralogs from soybean or peanut, with each set of orthologous proteins forming individual ortholog-based subclades. These results and approximate ratio of 2 in the numbers of CCTs from soybean and peanut relative to those in other legumes suggested that whole-genome duplication (WGD) was likely the cause of the striking increase in the number of CCTs in soybean or peanut. The possible synteny in genomic regions harboring GmCCT genes were further examined. The 69 GmCCTs were mapped to all 20 chromosomes and the majority were distributed in the distal telomeric regions. Chromosome 13 contains the maximum number of GmCCTs (7) followed by chromosomes 4, 6, and 8 each having 6 members. It was striking that 33 pairs of GmCCTs (66 of 69, 95.7%) were located within syntenic genomic regions. This result and high bootstrap values for the GmCCT pairs in the soybean phylogenetic tree collectively suggested that the paired GmCCTs were paralogs that have been retained from large-scale duplication events such as whole-genome duplication or segmental duplication. This notion should also be applicable to peanut because of the segmental allotetraploid in the peanut genome. In addition, two identified pairs of tandemly duplicated GmCCTs in the soybean genome (GmCCT9/10; GmCCT18/19) also fell within segmental duplicated regions between chromosomes 4 and 6, suggesting that the tandem duplication occurred prior to the soybean-specific WGD. These results well demonstrated that polyploidization, especially the lineage-specific tetraploid in soybean and peanut, was a major evolutionary driven force of CCT expansion.

[0314] To understand the evolution of CCT proteins in related legume species, the syntenic CCT-associated genes and genomic regions were analyzed among selected closely related legume species, including *Medicago*, pea, chickpea, cowpea, common bean, and soybean. The syntenic analysis among leguminous CCTs revealed that 58 (84%) of the GmCCTs have at least one syntenic CCTs in legume genomes (Table 19; FIG. 7). For most legume CCT proteins,

each corresponds to a pair of GmCCT paralogs, such as paralogs GmCCT12/21 in the syntenic regions of single CCT orthologous genes in five legumes (common bean, cowpea, chickpea, pea, *Medicago*) (FIG. 3A; Table 19; FIG. 7). This analysis also led to the identification of soybean-specific GmCCT without syntenic CCT homologs in other legumes, such as the pair of GmCCT34/67 (FIG. 3B; Table 19).

TABLE 19

List of legume CCTs syntenic with GmCCTs						
<i>Glycine</i> <i>Max</i> CCT gene IDs	<i>G. max</i>	Adzuki bean	chickpea Ca CDC	Common bean	Cowpea	Medicago
GmCCT01	Glyma.01G221100.1	Vang07g06400.1	Ca_02437.1	Phvul.002G158500.1	Vigun02g177100.1	Medtr5g010120.1
GmCCT02	Glyma.02G152900.1	Vang06g17920.2	.	Phvul.007G166100.1	Vigun07g116500.2	.
GmCCT03	Glyma.02G215600.1	Vang01g09310.1	.	Phvul.008G182900.1	Vigun08g109700.1	Medtr5g066510.1
GmCCT04	Glyma.02G223700.1	.	Ca_14236.1	Phvul.008G190200.1	.	Medtr5g069480.5
GmCCT05	Glyma.02G230800.1	Vang01g08550.1	Ca_17243.1	Phvul.008G257300.1	Vigun08g124000.1	Medtr5g072780.1
GmCCT06	Glyma.03G209800.1	Vang04g15980.1	Ca_00927.1	Phvul.001G203000.1	Vigun01g187100.1	Medtr7g108150.1
GmCCT07	Glyma.03G261300.1	Vang09g08740.1	Ca_01359.1	Phvul.009G258300.1	Vigun09g004700.1	Medtr7g118260.2
GmCCT08	Glyma.04G058900.1	Vang05g09750.1	Ca_01694.1	Phvul.009G085300.1	Vigun09g217500.1	Medtr3g105710.1
GmCCT09	Glyma.04G096200.1	Vang0017s00340.2	Ca_01471.1	Phvul.009G035400.1	Vigun09g180900.1	Medtr3g100050.1
GmCCT10	Glyma.04G236400.1	Vang0017s00360.1	Ca_03974.1	Phvul.009G035300.1	Vigun09g150200.1	Medtr3g100040.1
GmCCT11	Glyma.04G166300.1	Vang09g02620.1	Ca_18243.1	Phvul.009G185500.1	Vigun09g076300.1	Medtr3g037390.1
GmCCT12	Glyma.04G228300.1	Vang0084s00090.1	Ca_04056.1	Phvul.009G045000.1	Vigun09g140300.1	Medtr3g092780.2
GmCCT13	Glyma.04G236400.1	Vang0090s00480.2	Ca_03974.1	Phvul.009G053300.2	Vigun09g150200.1	Medtr3g091340.1
GmCCT14	Glyma.05G025000.1	Vang11g11310.1	Ca_13165.1	Phvul.003G188400.1	Vigun03g337800.1	Medtr4g108880.1
GmCCT15	Glyma.05G209800.1	Vang10g06540.1	Ca_05769.1	Phvul.002G291600.2	Vigun03g042600.1	Medtr8g098725.1
GmCCT16	Glyma.05G233700.1	Vang10g01650.1	.	Phvul.002G316100.1	Vigun03g017700.1	Medtr8g104190.1
GmCCT17	Glyma.06G059600.1	Vang05g09750.1	Ca_01694.1	Phvul.009G085300.1	Vigun09g217500.1	Medtr3g105710.1
GmCCT18	Glyma.06G097900.1	Vang0017s00340.2	Ca_01471.1	Phvul.009G035400.1	Vigun09g180900.1	Medtr3g100050.1
GmCCT19	Glyma.06G098000.1	Vang0017s00360.1	Ca_01470.1	Phvul.009G035300.1	Vigun09g180800.1	Medtr3g100040.1
GmCCT20	Glyma.06G127900.1	Vang0090s00480.2	Ca_03974.1	Phvul.009G053300.2	Vigun09g150200.1	Medtr3g091340.1
GmCCT21	Glyma.06G136600.1	Vang0084s00090.1	Ca_04056.1	Phvul.009G045000.1	Vigun09g140300.1	Medtr3g092780.2
GmCCT22	Glyma.06G196200.1	Vang09g02620.1	Ca_18243.1	Phvul.009G185500.1	Vigun09g076300.1	Medtr3g037390.1
GmCCT23	Glyma.07G049400.1	Vang0238s00030.1	.	Phvul.010G119700.2	Vigun10g152100.2	Medtr8g024260.1
GmCCT24	Glyma.07G091400.1	Vang02g04910.1	.	Phvul.004G157300.1	Vigun04g021700.1	.
GmCCT25	Glyma.07G182700.1	Vang0064ss00450.1	Ca_05292.1	Phvul.002G213800.2	Vigun03g124600.1	Medtr4g093730.1
GmCCT26	Glyma.08G016500.1	Vang10g06540.1	Ca_05769.1	Phvul.002G291600.2	Vigun03g042600.1	Medtr8g098725.1
GmCCT27	Glyma.08G041100.1	Vang10g01650.1	Ca_22732.1	Phvul.002G316100.1	Vigun03g017700.1	Medtr8g104190.1
GmCCT28	Glyma.08G067200.1	Vang0064ss00450.1	Ca_05292.1	Phvul.002G213800.2	Vigun03g124600.1	Medtr4g093730.1
GmCCT29	Glyma.08G168900.1	.	Ca_13576.1	.	.	Medtr2g068730.1
GmCCT30	Glyma.08G221800.1	Vang01g08240.1	Ca_17522.1	Phvul.008G254600.1	Vigun08g127200.1	Medtr4g008090.1
GmCCT31	Glyma.08G255200.1	Vang04g02640.1	.	Phvul.008G022800.1	Vigun05g024400.1	Medtr7g018170.1
GmCCT32	Glyma.09G184600.1	Vang02g04910.1	.	Phvul.004G157300.1	Vigun04g021700.1	.
GmCCT33	Glyma.10G021400.1	Vang06g17920.2	.	Phvul.007G166100.1	Vigun07g116500.2	.
GmCCT34	Glyma.10G048100.1	Vang0206s00130.1	Ca_14041.1	Phvul.007G162500.1	Vigun07g078600.2	Medtr1g067110.1
GmCCT35	Glyma.10G134400.1	.	.	.	Vigun07g011700.1	.
GmCCT36	Glyma.10G190300.1	Vang06g09620.1	.	Phvul.L001944.1	Vigun07g181300.1	Medtr1g073350.1
GmCCT37	Glyma.10G274300.1	Vang06g00840.1	Ca_05487.1	Phvul.007G027200.1	Vigun07g270500.1	Medtr1g110870.1
GmCCT38	Glyma.11G022600.1	Vang07g06400.1	Ca_02437.1	Phvul.002G158500.1	Vigun02g177100.1	.
GmCCT39	Glyma.12G074400.1	Vang0010ss00750.1	Ca_14630.1	Phvul.011G069800.1	Vigun11g150000.1	Medtr4g061823.1
GmCCT40	Glyma.12G196100.1	Vang09g03140.1	Ca_14148.1	Phvul.005G113300.1	Vigun05g225000.4	Medtr2g088900.1
GmCCT41	Glyma.13G009300.1	Vang03g16900.1	Ca_07641.1	Phvul.006G005200.3	.	Medtr3g082630.1
GmCCT42	Glyma.13G050300.1	Vang02g13360.1	.	Phvul.004G046601.1	Vigun04g068400.1	.
GmCCT43	Glyma.13G093800.1	Vang11g14640.1	Ca_03275.1	Phvul.003G149000.1	Vigun03g379600.1	Medtr4g128930.1
GmCCT44	Glyma.13G100700.1	Vang11g15310.1	Ca_03219.1	Phvul.003G141200.1	Vigun03g387700.1	Medtr4g127420.1
GmCCT45	Glyma.13G135900.1	Vang0206s00130.1	Ca_14041.1	Phvul.007G162500.1	Vigun07g078600.1	Medtr1g067110.1
GmCCT46	Glyma.13G306400.1	Vang09g03140.1	Ca_14148.1	Phvul.005G113300.1	Vigun05g225000.1	Medtr2g088900.1
GmCCT47	Glyma.13G331000.1	Vang0021ss01680.1	Ca_02579.1	Phvul.L002465.1	Vigun05g253600.1	Medtr2g096080.1
GmCCT48	Glyma.14G182800.1	Vang01g09310.1	Ca_14968.1	Phvul.008G182900.1	Vigun08g109700.1	Medtr5g066510.1
GmCCT49	Glyma.14G190400.1	Vang01g09960.2	Ca_14236.1	Phvul.008G190200.1	Vigun08g116900.1	Medtr5g069480.1
GmCCT50	Glyma.14G197900.1	Vang01g08550.1	Ca_17243.1	.	.	Medtr5g072780.1
GmCCT51	Glyma.15G219800.1	.	.	.	.	.
GmCCT52	Glyma.15G258200.1	Vang0022ss02670.1	Ca_13576.1	Phvul.005G049801.1	Vigun01g054600.1	Medtr2g068730.1
GmCCT53	Glyma.16G018000.1	.	.	Phvul.010G119700.1	Vigun10g152100.2	Medtr8g024260.1
GmCCT54	Glyma.16G050900.1	Vang04g13110.1	Ca_05985.1	Phvul.001G087300.1	Vigun01g075800.1	Medtr7g083540.1
GmCCT55	Glyma.16G067000.1	.	.	.	Vigun01g065500.1	Medtr7g032240.1
GmCCT56	Glyma.17G059100.1	Vang11g15310.1	Ca_03219.1	Phvul.003G141200.2	Vigun03g387700.1	Medtr4g127420.1
GmCCT57	Glyma.17G066600.1	Vang11g14640.1	Ca_03275.1	Phvul.003G149000.1	Vigun03g379600.1	Medtr4g128930.1
GmCCT58	Glyma.17G102200.1	Vang11g11310.1	Ca_13165.1	Phvul.003G188400.1	Vigun03g337800.1	Medtr4g108880.1
GmCCT59	Glyma.18G278100.1	Vang04g02640.1	Ca_22148.1	Phvul.008G022800.1	Vigun05g024400.1	Medtr7g018170.1
GmCCT60	Glyma.19G039000.1	Vang02g13360.1	.	Phvul.004G046601.1	Vigun04g068400.1	.

TABLE 19-continued

List of legume CCTs syntenic with GmCCTs						
GmCCT	Glyma.19G079600.1	Vang04g12460.1	.	Phvul.008G147400.1	Vigun01g065500.1	.
GmCCT63	Glyma.19G099700.1	Vang04g13110.1	Ca_05985.1	Phvul.001G087300.1	Vigun01g075800.1	Medtr7g083540.1
GmCCT64	Glyma.19G207100.1	Vang04g15980.1	Ca_00927.1	Phvul.001G203000.1	Vigun01g187100.1	Medtr7g108150.1
GmCCT65	Glyma.19G260400.1	Vang09g08740.1	Ca_01359.1	Phvul.009G258300.1	Vigun09g004700.1	Medtr7g118260.1
GmCCT66	Glyma.20G060400.1	.	Ca_07641.1	.	.	Medtr3g082630.1
GmCCT67	Glyma.20G085100.1	Vang06g25290.1	Ca_20458.1	Phvul.007G270600.1	Vigun07g011700.1	.
GmCCT68	Glyma.20G115600.1	Vang06g00840.1	Ca_05487.1	Phvul.007G027200.1	Vigun07g270500.1	Medtr1g110870.1
GmCCT69	Glyma.20G200400.1	Vang06g09620.1	.	Phvul.L001944.1	Vigun07g181300.1	Medtr1g073350.1
GmCCT38	Glyma.U033700.1	Vang0010ss00750.1	Ca_14630.1	Phvul.011G069800.1	Vigun11g150000.1	Medtr4g061823.1
<i>Glycine</i>						
Max CCT	Pea	Peanut	Pigeon	<i>G. max</i>	Arabidopsis	
gene IDs						
GmCCT01	Psat2g184400.1	arahy.B2EAID.1	C.cajan_13216.1	Glyma.11G022600.1	.	
GmCCT02	.	.	C.cajan_10458.1	Glyma.10G021400.1	AT2G47890.1	
GmCCT03	.	.	.	Glyma.14G182800.1	.	
GmCCT04	.	arahy.TFJ3CW.1	C.cajan_29262.1	Glyma.14G190400.1	.	
GmCCT05	Psat2g054040.2	arahy.XNQT3T.1	C.cajan_05381.1	Glyma.14G197900.1	.	
GmCCT06	Psat3g024360.1	arahy.7A5VUD.1	C.cajan_10578.1	Glyma.19G207100.1	.	
GmCCT07	.	arahy.Y6ILE8.1	C.cajan_10962.1	Glyma.19G260400.1	.	
GmCCT08	.	arahy.ZS9VWH.1	C.cajan_32168.1	Glyma.06G059600.1	.	
GmCCT09	.	.	C.cajan_17917.1	Glyma.06G097900.1	.	
GmCCT10	.	.	C.cajan_17916.1	Glyma.06G098000.1	.	
GmCCT11	Psat5g199360.1	.	.	Glyma.06G196200.1	.	
GmCCT12	Psat5g071920.1	arahy.A9G6ZW.1	.	Glyma.06G136600.1	AT5G24470.1	
GmCCT13	Psat5g077440.1	arahy.BVW291.1	C.cajan_08842.1	Glyma.06G127900.1	AT5G53420.1	
GmCCT14	Psat4g049800.1	arahy.2F1UPX.2	.	Glyma.17G102200.1	AT5G61380.1	
GmCCT16	Psat7g031600.1	arahy.BU1BMX.1	C.cajan_11568.1	Glyma.08G016500.1	AT5G41380.1	
GmCCT15	Psat7g011960.1	.	C.cajan_15626.1	Glyma.08G041100.1	.	
GmCCT17	.	.	C.cajan_32168.1	Glyma.04G058900.1	.	
GmCCT18	.	arahy.0U49HS.1	C.cajan_17917.1	Glyma.04G096200.1	.	
GmCCT19	.	.	C.cajan_17916.1	Glyma.04G096300.1	.	
GmCCT20	Psat5g077440.1	arahy.5WR99K.1	C.cajan_08842.1	Glyma.04G236400.1	AT5G53420.1	
GmCCT21	Psat5g071920.1	arahy.A9G6ZW.1	.	Glyma.04G228300.1	.	
GmCCT22	Psat5g199360.1	.	C.cajan_08715.1	Glyma.04G166300.1	.	
GmCCT23	Psat4g164800.1	.	C.cajan_39868.1	Glyma.16G018000.1	.	
GmCCT24	.	arahy.QP34EV.1	C.cajan_08639.1	Glyma.09G184600.3	.	
GmCCT25	Psat7g101680.1	.	C.cajan_22947.1	Glyma.08G067200.1	.	
GmCCT26	Psat7g031600.1	arahy.AGT1RS.1	C.cajan_11568.1	Glyma.05G209800.1	.	
GmCCT27	Psat7g011960.1	arahy.8MV22J.1	C.cajan_15626.1	Glyma.05G233700.1	.	
GmCCT28	Psat7g101680.1	.	C.cajan_22947.1	Glyma.07G182700.1	.	
GmCCT29	.	.	.	Glyma.15G258200.1	.	
GmCCT30	Psat7g260880.1	arahy.QG9XFC.1	C.cajan_32117.1	.	AT1G51600.1	
GmCCT31	Psat3g193520.1	arahy.S3IANQ.1	C.cajan_20131.1	Glyma.18G278100.1	.	
GmCCT46	.	arahy.WEZ503.1	C.cajan_08639.1	Glyma.07G091400.1	.	
GmCCT32	.	.	C.cajan_10458.1	Glyma.02G152900.1	.	
GmCCT33	Psat6g124840.1	.	C.cajan_41432.1	Glyma.13G135900.1	.	
GmCCT34	.	arahy.22FLX8.1	C.cajan_02757.1	Glyma.10G134400.1	.	
GmCCT35	.	arahy.EF48U6.1	C.cajan_22520.1	Glyma.20G200400.2	.	
GmCCT36	Psat6g228360.1	arahy.WVH2H8.1	C.cajan_31872.1	Glyma.20G115600.1	AT1G68520.1	
GmCCT37	.	.	C.cajan_13216.1	Glyma.01G221100.1	.	
GmCCT39	Psat7g221360.1	arahy.IR754M.1	C.cajan_21434.1	Glyma.U033700.1	.	
GmCCT40	Psat5g265480.1	.	C.cajan_36557.1	Glyma.13G306400.4	.	
GmCCT44	Psat5g105240.1	.	.	Glyma.13G009300.1	.	
GmCCT43	.	arahy.LV381L.1	C.cajan_26721.1	Glyma.19G039000.1	.	
GmCCT42	.	.	C.cajan_03622.1	Glyma.17G066600.1	AT5G57660.1	
GmCCT41	Psat4g013600.2	arahy.0QJD35.1	C.cajan_03709.1	Glyma.17G059100.1	.	
GmCCT45	Psat6g124840.1	.	.	Glyma.10G048100.1	AT5G02810.1	
GmCCT47	Psat5g265480.1	.	C.cajan_36557.1	Glyma.12G196100.1	.	
GmCCT48	Psat5g282000.1	arahy.KU7VJQ.1	C.cajan_13596.1	.	.	
GmCCT49	.	.	.	Glyma.02G215600.1	AT4G24470.1	
GmCCT50	Psat2g060680.2	arahy.NK4UTA.1	C.cajan_29262.1	Glyma.02G223700.1	.	
GmCCT51	Psat2g054040.1	arahy.XNQT3T.1	C.cajan_05381.1	Glyma.02G230800.2	AT1G07050.1	
GmCCT52	.	.	.	Glyma.15G219800.1	.	
GmCCT53	.	arahy.9Y5DYJ.1	.	Glyma.08G168900.1	.	
GmCCT54	.	arahy.PEG0RN.1	.	Glyma.07G049400.1	.	
GmCCT55	Psat3g094880.1	arahy.P8XRWE.1	C.cajan_17635.1	Glyma.19G099700.1	.	
GmCCT56	.	arahy.FCSG49.1	C.cajan_00921.1	.	AT5G14370.1	
GmCCT57	.	arahy.0QJD35.1	C.cajan_03709.1	Glyma.13G100700.1	AT5G57180.2	
GmCCT58	.	arahy.6J3MKV.1	C.cajan_03622.1	Glyma.13G093800.1	AT5G57660.1	
GmCCT59	Psat4g049800.1	arahy.D90B1M.1	.	Glyma.05G025000.1	AT5G61380.1	
GmCCT60	Psat3g193520.1	arahy.669CET.1	C.cajan_20131.1	Glyma.08G255200.1	.	
GmCCT61	.	.	C.cajan_26721.1	Glyma.13G050300.1	.	
GmCCT62	.	arahy.857KFG.1	.	.	.	

TABLE 19-continued

List of legume CCTs syntenic with GmCCTs					
GmCCT63	.	arahy.P8XRWE.1	C.cajan_17635.1	Glyma.16G050900.1	.
GmCCT64	Psat3g024360.1	arahy.7A5VUD.1	C.cajan_10578.1	Glyma.03G209800.1	AT2G47890.1
GmCCT65	.	.	C.cajan_10962.1	Glyma.03G261300.1	AT2G46790.1
GmCCT66	Psat5g105240.1	.	.	.	.
GmCCT67	.	.	C.cajan_02757.1	Glyma.10G134400.1	.
GmCCT68	Psat6g228360.1	arahy.WVH2H8.1	C.cajan_31872.1	Glyma.10G274300.1	.
GmCCT69	Psat6g131680.1	arahy.EF48U6.1	C.cajan_22520.1	Glyma.10G190300.1	AT2G33350.1
GmCCT38	Psat7g221360.1	arahy.IR754M.1	C.cajan_21434.1	Glyma.12G074400.1	.

D. Functional Insights into GmCCT Proteins

[0315] Given the conservation in CCT domain architecture and protein sequences within the clusters, a phylogenetic tree was reconstructed using the complete CCT proteins from four species (soybean, Arabidopsis, rice, maize) to infer functions since many CCT proteins from the latter three species have been functionally characterized. The phylogeny of the four-species CCT tree agrees with the global tree and soybean trees mentioned earlier. It was defined into six clusters (I to VI) by topology. CCT proteins with demonstrated roles in the regulation of flowering-related traits were clustered in different clusters, implying extensive roles of CCT proteins associated with flowering in soybean. These genes included ZmCCT and ZmCCT9, all five PRR proteins (PRR1, PRR 3, PRR 5, PRR 7, PRR 9) from Arabidopsis, two rice REC-CCT genes (Ghd7 and Ghd7.1), six COL family members (CO, COL1-5) associated with flowering time and shoot branching, Arabidopsis COL12 (BBX10) that associated with branching and flowering time and two COL9 (BBX7) with a role in flowering. Clustering of the CCT genes with different roles in some clusters inferred the implicit functions for the phylogenetically-clustered proteins, such as ASML2 involved in the induction of sugar-inducible genes and FITNESS and CIA associated with drought tolerance. This analysis indicated the function conservation and diversity of CCT families; whether the family possesses functions beyond those mentioned above merits more attention.

[0316] The above results suggested orthologous relationships for those closely clustered CCT genes. The syntenic relationship of the leguminous CCTs was investigated next. The syntenic analysis revealed that 58 (84%) of the GmCCTs had at least one syntenic CCTs in legume genomes (Table 3; FIG. 4A-4C). For most legume CCT proteins, each corresponded to a pair of GmCCT paralogs, such as paralogs GmCCT12/21 in the syntenic regions of single CCT orthologous genes in five legumes (common bean, cowpea, chickpea, pea, Medicago) (FIG. 3B; Table 3; FIG. 6). The numbers of CCTs in legumes that were in synteny with GmCCTs varied greatly, such as over 90% for adzuki bean and common bean, 72.72-76.19% for cowpea, *Medicago*, and pigeon pea, and 22.58-45.71% for pea, chickpea, and peanut. This observation was consistent with the phylogeny of these legumes (FIG. 2) where soybean, adzuki bean, and common bean are phylogenetically close and all are relatively far from Medicago and pigeon pea and chickpea, and much further from peanut (Wang et al. 2017), reflecting that evolutions of the CCT proteins in the individual species complied to respective genome evolution histories after legume divergence. The syntenic CCT proteins across legumes were likely originated from common ancestral CCT proteins. This analysis also led to the identification of

soybean-specific GmCCT without syntenic CCT homologs in legumes, such as the pair of GmCCT34/67 (FIG. 3B; Table 3).

E. Conservation and Diversification of CCT Domain Sequence

[0317] Phylogeny analyses in individual CCT trees and the global CCT tree indicated that domain architecture-oriented topology where CCT proteins carrying the same constituent domains were closely clustered. It was observed that six BBOX-CCT (GmCCT15, 27, 24, 46, 36, 68) and single CCT (GmCCT57, 41, 62, 56, 05, 51) proteins were respectively grouped closely and formed a strayed cluster that was separated from the main clusters consisting of the majority of other BBOX-CCT and single CCT proteins. It appeared that the existence of strayed clusters comprising single-CCT with BBOX-CCT proteins was common for all other investigated higher plant species, although they varied in topology. The phylogenetic relationship of the strayed members in plant species was determined. Intriguingly, in the global tree, those strayed single-CCT or BBOX-CCT proteins identified in the individual trees were clustered together and formed a large combined strayed cluster (FIG. 6). Within the large strayed cluster, single-CCT or BBOX-CCT proteins were separately clustered by monocots and dicots. These results suggested that these strayed CCT orthologs originated from common ancestral genes and have been retained following the divergence of rosids. Separation of those single-CCT or BBOX-CCT proteins from major clusters suggested possibly distinct roles from thus far known functions for most identified CCT genes such as photoperiod-associated flowering time control or yield-related traits. For example, BBX15/COL16 and FITNESS protein in this cluster were related to chlorophyll accumulation and H₂O₂-related defense and CIA2 involved in protein import into the chloroplast.

[0318] The global tree suggested the phylogenetic relationship of the entire CCT protein sequences, the phylogeny of CCT domains alone across the species was further investigated. Surprisingly, the topology was highly congruent between the protein and domain trees, including the strayed clusters (cluster III) and singletons (such as black dots, red dots in clusters I, IV, VI) dispersed in non-self subfamilies (FIG. 4A and FIG. 4B). This observation suggested that the CCT domains from the same cluster of domain tree were likely originated from common ancestral CCT domains and then co-evolved with respective protein sequences while remained diversified among clusters (I-VI) or subfamilies. With this observation, it was hypothesized that those protein singletons (black or red dots) were likely derived from phylogenetically-close members from the same domain cluster via addition or loss of one or two domains. For

example, the single-CCT proteins in cluster IV of the global tree were likely derived from the loss of REC domain in one of phylogenetically close REC-CCTs or common ancestral proteins. This analysis suggested phylogenetic diversification of CCT proteins in plant species which in part enriched CCT family diversity and explained the origin of a few CCT proteins.

[0319] Notably, single-CCT genes were found in all six clusters (FIG. 4A). In clusters I, II, IV, and VI, consist of only a few individual single-CCTs, likely representing recent deletions of the non-CCT domain in these genes. It is also likely that several 1×BBOX-CCTs in the two 2×BBOX-CCT clusters (I and II) likewise represent the deletion of a single BBOX domain. Cluster III, however, contains a large number of single-CCTs that form two clades in the domain phylogeny (FIG. 4B). These likely represent an ancient deletion of the BBOX domain in this clade prior to the origin of the angiosperms.

[0320] Interestingly, CCTs containing non-canonical domains were rare and dispersed across several clusters, likely representing singleton insertion events, for example, DUF740-CCT in Vang06g17920 (adzuki bean), Adaptin_N-CCT in Psat0s3732g0120 (pea), S_TKc-CCT in Ca.14621 (chickpea) (FIG. 4A, 4B). Non-typical CCT proteins were not identified in soybean and *Arabidopsis*. All identified CCT genes in this study were summarized in Table 1. HMM logos were next prepared, representing each cluster (I-VI) from the domain tree to analyze the amino acids across the clusters (FIG. 6C). Most of the amino acids were conserved in the CCT domain across the six clusters, with high conservation observed for seven amino acids (Arginine (R) 1, R15, Tyrosine (Y) 23, R26, Alanine (A) 30, R35, and Phenylalanine (F) 40). Also, cluster-specific conserved amino acids were identified. For example, F8 in clusters V and VI, while Lysine (K) 22 was highly conserved in IV, with some exceptions (FIG. 6C). These conserved amino acids across the clusters could likely represent the essential roles of CCT family genes in DNA binding or forming functional complexes. In contrast, the amino acids specific to one or certain clusters might associate with the DNA binding specificity representing functional variation in the CCT family. The results indicated that the CCT domain sequences are conserved in plant species with diversified function specificities plausibly facilitated by some uniquely conserved amino acids.

[0321] All these six groups were identified as angiosperms. To further investigate the origin of these clusters, their membership in a range of non-angiosperms were identified, including charophyte algae, mosses, ferns, and gymnosperms (Table 20). All six clusters could be identified in each of the land plant lineages; however, two groups (I and VI) were absent from all of the chlorophyte species. This indicates that most of these groups arose early in plant evolution except for one of the 2×BBOX-CCT groups (I) and the TIFY-CCT-Zn_GATA group (VI), which first appeared in the bryophytes. Additionally, within the chlorophytes, Cluster IV (REC-CCT) was missing from all species except *Chlamydomonas*, and Cluster III (1×BBOX) was missing from *Micromonas* and *Dusinelia*. These results indicated that individual chlorophyte lineages may have lost these genes or their sequences sufficiently diverged that the current search model could not identify them. Along with the increased number of CCTs from chlorophytes to bryophytes,

the CCT domain gene family is ancient and underwent substantial expansion and diversification in the land plant lineage.

TABLE 20

CCT Genes in Other Species							
Species	I	II	III	IV	V	VI	Unclustered
spikemoss	1	2	3	3	2	2	0
liverwort	0	1	2	2	1	2	0
Angiosperm	85	75	89	93	129	62	5
fern	3	3	4	6	6	1	1
gymnosperm	4	2	4	4	4	2	0
Bryophyte	9	7	24	10	10	10	0
chlorophyte	0	7	2	6	1	0	2

F. Function Diversification of CCT Proteins

[0322] Given the conservation in CCT domain architecture and protein sequences within the clusters, a phylogenetic tree was reconstructed using CCT proteins from four species (soybean, *Arabidopsis*, rice, maize) to infer functions since many CCT proteins from the latter three species have been functionally characterized. The phylogeny of the four-species CCT tree was in agreement with the global tree and soybean trees mentioned earlier, and was defined into six clusters (I to VI) by topology. Cluster I consisted of single-CCT proteins while rare of which were characterized. ZmCCT was located within a monocot-specific subcluster and it involved in maize adaptation from short-photoperiod tropical environments (Southern Mexico) to Northern long-day environments. Whether the phylogenetically close GmCCTs such as GmCCT38 possessed relevant roles warranted experimental determination. ASML2 was highly expressed in *Arabidopsis* stem and perhaps functioned as a transcriptional activator in the regulation of a subset of sugar-inducible genes, two homologs in soybean GmCCT29 and GmCCT53 were likely to have the similar function because both were found to be highly expressed in soybean stems (FIGS. 4A-4C and 8).

[0323] Cluster II represented REC-CCT domain proteins and the REC was conserved domains for PRR (PSEUDO-RESPONSE REGULATOR) proteins that were mainly studied in *Arabidopsis* but rarely in other species. All five PRR proteins (REC-CCT) from *Arabidopsis* (PRR1 (TOC1), PRR3, PRR5, PRR7, PRR9) were clustered in this cluster. The cluster also contained two rice REC-CCT proteins (Ghd7 and Ghd7.1) functioning in the regulation of flowering time (heading date)-associated adaptation with potential in enhancing yield potential (grain number) (Xue et al. 2008; Yan et al. 2013). These and other studies (Weller and Ortega 2015) suggested conserved functions of REC-CCTs in the regulation of circadian clock and light response-related flowering. With these results, it was deduced that many of those uncharacterized PRR proteins (REC-CCT) in legumes likely function related to circadian clock responses or photoperiodic flowering control. It was known that legumes originate differently, either temperate regions with long daylength (i.e. pea, chickpea) or lower latitudes with short-day photoperiod (i.e. soybean, cowpea, common bean). In the light of above results, the REC-CCT genes had the potential to widen the latitudinal range of legume cultivation (FIG. 6C).

[0324] Cluster III mainly comprised 2×BBOX-CCT proteins mixed with several other subfamily members. Six COL family members (CO, COL1-5) associated with flowering time, shoot branching and ZmCCT9 associated with high latitude adaptation were clustered in this cluster. In soybean, GmCCT proteins in this cluster except for GmCCT61 and GmCCT43 exhibited similar spatial expression patterns as *Arabidopsis* CO in floral bud, leaf, and stem, suggesting a possible conserved role in flowering time regulation. Proteins carrying single-CCT domain from cluster IV were phylogenetically close to FITNESS and CIA and might have functions relevant to chloroplast development or ROS homeostasis-associated drought tolerance. In cluster V, two BBOX-CCT proteins (GmCCT40 and GmCCT47) were phylogenetically closed to *Arabidopsis* COL12 (BBX10) that associated with branching and flowering time; and two 2×BBOX-CCT proteins (GmCCT44 and GmCCT66) were close to COL9 (BBX7) with a role in flowering. The four-species CCT tree provided an overall picture of functional diversity in plant species and insight into the putative role of GmCCTs.

G. GmCCTs Exhibited Expression Specificities to Circadian Clock, Environmental Stress, or Tissues

[0325] To gain more insight into the roles of GmCCT genes, the expression profiles in different conditions including circadian rhythm, abiotic stress (drought, Zn, low temperature, O₃), and biotic stress (cutworm, *F. graminearum*, reniform nematode, and aphid) were investigated. It was revealed that sixteen GmCCTs showed varying circadian clock responses in a Zeitgeber time (ZT) interval of 20 h, with four REC-CCTs and two single-CCT proteins highly expressed during ZT8-12 h and three pairs of BBOX-CCT paralogs exhibiting high expression during early and late ZT points of the period (FIG. 5), suggesting relevant roles in circadian rhythm and likely photoperiodic flowering time control. In addition, it was identified that GmCCT genes were responsive to the challenges of drought, salt, cutworm or *F. graminearum* (FIG. 5). Two pairs of GmCCTs (GmCCT34/67, GmCCT35/69) exhibited relative insensitivity to elevated temperature but were inducible to O₃ stress. Interestingly, the circadian clock responsive CCT genes were rarely identified to be responsive to abiotic and biotic stress and vice versa for the stress-responsive GmCCTs, implying functional specialization of the GmCCTs.

[0326] Given the agronomic importance of seed quality traits for soybean in current breeding programs, the analysis was extended to investigate the expression in different seed compartments (i.e. inner/outer integument of the seed coat, suspensor, endosperm) at different seed development stages (globular, heart, cotyledon, early-maturation) along with major vegetative organs (seedlings, leaves, floral bud, stem, root), aimed to understand whether GmCCTs correlate with seed compartment profiles. Overall, most of the single-CCT proteins were expressed in seed compartment tissues while 1-2×BBOX-CCT showed tissue-specific expressions in non-seed vegetative tissues, such as GmCCT02 preferentially expressed in stems (STEM) and GmCCT47 exclusively expressed in floral bud (FLUB) (FIG. 6). It was noteworthy that the four O₃-responsive GmCCT genes (GmCCT34/67, GmCCT35/69) were also preferentially expressed in the seed coat compartments (seed coat outer integument at the cotyledon stage (COT-OI) and seed coat parenchyma at the early maturation stage (EM_SC_PY)) (FIG. 6). The paren-

chyma cell was the innermost part of seed coat that was in direct contact with the embryo, and it contained components related to nutrient transport and metabolism to support embryo growth during seed filling. The four GmCCT genes may play roles associated with seed development or storage reserves accumulation, which was rarely reported in plants. GmCCT genes encoding TIFY-CCT-ZnF-GATA and REC-CCT did not appear obvious expression specificity in the tested seed compartment tissues.

[0327] It was also observed there was conserved and divergent expression for GmCCT paralogs across tissues, circadian clock response, and environmental stress. For example, seed coat-specific expressed paralogs GmCCT34/67 and circadian clock responsive gene pair GmCCT24/46 showed similar expression patterns, while GmCCT56/62 exhibited different circadian clock responses (FIGS. 8, 9). Similar expression pattern and conserved protein sequences in the paralogous GmCCT genes suggested that they preserved the ancient biological functions. They likely had retained the similar promoter elements essential for expression specificity. The divergent expression may lead to sub-functionalization or neo-functionalization in different tissues or environment responses, enriching the functional diversity of GmCCT family.

H. Exploring Natural Variation in GmCCTs and Co-Located QTLs

[0328] To explore the natural variation in GmCCT family, it was examined in the coding sequences within a panel of 1,556 soybean genomes from diverse genetic backgrounds. After investigation, four types of variants were identified that may cause amino acid changes in 58 (84.1%) of 69 GmCCTs. In total, 250 variants (minor allele frequency >0.01) were identified, including 214 non-synonymous SNPs, 5 SNPs causing alternative splicing, 30 indels ranging from 3-28 nucleotides, and 2 nonsense SNP mutations that caused premature proteins (Table 4). The variants that cause protein sequence changes might be responsible for morphological or physiological changes. For example, GmCCT67, also known as POWR1, contains a 321-bp indel in the CCT domain that likely abolishes the function. The indel in POWR1 accounts for the significant variation in seed protein and oil content. Other than this, GmCCT17 (2×BBOX-CCT) is phylogenetically close to COL3 and COL4 and associated with abiotic stress tolerance and flowering. The GmCCT17 carried an SNP in the 1st exon causing the premature stop codon in 28 diverse accessions, 22 of which (78.6%) originated from Northern China (north of the Shandong province (36.6° N)). It is intriguing to determine if the variant contributes to latitudinal adaption. The diverse variants revealed here can be valuable for gene functional characterization or breeding purposes.

[0329] Previously published QTLs were incorporated into the analysis and assessed co-location between GmCCT genes with the QTLs. Total of 66 (95.7%) of 69 GmCCTs reside in genomic regions were identified that had 680 non-redundant reported QTLs, including 220 for seed quality traits, 131 for seed set, 119 for abiotic/biotic stress tolerance, 51 for flowering time and maturity, and 158 for development-related traits (Tables 5-9). It has been demonstrated that many PRR homologous proteins under the major QTLs for heading time and grain yield. Seven (58.3%) of 12 REC-CCT proteins (PRRs) were situated in QTLs associated with the first flower, photoperiod sensitivity,

reproductive stage length, R8 full maturity and plant height or seed yield (one of Tables 5-9). Intriguingly, six of the 16 (37.5%) circadian clock-responsive GmCCTs (GmCCT11, 22, 05, 68, 43, 60) were located within the flowering-related QTLs (FIG. 5; one of Tables 5-9). All four proteins that were preferentially expressed in seed coat compartments were located within QTLs for seed quality or seed set traits (one of Tables 5-9).

[0330] Further, the genetic diversity (TT) for GmCCTs resident loci were analyzed and 12 GmCCTs (GmCCT06, GmCCT14, GmCCT20, GmCCT26, GmCCT32, GmCCT41, GmCCT42, GmCCT59, GmCCT61, GmCCT63, GmCCT64, GmCCT67) were located within selective sweep regions (Table 4). Many of these merited further attention, for example, GmCCT05 (a *FITNESS* homolog) co-located within a QTL associated with drought tolerance. The most noteworthy is GmCCT67, located within multiple QTLs, including four QTLs for protein, four for oil content, one for seed weight, and one for yield (Table 10). It was recently proven that the major QTL *cqPro20* controls protein, oil, and seed weight simultaneously and is subjected to strong artificial selection, which strongly supports the diversity analysis. Whether other QTL-colocalized genes carry advantageous mutations targeted by human selection deserves experimental determination.

I. GmCCT Genes are Stress-Responsive

[0331] CCT genes regulate a plethora of functions in plants. Expression profiles of GmCCT genes ($n=69$) were investigated in response to various abiotic and biotic signals, including circadian rhythm, abiotic stress (drought, Zn, low temperature, 03), and biotic stress (cutworm, *F. graminearum*, reniform nematode, and aphid). A set of sixteen GmCCTs showed varying circadian clock responses in a Zeitgeber time (ZT) interval of 20 h, including four REC-CCTs, two single-CCT proteins, and three pairs of BBOX-CCT paralogs (FIG. 5), suggesting relevant roles in circadian rhythm and likely photoperiodic flowering time control. In addition, GmCCT genes that were responsive to the challenges of drought, salt, cutworm, or *F. graminearum* (FIG. 5) were also identified, such as two pairs of GmCCTs (GmCCT34/67, GmCCT35/69) exhibiting relative insensitivity to elevated temperature but were inducible to 03 stress. Further, phylogenetically close genes were identified, particularly the paired GmCCT paralogs that retained similar expression patterns or exhibited divergent expression. For example, GmCCTs (64, 06, 63) showed similar expression responses to drought, and GmCCT56/62 exhibits different circadian clock responses (FIG. 5), which may enrich the functional diversity of the GmCCT family during evolution to cope with diverse environment responses.

J. GmCCT34 Involved in Seed Protein and Oil Content Accumulation

[0332] Considering the agronomic importance of soybean seed quality traits, the expression profiles of GmCCTs were investigated in different seed compartments (i.e., inner/outer integument, seed coat, suspensor, and cotyledon) and at different seed development stages (globular, heart, cotyledon, early-maturation). Also, the expression of these genes were analyzed in major vegetative organs (seedlings, leaves, floral bud, stem, root), aimed at additional GmCCTs involved in seed compartment profiles. Overall, a correlation

was observed between tree topology and expression profile, suggesting sequence co-evolution with spatial expression. Most of the single-CCT proteins were expressed in seed compartment tissues. In contrast, 1-2x BBOX-CCT showed tissue-specific expressions in non-seed vegetative tissues. For example, GmCCT02 was preferentially expressed in stems (STEM), and GmCCT47 exclusively expressed in the floral bud (FLUB) (FIG. 6). GmCCT genes encoding TIFY-CCT-ZnF-GATA and REC-CCT did not appear to have apparent expression specificity in the tested seed compartment tissues.

[0333] Remarkably, the cluster of four O3-responsive GmCCT genes (GmCCT34/67, GmCCT35/69) were preferentially expressed in the seed coat [seed coat outer integument at the cotyledon stage (COT-OI) and seed coat parenchyma at the early maturation stage] (FIG. 6). The parenchyma cells are the innermost part of seed coat that is in direct contact with the embryo. It contains nutrient transport and metabolism components to support embryo growth during seed filling. It was recently demonstrated that GmCCT67 (*POWR1*) regulates protein and oil accumulation, seed weight, and field yield. Given the conserved expression pattern in the seed coat, it was reasoned that the other three GmCCTs might function similarly in seed quality traits.

K. GmCCT34 Involved in Seed Protein and Oil Content Accumulation

[0334] A high expression of GmCCT34 was analyzed specific to seed coat tissues (FIG. 1A, FIGS. 10, 11A). To test if GmCCT, other than *POWR1*, is involved in the regulation of the seed composition in soybean, a fast neutron mutant FN0172932 was identified lacking a 1.3-Mb genomic region (Chr10: 35253890-36584337). Interestingly, the *GmCCT34* mutant (FN0172932) M4 seeds contain an average of ~5.5% less protein ($p<0.001$) and ~2.241% more oil content ($p<0.001$) than the wild-type (WT) seeds (FIG. 9E), suggesting its role involved in regulating protein and oil accumulation.

[0335] Additionally, to confirm if the absence of GmCCT34 causes the low protein-high oil phenotypic change in the fast neutron (FN) mutant, GmCCT34 knock-out lines were generated in soybean cv. Williams82 (Wm82) background using CRISPR/Cas9-mediated gene editing. Two mutant lines (*cct34-2*, *cct34-4*) homozygous for GmCCT34 with nucleotide deletions of varying lengths simultaneously at designed targeted sites in the 1st and 2nd exons were identified (FIG. 9B-9D). Consistent with the protein-oil content in FN mutant result, T2 seeds of homologous *cct34* edited lines showed significantly low protein and high oil accumulation. *GmCCT34* seeds contained ~7.86% ($p<0.001$) low protein content and ~2.85% ($p<0.001$) higher oil content than the WT Wm82 (FIG. 9E). Differing from GmCCT67 (*POWR1*) that CCT domain-truncated *POWR1* is responsible for greater 100-seed weight than the wild type, the 100-seed weight was reduced in both FN0172932 and *cct34* mutants, although the latter was not statistically significant. These results demonstrated and validated that the seed coat-preferentially expressed GmCCT34 regulates seed protein and oil accumulation in soybean.

[0336] Given the observed changes, it was reasoned that the CCT domain plays an important role for the GmCCT34's function. subcellular localization assays were next carried out of intact and fragmented GmCCT34 containing or

lacking the CCT domain. The analysis clearly illustrated that GmCCT34 carrying intact CCT domain (UBQ10:YFP-CCT34) and the CCT domain only (UBQ10:YFP-CCT) were located exclusively in the nucleus (FIGS. 2A and 2B), whereas, GmCCT34 lacking the CCT domain (UBQ10:YFP-CCT34ACCT), like the empty vector (UBQ10:YFP-Emptyvector), expressed in the nucleus and cytoplasm. This result indicated the essential role of CCT domain to direct GmCCT34 in the nucleus and that removal of CCT domain in GmCCT34 likely abolished the function in enhancing protein accumulation.

L. Arabidopsis CCT-Clade Protein Regulates Protein-Oil Content in Seeds

[0337] Beyond the four seed-coat GmCCTs from soybean, the phylogenetic analysis clustered a set of homologs from selected species with POWR1 and GmCCT34 into a distinct clade, it was asked whether those from non-legume plants remain similar function, the function of the Arabidopsis CCT gene, AT1G04500, was investigated for its involvement in regulating seed protein-oil composition. Two homozygous Arabidopsis T-DNA-insertion mutants were isolated as Atcct-1 and Atcct-2. The T-DNA insertion in these mutants occurred before and after the CCT domain, respectively, indicating that the CCT domain is dysfunctional (FIG. 10A-10C). Similar to GmCCT34, the seed composition analysis of the Atcct mutants revealed a higher oil and lower protein content compared with the wild type seeds (FIG. 10A-10B). These results suggest a conserved function of the CCTs between soybean and Arabidopsis in regulating protein and oil accumulation.

M. Arabidopsis CCT-Clade Protein Regulates Protein-Oil Content in Seeds

[0338] Function of the Arabidopsis CCT gene, AT1G04500, was investigated for its involvement in regulating seed oil composition. Like GmPOWR 1234 genes, there is only a single CCT domain found in Arabidopsis AT1G04500 gene (hence after AtPOWR1). The gene expression analysis showed that the AtPOWR1 is highly expressed in the seed coat tissues (FIG. 11A-11B, red color indicating the AtPOWR1 expression).

[0339] There is no information on the function(s) of AtPOWR1 concerning the regulation of seed protein-oil content. To know if this *Arabidopsis* gene also functions similarly to the GmPOWR genes, two homozygous T-DNA-insertion mutants were isolated (WiscDsLox297300_13A.1 and SALK_036731.1, (labeled as cct-1 and cct-2). The T-DNA insertion in these mutants occurred before and after the CCT domain, respectively, indicating that the CCT domain is dysfunctional. Similar to GmPOWR mutants, the seed composition analysis of the AtPOWR1 or Atcct mutants revealed a higher oil content compared with the wild type seeds. These results suggested a conserved function of the CCTs between soybean and *Arabidopsis* in regulating oil accumulation in seeds.

DISCUSSION

A. Evolution of CCT Proteins

[0340] The results indicated that CCT domain-containing proteins with four subfamilies were conserved in land plants, in both domain architecture and constitution. Three

of the four domain structures of CCT proteins can date back to an ancestral origin in chlorophyte species and were significantly expanded in land species after the long evolutionary divergence. The long evolutionary history after speciation has caused a great deal of sequence divergence because of mutation. Highly divergent domain sequences across the subfamilies suggested possibly distinct functions. “Membership” switch was also identified by observing possible addition or loss of domains between the subfamilies, which could be a result of domain shuffling due to unequal chromosomal crossover or transposon participation. The domain rearrangement might be a cause for the presence of all four subfamilies in land species compared with the absence of 1-3 subfamilies in chlorophyte species. Domain shuffling also increased the emergence of novel CCT genes with functional innovation, such as those uncanonical CCT proteins identified in this study.

[0341] It was also demonstrated that WGD during the plant evolution was the major driven force of CCT family expansion, which led to striking high numbers in higher plants compared to dramatic lower number chlorophyte species. CCT domain possessed a role in DNA binding by which the protein can affect downstream gene expression levels and associated functions. It appeared that CCT genes in soybean and peanut were retained after WGD based on the tree topology, which can be explained by previous studies demonstrating that functions of signaling or regulatory genes were more likely retained following WGD events relative to the genome-wide average. Most of the CCT paralogs remain similar expression patterns in tissues or environmental conditions after WGD, suggesting functions in common. On the other hand, approximately 50% of paralogs genome-wide in soybean are differentially expressed with possibly divergent functions. For example, ABSCISIC ACID INSENSITIVE 3a (ABI3a) retains functions associated with seed migration and dormancy while GmABI3b was neofunctionalized like GmLEC2 in modulating seed fatty acid biosynthesis in soybean. Similarly, many GmCCT paralogous pairs likely experienced expression divergence, suggesting they have undergone differentiation. Given the reported functions such photoperiod adaptation and drought tolerance, expansion of CCT genes with divergent functions may enable plants more resilient to the change of environmental factors such as latitudinal photoperiod or drought conditions.

B. More GmCCTs Might be Involved in Soybean Flowering Control

[0342] In general, legumes have their respective origins at lower (soybean, cowpea, pigeon pea, and common bean) or higher (such as chickpea, pea) latitudes, and their cultivation have been expanded to regions beyond the origins after domestication and modern improvement. The underlying mechanism was partially revealed in soybean by investigation of E series genes and Dof11/GmPRR37 that contribute to latitudinal adaptation. For example, GmPRR37 lacking the CCT domain (mutant allele in the Williams 82 genome) confers early flowering, which enables soybean to be adaptive at a higher latitude with the long-day condition (ref). However, the underlying mechanism remains largely unknown in legumes. The present study provides a list of GmCCTs with varying photoperiodic responses that may be involved in soybean latitudinal adaptation. The orthologs in Brassicaceae and Fabidae deserve further attention to

exploring the functions and potential for crop improvement extensively. Nevertheless, the analysis enables a comprehensive inventory of GmCCT genes with a variety of predicted functions, which is useful in improving the discovery of function for the syntenic orthologs in legumes, particularly in the circumstances that rare CCT underlying flowering-related QTLs and protein accumulation has been identified in legumes.

C. The Family Contains a Distinct CCT Clade with the Potential for Seed Traits Improvement

[0343] The demand for plant-based protein has been increasing worldwide because of rapid global population growth, and legumes are certainly the major providers of adequate plant-based protein worldwide. Finding genes that regulate seed protein levels is critical to leverage them for improvement. However, few genes controlling protein content have been identified in soybean, and much less in other legumes, hindering their use for substantial protein improvement.

[0344] Here, the instant systematic study identified four GmCCTs explicitly expressed in seed coat tissues wherein nutrient-transporting elements are active, suggesting a relevant role in nutrient accumulation. Indeed, both fast neutron and CRISPR-Cas9 mutation analyses showed that GmCCT34 reduced protein and increased oil content and seed weight. As additional strong evidence, the present contemporary assay demonstrated that GmCCT67 underlies the major QTL cqPro-20 controlling protein and oil levels in seeds. These results clearly demonstrate the role of both genes from the clade in regulating protein and oil content. Likewise, the other two seed-coat-specific GmCCTs (GmCCT35/GmCCT69) that are phylogenetically closest to GmCCT34/GmCCT67 likely function similarly. It is unexpected that mutation in the Arabidopsis ortholog AT1G04500 also affected protein and oil content, suggesting that the function is conserved between soybean and Arabidopsis, which diverged approximately 90 MYA (ref). In this context, legumes are much closer (~59 MYA) to soybean than Arabidopsis. Therefore, legume CCTs from this clade likely have similar functions associated with protein accumulation, although the spatial expression pattern and function need to be elucidated. Nevertheless, the present analysis identified this set of CCT genes, with two of which being functionally validated, may contribute to protein or oil content improvement by genetic engineering in legumes and other grain crops.

D. The Possible Mechanism for Regulating Seed Nutrient Accumulation

[0345] The four GmCCTs were highly expressed in developing seed coat tissues, such as parenchyma, during early and cotyledon stages. The parenchyma is the innermost part of the seed coat with direct contact with cotyledon. It contains transporters facilitating the nutrient transfer, such as a sugar transporter GmSWEET39, involved in sucrose transporting for oil and protein accumulation. The two stages represent the key period of seed filling when photosynthetic accumulates and is delivered from maternal tissues to filial cotyledon to support a developmental embryo. Therefore, relatively high expression of the four GmCCTs in the tissue at the stages suggests their stage and tissue-prominent function, which regulate biological processes associated with nutrient transport in the seed coat.

[0346] Studies in Arabidopsis demonstrated that CCT domains have DNA binding activity and are required for its interaction with COP1 or NF-YB2 in binding the promoter of FT to regulate flowering time. The present studies demonstrated that GmCCT34/GmCCT67 might function like transcription factors as knockout of the CCT domains abolished their exclusive expressions in the nucleus. Therefore, it is likely that the GmCCTs regulate an array of genes associated with nutrient transport as inferred by its primary expression in the seed coat. In addition, CCT genes are identified to activate the expression of a subset of sugar-inducible genes such as SUS2, and sugar can serve as the precursor for lipid biosynthesis. Overall, the present data suggested a plausible role of the CCTs affecting sugar transport capacity in the seed coat, which influences its supply for the biosynthesis of protein or oil in the cotyledon. Further studies must determine the detailed regulation mechanism and their roles in protein-oil negative correlation.

E. Functional Conservation and Mutation

[0347] Studies have shown that many CCT genes have conserved functions after specification in cereals and Arabidopsis, such as a role of photoperiod-associated flowering time control. In general, legumes had their respective origins at lower (soybean, cowpea, pigeon pea, and common bean) or higher (such as chickpea, pea) latitudes and their cultivation have been expanded to regions beyond the origins after domestication and modern improvement. In soybean, identification of flowering time controlling genes in legumes and soybeans such as E series genes and FT gene family provided one perspective of the mechanism of flowering time control, whereas the mechanism underlying latitudinal adaptation remained largely unclear. GmCCT genes played conserved roles in photoperiodic response as revealed in recent studies. For example, Gmpr37 lacking the CCT domain (mutant allele in the Williams 82 genome) conferred early flowering which enabled soybean to be adaptive at a higher latitude with the long-day condition. The instant study provided a list of GmCCTs with varying photoperiodic responses that may possibly be involved in soybean latitudinal adaptation, which needs experimental determination. Given the conserved function in affecting flowering time, the syntenic orthologs in legumes deserve further attention. Nevertheless, the analysis enabled a comprehensive inventory of GmCCT genes with a variety of predicted functions, which was useful in improving the discovery of function for the syntenic orthologs in legume, particularly in the circumstances that rare CCT underlying flowering-related QTLs and protein accumulation has been identified in legumes.

[0348] Mutations occurred in the CCT proteins caused substantial phenotypic changes. Thus far, the discovery of the agriculturally important CCTs in grain crops owes to the identification of the sequence variation in a natural population, such as Gmpr37 lacking the CCT domain, and transposable element interfered ZmCCT9. The current study identified truncated CCT proteins in the reference genomes of cultivated legumes and cereals and numerous variants in the soybean natural population with many of which were likely subjected to artificial selection. It was possible that many of these play roles in domestication syndrome traits. For example, seed coat-exclusively expressed GmCCT67 lacking the CCT motif was located within a sweep region and likely involved in seed quality traits in soybean, which

was supported by the alternative assay where the knockout of its syntenic gene GmCCT34 significantly increased oil accumulation while reduced protein content in soybean seeds. The mutated CCT identified in legumes and soybean natural population might under selective pressure for the responsible phenotypic variation and can be prioritized for further examination.

Gmct34 Possesses a New Role in Seed Composition Accumulation

[0349] Previous studies demonstrated that CCT domains had DNA binding activity and were required for its interaction with COP1 or NF-YB2 in binding the promoter of FT to regulate flowering time, and a CCT gene can also activate the expression of a subset of sugar-inducible genes such as SUS2. Sugar can serve as the precursor for lipid biosynthesis. On the other hand, the parenchyma was the innermost part of the seed coat with direct contact with cotyledon, and it contained transporters facilitating nutrients transfer, such as a sugar transporter GmSWEET39 involved in sucrose transporting for oil and protein accumulation. Considering these and the results presented herein, GmCCT34 perhaps associated with many genes involved in nutrients transport such as sucrose or amino acids into the cotyledon for storage reserves accumulation. The CCT domain might play a key role as disrupted CCT domain in cct34 might abolish its DNA binding function and associated biological pathways in oil and protein accumulation and seed weight. Seed oil often positively correlates with seed weight, an important yield component, while both negatively correlate with protein content in soybean, and the negative correlation poses a challenge for improving protein while maintaining satisfied yield. The synergistic changes in protein and seed weight in cct34 seeds may offer an opportunity to improve both traits simultaneously, although the mechanism remains to be uncovered. Further, GmCCT34 likely had no syntenic orthologs in legumes, therefore, the function involved in protein and oil accumulation might be lineage-specific to soybean.

CONCLUSIONS

[0350] Plant-specific CONSTANS, CONSTANS-LIKE, and TIMING OF CAB EXPRESSION1 (CCT) domain-containing proteins regulate diverse functions associated with plant growth, development, responses, and agronomic traits. The soybean genome contained 69 CCT-domain proteins preferentially retained after whole-genome duplication. Recently, the CCT-domain gene has been shown to regulate the protein content in soybean seeds. The present studies analyzed the role of four closely related CCT-family subcluster genes; GmCCT34, GmCCT35, GmCCT67, and GmCCT69. These genes were identified as highly conserved in seed coat and flower or reproductive tissues across plant species. Interestingly, the orthologues of these genes are present in early land plants and exhibit reproductive tissues-specific expression. The present disclosure evaluated the role of these four CCT-family subcluster genes in soybean seed protein-oil content. Notably, the GmCCT transgenic, gene-edited, and fast neutron mutant seed analysis showed that these genes contained significantly lower protein and higher oil content than the wild-type seeds. The present results provided deeper insight into the CCT gene family evolution, phylogeny, and functions. Overall, the present

disclosure showed that protein-oil-regulating CCT genes could be a potential source for seed quality improvement in soybean and other crops.

Example 2. POWR1, a Key Domestication Gene Pleiotropically Regulating Seed Quality and Yield in Soybean

[0351] Seed protein and oil content, weight and field yield were the major traits impacting the economic value of soybean. the present multidisciplinary study revealed that a CCT (CONSTANS, CO-like, and TOC1) gene, POWR1 (Seed Protein-Oil-Weight-Regulator 1), underlied a major QTL on chromosome 20, and pleiotropically regulated these important seed traits. A transposable element (TE) insertion truncated its CCT domain and altered its exclusive localization in the nucleus. The POWR1 was specifically expressed in the seed coat of developing seeds and preferentially regulated expression of nutrient transporting and lipid metabolism genes. Study revealed that a dynamic POWR1 allele transfer occurred post domestication. However, TE insertion was completely associated with the transition from *G. soja* to *G. max*. It was hypothesized that POWR1 was a key domestication gene and played an important role in pleiotropically regulating the seed quality and yielding traits likely through a seed-coat specific transcriptional regulatory program. Selection for larger seeds fixed POWR1_{+TE} allele in cultivated soybean and contributed to shaping cultivated soybean with higher seed yield/weight/oil content and relatively lower protein.

INTRODUCTION

[0352] Soybean [*Glycine max* (L.) Merr.] is one of the most important seed crops grown worldwide. It was domesticated from wild soybean (*G. soja* Sieb. & Zucc.) in East Asia about 6,000-9,000 years ago. Domestication and improvement have shaped soybean as the most important dual-function crop to provide both highly valuable seed protein and oil, which together account for almost all of soybean economic value.

[0353] Seed protein content, oil content and yield were considered as three of the most important traits in soybean improvement. On average, commodity-type soybean varieties contained about 40% seed protein and 20% seed oil. However, the three traits vary greatly in soybean nature population and often inter-relate with each other. Seed protein frequently showed a negative correlation with seed oil content and yield; however, its underlying genetic mechanism remain largely unknown. The complex correlation of the three important traits posed a great challenge in simultaneously improving both the soybean seed quality traits and yield to increase the overall economic value of soybean. In addition, cultivated soybean also contained a higher seed yield and oil content, but lower protein content than their ancestry wild soybean. It was important to illustrate the genetic and molecular basis underlying the three traits and their trait correlation, and to understand how those interrelated and important traits have been selected over the course of soybean domestication and improvement for soybean.

[0354] Through a combination of genomics, genetics, and molecular biology approaches, it was uncovered that a CCT-domain gene, POWR1 (Seed Protein-Oil-Weight-Regulator 1), underlied a large-effect protein and oil QTL on

chr20 that has been pursued for the past three decades. It was demonstrated in the current study that a TE (transposable element) insertion in the conserved CCT domain was the causative variant contributing to the large variation in seed protein and oil in soybean population. Expression of the high-protein POWR1 allele in soybean was supportive of its function and potential in present-day needed high protein breeding worldwide. The study provided an insight into the molecular and genetic basis underlying the important seed traits and their correlation, and its key role in soybean domestication and improvement.

Results

A. A 321-Bp TE Insertion is Likely the Causative Variant of a Major QTL on Chr20 Controlling Seed Oil and Protein Content and Seed Weight

[0355] Genome-wide association studies (GWASs) using GLM and MLM models with 38,066 genome-wide SNPs (Single Nucleotide Polymorphisms) identified three significant loci on chromosomes 10, 11, 20 for oil content with a values less than 0.05 in a panel of 278 diverse soybean accessions (FIGS. 13A and 14B). The most significant SNP (ss715637321 on chr20: 32,835,139) on the chr20 coincided with a genomic region where high-effect protein and oil QTLs have been repeatedly mapped to in the last three decades, but their underlying variant remains unknown. The current analysis was focused on the QTL on chr20 QTL and delimited the locus to an approximately 4-Mb region (chr20: 29,050,000-33,120,000) that expanded from the most significant SNP (FIGS. 11A and 11B). To uncover its underlying causative DNA variant, whole genome resequencing data of the 278 accessions was analyzed. The association study with the SNPs and InDels (Insertions and Deletions) present in the 4-Mb region identified a prominent cluster of 25 significant associations for oil that spanned a 154-kb region (chr20: 31,658,904-31,812,853) (FIG. 12A). Out of the 25 highly significant DNA variants (23 SNPs and 2 InDels with $p \leq 1 \times 10^{-17}$), a 321-bp InDel showed the most significant association ($p = 6.17 \times 10^{-24}$) (FIGS. 12A, 12B and 12C). The 321-bp InDel was also among the significant associations with protein content and 100-seed weight in the association analyses at a single nucleotide resolution (FIG. 12A; Table 10). None of these DNA variants located in coding regions of the 12 genes in the 154-kb region except for the 321-bp InDel present in Glyma.20G085100 (Table 10).

[0356] The seed oil and protein content, and seed weight were next examined in the panel of the accessions by splitting them into *G. max*-Del, *G. max*-Ins, and *G. soja*-Del. Interestingly, no *G. soja* accession containing the insertion allele was observed in the panel. However, both Del-carrying *G. soja* and *G. max* accessions were dramatically lower in oil (by 7.1 and 8.2%) and seed weight (by 14.0 and 14.59 g of 100-seed weight) and higher in protein (by 5.1 and 7.3%) than *G. max*-Ins accessions. In contrast, no or relatively small differences (1.5% for oil, 2.2% for protein, 0.2 g for 100-seed weight on average) were present between *G. soja*-Del and *G. max*-Del for the three seed traits, suggesting that the observed phenotypic differences were primarily contributed by the InDel allelic variation rather than the overall difference between *G. max* and *G. soja*. (FIG. 15D). These results further supported that the chr20 QTL is associated with seed oil, protein, and 100-seed weight, and

the 321-bp InDel in Glyma. 20G085100 was likely the causative variant for the chr20 QTL.

[0357] BLAST (Basic Local Alignment Search Tool) revealed that the 321-bp InDel sequence was highly homologous to the terminal sequence of a LINE (Long Interspersed Elements) transposon element (TE), which belongs to the Gm11 family. This gene was designated POWR1 for seed Protein, Oil, Weight Regulator 1. The POWR1 alleles with and without the 321-bp insertion were named POWR1_{+TE} and POWR1_{-TE}, respectively.

B. The TE Insertion Likely Underlies the High-Effect Protein and Oil QTLs on Chr20 in Multiple RIL Populations

[0358] A genotype analysis was conducted on a bi-parental population of 300 recombinant inbred lines (RILs) generated from Williams 82 (*G. max*, HOLP (High Oil, Low Protein)) and PI479752 (*G. soja*, LOHP (Low Oil, High Protein)) with the SoySNP50K array and further conducted both GWAS (GWAS_{RIL}) and linkage mapping. Linkage mapping identified two major QTLs on chr15 and chr20. The QTL on chr20 had a large effect and explained 21.9% of total oil variation and 23.4% of total protein variation. Both linkage mapping and GWAS revealed that the TE located in the most significant protein and oil QTL intervals on Chr20 (FIG. 13A). GWAS_{RIL} identified three adjacent SNPs on chr20 (ss715637271, ss715637273, ss715637274) that had the most significant, equal associations ($p = 1.19 \times 10^{-17}$) with oil and protein content. They were all located within the 154-kb region identified above in association analysis using the panel of 278 diverse soybean accessions (FIG. 12A, FIGS. 13B, 13C). The TE insertion was located between two of the three peak SNPs (ss715637273, ss715637274). Whole genome sequencing and PCR genotyping results verified the presence of the TE insertion in Williams82 and absence in PI479752 (FIGS. 12C, 12F) and showed 100% co-segregation of the TE insertion with HOLP in 30 selected RILs containing either high protein or high oil (Table 11). Consistent with its effects in natural population used for GWAS described above, RILs carrying TE insertion contained 5.2% higher oil ($p = 4.00 \times 10^{-13}$) and 6.2% lower protein ($p = 3.53 \times 10^{-10}$) than those RILs lacking the insertion with statistical significance (Table 11). GWAS_{RIL} and linkage mapping from the RIL population provided additional evidence supporting that the 321-bp insertion as the causative variant for the oil and protein QTL on Chr20.

[0359] Large-effect protein and/or oil QTLs have been identified in the genomic regions containing POWR1 in multiple bi-parental RIL mapping populations, but their causative variants have remained unknown. A genotype analysis was conducted on the TE in parents of 15 mapping populations previously used for protein or oil QTL mapping. The results revealed that parents of seven populations (3 *G. max* × *G. soja*, 4 *G. max* × *G. max*) were polymorphic for the TE, while parents of eight populations (*G. max* × *G. max*) were not (FIG. 12F; Table 9). Notably, the oil and/or protein QTL on the chr20 region was only identified in populations whose parents were polymorphic for the TE insertion but not in populations whose parents were not polymorphic. In all seven pairs of parental lines, the high-oil parent carried the TE insertion while the low-oil parent lacked it (Table 12). These results further supported that the TE variation was likely the DNA variant underlying these previously mapped QTLs on chr20.

C. POWR1 Associated with Seed Field Yield in Addition to Seed Weight, Protein, and Oil Content

[0360] Correlation of the TE allele and seed traits were further investigated by analyzing a set of near-isogenic *G. max* lines (NILs) at the QTL on chr20. The polymorphism of the TE insertion in NILs was experimentally confirmed, and the variation completely correlated with the phenotypic variation of seed protein and oil content and seed weight (FIG. 14). Consistently, NILs lacking the TE (POWR1_{-TE}) exhibited significantly 3.29% higher in seed protein ($p < 0.001$), 1.95% lower in seed oil ($p < 0.001$), and 1.04 g reduced 100-seed weight ($p < 0.001$) than those carrying the 321-bp insertion (POWR1_{+TE}) (FIG. 12E). Importantly, those POWR1_{+TE}-carrying lines had 150.3 kg/ha higher yield than POWR1_{-TE} lines ($p < 0.01$), suggesting that POWR1_{+TE} played an important role in increasing yield potential in addition to the three seed traits (oil, protein, and seed weight).

D. POWR1_{+TE} Encodes a Truncated CCT Domain Protein with Altered Nuclear Localization

[0361] POWR1_{-TE} encoded a protein containing a highly conserved CCT (CONSTANS, CO-like, and TOC1)-domain at the C-terminus. It was present in both dicot and monocot species, suggesting its ancient origin in plants (FIGS. 15A and 15D). POWR1_{-TE} in wild soybean PI479752 contained an intact CCT domain of 44 amino acids, whereas POWR1_{+TE} in cultivated soybean Williams 82 contained the TE insertion in Exon 4 encoding part of the CCT motif (FIGS. 15A, 15B and 15C). The LINE transposon in POWR1_{+TE} is 304 bp in size and generated a 17-bp target site duplication (SEQ ID NO: 24; GTATGCTTGCCGCAAAA) upon insertion (FIG. 15C). Consequently, this 304-bp insertion resulted in a reading frameshift and produced POWR1_{+TE} containing a truncated CCT domain (27 amino acids short) and a distinct amino acid sequence at the C-terminus relative to POWR1_{-TE} (FIG. 15B). LINE transposons did not require excision to replicate. The mutation generated by the insertion should be stable. None of its closely related CCT genes in examined legume species contained the TE insertion (FIGS. 15A, 15D), suggesting that the TE insertion occurred in soybean and had a lineage-specific role in soybean.

[0362] The TE insertion caused little overall structural change in the predicted 3D protein structure between POWR1_{+TE} and POWR1_{-TE} except for their C-terminal end harboring the CCT domain (FIG. 15E). The second half of the CCT-motif contained a putative nuclear localization signal. The subcellular localization of POWR1_{-TE} was examined and determined if the TE insertion altered subcellular localization of POWR1_{+TE}. Transient expression of the two protein alleles in tobacco (*Nicotiana benthamiana*) leaves revealed that POWR1_{-TE} was exclusively localized in the nucleus (FIG. 15G), suggesting that POWR1 is a transcription-associated factor, in consistency with the fact that many CCT-domain proteins are transcription co-factors. However, POWR1_{+TE}, like the empty vector, was localized in both nucleus and cytoplasm, implying that the CCT domain is a functional element in its subcellular localization, and the TE insertion might affect function of POWR1 through disrupting its subcellular localization pattern.

E. POWR1_{-TE} and POWR1_{+TE} Preferentially Expressed in Seed Coat and Flowers in a Similar Expression Pattern.

[0363] Gene expression analysis revealed that both POWR1 alleles preferentially expressed in flowers and developing seed coat at the early and middle maturation

stages. They also had a similar expression pattern in the tissues (FIG. 15H), suggesting that the TE insertion unlikely affected their expression pattern. Comparing transcriptomes of mid-maturation seeds from 132 soybean accessions containing POWR1_{+TE} and 40 containing POWR1_{-TE} revealed no significant expression difference between POWR1_{+TE} and POWR1_{-TE} (FIG. 15F). Sequence variation in the 2-kb promoter sequences was not observed between POWR1_{+TE} in Williams 82 and POWR1_{-TE} in PI479752, the RIL parental lines analyzed above (FIG. 16A-16B). Thus, both gene expression and sequence comparisons suggest that TE insertion cause variation of seed traits likely through altering protein activity, not gene expression. Preferential expression in seed coat implied a possible role of POWR1 in nutrient transport in seed coat, a major function of seed coat.

F. POWR1 Affects Genes and Pathways Involved in Seed Composition Traits and Seed Weight

[0364] To gain insight into molecular mechanism underlying how POWR1 regulates the seed traits, the transcriptomes of mid-maturation seeds were compared between four and six *G. max* accessions carrying POWR1_{-TE} and POWR1_{+TE}, respectively. As expected, the two genotypic groups had no significant difference in POWR1 expression (Table 13). The transcriptomic comparison identified a total of 1,163 differentially expressed genes (DEGs) associated with TE insertion. KEGG and GO terms related to metabolisms of fatty acid, lipid, and starch and sucrose, transmembrane transport, carbohydrate metabolism, regulation of transcription (biological process) and apoplast (cellular component) were significantly enriched for the DEGs (FIG. 15I). This result is consistent with the preferential expression of POWR1 in seed coat tissues that are mainly responsible for transporting multiple nutrients to support metabolic activities in cotyledon for seed development (FIG. 15H), as well as its pleiotropic effects on multiple seed traits including oil and protein content and seed weight.

[0365] Expression analysis revealed that a set of regulatory and metabolic genes involved in protein and oil production were differentially regulated in the seed coat and/or cotyledon of NILs for POWR1 (+TE/-TE) at the mid-maturation stage (FIG. 15J). For example, lipid biosynthesis genes (DGAT1, AAE, GAPT9) and sugar transporter genes (SUC2, SUS4) were significantly increased in POWR1_{+TE} relative to POWR1_{-TE} background in both seed coat and cotyledon tissues. The most striking increase was observed for BCAT2, which is involved in branched-chain amino acid metabolism, suggesting its contributing role for relatively lower protein content in POWR1_{+TE} than in POWR1_{-TE}. The regulators (WR11, ABI3b, and ABI5) involved in seed development and size, as well as oil accumulation, were also upregulated in POWR1_{+TE} relative to in POWR1_{-TE}, suggesting that these regulators might act downstream of POWR1. Differential expression of these genes in the NILs suggests that they are likely part of the transcriptional regulatory cascade underlying POWR1 regulation of the seed traits.

G. Expression of POWR1-TE in Transgenic Soybean Increased Protein and Reduced Oil Content and Seed Weight

[0366] To examine the function of POWR1_{-TE}, the intact POWR1_{-TE} cDNA was introduced driven by a strong and constitutive expressing Ubiquitin promoter and the 1.9-kb

POWR1 native promoter into POWR1_{+TE} *G. max* background (cultivars. Maverick and Williams 82, respectively). Two events overexpressing (OE) Ubiquitin promoter-driven POWR1 transgenic seeds (UbiOE1 and 2) were obtained, and qRT-PCR confirmed its high expression in OE plants (FIG. 18E). The UbiOE1 and UbiOE2 seeds contained significantly higher seed protein content ($p < 0.01$) by 2.50% and lower seed oil by 2.36% ($p < 0.05$) and 100-seed weight ($p < 0.05$) by 3.57 g compared with those in non-transgenic control seeds (FIG. 19A). Eighteen (18) independent T1 transgenic plants were analyzed, it was observed that soybean containing native promoter-driven POWR1_{-TE} (Nat-OE) contained significantly higher seed protein by 4.39% and significantly lower seed oil by 1.31%, but had no statistically significant change in seed weight (FIG. 19B). The results clearly supported that POWR1 controls seed oil and protein content and seed weight in soybean, and it can be manipulated to alter soybean protein, oil and seed weight for seed quality improvement.

H. POWR1 is a Domestication Gene

[0367] Next, the distribution of POWR1 alleles in an expanded soybean population consisting of 548 diverse accessions was evaluated. Principal component analysis (PCA) revealed that the majority of 150 *G. soja* accessions were clustered together as one group exterior to the group consisting of 398 *G. max* accessions (FIG. 20A). After allele assignment, it was found a nearly complete association of POWR1_{-TE} and POWR1_{+TE} alleles with *G. soja* and *G. max* populations, respectively, with a few exceptions. Specifically, 94.7% (377 of 398) of *G. max* possessed the POWR1_{+TE} allele, while all *G. soja* but one (149 of 150) carried the POWR1_{-TE} allele (FIG. 20A). In agreement with earlier results, the POWR1_{-TE} allele was associated with 4.47% lower oil and 5.73% higher protein contents, and 5.08 g lower seed weight than POWR1_{+TE} allele in *G. max* accessions significantly ($p < 0.001$). This pattern of allelic effects on the seed traits remained in *G. soja* groups (1.56% for oil, 3.65% for protein, 0.12 g for seed weight) (FIG. 20B). A genomic scan revealed that POWR1 was located within an approximately 520-kb selective sweep region (chr20: 31,641,057-32,160,913) as inferred by Tajima's D of < -2 (FIG. 20C) and high *G. soja*/*G. max* IT In-ratios (larger than 2.4) (chr20: 31,654,290-32,157,761) (FIG. 20D). These results indicated that POWR1 was a domestication gene contributing to the phenotypic variation of the seed traits, and that POWR1_{+TE} was subjected to artificial selection likely for higher seed weight and oil during soybean domestication.

I. Dynamic Interspecific Allele Transfer During Post-Domestication

[0368] It was observed that twenty-one *G. max*-POWR1_{-TE} accessions and one *G. soja*-POWR1_{+TE} accession had POWR1 alleles contrasting the majority of *G. max*-POWR1_{+TE} and *G. soja*-POWR1_{-TE} accessions (FIG. 20A). To learn about the origin of the unusual presence of POWR1 alleles in these exceptional accessions, a global phylogenetic tree was constructed using the genome-wide SoySNP50K SNPs and a local phylogenetic tree using the whole genome resequencing-generated SNPs in the 154-kb region for the 548 accessions (FIG. 12A). The global tree exhibited similarity to the PCA result (FIGS. 20A, 21A). All *G. max*

accessions (*G. max*-POWR1_{+TE} and *G. max*-POWR1_{-TE} (clusters 1.1, 1.2, 1.3, 2, 3)) clustered together and were separated from all *G. soja* accessions (*G. soja*-POWR1_{-TE} and *G. soja*-POWR1_{+TE} (singleton 4)), regardless of the TE variation (FIG. 21A). However, in the local phylogenetic tree, all *G. max*-POWR1_{-TE} accessions changed from the *G. max* cluster as seen in the global tree to the more diverse *G. soja* clusters (clusters 1, 2, 3) while the *G. soja*-POWR1_{+TE} accession (singleton 4) switched to the *G. max* cluster (FIG. 21B), indicating that transfers of POWR1 alleles occurred between *G. soja* and *G. max* after domestication and produced the *G. soja*-POWR1_{+TE} accession and the *G. max*-POWR1_{-TE} accessions. Without including these accessions with post-domestication allele transfer, all remaining *G. soja* accessions carried POWR1_{-TE} and all *G. max* accessions contained the POWR1_{+TE} allele. The complete association of POWR1_{+TE} with *G. max* and POWR1_{-TE} with *G. soja* and its function in controlling seed weight and yield, an important domestication syndrome, support that POWR1_{+TE} was subjected to strong and exclusive selection during the domestication and plays a key role in soybean domestication.

[0369] All *G. max*-POWR1_{-TE} were clearly clustered into three clusters (clusters 1, 2, 3) in *G. soja* clade of the local tree, while the accessions from each of these clusters were split into different, far related clusters in the global tree, such as cluster 1 to 1.1, 1.2, 1.3 and scattered distribution of cluster 3 accessions in *G. max* clade. This suggested that the fragments harboring POWR1_{-TE} were transferred into diverse *G. max* accessions likely from *G. soja* accessions, hence producing those *G. max*-POWR1_{-TE} accessions with diverse genetic backgrounds, as shown by their scattered distribution in the global tree (FIG. 21A). To gain insight into the allele transfer, the pairwise genetic distance across the 4.1-Mb region between each of the 21 *G. max*-POWR1_{-TE} accessions, and their phylogenetically closest *G. soja* accessions (PI464927A, PI578341, and Zj-Y188) in the local tree, was calculated and plotted to detect possible transferred regions harboring POWR1_{-TE} (FIG. 21C). Pairwise distance analysis showed diverse patterns of highly identical sequences with variable lengths within the region among the three clusters. Briefly, a region (roughly 1.2 Mb long) with high sequence identity with shared one end or both ends was identified in the cluster 1 while cluster 3 had the transferred fragments carrying the POWR1_{-TE} at variable lengths, and cluster 2 had the shortest transferred fragment containing the POWR1_{-TE} (~ 500 kb long). The results supported that the POWR1_{-TE} in those *G. max* accessions likely originated from post-domestication allele transfer events and went through multiple chromosomal crossovers. Next, these accessions were mapped to their geographic origins and revealed close geographic proximity of *G. max*-POWR1_{-TE} with their phylogenetically closest *G. soja*-POWR1_{-TE} (in the local tree) and *G. max*-POWR1_{+TE} (in the global tree) in multiple geographic locations (South Korea, Japan, China) of East Asia (FIG. 21E), implying that the allele transfers likely took place within these regions. Indeed, despite an average decrease of 2.7% oil content and 3.2 g 100-seed weight, those *G. max*-POWR1_{-TE} from East Asia contained 6.5% higher protein content than their closely related *G. max*-POWR1_{+TE} accessions (FIG. 21D, FIG. 17), in accordance with needs for high-protein soyfood in East Asia.

DISCUSSION

[0370] Significant efforts have been dedicated to identifying gene(s) and variant(s) causative for the QTL on chr20 in the past three decades, because of its strong association with multiple seed traits including seed weight, oil and protein content and yield, which represented the economically most important traits in soybean. Having taken advantage of whole genome re-sequencing data from 278 highly diverse accessions, the single nucleotide-resolution association analysis together with high-confident biparental genetic mapping, it was uncovered that a single gene, POWR1, underlies the QTL for the seed traits and a TE insertion in POWR1 was the causative allele (FIG. 12F). It is further supported by results from transgenic soybean experiments, subcellular localization studies of the two POWR1 alleles, and gene expression analyses. Association of POWR1_{+TE} with higher field yield in near isogenic lines is likely achieved through regulating seed weight because of their positive correlation. Although the pleiotropic effect of POWR1 on the important seed traits posed a challenge to improve all traits simultaneously, it offered an opportunity to use appropriate alleles for improving oil content, protein content or their balance. For example, soybean containing high protein content were developed by transferring POWR1_{-TE} from one of the few unusual *G. max* germplasms into elite lines carrying a POWR1_{+TE} allele.

[0371] It has been shown that CCT domain containing genes mainly function in photoperiod-related adaptation in Arabidopsis and cereals. However, the present study demonstrated that POWR1 regulates oil, protein and seed weight/yield in soybean. In consistence with its function, it was revealed that POWR1 was preferentially expressed in the coat, a tissue that played a key role in transporting nutrient into cotyledon in storage reserve production and seed filling. The TE insertion in the CCT domain disrupted the exclusive localization of POWR1 in the nucleus but caused little change in its expression in seeds and other seed compartments and tissues. Thus, TE insertion increased oil and seed weight likely through altering its protein function, not its expression. Given the role of CCT domain in DNA binding and protein interaction, the transcriptome and real-time RT-PCR showed that POWR1 is likely involved in regulating the expression of genes involved in oil and protein metabolism, nutrient transporting and regulating seed development. For example, ABI5 with a known role in determining seed size and BCAT2 with a function in protein degradation had significantly higher expression in a POWR_{+TE} background, in accordance with the result that seeds carrying POWR_{+TE} had lower protein content, higher oil content and larger seed weight. Based on the expression analysis, POWR1_{-TE} may act upstream of these metabolic genes, transporter genes and regulators (including WR1a, ABI5), which collectively affects the three seed traits.

[0372] Larger seed size was suggested as an earlier selected domestication trait for several cereal crops, and it was likely true for soybean as well. Recent archaeological studies suggested that arose of increased oil content in soybean seed might be no later than seed enlargement, suggesting that oil content increase might occur earlier than or simultaneously with seed size enlargement. Nearly complete fixation of POWR_{+TE} in *G. max* and the complete absence in *G. soja* in this 548-accession population was also identified in a larger population consisting nearly 4000 soybean accessions being sequenced recently (FIG. 22).

Thus, TE insertion in POWR1 should be among the key events during transitioning from *G. soja* to *G. max*. Selection for soybean with larger seeds and higher seed yield likely led to fix the POWR_{+TE} in modern *G. max*. However, it is unlikely that oil as a non-visible oil trait was the driving force for selection in early soybean domestication. Thus, oil increase could simply be the by-product since it is pleiotropically controlled by POWR1_{+TE} (FIG. 22). The resulting decrease in protein content in seed due to the preferential selection for POWR1_{+TE} might not have significant impact in ancient agriculture. However, it created present-day challenges for the animal feed industry and compromises seed protein content that was desired and increasingly demanded for human consumption. As the low-protein phenotype was fixed in *G. max*, transfer of the high-protein allele (POWR1_{-TE}) from *G. soja* into *G. max* may increase the seed protein content as needed. This represented a reversal of the domestication process, and introgression and selection for POWR1_{-TE} can be seen in Asian breeding programs which were likely driven by need for high protein soybean in Asia. A soybean accession with TE insertion that was annotated as *G. soja* was also observed. However, this accession was likely from a hybridization event between *G. max* and *G. soja*. Given an outcrossing rate of up to 19% for *G. soja* and up to 6% for *G. max*, natural gene flow and introgression from cultivated soybean to their wild relatives might be a common source such as arise of semi-wild soybean (FIG. 22).

[0373] The instant study provided strong evidence supporting that POWR1 played a key role in soybean domestication and pleiotropically regulates seed protein, oil, and weight, likely seed yield. However, many QTLs for the seed traits and several domestication genes including a recently identified GmSWEET gene underlying a QTL on Chr15 have been identified. Soybean seed oil, protein, seed weight and field yield phenotypic values were the accumulative effects of those QTLs across the soybean genome. It was still largely unknown about how POWR1 and other domestication genes were selected during soybean domestication in shaping modern cultivated soybean, and its interaction with other associated QTLs in determining the phenotypic value of those traits. This enabled better understanding of soybean domestication process and the molecular mechanism controlling those seed traits. A comprehensive investigation of these loci and their relationship with POWR1 may enable better understanding of soybean domestication process and their underlying molecular mechanism controlling those seed traits.

Materials and Methods

A. Plant Materials

[0374] A panel of 548 soybean accessions (398 cultivated soybean *G. max* and 150 wild soybean *G. soja* (Siebold & Zuccarini)) from the genetic resources information network (GRIN) database of U.S. National Plant Germplasm System (<https://npgsweb.ars-grin.gov/>) was used in this study. Out of 548 accessions, 278 accessions (116 *G. soja* and 162 *G. max*) with variations in seed oil (7.5-23.5%), seed protein (36.7-56.9%) and 100-seed weight (1.0-26.5 g) were used for association analysis (FIG. 13A-13C). An F_{6,7} population of 300 recombinant inbred lines (RILs) from a genetic cross between *G. max* cv. Williams 82 and *G. soja* PI479752 was used for genetic linkage mapping. Seed oil content among

the RILs varied from 9.82-20.47% and 37.64-47.99% for protein content. Seeds of the parents and RILs were planted at the USDA-ARS farms in Beltsville, Maryland, in 2012 and 2015 with two replications in a randomized block design. The highly homozygous (>99%) near-isogenic lines (NILs) were created from a F7 plant heterozygous for POWR1 from a cross of G03-3101×LD00-2817P. Plant growth and phenotype measurements were performed as previously described. The NILs homozygous at the POWR1 locus were planted in replicated field trials in nine environments (one in Arkansas, Missouri, North Carolina, and six in Tennessee) in 2016 and 2017 with randomized complete block design. The TE variations in NIL lines were validated by a PCR assay with a pair of PCR primers flanking the InDel. All soybean plants including the transgenic lines used for DNA genotyping and quantification of seed traits were grown in the Donald Danforth Plant Science Center greenhouses (St. Louis, MO, USA).

J. Measurement of Seed Traits

[0375] Phenotypic data including seed protein and oil content (%), 100-seed weight (g) for the panel of 548 accessions were acquired from the Germplasm Resources Information Network. Oil and protein content of the RIL population, the transgenic plants and all other soybean plants were measured using the near-infrared reflectance (NIR) spectroscopy using a DA 7250 NIR analyzer (Perten Instruments, Sweden) unless specified. Approximately 50 seeds per line were analyzed and measured twice. For NILs, approximately 20 g seeds were grounded to powder and also measured with Perten DA 7250 analyzer. Seed trait measurements were averaged over all replications and locations for both NIL groups and compared.

K. Sample Sequencing, Read Alignment, and Variant Calling

[0376] A total of 91 diverse *G. soja* accessions which represent over 90% diversity of wild soybeans in the US soybean collection were re-sequenced using the Illumina NextSeq500 sequencer. For the remaining 457 accessions in the association panel and newly published soybean re-sequencing data, raw sequencing reads were retrieved from the NCBI SRA database. All quality-controlled reads were aligned to the *G. max* reference genome (Williams 82.a2 v1) with BWA. DNA variants including SNPs and InDels were called using the GATKs pipeline. The resulting variants were filtered using GATKs VariantFiltration with following parameters: read depth ≥ 5 reads, SNP quality ≥ 50 , and at least 2 SNPs in a 10-bp window were allowed. Read alignments were visualized using the Integrative Genomics Viewer. The resulting 28,708 SNP and 131 InDel markers in a 4.1-Mb region (29-33.15 Mb) were used to carry out regional association analyses. Whole developing seeds at the mid-maturation stage were collected in environmental controlled greenhouses and multiple seeds per accession were pooled for transcriptome sequencing, as previously described. Transcriptome analysis was performed with TopHat and Cufflinks, and the FPKMs across samples were normalized with the quantile method in Cuffdiff.

L. Association and Linkage Mapping

[0377] DNA variants were quality controlled before being used for genome-wide or regional association analysis with

TASSEL5 with following criteria: a minimum minor SNP allele frequency of 0.05, a maximum proportion of heterozygous sites of 0.2, and a minimum number of accessions per site of 85%. Five principal components as determined in TASSEL5 were used for population structure (Q). Kinship (K) was calculated using centered IBS method in TASSEL5. GLM (general linear model) and MLM (mixed linear model) were used for genome-wide association mapping and regional association analysis, as implemented in TASSEL. For the RIL population, GLM without population structure Q, or GLM with Q, or MLM with Q and kinship K, returned almost identical mapping associations for oil and protein using 19,848 SNPs from the SoySNP50K-set. The Bonferroni-corrected genome-wide significance threshold was calculated as 0.05/SNP count. Linkage mapping was carried out using Windows QTL Cartographer v2.5 and QTLs were detected using the composite interval mapping with 1,000 permutations for each test as previously described.

M. Genetic Diversity Analyses

[0378] Principal Component Analysis (PCA) of the association panel was conducted in TASSEL using the SoySNP50K SNPs. The wild soybean and cultivated soybean accessions from the 548 accessions were used to calculate Tajima's D and the pairwise nucleotide diversity π was calculated in TASSEL5. Regions accounting for the top 15% In-ratios (which corresponds to an In-ratio threshold of about 2.4) or Tajima's D of < -2 were considered as domesticated.

N. Phylogenetic Tree and Sequence Alignment Analyses

[0379] The unrooted Neighbor-Joining phylogenetic tree was constructed with the 548 accessions using MEGA7 with the Maximum Likelihood method based on the Tamura-Nei model. A total of 19,284 genome-wide SNPs were used for the global tree and 1,023 SNPs within the 154-kb domestication region were used for the local tree. Multiple DNA and protein alignments were performed in Clustal Omega. Structures of the proteins were predicted by I-TASSER, were compared with RaptorX (TMscore 0.797), and visualized with iCn3D.

O. RNA Extraction and Expression Analyses

[0380] Soybean NILs for the POWR1 locus were used for expression analyses. Soybean leaves, roots, and stem tissues were collected at 4 weeks after planting. Fully-open flowers were collected after their emergence. Early maturation seeds (25–50 mg weight) and middle maturation seeds (100–125 mg weight) were collected, and half of them were dissected to obtain seed coat and cotyledon tissues separately. RNA was extracted as described previously. Expression levels of genes of interest were determined and normalized to that of *GmCYP2* (Glyma. 12G024700) with the BioRad CFX384 Real Time PCR System using SsoAdvanced Universal SYBR® Green Supermix. Primers for each gene are listed in Supplemental Table 11. Experiments were performed with both biological and technical triplicates. For POWR1 expression levels in soybean transgenic lines, seeds at early maturation (25–50 mg weight) were collected and used for RNA extraction. Transcriptome sequencing and analysis were performed as previously described.

P. DNA Vector Construction and Soybean Transformation

[0381] A vector (backbone pMU106) containing synthetic cDNA of POWR1_{-TE} allele from PI479752 driven by the Ubi917 promoter, pUbi: POWR1_{-TE} was constructed (FIG. 18A) and transformed into *G. max* cv. Maverick carrying POWR1_{+TE} using an improved *Agrobacterium* mediated transformation protocol as previously described. The presence of the construct in transgenic plants was confirmed by Basta leaf-painting (FIG. 18B) and PCR assay (FIGS. 18C, 18D). Expression level of POWR1 in transgenic plants was confirmed by qRT-PCR in developing seeds at the early maturation stage (FIG. 18E). With the same strategy, the cDNA of POWR1_{-TE} allele driven by its 1.9-kb native promoter sequence was cloned into a customized expression vector (backbone pAGM4673) and transformed into soybean using the *Agrobacterium* mediated transformation at Wisconsin Crop Innovation Center (Madison, WI) (FIG. 23A). The spectinomycin resistance was used as selection marker, followed by PCR (FIG. 23B) determination using the primers specific to the vector sequences were used to determine positive TO plants and the primers (F: TATC-CATATGACGTTCCAGATTACGCC (SEQ ID NO: 20); R: ACCTCAGAAATTTGCAGTGTGTGTG (SEQ ID NO: 21)) spanning the vector and CDS to identify T1 positive transformants. T1 seeds were used to measure protein, oil and weight. For transient expression, synthesized cDNAs of POWR1_{-TE} and POWR1_{+TE} were cloned into the Gateway entry vector pcr8/Topo. These constructs were moved into plant gateway expression vectors UBQ10:YFP-GW with LR clonease.

Q. Transient Expression and Microscopy Analyses

[0382] POWR1_{-TE} and POWR1_{+TE} expression localization were observed through transient expression in *N. benthamiana* using the method of Li. Briefly, UBQ10:YFP-POWR1_{-TE}, UBQ10:YFP-POWR1_{+TE}, and UBQ10:YFP in *A. tumefaciens* were infiltrated into young leaves of *N. benthamiana* plants (4–6 weeks) using a 3-ml syringe without the needle. Leaves were imaged 48 h after infiltration.

[0383] Confocal images were obtained with a Leica TCS SP8 confocal microscope using the 63× water immersion lens. Samples were excited with a 514-nm laser line and 649-nm laser line to detect YFP and chlorophyll signals, respectively. Fluorescence emission was collected for best signals of indicated fluorescent probes.

Example 3. Study Design

[0384] The current study aimed to comprehensively understand CCT domain-containing genes for their role in protein-oil accumulation in soybean seeds to facilitate functional genomics research and soybean improvement. The current study explored the evolution, expansion, and domain composition of CCTs by comparing them with a diverse range of plant species. The current study subsequently highlighted natural variation, overlap with known agriculturally important QTLs, and expression pattern diversity of

CCTs in soybean. To gain a comprehensive picture of CCT genes in soybean, the QTLs that were identified in the last three decades (1992-2018) were analyzed. These QTLs were reported for their involvement in controlling soybean agronomic traits, including seed set and quality, flowering time, and stress response regulation were incorporated into the QTL co-localization analysis. The expression profiles of these selected genes were assessed in developing seed tissues and their response to various abiotic and biotic stressors. It was further uncovered a set of GmCCT genes and proved their roles in regulating seed protein and oil accumulation. The results shed light on the evolution and potential functions of CCT genes in soybean. Moreover, the present study provided a set of genes to understand little-known mechanisms of protein regulation and improve protein content in soybean and other grain crops.

Fast-Neutron Mutant Identification

[0385] Fast neutron (FN) mutant line FN0172932 was selected from the M2 generation of irradiated elite line M92-220 in 2007 and further planted for homozygous mutants (FIG. 1). FN-induced genomic deletion region in M4-generation FN0172932 was previously determined using Comparative Genomic Hybridization (CGH) and further validated by whole genome sequencing (Illumine NovaSeq PE150, depth of coverage=16). The 1.3-Mb deletion region contains 52 gene models (Glyma. Wm82.a2) (Table 15). Plants were grown in the environment-controlled greenhouse in the Donald Danforth Plant Science Center with regular management (day 25° C./night 22° C., 40% humidity, 16 h/8 h day length for light/dark). Seed protein and oil content were measured on a pre-calibrated Perten DA 7250 analyzer (Perten Instruments, Inc., Springfield, IL, USA). Table 21 below provides details of POWR CCT-subfamily genes and their knockout and overexpression mutants. Table 22 provided field performance details about the POWR1 (CCT-subfamily gene) overexpression mutants.

TABLE 21

Details of POWR CCT-Subfamily Genes and Their Knockout and Overexpression Mutants				
Sr. No.	Gene ID	Gene Name	CRISPR-MEDIATED MUTATION COMBINATIONS	Oil Phenotype (%)
				(change)
1	Glyma.20G085100	POWR1	POWR1, 2, 3, 4	~3.78% ↑
			POWR1 OVER-EXPRESSION	~1.34% ↓
2	Glyma.10G134400	POWR2	POWR1, 2, 3	~3.91% ↑
			POWR2, 3	~3.44% ↑
			POWR2	~0.69% ↑
3	Glyma.10G190300	POWR3	POWR1, 3	~3.84% ↑
4	Glyma.20G200400	POWR4	POWR 1, 2, 4	~4.29% ↑

Note:
Arrows (↑ and ↓) indicate the increase or decrease in oil content in the seeds of the mutants grown in the greenhouse condition.

TABLE 22

Field performance details about the POWR1 (CCT-subfamily gene) overexpression mutants					
Sr.			Oil Phenotype	field performance (preliminary data)	
No.	Gene ID	Line Description	(% change)	Yield	Maturity
1	Glyma.20G085100 (POWR1)	Ubi promoter driving POWR1 expression transgenic line	~2.0% ↓	high	Early (~25 days)
		Native promoter driving POWR1 expression transgenic line	~2.00% ↓	high	Early (~21 days)

Generation of Gene-Edited Soybean Lines

[0386] Three guide-RNA (gRNA) sequences specific to the exons of GmCCT34, one for exon 1, and two for exon 2, were designed using the web tool CRISPOR. The gRNA sequences were synthesized and annealed to the CRISPR/Cas9 expression vector and transformed into soybean cv. Williams 82 by the Wisconsin Crop Innovation Center using an *Agrobacterium*-mediated transformation protocol. A pair of primers specific to the vector was used to confirm positive transformants via PCR amplification (Forward: CTGCTGTTGATGGAGGACTT SEQ ID NO: 22; Reverse: CTCCTGGAGAAGCAGAAAGTT SEQ ID NO: 23). T1 seeds from 10 independent TO plants were obtained and further grown in the environment-controlled greenhouse in the Donald Danforth Plant Science Center with the same condition as earlier mentioned. Unifoliate leaves were sampled from T1 plants to confirm the gene editing via PCR amplification followed by restriction enzyme digestion (BslI). Editing-generated deletion was further confirmed using Sanger sequencing. As mentioned above, T2 seeds from the two homologous cct34 mutants were used to measure the seed composition traits. The PCR and sequencing validation was repeated twice.

Subcellular Localization Analyses

[0387] The assay was performed through transient expression in *Nicotiana benthamiana* following a known method. The full-length CCT34 coding sequence (CDS), CCT34 lacking the CCT domain, and the CCT domain only were subcloned into the expression vector to generate UBQ10: YFP-CCT34, UBQ10: YFP-CCT34ΔCCT, and UBQ10: YFP-CCT, respectively. UBQ10: YFP was used as the empty vector. The vectors were individually transformed into *Agrobacterium tumefaciens*, and cultures of each construct were infiltrated into young leaves of *N. benthamiana* plants (4-6 weeks) using a 3-mL syringe without the needle. Leaves were imaged 48 h after infiltration. Imaging was carried out a Leica TCS SP8 confocal microscope using the 63× water immersion lens. Samples were excited with a 514-nm laser line and 649-nm laser line to detect YFP and chlorophyll signals, respectively. Fluorescence emission was collected for best signals of indicated fluorescent probes. This experiment was repeated twice.

Arabidopsis Mutant Analysis

[0388] Two independent T-DNA insertion mutant lines (WiscDsLox297300_13A.1 (cct1) and SALK_036731.1

(cct2)) were obtained from ARBC (Arabidopsis Biological Resource Center). These two T-DNA insertion regions lie with different sites of the 3'end of the CDS of AT1G04500 (FIG. 25A), the closest homolog of soybean GmCCT67 (POWR1) and GmCCT34. The homozygous mutants were identified by PCR with specific primer sets listed in Table 18.

Example 4. Results

Cct Domains are Ancient and Diverse Across Plant Species

[0389] The CCT domain is a highly conserved basic module with ~43 amino acids at the protein's C-terminus. The Hidden Markov Model (HMM) and the CCT domain (Pfam ID-PF06203) were used to search for the CCT proteins in selected plant species covering all members of the plant kingdom, including algae, mosses, ferns, conifers, and flowering plants. A set of 543 CCTs across the 24 plant species were identified (Table 19), including 69 soybean CCT domain-containing proteins (FIG. 2A, Table 21) and a range from 33 to 62 in other legumes, 40 and 52 CCT proteins, respectively, in the cereal crops rice and maize, and 13 to 29 in non-angiosperm land plants. (FIG. 2A). Traditionally, CCT proteins are classified into three subfamilies according to their constituent domains: single CCT (CCT Motif Family (CMF)), 1-2×BBOX-CCT (CONSTANTlike (COL) Family), and REC-CCT (Pseudo-Response Regulator Family). The present disclosure identified an additional protein group that carries the CCT domain, TIFY-CCT-ZnF_GATA. In these proteins, the CCT domain was located between two different domains, TIFY and ZnF_GATA. It is irrational to exclude the possibility that the CCT domain is involved in the function. Therefore, TIFY-CCT-ZnF_GATA was included in the analysis (FIG. 2B). The numbers of CCT protein genes in the tetraploids soybean and peanut were nearly doubled those in other diploid legumes. The CCT genes identified in *Arabidopsis* and the two cereal crops were generally more than those in legumes except for common bean and peanut. A small number of CCT genes (2-8) were present in chlorophyte species.

[0390] Phylogenetic analysis and phylogenetic trees generated from the CCT domain sequence identified six distinct clusters (FIG. 4A, 4B). These six clusters often, but not always, reflected the traditional domain-based classification system. Clusters I-III contained all of the members of the 1-2×BBOX-CCT subfamily, with Clusters I and II almost exclusively comprised of 2×BBOX-CCT genes and Cluster III containing the majority of 1×BBOX-CCTs. Clusters IV,

V, and VI almost exclusively contained REC-CCT, single-CCT, and TIFY-CCT-Zn_GATA genes, respectively.

[0391] Notably, single-CCT genes were found in all six clusters (FIG. 4A). In clusters I, II, IV, and VI, consist of only a few individual single-CCTs, likely representing recent deletions of the non-CCT domain in these genes. It is also likely that several 1xBBX-CCTs in the two 2xBBX-CCT clusters (I and II) likewise represent the deletion of a single BBX domain. Cluster III, however, contains a large number of single-CCTs that form two clades in the domain phylogeny (FIG. 4B). These likely represent an ancient deletion of the BBX domain in this clade prior to the origin of the angiosperms.

[0392] Interestingly, CCTs containing non-canonical domains were rare and dispersed across several clusters, likely representing singleton insertion events, for example, DUF740-CCT in Vang06g17920 (adzuki bean), Adaptin_N-CCT in Psat0s3732g0120 (pea), S_TKc-CCT in Ca.14621 (chickpea) (FIG. 4A, 4B). Non-typical CCT proteins were not identified in soybean and Arabidopsis. All identified CCT genes in this study were summarized in Table 20. HMM logos were next prepared, representing each cluster (I-VI) from the domain tree to analyze the amino acids across the clusters (FIG. 4C). Most of the amino acids were conserved in the CCT domain across the six clusters, with high conservation observed for seven amino acids (Arginine (R) 1, R15, Tyrosine (Y) 23, R26, Alanine (A) 30, R35, and Phenylalanine (F) 40). Also, cluster-specific conserved amino acids were identified. For example, F8 in clusters V and VI, while Lysine (K) 22 was highly conserved in IV, with some exceptions (FIG. 4A and FIG. 4B). These conserved amino acids across the clusters could likely represent the essential roles of CCT family genes in DNA binding or forming functional complexes. In contrast, the amino acids specific to one or certain clusters might associate with the DNA binding specificity representing functional variation in the CCT family. The results indicated that the CCT domain sequences are conserved in plant species with diversified function specificities plausibly facilitated by some uniquely conserved amino acids.

[0393] All these six groups were identified as angiosperms. To further investigate the origin of these clusters, their membership in a range of non-angiosperms were identified, including charophyte algae, mosses, ferns, and gymnosperms (Table 7). All six clusters could be identified in each of the land plant lineages; however, two groups (I and VI) were absent from all of the chlorophyte species. This indicates that most of these groups arose early in plant evolution except for one of the 2xBBX-CCT groups (I) and the TIFY-CCT-Zn_GATA group (VI), which first appeared in the bryophytes. Additionally, within the chlorophytes, Cluster IV (REC-CCT) was missing from all species except Chlamydomonas, and Cluster III (1xBBX) was missing from Micromonas and Dusiella. These results indicate that individual chlorophyte lineages may have lost these genes or their sequences sufficiently diverged that the present search model could not identify them. Along with the increased number of CCTs from chlorophytes to bryophytes, the CCT domain gene family is ancient and underwent substantial expansion and diversification in the land plant lineage.

TABLE 23

CCT Genes in Other Species							
Species	I	II	III	IV	V	VI	Unclustered
spikemoss	1	2	3	3	2	2	0
liverwort	0	1	2	2	1	2	0
Angiosperm	85	75	89	93	129	62	5
fern	3	3	4	6	6	1	1
gymnosperm	4	2	4	4	4	2	0
Bryophyte	9	7	24	10	10	10	0
chlorophyte	0	7	2	6	1	0	2

Soybean CCT Gene Family

[0394] The 69 soybean CCT-containing genes identified here were designated as GmCCT01 to GmCCT69 based on the chromosomal coordinates. The 69 GmCCTs were mapped to all 20 chromosomes, and the majority were distributed in the distal telomeric regions (Table 21). Chromosome 13 contains the maximum number of GmCCTs (7) followed by chromosomes 4, 6, and 8, each having six members. Interestingly, 33 pairs of GmCCTs (66 of 69, 95.7%) were located within syntenic genomic regions. Additionally, the high bootstrap values for the GmCCT pairs in the soybean phylogenetic tree (FIG. 3) suggest that the paired GmCCTs are paralogs that have been retained from large-scale duplication events such as whole-genome duplication (WGD) or segmental duplication. This notion should also apply to peanut CCT genes because of the segmental allotetraploid in the peanut genome. In addition, two pairs of tandemly duplicated GmCCTs in the soybean genome (GmCCT9/10; GmCCT18/19) that were also fell within segmentally duplicated regions between chromosomes 4 and 6, suggesting that the tandem duplication occurred prior to the soybean specific WGD. These results showed that polyploidization, especially the lineage-specific tetraploid in soybean, is a major evolution-driven force of CCT expansion.

[0395] To understand the evolution of CCT proteins in related legume species, the syntenic CCT-associated genes and genomic regions were analyzed among selected closely related legume species, including *Medicago*, pea, chickpea, cowpea, common bean, and soybean. The syntenic analysis among leguminous CCTs revealed that 58 (84%) of the GmCCTs have at least one syntenic CCTs in legume genomes (Table 23; FIG. 7). For most legume CCT proteins, each corresponds to a pair of GmCCT paralogs, such as paralogs GmCCT12/21 in the syntenic regions of single CCT orthologous genes in five legumes (common bean, cowpea, chickpea, pea, Medicago) (FIG. 3A; Table 23; FIG. 7). This analysis also led to the identification of soybean-specific GmCCT without syntenic CCT homologs in other legumes, such as the pair of GmCCT34/67 (FIG. 3B; Table 23).

[0396] The frequency of POWR1 alleles in a diverse population consisting of 3,956 accessions and the allele effects on protein, oil and seed weight from analyzing their whole genome resequencing data (FIG. 24). The subcellular localization of GmCCT34 was shown in FIG. 25A and FIG. 25B. Function of the Arabidopsis CCT gene, AT1G04500, was investigated for its involvement in regulating seed oil composition. Like GmPOWR 1234 genes, there is only a single CCT domain found in Arabidopsis AT1G04500 gene (hence after AtPOWR1). The gene expression analysis

showed that the AtPOWR1 is highly expressed in the seed coat tissues (FIG. 26 and FIG. 27, red color indicating the AtPOWR1 expression).

[0397] There is no information on the function(s) of AtPOWR1 concerning the regulation of seed protein-oil content. To know if this Arabidopsis gene also functions similarly to the GmPOWR genes, two homozygous T-DNA-insertion mutants were isolated (WiscDsLox297300_13A.1 and SALK_036731.1, (labeled as cct-1 and cct-2). The T-DNA insertion in these mutants occurred before and after

the CCT domain, respectively, indicating that the CCT domain is dysfunctional. Similar to GmPOWR mutants, the seed composition analysis of the AtPOWR1 or Atcct mutants revealed a higher oil content compared with the wild type seeds. These results suggested a conserved function of the CCTs between soybean and Arabidopsis in regulating oil accumulation in seeds (FIG. 26 and FIG. 27).

Use of Cct-Domain Proteins to Improve Agronomic Traits of Plants

TABLE 1

Information of 69 CCT domain-containing proteins in soybean genome						
Name	Gmax ID (v2.0)	Domain organization	Protein aa	Genomic coordinates	Best match Ara orthologs	Ara symbol
GmCCT01	Glyma.01G221100	CCT	280	Chr01: 55018467-55020621	AT5G41380	
GmCCT02	Glyma.02G152900	BBOX-BBOX-CCT	341	Chr02: 15702105-15705227	AT2G47890	
GmCCT03	Glyma.02G215600	TIFY-CCT-ZnF_GATA	311	Chr02: 40195550-40200143	AT4G24470	ZIM
GmCCT04	Glyma.02G223700	BBOX-CCT	406	Chr02: 41140728-41148151	AT5G48250	
GmCCT05	Glyma.02G230800	CCT	231	Chr02: 41829531-41831187	AT1G07050	
GmCCT06	Glyma.03G209800	BBOX-BBOX-CCT	350	Chr03: 41649706-41654690	AT2G47890	
GmCCT07	Glyma.03G261300	REC-CCT	693	Chr03: 45482095-45487185	AT5G24470	PRR5
GmCCT08	Glyma.04G058900	BBOX-BBOX-CCT	310	Chr04: 4830133-4831900	AT5G24930	COL4
GmCCT09	Glyma.04G096200	TIFY-CCT-ZnF_GATA	310	Chr04: 8627545-8632713	AT4G24470	ZIM
GmCCT10	Glyma.04G096300	TIFY-CCT-ZnF_GATA	351	Chr04: 8635553-8654734	AT3G21175	ZML1
GmCCT11	Glyma.04G166300	REC-CCT	561	Chr04: 41751974-41757731	AT5G61380	TOC1
GmCCT12	Glyma.04G228300	REC-CCT	692	Chr04: 49758484-49763637	AT5G24470	PR
GmCCT13	Glyma.04G236400	CCT	282	Chr04: 50475854-50481115	AT5G53420	
GmCCT14	Glyma.05G025000	REC-CCT	576	Chr05: 2174491-2182185	AT5G61380	TO
GmCCT15	Glyma.05G233700	BBOX-CCT	366	Chr05: 41087670-41089500	AT1G25440	
GmCCT16	Glyma.05G209800	CCT	312	Chr05: 39176944-39179334	AT5G41380	
GmCCT17	Glyma.06G059600	BBOX-BBOX-CCT	311	Chr06: 4519731-4521199	AT5G24930	CO
GmCCT18	Glyma.06G097900	TIFY-CCT-ZnF_GATA	305	Chr06: 7769865-7775571	AT4G24470	ZI2
GmCCT19	Glyma.06G098000	TIFY-CCT-ZnF_GATA	352	Chr06: 7777860-7785228	AT3G21175	ZM
GmCCT20	Glyma.06G127900	CCT	279	Chr06: 10510305-10515402	AT5G53420	
GmCCT21	Glyma.06G136600	REC-CCT	701	Chr06: 11182550-11187451	AT5G24470	PR
GmCCT22	Glyma.06G196200	REC-CCT	559	Chr06: 17608368-17614484	AT5G61380	TO
GmCCT23	Glyma.07G049400	REC-CCT	723	Chr07: 4201367-4206174	AT5G24470	PR
GmCCT24	Glyma.07G091400	BBOX-CCT	428	Chr07: 8539100-8549598	AT1G25440	
GmCCT25	Glyma.07G182700	TIFY-CCT-ZnF_GATA	356	Chr07: 35050481-35056486	AT1G51600	ZM
GmCCT26	Glyma.08G016500	CCT	305	Chr08: 1311495-1313676	AT5G41380	
GmCCT27	Glyma.08G041100	BBOX-CCT	372	Chr08: 3254251-3256417	AT1G25440	
GmCCT28	Glyma.08G067200	TIFY-CCT-ZnF_GATA	359	Chr08: 5163171-5168689	AT3G21175	ZML1
GmCCT29	Glyma.08G168900	CCT	285	Chr08: 13433813-13436662	AT3G12890	ASML2
GmCCT30	Glyma.08G221800	TIFY-CCT-ZnF_GATA	335	Chr08: 18006457-18012023	AT1G51600	ZML2
GmCCT31	Glyma.08G255200	BBOX-BBOX-CCT	349	Chr08: 22599513-22602226	AT3G02380	COL2
GmCCT32	Glyma.10G021400	BBOX-BBOX-CCT	349	Chr10: 1857312-1860298	AT2G47890	
GmCCT33	Glyma.10G048100	REC-CCT	754	Chr10: 4315721-4327545	AT5G02810	PRR7
GmCCT34	Glyma.10G134400	CCT	424	Chr10: 36034798-36039695	AT1G04500	
GmCCT35	Glyma.10G190300	CCT	420	Chr10: 42314396-42318797	AT1G04500	
GmCCT36	Glyma.10G274300	BBOX-CCT	420	Chr10: 49716240-49718424	AT1G25440	
GmCCT37	Glyma.11G022600	CCT	293	Chr11: 1624015-1626279	AT5G41380	
GmCCT38	Glyma.U033700	CCT	298	scaffold_32: 104499-107620	AT5G59990	
GmCCT39	Glyma.12G074400	CCT	280	Chr12: 5601541-5604778	AT5G59990	
GmCCT40	Glyma.12G196100	BBOX-CCT	424	Chr12: 35736690-35741033	AT3G21880	
GmCCT41	Glyma.13G100700	CCT	400	Chr13: 21588820-21592548	AT5G57180	CIA
GmCCT42	Glyma.13G093800	BBOX-BBOX-CCT	366	Chr13: 20903248-20905141	AT5G57660	CO
GmCCT43	Glyma.13G050300	BBOX-BBOX-CCT	362	Chr13: 14630393-14632571	AT3G02380	CO
GmCCT44	Glyma.13G009300	BBOX-BBOX-CCT	410	Chr13: 2573536-2580122	AT3G07650	CO
GmCCT45	Glyma.13G135900	REC-CCT	756	Chr13: 24839736-24851002	AT5G02810	PRI
GmCCT46	Glyma.09G184600	BBOX-CCT	481	Chr09: 40986139-40988677	AT1G25440	
GmCCT47	Glyma.13G306400	BBOX-CCT	424	Chr13: 40256983-40261589	AT3G21880	
GmCCT48	Glyma.13G331000	CCT	244	Chr13: 42560606-42563362	AT5G59990	
GmCCT49	Glyma.14G182800	TIFY-CCT-ZnF_GATA	308	Chr14: 44629125-44634088	AT4G24470	ZI2
GmCCT50	Glyma.14G190400	BBOX-BBOX-CCT	412	Chr14: 45536718-45544087	AT5G48250	
GmCCT51	Glyma.14G197900	CCT	221	Chr14: 46327071-46328700	AT1G07050	
GmCCT52	Glyma.15G219800	CCT	250	Chr15: 38464689-38467299	AT5G59990	
GmCCT53	Glyma.15G258200	CCT	279	Chr15: 48862806-48865252	AT3G12890	ASI
GmCCT54	Glyma.16G018000	REC-CCT	707	Chr16: 1601747-1608339	AT5G24470	PRI
GmCCT55	Glyma.16G050900	BBOX-CCT	382	Chr16: 4874633-4878624	AT1G28050	
GmCCT56	Glyma.16G067000	CCT	355	Chr16: 6640310-6646267	AT5G14370	
GmCCT57	Glyma.17G059100	CCT	399	Chr17: 4486592-4490466	AT5G57180	CIA2
GmCCT58	Glyma.17G066600	BBOX-BBOX-CCT	375	Chr17: 5146336-5148259	AT5G57660	COL5
GmCCT59	Glyma.17G102200	REC-CCT	580	Chr17: 8018189-8023260	AT5G61380	TOC1

TABLE 1-continued

Information of 69 CCT domain-containing proteins in soybean genome						
Name	Gmax ID (v2.0)	Domain organization	Protein aa	Genomic coordinates	Best match Ara orthologs	Ara symbol
GmCCT60	Glyma.18G278100	BBOX-BBOX-CCT	353	Chr18: 55964514-55967181	AT3G02380	COL2
GmCCT61	Glyma.19G039000	BBOX-BBOX-CCT	367	Chr19: 5467685-5469895	AT5G15840	CO
GmCCT62	Glyma.19G079600	CCT	355	Chr19: 28916115-28930778	AT5G14370	
GmCCT63	Glyma.19G099700	BBOX-CCT	386	Chr19: 34682554-34686379	AT2G33500	
GmCCT64	Glyma.19G207100	BBOX-BBOX-CCT	352	Chr19: 46241151-46245817	AT2G47890	
GmCCT65	Glyma.19G260400	REC-CCT	686	Chr19: 50364717-50369677	AT5G24470	PRR5
GmCCT66	Glyma.20G060400	BBOX-BBOX-CCT	410	Chr20: 18643038-18650833	AT3G07650	COL9
GmCCT67	Glyma.20G085100	CCT	444	Chr20: 31774769-31779804	AT1G04500	
②mCCT68	Glyma.20G115600	BBOX-CCT	419	Chr20: 35746759-35748974	AT1G68520	
GmCCT69	Glyma.20G200400	CCT	422	Chr20: 43781164-43785175	AT1G04500	

② indicates text missing or illegible when filed

TABLE 2

List of all CCT-domain-containing proteins identified in this study				
Species	gene ID	domain1	domain2	domain3
adzukibean	Vang0010ss00750.1			CCT
adzukibean	Vang0010ss00800.1			CCT
adzukibean	Vang0017s00340.1	TIFY	CCT	ZnF_GATA
adzukibean	Vang0017s00360.1			CCT
adzukibean	Vang0021ss01680.1			CCT
adzukibean	Vang0022ss02670.1			CCT
adzukibean	Vang0064ss00450.1	TIFY	CCT	ZnF_GATA
adzukibean	Vang0070s00170.1	BBOX		CCT
adzukibean	Vang0084s00090.1	REC		CCT
adzukibean	Vang01g08240.1	TIFY	CCT	ZnF_GATA
adzukibean	Vang01g09310.1	TIFY	CCT	ZnF_GATA
adzukibean	Vang01g09960.1	BBOX	BBOX	CCT
adzukibean	Vang01g10650.1			CCT
adzukibean	Vang0238s00030.1	REC		CCT
adzukibean	Vang02g04910.1	BBOX		CCT
adzukibean	Vang02g13360.1	BBOX	BBOX	CCT
adzukibean	Vang03g16900.1	BBOX	BBOX	CCT
adzukibean	Vang04g02640.1			CCT
adzukibean	Vang04g13110.1	BBOX		CCT
adzukibean	Vang04g15980.1	BBOX	BBOX	CCT
adzukibean	Vang05g09750.1	BBOX	BBOX	CCT
adzukibean	Vang06g00840.1			CCT
adzukibean	Vang06g09620.1			CCT
adzukibean	Vang06g17920.1	DUF740	DUF740	CCT
adzukibean	Vang06g25290.1			CCT
adzukibean	Vang09g02620.1	REC		CCT
adzukibean	Vang09g03140.1	BBOX		CCT
adzukibean	Vang10g01650.1	BBOX		CCT
adzukibean	Vang10g06540.1			CCT
adzukibean	Vang11g11310.1	REC		CCT
adzukibean	Vang11g14640.1	BBOX		CCT
adzukibean	Vang11g15310.1			CCT
adzukibean	Vang11g15630.1		TIFY	CCT
<i>arabidopsis</i>	AT1G04500.1			CCT
<i>arabidopsis</i>	AT1G05290.1			CCT
<i>arabidopsis</i>	AT1G07050.1			CCT
<i>arabidopsis</i>	AT1G25440.1	BBOX		CCT
<i>arabidopsis</i>	AT1G28050.1	BBOX	BBOX	CCT
<i>arabidopsis</i>	AT1G49130.1	BBOX		CCT
<i>arabidopsis</i>	AT1G51600.1	TIFY	CCT	GATA
<i>arabidopsis</i>	AT1G63820.1			CCT
<i>arabidopsis</i>	AT1G68520.1	BBOX		CCT
<i>arabidopsis</i>	AT1G73870.1	BBOX		CCT
<i>arabidopsis</i>	AT2G24790.1	BBOX	BBOX	CCT
<i>arabidopsis</i>	AT2G32310.1			CCT
<i>arabidopsis</i>	AT2G33350.1			CCT
<i>arabidopsis</i>	AT2G33500.1	BBOX	BBOX	CCT
<i>arabidopsis</i>	AT2G46670.1			CCT
<i>arabidopsis</i>	AT2G46790.1	REC		CCT
<i>arabidopsis</i>	AT2G47890.1	BBOX	BBOX	CCT
<i>arabidopsis</i>	AT3G02380.1	BBOX	BBOX	CCT

TABLE 2-continued

List of all CCT-domain-containing proteins identified in this study				
Species	gene ID	domain1	domain2	domain3
<i>arabidopsis</i>	AT3G07650.1	BBOX	BBOX	CCT
<i>arabidopsis</i>	AT3G12890.1			CCT
<i>arabidopsis</i>	AT3G21175.1	TIFY	CCT	GATA
<i>arabidopsis</i>	AT3G21880.1	BBOX		CCT
<i>arabidopsis</i>	AT4G15250.1	BBOX		CCT
<i>arabidopsis</i>	AT4G24470.3	TIFY	CCT	ZnF_GATA
<i>arabidopsis</i>	AT4G25990.2			CCT
<i>arabidopsis</i>	AT4G27900.1			CCT
<i>arabidopsis</i>	AT5G02810.1	REC		CCT
<i>arabidopsis</i>	AT5G14370.1			CCT
<i>arabidopsis</i>	AT5G15840.1	BBOX	BBOX	CCT
<i>arabidopsis</i>	AT5G15850.1	BBOX	BBOX	CCT
<i>arabidopsis</i>	AT5G24470.1		REC	CCT
<i>arabidopsis</i>	AT5G24930.1	BBOX	BBOX	CCT
<i>arabidopsis</i>	AT5G41380.1			CCT
<i>arabidopsis</i>	AT5G48250.1	BBOX	BBOX	CCT
<i>arabidopsis</i>	AT5G53420.1			CCT
<i>arabidopsis</i>	AT5G57180.2			CCT
<i>arabidopsis</i>	AT5G57660.1	BBOX	BBOX	CCT
<i>arabidopsis</i>	AT5G59990.1			CCT
<i>arabidopsis</i>	AT5G60100.2	REC		CCT
<i>arabidopsis</i>	AT5G61380.1	REC		CCT
<i>chickpea</i> CDC	Ca_00927.1	BBOX	BBOX	CCT
<i>chickpea</i> CDC	Ca_01359.1	REC		CCT
<i>chickpea</i> CDC	Ca_01470.1	TIFY	CCT	ZnF_GATA
<i>chickpea</i> CDC	Ca_01471.1	TIFY	CCT	ZnF_GATA
<i>chickpea</i> CDC	Ca_01694.1	BBOX	BBOX	CCT
<i>chickpea</i> CDC	Ca_02437.1			CCT
<i>chickpea</i> CDC	Ca_02579.1			CCT
<i>chickpea</i> CDC	Ca_03219.1			CCT
<i>chickpea</i> CDC	Ca_03275.1	BBOX	BBOX	CCT
<i>chickpea</i> CDC	Ca_03974.1			CCT
<i>chickpea</i> CDC	Ca_04056.1	REC		CCT
<i>chickpea</i> CDC	Ca_05292.1	TIFY	CCT	ZnF_GATA
<i>chickpea</i> CDC	Ca_05487.1	BBOX		CCT
<i>chickpea</i> CDC	Ca_05769.1			CCT
<i>chickpea</i> CDC	Ca_05985.1	BBOX		CCT
<i>chickpea</i> CDC	Ca_07641.1	BBOX		CCT
<i>chickpea</i> CDC	Ca_09110.1	BBOX	BBOX	CCT
<i>chickpea</i> CDC	Ca_13165.1	REC		CCT
<i>chickpea</i> CDC	Ca_13576.1			CCT
<i>chickpea</i> CDC	Ca_14041.1			CCT
<i>chickpea</i> CDC	Ca_14148.1			CCT
<i>chickpea</i> CDC	Ca_14236.1	BBOX	BBOX	CCT
<i>chickpea</i> CDC	Ca_14621.1	REC	CCT	S_TKc
<i>chickpea</i> CDC	Ca_14630.1			CCT
<i>chickpea</i> CDC	Ca_14968.1	TIFY	CCT	ZnF_GATA
<i>chickpea</i> CDC	Ca_16804.1	TIFY	CCT	CCT
<i>chickpea</i> CDC	Ca_17243.1			CCT
<i>chickpea</i> CDC	Ca_17522.1	TIFY	CCT	GATA
<i>chickpea</i> CDC	Ca_18243.1	REC		CCT
<i>chickpea</i> CDC	Ca_20458.1	CCT		
<i>chickpea</i> CDC	Ca_22148.1	BBOX	BBOX	CCT
<i>chickpea</i> CDC	Ca_22571.1			CCT
<i>chickpea</i> CDC	Ca_22631.1			CCT
<i>chickpea</i> CDC	Ca_22732.1	BBOX		CCT
<i>chickpea</i> CDC	Ca_27452.1			CCT
<i>chickpea</i> CDC	Ca_28031.1			CCT
<i>commonbean</i>	Phvul.001G074900.1			CCT
<i>commonbean</i>	Phvul.001G087300.1	BBOX	BBOX	CCT
<i>commonbean</i>	Phvul.001G137925.1			CCT
<i>commonbean</i>	Phvul.001G203000.1	BBOX	BBOX	CCT
<i>commonbean</i>	Phvul.002G158500.1			CCT
<i>commonbean</i>	Phvul.002G213800.1	TIFY	CCT	ZnF_GATA
<i>commonbean</i>	Phvul.002G291600.2			CCT
<i>commonbean</i>	Phvul.002G316100.1	BBOX		CCT
<i>commonbean</i>	Phvul.003G036000.1	BBOX		CCT
<i>commonbean</i>	Phvul.003G125700.1		TIFY	CCT
<i>commonbean</i>	Phvul.003G141200.1			CCT
<i>commonbean</i>	Phvul.003G149000.1	BBOX	BBOX	CCT
<i>commonbean</i>	Phvul.003G188400.1	REC		CCT
<i>commonbean</i>	Phvul.004G046601.1			CCT
<i>commonbean</i>	Phvul.004G157300.1	BBOX		CCT

TABLE 2-continued

List of all CCT-domain-containing proteins identified in this study				
Species	gene ID	domain1	domain2	domain3
commonbean	Phvul.005G049801.1			CCT
commonbean	Phvul.005G113300.1	BBOX		CCT
commonbean	Phvul.006G005200.2	BBOX	BBOX	CCT
commonbean	Phvul.007G027200.1	BBOX		CCT
commonbean	Phvul.007G162500.1	REC		CCT
commonbean	Phvul.007G166100.1	BBOX	BBOX	CCT
commonbean	Phvul.007G270600.1			CCT
commonbean	Phvul.008G022800.1	BBOX	BBOX	CCT
commonbean	Phvul.008G022925.1			CCT
commonbean	Phvul.008G147400.1			CCT
commonbean	Phvul.008G182900.1	TIFY	CCT	ZnF_GATA
commonbean	Phvul.008G190200.1	BBOX	BBOX	CCT
commonbean	Phvul.008G254600.1	TIFY	CCT	ZnF_GATA
commonbean	Phvul.008G257300.1			CCT
commonbean	Phvul.009G035300.1	TIFY	CCT	ZnF_GATA
commonbean	Phvul.009G035400.1	TIFY	CCT	ZnF_GATA
commonbean	Phvul.009G045000.1	REC		CCT
commonbean	Phvul.009G053300.2			CCT
commonbean	Phvul.009G085300.1	BBOX	BBOX	CCT
commonbean	Phvul.009G118600.1		CCT	ZnF_GATA
commonbean	Phvul.009G185500.1	REC		CCT
commonbean	Phvul.009G258300.1	REC		CCT
commonbean	Phvul.010G119700.1	REC		CCT
commonbean	Phvul.011G069800.1			CCT
commonbean	Phvul.011G070600.1			CCT
commonbean	Phvul.L001944.1			CCT
commonbean	Phvul.L002465.1			CCT
cowpea	Vigun01g054600.1			CCT
cowpea	Vigun01g065500.1			CCT
cowpea	Vigun01g075800.1	BBOX		CCT
cowpea	Vigun01g187100.1	BBOX	BBOX	CCT
cowpea	Vigun02g042100.1	BBOX		CCT
cowpea	Vigun02g177100.1			CCT
cowpea	Vigun03g017700.1	BBOX		CCT
cowpea	Vigun03g042600.1			CCT
cowpea	Vigun03g124600.1	TIFY	CCT	ZnF_GATA
cowpea	Vigun03g337800.1	REC		CCT
cowpea	Vigun03g379600.1	BBOX	BBOX	CCT
cowpea	Vigun03g387700.1			CCT
cowpea	Vigun04g021700.1	BBOX		CCT
cowpea	Vigun04g068400.1	BBOX	BBOX	CCT
cowpea	Vigun05g225000.1	BBOX		CCT
cowpea	Vigun05g253600.1			CCT
cowpea	Vigun06g006800.1	BBOX		CCT
cowpea	Vigun07g011700.1			CCT
cowpea	Vigun07g078600.1	REC		CCT
cowpea	Vigun07g116500.1	BBOX	BBOX	CCT
cowpea	Vigun07g181300.1			CCT
cowpea	Vigun07g270500.1	BBOX		CCT
cowpea	Vigun08g080300.1			CCT
cowpea	Vigun08g109700.1	TIFY	CCT	ZnF_GATA
cowpea	Vigun08g116900.1	BBOX	BBOX	CCT
cowpea	Vigun08g124000.1			CCT
cowpea	Vigun08g127200.1	TIFY	CCT	ZnF_GATA
cowpea	Vigun09g004700.1	REC		CCT
cowpea	Vigun09g076300.1	REC		CCT
cowpea	Vigun09g140300.1	REC		CCT
cowpea	Vigun09g150200.1			CCT
cowpea	Vigun09g180800.1	TIFY	CCT	ZnF_GATA
cowpea	Vigun09g180900.1	TIFY	CCT	ZnF_GATA
cowpea	Vigun09g217500.1	BBOX	BBOX	CCT
cowpea	Vigun09g276400.1		CCT	ZnF_GATA
cowpea	Vigun10g152100.1	REC		CCT
cowpea	Vigun11g149100.1	REC		CCT
cowpea	Vigun11g150000.1			CCT
maize	AC215811.3_FGP003			CCT
maize	AC233888.1_FGP002	BBOX	BBOX	CCT
maize	AC233946.1_FGP004			CCT
maize	GRMZM2G001447_P01			CCT
maize	GRMZM2G004483_P01			CCT
maize	GRMZM2G005732_P03			CCT
maize	GRMZM2G008482_P01			CCT
maize	GRMZM2G012717_P01	BBOX		CCT

TABLE 2-continued

List of all CCT-domain-containing proteins identified in this study				
Species	gene ID	domain1	domain2	domain3
maize	GRMZM2G013398_P04	BBOX	BBOX	CCT
maize	GRMZM2G013913_P01			CCT
maize	GRMZM2G020081_P01	REC		CCT
maize	GRMZM2G021777_P01	BBOX	BBOX	CCT
maize	GRMZM2G033962_P01	REC		CCT
maize	GRMZM2G038783_P01	BBOX	BBOX	CCT
maize	GRMZM2G041991_P01	BBOX		CCT
maize	GRMZM2G042198_P01	BBOX	BBOX	CCT
maize	GRMZM2G057529_P01			CCT
maize	GRMZM2G058479_P01	TIFY	CCT	ZnF_GATA
maize	GRMZM2G062218_P01			CCT
maize	GRMZM2G062885_P01			CCT
maize	GRMZM2G065896_P01	TIFY	CCT	ZnF_GATA
maize	GRMZM2G075562_P01	BBOX	BBOX	CCT
maize	GRMZM2G080509_P02	TIFY	CCT	ZnF_GATA
maize	GRMZM2G092363_P01	BBOX		CCT
maize	GRMZM2G095598_P01	BBOX	BBOX	CCT
maize	GRMZM2G095727_P05	REC		CCT
maize	GRMZM2G097454_P01			CCT
maize	GRMZM2G106108_P01			CCT
maize	GRMZM2G107595_P01			CCT
maize	GRMZM2G107886_P01	BBOX		CCT
maize	GRMZM2G114137_P01	BBOX		CCT
maize	GRMZM2G123550_P01			CCT
maize	GRMZM2G127426_P01			CCT
maize	GRMZM2G133555_P01			CCT
maize	GRMZM2G134066_P01			CCT
maize	GRMZM2G134671_P01	BBOX		CCT
maize	GRMZM2G135446_P01			CCT
maize	GRMZM2G144346_P01	BBOX		CCT
maize	GRMZM2G148453_P01			CCT
maize	GRMZM2G148772_P01	BBOX	BBOX	CCT
maize	GRMZM2G155370_P01			CCT
maize	GRMZM2G159996_P01	BBOX		CCT
maize	GRMZM2G167560_P01	BBOX	BBOX	CCT
maize	GRMZM2G176173_P01	BBOX		CCT
maize	GRMZM2G179024_P01	REC		CCT
maize	GRMZM2G367834_P01	REC		CCT
maize	GRMZM2G381691_P01			CCT
maize	GRMZM2G405368_P01	BBOX	BBOX	CCT
maize	GRMZM2G414423_P02			CCT
maize	GRMZM2G474258_P01			CCT
maize	GRMZM5G866699_P02	CBFG-NFYB-HMF		CCT
maize	GRMZM5G868285_P02			CCT
medicago	Medtr1g008220.1			CCT
medicago	Medtr1g013450.1	BBOX	BBOX	CCT
medicago	Medtr1g044785.1			CCT
medicago	Medtr1g067110.1	REC		CCT
medicago	Medtr1g073350.1			CCT
medicago	Medtr1g110870.1	BBOX		CCT
medicago	Medtr2g068730.1			CCT
medicago	Medtr2g088900.1	BBOX		CCT
medicago	Medtr2g096080.1			CCT
medicago	Medtr3g037390.1	REC		CCT
medicago	Medtr3g082630.1	BBOX	BBOX	CCT
medicago	Medtr3g091340.1			CCT
medicago	Medtr3g092780.1	REC		CCT
medicago	Medtr3g100040.1	TIFY	CCT	ZnF_GATA
medicago	Medtr3g100050.1	TIFY	CCT	ZnF_GATA
medicago	Medtr3g105710.1	BBOX	BBOX	CCT
medicago	Medtr4g008090.1			CCT
medicago	Medtr4g061360.1	REC		CCT
medicago	Medtr4g061823.1			CCT
medicago	Medtr4g061910.1			CCT
medicago	Medtr4g093730.1	TIFY	CCT	ZnF_GATA
medicago	Medtr4g108880.1	REC		CCT
medicago	Medtr4g127420.1			CCT
medicago	Medtr4g128930.1	BBOX	BBOX	CCT
medicago	Medtr5g010120.1			CCT
medicago	Medtr5g066510.1	TIFY	CCT	ZnF_GATA
medicago	Medtr5g069480.1	BBOX	BBOX	CCT
medicago	Medtr5g072780.1			CCT

TABLE 2-continued

List of all CCT-domain-containing proteins identified in this study				
Species	gene ID	domain1	domain2	domain3
<i>medicago</i>	Medtr7g018170.1	BBOX	BBOX	CCT
<i>medicago</i>	Medtr7g032240.1			CCT
<i>medicago</i>	Medtr7g083540.1	BBOX		CCT
<i>medicago</i>	Medtr7g108150.1	BBOX	BBOX	CCT
<i>medicago</i>	Medtr7g118260.1	REC		CCT
<i>medicago</i>	Medtr8g024260.1	REC		CCT
<i>medicago</i>	Medtr8g098725.1			CCT
<i>medicago</i>	Medtr8g104190.1	BBOX		CCT
<i>pea</i>	Psat0s1849g0080.1	BBOX	BBOX	CCT
<i>pea</i>	Psat0s3732g0120.1	Adaptin_N		CCT
<i>pea</i>	Psat0s3732g0240.1			CCT
<i>pea</i>	Psat1g004200.1			CCT
<i>pea</i>	Psat2g054040.1			CCT
<i>pea</i>	Psat2g060680.1	BBOX	BBOX	CCT
<i>pea</i>	Psat2g184400.1			CCT
<i>pea</i>	Psat3g024360.1	BBOX	BBOX	CCT
<i>pea</i>	Psat3g094880.1	BBOX		CCT
<i>pea</i>	Psat3g146040.1		CCT	H2A
<i>pea</i>	Psat3g193520.1	BBOX	BBOX	CCT
<i>pea</i>	Psat4g013600.2			CCT
<i>pea</i>	Psat4g049800.1	REC		CCT
<i>pea</i>	Psat4g159360.1			CCT
<i>pea</i>	Psat5g032240.1	BBOX	BBOX	CCT
<i>pea</i>	Psat5g047080.1	TIFY	CCT	ZnF_GATA
<i>pea</i>	Psat5g071920.1	REC		CCT
<i>pea</i>	Psat5g077440.1			CCT
<i>pea</i>	Psat5g105240.1	BBOX	BBOX	CCT
<i>pea</i>	Psat5g199360.1	REC		CCT
<i>pea</i>	Psat5g200320.1	REC		CCT
<i>pea</i>	Psat5g265480.1	BBOX		CCT
<i>pea</i>	Psat5g282000.1			CCT
<i>pea</i>	Psat6g004440.1			CCT
<i>pea</i>	Psat6g025680.1	BBOX	BBOX	CCT
<i>pea</i>	Psat6g107360.1			CCT
<i>pea</i>	Psat6g124840.1	REC		CCT
<i>pea</i>	Psat6g131680.1			CCT
<i>pea</i>	Psat6g228360.1	BBOX		CCT
<i>pea</i>	Psat7g011960.1	BBOX		CCT
<i>pea</i>	Psat7g031600.1			CCT
<i>pea</i>	Psat7g037680.1			CCT
<i>pea</i>	Psat7g101680.1	TIFY	CCT	ZnF_GATA
<i>pea</i>	Psat7g221720.1	REC		CCT
<i>pea</i>	Psat7g260880.1	TIFY	CCT	ZnF_GATA
<i>peanut</i>	arahy.Tifrunner.gnm1.ann1.0QJD35.1			CCT
<i>peanut</i>	arahy.Tifrunner.gnm1.ann1.0U49HS.1	TIFY	CCT	ZnF_GATA
<i>peanut</i>	arahy.Tifrunner.gnm1.ann1.129W39.1	TIFY	CCT	ZnF_GATA
<i>peanut</i>	arahy.Tifrunner.gnm1.ann1.1K6ZHH.1	REC		CCT
<i>peanut</i>	arahy.Tifrunner.gnm1.ann1.22FLX8.1			CCT
<i>peanut</i>	arahy.Tifrunner.gnm1.ann1.2F1UPX.2	REC		CCT
<i>peanut</i>	arahy.Tifrunner.gnm1.ann1.2L20XB.2	TIFY	CCT	ZnF_GATA
<i>peanut</i>	arahy.Tifrunner.gnm1.ann1.3QBB6E.1			CCT
<i>peanut</i>	arahy.Tifrunner.gnm1.ann1.5WR99K.1			CCT
<i>peanut</i>	arahy.Tifrunner.gnm1.ann1.669CET.1	BBOX	BBOX	CCT
<i>peanut</i>	arahy.Tifrunner.gnm1.ann1.6J3MKV.1	BBOX	BBOX	CCT
<i>peanut</i>	arahy.Tifrunner.gnm1.ann1.7A5VUD.1	BBOX	BBOX	CCT
<i>peanut</i>	arahy.Tifrunner.gnm1.ann1.857KFG.1			CCT
<i>peanut</i>	arahy.Tifrunner.gnm1.ann1.86LJVS.1	TIFY	CCT	ZnF_GATA
<i>peanut</i>	arahy.Tifrunner.gnm1.ann1.8MV22J.1	BBOX		CCT
<i>peanut</i>	arahy.Tifrunner.gnm1.ann1.9Y5DYJ.2			CCT
<i>peanut</i>	arahy.Tifrunner.gnm1.ann1.A77B5H.1	REC		CCT
<i>peanut</i>	arahy.Tifrunner.gnm1.ann1.A9G6ZW.1	REC		CCT
<i>peanut</i>	arahy.Tifrunner.gnm1.ann1.AE0X9Y.1			CCT
<i>peanut</i>	arahy.Tifrunner.gnm1.ann1.AGT1RS.1			CCT
<i>peanut</i>	arahy.Tifrunner.gnm1.ann1.AY816Y.1	BBOX		CCT
<i>peanut</i>	arahy.Tifrunner.gnm1.ann1.B2EAID.1			CCT
<i>peanut</i>	arahy.Tifrunner.gnm1.ann1.B9QMIW.1	BBOX	BBOX	CCT
<i>peanut</i>	arahy.Tifrunner.gnm1.ann1.BQXP8E.2	BBOX		CCT
<i>peanut</i>	arahy.Tifrunner.gnm1.ann1.BU1BMX.1			CCT
<i>peanut</i>	arahy.Tifrunner.gnm1.ann1.BVW291.1			CCT
<i>peanut</i>	arahy.Tifrunner.gnm1.ann1.BXJN39.1			CCT
<i>peanut</i>	arahy.Tifrunner.gnm1.ann1.D90B1M.1	REC		CCT
<i>peanut</i>	arahy.Tifrunner.gnm1.ann1.E88P3Q.1	BBOX	BBOX	CCT
<i>peanut</i>	arahy.Tifrunner.gnm1.ann1.EF48U6.1			CCT

TABLE 2-continued

List of all CCT-domain-containing proteins identified in this study				
Species	gene ID	domain1	domain2	domain3
peanut	arahy.Tifrunner.gnm1.ann1.EI5605.1	TIFY	CCT	ZnF_GATA
peanut	arahy.Tifrunner.gnm1.ann1.FA5JII.1			CCT
peanut	arahy.Tifrunner.gnm1.ann1.FCSG49.1			CCT
peanut	arahy.Tifrunner.gnm1.ann1.FQS251.1			CCT
peanut	arahy.Tifrunner.gnm1.ann1.GAW9KQ.2	BBOX		CCT
peanut	arahy.Tifrunner.gnm1.ann1.GMWG2V.1	BBOX	BBOX	CCT
peanut	arahy.Tifrunner.gnm1.ann1.IR754M.1			CCT
peanut	arahy.Tifrunner.gnm1.ann1.KU7VJQ.1			CCT
peanut	arahy.Tifrunner.gnm1.ann1.LV381L.1	BBOX	BBOX	CCT
peanut	arahy.Tifrunner.gnm1.ann1.N2A79V.1	BBOX	BBOX	CCT
peanut	arahy.Tifrunner.gnm1.ann1.NK4UTA.3	BBOX	BBOX	CCT
peanut	arahy.Tifrunner.gnm1.ann1.P8XRWE.1	BBOX	BBOX	CCT
peanut	arahy.Tifrunner.gnm1.ann1.PDCB5X.1	REC		CCT
peanut	arahy.Tifrunner.gnm1.ann1.PEGORN.2	REC		CCT
peanut	arahy.Tifrunner.gnm1.ann1.QG9XFC.1	TIFY	CCT	ZnF_GATA
peanut	arahy.Tifrunner.gnm1.ann1.S31ANQ.1	BBOX	BBOX	CCT
peanut	arahy.Tifrunner.gnm1.ann1.TFJ3CW.1	BBOX	BBOX	CCT
peanut	arahy.Tifrunner.gnm1.ann1.UNUG60.1			CCT
peanut	arahy.Tifrunner.gnm1.ann1.V64N20.2	REC		CCT
peanut	arahy.Tifrunner.gnm1.ann1.V8BXHT.1			CCT
peanut	arahy.Tifrunner.gnm1.ann1.V960AN.1	REC		CCT
peanut	arahy.Tifrunner.gnm1.ann1.WEZ503.1	BBOX		CCT
peanut	arahy.Tifrunner.gnm1.ann1.WVH2H8.1	BBOX		CCT
peanut	arahy.Tifrunner.gnm1.ann1.X5M96P.1			CCT
peanut	arahy.Tifrunner.gnm1.ann1.XNQT3T.1			CCT
peanut	arahy.Tifrunner.gnm1.ann1.Y6ILE8.1	REC		CCT
peanut	arahy.Tifrunner.gnm1.ann1.Y6Q3ES.1	REC		CCT
peanut	arahy.Tifrunner.gnm1.ann1.YRC4PH.1			CCT
peanut	arahy.Tifrunner.gnm1.ann1.Z8TZMB.1	REC		CCT
peanut	arahy.Tifrunner.gnm1.ann1.ZRH25X.1			CCT
peanut	arahy.Tifrunner.gnm1.ann1.ZS9VWH.1	BBOX	BBOX	CCT
peanut	arahy.Tifrunner.gnm1.ann1.ZUL46H.1	BBOX		CCT
pigeon	<i>C.cajan</i> _00921.1			CCT
pigeon	<i>C.cajan</i> _02757.1			CCT
pigeon	<i>C.cajan</i> _03622.1	BBOX	BBOXBBOX	CCT
pigeon	<i>C.cajan</i> _03709.1			CCT
pigeon	<i>C.cajan</i> _04552.1			CCT
pigeon	<i>C.cajan</i> _05381.1			CCT
pigeon	<i>C.cajan</i> _08639.1	BBOX		CCT
pigeon	<i>C.cajan</i> _08715.1	REC		CCT
pigeon	<i>C.cajan</i> _08842.1			CCT
pigeon	<i>C.cajan</i> _10458.1	BBOX	BBOX	CCT
pigeon	<i>C.cajan</i> _10578.1	BBOX	BBOX	CCT
pigeon	<i>C.cajan</i> _10962.1	REC		CCT
pigeon	<i>C.cajan</i> _11568.1			CCT
pigeon	<i>C.cajan</i> _13216.1			CCT
pigeon	<i>C.cajan</i> _13596.1			CCT
pigeon	<i>C.cajan</i> _15626.1	BBOX		CCT
pigeon	<i>C.cajan</i> _17635.1	BBOX		CCT
pigeon	<i>C.cajan</i> _17916.1	TIFY	CCT	ZnF_GATA
pigeon	<i>C.cajan</i> _17917.1		CCT	ZnF_GATA
pigeon	<i>C.cajan</i> _20131.1	BBOX	BBOX	CCT
pigeon	<i>C.cajan</i> _21434.1			CCT
pigeon	<i>C.cajan</i> _21442.1	REC		CCT
pigeon	<i>C.cajan</i> _22520.1			CCT
pigeon	<i>C.cajan</i> _22947.1	TIFY	CCT	ZnF_GATA
pigeon	<i>C.cajan</i> _24786.1			CCT
pigeon	<i>C.cajan</i> _26721.1	BBOX	BBOX	CCT
pigeon	<i>C.cajan</i> _29262.1	BBOX	BBOX	CCT
pigeon	<i>C.cajan</i> _31872.1	BBOX		CCT
pigeon	<i>C.cajan</i> _32117.1	TIFY	CCT	ZnF_GATA
pigeon	<i>C.cajan</i> _32168.1	BBOX	BBOX	CCT
pigeon	<i>C.cajan</i> _36557.1	BBOX		CCT
pigeon	<i>C.cajan</i> _39146.1	REC		CCT
pigeon	<i>C.cajan</i> _39868.1	REC		CCT
rice	LOC_Os01g61900.1			CCT
rice	LOC_Os02g01990.1			CCT
rice	LOC_Os02g05470.1			CCT
rice	LOC_Os02g05510.1	TIFY	CCT	ZnF_GATA
rice	LOC_Os02g08150.1	BBOX	BBOX	CCT
rice	LOC_Os02g39710.1	BBOX	BBOX	CCT
rice	LOC_Os02g40510.1	REC		CCT
rice	LOC_Os02g49230.2	BBOX	BBOX	CCT

TABLE 2-continued

List of all CCT-domain-containing proteins identified in this study				
Species	gene ID	domain1	domain2	domain3
rice	LOC_Os02g49880.1	BBOX		CCT
rice	LOC_Os03g04620.1			CCT
rice	LOC_Os03g17570.3	REC		CCT
rice	LOC_Os03g22770.1	BBOX		CCT
rice	LOC_Os03g47970.1	TIFY	CCT	ZnF_GATA
rice	LOC_Os03g50310.1	BBOX		CCT
rice	LOC_Os03g52450.1	TIFY	CCT	ZnF_GATA
rice	LOC_Os04g42020.1	BBOX	BBOX	CCT
rice	LOC_Os05g38990.1			CCT
rice	LOC_Os05g51690.1			CCT
rice	LOC_Os06g01340.1			CCT
rice	LOC_Os06g15330.1	BBOX		CCT
rice	LOC_Os06g16370.1	BBOX	BBOX	CCT
rice	LOC_Os06g19444.1	BBOX	BBOX	CCT
rice	LOC_Os06g44450.1	BBOX	BBOX	CCT
rice	LOC_Os06g48534.1	TIFY	CCT	ZnF_GATA
rice	LOC_Os06g48610.1			CCT
rice	LOC_Os07g15770.1			CCT
rice	LOC_Os07g47140.1	BBOX		CCT
rice	LOC_Os07g49460.1	REC		CCT
rice	LOC_Os08g15050.1	BBOX		CCT
rice	LOC_Os08g42440.1	BBOX		CCT
rice	LOC_Os09g06464.1	BBOX		CCT
rice	LOC_Os09g33550.1	BBOX		CCT
rice	LOC_Os09g36220.1	REC		CCT
rice	LOC_Os10g32900.1			CCT
rice	LOC_Os10g41100.1			CCT
rice	LOC_Os11g01100.1			CCT
rice	LOC_Os11g05930.1	REC		CCT
rice	LOC_Os12g01080.1			CCT
rice	LOC_Os12g01100.1			CCT
rice	LOC_Os12g16160.1			CCT
soybean	Glyma.01G221100.1.p			CCT
soybean	Glyma.02G152900.1.p	BBOX	BBOX	CCT
soybean	Glyma.02G215600.1.p	TIFY	CCT	ZnF_GATA
soybean	Glyma.02G223700.1.p	BBOX		CCT
soybean	Glyma.02G230800.1.p			CCT
soybean	Glyma.03G209800.1.p	BBOX	BBOX	CCT
soybean	Glyma.03G261300.1.p	REC		CCT
soybean	Glyma.04G058900.1.p	BBOX	BBOX	CCT
soybean	Glyma.04G096200.1.p	TIFY	CCT	ZnF_GATA
soybean	Glyma.04G096300.1.p	TIFY	CCT	ZnF_GATA
soybean	Glyma.04G166300.1.p	REC		CCT
soybean	Glyma.04G228300.1.p	REC		CCT
soybean	Glyma.04G236400.1.p			CCT
soybean	Glyma.05G025000.1.p	REC		CCT
soybean	Glyma.05G209800.1.p			CCT
soybean	Glyma.05G233700.1.p	BBOX		CCT
soybean	Glyma.06G059600.1.p	BBOX	BBOX	CCT
soybean	Glyma.06G097900.1.p	TIFY	CCT	ZnF_GATA
soybean	Glyma.06G098000.1.p	TIFY	CCT	ZnF_GATA
soybean	Glyma.06G127900.1.p			CCT
soybean	Glyma.06G136600.1.p	REC		CCT
soybean	Glyma.06G196200.1.p	REC		CCT
soybean	Glyma.07G049400.1.p	REC		CCT
soybean	Glyma.07G091400.1.p	BBOX		CCT
soybean	Glyma.07G182700.1.p	TIFY	CCT	ZnF_GATA
soybean	Glyma.08G016500.1.p			CCT
soybean	Glyma.08G041100.1.p	BBOX		CCT
soybean	Glyma.08G067200.1.p	TIFY	CCT	ZnF_GATA
soybean	Glyma.08G168900.1.p			CCT
soybean	Glyma.08G221800.1.p	TIFY	CCT	ZnF_GATA
soybean	Glyma.08G255200.1.p	BBOX	BBOX	CCT
soybean	Glyma.09G184600.1.p	BBOX		CCT
soybean	Glyma.10G021400.1.p	BBOX	BBOX	CCT
soybean	Glyma.10G048100.1.p	REC		CCT
soybean	Glyma.10G134400.1.p			CCT
soybean	Glyma.10G190300.1.p			CCT
soybean	Glyma.10G274300.1.p	BBOX		CCT
soybean	Glyma.11G022600.1.p			CCT
soybean	Glyma.12G074400.1.p			CCT
soybean	Glyma.12G196100.1.p	BBOX		CCT
soybean	Glyma.13G009300.1.p	BBOX	BBOX	CCT

TABLE 2-continued

List of all CCT-domain-containing proteins identified in this study				
Species	gene ID	domain1	domain2	domain3
soybean	Glyma.13G050300.1.p	BBOX	BBOX	CCT
soybean	Glyma.13G093800.1.p	BBOX	BBOX	CCT
soybean	Glyma.13G100700.1.p			CCT
soybean	Glyma.13G135900.1.p	REC		CCT
soybean	Glyma.13G306400.1.p	BBOX		CCT
soybean	Glyma.13G331000.1.p			CCT
soybean	Glyma.14G182800.1.p	TIFY	CCT	ZnF_GATA
soybean	Glyma.14G190400.1.p	BBOX	BBOX	CCT
soybean	Glyma.14G197900.1.p			CCT
soybean	Glyma.15G219800.1.p			CCT
soybean	Glyma.15G258200.1.p			CCT
soybean	Glyma.16G018000.1.p	REC		CCT
soybean	Glyma.16G050900.1.p	BBOX		CCT
soybean	Glyma.16G067000.1.p			CCT
soybean	Glyma.17G059100.1.p			CCT
soybean	Glyma.17G066600.1.p	BBOX	BBOX	CCT
soybean	Glyma.17G102200.1.p	REC		CCT
soybean	Glyma.18G278100.1.p	BBOX	BBOX	CCT
soybean	Glyma.19G039000.1.p	BBOX	BBOX	CCT
soybean	Glyma.19G079600.1.p			CCT
soybean	Glyma.19G099700.1.p	BBOX		CCT
soybean	Glyma.19G207100.1.p	BBOX	BBOX	CCT
soybean	Glyma.19G260400.1.p	REC		CCT
soybean	Glyma.20G060400.1.p	BBOX	BBOX	CCT
soybean	Glyma.20G085100.1.p			CCT
soybean	Glyma.20G115600.1.p	BBOX		CCT
soybean	Glyma.20G200400.1.p			CCT
soybean	Glyma.U033700.1.p			CCT
<i>Creinhardtii</i>	Cre06.g278200.t1.1	BBOX		CCT
<i>Creinhardtii</i>	Cre06.g278159.t1.1	BBOX	BBOX	CCT
<i>Creinhardtii</i>	Cre02.g094150.t1.2	REC		CCT
<i>Creinhardtii</i>	Cre07.g351550.t1.1			CCT
<i>Creinhardtii</i>	Cre15.g635100.t1.2			CCT
<i>Creinhardtii</i>	Cre12.g534100.t1.1			CCT
<i>Creinhardtii</i>	Cre16.g676421.t1.1	REC		CCT
<i>Creinhardtii</i>	Cre16.g675100.t1.1			CCT
<i>Csubellipsoidea</i>	64395	BBOX		CCT
<i>Csubellipsoidea</i>	65695	IF-2B	REC	CCT
<i>Csubellipsoidea</i>	58803			CCT
<i>Csubellipsoidea</i>	55541			CCT
<i>Csubellipsoidea</i>	48136			CCT
<i>Dsalina</i>	Dusal.0007s00046.1.p	BBOX	BBOX	CCT
<i>Dsalina</i>	Dusal.0068s00019.1.p	REC		CCT
<i>Mpusilla</i> CCMP1545	56149	BBOX		CCT
<i>Mpusilla</i> CCMP1546	48139	REC		CCT
<i>Mpusilla</i> CCMP1547	195113	BBOX		CCT
MspRCC299	101672	BBOX	BBOX	CCT
MspRCC299	63995	BBOX		CCT
MspRCC299	59410	PFAM		CCT
MspRCC299	57487	BBOX	BBOX	CCT
<i>Olucimarinus</i>	37312	BBOX		CCT
<i>Olucimarinus</i>	33434	BBOX		CCT
<i>Olucimarinus</i>	24762			CCT
<i>Olucimarinus</i>	29755	REC		CCT
<i>Olucimarinus</i>	41670	REC		CCT

TABLE 3

Syntenic CCTs in legume genomes					
Soybean	adzukibean	chickpeaCDC	commonbean	cowpea	medicago
Glyma.01G221100.1	Vang07g06400.1	Ca_02437.1	Phvul.002G158500.1	Vigun02g177100.1	Medtr5g010120.1
Glyma.02G152900.1	Vang06g17920.2	.	Phvul.007G166100.1	Vigun07g116500.2	.
Glyma.02G215600.1	Vang01g09310.1	.	Phvul.008G182900.1	Vigun08g109700.1	Medtr5g066510.1
Glyma.02G223700.1	.	Ca_14236.1	Phvul.008G190200.1	.	Medtr5g069480.5
Glyma.02G230800.1	Vang01g08550.1	Ca_17243.1	Phvul.008G257300.1	Vigun08g124000.1	Medtr5g072780.1
Glyma.03G209800.1	Vang04g15980.1	Ca_00927.1	Phvul.001G203000.1	Vigun01g187100.1	Medtr7g108150.1
Glyma.03G261300.1	Vang09g08740.1	Ca_01359.1	Phvul.009G258300.1	Vigun09g004700.1	Medtr7g118260.2
Glyma.04G058900.1	Vang05g09750.1	Ca_01694.1	Phvul.009G085300.1	Vigun09g217500.1	Medtr3g105710.1

TABLE 3-continued

Syntenic CCTs in legume genomes					
Glyma.04G096200.1	Vang0017s00340.2	Ca_01471.1	Phvul.009G035400.1	Vigun09g180900.1	Medtr3g100050.1
Glyma.04G096300.1	Vang0017s00360.1	Ca_01470.1	Phvul.009G035300.1	Vigun09g180800.1	Medtr3g100040.1
Glyma.04G166300.1	Vang09g02620.1	Ca_18243.1	Phvul.009G185500.1	Vigun09g076300.1	Medtr3g037390.1
Glyma.04G228300.1	Vang0084s00090.1	Ca_04056.1	Phvul.009G045000.1	Vigun09g140300.1	Medtr3g092780.2
Glyma.04G236400.1	Vang0090s00480.2	Ca_03974.1	Phvul.009G053300.2	Vigun09g150200.1	Medtr3g091340.1
Glyma.05G025000.1	Vang11g11310.1	Ca_13165.1	Phvul.003G188400.1	Vigun03g337800.1	Medtr4g108880.1
Glyma.05G209800.1	Vang10g06540.1	Ca_05769.1	Phvul.002G291600.2	Vigun03g042600.1	Medtr8g098725.1
Glyma.05G233700.1	Vang10g01650.1	.	Phvul.002G316100.1	Vigun03g017700.1	Medtr8g104190.1
Glyma.06G059600.1	Vang05g09750.1	Ca_01694.1	Phvul.009G085300.1	Vigun09g217500.1	Medtr3g105710.1
Glyma.06G097900.1	Vang0017s00340.2	Ca_01471.1	Phvul.009G035400.1	Vigun09g180900.1	Medtr3g100050.1
Glyma.06G098000.1	Vang0017s00360.1	Ca_01470.1	Phvul.009G035300.1	Vigun09g180800.1	Medtr3g100040.1
Glyma.06G127900.1	Vang0090s00480.2	Ca_03974.1	Phvul.009G053300.2	Vigun09g150200.1	Medtr3g091340.1
Glyma.06G136600.1	Vang0084s00090.1	Ca_04056.1	Phvul.009G045000.1	Vigun09g140300.1	Medtr3g092780.2
Glyma.06G196200.1	Vang0017s00360.1	Ca_18243.1	Phvul.009G185500.1	Vigun09g076300.1	Medtr3g037390.1
Glyma.07G049400.1	Vang0238s00030.1	.	Phvul.010G119700.2	Vigun10g152100.2	Medtr8g024260.1
Glyma.07G091400.1	Vang02g04910.1	.	Phvul.004G157300.1	Vigun04g021700.1	.
Glyma.07G182700.1	Vang0064ss00450.1	Ca_05292.1	Phvul.002G213800.2	Vigun03g124600.1	Medtr4g093730.1
Glyma.08G016500.1	Vang10g06540.1	Ca_05769.1	Phvul.002G291600.2	Vigun03g042600.1	Medtr8g098725.1
Glyma.08G041100.1	Vang10g01650.1	Ca_22732.1	Phvul.002G316100.1	Vigun03g017700.1	Medtr8g104190.1
Glyma.08G067200.1	Vang0064ss00450.1	Ca_05292.1	Phvul.002G213800.2	Vigun03g124600.1	Medtr4g093730.1
Glyma.08G168900.1	.	Ca_13576.1	.	.	Medtr2g068730.1
Glyma.08G221800.1	Vang01g08240.1	Ca_17522.1	Phvul.008G254600.1	Vigun08g127200.1	Medtr4g008090.1
Glyma.08G255200.1	Vang04g02640.1	.	Phvul.008G022800.1	Vigun05g024400.1	Medtr7g018170.1
Glyma.09G184600.1	Vang02g04910.1	.	Phvul.004G157300.1	Vigun04g021700.1	.
Glyma.10G021400.1	Vang06g17920.2	.	Phvul.007G166100.1	Vigun07g116500.2	.
Glyma.10G048100.1	Vang0206s00130.1	Ca_14041.1	Phvul.007G162500.1	Vigun07g078600.2	Medtr1g067110.1
Glyma.10G134400.1	.	.	.	Vigun07g011700.1	.
Glyma.10G190300.1	Vang06g09620.1	.	Phvul.L001944.1	Vigun07g181300.1	Medtr1g073350.1
Glyma.10G274300.1	Vang06g00840.1	Ca_05487.1	Phvul.007G027200.1	Vigun07g270500.1	Medtr1g110870.1
Glyma.11G022600.1	Vang07g06400.1	Ca_02437.1	Phvul.002G158500.1	Vigun02g177100.1	.
Glyma.12G074400.1	Vang0010ss00750.1	Ca_14630.1	Phvul.011G069800.1	Vigun11g150000.1	Medtr4g061823.1
Glyma.12G196100.1	Vang09g03140.1	Ca_14148.1	Phvul.005G113300.1	Vigun05g225000.4	Medtr2g088900.1
Glyma.13G009300.1	Vang03g16900.1	Ca_07641.1	Phvul.006G005200.3	.	Medtr3g082630.1
Glyma.13G050300.1	Vang02g13360.1	.	Phvul.004G046601.1	Vigun04g068400.1	.
Glyma.13G093800.1	Vang11g14640.1	Ca_03275.1	Phvul.003G149000.1	Vigun03g379600.1	Medtr4g128930.1
Glyma.13G100700.1	Vang11g15310.1	Ca_03219.1	Phvul.003G141200.1	Vigun03g387700.1	Medtr4g127420.1
Glyma.13G135900.1	Vang0206s00130.1	Ca_14041.1	Phvul.007G162500.1	Vigun07g078600.1	Medtr1g067110.1
Glyma.13G206400.1	Vang09g03140.1	Ca_14148.1	Phvul.005G113300.1	Vigun05g225000.1	Medtr2g088900.1
Glyma.13G331000.1	Vang0021ss01680.1	Ca_02579.1	Phvul.L002465.1	Vigun05g253600.1	Medtr2g096080.1
Glyma.14G182800.1	Vang01g09310.1	Ca_14968.1	Phvul.008G182900.1	Vigun08g109700.1	Medtr5g066510.1
Glyma.14G190400.1	Vang01g09960.2	Ca_14236.1	Phvul.008G190200.1	Vigun08g116900.1	Medtr5g069480.1
Glyma.14G197900.1	Vang01g08550.1	Ca_17243.1	.	.	Medtr5g072780.1
Glyma.15G219800.1	.	.	.	.	.
Glyma.15G258200.1	Vang0022ss02670.1	Ca_13576.1	Phvul.005G049801.1	Vigun01g054600.1	Medtr2g068730.1
Glyma.16G018000.1	.	.	Phvul.010G119700.1	Vigun10g152100.2	Medtr8g024260.1
Glyma.16G050900.1	Vang04g13110.1	Ca_05985.1	Phvul.001G087300.1	Vigun01g075800.1	Medtr7g083540.1
Glyma.16G067000.1	.	.	.	Vigun01g065500.1	Medtr7g032240.1
Glyma.17G059100.1	Vang11g15310.1	Ca_03219.1	Phvul.003G141200.2	Vigun03g387700.1	Medtr4g127420.1
Glyma.17G066600.1	Vang11g14640.1	Ca_03275.1	Phvul.003G149000.1	Vigun03g379600.1	Medtr4g128930.1
Glyma.17G102200.1	Vang11g11310.1	Ca_13165.1	Phvul.003G188400.1	Vigun03g337800.1	Medtr4g108880.1
Glyma.18G278100.1	Vang04g02640.1	Ca_22148.1	Phvul.008G022800.1	Vigun05g024400.1	Medtr7g018170.1
Glyma.19G039000.1	Vang02g13360.1	.	Phvul.004G046601.1	Vigun04g068400.1	.
Glyma.19G079600.1	Vang04g12460.1	.	Phvul.008G147400.1	Vigun01g065500.1	.
Glyma.19G099700.1	Vang04g13110.1	Ca_05985.1	Phvul.001G087300.1	Vigun01g075800.1	Medtr7g083540.1
Glyma.19G207100.1	Vang04g15980.1	Ca_00927.1	Phvul.001G203000.1	Vigun01g187100.1	Medtr7g108150.1
Glyma.19G260400.1	Vang09g08740.1	Ca_01359.1	Phvul.009G258300.1	Vigun09g004700.1	Medtr7g118260.1
Glyma.20G060400.1	.	Ca_07641.1	.	.	Medtr3g082630.1
Glyma.20G085100.1	Vang06g25290.1	Ca_20458.1	Phvul.007G270600.1	Vigun07g011700.1	.
Glyma.20G115600.1	Vang06g00840.1	Ca_05487.1	Phvul.007G027200.1	Vigun07g270500.1	Medtr1g110870.1
Glyma.20G200400.1	Vang06g09620.1	.	Phvul.L001944.1	Vigun07g181300.1	Medtr1g073350.1
Glyma.U033700.1	Vang0010ss00750.1	Ca_14630.1	Phvul.011G069800.1	Vigun11g150000.1	Medtr4g061823.1
Soybean	pea	peanut	pigeon	soybean-copy	arabidopsis
Glyma.01G221100.1	Psat2g184400.1	arahy.B2EAID.1	C.cajan_13216.1	Glyma.11G022600.1	.
Glyma.02G152900.1	.	.	C.cajan_10458.1	Glyma.10G021400.1	AT2G47890.1
Glyma.02G215600.1	.	.	.	Glyma.14G182800.1	.
Glyma.02G223700.1	.	arahy.TFJ3CW.1	C.cajan_29262.1	Glyma.14G190400.1	.
Glyma.02G230800.1	Psat2g054040.2	arahy.XNQT3T.1	C.cajan_05381.1	Glyma.14G197900.1	.
Glyma.03G209800.1	Psat3g024360.1	arahy.7A5VUD.1	C.cajan_10578.1	Glyma.19G207100.1	.
Glyma.03G261300.1	.	arahy.Y6ILE8.1	C.cajan_10962.1	Glyma.19G260400.1	.
Glyma.04G058900.1	.	arahy.ZS9VWH.1	C.cajan_32168.1	Glyma.06G059600.1	.
Glyma.04G096200.1	.	.	C.cajan_17917.1	Glyma.06G097900.1	.
Glyma.04G096300.1	.	.	C.cajan_17916.1	Glyma.06G098000.1	.

TABLE 3-continued

Syntenic CCTs in legume genomes									
Glyma.04G166300.1	Psat5g199360.1	.	.	Glyma.06G196200.1	.	.	Glyma.06G136600.1	.	AT5G24470.1
Glyma.04G228300.1	Psat5g071920.1	arahy.A9G6ZW.1	.	Glyma.06G136600.1	.	.	Glyma.06G127900.1	.	AT5G53420.1
Glyma.04G236400.1	Psat5g077440.1	arahy.BVW291.1	C.cajan__08842.1	Glyma.06G127900.1	.	.	Glyma.17G102200.1	.	AT5G61380.1
Glyma.05G025000.1	Psat4g049800.1	arahy.2F1UPX.2	.	Glyma.08G016500.1	.	.	Glyma.08G041100.1	.	AT5G41380.1
Glyma.05G209800.1	Psat7g031600.1	arahy.BU1BMX.1	C.cajan__11568.1	Glyma.08G041100.1	.	.	Glyma.04G058900.1	.	.
Glyma.05G233700.1	Psat7g011960.1	.	C.cajan__15626.1	Glyma.04G058900.1	.	.	Glyma.04G096200.1	.	.
Glyma.06G059600.1	.	.	C.cajan__32168.1	Glyma.04G096200.1	.	.	Glyma.04G096300.1	.	.
Glyma.06G097900.1	.	arahy.0U49HS.1	C.cajan__17917.1	Glyma.04G096300.1	.	.	Glyma.04G236400.1	.	AT5G3420 ^②
Glyma.06G098000.1	.	.	C.cajan__17916.1	Glyma.04G236400.1	.	.	Glyma.04G228300.1	.	.
Glyma.06G127900.1	Psat5g077440.1	arahy.5WR99K.1	C.cajan__08842.1	Glyma.04G228300.1	.	.	Glyma.04G166300.1	.	.
Glyma.06G136600.1	Psat5g071920.1	arahy.A9G6ZW.1	.	Glyma.04G166300.1	.	.	Glyma.16G018000.1	.	.
Glyma.06G196200.1	Psat5g199360.1	.	C.cajan__08715.1	Glyma.16G018000.1	.	.	Glyma.09G184600.3	.	.
Glyma.07G049400.1	Psat4g164080.1	.	C.cajan__39868.1	Glyma.09G184600.3	.	.	Glyma.08G067200.1	.	.
Glyma.07G091400.1	.	arahy.QP34EV.1	C.cajan__08639.1	Glyma.08G067200.1	.	.	Glyma.05G209800.1	.	.
Glyma.07G182700.1	Psat7g101680.1	.	C.cajan__22947.1	Glyma.05G209800.1	.	.	Glyma.05G233700.1	.	.
Glyma.08G016500.1	Psat7g031600.1	arahy.AGT1RS.1	C.cajan__11568.1	Glyma.05G233700.1	.	.	Glyma.07G182700.1	.	.
Glyma.08G041100.1	Psat7g011960.1	arahy.8MV22J.1	C.cajan__15626.1	Glyma.07G182700.1	.	.	Glyma.15G258200.1	.	.
Glyma.08G067200.1	Psat7g101680.1	.	C.cajan__22947.1	Glyma.15G258200.1	.	.	.	.	AT1G1600 ^②
Glyma.08G168900.1	.	.	.	.	.	.	Glyma.18G278100.1	.	.
Glyma.08G221800.1	Psat7g260880.1	arahy.QG9XFC.1	C.cajan__32117.1	Glyma.18G278100.1	.	.	Glyma.07G091400.1	.	.
Glyma.08G255200.1	Psat3g193520.1	arahy.S3IANQ.1	C.cajan__20131.1	Glyma.07G091400.1	.	.	Glyma.02G152900.1	.	.
Glyma.09G184600.1	.	arahy.WEZ503.1	C.cajan__08639.1	Glyma.02G152900.1	.	.	Glyma.13G135900.1	.	.
Glyma.10G021400.1	.	.	C.cajan__10458.1	Glyma.13G135900.1	.	.	Glyma.10G134400.1	.	.
Glyma.10G048100.1	Psat6g124840.1	.	C.cajan__41432.1	Glyma.10G134400.1	.	.	Glyma.20G200400.2	.	.
Glyma.10G134400.1	.	arahy.22FLX8.1	C.cajan__02757.1	Glyma.20G200400.2	.	.	Glyma.20G115600.1	.	AT1G68520.1
Glyma.10G190300.1	.	arahy.EF48U6.1	C.cajan__22520.1	Glyma.20G115600.1	.	.	Glyma.01G221100.1	.	.
Glyma.10G274300.1	Psat6g228360.1	arahy.WVH2H8.1	C.cajan__31872.1	Glyma.01G221100.1	.	.	Glyma.U033700.1	.	.
Glyma.11G022600.1	.	.	C.cajan__13216.1	Glyma.U033700.1	.	.	Glyma.13G306400.4	.	.
Glyma.12G074400.1	Psat7g221360.1	arahy.IR754M.1	C.cajan__21434.1	Glyma.13G306400.4	.	.	Glyma.13G009300.1	.	.
Glyma.12G196100.1	Psat5g265480.1	.	C.cajan__36557.1	Glyma.13G009300.1	.	.	Glyma.19G039000.1	.	.
Glyma.13G009300.1	Psat5g105240.1	.	.	Glyma.19G039000.1	.	.	Glyma.17G066600.1	.	AT5C ^② 7660 ^②
Glyma.13G050300.1	.	arahy.LV381L.1	C.cajan__26721.1	Glyma.17G066600.1	.	.	Glyma.17G059100.1	.	.
Glyma.13G093800.1	.	.	C.cajan__03622.1	Glyma.17G059100.1	.	.	Glyma.10G048100.1	.	AT5C ^② 2810 ^②
Glyma.13G100700.1	Psat4g013600.2	arahy.0QJD35.1	C.cajan__03709.1	Glyma.10G048100.1	.	.	Glyma.12G196100.1	.	.
Glyma.13G135900.1	Psat6g124840.1	.	.	Glyma.12G196100.1	.	.	Glyma.02G215600.1	.	AT4G24470.1
Glyma.13G306400.1	Psat5g265480.1	.	C.cajan__36557.1	Glyma.02G215600.1	.	.	Glyma.02G223700.1	.	.
Glyma.13G331000.1	Psat5g282000.1	arahy.KU7VJQ.1	C.cajan__13596.1	Glyma.02G223700.1	.	.	Glyma.02G230800.2	.	AT1G07050.1
Glyma.14G182800.1	.	.	.	Glyma.02G230800.2	.	.	Glyma.15G219800.1	.	.
Glyma.14G190400.1	Psat2g060680.2	arahy.NK4UTA.1	C.cajan__29262.1	Glyma.15G219800.1	.	.	Glyma.08G168900.1	.	.
Glyma.14G197900.1	Psat2g050400.1	arahy.XNQT3T.1	C.cajan__05381.1	Glyma.08G168900.1	.	.	Glyma.07G049400.1	.	.
Glyma.15G219800.1	.	.	.	Glyma.07G049400.1	.	.	Glyma.19G099700.1	.	.
Glyma.15G258200.1	.	arahy.9Y5DYJ.1	.	Glyma.19G099700.1	.	.	.	.	AT5C ^② 4370 ^②
Glyma.16G018000.1	.	arahy.PEG0RN.1	.	.	.	.	Glyma.13G100700.1	.	AT5C ^② 7180 ^②
Glyma.16G050900.1	Psat3g094880.1	arahy.P8XRWE.1	C.cajan__17635.1	Glyma.13G100700.1	.	.	Glyma.13G093800.1	.	AT5C ^② 7660 ^②
Glyma.16G067000.1	.	arahy.FCSG49.1	C.cajan__00921.1	Glyma.13G093800.1	.	.	Glyma.05G025000.1	.	AT5G61380.1
Glyma.17G059100.1	.	arahy.0QJD35.1	C.cajan__03709.1	Glyma.05G025000.1	.	.	Glyma.08G255200.1	.	.
Glyma.17G066600.1	.	arahy.6J3MKV.1	C.cajan__03622.1	Glyma.08G255200.1	.	.	Glyma.13G050300.1	.	.
Glyma.17G102200.1	Psat4g049800.1	arahy.D90B1M.1	.	Glyma.13G050300.1	.	.	.	.	.
Glyma.18G278100.1	Psat3g193520.1	arahy.669CET.1	C.cajan__20131.1	.	.	.	Glyma.16G050900.1	.	.
Glyma.19G039000.1	.	.	C.cajan__26721.1	Glyma.16G050900.1	.	.	Glyma.03G209800.1	.	AT2G47890.1
Glyma.19G079600.1	.	arahy.857KFG.1	.	Glyma.03G209800.1	.	.	Glyma.03G261300.1	.	AT2C ^② 6790 ^②
Glyma.19G099700.1	.	arahy.P8XRWE.1	C.cajan__17635.1	Glyma.03G261300.1	.	.	.	.	.
Glyma.19G207100.1	Psat3g024360.1	arahy.7A5VUD.1	C.cajan__10578.1	.	.	.	Glyma.10G134400.1	.	.
Glyma.19G260400.1	.	.	C.cajan__10962.1	Glyma.10G134400.1	.	.	Glyma.10G274300.1	.	.
Glyma.20G060400.1	Psat5g105240.1	.	.	Glyma.10G274300.1	.	.	Glyma.10G190300.1	.	AT2C ^② 3350 ^②
Glyma.20G085100.1	.	.	C.cajan__02757.1	Glyma.10G190300.1	.	.	Glyma.12G074400.1	.	.
Glyma.20G115600.1	Psat6g228360.1	arahy.WVH2H8.1	C.cajan__31872.1	Glyma.12G074400.1	.	.	.	.	.
Glyma.20G200400.1	Psat6g131680.1	arahy.EF48U6.1	C.cajan__22520.1	.	.	.	.	.	.
Glyma.U033700.1	Psat7g221360.1	arahy.IR754M.1	C.cajan__21434.1	.	.	.	.	.	.

^② indicates text missing or illegible when filed

TABLE 4

Single nucleotide polymorphism				
Gene	non-Synonymous	splice__site	INDEL	termination__codon_snps
Glyma.01G221100	4	0	0	0
Glyma.02G152900	6	0	0	0
Glyma.02G215600	0	0	0	0
Glyma.02G223700	2	0	0	0

TABLE 4-continued

Single nucleotide polymorphism				
Gene	non-Synonymous	splice__site	INDEL	termination__codon_snps
Glyma.02G230800	2	0	0	0
Glyma.03G209800	0	0	0	0
Glyma.03G261300	1	0	0	0
Glyma.04G058900	2	0	0	0

TABLE 4-continued

Single nucleotide polymorphism				
Gene	non-Synonymous	splice__site	INDEL	termination__codon__snps
Glyma.04G096200	19	0	0	0
Glyma.04G096300	3	0	0	0
Glyma.04G166300	1	1	0	0
Glyma.04G228300	1	0	0	0
Glyma.04G236400	1	0	0	0
Glyma.05G025000	2	0	0	0
Glyma.05G209800	2	0	0	0
Glyma.05G233700	2	0	1	0
Glyma.06G059600	14	0	1	2
Glyma.06G097900	4	0	1	0
Glyma.06G098000	4	1	2	0
Glyma.06G127900	0	0	0	0
Glyma.06G136600	5	0	0	0
Glyma.06G196200	4	0	1	0
Glyma.07G049400	7	0	0	0
Glyma.07G091400	7	0	0	0
Glyma.07G182700	1	0	0	0
Glyma.08G016500	4	0	0	0
Glyma.08G041100	2	0	0	0
Glyma.08G067200	1	0	0	0
Glyma.08G168900	2	0	0	0
Glyma.08G221800	2	0	0	0
Glyma.08G255200	3	1	1	0
Glyma.09G184600	2	1	2	0
Glyma.10G021400	7	0	0	0
Glyma.10G048100	9	0	0	0
Glyma.10G134400	1	0	0	0
Glyma.10G190300	6	0	0	0
Glyma.10G274300	1	0	1	0
Glyma.11G022600	1	0	0	0
Glyma.12G074400	4	0	1	0
Glyma.12G196100	3	0	0	0
Glyma.13G009300	0	0	0	0
Glyma.13G050300	1	0	0	0
Glyma.13G093800	1	0	0	0
Glyma.13G100700	5	0	0	0
Glyma.13G135900	6	0	0	0
Glyma.13G306400	4	0	0	0
Glyma.13G331000	0	0	0	0
Glyma.14G182800	0	0	0	0
Glyma.14G190400	4	0	0	0
Glyma.14G197900	0	0	0	0
Glyma.15G219800	2	0	1	0
Glyma.15G258200	0	0	0	0
Glyma.16G018000	13	0	1	0
Glyma.16G050900	4	0	0	0
Glyma.16G067000	2	1	7	0
Glyma.17G059100	0	0	0	0
Glyma.17G066600	0	0	0	0
Glyma.17G102200	3	0	0	0
Glyma.18G278100	3	0	1	0
Glyma.19G039000	2	0	0	0
Glyma.19G079600	0	0	2	0
Glyma.19G099700	5	0	1	0
Glyma.19G207100	1	0	0	0
Glyma.19G260400	8	0	0	0
Glyma.20G060400	1	0	0	0
Glyma.20G085100	1	0	3	0
Glyma.20G115600	3	0	2	0
Glyma.20G200400	3	0	0	0
total	214	5	29	2

TABLE 5

Agronomic trait being seed quality and genes encoding CCT proteins	
gene	QTL TraitName
Glyma.01G221100	Seed oil content
Glyma.01G221100	Seed oil content
Glyma.01G221100	Seed protein content
Glyma.01G221100	Seed glycine content
Glyma.01G221100	Seed threonine content
Glyma.01G221100	Seed glycine content
Glyma.02G152900	Seed length to width ratio
Glyma.02G215600	Seed protein content
Glyma.02G215600	Seed isoflavone content
Glyma.02G215600	Seed oil content
Glyma.02G215600	Seed sucrose content
Glyma.02G223700	Seed glycine content
Glyma.03G209800	Seed linoleic acid content
Glyma.03G209800	Seed oleic acid content
Glyma.03G209800	Seed oil content
Glyma.03G209800	Seed protein content
Glyma.03G209800	Seed oil content
Glyma.03G209800	Seed genistein content
Glyma.03G209800	Seed oil content
Glyma.03G209800	Seed protein content
Glyma.03G209800	Seed stearic acid content
Glyma.03G209800	Seed linolenic acid content
Glyma.03G209800	Seed oil content
Glyma.04G058900	Seed isoflavone content
Glyma.04G058900	Seed genistein content
Glyma.04G058900	Seed protein content
Glyma.04G228300	Seed daidzein content
Glyma.04G228300	Seed isoflavone content
Glyma.04G228300	Seed protein content
Glyma.04G228300	Seed oil content
Glyma.04G228300	Seed protein content
Glyma.04G228300	Seed protein content
Glyma.04G236400	Flooding Tolerance
Glyma.05G025000	Seed cysteine content
Glyma.05G025000	Seed methionine content
Glyma.05G209800	Seed oil content
Glyma.05G209800	Seed oil content
Glyma.05G209800	Seed oil content
Glyma.05G233700	Pectin content
Glyma.05G209800	Seed oil content
Glyma.05G209800	Seed linoleic acid content
Glyma.05G233700	Seed oil content
Glyma.05G233700	Seed oleic acid content
Glyma.05G209800	Seed arabinose-galactose content
Glyma.05G233700	Seed oil content
Glyma.05G233700	Seed daidzein content
Glyma.05G233700	Seed isoflavone content
Glyma.05G233700	Seed oil content
Glyma.05G233700	Seed protein content
Glyma.05G233700	Seed daidzein content
Glyma.05G233700	Seed isoflavone content
Glyma.05G233700	Seed glycine content
Glyma.05G233700	Seed genistein content
Glyma.05G233700	Seed genistein content
Glyma.05G233700	Seed linoleic acid content
Glyma.05G233700	Seed oil content
Glyma.05G233700	Seed protein content
Glyma.05G233700	Seed galactose content
Glyma.05G233700	Seed oil content
Glyma.05G233700	Seed oil content
Glyma.05G233700	Seed stearic acid content
Glyma.05G233700	Seed daidzein content
Glyma.05G233700	Seed isoflavone content
Glyma.05G233700	Seed daidzein content
Glyma.05G233700	Seed genistein content
Glyma.05G233700	Seed linoleic acid content
Glyma.05G233700	Seed oleic acid content
Glyma.06G097900	Seed oil content
Glyma.06G097900	Seed isoflavone content
Glyma.06G098000	Seed isoflavone content

TABLE 5-continued

Agronomic trait being seed quality and genes encoding CCT proteins	
gene	QTL TraitName
Glyma.06G196200	Seed glycitein content
Glyma.06G196200	Seed glycinin and beta-conglycinin content
Glyma.06G196200	Seed glycinin content
Glyma.06G196200	Seed stearic acid content
Glyma.06G196200	Seed oil plus protein content
Glyma.06G196200	Seed methionine content
Glyma.06G196200	Seed oil content
Glyma.06G196200	Seed oil content
Glyma.06G196200	Seed protein content
Glyma.06G196200	Seed protein content
Glyma.07G049400	Seed oil content
Glyma.07G049400	Seed protein content
Glyma.07G049400	Seed glycitein content
Glyma.07G091400	Seed protein content
Glyma.07G091400	Seed isoflavone content
Glyma.07G091400	Seed daidzein content
Glyma.07G091400	Seed isoflavone content
Glyma.07G091400	Seed oil content
Glyma.07G091400	Seed protein content
Glyma.07G091400	Seed oil content
Glyma.08G168900	Seed protein content
Glyma.08G168900	Seed oil content
Glyma.08G168900	Seed oil content
Glyma.08G168900	Seed oil content
Glyma.08G221800	Seed fucose content
Glyma.08G221800	Seed palmitic acid content
Glyma.09G184600	Seed oil content
Glyma.09G184600	Seed palmitic acid content
Glyma.09G184600	Seed linolenic acid content
Glyma.09G184600	Seed palmitic acid content
Glyma.09G184600	Seed daidzein content
Glyma.09G184600	Seed genistein content
Glyma.09G184600	Seed isoflavone content
Glyma.09G184600	Seed stearic acid content
Glyma.09G184600	Seed daidzein content
Glyma.09G184600	Seed protein content
Glyma.09G184600	Seed protein content
Glyma.10G021400	Seed glycinin and beta-conglycinin content
Glyma.10G048100	Seed oil content
Glyma.10G048100	Seed protein content
Glyma.10G190300	Seed protein content
Glyma.10G190300	Seed protein content
Glyma.10G190300	Seed glycitein content
Glyma.10G190300	Seed tocopherol content
Glyma.10G274300	Seed linoleic acid content
Glyma.10G274300	Seed linolenic acid content
Glyma.10G274300	Seed glucose content
Glyma.12G074400	Seed protein content
Glyma.12G074400	Seed daidzein content
Glyma.12G074400	Seed genistein content
Glyma.12G074400	Seed isoflavone content
Glyma.12G196100	Seed oil content
Glyma.12G196100	Seed protein content
Glyma.12G196100	Seed oil content
Glyma.12G196100	Seed oil content
Glyma.12G196100	Seed protein content
Glyma.12G196100	Seed protein content
Glyma.13G009300	Seed protein content
Glyma.13G050300	Seed isoflavone content
Glyma.13G093800	Seed genistein content
Glyma.13G093800	Seed linoleic acid content
Glyma.13G093800	Seed oleic acid content
Glyma.13G135900	Seed linoleic acid content
Glyma.13G135900	Seed oleic acid content
Glyma.13G135900	Seed glycitein content
Glyma.13G135900	Seed oil content
Glyma.13G306400	Seed isoflavone content
Glyma.13G331000	Seed protein content
Glyma.13G331000	Seed stearic acid content
Glyma.14G182800	Seed isoflavone content

TABLE 5-continued

Agronomic trait being seed quality and genes encoding CCT proteins	
gene	QTL TraitName
Glyma.14G182800	Seed oil content
Glyma.14G190400	Seed oil content
Glyma.14G197900	Seed linoleic acid content
Glyma.14G197900	Seed linoleic acid content
Glyma.14G197900	Seed genistein content
Glyma.14G197900	Seed linolenic acid content
Glyma.16G018000	Seed isoflavone content
Glyma.16G018000	Seed protein content
Glyma.16G018000	Seed isoflavone content
Glyma.16G018000	Seed sucrose content
Glyma.16G018000	Seed soluble oligosaccharides
Glyma.16G050900	Seed palmitic acid content
Glyma.16G050900	Seed protein content
Glyma.16G050900	Seed oil content
Glyma.16G050900	Seed oil content
Glyma.16G050900	Seed glycitein content
Glyma.16G067000	Seed soluble oligosaccharides
Glyma.17G059100	Seed linolenic acid content
Glyma.17G059100	Seed genistein content
Glyma.17G059100	Starch content
Glyma.17G066600	Seed palmitic acid content
Glyma.17G102200	Seed oil content
Glyma.17G102200	Seed beta-conglycinin beta-subunit protein content
Glyma.17G102200	Seed glycinin and beta-conglycinin content
Glyma.17G102200	Seed isoflavone content
Glyma.17G102200	Seed leucine content
Glyma.17G102200	Seed methionine and cysteine content
Glyma.17G102200	Seed palmitic + stearic acid content
Glyma.17G102200	Seed glycitein content
Glyma.17G102200	Seed palmitic acid content
Glyma.17G102200	Seed linoleic acid content
Glyma.18G278100	Seed stearic acid content
Glyma.19G039000	Seed oil content
Glyma.19G039000	Seed oil content
Glyma.19G039000	Seed protein content
Glyma.19G039000	Seed palmitic + stearic acid content
Glyma.19G039000	Seed soluble oligosaccharides
Glyma.19G039000	Seed sucrose content
Glyma.19G079600	Seed oil content
Glyma.19G099700	Seed oil content
Glyma.19G099700	Seed sucrose content
Glyma.19G099700	Seed isoleucine content
Glyma.19G099700	Seed soluble oligosaccharides
Glyma.19G099700	Seed tyrosine content
Glyma.19G099700	Seed sucrose content
Glyma.19G099700	Seed genistein content
Glyma.19G099700	Seed genistein content
Glyma.19G099700	Seed isoflavone content
Glyma.19G207100	Seed palmitic acid content
Glyma.19G207100	Seed daidzein content
Glyma.19G207100	Seed genistein content
Glyma.19G207100	Seed isoflavone content
Glyma.19G207100	Seed oil content
Glyma.19G207100	Seed protein content
Glyma.19G207100	Seed hardness
Glyma.19G207100	Seed oil content
Glyma.19G207100	Seed oil content
Glyma.19G260400	Seed daidzein content
Glyma.19G260400	Seed protein content
Glyma.20G085100	Seed lysine content
Glyma.20G085100	Seed methionine content
Glyma.20G085100	Seed protein content
Glyma.20G085100	Seed threonine content
Glyma.20G085100	Seed oil content
Glyma.20G085100	Seed protein content
Glyma.20G085100	Seed protein content

TABLE 5-continued

Agronomic trait being seed quality and genes encoding CCT proteins	
gene	QTL TraitName
Glyma.20G085100	Seed oil content
Glyma.20G085100	Seed calcium content
Glyma.20G085100	Seed oil content
Glyma.20G085100	Seed protein content
Glyma.20G085100	Seed linoleic acid content
Glyma.20G085100	Seed oleic acid content
Glyma.20G085100	Seed oil content
Glyma.20G085100	Seed protein content
Glyma.20G115600	Seed isoflavone content
Glyma.20G115600	Seed delta-tocopherol content
Glyma.20G200400	Seed genistein content
Glyma.20G200400	Seed oleic acid content

TABLE 6

Agronomic trait being seed set and genes encoding CCT proteins	
gene	QTL TraitName
Glyma.01G221100	Seed weight
Glyma.01G221100	Seed weight
Glyma.02G223700	Seed Set
Glyma.03G209800	Pod weight ratio
Glyma.03G209800	Seed height
Glyma.03G209800	Seed Number
Glyma.03G209800	Seed weight
Glyma.03G209800	Seed width
Glyma.04G058900	Seed weight
Glyma.04G058900	Seed height
Glyma.04G058900	Seed size
Glyma.04G058900	Seed width
Glyma.04G058900	Seed Set
Glyma.04G058900	Seed Set
Glyma.04G058900	Seed weight
Glyma.04G058900	Seed weight
Glyma.04G228300	Seed height
Glyma.04G228300	Seed weight
Glyma.04G228300	Seed width
Glyma.04G228300	Seed Set
Glyma.04G236400	Seed Number
Glyma.04G236400	Seed weight
Glyma.05G209800	Seed height
Glyma.05G209800	Seed size
Glyma.05G209800	Seed weight
Glyma.05G233700	Seed yield
Glyma.05G233700	Seed weight
Glyma.06G059600	Seed size
Glyma.06G059600	Seed width
Glyma.06G059600	Seed yield
Glyma.06G059600	Seed weight
Glyma.06G097900	Seed weight
Glyma.06G097900	Seed weight
Glyma.06G097900	Seed Number
Glyma.06G098000	Seed weight
Glyma.06G097900	Seed width
Glyma.06G097900	Seed height
Glyma.06G097900	Seed width
Glyma.06G196200	Seed yield
Glyma.06G196200	Seed Number
Glyma.06G196200	Seed weight
Glyma.06G196200	Pods Per Node
Glyma.06G196200	Seed yield
Glyma.06G196200	Seed Set
Glyma.06G196200	Seed width
Glyma.06G196200	Pods per plant
Glyma.06G196200	Seed Set
Glyma.06G196200	Seed weight
Glyma.06G196200	Seed weight
Glyma.06G196200	Seed weight

TABLE 6-continued

Agronomic trait being seed set and genes encoding CCT proteins	
gene	QTL TraitName
Glyma.06G196200	Seed yield
Glyma.06G196200	Seed yield
Glyma.06G196200	Seed yield
Glyma.06G196200	Seed yield
Glyma.06G196200	Seed yield
Glyma.07G049400	Seed width
Glyma.07G049400	Seed height
Glyma.07G049400	Seed size
Glyma.07G049400	Seed width
Glyma.07G049400	Seed yield
Glyma.07G049400	Seed weight
Glyma.07G049400	Seed yield
Glyma.07G091400	Seed yield
Glyma.07G091400	Seed height
Glyma.07G182700	Seed weight
Glyma.08G016500	Seed weight
Glyma.08G016500	Seed weight
Glyma.08G067200	Seed Set
Glyma.08G067200	Seed width
Glyma.09G184600	Seed Set
Glyma.09G184600	Seed weight
Glyma.09G184600	Seed weight
Glyma.09G184600	Seed weight
Glyma.09G184600	Seed weight
Glyma.09G184600	Seeds per plant
Glyma.09G184600	Seed yield
Glyma.10G134400	Seed Set
Glyma.10G190300	Seed weight
Glyma.10G190300	Seed length to width ratio
Glyma.10G190300	Seed width
Glyma.11G022600	Seed Set
Glyma.11G022600	Seed Number
Glyma.11G022600	Seed weight
Glyma.11G022600	Seed height
Glyma.11G022600	Seed weight
Glyma.12G074400	Seed weight
Glyma.12G196100	Seed weight
Glyma.12G196100	Seed weight
Glyma.12G196100	Seed yield
Glyma.13G093800	Seed Set
Glyma.13G093800	Seed weight
Glyma.13G135900	Seed weight
Glyma.13G135900	Seed weight
Glyma.14G182800	Seed yield
Glyma.14G182800	Seed yield
Glyma.15G219800	Seed weight
Glyma.16G018000	Seed yield
Glyma.16G018000	Seed yield/Plant height
Glyma.16G050900	Seed weight
Glyma.16G050900	Seed weight
Glyma.16G050900	Seed yield
Glyma.17G059100	Seed Set
Glyma.17G059100	Seed weight
Glyma.17G066600	Seed weight
Glyma.17G066600	Seed weight
Glyma.17G066600	Seed weight
Glyma.17G102200	Seed weight
Glyma.17G102200	Seed yield
Glyma.17G102200	Seed yield
Glyma.17G102200	Seed weight
Glyma.17G102200	Seed yield/Plant height
Glyma.17G102200	Seeds per plant
Glyma.17G102200	Seed weight
Glyma.17G102200	Seed height
Glyma.18G278100	Seed weight
Glyma.18G278100	Seed yield
Glyma.19G039000	Seed Set
Glyma.19G099700	Seed Set
Glyma.19G099700	Seed weight
Glyma.19G099700	Seed yield
Glyma.19G207100	Seed weight
Glyma.19G207100	Seed yield/Plant height

TABLE 6-continued

Agronomic trait being seed set and genes encoding CCT proteins	
gene	QTL TraitName
Glyma.19G207100	Seed Number
Glyma.19G207100	Seed weight
Glyma.19G260400	Seed yield/Plant height
Glyma.20G085100	Seed weight
Glyma.20G085100	Seed yield
Glyma.20G115600	Seed Set
Glyma.20G200400	Seed weight
Glyma.20G200400	Seed weight

TABLE 7

Agronomic traits being abiotic/biotic stress tolerance and genes encoding CCT proteins	
gene	QTL TraitName
Glyma.01G221100	Reaction to Heterodera glycines
Glyma.01G221100	Reaction to Sclerotinia sclerotiorum infection
Glyma.02G152900	Reaction to Phytophthora sojae infection
Glyma.03G209800	Reaction to Bemisia tabaci
Glyma.04G058900	Reaction to Heterodera glycines
Glyma.04G058900	Reaction to Heterodera glycines
Glyma.04G096300	Ozone sensitivity, leaflet
Glyma.04G228300	Reaction to Fusarium solani f sp glycines infection
Glyma.04G228300	Phosphorus content
Glyma.04G228300	Phosphorus use efficiency
Glyma.04G228300	Reaction to Soybean Mosaic Virus Infection (SMV)
Glyma.04G236400	Flooding Tolerance
Glyma.04G236400	Lodging
Glyma.05G025000	Reaction to Heterodera glycines
Glyma.05G209800	Reaction to Sclerotinia sclerotiorum infection
Glyma.05G209800	Lodging
Glyma.05G233700	Reaction to Bemisia tabaci
Glyma.06G059600	Reaction to Fusarium solani f sp glycines infection
Glyma.06G059600	Seed glycinin and beta-conglycinin content
Glyma.06G098000	Manganese induced root necrosis
Glyma.06G098000	Reaction to Fusarium solani f sp glycines infection
Glyma.06G127900	Embryo Abortion
Glyma.06G136600	Manganese induced root necrosis
Glyma.06G196200	Reaction to Sclerotinia sclerotiorum infection
Glyma.06G196200	R/V photothermal sensitivity
Glyma.06G196200	Reaction to Aulacorthum solani, choice
Glyma.06G196200	Reaction to Phytophthora sojae infection
Glyma.06G196200	Lodging
Glyma.06G196200	Lodging
Glyma.07G049400	Reaction to Aulacorthum solani, choice
Glyma.07G091400	Reaction to Spodoptera litura
Glyma.07G091400	Reaction to Phytophthora sojae infection
Glyma.07G091400	Row spacing response
Glyma.07G091400	Reaction to Helicoverpa zea
Glyma.08G016500	Reaction to Fusarium solani f sp glycines infection
Glyma.08G041100	Reaction to Fusarium solani f sp glycines infection
Glyma.08G067200	Reaction to Fusarium solani f sp glycines infection
Glyma.08G067200	Reaction to Fusarium solani f sp glycines infection
Glyma.08G067200	Reaction to Heterodera glycines
Glyma.08G067200	Reaction to Heterodera glycines
Glyma.08G168900	Acid phosphatase activity
Glyma.08G168900	Phosphorus content
Glyma.08G168900	Phosphorus use efficiency
Glyma.08G255200	Reaction to Heterodera glycines
Glyma.08G221800	Reaction to Sclerotinia sclerotiorum infection
Glyma.08G221800	Lodging
Glyma.08G255200	Reaction to Sclerotinia sclerotiorum infection
Glyma.08G221800	Reaction to Phytophthora sojae infection
Glyma.09G184600	Flooding Tolerance
Glyma.10G021400	Reaction to Fusarium solani f sp glycines infection
Glyma.10G021400	Reaction to Meloidogyne incognita
Glyma.10G048100	Reaction to Fusarium solani f sp glycines infection
Glyma.10G048100	Reaction to Meloidogyne incognita
Glyma.10G048100	Reaction to Meloidogyne incognita

TABLE 7-continued

Agronomic traits being abiotic/biotic stress tolerance and genes encoding CCT proteins	
gene	QTL TraitName
Glyma.10G134400	Reaction to Heterodera glycines
Glyma.10G190300	Drought Tolerance
Glyma.10G190300	Reaction to Lamprosema indicata
Glyma.10G190300	Reaction to Sclerotinia sclerotiorum infection
Glyma.10G190300	Probability an electron reduces an end acceptor at PSI electron acceptor side
Glyma.10G190300	Reaction to Heterodera glycines
Glyma.11G022600	Reaction to Fusarium solani f sp glycines infection
Glyma.12G074400	Reaction to Phytophthora sojae infection
Glyma.12G074400	Reaction to Heterodera glycines
Glyma.12G196100	Reaction to Helicoverpa zea
Glyma.12G196100	Drought Tolerance
Glyma.12G196100	Drought Susceptibility Index
Glyma.13G009300	Reaction to Phytophthora sojae infection
Glyma.13G050300	Reaction to Aphis glycines
Glyma.13G050300	Reaction to Phytophthora sojae infection
Glyma.13G093800	Reaction to Fusarium solani f sp glycines infection
Glyma.13G093800	Flooding Tolerance
Glyma.13G093800	Reaction to Heterodera glycines
Glyma.13G093800	Reaction to Meloidogyne javanica
Glyma.13G093800	Reaction to Phytophthora sojae infection
Glyma.13G135900	Reaction to Phytophthora sojae infection
Glyma.13G331000	Reaction to Helicoverpa zea
Glyma.14G190400	Iron Efficiency
Glyma.14G197900	Iron Efficiency
Glyma.15G219800	Reaction to Heterodera glycines
Glyma.15G258200	Lodging
Glyma.16G018000	Drought Susceptibility Index
Glyma.16G018000	Reaction to Helicoverpa zea
Glyma.16G067000	Reaction to Soybean Mosaic Virus Infection (SMV)
Glyma.16G067000	Reaction to Bemisia tabaci
Glyma.17G066600	Phosphorus content
Glyma.17G066600	Reaction to Heterodera glycines
Glyma.17G059100	Reaction to Sclerotinia sclerotiorum infection
Glyma.17G066600	Reaction to Sclerotinia sclerotiorum infection
Glyma.17G066600	Root phosphorus content
Glyma.17G066600	Shoot phosphorus content
Glyma.17G102200	Drought Susceptibility Index
Glyma.17G102200	Manganese induced leaflet necrosis
Glyma.17G102200	Reaction to Sclerotinia sclerotiorum infection
Glyma.17G102200	Reaction to Sclerotinia sclerotiorum infection
Glyma.18G278100	Reaction to Aulacorthum solani, choice
Glyma.18G278100	Reaction to Heterodera glycines
Glyma.18G278100	Reaction to Phytophthora sojae infection
Glyma.18G278100	Reaction to Helicoverpa zea
Glyma.18G278100	Reaction to Heterodera glycines
Glyma.19G039000	Reaction to Sclerotinia sclerotiorum infection
Glyma.19G039000	UV-B induced leaflet shape response
Glyma.19G039000	Chlorimuron Ethyl Sensitivity
Glyma.19G099700	Chlorimuron Ethyl Sensitivity
Glyma.19G207100	Reaction to Fusarium solani f sp glycines infection
Glyma.19G207100	Lodging
Glyma.19G207100	Lodging
Glyma.19G207100	Reaction to Fusarium solani f sp glycines infection
Glyma.19G207100	Flooding Tolerance
Glyma.19G207100	Reaction to Fusarium infection
Glyma.20G085100	Reaction to Heterodera glycines
Glyma.20G085100	Ozone sensitivity, leaflet
Glyma.20G115600	Reaction to Phytophthora sojae infection
Glyma.20G115600	Reaction to Fusarium solani f sp glycines infection
Glyma.20G200400	Reaction to Fusarium solani f sp glycines infection
Glyma.20G200400	Reaction to Popilliae japnoica
Glyma.20G200400	Reaction to Heterodera glycines
Glyma.20G200400	Reaction to Heterodera glycines
Glyma.20G200400	Reaction to Heterodera glycines
Glyma.20G200400	Reaction to Heterodera glycines

TABLE 8

Agronomic trait being flowering time and genes encoding CCT protein	
gene	QTL TraitName
Glyma.02G152900	First flower
Glyma.02G215600	First flower
Glyma.02G223700	First flower
Glyma.04G166300	Reproductive stage length
Glyma.04G228300	R8 Full Maturity
Glyma.04G228300	R8 Full Maturity
Glyma.04G236400	First flower
Glyma.04G236400	R8 Full Maturity
Glyma.06G059600	R8 Full Maturity
Glyma.06G097900	R8 Full Maturity
Glyma.06G196200	First flower
Glyma.06G196200	Photoperiod sensitivity
Glyma.06G196200	R3 Beginning pod
Glyma.06G196200	R7 Beginning Maturity
Glyma.06G196200	First flower
Glyma.06G196200	First flower
Glyma.06G196200	R5 Beginning Seed
Glyma.06G196200	R8 Full Maturity
Glyma.06G196200	R8 Full Maturity
Glyma.06G196200	First flower
Glyma.06G196200	First flower
Glyma.07G049400	R8 Full Maturity
Glyma.07G091400	First flower
Glyma.07G091400	R8 Full Maturity
Glyma.07G091400	Seed filling period
Glyma.10G190300	Reproductive stage length
Glyma.10G190300	R8 Full Maturity
Glyma.12G074400	First flower
Glyma.12G074400	R8 Full Maturity
Glyma.12G196100	R8 Full Maturity
Glyma.13G009300	R8 Full Maturity
Glyma.13G050300	R8 Full Maturity
Glyma.13G135900	Photoperiod sensitivity
Glyma.13G135900	Reproductive stage length
Glyma.13G135900	R8 Full Maturity
Glyma.14G182800	Reproductive period
Glyma.16G018000	R8 Full Maturity
Glyma.16G018000	R8 Full Maturity
Glyma.16G018000	Reproductive stage length
Glyma.18G278100	R8 Full Maturity
Glyma.18G278100	R8 Full Maturity
Glyma.19G207100	First flower
Glyma.19G207100	R8 Full Maturity
Glyma.19G207100	First flower
Glyma.19G260400	First flower
Glyma.19G207100	Flower form
Glyma.19G260400	First flower
Glyma.19G260400	First flower
Glyma.19G260400	R8 Full Maturity
Glyma.19G260400	Reproductive stage length
Glyma.20G115600	R8 Full Maturity

TABLE 9

Agronomic trait being development-related traits and genes encoding CCT proteins	
gene	QTL TraitName
Glyma.01G221100	Pubescence Density
Glyma.02G223700	Canopy wilt
Glyma.02G215600	Shattering
Glyma.02G230800	Internode length
Glyma.02G215600	Node Number
Glyma.02G215600	Canopy wilt
Glyma.03G209800	Leaflet chlorophyll content
Glyma.03G209800	Leaflet Width
Glyma.03G209800	Petiole color
Glyma.03G209800	UV-B induced leaflet shape response

TABLE 9-continued

Agronomic trait being development-related traits and genes encoding CCT proteins	
gene	QTL TraitName
Glyma.03G209800	UV-B induced plant damage
Glyma.03G209800	Plant Height
Glyma.04G058900	Canopy wilt
Glyma.04G058900	Branching
Glyma.04G228300	Plant Height
Glyma.04G228300	Seed coat cracking
Glyma.04G228300	Lodging
Glyma.04G228300	Plant Height
Glyma.04G228300	Root weight, dry
Glyma.04G236400	Leaflet shape
Glyma.04G236400	Leaflet Width
Glyma.05G233700	Plant Height
Glyma.05G209800	Plant Height
Glyma.05G209800	Cell wall polysaccharide composition
Glyma.05G209800	Root Nodule Size
Glyma.05G233700	Leaflet Length
Glyma.05G233700	Leaflet Width
Glyma.06G097900	shoot weight, fresh
Glyma.06G059600	Flower form
Glyma.06G098000	Plant weight
Glyma.06G098000	shoot weight, dry
Glyma.06G097900	Root nodule number
Glyma.06G097900	Root nodule weight
Glyma.06G097900	shoot weight, dry
Glyma.06G136600	Node Number
Glyma.06G196200	Plant Height
Glyma.06G196200	Root length, coarse
Glyma.06G196200	Root volume
Glyma.06G196200	Plant Height
Glyma.07G049400	Plant Height
Glyma.07G049400	Seed height
Glyma.07G091400	Seed height
Glyma.07G091400	Canopy Height
Glyma.07G091400	Leaflet area
Glyma.07G091400	Root nodule weight, dry
Glyma.07G091400	Leaflet chlorophyll content
Glyma.07G091400	Internode length
Glyma.07G091400	Pods Per Node
Glyma.07G182700	Plant Height
Glyma.08G016500	Seed delta-tocopherol content
Glyma.08G016500	Seed gamma-tocopherol content
Glyma.08G016500	Pods per plant
Glyma.08G041100	Stem strength
Glyma.08G067200	Canopy wilt
Glyma.08G067200	Pubescence Density
Glyma.08G067200	Pubescence Density
Glyma.08G067200	Seed coat cracking
Glyma.08G067200	Leaflet area
Glyma.08G067200	Canopy wilt
Glyma.09G184600	Pods per plant
Glyma.10G021400	Root dry weight to shoot dry weight ratio
Glyma.10G048100	Row spacing response
Glyma.10G048100	Seed length to height ratio
Glyma.10G048100	Seed length to height ratio
Glyma.10G134400	Net photosynthetic rate
Glyma.10G190300	Pods per plant
Glyma.10G190300	Total growth duration
Glyma.10G190300	Root surface area, medium
Glyma.10G190300	Root volume
Glyma.10G190300	Root weight, dry
Glyma.10G190300	Branching
Glyma.10G190300	Leaflet shape
Glyma.10G190300	Leaflet Width
Glyma.10G190300	Plant Height
Glyma.11G022600	Hypocotyl Length
Glyma.12G074400	Leaflet area
Glyma.12G074400	Plant Height
Glyma.12G074400	Total growth duration
Glyma.12G074400	Branching
Glyma.12G196100	Neutral detergent fiber content
Glyma.12G196100	Plant Height
Glyma.12G196100	Pubescence Density

TABLE 9-continued

Agronomic trait being development-related traits and genes encoding CCT proteins	
gene	QTL TraitName
Glyma.12G196100	Plant Height
Glyma.12G196100	Canopy wilt
Glyma.13G009300	Leaf phosphorus content
Glyma.13G050300	Node Number
Glyma.13G050300	Leaflet chlorophyll content
Glyma.13G093800	Seed coat cracking
Glyma.13G093800	Canopy Height
Glyma.13G093800	Embryo Abortion
Glyma.13G093800	Leaflet Width
Glyma.13G093800	Hypocotyl Length
Glyma.13G093800	Plant Height
Glyma.13G306400	Plant Height
Glyma.13G306400	Neutral detergent fiber content
Glyma.13G306400	Plant Height
Glyma.13G331000	Acid detergent fiber content
Glyma.13G331000	Plant Height
Glyma.14G197900	Leaflet Length
Glyma.14G197900	Somatic embryos per explant
Glyma.16G018000	Plant Height
Glyma.16G018000	Leaflet area
Glyma.16G018000	Shattering
Glyma.16G050900	Leaflet shape
Glyma.16G050900	Plant Height
Glyma.16G050900	Shattering
Glyma.16G050900	shoot weight, dry
Glyma.16G050900	Leaflet trigonelline content
Glyma.16G050900	Seed viability
Glyma.17G066600	Plant weight
Glyma.17G066600	Plant weight, dry
Glyma.17G059100	Canopy wilt
Glyma.17G066600	Root Weight
Glyma.17G066600	Root weight, dry
Glyma.17G066600	shoot weight, dry
Glyma.17G102200	Node Number
Glyma.17G102200	Canopy wilt
Glyma.17G102200	Canopy wilt
Glyma.17G102200	Canopy wilt
Glyma.17G102200	Canopy wilt

TABLE 9-continued

Agronomic trait being development-related traits and genes encoding CCT proteins	
gene	QTL TraitName
Glyma.18G278100	Plant Height
Glyma.18G278100	Plant Height
Glyma.18G278100	Plant Height
Glyma.19G039000	Shattering
Glyma.19G039000	Leaflet Length
Glyma.19G039000	Leaflet shape
Glyma.19G039000	Leaflet Width
Glyma.19G099700	Plant Height
Glyma.19G099700	Seed length to width ratio
Glyma.19G099700	Plant Height
Glyma.19G099700	Leaflet Length
Glyma.19G099700	Leaflet shape
Glyma.19G099700	Leaflet Length
Glyma.19G099700	Leaflet Width
Glyma.19G207100	Pods per plant
Glyma.19G207100	Canopy wilt
Glyma.19G207100	Canopy wilt
Glyma.19G207100	Plant Height
Glyma.19G207100	Height/Lodging
Glyma.19G207100	Leaflet Length
Glyma.19G207100	Plant Height
Glyma.19G207100	Branching
Glyma.19G207100	Canopy wilt
Glyma.19G207100	Leaflet shape
Glyma.19G207100	Leaflet Width
Glyma.19G207100	Seed Coat Hardness
Glyma.19G260400	Leaflet Length
Glyma.19G260400	Leaflet shape
Glyma.19G260400	Leaflet Width
Glyma.19G260400	Plant Height
Glyma.19G260400	Plant Height
Glyma.20G085100	Internode length
Glyma.20G115600	Pods per plant
Glyma.20G115600	Leaflet Width
Glyma.20G115600	Specific leaflet weight
Glyma.20G200400	Canopy Width
Glyma.20G200400	Root nodule number

TABLE 10

SNPs significantly associated with seed protein and oil content and 100-seed weight. Significant variants								
Trait/ gene model	Chr	Position	Oil p-value	Protein p-value	Seed weight p-value	SNP/gene Annotation	SNP Distance to 5' flanking gene lhpl	SNP Distance to 3' flanking gene lhpl
Glyma.20G084400	20	31,650,003 start 31,650,419 stop				N/A		
SNP1	20	31,656,273		4.91E-08		intergenic	5,854	9,917
SNP2	20	31,658,904	2.41E-22	7.25E-08		intergenic	8,485	7,286
SNP3	20	31,658,920		1.82E-07		intergenic	8,501	7,270
SNP4	20	31,661,646		1.98E-07		intergenic	11,227	4,544
SNP5	20	31,664,872	4.91E-22	6.00E-08	4.38NE-14	intergenic	14,453	1,318
Glyma.20G084500	20	31,666,190 stop 31,674,959 start				MOS4- associated complex 3B		
Glyma.20G084600	20	31,668,842 start 31,669,146 stop				N/A		
SNP6	20	31,680,622	2.56E-21	8.96E-08		intergenic	5,663	6,191
SNP7	20	31,682,844		4.60E-07		intergenic	7,885	3,969
SNP8	20	31,685,116		2.95E-07		intergenic	10,157	1,697
Glyma.20G084700	20	31,686,813 stop 31,687,483 start				unknown protein		
Glyma.20G084800	20	31,689,766 start 31,690,181 stop				N/A		

TABLE 10-continued

SNPs significantly associated with seed protein and oil content and 100-seed weight. Significant variants								
Trait/ gene model	Chr	Position	Oil p-value	Protein p-value	Seed weight p-value	SNP/gene Annotation	SNP Distance to 5' flanking gene lhpl	SNP Distance to 3' flanking gene lhpl
SNP9	20	31,691,428	2.72E-22	2.88E-07		intergenic	1,247	15,919
SNP10	20	31,692,813	3.96E-20	6.63E-08		intergenic	2,632	14,534
SNP11	20	31,693,770		5.05E-08		intergenic	3,589	13,577
SNP12	20	31,696,275	1.86E-19	8.61E-08	1.03E-12	intergenic	6,094	11,072
Glyma.20G084900	20	31,707,347 stop 31,709,780 start				hAT dimerisation domain- containing protein		
SNP13	20	31,713,462	8.25E-20	3.45E-07		intergenic	3,682	32,024
SNP14	20	31,718,263	6.74E-19	3.26E-08		intergenic	8,483	27,223
SNP15	20	31,718,476		1.45E-07		intergenic	8,696	27,010
SNP16	20	31,724,545	4.71E-20			intergenic	14,765	20,941
SNP17	20	31,727,573	1.40E-20	4.49E-08		intergenic	17,793	17,913
SNP18	20	31,730,753	4.78E-21	3.24E-08		intergenic	20,973	14,733
SNP19	20	31,731,003	1.56E-20	6.28E-08		intergenic	21,223	14,483
SNP20	20	31,733,134		2.79E-07		intergenic	23,354	12,352
SNP21	20	31,737,391	1.37E-22	1.84E-07		intergenic	27,611	8,095
SNP22	20	31,738,091				intergenic	28,311	7,395
Glyma.20G085000	20	31,745,486 stop				Conversed oligomeric complex COG6		
InDell	20	31,746,064	4.25E-21	4.87E-12	4.87E-12	intron 2	N/A	
Glyma20G085000	20	31,748,473 start				intergenic	3,087	23,210
SNP23	20	31,751,560 2-bp ins	2.253E-23			intergenic	6,565	19,732
SNP24	20	31,755,038	2.78E-18			intergenic	10,686	15,611
SNP25	20	31,759,159	4.68E-20			intergenic	11,820	14,477
SNP26	20	31,760,293	1.40E-18			intergenic	3,087	23,210
SNP27	20	31,761,452	1.71E-18	2.59E-07		intergenic	12,979	13,318
Glyma.20G085100	20	31,774,770 start				CCT motif family protein		
InDel2	20	31,778,798- 31,779,118	6.17E-24	2.69E-08	2.90E-1	exon 4	N/A	N/A
Glyma.20G085100	20	31,779,804 stop						
Glyma.20G085200	20	31,784,964 stop 31,786,969 start				GATA type zinc finger transcription factor family protein		
SNP28	20	31,791,856	7.92E-22		IP	intergenic	4,887	31,317
SNP29	20	31,793,219	3.45E-21	6.44E-08		intergenic	6,250	29,954
SNP30	20	31,795,724	1.59E-18			intergenic	8,755	27,449
SNP31	20	31,801,624	8.03E-20	1.16E-07	126E-12	intergenic	14,655	21,549
SNP32	20	31,812,853	7.85E-19			intergenic	25,884	10,320
Glyma.20G085200	20	31,823,173 start 31,827,117 stop				protein of unknown function (DUF3527)		

TABLE 11

Genotyping analysis of the 321-bp InDel in 30 RILs and their parental lines.			
Common Name	TE insertion	Oil content [%]	Protein content 1% 1
P1479752	no	9.6	44.4
I1376	no	11.3	46.0
I628	no	11.5	44.2
I976	no	11.5	48.0
I695	no	11.9	46.5
I1338	no	12.0	46.9
I828	no	12.1	46.8

TABLE 11-continued

Genotyping analysis of the 321-bp InDel in 30 RILs and their parental lines.			
Common Name	TE insertion	Oil content [%]	Protein content 1% 1
I668	no	12.2	44.8
I1375	no	12.3	46.7
I680	no	12.3	44.2
I1530	no	12.5	45.6
I144	no	13.2	46.4
I1593	no	13.6	46.2
I665	no	13.6	46.1

TABLE 11-continued

Genotyping analysis of the 321-bp InDel in 30 RILs and their parental lines.			
Common Name	TE insertion	Oil content [%]	Protein content 1% 1
I1630	no	13.7	46.4
I200	no	15.3	38.8
I306	yes	15.9	39.0
I646	yes	16.1	38.4
I1652	yes	16.3	38.8
I1342	yes	16.7	38.9
I1490	yes	17.1	38.7
I684	yes	17.5	41.2
I477	yes	17.6	39.9
I32	yes	17.6	40.6
I1234	yes	17.6	39.0
I347	yes	17.7	41.3

TABLE 11-continued

Genotyping analysis of the 321-bp InDel in 30 RILs and their parental lines.			
Common Name	TE insertion	Oil content [%]	Protein content 1% 1
I333	yes	17.8	38.6
I1596	yes	18.2	37.9
I259	yes	18.3	39.4
I113	yes	18.3	39.7
I581	yes	18.6	37.6
Williams82	yes	20.5	39.3
Average	no	12.4	45.5
Average	Yes	17.6	39.3
Difference between average contents	—	5.2	6.2
p-value if-test)	—	4.00E-13	3.53E-10

TABLE 12

TE genotypes of accessions used for biparental QTL mapping experiments.													
high oil/low protein parent						low oil/high protein parent					Chr20		
P1	Name	Species	TE insertion	Oil Content [%]	Protein Content [%]	PI	Name	Species	TE insertion	Oil Content [%]	Protein Content [%]	oil and protein QTL	Reference
P1518671	Williams 82	<i>Glycine max</i>	yes	20.3	41.5	P1479752	GD 50388-2	<i>Glycine soja</i>	no	9.4	46.4	yes	this study
P1562374	Parker	<i>Glycine max</i>	yes	22.2	42.6	P1468916		<i>Glycine soja</i>	no	10.1	44	yes	Sebolt et al. (2000)
P1518671	Williams 82	<i>Glycine max</i>	yes	20.3	41.5	P1483460B		<i>Glycine soja</i>	no	8.9	48	yes (protein)	Patil et al. 2018
	R05-638	<i>Glycine max</i>	yes	23 ^a	40.8 ^a	P1659697	R05-1415	<i>Glycine max</i>	no	18.4 ^a	48.3 ^a	yes	Wang et al. (2015a), Wang et al. (2015b)
	R05.4256	<i>Glycine max</i>	yes	23 ^a	40.8 ^a		V97-1346	MIX	no	18.4 ^a	48.3 ^a	yes	Wang et al. (2015a), Wang et al. (2015b)
P1595645	Benning	<i>Glycine max</i>	yes	18.4	42 ^b	P1619083	Danbaekkong	<i>Glycine max</i>	no	N/A	51 ^b	yes (protein)	Warring ^② (2015)
P1556814	A3733	<i>Glycine max</i>	yes	20.5 ^c	42 ^c	P1437088A	DV-147	<i>Glycine max</i>	no	16 ^c	48 ^c	yes	Chung et al. (2003)
P1556536	Coker 237	<i>Glycine max</i>	yes	20 ^d	42.6 ^d	P197100	1249	<i>Glycine max</i>	yes	18.6 ^d	45.1 ^d	no	Fasoula ^② (2004), (Lee et al. (1996))
P1595362	Magellan	<i>Glycine max</i>	yes	20.2 ^c	38 ^c	P1567516C	Ba yue zha	<i>Glycine</i>	yes	17.8	42.6	no	Pathan et al. (2013)
P1508266	Young	<i>Glycine max</i>	yes	19.4	44.2	P1416937	Houjaku Kuwazu	<i>Glycine max</i>	yes	15.6	47.1	no	Fasoula et al. (2004) (Lee et al. (1996))
P1548631	Williams	<i>Glycine max</i>	yes	19.8	45.6	P1548667	Essex	<i>Glycine max</i>	yes	20	43.9	no	Hyten et al. (2004)

TABLE 12-continued

TE genotypes of accessions used for biparental QTL mapping experiments.													
high oil/low protein parent						low oil/high protein parent						Chr20	
P1	Name	Species	TE insertion	Oil Content [1%]	Protein Content [1%]	PI	Name	Species	TE insertion	Oil Content [1%]	Protein Content [1%]	oil and protein QTL	Reference
P1548389	Minsoy	<i>Glycine max</i>	yes	19.7	39.4	P1290136	Noir1	<i>Glycine max</i>	yes	19.8	42.9	no	Orf et al. (1999), (Manst ^② (1993), Mansu ^② (1996), Specht ^② (2001))
P1546487	Archer	<i>Glycine max</i>	yes	21	43.3	P1290136	Noir 1	<i>Glycine max</i>	yes	19.8	42.9	no	Orf et al. (1999)
P1546487	Archer	<i>Glycine max</i>	yes	21	43.3	P1548389	Minsoy	<i>Glycine max</i>	yes	19.7	39.4	no	Orf et al. (1999),
P1548667	Essex	<i>Glycine max</i>	yes	20	43.9	P1548402	Peking	<i>Glycine max</i>	yes	18.5	38.2	no	Qiu et al. (1999),

For oil and protein content:

^afrom Wang et al. 2015a and b, converted from 13% moisture content to dry weight;^bfrom Warrington et al. 2015^cfrom Chung et al. 2003^dfrom Lee et al. 1996^efrom Pathan et al. 2013

all other phenotype values from GRIN.

^② indicates text missing or illegible when filed

TABLE 13

Information of accessions used for RNA-Seq and POWR1 expression levels.						
P1	Common Name	Species Subspecies	TE insertion	FPICM	AverageFPKM	Stdev
P1518671	Williams 82	<i>Glycine max</i>	yes	1.010692	0.6679555	0.18984621
P1540666	Jack	<i>Glycine max</i>	yes	0.724321		
P1548348	^②	<i>Glycine max</i>	yes	0.508646		
P154863 1	Williams	<i>Glycine max</i>	yes	0.673904		
P1548668	Lee	<i>Glycine max</i>	yes	0.589683		
P1598124	Maverick	<i>Glycine max</i>	yes	0.502487		
P1647962	R95-1705	<i>Glycine max</i>	no	0.847271	0.9701485	0.28815252
.	JL275	<i>Glycine max</i>	no	0.632052		
.	LLL05-14	<i>Glycine max</i>	no	1.282608		
.	S09-2902-145	<i>Glycine max</i>	no	1.118663		

^② indicates text missing or illegible when filed

TABLE 14

Agronomic traits that can be improved using compositions	
QTL	TraitName
	Acid detergent fiber content
	Acid phosphatase activity
	Branching
	Canopy Height
	Canopy Width
	Canopy wilt
	Cell wall polysaccharide composition
	Chlorimuron Ethyl Sensitivity
	Drought Susceptibility Index
	Drought Tolerance
	Embryo Abortion
	First flower
	Flooding Tolerance
	Flower form
	Height/Lodging

TABLE 14-continued

Agronomic traits that can be improved using compositions	
QTL	TraitName
	Hypocotyl Length
	Internode length
	Iron Efficiency
	Leaf phosphorus content
	Leaflet area
	Leaflet chlorophyll content
	Leaflet Length
	Leaflet shape
	Leaflet trigonelline content
	Leaflet Width
	Lodging
	Manganese induced leaflet necrosis
	Manganese induced root necrosis
	Net photosynthetic rate
	Neutral detergent fiber content

TABLE 14-continued

Agronomic traits that can be improved using compositions
QTL TraitName
Node Number
Ozone sensitivity, leaflet
Pectin content
Petiole color
Phosphorus content
Phosphorus use efficiency
Photoperiod sensitivity
Plant Height
Plant weight
Plant weight, dry
Pod weight ratio
Pods Per Node
Pods per plant
Probability an electron reduces an end acceptor at PSI electron acceptor side
Pubescence Density
R/V photothermal sensitivity
R3 Beginning pod
R5 Beginning Seed
R7 Beginning Maturity
R8 Full Maturity
Reaction to Aphis glycines
Reaction to Aulacorthum solani, choice
Reaction to Aulacorthum solani, no choice
Reaction to Bemisia tabaci
Reaction to Fusarium infection
Reaction to Fusarium solani f sp glycines infection
Reaction to Helicoverpa zea
Reaction to Heterodera glycines
Reaction to Lamprosema indicata
Reaction to Meloidogyne incognita
Reaction to Meloidogyne javanica
Reaction to Phytophthora sojae infection
Reaction to Popilliae japnoica
Reaction to Sclerotinia sclerotiorum infection
Reaction to Soybean Mosaic Virus Infection (SMV)
Reaction to Spodoptera litura
Reproductive period
Reproductive stage length
Root dry weight to shoot dry weight ratio
Root length, coarse
Root length, fine
Root length, medium
Root nodule number
Root Nodule Size
Root nodule weight
Root nodule weight, dry
Root phosphorus content
Root Surface Area
Root surface area, coarse
Root surface area, fine
Root surface area, medium
Root volume
Root volume, coarse
Root volume, fine
Root volume, medium
Root Weight
Root weight, dry
Row spacing response
Seed arabinose-galactose content
Seed beta-conglycinin beta-subunit protein content
Seed calcium content
Seed coat cracking
Seed Coat Hardness

TABLE 14-continued

Agronomic traits that can be improved using compositions
QTL TraitName
Seed cysteine content
Seed daidzein content
Seed delta-tocopherol content
Seed filling period
Seed fucose content
Seed galactose content
Seed gamma-tocopherol content
Seed genistein content
Seed glucose content
Seed glycinin and beta-conglycinin content
Seed glycinin content
Seed hardness
Seed height
Seed isoflavone content
Seed isoleucine content
Seed length to height ratio
Seed leucine content
Seed linoleic acid content
Seed linolenic acid content
Seed lysine content
Seed methionine and cysteine content
Seed methionine content
Seed Number
Seed oil content
Seed oil plus protein content
Seed oleic acid content
Seed palmitic acid content
Seed palmitic + stearic acid content
Seed protein content
Seed Set
Seed size
Seed soluble oligosaccharides
Seed stearic acid content
Seed sucrose content
Seed threonine content
Seed tyrosine content
Seed viability
Seed weight
Seed width
Seed yield
Seed yield/Plant height
Seeds per plant
Shattering
Shoot phosphorus content
shoot weight, dry
shoot weight, fresh
Somatic embryos per explant
Specific leaflet weight
Starch content
Stem strength
Total growth duration
UV-B induced leaflet shape response
UV-B induced plant damage

TABLE 15

List of QTLs co-located with GmCCTs and Diversity Pi values for GmCCTs								
Gene Name	ID	gene	locus	QTL chr	QTL start	QTL stop	interval	QTL Trait Name
POWR2	GmCCT34	Glyma.10G134400	Chr10: 36034798-36039695	Chr10	33,311,029	38,033,182	4,722,153	Net photosynthetic rate
				Chr10	29,480,120	36,986,811	7,506,691	Reaction to Heterodera glycines
				Chr10	29,427,575	38,574,461	9,146,886	Seed Set 0
POWR3	GmCCT35	Glyma.10G190300	Chr10: 42314396-42318797	Chr10	41,204,461	44,713,925	3,509,464	Branching
				Chr10	37,300,089	43,004,879	5,704,790	Drought Tolerance
				Chr10	42,295,191	44,713,925	2,418,734	Leaflet shape
				Chr10	42,295,191	44,713,925	2,418,734	Leaflet Width
				Chr10	42,295,191	44,713,925	2,418,734	Plant Height
				Chr10	36,986,811	44,713,925	7,727,114	Pods per plant
				Chr10	41,204,461	43,788,256	2,583,795	Probability an electron reduces an end acceptor at PSI electron acceptor side
				Chr10	42,295,191	44,713,925	2,418,734	R8 Full Maturity
				Chr10	42,295,191	46,492,285	4,197,094	Reaction to Heterodera glycines
				Chr10	38,957,017	43,004,879	4,047,862	Reaction to Lamprosema indicata
				Chr10	40,306,793	43,004,879	2,698,086	Reaction to Sclerotinia sclerotiorum infection
				Chr10	41,204,461	44,713,925	3,509,464	Reproductive stage length
				Chr10	40,306,793	48,423,831	8,117,038	Root surface area, medium
				Chr10	40,306,793	48,423,831	8,117,038	Root volume
				Chr10	40,306,793	48,423,831	8,117,038	Root volume, medium
				Chr10	40,306,793	48,423,831	8,117,038	Root weight, dry
				Chr10	42,295,191	44,713,925	2,418,734	Seed glycitein content
				Chr10	41,204,461	43,509,732	2,305,271	Seed length to width ratio
				Chr10	36,986,811	43,004,879	6,018,068	Seed protein content
				Chr10	36,986,811	44,713,925	7,727,114	Seed protein content
				Chr10	42,295,191	44,713,925	2,418,734	Seed tocopherol content
				Chr10	36,986,811	43,004,879	6,018,068	Seed weight
				Chr10	41,204,461	43,509,732	2,305,271	Seed width
				Chr10	39,108,281	43,004,879	3,896,598	Total growth duration 0
POWR1	GmCCT67	Glyma.20G085100	Chr20: 31774769-31779804	Chr20	25,275,083	34,052,339	8,777,256	Internode length
				Chr20	25,693,363	34,302,228	8,608,865	Ozone sensitivity, leaflet
				Chr20	25,275,083	34,302,228	9,027,145	Reaction to Heterodera glycines
				Chr20	25,693,363	34,052,339	8,358,976	Seed calcium content
				Chr20	25,727,028	35,362,576	9,635,548	Seed linoleic acid content
				Chr20	25,275,083	34,052,339	8,777,256	Seed lysine content
				Chr20	25,275,083	34,052,339	8,777,256	Seed methionine content
				Chr20	27,664,504	33,590,931	5,926,427	Seed oil content
				Chr20	25,727,028	34,302,228	8,575,200	Seed oil content
				Chr20	25,275,083	34,302,228	9,027,145	Seed oil content

TABLE 15-continued

List of QTLs co-located with GmCCTs and Diversity Pi values for GmCCTs								
Gene Name	ID	gene	locus	QTL chr	QTL start	QTL stop	interval	QTL Trait Name
POWR4	GmCCT69	Glyma.20G200400	Chr20: 43781164-43785175	Chr20	25,674,167	35,362,576	9,688,409	Seed oil content
				Chr20	25,674,167	35,362,576	9,688,409	Seed oil content
				Chr20	25,727,028	35,362,576	9,635,548	Seed oleic acid content
				Chr20	27,664,504	33,590,931	5,926,427	Seed protein content
				Chr20	25,727,028	34,302,228	8,575,200	Seed protein content
				Chr20	25,275,083	34,052,339	8,777,256	Seed protein content
				Chr20	25,275,083	34,302,228	9,027,145	Seed protein content
				Chr20	25,498,552	35,362,576	9,864,024	Seed protein content
				Chr20	25,275,083	34,052,339	8,777,256	Seed threonine content
				Chr20	25,727,028	34,052,339	8,325,311	Seed weight
				Chr20	27,664,504	33,590,931	5,926,427	Seed yield
							0	
				Chr20	43,382,152	45,170,799	1,788,647	Canopy Width
				Chr20	43,634,534	44,710,552	1,076,018	Reaction to Heterodera glycines
				Chr20	42,426,240	45,134,969	2,708,729	Reaction to Heterodera glycines
				Chr20	43,198,876	47,757,190	4,558,314	Reaction to Heterodera glycines
				Chr20	41,415,974	46,787,225	5,371,251	Reaction to Heterodera glycines
				Chr20	40,623,814	46,787,225	6,163,411	Reaction to Popilliae japnoica
				Chr20	43,382,152	45,170,799	1,788,647	Root nodule number
				Chr20	42,798,354	44,710,552	1,912,198	Seed genistein content
				Chr20	42,953,527	44,710,552	1,757,025	Seed oleic acid content
				Chr20	40,623,814	44,710,552	4,086,738	Seed weight
				Chr20	42,165,892	46,678,762	4,512,870	Seed weight
				Chr20	38,172,219	46,678,762	8,506,543	Reaction to Fusarium solani f sp glycines infection

SEQUENCES			
SEQ ID NO.	Organism (Latin name) OR Synthetic	Sequence type DNA; Protein; RNA or DNA/RNA	Sequence
1	WT >POWR1-TE PROTEIN SEQ	Protein	MLQDVVHPSTPAEQLPIDEISGPISARIFELCDPDPFFPHTLQNSEV TSSSNCCHEEKSSYATTISPPLDVVDDNNKFNINSNSNSNIVTTSST TTTTSTTTNNNNNATNGNNLSIFFDQDEIDNDISASIDFSSSPF VVPPLLPISTQQDQDFDPSAQPOVQLSTAAGSILTGLSHYPTDPVI APLIGAPLPSVFDDDCISSIPSYVPLNPSSPSCSYLSPGIGVYMPP PGLNLTALSDSSGLFGGNILLGSELQAHELDYQGENGGMCTDSI QRFVNSPDLQALGNESQKLAVAGGSATLAPEISHLEDSTLKVGKL SVEQRKEKIHRYMKKRNERNFSSKKIKYACRKTLDASRPVRGRFAK NDDFGESHKQGSNNHEDDDEEIVKEDDDMVDSSDI FAHISGVNSF KCNYSIQSLI

-continued

SEQUENCES			
2	WT >POWR1- TE NUCLEOTIDE ACID SEQ	<u>DNA</u>	ATGTTGCAGGATGTTGTCCACCCCTCAACACCGGCTGAGCAACTCC CCATTGATGAGATTTTCAGGCCGATTAGTGCTCGAATTTTCGAACT TTGCGACCCCGATTTCTCCACACACACTGCAAAATTTGAGGTT ACCTCCAGCTCAAATGTTGCCATGAAGAGAAGTCTCATATGCCA CAACCATATCTCCACCTTTAGATGTAGTAGACAACAATAAGTTCAA TATCAATAGCAATAGCAGCAACATAGTCACCACTACCTCATCTAGC ACTACCACAACCAGCACCACAACCAACAACAACAACGCAACGA ACGGCAATAATCTTTCTATCTTCTTTGACACTCAAGATGAAATTGA CAATGACATCTCAGCCTCCATAGACTTCTCATCTCTCCATCTTTT GTCGTTCCACCACTTCTCCCAATCTCAACTCAGCAGGATCAGTTTG ATTTCCCTTCAGCTCAGCCACAGGTGCAACTATCAACAGCAGCAGG TTCAATTTTGACGGGCTCTCTCACTACCTTACAGATCCTGTGATT GCACCCCTTATTGGAGCTCCGTTACCATCTGTTTTTGATGATGATT GCATATCTTCCATCCCTTCTTATGTGCCTCTCAACCCCTTCATCACC CTCTTGCTCTTATCTCAGTCCTGGCATAGGAGTGATACATGCCACCT CCTGGTTCCCTTAACACTGCCTTATCTGCTGACAGTTCTGGATTGT TTGGTGGGAACATTCTACTGGGGTCTGAACTGCAGGCACATGAATT GGACTACCGGGAGAAAATGGTGGAATGTATTGTACAGATTCAATT CAAAGGGTGTTTAACTCCCAGACCTTCAGGCACCTGGTAATGAGA GTCAGAAACTTGTAGCTGGGGCTGGAAGCTCTGCCACTTTGGCACC AGAAATCTCACACTTGGAGGACTCTACCTTGAAGGTGGAAAACCTC TCTGTTGAGCAGAGGAAGGAAAAGATTATAGATACATGAAGAAGA GAAACGAAAGAAATTTAGCAAGAAATCAAGTATGCTTGCCGCAA AACTTTGGCGGATAGCCGGCCCGGGTTAGAGGAAGGTTTGCAAAG AATGATGACTTTGGAGAGAGCCATAAACAAGGAAGTAGCAATCATG AAGATGATGATGAAGAGATAATTGTGAAAGAAGATGATGATATGGT TGATTCCCTCAGATATCTTTGCACATATCAGCGGAGTGAACCTTTTC AAATGCAACTATTCCATCCAGTCCTTGATTGA
3	Mutant >POWR1_w_ TE_truncated CCT_CDS_ CCT67 >POWR1,TE (Glyma. 20G085100) Genomic DNA (nucleotide) sequence: 17 bp duplications are bolded and underlined	Dna	Chr20:31774769..31779804-Forward: key: 5'UTR CDS 3'UTR CATCATCTTAGCTAACACAACACAGCATACACAACCTTCTCTCTCTC TACTACTTTCTTTTATGTACCTTCTTGATGTCTTAAATTGTTTT TCTTTTTTAGAAAAAATTAATCCTTTTAAGGCTTTAATTTCCTA AGCATGCAATATTATTGTTTTTAGTCACCTTCAAGTTGAGGAACAT ATACACGTTTGTGACCACACCAAAATCCACTTCTTTCTAAGTGTGT GTACATAGATTTTGTTTTATATATATTGAGGTTTCAACCTTCATTG CCTCCTTAATTCTGTGCAAAAGAGGATTATCACCACCATTGTGCA GGATGTTGTCCACCCCTCAACACCGGCTGAGCAACTCCCCATTGTA ATCACCTCAAGCTTGTACATTTTATATCTCGATCATGTATATTTA GTGTTAATGCATGTATTGGAACAAAGCTATATATGTATGTGTGTAT GTATCTTTGTGTGTCATGCAATATTTATGACCACAGCTAAATGGTTTC CTTCGGGTTTTGTGATCGTGTGTAACCTTGACAAGTTTGTGTTGGAT CAAGGGCTTGTTTTTTCTTTCAAATAATGATTTTCACTCCAAG AGTGTGTAAACCTTGGGAGGGAAGAGAAGACAGGAAAGAGACCCC AGAAAAGAACAAAAGAAAGTCTAAATTTGATGTACGGATTAAATAGCG TGTTTGAAAACCCATTGAGAAGTAAGAAAATCTCATCTGAGATATG AAAGTTATGACTATTAACCTTCTTAATAAAGCTCAAGAAATCATCT GAGATGTGAACCAACATGCTCTTACGAATTGTAGAAATTTTGAT GGTTTTTCTCTGGGGTACTCTTTCTATATTTTGAGGCCAATTATT ACTAATTTCTCTAGGAATTATTCGATCAATTGCTCCATCTCTGCAT TTAATTCTTTCCATGCATGAATCACGAATATGGTTTTGTTAATGT TTGTTGGCATGTTAATTAATTTTATCCTAATTAGTTCTCTGAGAAA CCTGAACAATATTTCCCTTATTGGTTATATTTCTGGAGCTAAGTG GAAGCTGCTAGTGCTATTAGCCATCTTAGAAAACCCACAAGCCATG CTCACTATTTGTGGGACGGAACGTTTTCTTTCTCTATTTTATATGG TTGGAGAGCATAACGTAATTTTATTGACCGAACAATAGGAAGGAAT CAAAGACTCCGTGACCGTGTGTCAACTCTTTTGATATTTATTATC ATCCAATGGGCCCTACCACCAAAATGTTTAATGTCAATTTTCAAGG GGCAAAGTTGTAATAATGATGTAACATTTCCGTGATAGTAATTTAG ACATAGCTTTCCGTTCTGTAGAATATATATATCTCTTTGGGGGCT TGTTGTGTAAGCTTTGCTAATTTTGTATATCTTTTATTCAAGTTAGA TAGGGAGAAGAGGTCTTAGTGACAAACATTCATATGGATAGCATA TATAGCATATGGGGTTCTCTTCAAGTGGCTTTCAATTCCCATTTA AGTTCAAATTTTCCACTTAAATGAACAAAAATAACATAGCATGAG GTTCTTGTGTTGTTTTGTCATCTTGTATTTTCACTAATCATTTGGC TACCTTGGCTCCTACCTGTCAATGAAGGCATAGATGCACATCACT TTCACCAATAAATAACCAACATGCAACCCCTTTAGGAAAATTTTC TCTTCAGTCTATTACCAATATGGACAAAACCTGACTAGACTATGCA ACCAACCACAGTCATATGTAGAATTTCTCGTTGGGGCCTTTGCACA TCGATTTATATTTGTTTCTCTTGATAAAGAAGGAAGGGGTGTTG TGGCAGATACATTTATATCATTTGGCATGTCTTTCACTTTTGAATG CCCCTACATCTACAAGCCAAAGCTCAATTGATTAGAGAATATTA

-continued

SEQUENCES

GCTTTTGAATTTCTTTTCTGTTCTAAGTGATCAGATCAGACTCATG
TCTATCAACAAAAAGAGGAATTGGATTTCAGATTTCAAACCCATCAT
AATGAAGAAAAAATAAAAAAATAAATATAGAGAATCACGTTTC
CTAGATTGTTTATAAAGAAGAGGCTTTGTTAAGACTATTCTCAGGC
TGTGTTTTAGTAGTTTATGGTAACATGTTGTTTACAATGCTTATA
TGTCACCACTCATTGCAGGATGAGATTTTCAGGCCGATTAGTGCTC
GAATTTTCGAACTTTGCAGCCCGATTTCCTCCACACACACTGCA
AAATTCAGAGGTACCTCCAGCTCAAATGTTGCCATGAAGAGAAG
TCCTCATATGCCACAACCATATCTCCACCTTTAGATGTAGTAGACA
ACAATAAGTTCATATCAATAGCAATAGCAGCAACATAGTCACCAC
TACCTCATCTAGCACTACCACAACCAGCACCACAACCAACAAC
AACAACGCAACGAACGGCAATAATCTTTCTATCTCTTTGCACTC
AAGATGAAATTGACAATGACATCTCAGCCTCCATAGACTTCTCATC
ATCTCCATCTTTTGTGCTTCCACACTTCTCCCAATCTCAACTCAG
CAGGATCAGTTGATTTCCCTTCAGCTCAGCCACAGGTGCAACTAT
CAACAGCAGCAGGTTCAATTTTGACGGGCTCTCTCACTACCTTAC
AGATCCTGTGATTGCACCCCTTATTGGAGCTCCGTTACCATCTGTT
TTGATGATGATTGCATATCTTCCATCCCTTCTTATGTGCTCTCA
ACCTTCATCACCCCTCTTGCTCTTATCTCAGTCTGGCATAGGAGT
GTACATGCCACCTCCTGGTTCCCTTAACACTGCCTTATCTGCTGAC
AGTTCTGGATTGTTTGGTGGGAACATTCTACTGGGGTCTGAAGTGC
AGGCACATGAATTGGACTACCAGGGAGAAAAATGGTGAATGTATTG
TACAGATTCAATCAAAGGGTGTTAACCTCCGAGACCTTCAGGTA
TGTGCAATTTGCAAGCAATTAGAGTTTAATAGACATTCATTGTC
TGGTATAAAGTTTTTACATTATCAATCAATCAGATACCATGTTG
ATATAAATTTTAAAAAATTTGTTATAATAATTAATAATTTAATGTA
CTTGATAATTTGTGATTTGATAATAATATAAAAAAATTTACACTG
CATTATATTTATTTTCTGTCGACTGTCAATAAACTAAATGAAA
ATTTTCAGTTCCAATGTTCAATGTGTTTGAATAAGGAAAAAGAA
TTTAATAATGCTGCAAGGTTACTATACCTTGCAGCAGTGAAGTTT
TTATTTTAAAAATAGAAGAGGCTTTATCAGAGGTGGACTTTTGGGG
AAAGCTCAGGGTCCACAAATCTCTAACTATAAACTCATAGGTGCC
CCATGACCATCAAATAGTAGGTAGCACAGATATGAGTCCATTTAT
AAAGTCACATGCATTAATAAATACTATAAATTTGGCTAGCAAGAA
GGAAGAACCCTTTTATCCAAAAGAAAAATAGAAAAAAGGATAATA
AACTGTAGCATCATTAGATAGAAAGACCACTTCAAGGGTGGCAGT
GTTATATCTCTTTCTACAGTCTATAAAGTTAATGTGACAGTTTAT
TGAATAAGTAAGAAATGATCTTTAATTATAATTTCTCTCTCAGGC
ACTTGGTAATGAGAGTCAGAACTTGTAGCTGGGGCTGGAAGCTCT
GCCACTTTGGCACCAGAAATCTCACACTTGAGGACTCTACCTTGA
AGGTTGGAAGAACTCTCTGTTGAGCAGAGGAAGGAAAGATTATAG
ATACATGAAGAAGAGAAACGAAAGAAATTTGAGCAAGAAATCAAG
GTACTACATCTGAACACCAACATTAAACAAACAAATTTCAATCTTA
TACTGTTTTACATGATTTCCAATCTACTGCATCAACCAAGCCTTAT
GCATATTTTCAAAATTCAACTAATGATGCAATTTTTTTTATATAAA
AAAAATGCAGTATGCTTGC CGCAAAATTAGAGAGAGGGTTGGAGTA
GCGCCTATTGTAGAGAAGATGGTGGAAATAGACTTAGGTGGTTTG
GGCATGTAGAGAGAAGACCGGTAGACTCTGTAGTGAGGAGAGTAGA
CCAGATGGAGAGAAGACAAACAAATTCGAGGCAGAGGAAGACCCAAA
AAGACTATAAGAGAGGTTATAAAAAAGGATCTCGAAATTAATGGTT
TGGATAGAAGTATGGTACTTGATAGAACATTATGGCGGAAGTTGAT
CCATGTAGCCGACCCACCTAGTGGGATAAGGCGTTGTTGTTGTTG
TTGTTGTTGATGCTTGCCGCAAACTTTGGCGGATAGCCGGCCCC
GGGTTAGAGGAAGGTTTGCAAGAATGATGACTTTGGAGAGAGCCA
TAAACAAGGAAGTAGCAATCATGAAGATGATGATGAAGAGGTAAGA
TTCCCTTAATCGGATACTGTTGTTCAACTTGCCTTAGTCTAAAAAT
TAAAAATACAAAAAATTCCTGATCACTTTTACCTTTTCAATATTT
GATGGCATAATTCCTTGATGTTATATTCCTTCCATTTTTTGTACTT
GCAGATAATTTGAAAGAAGATGATGATATGGTTGATTCTTCAGAT
ATCTTTGCACATATCAGCGGAGTGAACCTTTCAAATGCAACTATT
CCATCCAGTCCTTGATTTGAATTAATTTATAGTTTGACTAGTGAA
AGCTTATTTATATAATTAGCTTCTGTAGATTAAATTTGGTAGGACA
CTTTTCCCATCCCGGTTCTCTAAAATCCGGGTTTAGTGGTTGAGT
AAACTGAATAAATGGGGTCAAAATAAATATACCAATAAGTTAAGTG
AGTTAGAAACGTACAGAAATGGAACGTATACATTTTTTGAGAT
ATATATTATCTTTTCAATTAAGTTGTACCAGAACATGGAGTTGTGT
TAACCAAGAAAAATTCAGTTACCCCATCCAAGACTGATGTAACC
AATTGATGTAGCTTCTTTTAT

-continued

SEQUENCES			
5	WT >POWR2_ CCT34_Glyma. 10G134400_ protein	Protein	MLQDVIHPSTPAEQLPIDEISSPISARIFELCEPDFFPDTLQNSDV TSSSNCCHEEKSSYATTISPPDLVDNKNINNNNSNIVTTSSSTT TTSTTTNNNNNTTNSNLSILFDTQDEIDNDISASIDFSSCRSLV VPPLLSISTQQQDFDSSAQPVQLSAAAGSVLKGLSHYPTDHWIA PLIGSPLPSVDFEDCISSIPSYVPLNPSSPSCSYLSPGIGVYMPPP GSLNTALSADSSGLFGGNILLGSELQAHELDYQGENGGIFCTDSIQ RVFNPPDLQALGTETQKLVAGAGSSATLTPEISHLEDSNLKVGKLS VEQRKEKIHRYMKRNRNRFSSKKIKYACRKTLDASRPVRVGRFAKN DEFGESHRRQSSNHEEDDEEIVKEDDDMVDSSDIFAHISGVNSFK CNYSIQSLI
6	WT >POWR2_CCT34_ Glyma. 10G134400_ CDS	DNA	ATGTTGCAGGATGTTATCCACCCCTCAACACCGGCTGAGCAACTCC CCATTGATGAGATTTCAAGCCCGATTAGTGCTCGAATTTTCGAAC TTGCGAGCCTGATTCTTCCCAGACACTGCAAAATTCAGATGTT ACTTCCAGCTCAAATGTTGCCATGAAGAGAAGTCCTCATATGCCA CAACCATACTCCACCTTAGATTTAGTAGACAACAAGATCAATAT CAATAACAATAGCAACATAGTCACTACTACCTCATCTAGCACTACC ACAACCAGCACCACAACCAACAACAACAACAACAACAACGAACA GCAATAACCTGTCCATCCTCTTGACACTCAAGATGAAATTGACAA TGACATCTCAGCCTCCATAGACTTCTCATCATGTGATCTTTAGTT GTTCCACCACTTCTCTCAATCTCAACTCAGCAGGATCAGTTTGATT TCTCTTCAGCTCAGCCACAGGTGCAACTATCAGCAGCAGCAGGTT AGTTTGAAGGCCTCTCTCACTACCCACAGATCATGTGATTGCA CCCCCTATTGGATCTCCGTTACCATCTGTTTTTGATGAAGATTGCA TATCTTCCATCCCTTCTTATGTGCCTCTCAACCCATCATCACCTC TTGCTCTTATCTCAGTCTGGCATAGGAGTGACATGCCTCCTCCT GGTCCCTTAACACTGCCTTATCTGCTGACAGTTCTGGATTGTTTG GTGGGAACATTCTACTGGGGTCTGAACTGCAGGCACATGAATTGGA CTATCAGGGAGAAAAATGGTGGAATATTTTGACAGATTCAATTGAG AGGGTGTTTAACCCCCAGATCTTCAGGCACTTGGTACTGAGACTC AGAACTTGTAGCTGGGGCTGGAAGTTCTGCCACTTTGACACCAGA AATCTCACACTTGGAGGACTCTAACTTGAAAGTTGAAAACTCTCT GTTGAGCAGAGGAAGGAAAAGATTCATAGATACATGAAGAAGAGAA ATGAAAGAAATTTAGCAAGAAAAATCAAGTATGCTTGCCGCAAAAC TTTAGCAGATAGCCGGCCCCGGGTTAGAGGAAGTTTGCAAGAAT GATGAGTTTGGAGAGAGCCATAGACAAGGAAGTAGCAATCATGAAG AAGATGATGAAGAAATAATTGTGAAGAAAGATGATGATATGGTTGA TTCTCTCAGATATCTTTGCACATATCAGTGGAGTGAACCTTTTCAA TGCAACTATTCCATCCAGTCTCTGATTTGA
8	cct34-2-2 Shown in FIG. 12D	DNA	ATGTTGCAGGATGTTATCCACCCACACCGGCTGAGC
9	cct34-2-2 Shown in FIG. 12D	DNA	ATCATGTGATTGCACCCCTTATCTCCGTTACC
10	cct34-2-2 Shown in FIG. 12D	DNA	CTCAGTCTGGCATAGGAGTGACATGCCTTCTGGTCCCTTAAC ACTG
11	cct34-4-5 Shown in FIG. 12D	DNA	ATGTTGCAGGATGTTATCCACCCACCGGCTGAGC
12	cct34-4-5 Shown in FIG. 12D	DNA	ATCATGTGATTGCACCCCTTATCTCCGTTACC
13	cct34-4-5 Shown in FIG. 12D	DNA	CTCAGATGTTACTTCTGGTTCCTTAACACTG
14	cct34-4-7 Shown in FIG. 12D	DNA	ATGTTGCAGGATGTTATCCACCTCAACACCGGCTGAGC
15	cct34-4-7 Shown in FIG. 12D	DNA	ATCATGTGATTGCACCCCTTATCTCCGTTACC

-continued

SEQUENCES			
16	cct34-4-7 Shown in FIG. 12D	DNA	CTCAGTCCTGGCATAGGAGTGACATGTCTGGTTCCCTTAACACT G
17	gRNA1	DNA	CCACCCCTCAACACCGGCTGAGC
18	gRNA2	DNA	CCCCTTATTGGATCTCCGTTACC
19	gRNA3	DNA	GGAGTGATACATGCCTCCTCCTGG
20	PCR primer	DNA	TATCCATATGACGTTCAGATTACGCC
21	PCR primer	DNA	ACCTCAGAATTTTGCAGTGTGTGTG
22	PCR primer	DNA	CTGCTGTTGATGGAGGACTT
23	PCR primer	DNA	CTCCTGGAGAAGCAGAAGTT
24	A 17-bp target site duplication upon insertion of LINE transposon (in POWR1+TE	DNA	GTATGCTTGCCGCAAAA
25	WT Protein sequence of POWR3 >G.maxWm82. a2.v1 Glyma. 10G190300. 1.p	Protein	MLQDVFDQPLPPKAEPQQQHP IWLQDEISSPLSARIFELCNTDFFP ETLPNSEVTSSNCCYEENSSYATTNISVTVDVDNKLNSNSNTVTT PTSTTTTNTNNNTTNSNLSIIFDSQEEIDNDISASIDFSLSPSPFN VPPFLPATSQQEQDFSSVQHQVQLAACSVVEGFSQYPNDVSVAPPF MGAPLPSVFEEDCISSVPSYMALNPSSPCTYLSPGMPYPMPHGPLT TALSTGSSGIFGGNILLGSELQAQLEYQGNGRMYCTDSIQRVFN PPDLQALGTESQQLVPGAGSSASLPEISNLEDSSFKVGKLSVEQRK EKINRYMKRNERNFSSKKIKYACRKTLDASRPVRGRFAKNDDFGE TNRTTSSNHEEDDEEVVVKDEDDMVDSSDIFAHISGVNSFKCNYS PQSWI*
26	WT >G.maxWm82. a2.v1 Glyma. 10G190300 Chr10: 42314396.. 42318797 reverse key: 5'UTR CDS 3'UTR CDS of POWR3	DNA	GTCAACCCAAAATCCAACCTCAAGACAAGGAGAGTGGCATTGGCAGA TGGCACCCGAGAGCCAAAGAGGGTTTGAGTGAATGAATGGCCCACC TTATGTCTCGCTTTGACGCCAGCTTGACGCCCTTAGGGAGAACCCC TTTGCCCTATATATGAACATAAATAATCTTTGAGTCTTTGACAT CCTCATACTCTCCATCATCTTAGTAACTACTCATCATCATATTCA TAACTTTCCCCCTCTTTCTCTCTCTTTTCCCTATGGAGTGTCTT TGAGCACATGTTTTAAGTCACCAACACACGTGTGAATTTCCACAT TTTGCACTCCTATATAGTACTCTCTTTCTTTCTAGATATAGCTTTA ATTTTGTGTACATTTTTTTTGTATTAAAGTTTCTGGGGTGCCTTGA TTGCCCTCGTGCAAAGGGAACATACATACATACACCATGTTCGAGG ATGTTTTTGATCAGCCACTGCCACCAAGGCCGAACCAACAGCA ACATCCCATTGTAATCACCTCATCTTGCCAGCAGTCTCAATATTCA TCTATTATGTTAATGCATACATGCATGTGTTCTTCTACCTTAATG ACATAGTGTAATCCTAACTCAATATTAATTTTCATGAAAACCTTACA AGTTTGTTAATTGTTTCCATTTTGAACGAATCTCTTCTCATATG TTTGAATATATTCAAATCTAAGGCTAAGTTGTTGTTTACAGTTGT CTCGTGATATTGCCCCACTTGTGATCTCTCTTTTCTTTTTCATGCAT GCATGCGTGAATGATATATATATATATATATATATATATATATATT ATTGTTTGTGTTTTATGTTTGTGCAAGTTTGATGATATTATTGCC AGTGCTAATAGAATTAAAGCCATTCTCAAATCTTTTCCAGGCTTC CCTGTCAATTTGCATGGAAGAACGGGTCTTTTCTCTCAATTTATA TAGTTGGGGAGCATAATGCTTATTGACTAGCAATAAGAAAGAACGT ATTAGAGATTCCGCAATAACTATTAGACTTTGTCTCTTTTATATA TTATCCTATCCTAACCAAGACAATGGCATTCAACACAATGTCATT TCTGAGGGCAACATTGTAACTGCAAAGTGATATCAATTTCTACAT GATTTCCAGTGTGATAGCTAGCCATTTAAACGTAGCTTTTCGATTAT TGTGGAATATATATCTTCTGTGTTTGTCTATTGCCAATTTTGAT ATTCTTATTCAATTTAGAGAGGGGAGAGGAGACTTGAGGAACACTTC ACATGTACAGCATTATGTGTTCTTTAAGTGCCTTTTAATTCCTAT TAAATCAGCTTCCATACACTGAACAAACACAACGTAGCATGGCATG AGATTCCTTCTTTGTTACAAACATACAAGTAACATTGTCACATTAA GGCATAAATGCCCATCTTTTCAATTTCTTAAAAAGCCAACCTTC

-continued

SEQUENCES

TTGTTGAATTTGTTATCTGTACATGACAAAATATAAACACAACCTGA
CTAGAGGACACAGCCAACCAAATATGTATTTGAAGTTCCTCATGAT
GGGGTATTCTACATTGATTATTTTCGTTGATTTCTAGACAAAGGGT
TTGTGTAGTGATGAACATAACTGATGCTTATAGGCGTGTTACTTAC
TCACTTTAGAATGCTCACTATTTGATGAAGACATGCAATAAACCAA
AGTTTAGACTGATAATAACAAGAGAATATTATCTTTTGAAAGTTTA
GACTCTGATCATTATAGGCATGGTCCCAATTTCTATTTTGCTGATA
GAAACGAGTCTGATAAGATCATATGGCCATCCCATAGGGAAAAATA
AATGGTGTAGAATCAGTCTTCTACCTGTTTATAAAGGGTCTATCAA
AATTCAAAATGTTATATCAATTTGTTTAGTAGTTATTTTGGTAG
TCATGATGCTTACATTGTTTTATATCAGTGGCTGCAGGATGAAAT
TTCAAGCCCTCTAGTGCTCGAATTTTCGAACTTTGTAACACTGAT
TTTTTCCCAGAGACTTTGCCAAATTCAGGGTACTTCCAGCTCAA
ACTGTTGTTATGAGGAGAACTCCTCATATGCCACAACAAATATATC
TGTAAGTGTAGATGTAGACAATAAGTTGAATAGCAATAGCAACACA
GTCACCACCCCACTAGCACTACCAACCAACCAACCAACCAACA
CAACCAACAGTAGTAACCTGTCTATTATCTTTGACTCCCAAGAAGA
AATTGACAATGACATCTCTGCCTCCATAGACTTCTCACTATCTCCA
TCTTTCAATGTTCCCCATTTCTCCAGCCACATCCAGCAAGAAC
AGTTTGATTCTCTCTGTGCAGCATCAGGTCCAACCTAGCAGCATG
TTCAGTTGTGGAGGGCTTCTCTCAGTATCCCAATGACTCTGTTGCA
CCTCCTTTTATGGGGCTCCATTACCTCTGTTTTTGAGAGGACT
GCATATCTCTGTCCCTCTTATATGGCTCTTAACCCCTCATCTCC
TTGCACCTTATCTCAGTCTGGCATGCCACCATACATGCCTCATGGC
CCCCTTACTACTGCCTTATCAACTGGTAGTTCTGGAATATTTGGTG
GGAACATTCTTCTGGTCTGAACTTCAGGCACAAGAATTGGAATA
CCAAGGAGAAAATGGAAGAATGATTTGTACAGATTCAATTCAGCGG
GTGTTTAACCTCCAGACCTTCAGGTATGGATGATTTTCAAACATAT
TTTGTTAAGTCATCATTGCCTTTTGTAGTTTGTGTTAACAATACCAT
CAAAATGAAGTGTTCAAGTTCAAATGTTCAATATGCTCAGTAATGAG
AGAAAAGAAGTCCATTGTCCAAGGATTTTTTATTTCTAATGATG
CTAAGCTGTTGTTGAAGCTTGCTCTTTTCAGGTTTCTAGTCTGAAA
GGTTGCAGCAGTAAGGTTCTTGACAATAAAGAGTGACTTTACAAGA
GGTGAAAAGTTTGCAAAAGATAACGGCCCACAAATTTCTAAGAAGT
TTATATGCCCCATAGAATATATGCCCATATCCATCAAACCAGTAGC
TAGCATATCAACCCCTTTTATAATCTCATTCTCACCTGTAAAAAGT
AGCATAAAATCTGAGCTAGCTAGTAGCAGGAAGGAATCACAACTTT
CCACAAAAATGATAACAATCATCATCAGAAAAGACCCACTTCATG
AGTGGTTCTTTTGCATATCGCTTTCCGGAACCTCTAACTAAAGCT
AATGACAGTTTGTATGACCAATTTTCCTTTGCGTAGCAAGAAAAGT
AACTGATATTGTTGTGATGTCTTTCTAATTCCTAGGCTCTTGGC
ACTGAGAGTCAGCAACTAGTACCAGGGGCTGGAAGTTCTGCCAGTC
TACCAGAAATCTCAAACCTTAGAGGACTCCTCCTTCAAGGTTGAAAA
ACTTCTGTTGAGCAAAGGAAAGAAAAGATCAATAGATACATGAAG
AAAAGAAATGAAAGAACTTCAGCAAGAAAATTAAGGTACACTCAA
GCTTCTTCCCATCTTCAAACAGCCATATCATGTTACATATGCATA
TGCTTATAAAACAACCACTTTTATTTGCATACTTTCAAGTTCAACTA
ATGATGAAACCAAATATGCAGTATGCCTGCCGCAAACTCTGGCAG
ATAGCCGACCTCGAGTCAGAGGAAGGTTTGCAAAGAATGATGACTT
TGGGGAAACCAATAGAACTACAAGTAGCAATCATGAAGAAGATGAT
GAAGAAGAAGTAAGATTTCCAGGGCTCATTCAATTTATTAACACTT
AACAGGACATTTTATTCAACTTGCCCAAGTAAAAAATAAATC
CCTATCACTTGTACCTTGTCAATTATTTGCTGATATAGTTTTTAA
TCCTGCAGGTAGTTGTGAAGATGAAGATGATATGGTTGATTCCTC
AGATATCTTTGCTCATATCAGTGGGGTGAACCTCCTTCAAATGCAAC
TATTCCTTCCAGTCTGGATTTGACTTAATAGTTTGTATCAGTATAC
CTACATTTAGCTTGTGTAGATTTTGTGATGGCCATAGACCACCC
AGTTTTCTCATATCTGGTTTTGATGGATGAGGAACTGAATAATTA
GGGTCTTAATACTATACAATAAGTGAATGAAGAACATTTAGAAAAGA
AACTGTATACATTTTTTGAGACACTATTTCTTACTACTTAAATGT
ATTAAGAATCAGATGTTGTATTAACCTTTGTCTCCATGTACTAGTGG
AGCTCATTATGTAAGTCTCTTTTATAATATAGTGATGAGGAAGTAG
TTTTGGTTACATTCAGTGCTTAAAGCTTAAAGCCCAAGGTTTAAA
ATATTTTACTGGCTTTGAACTTTTTTTTTT

-continued

SEQUENCES			
27	Mutant POWR3- Glyma. 10G190300- CDS sequence Mutant comprising a 4 bp (GGGC) deletion on exon 3	DNA	ATGTTGCAGGATGTTTTTGTATCAGCCACTGCCACCAAAGGCCGAAC CACAACAGCAACATCCCATTGGCTGCAGGATGAAATTTCAAGCCC TCTTAGTGCTCGAATTTTCGAACCTTGTAACTGATTTTTTCCCA GAGACTTTGCCAAATCTCGAGGTACTTCCAGCTCAAAGTGTGTT ATGAGGAGAACTCCTCATATGCCACAACAAATATATCTGTAAGTGT AGATGTAGACAATAAGTTGAATAGCAATAGCAACACAGTCACCACC CCAACTAGCACTACCACCACCACCAACAACAACAACAACAACA GTAGTAACCTGTCTATTATCTTTGACTCCCAAGAAGAAATTGACAA TGACATCTCTGCCTCCATAGACTTCTCACTATCTCCATCTTCAAT GTTCCCCCATTTCTCCAGCCACATCCAGCAAGAACAGTTTGATT TCTCTTCTGTGCAGCATCAGGTCCAACTAGCAGCATGTTGAGTTGT GGAGGGCTTCTCTCAGTATCCCAATGACTCTGTTGACCTCTCTTT ATGGGGGCTCCATTACCTTCTGTTTTTGAAGAGGACTGCATATCTT CTGTCCCTTCTTATATGGCTCTTAACCCCTCATCTCCTTGCACCTTA TCTCAGTCTCGCATGCCACCATACATGCCTCATGGCCCCCTTACT ACTGCCTTATCAACTGGTAGTTCTGGAATATTGTTGGGAACATTCT TTCTTGGTCTGAACCTCAGGCACAAGAATTGGAATACCAAGGAGA AAATGGAAGAATGTATTGTACAGATTCAATTGAGCGGGTGTAAAC CCTCCAGACCTTCAGGCTCTTGGCACTGAGAGTCAGCAACTAGTAC CAGGGGCTGGAAGTCTGCGCAGTCTACCAGAAATCTCAAAGTTAGA GGACTCTCTCTCAAGGTTGGAACCTTCTGTTGAGCAAGGAAA GAAAAGATCAATAGATACATGAAGAAAAGAAATGAAAGAACTTCA GCAAGAAAATTAAGTATGCCTGCCGCAAACTCTGGCAGATAGCCG ACCTCGAGTCAGAGGAAGGTTTGCAAGAATGATGACTTTGGGGAA ACCAATAGAACTACAAGTAGCAATCATGAAGAAGATGATGAAGAAG AAGTAGTTGTGAAAGATGAAGATGATATGGTTGATTCTCAGATAT CTTTGCTCATATCAGTGGGGTGAACCTCTTCAAATGCAACTATTCC TTCCAGTCCTGGATTGTA
28	WT Protein sequence of POWR4- Glyma. 20G200400 >G.maxWm82. a2.v1 Glyma. 20G200400. 1.p	Protein	MIPISLLLIETSLIRSHSPRKGKQMDIEISSPLSARIFELCNTDFP PEALQNSEVTSSNCCYEENS SYATNISVTVDVNDKLNNSNTVT TPTSTTTNTNNTNNSNLSIFPDSQEEIDNDISASIDFSLSPSF NVPPFLPVSSQQEQDFSSVQHQVQLAACSVVEGFTQYPTDAVAPP FMGAPLPVFEEDCISVPSYMLNPFSSPCTYLSPGMPYPMPHGL TTALSTDSSGIFGNNILLGSELQTQLELYQGENGRMYCTDSIQRVF NPPDLQALGTESQQLVPGAGGSGTLTPEISNLEDSFKVGLKLSVEQ RKEKINRYMKRNRNRFKKIKYACRKTLDSPRVRGRFAKNDDF GETNRTTSSNHEEDDEEVVMKDEDDMDVSSDIFAHISGLNSFKCN YSIQSWI*
29	WT CDS of POWR4 >G.maxWm82. a2.v1 Glyma. 20G200400 Chr20: 43781164.. 43785175 forward key: 5'UTR CDS 3'UTR	DNA	ACACCCCTCATACTCTCCATCATCCTAGCTAACTCATCATCATATTC ATAACCTCCCCCCTCTCTCTCTCTCTCTCTCTCTCTCTCTCTCT GAGTGCCTTGAGCAGATGTTTTTAGTCACCAACACACAGTGTGAAT TTTCCACATTTTGCACTTCCATATAGCTCTTTCTGGGGTGCCCTT GATTGCCTCGTGCAAGGGGATACATACACACACACACACACCATGT TGCAGGATGTTTTGTATCAGCCACTGCCACCAAGGCCGAACCCACA ACAGCAACTTCCCCTTGAATAATCTCCAGCCAGCGGTCTCATTCA TATCTTCCATGTTAATGCATGCATGTGTTCTGCCTACCTTAATTAA TTAGTTAATTGTGTCATGACTCATGACATGGTGTGATCTTAACCC AAAATTATGAAACTTTACAAGTTGTTGTTTCCATTTTGGAAATC TCTTCTCAGGAATATATTCAAGTCTAAGGCTAAGTTGTTTACAAT TGCTCTCGTGATATTGGACCACTTGATGATTCATTTTTCATGCATG CATGAATGATATATATGGTTGTTTTATTTTTATGTTTTTTTGAAG TTTGATATATATTTCCAGTGCTAATAGAATTAAAGCCATCTCTCA AATTCTTTTCCAGGCTTCTCTGGCAATTGCAAGGAGGAACGGGT ATTTCTCTCAATTTATAGAGTTGGGGGGGGAGGGGGGGGGGGCA TAATGCTTATTGACTAGCAATAAGAAAGAACGTATTAGAGATTCCG CAATAATTATTAGTCTTTGTCTCTTTTCTCTCTATATTATCCTA TCCAATCCAAGACAATGCCATTTACCAAGATGTCATTTTCTGAGG GCAACATTGTAAGTCAAGTATATCAATTTCTACATGATTTTCA GTGTGATAGCTAGCCATTTAAACATAGCTTTCGATTGTTGTGGAAT ATATATCTTTCTGTGGTTGCCATTTTGCCAATTTTGATATTCTTAT TCATTTAGAGAGGGGAGAGGAGACTTGAGGAACACTTCACATATAT AGCATTATGTGTTCTTTAAGTGCCTTTTAATTTAATTCCTATTAA ATCAGTTTCCATACCTAAACAAACACACAGTAGCATGGCATGAGG TTCTTCTTGTGTACAAACATACAAGTAACATGTCACGCTAAGGC ATAAATGCCCATCTTTTCAATTTCTTAAAAAACCACCCCTCTTG TAGAATTTATCTGTACATGACCAAAATATAAGCACAACTGATAGAG ACACAACCAACCTAATCTGTATTGAGGTTCTCATGATGGGGTAT TGTACATTGATTATTTGTTGATTCTAGACAAAGGGCTTGTAGT GATGAACATAACTGATGCTTATAGGCGTGTACTCACTATTGATG AAGACACGCAATAAACCAAGTTTAGACTGATAATAACAAGAGAAT ATTATTATCTTTGAAAGTTTAGACTCTGATCATATTAGGCATGAT

-continued

SEQUENCES

			CCCAATTTCTCTTTTGCTGATAGAAACGAGTCTGATAAGATCACAC AGCCATCCCCGTAAGGGAAAACAAATGGTGTAGAATCAGTCTTCTA CCTGTTTATAAAGGGTCTATCAAAATGTTATCAATATGTGTCTAAT AGTTTTTTTGGTAGTCATGATGCTTACATTGTTTTTCATATCACTG GCTGCAGGATGAAATTTCAAGCCCTCTTAGTGCTCGAATCTTCGAA CTTTGTAACACTGATTTTTTCCAGAGGCTTGCAGAATTCTGAGG TTACTTCCAGCTCAAAGTGTGTACGAGGAGAACTCCTCATATGC CACAACAAATATATCTGTAACGTGTAGATGTAGACAATAAGTTGAAT AGCAATAGCAACACAGTCACCACCCCACTAGCACTACCACCACCA ACACCAACAACAACAACCAACAGTAGTAACCTGTCTATTTCTT TGACTCCCAAGAAGAGATTGACAATGACATTTCTGCCTCCATAGAC TTCTCACTCTCTCCATCTTTCAATGTTCCCCCATTCTCCCAGTCT CATCCAGCAAGAACAGTTTGATTTCTCTTCTGTGCAGCATCAGT CCAACTAGCAGCATGTTGCGTTGTGGAGGGCTTCACTCAGTATCCC ACTGACGCTGTTGCACCTCCTTTTATGGGGCTCCATTACCTTCTG TTTTGAAGAGGACTGCATATCTCAGTCCCTTCTTATATGCCTCT TAACCCCTCATCTCCTTGCACTTATCTCAGTCTGGCATGCCACCA TACATGCCCTCATGGCCCCCTTACTACTGCCCTTATCACTGATAGT CTGGAATATTGGTGGGAACATTTCTTGGTCTGAACTTCAGAC ACAAGAATTGGAATACCAAGGAGAAAATGGAAGAATGTATTGTACA GATTCAATTCAGCGGGTGTAAACCTCCAGACCTTCAGGTATGAA TGATTTTCAAATATTTAGTTAAGTTCTCCATCATTGCCTTTGTTA ACAATACCATCCTAATGAAAAGTGTTCGGTTCGTTCAATATGCTCA GTAACGAGAGAAAAAGAAGTCTTGTCCAAGGATTTTTTTTTTTTA TTCTTATGATGCTAAGCTGTGTGTGTAGCTTGCTCTTTCAGGTT TCTAGTCTGAAAGGTGCAAGCAGTAAGGTTCTTGACTATAAAGAGT GGCTTTACAAGAGGTGAAAAATCTGCAAAAGATAACGGTCCACAAA TTCTTAAGAAGCTGATATGCCCATAGAAATATATGCCATATCCAT CCATACAGTAGCTAGCATATCAACCCCTTTTATAATCTCATTCTCA CCTGTAAAAGTAGCATAAAACTGAGCTAGCTAGTAGCAGGAAGG AATCACAACATTCACAAAAATGATAACGAACAACATCATCATCA GAAAAGACCCACTTCATGACAGTTGTGTATGATCAATTTTCCTTG CGTAGCAAGAAAGTAAGTATTTGTTTGCAATTTCTTTCTAGGC ACTTGGTACTGAGAGCCAGCAACTAGTACCGGGGCTGGTGGTTCT GGCACCTTAACACCAGAAATCTCAAACTTAGAGGACTCCTCCTTCA AGGTTGGAAGCTTTCTGTGAGCAAGGAAAGAAAGATCAATAG ATACATGAAGAAAGAAATGAAAGAAATTCAGCAAGAAATTAAG GTACAGTCAAGCTCCTTCCATCTTCAAAACAGCCACATCATCAAA CAAAATTGAAATCTTATCATGTTACATATGCATATGCAAGGTATTC CTGTCTTAAACAACCACTTGTATTGTCATACTTTCAAGTCAACT TAATGATGCAACCAATATGCAGTATGCCTGCCGCAAACTCTGGC AGATAGCCGACCTCGAGTTAGAGGAAGGTTGCAAGAAATGATGAC TTTGGAGAAACCAATAGAAGTACTAGTAGCAATCATGAAGAAGATG ATGAAGAAGAAGTAAGATTCCAGGGTTCAATTCATTATTAAACAC TTAACAGGACGCTTTTATTTCAACTATTTGTTGATATAGTTTTTTA ACCCTGCAGGTAGTTATGAAAGATGAAGATGATATGGTTGATTCCT CAGATATCTTTGCTCATATCAGTGGCTGAACTCCTTCAATGCAAA CTATTCCATCCAGTCTCGATTGACTTAATAGTTTGATCAGTATA CCTACATTTAGCCTCTGTAGATTTTTCATGGCCATATACACCCC CAGATTTCTCCTATCTGTTTTGATGGATGAGGAACTGAATAATT GGGGTCCTAATACTATACAATAAGTGAATGAAAAATATATAGAAAA AAAACTGTATACATTTTTCAGACACTACCTTTCTTACTACTTGT ATGTACTAAGAATCACATGTTGTATTAACTCAAACGATTTTCATTTG TCTCCATGTACTAGTGGAGCTCATTATGTACTCTCTTTTATAATA TAGTGATAA
30	Mutant CDS sequence of POWR4 protein	DNA	ATGATCCCAATTTCTCTTTTGCTGATAGAAACGAGTCTGATAAGAT CACACAGCCATCCCCGTAAGGGAAAACAAATGGATGAAATTTCAAG CCCTCTTAGTGCTCGAATCTTCGAACCTTGTAACACTGATTTTTTT CCAGAGGCTTTCGAGAATTCTGAGGTTACTTCCAGCTCAAACGTGT GTTACGAGGAGAACTCCTCATATGCCACAACAAATATATCTGTAAC TGTAGATGTAGACAATAAGTTGAATAGCAATAGCAACACAGTCACC ACCCAACTAGCACTACCACCACCAACACCAACAACAACAACCA ACAGTAGTAACCTGTCTATTTCTTTGACTCCCAAGAAGAGATTGA CAATGACATTTCTGCCTCCATAGACTTCTCACTCTCTCCATCTTTT AATGTTCCCCCATTTCTCCAGTCTCATCCGCAAGAACAGTTTG ATTTCTCTTCTGTGCAGCATCAGGTCCTAGCAGCATGTTGCGT TGTGGAGGGCTTCACTCAGTATCCCACTGACGCTGTGCACCTCCT TTTATGGGGCTCCATTACCTTCTGTTTTGAAGGAGCTGCATAT CTTAGTCCCTTCTTATATGCCTCTTAACCCCTCATCTCCTTGCAC TTATCTCAGTCTGGCATGCCACCATACATGCCTCATGGCCCCCTT ACTACTGCCTTATCAACTGATAGTTCTGGAATATTTGGTGGGAACA TTCTTCTTGGTTCTGAACCTCAGACACAAGAATTGGAATACCAAGG AGAAAATGGAAGAATGTATTGTACAGATTCAATTCAGCGGGTGT

-continued

SEQUENCES			
			AACCCTCCAGACCTTCAGGCACTTGGTACTGAGAGCCAGCAACTAG TACCCGGGGCTGGTGGTTCTGGCACTCTAACCCAGAAATCTCAAA CTTAGAGGACTCCTCCTCAAGGTTGGAAGAACTTTCTGTTGAGCAA AGGAAAGAAAAGATCAATAGATACATGAAGAAAAGAAATGAAAGAA ACTTCAGCAAGAAAATTAGTATGCCTGCCGAAAACCTCTGGCAGA TAGCCGACCTCGAGTTAGAGGAAGGTTTGCAAGAATGATGACTTT GGAGAAACCAATAGAACTACTAGTAGCAATCATGAAGAAGATGATG AAGAAGAAGTAGTTATGAAGATGAAGATGATATGTTGATTCTCTC AGATATCTTTGCTCATATCAGTGGGCTGAACCTCTTCAAATGCAAC TATTCATCCAGTCTTGATTTGA
3.1	AtPOWR1 full- length genomic DNA sequence	DNA	TATAATAAAATTGGTTACGATAACTTCCATAACCTTCATTAGTTTC TCCAGAAGAAACCTGTAAGAAATTTCTTCTCTTCTCTGCTTTTAA ACAGATTTTTATTCTGTTCTGTTTGATTAATTATTATCTCTTTTGT TCCAGTATTTTAATTCTTTAGATTTTCTTGTTTCCAGACATCACTAGC TCCACTACCACATTCAGGAAGCCTCCTTCAATAATCTTCCCTTTT CCTGGGGAATTTTCTCTGATTCATTCCCGGAAAATCATTCA ATTTGGAGACCATAAAGAACACATAAACATGTCGAGGAGCTAATT TCGTCTCATGAGCAGCTCTCCATGGTACGTAATAATCCGTGCAATG TTTATGTTTGTCTCGAATCTGTTTATGCCCCGAAATCTGATGTTT TCTTCTTTGGGTTTCTCTTTTCTATCATACATTGCTTAAATGATT AAAAGCATGAACCTTTTTTCTGAAAATTTTGCAATGATATATCT ATGATTTATTTCGAAGGAATAAAATTTTCTGAAATCTTTGTAT TTGTAGGTAACATCGTTTCTTAATTTTCTTAGTTATTTCGATTTT ATCTTTTGTCAAATGCTTCATTCTTTTACTTTCTTTTCTGTTTC TGGTTATTCTAGATACAATTCAGTTTCAATTTCCAAAAGTCTGAAAC GATAATTAGTAAAGCAAACTTTTAGTGCTAACTCTTTGTACGGT GTTTATACGTAGTTTACGTACCTTCTCGTAGCCAAACCAAAATCGT CACAACTTAAATCAGATAGAATAGAACAGTTTATCAGAATTCATA AGATGCTACATCTCTCTTTTAGTAATTCAGTAATTAATTTTGA ACTTTTATTATTAATCTTTCGAAATAGAAATCACAGAAATTTTATTAGA AATATATAATGAGTGTATCAAAAGCAATGATTCTTGCTCTTACCTC GGCCTTTATTGTTGTTAAAAAATCAGATGTTAGTTTAAATAGGG AACGTTTGGTTCTAAAGAAACGCAAGAAATCCAAAACCTTTCTGT TATGGAACCTCTCCGCTTATCGTTAATTTCTTCTAGATTCCC TCTGATGAAAACCATAGAAGGAACAGATCTAATCGTATAAAT ACATAGTTCTCTTAAACGACTGTGTTAAGTTAAAACTAACCTTAG GTTTATTCAAAGTTATTGTTTCTGTTTCAAGAAAATCCCAACA ATTAGCTTTTATGCAAGCCTTCAACGTATGCAATAAACAAACAA TTAGATGCATATATTACTTGTCTATTCTCTGCTAATTTGATCTT TCTCTGCTGTTGTTATTGTTGTTAGGATGAGATCACAAGTCTCTCA CCGCACAGATCTTCGATTTCTGCGATTACAGTTATTTCAGAAAC CTTCAATCAAACCTTCTGAAGTAACCTCTGCTTCAAACGGTTGTGGC TATGTCGAAAACAAACAACCAACAACCTTCCAGACAAATCCAACA GTGGTTTCAATCAAGATCATGAAGACAACAACGACAACGCTGATTT ATCAATCATATTTCGATTCAAGATGATTTTCGATAACGATATCACA GCTTCTATCGATTCTCTTCAATCCATCCAGTTCCAGCCTCCGATC AACTCCAAGAGCAGTTTGATTTTCAACCGAATTCACCTTCAACAGCC TCCGAACACTCTCTACTCTTCTCTCTGCTGACCTTCTGCTCCG CCTTTATCGGTTTTCGAAGAAGATTGCTTTCTTCTGTCGCAAGTT ACAATCTTGGTTCCATAAACCTTCTTCTCTCTTCTGCTCTTTT GGGTAAATCCGTTTACCGACTTACATGACCGTAACCGGAATATG ATGAACACCGTTTAGGATCTGTTTCTTCTCGGGAATATCCATC TCGGTTCTGATTTCAAACCATCATATGATCAATTGATGGAATTC AGCTGATAATGGTGGATTGTTCTGTCCAGATCCGATTAAACCCATC TTAATCCAGGAGATCATCATCTCCAGGTATATATTCAAACCTTA TATATAAATAATGAAAGGTATTGCTGGCAACAAATTTGTGAATGA AACCAGATTAGTAAACCGGTCTGATACTAACCGGTTATGGGGTTT TTTTCTTAATAGGACTTGACGGTGTGAGAACCAGAACCATG TGGCACAGCCGGTTCTCCGCGAGTTAGGGACGAGATAACCGGTT AGACGACCCGAGTTTCAACAAAGTCGGAACCTCTCGGCAGAGCAG AGAAAAGAGAAGATTATAGGTATATGAAGAAGAGGAACGAGAGGA ATTTCAAGCAAGAAAATCAAGGCTTTGTGCTTAAACATCATAACCG GATTAAACAGTGGTTATCCATATACCGATTGTTTCAATGATTTTT TTTGTATATTTTGTGTAGTATGATGATGAGGAAGAGTTGGCAGA TAGTCGTCCAAGAGTTAGAGGAAGATTGCAAGAATGATGAATTT GGTGAACCAATAGACAAGCCTGTTCAAGCCATCATGAAGATGATG ATGATGATGTAAGATTAATCTGTTTGAATTAACAAATGAAAAACAGA TTAGTTTTTGTGTTGATATAATAAGCGTTTCTAAACGGGGTGTGG TTTGCAGGTGGGGTAAAGGAAGAAGAACAGTTAGTTGATTCTTCG GACATATTTTCAACATAAGTGGTGTGAACCTTTCAAATGCAATT ATCCAATCCAGTCTTGGATTGAGTTGGATTACTTACTTGGTCTTA TGATTTACAAAATAAAACAAAATAGTATTGTTATAGAGGTCGT AATTGTATATTTATTAGTACATTTTTTGGCCATGACTCATAGTTT

-continued

SEQUENCES			
			GTCGAATAAGTAGAACATAAGAAAAGAGAGAGATGAATAATGGGTG TAGGCAGGTAATCATGAATAAAATTGTAGTGAGTGAGAGGGGTGT TTATGTTACATAGAACATTACAAAATTTGTAGATATATATTATAT GGTATGATAATTATGATGTCTATTAATGTATTGATGGGTGTTGTA TAAATAATAATCATTGAACATTATAAATAAGCAGTTGTGTTCAAG T
32	The AtPOWR1 full- length cDNA sequence	DNA	ATGTCGCAGGACGTAATTCGTCTCATGAGCAGCTCTCCATGGATG AGATCACAAAGTCTCTCACCGCACAGATCTTCGATTCTGCGATTCT ACAGTTATTTCAAGAAACCTTCAATCAAACCTCTGAAGTAACCTCT GCTTCAAACGGTTGTGGCTATGTCGAAAACAACACCAACAACCT TTCCAGACAAATCCAACAGTGGTTCGAATCAAGATCATGAAGACAA CAACGACAACGCTGATTTATCAATCATATTCGATTCAAGATGAT TTCGATAACGATATCACAGCTCTCATCGATTCTCTTCATCCATCC AGTCCCAGCCTCCGATCAACTCCAAGAGCAGTTTGATTTCACCGG AATTCAACTTCACCGCCTCCGAACACTCTACTCTTCTCTCTCT GGTGACCTTCTGCCTCCGCCTTTATCGGTTTTCGAAGAAGATTGTC TTCTTCTGTGCCAAGTTACAATCTGGTTCATAAAACCTTCTTCT TCCTTCTGTCTCTTTTTGGGTAATACCGGTTTACCGACTTACATG ACCGTAACCGGAATATGATGAACACCGGTTTAGGATCTGGTTTTT ACTCCGGAATATCCATCTCGGTTCTGATTTCAAACCATCACATGA TCAATTGATGGAATTCAGCTGATAATGGTGGATTGTTCTGTCCA GATCCGATTAAACCCATCTTAAATCCAGGAGATCATCATCTCCAGG GACTTGACGGTGTGAGAACCAGAACCACATGGTGGCACAGCCGGT TCTCCCGCAGTTAGGGACGGAGATAACCGGTTTAGACGACCCGAGT TTCAACAAAGTCGGAACCTCTCGGCAGAGCAGAGAAAGAGAAGA TTCATAGGTATATGAAGAAGAGGAACGAGAGGAATTCAGCAAGAA AATCAAGTATGCATGTAGGAAGACGTTGGCAGATAGTCGTCCAAGA GTTAGAGGAAGATTTGCAAGAATGATGAATTTGGTGAACCAATA GACAAGCCTGTTCAAGCCATCATGAAGATGATGATGATGTGGG GGTAAAGGAAGAAGACAGTTAGTTGATTCTTCGGACATATTTTCA CACATAAGTGGTGTGAACCTCTTCAAATGCAATTATCCAATCCAGT CTTGGATTGA
33	The AtPOWR1 full- length protein sequence	protein	MSQDVISSHEQLSMDEITSPLTAQIFDFCDSQLFQETFPNQTSEVTS ASNGCGYVENNNTNINFPDKSNSGSNQDHEDNNDNADLSIIFDSQDD FDNDITASIDFSSSIQFPASDQLQEQDFDTGIQLHQPPNTLYSSSS GDLLPPPLSVFEEDCLSVPSYNLGSINPSSPSCSFLGNTGLPTYM TVTGNMNTGLGSGFYSGNIHLGSDFKPSHDQLMEIQADNGGLFCP DPIKPIFNPGDHHLQGLDGVENQNHMVAQVLPQLGTEITGLDDPS FNKVGKLSAEQRKEKIHRYMKKRNERNFSSKKIKYACRKTLDASRPR VRGRFAKNDEFGEPNRQACSSHHEDDDDDVGVKEEQQLVDSIDIFS HISGVNSFKCNYPIQSWI
34	Primer		ctggcagaacttcagcccc
35	Primer		ccaggactgagataaagtga
36	Primer		ccaggactgagataaagtga
37	Primer		ATTTATCCCGAATGGTAACGC
38	Primer		TACCGGAGATGTGTTGAAGG
39	Primer		AACGTGAAACGGATTGTGTCAG
40			AAATTTGGTGAAACCAAGACC
41			ATTTTGCCGATTTCGGAAC
42			ATTTATCCCGAATGGTAACGC
43			TACCGGAGATGTGTTGAAGG
44			AACGTGAAACGGATTGTGTCAG

-continued

SEQUENCES

45 AAATTGGTGAAACCAAGACC

46 ATTTGCCGATTTCGGAAC

(DNA Synthetic)

SEQ ID NO: 4

1 gtgccgaatt cggatccagc gatctggatt ttagtactgg attttggttt taggaattag
 61 aaattttatt gatagaagta ttttacaat acaatacat actaagggtt tcttatatgc
 121 tcaacacatg agcgaaaccc tataggaacc ctaattccct tatctgggaa ctactcacac
 181 attattatgg agaaactcga gcttgtcgat cgactctagc tagagaagct cactttccaa
 241 caaccttagt aattttctcc ttaacataat gaacaaattc ttcaagtga tcagctctag
 301 aagcaagtct atcttcttct tgtccaagat aagcttgtct agcttcaaga ataactggtt
 361 gatattgagc tggaagtctt tccatagccc aatcagcagc aacatccttt ggagcaatct
 421 ttccagtaac agcagaatac caaattctag aaagagtaag aacaacattt ctttcatctc
 481 cagcccaatc tgggtggagaa ttccaaagag taagagtctt attaagagct tcaaaaagat
 541 cttgttctgg aactggatca aaaagtctt cagcagctgg tccaacaaga gcaacagaat
 601 gttctctagc cttagtaaga agaatagcaa gatcaatct aatagtagct ggttcaaaaa
 661 ttccagcaag aatatcattt ctttgccatt ctccaaattg aagtctctct ttagctggat
 721 atctccatgg aataatatca tcatgaacaa caatagtaac ttcaacagct ctaagaattt
 781 cagattctcc tggagaagca gaagtttcaa gaagatcatt aataagagct cttctagtag
 841 tttcatcaag tctaacagta acagtaacaa gaagatcaat atcagaatgt ggcttaagtc
 901 ctccatcaac agcagatcca taaagatgaa cagcaagaag agtaggttca agatgtcttt
 961 caataactcc aacaacttca gaaagttag tagaaaactc agcaataaca gcttctctca
 1021 tgcaagcagt agcaacagaa gcagaaattc tgaaagaaca gaacttttgc atgaaaatag
 1081 aatccttctt aagaacaagc atagaattag cagagtctct aagcttctta gatccgaaaa
 1141 caagaaaaga agatgactta ggaacttggt gcttatggaa attagagtta ggattaagag
 1201 tttgaattcc ttgagccata ttgttaatt gagccattgt atcgataatt gtaaatgtaa
 1261 ttgtaatggt gtttgttgtt tgttgttgtt ggtaattggt gtaaaaatga gctcttatac
 1321 tcgagcgtgt cctctccaaa tgaaatgaac ttccttatat agaggaagggt tcttgccgaa
 1381 gatagtggga ttgtgcgtca tcccttacgt cagtggagat gtcacatcaa tccacttgct
 1441 ttgtagacgt ggttgaagc tcttctttt ccacgatgct cctcgtgggt ggggttccat
 1501 ctttgggacc actgtcggca gagagatctt gaatgatagc ctttccttta tcgcaatgat
 1561 ggcatttgta ggagccacct tccctttcta ctgtccttc gatgaagtga cagatagctg
 1621 ggcaatggaa tccgaggagg tttcccgaaa ttatccttg ttgaaaagtc tcaatagccc
 1681 tttgatcttc tgagactgta tctttgacat ttttgagta gaccagagtg tcgtgctcca
 1741 ccatgttatc acatcaatcc acttgctttg tagacgtggt tggaacctct tctttttcca
 1801 cgatgctcct cgtgggtggg ggtccatctt tgggaccact gtcggcagag agatcttgaa
 1861 tgatagcctt tctttatcg caatgatggc attttagga gccaccttc ttttctactg
 1921 tcttttcgat gaagtgcagc atagctgggc aatggaatcc gaggagggtt cccgaaatta
 1981 tctttgttg aaaaagtcca atagccctt gatcttctga gactgtatct ttgacatttt

-continued

SEQUENCES

2041 tggagtagac cagagtgtcg tgetccacca tgttgacctc cgcaagaatt caagcttagc
 2101 gttcctaaca cctggagaac cttttatgta cttcacaacc ctcaatgctg cttccagggtg
 2161 tgactgtttg ggttgcgtga tgaattgact cagcacctgc actgcaaagc tgatatctgg
 2221 ccttgtgatt gtcaagtaga gaagctttcc caccagtctc tgataggatc ccacatcctc
 2281 taactctgca tcatcagtct ttccctaagtg cttgtcatat tctacagtag tcagcctttg
 2341 gttctgttcc attgggggtg aactgggtct gcagcctccc agaccaaacac ctgatatcag
 2401 ttccagtgca tacttcctct ggttcagtag gattcccttt tctgatctca gcacctcaat
 2461 gcctaagaag tatttttagt ctcccaaatc tttcatcttg aaatgctgat gcagggttgc
 2521 ctttgcctct gaaatcaaaa cattgctgct gcctgttatt aacagatcat ccacataaat
 2581 caggattatg acaaggtcag tcccctctct tttggtaaac aaggagtgat cataagcact
 2641 ttgcataaaa ccagcctgca taaggacagt ggtaagcttg atgttccact gccttgatgc
 2701 ttgctttaaa ccatagaggg attcaacagc ctgcacactt tggtttcccc ttggctgtga
 2761 aaacctgag gcagagacat ataaacttct tccatgaggt caccctgtag aaaagcattg
 2821 ttgacatcca tctggaaaag gaaccagccc ttggaagcag caacagatat gacagctttt
 2881 acagtgacca ttttgccac tggagaaaaa gtttcatggt agtcaaggcc ttcttgcgtga
 2941 gtgtatccct tggccactag ccttgctta aacctgtcaa cttcaccatt agctttgtat
 3001 ttaattttgt acacccttt ggacctata ggctgtttac cagggggtaa agggacaatc
 3061 tcccagggtt tattatcctc aagagcctgt atctcaaggg acatggctc catccatttc
 3121 tcatcttgag ctgcttcttt gaaggattta gggtcagtat aagtggagaa agcactcaaa
 3181 taagcttgat agtgaacagg caagtgatca taggaaacat agttggctat agggtatgga
 3241 acatccctag agcctttggt cagtgtcaca aagtcctga gccagatggg aggaacctgca
 3301 ttgcgtttag gtctagtgtg cagggttgct ggaactagag agggatcagc tacagcagta
 3361 tgggtgctca attcagcatt tgctaggtea ggctcagctg actcaactga ctctgcagga
 3421 gcatgcaggt ggctgaagg tgcagcatca gctgaagtga tatgattttg ggtagaatga
 3481 gtcccaaaag tgatatectc attggcctct acaatgtcag gaataagaac tggagtatca
 3541 tcatcatcag tatgaaagct ggaaggaaaa atatcattat acactagctg caaagctgta
 3601 tcttcagagc taagtgaacc tgctcgagtg aacatatcag gttcatggga gatggactct
 3661 ttgaaaggga actgaaactc tctgaaaact acatccctgc tcactatgat caccttatta
 3721 tccaagtcac acaacctgta acccttttga gtctcagaat aaccaatgaa gatggttcgc
 3781 ttggctctag gtgcaagttt gtcacctttg ggcagtgttg ctgcaaaagc aagacacca
 3841 aacactctca agtgatcaag cttggcatgt ttctgataga gtagttcata tggacatttg
 3901 ccttgtaaga ttggagtagg gagtctattg atcaggtata caacagtctt gacacagtct
 3961 ccccaaaacc tggtaggtac accactctga aacttaagtg cccttgccat ctcaaggatg
 4021 tgtctgtgct ttctctccac aacaccattc tgttgtgggtg tgtagggaca gctactttga
 4081 tgaacaatcc caagagaggc cagcaactca ttacaacttc tccactagaa ttcgagctcg
 4141 gagctgtagg ccggcctagc ttattctcac ccctaattaa gacaagctgt aggtcggccc
 4201 ctgttcatgt tgttcatcaa ttattgttta accgagttaa aggagatggg caattttaca

-continued

SEQUENCES

4261 gtagtaaata tgacaagaaa tacacgaaat ttgtgtttaa gtattgaagg acttccattt
 4321 ttttgaatcc accatcctaa aactaatagt gtctaatttt cccgtaacaa tttcttttgt
 4381 ataacattca aataaatgtg gttgacgata tatatttagc ttgcaaattt cctaaaaatt
 4441 attgctccaa acttagtatg cacttaggat tggatttgtt cattattgta cttggcaagg
 4501 gtacatgtgt taattatatg cacttaagcg tgcatttgta cttggctact taaaattgtt
 4561 ttttgttctt cggatgatgt tgatgagtta taaaaagaa atatatgcct tgaaggaaag
 4621 aaagtttatt ttacgggtt atgttaaaat acaatacttt caactatgta gattgtggag
 4681 caaacaagct ttcagtttct ctatcgtgtt ccgatgacac gtaataatg aacttttctt
 4741 ataacatca attagaagat gttgattttg ttggtcaatt ggtgttattg ttgtgactac
 4801 tttatatttt cgtagattc caagcattat tccgctctat agggtttgct cttaactgat
 4861 tgttttgagt tattaattat tcaagttgat ttgattgga tagtaaatct ggatctatta
 4921 tgtaatatat tagggacgaa atcttgacac ttgagaatga ttagtctctc tcaaaatttt
 4981 gataaccatt gttttcgagt gatagatgtt gtctatgtca aggttgctga tgaaattagc
 5041 aatgatggac agactttgtc gaacaagttt tgtaatgcca tcacagaaaa gcatctagaa
 5101 gtactgaaaa acattttttt taacatatac ttttcaaaa tcaagagaat gaataattaa
 5161 atatgaaata agaattacaa aaattttata tttttttaat aaatttcaaa tcaataataa
 5221 gaaagaatgg gataaaaatta aagaatatgt tggaagatgt gttattagta tttctgtgca
 5281 taagaatgct tccaaaatta aactctttcc ccaaaattgt atgagtaaaa tttcaaattg
 5341 ctctgtttta atataacata agctaaaatta aaccagctt agtaaaaaaa atgcatttcc
 5401 gtaaatcttg atgtccaaac aattgtaatg acttaaacct agaacaaaac ctatttttat
 5461 accatgtaaa ttcttacaca ttataatcgt atgtttatcc atccatgcac cccaacaaat
 5521 gaagaaaagg aataaaaaaa aaggtgaatt tgccaaaaac aaattaatgg gatgctaaca
 5581 ttaacataag gcggcatagg gcatcattta attgtgaata aaaaggcttg gtcaacccaa
 5641 tttcttctct tagttactag ttattaacca tgtcatttgg catatatata ataaccata
 5701 tccatataaa aatacatgtc ccaaaaattg aattccacta cccccacca aagtggtgaa
 5761 ttaatggcca aacgccactc tcgcgttgac acccgcgagt gccaggcttg acgactctac
 5821 gtacaacctc atcaattcct ctatatatgc acataactaa ctttctctgt ttttgacacc
 5881 ctcatacacc ccatcatcct agctaacaca acacagcata cacaacttct ctctctctac
 5941 tactttcttt ttatgtcacc ttcttgatgt ctttaatttg ttttcttttt tagaaaaaaa
 6001 ttaatccttt ttaaggcttt aatttcttaa gcatgcaata ttattgtttt tagtcacctt
 6061 caagttgagg aacatatata cgtttgtgac cacaccaaatt ccacttctt tctaagtggtg
 6121 tgtacataga ttttgtttta tatatatttg aggttcaccc ttcatgacct ccttaattct
 6181 gtgcaaaaga ggattcatca ccaccccatg taccatagc atgttctga ctatgogggc
 6241 tatccctatg acgtcccgga ctatgcagga ttgtatccat atgacgttcc agattacgcc
 6301 actagagctg cttaccata cgatgttctt gactatgagg gctatcccta tgacgtcccg
 6361 gactatgcag gattgtatcc atatgacgtt ccagattacg caatgttgca ggatgttgct
 6421 caccctcaa caccggtga gcaactcccc attgatgaga ttccaggccc gattagtgtc

-continued

SEQUENCES

6481 cgaattttcg aactttgcca ccccgatttc tccccacaca cactgcaaaa ttctgaggtt
 6541 acctccagct caaattgttg ccatgaagag aagtcctcat atgccacaac catatctcca
 6601 cctttagatg tagtagacaa caataagttc aatatcaata gcaatagcag caacatagtc
 6661 accactacct catctagcac taccacaacc agcaccacaa ccaacaacaa caacaacgca
 6721 acgaacggca ataattcttc tatcttcttt gacactcaag atgaaattga caatgacatc
 6781 tcagcctcca tagacttctc atcatctcca tcttttgctg ttcaccact tctcccaatc
 6841 tcaactcagc aggatcagtt tgatttcctc tcagctcagc cacaggtgca actatcaaca
 6901 gcagcaggtt caattttgac gggcctctct cactacccta cagatcctgt gattgcaccc
 6961 cttattggag ctccgttacc atctgttttt gatgatgatt gcatatcttc catccctctc
 7021 tatgtgcctc tcaacccttc atcaccctct tgctcttctc tcagtcctgg cataggagtg
 7081 tacatgccac ctctcggttc ccttaacact gccttatctg ctgacagttc tggattgttt
 7141 ggtgggaaca ttctactggg gtctgaactg caggcacatg aattggacta ccaggagaaa
 7201 aatggtggaa tgtattgtac agattcaatt caaagggtgt ttaactccc agaccttcag
 7261 gcacttggtg atgagagtca gaaacttgta gctggggctg gaagctctgc cactttggca
 7321 ccagaaatct cacacttgga ggactctacc ttgaaggttg gaaaactctc tgttgagcag
 7381 aggaaggaaa agattcatag atacatgaag aagagaaacg aaagaaattt cagcaagaaa
 7441 atcaagtatg ctgcccga aactttggcg gatagccggc cccgggttag aggaaggttt
 7501 gcaaagaatg atgactttgg agagagccat aaacaaggaa gtagcaatca tgaagatgat
 7561 gatgaagaga taattgtgaa agaagatgat gatatggttg attcctcaga tatctttgca
 7621 catatcagcg gagtgaactc tttcaaatgc aactattcca tccagtcctt gatttgagct
 7681 tattaaatta ttagtttgac tagtgaaagc ttatttatat aattagcttc tgtagattaa
 7741 ttttgtagg acacttttcc catcccggtt ctctaaaatc cgggtttagt ggtttgagta
 7801 aactgaataa atggggtcaa aataaatata ccaataagtt aagtgaagta gaaacgtaca
 7861 gaaattggaa actgtatata tttttgcaga tatatattat ctttttcatt aagttgtacc
 7921 agaacatgga gttgtgttaa ccaagaaaat ttccagttac ccccatccaa gactgatgta
 7981 accaattgat gtagcttctt ttatggtagt caagcagatc gttcaaacat ttggcaataa
 8041 agtttcttaa gattgaatcc tgttgccggt cttgcgatga ttatcatata atttctgttg
 8101 aattacgta agcatgtaat aattaacatg taatgcatga cgttatattat gagatgggtt
 8161 tttatgatta gagtcccgca attatacatt taatacgcga tagaaaaaa aatatagcgc
 8221 gcaaaactagg ataaattatc gcgcgcggtg tcacttatgt tactagatcg acgctttacg
 8281 aggatgcaca tgtgaccgag ggacacgaag tgatccgttt aaactatcag tgtttgacag
 8341 gatataattg cgggtaaac taagagaaaa gagcgtttat tagaataatc ggatatttaa
 8401 aaggcggtga aaaggtttat ccgttcgtcc atttgatgt gccagcgcgc tttgcgacgc
 8461 tcaccgggct ggttgccctc gccgctgggc tggcggcgt ctatggccct gcaaacgcgc
 8521 cagaaacgcc gtcaagccg tgtgcgagac accgcggccg ccggcggttg ggatacctcg
 8581 cggaaaactt ggcctcact gacagatgag gggcgacgt tgacacttga gggcccgact
 8641 caccggcgcc ggcgttgaca gatgaggggc aggtctgatt tcggccggcg acgtggagct

-continued

SEQUENCES

8701 ggccagcctc gcaaatcggc gaaaacgcct gattttacgc gagtttccca cagatgatgt
 8761 ggacaagcct ggggataagt gccctgcggt attgacactt gaggggcgcg actactgaca
 8821 gatgaggggc gcgactcctt acacttgagg ggcagagtgc tgacagatga ggggcgcacc
 8881 tattgacatt tgaggggctg tccacaggca gaaaatccag catttgcaag ggtttccgcc
 8941 cgtttttcgg ccaccgctaa cctgtctttt aacctgcttt taaaccaata ttataaaacc
 9001 ttgtttttaa ccagggtgc gccctgtgag cgtgaccgcg cagccgaag gggggtgcc
 9061 ccccttctcg aacctcccg gcccgctaac gcgggcctcc catccccca ggggctgcgc
 9121 ccctcgccg cgaacggcct cccccaaaa atggcagcgc tggccaattc cagagtgcgc
 9181 ggaaccctta ttgtttatt ttctaaata cattcaaata tgtatccgct catgagacaa
 9241 taacctgat aaatgcttca ataatttga aaaaggaaga gtatggctaa aatgagaata
 9301 tcaccggaat tgaaaaaact gatcgaaaaa taccgtgcgc taaaagatac ggaaggaaatg
 9361 tctcctgcta aggtatataa gctggtggga gaaaatgaaa acctatattt aaaaatgacg
 9421 gacagccggt ataaaggac cacctatgat gtggaacggg aaaaggacat gatgctatgg
 9481 ctggaaggaa agctgcctgt tccaaaggtc ctgcactttg aacggcatga tggctggagc
 9541 aatctgctca tgagtgggc cgatggcgtc ctttgcctcg aagagtatga agatgaacaa
 9601 agccctgaaa agattatcga gctgtatgcg gagtgcacga ggctctttca ctccatcgac
 9661 atatcggtt gtccctatac gaatagctta gacagccgct tagccgaatt ggattactta
 9721 ctgaataacg atctggccga tgtggattgc gaaaactggg aagaggacac tccatttaaa
 9781 gatccgcgcg agctgtatga ttttttaaag acggaaaagc ccgaagagga acttgtcttt
 9841 tcccacggcg acctgggaga cagcaacatc tttgtgaaag atggcaaatg aagtggcttt
 9901 attgatcttg ggagaagcgg cagggcggac aagtggatg acattgcctt ctgcgtccgg
 9961 tcgctcaggg aggatatcgg ggaagaacag tatgtcgagc tattttttga cttactgggg
 10021 atcaagcctg attggggaaa aataaaatat tatattttac tggatgaatt gttttagctg
 10081 tcagaccaag ttactcata tatactttag attgatttaa aacttcattt ttaatttaaa
 10141 aggatctagg tgaagacct ttttgataat ctcatgacca aaatccctta acgtgagttt
 10201 tcgtccact gagcgtcaga ccccgtagaa aagatcaaag gatctctctg agatcctttt
 10261 tttctgcgcg taatctgctg cttgcaaaca aaaaaaccac cgtaccagc ggtggtttgt
 10321 ttgccggatc aagagctacc aactcttttt ccgaaggtaa ctggcttcag cagagcgcag
 10381 ataccaaata ctgtccttct agtgtagccg tagttaggcc accacttcaa gaactctgta
 10441 gcaccgccta catacctcgc tctgctaata ctgttaccag tggctgctgc cagtggcgat
 10501 aagtcgtgtc ttaccgggtt ggactcaaga cgatagttac cggataaggc gcagcggctg
 10561 ggctgaacgg ggggttcgtg cacacagccc agcttgagc gaacgaacta caccgaactg
 10621 agatacctac agcgtgagct atgagaaaag gccacgcttc ccgaaggag aaaggcggac
 10681 aggtatccgg taagcggcag ggtcggaaaca ggagagcgca cgagggagct tccaggggga
 10741 aacgcctggt atctttatag tctgtcggg tttcgccacc tetgacttga gcgtcgattt
 10801 ttgtgatgct cgtcaggggg gcggagccta tggaaaaacg ccagcaacgc ggccttttta
 10861 cggttcctgg cagatcctag atgtggcgca acgatgccg cgacaagcag gagcgcaccg

-continued

SEQUENCES

10921 acttcttccg catcaagtgt ttggtcttc aggcgaggc ccacggcaag tatttgggca
 10981 aggggtcgct ggtattcgct cagggaaga ttcggaatac caagtacgag aaggacggcc
 11041 agacggtcta cgggaccgac ttcatgccc ataaggtgga ttatctggac accaaggcac
 11101 caggcgggtc aaatcaggaa taagggcaca ttgccccggc gtgagtcggg gcaatccgc
 11161 aaggagggtg aatgaatcgg acgtttgacc ggaaggcata caggcaagaa ctgatcgacg
 11221 cgggggtttc cggcaggat gccgaaacca tcgcaagccg caccgtcatg cgtgcgccc
 11281 gcgaaacctt ccagtcgctc ggctcgatgg tccagcaagc tacggccaag atcgagcgcg
 11341 acagcgtgca actggctccc cctgcctgc ccgcgccatc ggccgcgctg gagcgttcgc
 11401 gtctgtctga acaggaggcg gcaggtttgg cgaagtcgat gaccatcgac acgcgaggaa
 11461 ctatgacgac caagaagcga aaaaccgccc gcgaggacct ggcaaacag gtcagcgagg
 11521 ccaagcagcg cgcgttgctg aaacacacga agcagcagat caaggaaatg cagctttcct
 11581 tgttcgatat tgcgcgctgg ccggacacga tgcgagcgat gccaaacgac acggcccgt
 11641 ctgcctgttt caccacgcgc aacaagaaaa tcccgcgcga ggcgctgcaa aacaaggtea
 11701 ttttccacgt caacaaggac gtgaagatca cctacaccgg cgtcgagctg cgggcccagc
 11761 atgacgaact ggtgtggcag caggtgttgg agtacgcgaa gcgcacctt atcggcgagc
 11821 cgatcacctt cagttctac gagctttgcc aggaacctgg ctggtcgatc aatggccggt
 11881 attacacgaa ggccgaggaa tgcctgtcgc gcctacaggc gacggcgatg ggcttcacgt
 11941 ccgaccgcgt tgggcacctg gaatcgggtg cgctgctgca ccgcttcgc gtcttggaac
 12001 gtggcaagaa aacgtcccgt tgccaggtec tgatcgacga ggaaatcgtc gtgctgtttg
 12061 ctggcgacca ctacacgaaa ttcatatggg agaagtaccg caagctgtcg ccgacggccc
 12121 gacggatggt cgactatttc agctcgcacc gggagccgta ccgctcaag ctggaaacct
 12181 tccgctcat gtgcggatcg gattccacc gcgtgaagaa gtggcgcgag caggtcggcg
 12241 aagcctgcga agagttgcga ggcagcgccc tgggtggaaca ccctgggtc aatgatgacc
 12301 tgggtgcatg caaacgctag ggccttgctg ggtcagttcc ggtgggggt tcagcagccc
 12361 ctgctcgat ctgttgacc ggacagtagt catggtgat ggcctgctg tatcgagtgg
 12421 tgattttgtg ccgagctgcc ggtcggggag ctgttgctg gctggtggca ggatatattg
 12481 tgggtgtaaac aaattgacgc ttagacaact taataacaca ttgcggacgt ttttaattga
 12541 ctgggggtga acactct

(DNA Synthetic)

SEQ ID NO: 7

1 gtgccgaatt cggatccagc gatctggatt ttagtactgg attttgggtt taggaattag
 61 aaattttatt gatagaagta ttttacaat acaatacat actaagggtt tcttatatgc
 121 tcaacacatg agcgaacccc tataggaacc ctaattccct tatctgggaa ctactcacac
 181 attattatgg agaaactcga gcttgctgat cgactctagc tagagaagct cactttccaa
 241 caaccttagt aattttctccc ttaacataat gaacaaatc ttcaagttga tcagctctag
 301 aagcaagtct atcttctctt tgtccaagat aagcttgctt agcttcaaga ataactggtt
 361 gatattgagc tggaagtctt tccatagccc aatcagcagc aacatccttt ggagcaatct
 421 ttccagtaac agcagaatac caaattctag aaagagtaag aacaacattt ctttcattct

-continued

SEQUENCES

481 cagcccaatc tggtagagaa ttccaaagag taagagtttc attaagagct tcaaaaagat
 541 cttgttctgg aactggatca aaaagttctt cagcagctgg tccaacaaga gcaacagaat
 601 gttctctagc cttagtaaga agaataagcaa gatcaatata aatagtagct ggttcaaaaa
 661 ttccagcaag aatatcattt ctttgccatt ctccaaattg aagtctcttc ttagctggat
 721 atctccatgg aataatatca tcatgaacaa caatagtaac ttcaacagct ctaagaattt
 781 cagattcttc tggagaagca gaagtttcaa gaagatcatt aataagagct cttctagtag
 841 tttcatcaag tctaacagta acagtaacaa gaagatcaat atcagaatgt ggcttaagtc
 901 ctccatcaac agcagatcca taaagatgaa cagcaagaag agtaggttca agatgtcttt
 961 caataactcc aacaacttca gaaagttgag tagaaacttc agcaataaca gcttctctca
 1021 tgcaagcagt agcaacagaa gcagaaattc tgaaagaaca gaacttttgc atgaaaatag
 1081 aatccttctt aagaacaagc atagaattag cagagttctt aagcttctta gatccgaaaa
 1141 caagaaaaga agatgactta ggaacttggt gcttatggaa attagagtta ggattaagag
 1201 tttgaattcc ttgagccata ttgttaattt gagccattgt atcgataatt gtaaatgtaa
 1261 ttgtaattgt gtttgttgtt tgttgttgtt ggtaattgtt gtaaaaatga gctcttatcc
 1321 tcgagcgtgt cctctccaaa tgaaatgaac ttccttatat agaggaaggg tcttgcgaag
 1381 gatagtgagg ttgtgcgtca tcccttacgt cagtggagat gtcacatcaa tccacttgct
 1441 ttgtagacgt ggttggaacc tcttcttttt ccacgatgct cctcgtgggt gggggtccat
 1501 ctttgggacc actgtcggca gagagatctt gaatgatagc ctttcttcta tcgcaatgat
 1561 ggcattttga ggagccacct tcccttttcta ctgtccttcc gatgaagtga cagatagctg
 1621 ggcaatggaa tccgaggagg tttcccgaaa ttatcctttg ttgaaaagtc tcaatagccc
 1681 tttgatcttc tgagactgta tctttgacat ttttgagta gaccagagtg tcgtgctcca
 1741 ccatgttata acatcaatcc acttgctttg tagacgtggg tggaacctct tctttttcca
 1801 cgatgctcct cgtgggtggg ggtccatctt tgggaccact gtcggcagag agatcttgaa
 1861 tgatagcctt tcttttatcg caatgatggc atttgtagga gccaccttcc ttttctactg
 1921 tcccttcgat gaagtgcagc atagctgggc aatggaatcc gaggaggttt ccgaaatta
 1981 tccctttgtg aaaagtctca atagcccttt gatcttctga gactgtatct ttgacatttt
 2041 tggagtagac cagagtgtcg tgctccacca tgttgacctc cgcaagaatt caagcttagc
 2101 gcctgtgtgg atcatctaaa tgtgcattcc gcaagccaat taatgttcca actgaagcag
 2161 attgcttaca taatttatat acgataccaa atatgcttca cgagcataat ttttaagtgt
 2221 ctacctcttg gtaaggggca gctaattgaa actagggatc atgatctccg tttgttcaaa
 2281 aatccttttg tgtgttttca aaatatcact ttagaatcta atccaagagt tctccctacc
 2341 taataaatat ctttaacctt taataccaaa atctaataat tcgtgggatg atcaccatcc
 2401 ttcacacaac tcatccctgc ttttctccg tgataagcat gaagaataaa aaaccacta
 2461 atatacgcca attcactgga ttgagaacaa aattcaacaa caatatgcca gcatacagtt
 2521 gataattata tatcagctcc acaaattgat aacaaacagt agtaacatta atgggacaat
 2581 agagttctac acaaaccaaa cttattttca aaggtcatta cagacctatt acacatcaca
 2641 tttcttctta aaatgttttg gaaaatccta catcgaacat cacacagaaa tcagctccag

-continued

SEQUENCES

2701 caagctcacc ctcccacatt tctctcttc tttggttgca gatctcctcc cagttgggac
 2761 agatcgattc tagtctcgta cagtccagtg atggattggg ggatcagagt agcatccagc
 2821 acctccttag tggaagtgtg tctctttcta tcgatatag tagtgaagta cttgaaagca
 2881 gctggagctc ccaggtagt cagagtgaac aggtggatga tgttctcagc ttgctctctg
 2941 attggcttat ctctgtgctt gttgtaagcg gacagcacct tatccagggt agcatcagcc
 3001 aggatcactc tcttgagaa ctccgagatt tgcctgatga tctcatccag gtagtgcttg
 3061 tgttgctcca cgaacagttg cttttgctcg ttatcctctg gggatccctt cagcttctcg
 3121 tagtgggaag ccaggtagc gaagttcacg tacttggttg gcagagccag ctcgtttccc
 3181 ttttgagctc ctccagcgga agccagcatt ctctttcttc cgttctccag ctggaacagg
 3241 gagtactttg gcagcttgat gatcagatcc ttcttcacct ccttgatccc cttagcctcc
 3301 aggaatcgga ttgggttctt ctogaaggag gatctctcca tgatagtgat tcccagcagc
 3361 tccttcacgg acttcagctt cttggacttt ccctctccca ccttagccac caccagcagc
 3421 gagtaagcca cagttgggga atcgaaacct ccgtacttct ttggatccca atccttcttt
 3481 ctacgcatga gcttatcgga gtttctcttt ggcaggatgg actccttgga gaatcctcca
 3541 gtttgccact cagtcttttt cagcatgttc acttggtggc tggacagcac ctttctcaca
 3601 gtagcgaaat ctcttccctt atcccacagc atctctccag tctctccgtt agtctcgatc
 3661 agtgggtctt ttctgatctc tccgttagcc agagtgatct cagtcttgaa gaagtccatg
 3721 atgttgagtg agaagaagta cttagcagta gcctttccga tctcttgctc ggaactagcg
 3781 atcatctttc tcacatcgta caccctgtaa tctccgtaca cgaactcgga ctccagcttt
 3841 ggggtactct tgatcagagc agttcccacc acagcgttca ggtaagcatc gtgagcgtgg
 3901 tggtagttgt tgatctctct caccctgtag aattggaaat cctttctgaa atcggaaccc
 3961 agcttggaat tcagagtgat caccctcacc tctctgatca gcttatcggt ctcatcgtag
 4021 ttagtggtca ttctggaatc caggatttga gccacgtgct tagtgatttg tctagtctcc
 4081 accagttgtc tcttgatgaa tccagcctta tccagctcgg acagtcctcc tctctcagcc
 4141 ttagtcaggt tatogaactt tctttgagtg atcagcttag cgttcagcag ttgtctccag
 4201 tagttcttca tcttcttcac caccctctcg gatggcacgt tatcggaact tctctgttc
 4261 ttatcggatc tagtcagcac cttgttatcg atggaatcat ccttcaggaa ggattgtggc
 4321 acgatgtgat ccacatcgta atcgacagc ctgttgatat ccagctcttg atccacgtac
 4381 atatctcttc cgttttgtag gtagtacagg tacagcttct cgttttgtag ttgagtgttc
 4441 tccactgggt gctccttcag gatttgggat cccagctcct tgattccctc ctgattcttc
 4501 ttcattctct ctctggagtt cttttgtccc ttttgagtag ttggttctc tctagccatc
 4561 tcgatcacga tgttctctgg cttgtgtcta cccatcaact tcaccagctc atccaccacc
 4621 ttcacagttt gcaggattcc cttcttgata gctggggatc cagccagggt agcgatgtgc
 4681 tcgtgcaggg aatctccttg tccggacact tgagcctttt ggatatactc cttgaaagtc
 4741 agggaaatcat cgtggatcag ttgcataag tttctgttag cgaatccatc ggaattcagg
 4801 aaatccagga tagtctttcc ggattgttta tctctgatcc cgttgatcag ctttctggac
 4861 agtctacccc atccagtgtg tcttcttctc ttcagttgct tcatacactt atcatgaac

-continued

SEQUENCES

4921 aggtgagcgt aagtcttaag tctctctcgc atcatctctc tatectcgaa cagagtcaga
 4981 gtcagcacga tatctctcag gatctctcgc ttctctcgt tatccaggaa atccttatcc
 5041 ttgatgatct tcagcagatc gtggtaagtt cccagggaag cgttgaatct atcctccact
 5101 ccggagatct ccacggaatc gaagcactcg atcttcttga agtaatctc cttcagttgc
 5161 ttcacagtca cctttctggt agtcttgaac agcagatcca cgatagcctt cttttgctct
 5221 ccggacagga aagctggctt tctcattccc tcagtcacgt acttcacctt agtcagctcg
 5281 ttgtacacag tgaagtactc gtacagcagg gagtgctttg gcagcacctt ctcgtttggc
 5341 aggttcttat cgaagttagt cattctctcg atgaaggatt gagcggaaac tcccttatcc
 5401 accacctgca catcaacaaa ttttggtcac atattagaaa agttataaat taaaatatac
 5461 acacttataa actacagaaa agcaattgct atatactaca ttcttttatt ttgaaaaaaa
 5521 tatttgaat atttatattac tactaattaa tgataattat tatatatata tcaaaggtag
 5581 aagcagaaac ttacctctc gaagttccat ggagtgatag tctctcggga ctttctagtc
 5641 atccaagcga atctggagtt tcctctagcc agtggtccca cgtagtatgg gattctgaaa
 5701 gtcaggatct tctcgatctt ctctctgtta tccttcagga atgggtagaa atcctctgtt
 5761 ctctcagga tagcgtgcag ctctcccagg tggatttgggt gtgggatgga tccgttatcg
 5821 aaagtctttt gctttctcag cagatctct ctgttcagct tcaccagcag ctctcagtt
 5881 ccatccatct tctccaggat tggcttgatg aacttgtaga actcctcttg ggaagctcct
 5941 ccatcgatgt atccagcgta tccgttcttg gattgatcga agaagatct cttgtacttc
 6001 tctggcagtt gttgtctcac cagagccttc agcagagtca gatcttgggt gtgctcatcg
 6061 tatctcttga tcatggaagc ggacagtgga gccttagtga tctcagtgtt cactctcagg
 6121 atatcggaac gcaggatagc atcggacagg ttcttagcag ccagggaacag atcagcgtat
 6181 tgatctccga ttgagccag caggttatcc agatcatcat cgtaagtatc cttggacagt
 6241 tgcagcttag catcctcagc cagatcgaag ttggacttga agtttggagt cagtcccagg
 6301 gacagagcga tcaggtttcc gaacagtccg ttcttcttct ctctggcag ttgagcgatc
 6361 aggttctcca gtctcctgga cttggacagt ctagcggaac ggatagcctt agcatccact
 6421 ccggaagcgt tgattgggtt ctctcgaac agttgggtgt aagtttgac cagttggatg
 6481 aacagcttat ccacatcgga gttatctggg ttcatgctc cctcgatcag gaagtgtcct
 6541 ctgaacttga tcatgtgagc cagagccagg tagatcagtc tcagatcagc cttatcagtg
 6601 gaatccacca gcttctttct cagggtgtag atagttgggt acttctcgtg gtaagccacc
 6661 tcatccacga tgtttccgaa gattgggtgt ctctcgtgct tcttatectc ctccaccagg
 6721 aaggactcct ccagtctgtg gaagaaggaa tcatccacct tagccatctc gttggagaag
 6781 atctcttgca ggtagcagat tctgttcttt cttctagtgt atcttcttct agcagttctc
 6841 ttcagtctag tagcctcagc agtctctccg gaatcgaaca gcagagctcc gatcaggttc
 6901 ttcttgatgg agtgtctatc agtggttccc agcaccttga acttcttgga tggcaccttg
 6961 tactcatcag tgatcacagc ccatccacag gagttagttc cgatatccag tccgatggag
 7021 tacttcttat ccattctgtc gagtcaacaa tcacagataa atcagaatca aaatccatta
 7081 cgtaatctta agtacggatc tgaagcaatc tcgtgaaaaa aaacccta atccacggcaat

-continued

SEQUENCES

```

7141  agaatcaaca aacaaggcag ttgctaacag atctgagcaa tatctctcga aaaacgatat
7201  agaatcaacg aataaggcctt ttgtcaacg gatctgaaca atttcttgaa aaaccacaga
7261  agtataaaaa ataaggctgt tgtttaacgg atctgaagta attcatgaaa aagccctgat
7321  ctatggttac gtttaacgga tctgaagtaa tcaatgaaaa aagccctaatt ctacgattaa
7381  tgaatcaaca attaaggcctt gttgtaaagg gatctgacat gaaaaagctc taattagagc
7441  aaatctacga ttaagaatc atcaaattaa ggacgttggt gtaaagataa taaaagctga
7501  atctgattat gaaaattgaa gatcgatgaa accaaacaaa acacaatcaa atcaacatcg
7561  tattgatcaa agaacataca caattcatgg aaagaacgaa gaagaaaaac agaagaaatc
7621  gactaacctt gaaggagag gtagggaat acacagagag aatggaatat attggaattg
7681  aactgcgaag gaaggaggag ggggtgtggg cgttttatag cataagggtg ggaacccttc
7741  tccagtcata ttgtgacgcg tgttcatgta tgggtgcact tgggtgcctat gacacgggta
7801  atatttggtta ttggatgcat ttctggagag tccttctagg aaattgagaa ttgcaatttg
7861  cgtggtagtt ggcggtttcg taatcttgac ggagtcacat gcacctcaa gcttcgcttt
7921  atctgttacg acgtcgttgt tatattgggc ccacccggt gcaagtggag cgtccgaagt
7981  tgatacgggc cccactatcc attgcctgaa gtcggtgct tttacattat taaaaatcta
8041  tggaaaagct gtcatgctag atttttttta ttatggacca gaaaagaata attgaaaagt
8101  tttcttatta tggagcaaaa aaatctttaa taattatttt ttaataaaaa tcataattat
8161  tccactaaat tctttcttat ttaatttcac cttgtaagat atattttcag tagatttttc
8221  aagtattttt ttaacataca ttacttgtat acatttagtt acagtttata taaaattatt
8281  catttaaaaa atatttagat atttttttaa aaataattca aaatattttt tttaatgaac
8341  tttccttttt cctaattttt ctttcttatt tcttattaga tgtttttact accaatcggt
8401  tagggaagaa tttttttctc taatgaggag atgtgtttaa tgatgcaact taaggacaga
8461  ttcaccaata acataagact aaaagatagc tcattgacaa aaacaataat gatgttaaag
8521  aaaacgaatc catttccatt cggataagcc acaccattta cgaacaagtt gctactaat
8581  aatatatggt gtcagttttt tttttttttg ggctgaattg gaggggaaat tatcccaaaa
8641  gtttgattaa aaatcacgcg acaaagtaac aaaactttta ttccgctaaa aaaattagtc
8701  aaagtcaatt aaatctggtta ttagaattac ttttcgcta atgatatatc cacaataaca
8761  atgtattaat attgaaaaaa ataaagtta tagagtgtga cttttctttt catttttacg
8821  taaattgaaa aaaaaatcaa gcctaataata taaactttca ttgatcctca cctaatacct
8881  aaagtctctg agcccaatta aaaaacttac tggaggataa aggcccaaac ttcttttttt
8941  ttttaggaac agagcatgta atgggtaatg aggcagccta agcccataac catgttagcc
9001  cgtatatatt ttgttaatt agatttcggg tttcttattt ttttttagat ttcagttttt
9061  gtttctcaaa aaaaaaaat aacttttagt gctgtattaa ctttctgtaa ataattttag
9121  tctcaataaa gattaattct gaataataaa aataaacgag gacaattttt ttcacaactt
9181  ttaattcttt atttattttt tacacttaat tttgggaaac atggacctag tgtcaaagaa
9241  ctcaaaacta acattacaaa aacttactac atattcaagg atcatggctt gaactaatct
9301  ccacataaca tataatcaca tttttcaatt aaaaagaaga attaataatg tattaatga

```

-continued

SEQUENCES

9361 tgategacaa gatcacatga tttagagattt tttttatga tgggaaaaat atgatttgag
 9421 attaggtaac aatctttcca ttttagatag acttatgtct ccactagaat tcgagctcgg
 9481 aggacgtcgt tgtggttggg gctttcctta cattctgagc ctctttcctt ctaatccact
 9541 catctgcac tctttgtgtc cttactaata cctcattggg tccaaattcc ctccctttaa
 9601 gcaccagctc gtttctgttc ttccacagcc tccaagtat ccaagggact aaagcctcca
 9661 cattcttcag atcaggatat tcttgtttaa gatgttgaac tctatggagg tttgtatgaa
 9721 ctgatgatct aggaccggat aagttccctt cttcatagcg aacttattca aagaatgttt
 9781 tgtgtatcat tcttgttaca ttgttattaa tgaaaaata ttattggtca ttggactgaa
 9841 cagcagtggt aaatatggac caggccccaa ataagatcca ttgatatatg aattaaataa
 9901 caagaataaa tcgagtcacc aaaccacttg ccttttttaa cgagacttgt tcaccaactt
 9961 gatacaaaag tcattatcct atgcaaatca ataatcatac aaaaatatcc aataacacta
 10021 aaaaaattaa aagaatgga taatttcaca atatgttata cgataaagaa gttacttttc
 10081 caagaaattc actgatttta taagccact tgcattagat aaatggcaaa aaaaaacaaa
 10141 aaggaaaaga aataaagcac gaagaattct agaaaatcag aaatacgtt caatgcagtg
 10201 ggacccacgg ttcaattatt gccaattttc agctccaccg tatattttaa aaataaaacg
 10261 ataatgctaa aaaaatataa atcgtaacga tcgttaaata tcaacggctg gatcttatga
 10321 cgaccgttaa ggaaattgtg gttgtcggac gaagtccagt aataaacggc gtcaaagtgg
 10381 ttgcagccgg cacacacgag tcgtgtttat caactcaaag cacaataact tttcctcaac
 10441 ctaaaaataa ggcaattagc caaaaacaac tttgcgtgta aacaacgctc aatacacgtg
 10501 tcattttatt attagctatt gcttcaccgc cttagcttcc tcgtgacctt gtcgtcctcg
 10561 tctttttctt tcttcttctt ataaaacaat acccaaagag ctcttcttct tcacaattca
 10621 gatttcaatt tctcaaatc ttaaaaactt tctctcaatt ctctctaccg tgatcaaggt
 10681 aaatttctgt gttccttatt ctctcaaat cttcgatttt gtttctgttc gatcccaatt
 10741 tcgtatatgt tctttgggtt agattctgtt aatcttagat cgtagacgat tttctgggtt
 10801 tgategttag atatcatctt aattctcgat tagggtttca tagatatcat ccgatttggt
 10861 caaataattt gagttttgtc gaataattac tcttcgattt gtgatttcta tctagatctg
 10921 gtgttagttt ctagtgtgtg cgatcgaatt tgcgattaa tctgagttt tctgattaac
 10981 agatgcagat cttaatgaca aagcaccagt ggtctagtgg tagaatagta ccctgccacg
 11041 gtacagaccc gggttcgatt cccggctggg gcagctcagc cgggtgttag ggggttttag
 11101 agctagaaat agcaagttaa aataaggcta gtcggttacc aacttgaaaa agtggcaccg
 11161 agtcggtgca acaaagcacc agtgggtctag tggtagaata gtaccctgcc acggtacaga
 11221 cccgggtctg attcccggtt ggtgcaggta acggagatcc aataagggtt tagagctaga
 11281 aatagcaagt taaaataagg ctagtccgtt atcaactga aaaagtggca ccgagtcggt
 11341 gcaacaaagc accagtggtc tagtggtaga atagtaccct gccacggtac agaccgggt
 11401 tcgattcccc gctgggtcag gagtgtacat gcctcctccg ttttagagct agaaatagca
 11461 agttaaataa aggctagtcc gttatcaact tgaaaaagtg gcaccgagtc ggtgccaaca
 11521 aagcaccagt ggtctagtgg tagaatagta ccctgccacg gtacagaccc gggttcgatt

-continued

SEQUENCES

11581 cccggctggg gcagcttctc tagctagagt cgategacaa gctcgagttt ctccataata
 11641 atgtgtgagt agttcccaga taagggaatt aggggttccta tagggtttcg ctcatgtgtt
 11701 gagcatataa gaaaccotta gtatgtattt gtatttgtaa aatacttcta tcaataaaat
 11761 ttctaattcc taaaaccaa atccagtact aaaatccaga tcgctttacg aattcccatg
 11821 gggagggtcaa catgggtggag cacgacactc tggctactc caaaaatgtc aaagatacag
 11881 tctcagaaga tcaagggtt attgagactt ttcaacaaag gataatttcg ggaaacctcc
 11941 tcggattcca ttgccagct atctgtcact tcatcgaaag gacagtagaa aaggaagggtg
 12001 gctcctacaa atgccatcat tgcgataaag gaaaggctat cattcaagat ctctctgccg
 12061 acagtgggtc caaagatgga cccccacca cgaggagcat cgtggaaaaa gaagagggtc
 12121 caaccacgtc tacaagcaa gtggattgat gtgataacat ggtggagcac gacactctgg
 12181 tctactccaa aaatgtcaaa gatacagtct cagaagatca aagggtctatt gagacttttc
 12241 aacaaaggat aatttcggga aacctctcg gattccattg cccagctatc tgtcacttca
 12301 tcgaaaggac agtagaaaag gaaggtggct cctacaaatg ccatcattgc gataaaggaa
 12361 aggtatcat tcaagatctc tctgcccaga gtggtcccaa agatggacct caccaccaga
 12421 ggagcatcgt ggaaaaagaa gaggttccaa ccacgtctac aaagcaagtg gattgatgtg
 12481 acatctccac tgacgtaagg gatgaacac aatcccacta tccttcgcaa gaccttcct
 12541 ctatataagg aagtctctt catttgaga ggacacgtc gagtataaga gctcattttt
 12601 acaacaatta ccaacaacaa caaacaacaa acaacattac aattacattt acaattatcg
 12661 atacaatggt caagactaat ctttctctt tcttcattt ctgcttctc ctctcacttt
 12721 ccagcgcgga catgggtgtc aagggcgaag aggtgattaa agagttcatg aggttcaaa
 12781 tgagaatgga gggcagtatg aacggccatg agttcgagat cgaagggtgaa ggccaaggta
 12841 gaccatacga gggcaccag acggcgaaac tgaaagtac gaaggcggc cactgccct
 12901 tcgcctggga cattctgagt ccacaattta tgtacggttc caaggcatac gtttaagcacc
 12961 ccgcccagat ccctgattac aaaaagctgt cgttcccga ggggtttaaa tgggaaagag
 13021 tgatgaattt tgaggacgga gggctcgtca ctgtgactca agactcctca cttcaagacg
 13081 gcacccta at ttacaagggt aagatgaggg gaacaaattt ccccccctgat ggccctgtta
 13141 tgcagaaaaa aacaatggga tgggaggcat ccaccgagag actttaccg cgcgacggag
 13201 tgctcaaagg cgaaatccat caggcactga aactaaagga cgcgggccac tacctgggtg
 13261 agttcaaaac tatatatatg gccaaagaa cgtccagtt gccgggatat tactatgtgg
 13321 acacaaaact ggatataacg tcccacaacg aagattacac catagtggaa cagtatgagc
 13381 gcagtgaggg tcgtcaccat ctattctctg gacatggcac gggttccacg ggctcgggtt
 13441 ccagcggcac ggcaagctcg gaggacaaca acatggcggg gatcaaagag ttcattgcgtt
 13501 ttaaagtcg tatggagggc agcatgaatg ggcacgaatt tgaaattgag ggagaaggcg
 13561 agggccgacc atatgagggg acacaaaccg cgaaactaaa agtcacaaaa ggtgggccc
 13621 ttctttttg ctgggacatt ctctctctc agttcatgta tggctccaag gctacgtta
 13681 aacatccggc tgatatcccc gattataaga aattgtctt cccggagggg tttaaatggg
 13741 agcgggttat gaatttcgag gatgtggac tggttactgt tacacaagat tcgagtctgc

-continued

SEQUENCES

13801 aggatggcac attgatatac aaagtcaaaa tgaggggtac gaattttccc ccggacggcc
 13861 cgggtgatgca gaaaaaaact atgggctggg aggccagcac ggagaggctg tacccgaggg
 13921 acgggggtttt gaaaggggag atacaccagg cgcttaaact gaaggacggg gggcactatc
 13981 tgggtggaatt taagacgata tatatggcga aaaagcccggt gcagctacct ggctactact
 14041 acgtcgatac caaactcgat attacctctc ataacgagga ctacactatc gtggagcaat
 14101 atgaaagatc agaaggacgt caccaccttt ttctatacgg aatggacgag ctttacaaaa
 14161 aggacgagtt gtaagcttgt cctgctttaa tgagatatgc gagaagccta tgatcgcatg
 14221 atatttgctt tcaattctgt tgtgcacgtt gtaaaaaacc tgagcatgtg tagctcagat
 14281 ccttaccgcc gggttcgggt cattctaattg aatatatcac ccgttactat cgtattttta
 14341 tgaataatat tctccgttca atttactgat tgtaccctac tacttatatg tacaatatta
 14401 aaatgaaac aatataattgt gctgaatagg tttatagcga catctatgat agagcgccac
 14461 aataacaaac aattgcgttt tattattaca aatccaattt taaaaaagc ggcagaacgg
 14521 gtcaaaccta aaagactgat tacataaatc ttattcaaat ttcaaaagtg ccccgagggc
 14581 tagtatctac gacacaccga gcggcgaaact aataacgctc actgaaggga actccgggtc
 14641 cccgccggcg cgcatgggtg agattccttg aagttgagta ttggccgtcc gctctaccga
 14701 aagttacggg caccattcaa cccggtccag cacggcggcc gggtaacgga cttgctgccc
 14761 cgagaattat gcagcatttt ttgtgtgtat gtgggcccga aatgaagtgc aggtcaaaac
 14821 ttgacagtga cgacaaatcg ttgggcggggt ccaggcgcaa ttttgcgaca acatgtcgag
 14881 gctcagcagg accgctcaga gaggatgcac atgtgaccga gggacacgaa gtgatccgtt
 14941 taaactatca gtgtttgaca ggatatattg gcgggtaaac ctaagagaaa agagcgttta
 15001 ttagaataat cggatattta aaaggcggtg aaaagggtta tccgttcgtc catttgtatg
 15061 tgccagccgc ctttgcgacg ctacccgggc tggttgccct cgccgctggg ctggcgggcg
 15121 tctatggccc tgcaaacgcg ccagaaacgc cgtcgaagcc gtgtgcgaga cacccgggcc
 15181 gccggcggtt tggatacctc gcggaaaact tggccctcac tgacagatga gggcgggacg
 15241 ttgacacttg aggggcccgc tcaccggcg cggcgttgac agatgagggg caggctcgat
 15301 ttcggccggc gacgtggagc tggccagcct cgcaaatcgg cgaaaacgcc tgattttacg
 15361 cgagtttccc acagatgatg tggacaagcc tggggataag tgccctgagg tattgacact
 15421 tgagggcgcg gactactgac agatgagggg cgcgatcctt gacacttgag gggcagagtg
 15481 ctgacagatg agggcgccac ctattgacat ttgaggggct gtccacaggc agaaaaacca
 15541 gcatttgcaa gggtttccgc ccgtttttcg gccaccgcta acctgtcttt taacctgctt
 15601 ttaaaccaat atttataaac cttgttttta accagggtg cgccctgtgc gcgtgaccgc
 15661 gcacgcggaa ggggggtgcc ccccttctc gaacctccc gggccgctaa cgcgggcctc
 15721 ccatccccc aggggtgctg cccctcgcc gcgaacggcc tcacccaaa aatggcagcg
 15781 ctggccaatt cccgagtgcg cggaaccctt atttgtttat ttttctaaat acattcaaat
 15841 atgtatccgc tcatgagaca ataacctga taaatgcttc aataatattg aaaaaggaag
 15901 agtatggcta aaatgagaat atcaccggaa ttgaaaaaac tgatcgaaaa ataccgctgc
 15961 gtaaaagata cggaaggaaat gtctcctgct aaggtatata agctggtggg agaaaaagaa

-continued

SEQUENCES

16021 aacctatatt taaaaatgac ggacagccgg tataaaggga ccacctatga tgtggaacgg
 16081 gaaaaggaca tgatgctatg gctggaagga aagctgcctg ttccaaaggt cctgcacttt
 16141 gaacggcatg atggctggag caatctgctc atgagtgagg ccgatggcgt cctttgctcg
 16201 gaagagtatg aagatgaaca aagccctgaa aagattatcg agctgtatgc ggagtgcac
 16261 aggctctttc actccatcga catatcggat tgtccctata cgaatagctt agacagccgc
 16321 ttagccgaat tggattactt actgaataac gatctggccg atgtggattg cgaaaactgg
 16381 gaagaggaca ctccatttaa agatccgcgc gagctgtatg attttttaa gacgaaaag
 16441 cccgaagagg aacttgtctt ttcccacggc gacctgggag acagcaacat ctttgtgaaa
 16501 gatggcaaa taagtggctt tattgatctt gggagaagcg gcagggcgga caagtggat
 16561 gacattgect tctgcgtccg gtcgctcagg gaggatcgc ggaagaaca gtatgtcgag
 16621 ctatcttttg acttactggg gatcaagcct gattgggaga aaataaaata ttatatttta
 16681 ctggatgaat tgttttagct gtcagaccaa gtttactcat atatacttta gattgattta
 16741 aaacttcatt ttaatttaa aaggatctag gtgaagatcc ttttgataa tctcatgacc
 16801 aaaatccctt aacgtgagtt ttcgttcac tgagcgtcag accccgtaga aaagatcaaa
 16861 ggatctctct gagatccttt tttctgcgc gtaatctgct gcttgcaaac aaaaaacca
 16921 ccgtaccag cgggtggttg tttgcggat caagagctac caactcttt tccgaaggtta
 16981 actggcttca gcagagcgca gataccaaat actgtccttc tagttagacc gtagttaggc
 17041 caccacttca agaactctgt agcacgcct acatacctcg ctctgtaat cctgttacca
 17101 gtggtgctg ccagtggcga taagtcgtgt cttaccgggt tggactcaag acgatagtta
 17161 ccgataaag ccgagcggc gggctgaacg ggggggtcgt gcacacagcc cagcttgagg
 17221 cgaacgacct acaccgaact gagataccta cagcgtgagc tatgagaaa cgcacgctt
 17281 cccgaaggga gaaaggcgga caggatccg gtaagcggca gggtcggaac aggagagcgc
 17341 acgagggagc ttccaggggg aaacgcctgg tatctttata gtctgtcgg gtttcgccac
 17401 ctctgacttg agcgtcgatt tttgtgatgc tcgtcagggg ggcggagcct atggaaaaac
 17461 gccagcaacg cggccttttt acggttcctg gcagatccta gatgtggcgc aacgatgccg
 17521 gcgacaagca ggagcgcacc gacttcttcc gcataaagt ttttggtctt caggccgagg
 17581 cccacggcaa gtatttgggc aaggggtcgc tggattcgt gcagggcaag attcggaata
 17641 ccaagtacga gaaggacggc cagacggtct acgggaccga cttcattgcc gataaggtgg
 17701 attatctgga caccaaggca ccaggcgggt caaatcagga ataagggcac attgccccgg
 17761 cgtgagtcgg ggcaatcccc caaggagggt gaatgaatcg gacgtttgac cggaaggcat
 17821 acaggcaaga actgatcgac gcgggggttt ccgcccagga tgccgaaacc atcgcaagcc
 17881 gcacgctcat gcgtgcgccc cgcgaaacct tcagtcctg cggtcgtatg gtccagcaag
 17941 ctacggccaa gatcgagcgc gacagcgtgc aactggctcc cctgcccctg cccgcgccat
 18001 cggccgcctg ggagcgttcg cgtcgtcttg aacaggaggc ggcaggtttg gcgaagtcga
 18061 tgaccatcga cagcgaggga actatgacga ccaagaagcg aaaaaccgcc ggcgaggacc
 18121 tggcaaaaca ggtcagcgag gccaaagcgg ccgcgttgct gaaacacacg aagcagcaga
 18181 tcaaggaaat gcagctttcc ttgttcgata ttgcgcgtg gccggacacg atgcgagcga

-continued

SEQUENCES

18241 tgccaaacga caggcccg cctgcccgtg tcaccacg cg caacaagaaa atcccgcgcg
 18301 aggcgcgtga aaacaaggtc attttccacg tcaacaagga cgtgaagatc acctacaccg
 18361 gcgtcgagct gcgggcccgc gatgacgaac tgggtgggca gcagggtgtg gagtacgcga
 18421 agcgcacccc tatcgccgag ccgatcacct tcacgttcta cgagctttgc caggacctgg
 18481 gctggtcgat caatggccgg tattacacga aggcggagga atgcctgtcg cgcctacagg
 18541 cgacggcgat gggcttcacg tccgaccg cg tgggcacct ggaatcggtg tcgctgctgc
 18601 accgcttcgg cgtcctggac cgtggcaaga aaacgtcccg ttgccagggtc ctgatcgacg
 18661 aggaaatcgt cgtgctgttt gctggcgacc actacacgaa attcatatgg gagaagtacc
 18721 gcaagctgtc gccgacggcc cgacggatgt tcgactatct cagctcgac cgggagccgt
 18781 acccgctcaa gctggaaacc ttccgctca tgtgcggatc ggattccacc cgcgtgaaga
 18841 agtggcgcca gcaggtcggc gaagcctgcg aagagttgcg aggcagcggc ctggtggaac
 18901 acgcctgggt caatgatgac ctggtgcatt gcaaacgcta gggcctgtg gggtcagttc
 18961 cggtggggg ttcagcagcc cctgctcgga tctgttgga cggacagtag tcatggttga
 19021 tgggctgctt gtatcgagt gtgattttgt gccgagctgc cggtcgggga gctgttggt
 19081 ggctggtggc aggatattt gtggtgtaaa caaattgacg cttagacaac ttaataacac
 19141 attgcggacg tttttaatgt actgggggtg aacactct

SEQUENCE LISTING

Sequence total quantity: 46

SEQ ID NO: 1 moltype = AA length = 424
 FEATURE Location/Qualifiers
 source 1..424
 mol_type = protein
 organism = Glycine max

SEQUENCE: 1

MLQDVVHPST PAEQLPIDEI SGPISARIFE LCDPDPFPHT LQNSEVTSSS NCCHEEKSSY 60
 ATTISPPLDV VDNKFNINS NSSNIVTTS SSTTTTSTT NNNNNATNGN NLSIFFDTQD 120
 EIDNDISASI DFSSSPSFVV PPLLPISTQQ DQDFPSPAQP QVQLSTAAGS ILTGLSHYPT 180
 DPVIAPLIGA PLPSVFDDEC ISSIPSYVPL NPSSPSCSYL SPGIGVYMPP PGSINTALSA 240
 DSSGLFGGNI LLGSELQAHE LDYQGENGGM YCTDSIQRVF NSPDLQALGN ESQKLAVAGAG 300
 SSATLAPEIS HLEDSTLKV G KLSVEQRKEK IHRYMKRNE RNFSKKIKYA CRKTLADSRP 360
 RVRGRPAKND DFGESHKQGS SNHEDDDEEI IVKEDDDMVD SSDIFAHISG VNSFKCNYSI 420
 QSLI 424

SEQ ID NO: 2 moltype = DNA length = 1275
 FEATURE Location/Qualifiers
 source 1..1275
 mol_type = genomic DNA
 organism = Glycine max

SEQUENCE: 2

atgttgacag atgtgtgtcca cccctcaaca ccggctgagc aactcccat tgatgagatt 60
 tcaggcccca ttagtgctcg aattttcgaa ctttgcgacc ccgatttctt cccacacaca 120
 ctgcaaaatt ctgaggttac ctccagctca aattgttgcc atgaagagaa gtctcatat 180
 gccacaacca tatctccacc tttagatgta gtagacaaca ataagttcaa tatcaatagc 240
 aatagcagca acatagtcac cactacotca tctagcacta ccacaaccag caccacaacc 300
 aacaacaaca acaacgcaac gaacggcaat aatctttcta tcttctttga cactcaagat 360
 gaaattgaca atgacatctc agcctccata gaattctcat catctccatc tttgtcgtt 420
 ccaccacttc tcccaatctc aactcagcag gatcagtttg atttcccttc agctcagcca 480
 cagggtgcaac tatcaacagc agcaggttca attttgacgg gcctctctca ctaccctaca 540
 gatcctgtga ttgcaccctt tattggagct ccgttaccat ctgtttttga tgatgattgc 600
 atatcttcca tcccttctta tgtgctctc aacccttcat caccctcttg ctcttatctc 660
 agtctggcca taggagtgta catgccacct cctggttccc ttaacactgc cttatctgct 720
 gacagtctct gattgtttgg tgggaacatt ctactggggg ctgaactgca ggcacatgaa 780

-continued

ttggactacc	agggagaaaa	tgggtggaatg	tattgtacag	attcaattca	aagggtgttt	840
aactccccag	accttcaggc	acttggtaat	gagagtcaga	aacttgtagc	tggggctgga	900
agctctgccca	ctttggcacc	agaaatctca	cacttggagg	actctacctt	gaagggttga	960
aaactctctg	ttgagcagag	gaagggaaaag	attcatagat	acatgaagaa	gagaaaacgaa	1020
agaaatttca	gcaagaaaaat	caagtatgct	tgccgcacaaa	ctttggcgga	tagccggccc	1080
cgggttagag	gaagggttgc	aaagaatgat	gacttggag	agagccataa	acaaggaagt	1140
agcaatcatg	aagatgatga	tgaagagata	attgtgaaa	aagatgatga	tatgggtgat	1200
tcttcagata	tctttgcaca	tatcagcgga	gtgaactctt	tcaaatgcaa	ctattccatc	1260
cagtccttga	tttga					1275

SEQ ID NO: 3 moltype = DNA length = 5035
 FEATURE Location/Qualifiers
 source 1..5035
 mol_type = genomic DNA
 organism = Glycine max

SEQUENCE: 3

catcatctta	gctaacacaa	cacagcatac	acaacttctc	tctctctact	actttctttt	60
tatgtcacct	tcttgatgtc	tttaatttgt	tttctttttt	agaaaaaaat	taatcctttt	120
taaggcttta	atttcttaag	catgcaatat	tattgttttt	agtcaccttc	aagttgagga	180
acataatac	gtttgtgacc	acaccaaatt	ccacttcttt	ctaagtgtgt	gtacatagat	240
tttgttttat	atatatttga	gggtccaccct	tcatgtccctc	cttaattctg	tgcaaaagag	300
gattcatcac	caccatgttg	caggatgttg	tccacccctc	aacacgggt	gagcaactcc	360
ccattgtaat	cacctcaagc	ttgttacatt	tcatatctcg	atcatgtata	tttagtgta	420
atgcattgat	tggaaactaag	ctatatatgt	atgtgtgtat	gtatctttgt	gtcatgcaat	480
atttatgacc	acagctaaat	ggtttccttc	tgggttttgt	gatcgtgtgt	aacttgacaa	540
gtttgtttgg	atcaagggtc	tttctttttc	tttcaaaata	atgatttcac	actccaagag	600
gtgttaaac	ttgggaggga	agagaagac	aggaagaga	cccagaaaa	gaacaaaaga	660
agttctaaat	tgatgtacgg	attaatagcg	tgtttgaaaa	ccatttgaga	agtaagaaaa	720
tctcatctga	gatatgaaag	ttatgactat	taacttctca	ataaactcaa	gaaatcacat	780
ctgagatgtg	aaaccaaaca	tgctcttacg	aattgtagaa	attttgatgg	ttttctctct	840
gggtactctt	tcttatattt	tgaggccaat	tattactaat	ttctctagga	attattcgat	900
caattgtctc	attcctgcat	tttaattctt	tccatgcatt	aatcacgaat	atgggtttgt	960
taatgtttgt	tggcatgtta	attaatttca	tctcaattag	ttctctgaga	aacctgaaca	1020
atatttccct	tatttggtta	tattctctga	gctaagtggg	agctgctagt	gctattagcc	1080
atcttagaaa	accccaagc	catgctcact	atttgtggga	cggaaacgtt	tctttctctc	1140
atttatattg	ttggagagca	taacgtaatt	ttattgaccg	aacaataagg	aggaatcaaa	1200
gactccgtga	ccgtgctgtc	aactcctttg	atattttatta	tcataccaatg	ggccctacca	1260
ccaaaatgtt	taatgtcatt	ttccaagggg	caaagtgtga	aaaatgatgt	aacatttccg	1320
tgatagtaat	ttagacatag	cttccgggtc	ttgtagaata	tatatatctc	tttgggggct	1380
tgttgtgtaa	gctttgttaa	tttttgatat	tctttattca	gttagatagg	gagaagaggt	1440
cttagtgaca	aacaattcat	atggatagca	tatatagcat	tatgggggtc	tcttcaagt	1500
gctttcaatt	cccatttaag	ttcaaaattt	ccacttaaaa	tgaacaaaaa	taacatagca	1560
tgaggttcct	tgattgtttt	tgcatacttg	tatttcatat	taacattggc	tacottggct	1620
cctacctgtc	acatgaaggc	atagatgcac	atcactttca	ccaataaata	acccaacaat	1680
gcaaacctct	taggaaaatt	tctcttcagt	ctattaccaa	atatggacaa	aactgactag	1740
actatgcaac	caaccacagt	catatgtaga	attcttcgtt	ggggcctttg	cacatcgatt	1800
tatatgtttt	tctctctgat	aaagaaggaa	gggggtgttg	tggcagatac	tattatatca	1860
tttggcatgt	cttccacttt	tgaatgccc	ctacatctac	aagccaaagc	ttcaattgat	1920
tagagaatat	tagcttttga	atttcttttc	tgttctaagt	gatcagatca	gactcatgtc	1980
tatcaacaaa	aagaggaatt	ggattcagat	ttcaaaccca	tcataatgaa	gaaaaaaata	2040
aaaaaaataa	actatagaga	atcacgtttc	ctagattgtt	tataaagaag	aggcttttgt	2100
aagacttatt	tcaggctgtg	tttttagtagt	ttatggtaaa	catgttgttc	acaatgotta	2160
tatgtcacca	ctcattgcag	gatgagattt	caggcccgat	tagtgctcga	attttcgaac	2220
tttgcgaccc	cgattttctc	ccacacacac	tgcacaaatc	tgagggtacc	tcagctcaa	2280
attgttgcca	tgaagagaa	tctctatag	ccacaacccat	atctccacct	ttagatgtag	2340
tagacaacaa	taagtccaat	atcaatagca	atagcagcaa	catagtcacc	actacctcat	2400
ctagcactac	cacaaccagc	accacaacca	acaacaacaa	caacgcaacg	aacggcaata	2460
atctttctat	cttcttttga	actcaagatg	aaattgacaa	tgacatctca	gctccatag	2520
acttctctac	atctccatct	tttgtcgttc	caccacttct	cccaatctca	actcagcagg	2580
atcagtttga	tttcccttca	gtccagccac	agggtgcaact	atcaacagca	gcagggtcaa	2640
ttttgacggg	cctctctcac	taccctacag	atcctgtgat	tgacccctct	attggagctc	2700
cgttaccatc	tgtttttgat	gatgattgca	tatcttccat	cccttcttat	gtgctctca	2760
acccttcac	accctcttgc	tcttatctca	gtcctggcat	aggagtgtac	atgccacctc	2820
ctgggttcct	taaacatgcc	ttatctgctg	acagttctgg	attgttttgt	gggaacattc	2880
tactgggggt	tgaactgcag	gcacatgaat	tggactacca	gggagaaaa	gggtggaatg	2940
attgtacaga	ttcaattcaa	agggtgttta	actccccaga	ccttcaggta	tgtgcaattt	3000
cgcaagccaa	ttagagttta	atagacattc	attgtctggt	ataaaagtgt	ttacattatc	3060
aatcaatcag	ataccattgt	tgatataaat	tttaaaataa	ttgttataat	aattaataat	3120
ttaatgtact	tgataatttg	tgatttgata	ataatataaa	aaaaatttac	actgcattat	3180
tatttatttt	tctctgtcag	gtccaataaa	ctaaatgaaa	attttcagtt	ccaatgttca	3240
attgtgtcag	ataagggaaa	aagaagttta	ataatgctgc	aaagggtact	atacctgca	3300
gcagtgaagt	ttttatttta	aaatagaaga	ggctttatca	gaggtggact	tttgggggaa	3360
agctcagggt	ccacaaatct	ctaaactata	aactcatagg	tgccccatga	ccatcaaaata	3420
gtaggtagca	caagatatga	gtccatttat	aaagtcacat	gcattaaaaa	atactataaa	3480
tttggcctag	caagaaggaa	gaaccacttt	catccaaaag	aaaaatagaa	aaaaggataa	3540
taaactgtag	catcattaga	tagaaagacc	cacttcaagg	gtggcagttg	tatatctctt	3600

-continued

tctacagtct	ataaagttaa	tgtgcagttt	ttattgaata	agtaagaat	tgatctttaa	3660
ttataatttc	tctctcaggc	acttggtaat	gagagtcaga	aacttgtagc	tggggctgga	3720
agctctgcc	ctttggcacc	agaaatctca	cacttggagg	actctacctt	gaaggttgga	3780
aaactctctg	ttgagcagag	gaaggaaaag	atccatagat	acatgaagaa	gagaaacgaa	3840
agaaatttca	gcaagaaaat	caaggtaacta	catctgaaca	ccaacattaa	caaacaaatt	3900
tcaaatctta	tactgtttta	catgatttcc	aatctactgc	atcaaccaag	ccttatgcat	3960
attttcaaaa	ttcaactaat	gatgcaattt	tttttatata	aaaaaaatgc	agtatgcttg	4020
ccgcaaaatt	agagagaggg	ttggagtagc	gcctattgta	gagaagatgg	tggaataatg	4080
acttaggttg	tttgggcatg	tagagagaag	accggtagac	tctgtagtga	ggagagtaga	4140
ccagatggag	agaagacaaa	caattcgagg	cagaggaaga	cccaaaaaga	ctataagaga	4200
ggttataaaa	aaggatctcg	aaattaatgg	tttggataga	agtatggtag	ttgatagaac	4260
attatggcgg	aagttgatcc	atgtagccga	ccccacctag	tggtgataag	cgttgttgtt	4320
gttgttgttg	ttgtatgctt	gccgcaaaac	tttggcggat	agccggcccc	gggttagagg	4380
aagggtttga	agaaatgatg	actttggaga	gagccataaa	caagggaagt	gcaatcatga	4440
agatgatgat	gaagaggtta	gattccctta	atcggatact	gttgttcaac	ttgccttagt	4500
ctaaaaatta	aaatacaaaa	aaattcccca	tcacttttac	ctttcaatt	atttgatggc	4560
ataattcctt	gatgttatat	tccttccatt	ttttgtactt	gcagataaatt	gtgaaagaag	4620
atgatgatat	gggtgatctc	tcagatatct	ttgcacatat	cagcggagtg	aactctttca	4680
aatgcaacta	ttccatccag	tccttgattt	gaattaaatt	attagtttga	ctagtgaag	4740
cttatttata	taattagctt	ctgtagatta	attttggtag	gacacttttc	ccatcccggt	4800
tctctaaaa	ccgggttttag	tggtttgagt	aaactgaata	aatgggttca	aaataaatat	4860
accaataagt	taagttagtt	agaaacgtac	agaaattgga	aactgtatc	attttgtag	4920
atataatata	tacttttcat	taagtgtgac	cagaacatgg	agttgtgtta	accaagaaaa	4980
tttcagttta	cccccatcca	agactgatgt	aaccaattga	tgtagcttct	tttat	5035

SEQ ID NO: 4

FEATURE

source

moltype = DNA length = 12557

Location/Qualifiers

1..12557

mol_type = other DNA

organism = synthetic construct

SEQUENCE: 4

gtgccgaatt	cggatccagc	gatctggatt	ttagtactgg	attttggttt	taggaattag	60
aaattttatt	gatagaagta	ttttacaaat	acaaatacat	actaagggtt	tcttatatgc	120
tcaacacatg	agcgaaaccc	tataggaacc	ctaattccct	tatctgggaa	ctactccac	180
attattatgg	agaaactcga	gcttgtcgat	cgactctagc	tagagaagct	cactttccaa	240
caaccttagt	aatttctccc	ttaacataat	gaacaaattc	ttcaagttag	tcagctctag	300
aagcaagtct	atcttctctt	ttccaaagat	aagcttgtct	agcttcaaga	ataactgggt	360
gatattgagc	tggaagtctt	ttccatagccc	aatcagcagc	aacatccctt	ggagcaatct	420
ttccagtaac	agcagaatac	caaattctag	aaagagtag	aacaacattt	ctttcatctc	480
cagcccaatc	tggtggagaa	ttccaaagag	taagagtttc	attaagagct	tcaaaaagat	540
cttgttctgg	aactggatca	aaaagttctt	cagcagctgg	tccaacaaga	gcaacagaat	600
gttctctagc	cttagtaaga	agaatagcaa	gatcaatctc	aatagtagct	gggttcaaaa	660
ttccagcaag	aatatccatt	ctttgccatt	ctccaaattg	aagttctctc	ttagctggat	720
atctccatgg	aataatatca	tcatagaaca	caatagtaac	ttcaacagct	ctaagaattt	780
cagattctcc	tggaagagca	gaagtttcaa	gaagatcatt	aataagagct	cttctagtag	840
tttcatcaag	tctaacagta	acagtaacaa	gaagatcaat	atcagaatgt	ggcttaagtc	900
ctccatcaac	agcagatcca	taaaagtgaa	cagcaagaag	agtaggttca	agatgtcttt	960
caataactcc	aacaacttca	gaaagttgag	tagaaacttc	agcaataaca	gcttctctca	1020
tgcaagcagt	agcaaacatc	gcagaaattc	tgaaagaaca	gaacttttgc	atgaaaatag	1080
aatctctctt	agaacacagc	atagaattag	cagagttctt	aagcttctta	gatccgaaaa	1140
caagaaaaag	agatgactta	ggaactttgt	gcttatggaa	attagagtta	ggattaaag	1200
tttgaattcc	ttgagccta	ttgttaattt	gagccattgt	atcgataatt	gtaaatgtaa	1260
ttgtaattgt	gtttgtgtgt	tggtgtgtgt	ggtaattgtt	gtaaaaatga	gctcttatac	1320
tcgagcgtgt	cctctccaaa	tgaatatgaac	ttccttatat	agaggaaggg	tcttgcgaag	1380
gatagtggga	ttgtgcgtca	tcctttacgt	cagtgaggat	gtcacatcaa	tccacttgct	1440
ttgtagacgt	gggtggaacc	tcttcttttt	ccacgatgct	cctcggtggg	gggggtccat	1500
ctttgggacc	actgtcggca	gagagatctt	gaatgatagc	ctttccttta	tcgcaatgat	1560
ggcattttga	ggagccacct	tcttttctta	ctgtccttct	gatgaagtga	cagatagctg	1620
ggcaatggaa	ttccaggagg	tttcccga	ttatcctttg	ttgaaaagtc	tcaatagccc	1680
tttgatcttc	tgagactgta	tctttgacat	ttttggagta	gaccagagtg	tcgtgtctca	1740
ccatgttatc	acatcaatcc	acttgccttg	tagacgtggg	tggaacctct	tctttttcca	1800
cgatgctcct	cgtgggtggg	gggtccatct	tggtgacctc	gtcggcagag	agatcttgaa	1860
tgatagcctt	tcctttatcg	caatgatggc	attttagtag	gccaccttcc	ttttctactg	1920
tcctttcgat	gaagtgcag	atagctgggc	aatggaatcc	gaggaggttt	cccgaatta	1980
tcctttgttg	aaaagtctca	atagcccttt	gatcttctga	gactgtatct	ttgacatttt	2040
tgagtagaac	cagagtgtcg	tgctccacca	tggtgacctc	cgcaagaatt	caagcttagc	2100
gttccataca	cctggagaac	cttttatgta	cttcacaacc	ctcaatgctg	cttccagggt	2160
tgactgtttg	gggtgctgca	tgaattgact	cagcacctgc	actgcaaaag	tgatatctgg	2220
ccttgtgatt	gtcaagtaga	gaagctttcc	caccagcttc	tgataggatc	ccacatcttc	2280
taactctgca	tcacacgtct	ttcctaagtg	cttgtcataat	tctacagtag	tcagcctttg	2340
gttctgttcc	attgggtgtc	acactgggtc	gcagcctccc	agaccaacac	ctgatcatag	2400
ttccagtgca	tacttctctc	gggttcagt	gattcccttt	tctgatctca	gcacctcaat	2460
gcctaagaag	tattttagtt	ctcccaaatc	tttcatcttg	aaatgctgat	gcagggttgc	2520
ctttgcttct	gaaatcaaaa	cattgtctgt	gcctgttatt	aacagatcat	ccacataaat	2580
caggattatg	acaaggtcag	tcctctctct	tttggtaaac	aaggagtgat	cataagcact	2640
ttgcataaaa	ccagcctgca	taaggacagt	ggtaagcttg	atgttccact	gccttgatgc	2700

-continued

ttgctttaaa	ccatagaggg	attcaacagc	ctgcacactt	tgggttcccc	ttggtgtgta	2760
aaaccctgag	gcagagacat	ataaacttct	tccatgaggt	caccttgtag	aaaagcattg	2820
ttgacatcca	tctggaaaag	gaaccagccc	ttggaagcag	caacagatat	gacagctttt	2880
acagtgacca	ttttggccac	tggaagaaaa	gtttcatggg	agtcagggcc	ttcttgctga	2940
gtgtatccct	tgggccactag	ccttgcccta	aacctgtcaa	cttcaccatt	agctttgtat	3000
ttaaatttgt	acacccattt	ggacccctata	ggctgtttac	cagggggtaa	aggggacaatc	3060
tcccaggtgt	tattatcctc	aagagcctgt	atctcaaggg	acatggcctc	catccatttc	3120
tcatcttgag	ctgcttcttt	gaaggattta	gggtcagtat	aagtgaggaa	agcactcaaa	3180
taagcttgat	agtgaacagg	caagtgatca	taggaaacat	agttggctat	agggtatgga	3240
acatccctag	agcctttgtt	cagtgtcaca	aagtccttga	gccagatggg	aggacctgca	3300
ttgctgttag	gtctagtggt	cagggtttgt	ggaactagag	agggatcagc	tacagcagta	3360
tgggtgctcaa	attcagcatt	tgctaggtca	ggctcagctg	actcaactga	ctctgcagga	3420
gcctgcaggt	ggcctgaagg	tgacgcatca	gctgaagtga	tatgattttg	ggtagaatga	3480
gtccccaag	tgatatccct	attggcctct	acaatgtcag	gaataagaac	tgagtatca	3540
tcatcatcag	tatgaaagct	ggaaggaaaa	atatcattat	acactagctg	caaagctgta	3600
tcttcagagc	taagtgaacc	tgctcgagtg	aacatatacag	gttcattggga	gatggactct	3660
ttgaaaggga	actgaaactc	tctgaaaact	acatccctgc	tcactatgat	caccttatta	3720
tccaagtcat	acaacctgta	acccttttga	gtctcagaat	aaccaatgaa	gatggttcgc	3780
ttggctctag	gtgcaagttt	gtcacctttg	ggcagtggtg	ctgcaaaagc	aagacacca	3840
aaactctca	agtgatcaag	cttggcatgt	ttctgataga	gtagttcata	tggaatttg	3900
ccttgtaaga	ttggagtagg	gagttctattg	atcagggtata	caacagctct	gacacagctc	3960
ccccaaaaac	ttgtaggtta	acactctga	aaactaagtg	ccctggccat	ctcaaggatg	4020
tgtctgtgct	ttctctccac	aaacaccattc	tgttgttggtg	tgtagggaca	gctactttga	4080
tgaacaatcc	caagagaggg	cagcaactca	ttacaacttc	tccactagaa	ttcgagctcg	4140
gagctgtagg	cggcgctagc	ctattctcac	ccctaattaa	gacaagctgt	aggctcgccc	4200
ctgttcctgt	tggtcatcaa	ttattgttta	accgagttaa	aggagatggg	caattttaca	4260
gtagtaataa	tgacaagaaa	tacacgaaat	ttgtgtttaa	gtattgaagg	acttccattt	4320
ttttgaatcc	accatctcaa	aactaatagt	gtctaatttt	cccgtaacaa	ttctttttgt	4380
ataaacattca	ataaatgtg	gttgacgata	tatatattagc	ttgcaaaatt	cctaaaaatt	4440
attgtcccaa	acttagtatt	cacttaggat	tggatttgtg	cattattgta	cttggcaagg	4500
gtacatgtgt	taatttatag	cacttaagcg	tgctattgta	cttggctact	taaaattggt	4560
ttttgttctc	cggatgatgt	tgatgagtta	taaaaaagaa	atatatgctc	tgaaggaaaag	4620
aaagtttatt	tttacggttt	atgttaaaat	acaatacttt	caactatgta	gattgtggag	4680
caaaacaagc	ttcagtttat	ctatcgtggt	ccgatgacac	gttaataatg	aacttttctc	4740
ataaacatca	attagaagat	gttgattttg	ttgggtcaatt	gggtgttattg	ttgtgactac	4800
tttatatttt	cgtagatttc	caagcattat	ttccgtctat	agggtttgct	cttaactgat	4860
tgttttgagt	tattaatatt	tcaagttgat	ttgattggaa	tagtaaatct	ggatctatta	4920
tgtaatatat	tagggagcaa	atcttgacac	ttgagaatga	tgtagtctcc	tcaaaatttt	4980
gataaccatt	gttttcgagt	gatagatgtt	gtctatgtca	aggttgtcga	tgaatttagc	5040
aatgatggac	agactttgtc	gaacaagttt	tgtaatgcc	tcacagaaaa	gcactagaa	5100
gtactgaaaa	acattttttt	taacatatac	ttttacaaaa	tcaagagaat	gaataattaa	5160
atatgaaata	agaattacaa	aaattttata	tttttttaat	aaatttcaaa	tcaataataa	5220
gaaagaatgg	tgaaaaatta	aagaatatgt	tggaagatgt	gttattagta	ttctgtgtga	5280
taagaatgct	tccaaaatta	aaactcttcc	ccaaaaatgt	atgagtaaaa	tttcaaatgt	5340
ctcgttttaa	atataacata	agctaaatta	aaccagctt	agtaaaaaaa	atgcattttc	5400
gtaaatttcg	atgtccaaac	aattgtaatg	acttaaaact	agaacaaaa	ctatttttat	5460
accatgtaaa	ttcttacaca	ttataatcgt	atgtttatcc	atccatgcac	cccaacaaat	5520
gaagaaaagg	aataaaaaaa	aaggtgaatt	tgccaaaaac	aaattaatgg	gatgctaaca	5580
ttacataaag	gcgccatagg	gcatacttta	attgtgaata	aaaaggcttg	gtcaacccaa	5640
ttttctctcc	tagttactag	ttattaacca	tgctatttgg	catatatata	ataatccata	5700
tccatataaa	aatacatgtc	ccaaaaattg	aattccacta	cacccaccca	aagtgttgaa	5760
ttaatggcca	aacgcacact	tcgcgttgac	accgcgagtg	gccaggcttg	acgactctac	5820
gtacaacactc	atcaattctc	ctatatatgc	acataactaa	ctttctctgt	ttttgacacc	5880
ctcatacacc	ccatcatctt	agctaacaca	acacagcata	cacaactctc	ctctctctac	5940
tactttcttt	ttatgtcaac	ttcttgatgt	ctttaatttg	ttttcttttt	tagaaaaaaa	6000
ttaatctctt	ttaaggcttt	aatttctctaa	gcattgcaata	ttattgtttt	tagtcacctt	6060
caagttgagg	aacatataca	cgtttgtgac	cacacccaaat	tccacttctt	tctaagtgtg	6120
tgtaacataga	ttttgtttta	tatatatttg	agggttcaccc	ttcattgcct	ccttaattct	6180
tgcaaaaaga	ggattcatca	ccaccccatg	taccatcacg	atgttctctga	ctatgcgggc	6240
tatccctatg	acgtcccgga	ctatgcagga	ttgtatccat	atgacgttcc	agattacgcc	6300
actagagctg	cttacccata	cgatgttctc	gactatgcgg	gctatcccta	tgacgtcccg	6360
gactatgcag	gattgtatcc	atatgacgtt	ccagattacg	caatgttgca	ggatgttgct	6420
cacccctcaa	caccggctga	gcaactcccc	attgatgaga	tttcaggccc	gatttagtgt	6480
cgaattttcg	aactttgcca	ccccgatttc	ttcccacaca	cactgcaaaa	ttctgaggtt	6540
acctccagct	caaattgttg	ccatgaagag	aagtcctcat	atgccacaac	catatctcca	6600
ccttttagatg	tagtagacaa	caataagttc	aatatcaata	gcaatagcag	caacatagtc	6660
accactacct	catctagcac	taccacaacc	agcaccacaa	ccaacacaaa	caacaacgca	6720
acgaacggca	ataacttttc	tacttctttt	gacactcaag	atgaatttga	caatgacatc	6780
tcagcctcca	tagactttctc	atcatctcca	tcttttgtcg	ttccaccact	tctcccaatc	6840
tcaactcagc	aggatcagtt	tgatttcccc	tcagctcagc	cacaggtgca	actatcaaca	6900
gcagcaggtt	caattttgac	gggcctctct	cactacccta	cagatcctgt	gattgacacc	6960
cttattggag	ctccgttacc	atctgttttt	gatgatgatt	gcatactctt	catcccttct	7020
tatgtgcctc	tcaacccctc	atcaccctct	tgctcttctc	tcagtcctgg	cataggagtg	7080
tacatgccac	ctcctgggtc	ccttaacact	gccttatctg	ctgacagtgc	tggaattgtt	7140
ggtgggaaca	ttctactggg	gtctgaactg	caggcacatg	aattggacta	ccaggagaaa	7200
aatggtggaa	tgtattgtac	agattcaatt	caaagggtgt	ttaactcccc	agaccttcag	7260

-continued

gcacttggta	atgagagtc	gaaacttcta	gctggggctg	gaagctctgc	cactttggca	7320
ccagaaatct	cacacttgg	ggactctacc	ttgaagggtg	gaaaactctc	tgttgagcag	7380
aggaaggaaa	agattcatag	atcacatgaag	aagagaaaacg	aaagaaattt	cagcaagaaa	7440
atcaagtatg	cttgccgcaa	aactttggcg	gatagccggc	cccgggttag	aggaagggtt	7500
gcaaagaatg	atgactttgg	agagagccat	aaacaaggaa	gtagcaatca	tgaagatgat	7560
gatgaagaga	taatttgtga	agaagatgat	gatatgggtg	attcctcaga	tatccttgca	7620
catatcagcg	gagtgaactc	ttccaatgc	aactattcca	tccagtcctt	gatttgagct	7680
tattaaatta	ttagtttgac	tagtgaaagc	ttatttatat	aattagcttc	tgtagattaa	7740
tttttgtagg	acacttttcc	catcccggtt	ctctaaaatc	cggttttagt	ggtttgagta	7800
aactgaataa	atgggggtcaa	ataaatata	ccaataagtt	aagttagtta	gaaacgtaca	7860
gaaattggaa	actgtataca	tttttgcaga	tatatattat	ctttttcatt	aagttgtacc	7920
agaacatgga	gttgtgttaa	ccaagaaaat	ttccagttac	ccccatccaa	gactgatgta	7980
accaattgat	gtagcttctt	ttatggtagt	caagcagatc	gttcaaacat	ttggcaataa	8040
agtttcttaa	tgtgaatcc	tgttgccgtt	cttgcgatga	ttatcatata	atttctgttg	8100
aattacgtta	agcatgtaat	aataacatg	taatgcatga	cgttatttat	gagatgggtt	8160
tttatgatta	gagtcccgca	attatacatt	taatacgcga	tagaaaacaa	aatatagcgc	8220
gcaaactagg	ataaattatc	gcgcgcgggt	tcatctatgt	tactagatcg	acgctttacg	8280
aggatgcaca	tgtgaccgag	ggacacgaag	tgtatccgtt	aaactatcag	tgtttgacag	8340
gatatattgg	cggtgtaaac	taagagaaaa	gagcgtttat	tagaataatc	ggatatttaa	8400
aagggcgtga	aaagggttat	ccgttcgtcc	atttgtatgt	gccagccgcc	tttgcgacgc	8460
tcaccgggct	ggttgccctc	gccgctgggc	tggcggccgt	ctatggccct	gcaaacgcgc	8520
cagaaacgcc	gtcgaagccg	tgtgcgagac	accgcggccg	ccggcgttgt	ggataacctcg	8580
cggaaaaact	ggccctcact	gacagatgag	gggcggacgt	tgacacttga	ggggccgact	8640
caccgcggcg	ggcggtgaca	gatgaggggc	aggctcgatt	tcggccggcg	acgtggagct	8700
ggccagcctc	gcaaatccgc	gaaaacgcct	gattttacgc	gagtttccca	cagatgatgt	8760
ggcaagcct	ggggataagt	gccctgcggt	attgacactt	gagggcgccg	actactgaca	8820
gatgagggcg	gcgatccctg	acacttgagg	ggcagatgct	tgacagatga	ggggcgccacc	8880
tattgacatt	tgaggggctg	tcacacggca	gaaaatccag	catttgcaag	ggtttccgcc	8940
cgtttttcgg	ccacccgtaa	cctgtctttt	aaactgcttt	taaaaccaata	tttataaac	9000
ttgtttttaa	ccaggggtgc	gccctgtgcg	cgtgaccgcg	cacgcgcaag	gggggtgccc	9060
cccttctcgc	aacctcccgc	gcccgctaac	gcgggcctcc	catcccccca	ggggctgcgc	9120
ccctcgccgc	gcaacggcct	cccccaaaa	atggcagcgc	tggccaattc	ccgagtgccg	9180
ggaaccccta	tttgtttatt	ttctaaaata	cattcaaaata	tgtatccgct	catgagacaa	9240
taacctgat	ataatgttca	ataatattga	aaaaggaaga	gtatggctaa	aatgagaata	9300
tcaccggaat	tgaaaaaact	gatcgaaaaa	taccgctgcg	taaaagatac	ggaaggaatg	9360
tctctgcta	aggtatataa	gctggtggga	gaaaatgaaa	acctatattt	aaaaatgacg	9420
gacagccggt	ataaaggagc	cacctatgat	gtggaaacgg	aaaaggacat	gatgctatgg	9480
ctgggaaggaa	agctgcctgt	tcctaaaggt	ctgcactttg	aacggcatga	tggctggagc	9540
aatctgctca	tgagttaggc	cgtatggcgt	ctttgtcctg	aagagatga	agatgaacaa	9600
agcctgaaaa	agattatcga	gctgtatgcg	gagtgcatac	ggctctttca	ctccatcgac	9660
atatcggtat	gtccctatag	gaatagctta	gacagccgct	tagccgaatt	ggattactta	9720
ctgaataacg	atctggccga	tgtggattgc	gaaaactggg	aagaggacac	tcattttaaa	9780
gatccgcgcg	agctgtatga	ttttttaaag	acggaaaaagc	ccgaagagga	acttgtcttt	9840
tcccacggcg	acctggggaga	cagcaacatc	tttgtgaaa	atggcacaag	aagtggcttt	9900
attgatcttg	ggagaagcgg	cagggcgagc	aagtggtatg	acattgcctt	ctgcgtccgg	9960
tcgctcaggg	aggatatccg	ggaagaacag	tatgtcgagc	tattttttga	cttactgggg	10020
atcaagcctg	attgggagaa	aataaaaat	tatatattac	tggatgaatt	gttttagctg	10080
tcagaccaag	ttactcata	tatactttag	attgatttaa	aaactcattt	ttattttaaa	10140
aggatctagg	tgaagatcct	tttgtataat	ctcatgacca	aaatccctta	acgtgagttt	10200
tcgttccact	gagcgtcaga	ccccgtagaa	aagatcaaa	gatcttcttg	agatcccttt	10260
ttcttcgcgc	taactctgct	cttgcaaaaca	aaaaaacacc	cgctaccagc	ggtggtttgt	10320
ttgcgggcat	aagagctacc	aactcttttt	ccgaaggtaa	ctggcttcag	cagagcgacg	10380
ataccaaaata	ctgtccttct	agtgtagccg	tagttaggcc	accacttcaa	gaactctgta	10440
gcaccgccta	catacctcgc	tctgctaata	ctgttaccag	tggctgctgc	cagtggcgat	10500
aagtctgtgc	ttaccgggtg	ggactcaaga	cgatagttac	cggataaagg	gcagcggctg	10560
ggctgaacgg	gggggttcgt	cacacagccc	agcttggagc	gaacgacctc	caccgaaact	10620
agataccctac	agcgtgagct	atgagaaaag	gccacgcttc	ccgaagggag	aaaggcgagc	10680
aggatccggg	taagcgccag	ggtcggaaaca	ggagagcgca	cgaaggagct	tccaggggga	10740
aacgcctggt	atctttatag	tctgtcggg	tttcgccacc	tctgacttga	gcgtcgattt	10800
ttgtgatgct	cgtcaggggg	gcggagcccta	tggaaaaacg	ccagcaacgc	ggccttttta	10860
cgggttctgg	cagatcctag	atgtggcgca	acgatgcggg	cgacaagcag	gagcgcacgc	10920
acttcttccg	catcaagtgt	tttggtcttc	aggccgaggc	ccacggcaag	tatttgggca	10980
aggggtcgct	ggtattcggt	cagggcaaga	ttcggaatac	caagtacgag	aaggacggcc	11040
agacgggtcta	cgggaccagc	ttcattgccc	ataaggtgga	ttatctggac	accaaggcac	11100
caggcggttc	aaatcaggaa	taagggcaca	ttgccccggc	gtgagtcggg	gcaatcccgc	11160
aaggagggtg	aatgaatcgg	acgtttgacc	ggaaggcata	caggcaagaa	ctgatcgacg	11220
cgggtttttc	cgccgaggat	gcgaaacca	tcgcaagccg	caccgtcatg	cgtgcgcccc	11280
gcgaaacctt	ccagtcctgc	ggctcgatgg	tccagcaagc	tacggccaag	atcgagcgcg	11340
acagcgtgca	actggctccc	cctgccctgc	ccgcgccatc	ggccgcctgt	gagcgttcgc	11400
gtcgtcttga	acaggagggc	gcaggtttgg	cgaagtctgat	gaccatcgac	acgcgaggaa	11460
ctatgacgac	caagaagcga	aaaaccgcgc	gcgaggacct	ggcaaaaacg	gtcagcgagg	11520
ccaagcagcg	cgcgttgtgt	aaacacacga	agcagcagat	caaggaaatg	cagctttcct	11580
tgttcgatat	tgcgccgtgg	ccggacacga	tgcgagcgat	gccaaacgac	acggcccgct	11640
ctgcctggtt	caccacgcgc	aacaagaaaa	tcccgcgcga	ggcgctgcaa	aacaagggtca	11700
ttttccagct	caacaaggac	gtgaagatca	cctacacggg	cgtcgagctg	cgggcgcgacg	11760
atgacgaact	ggtgtggcag	caggtgttgg	agtacgcgaa	gcgcacccct	atcggcgagc	11820

-continued

```

cgatcacctt cactgttctac gagctttgcc aggacctggg ctggtcgatc aatggccggg 11880
attacacgaa ggccgaggaa tgccctgtcgc gcctacaggc gacggcgatg ggettcacgt 11940
ccgaccgcgt tgggcacactg gaatcgggtgt cgcctgtgca ccgcttcgcg gtccctggacc 12000
gtggcaagaa aacgtcccggt tgccagggtcc tgatcgacga ggaaatcgtc gtgctgtttg 12060
ctggcgacca ctacacgaaa ttcatatggg agaagtaccg caagctgtcg ccgacggccc 12120
gacggatggt cgactatttc agctcgcacc gggagccgta cccgctcaag ctggaaacct 12180
tccgcctcat gtgcgggatcg gattccaccc gcgtgaagaa gtggcgcgag caggtcggcg 12240
aagcctgcga agagttgcga ggcagcggcc tgggtggaaca cgctgggtc aatgatgacc 12300
tggtgcattg caaacgctag ggccctgttg ggctcagttcc ggctgggggt tcagcagccc 12360
ctgctcggat ctggttgacc ggacagtagt catggttgat gggtgcctg tatcgagtgg 12420
tgattttgtg ccgagctgcc ggtcggggag ctggtggctg gctgggtggca ggataattg 12480
tggtgtaaac aaattgacgc ttagacaact taataacaca ttgcggacgt ttttaatgta 12540
ctgggggtga acactct 12557

```

```

SEQ ID NO: 5      moltype = AA length = 423
FEATURE
source           Location/Qualifiers
                 1..423
                 mol_type = protein
                 organism = Glycine max

```

```

SEQUENCE: 5
MLQDVIHPST PAEQLPIDEI SSPISARIFE LCEPDPFFPD LQNSDVTSSS NCCHEEKSSY 60
ATTISPPPLDL VDNKINNNNN SNIVTTTSSS TTTTSTTNN NNNNTTNSNN LSILFDTQDE 120
IDNDISASID FSSCRSLVVP PLLSISTQQD QPFDSSAQPO VQLSAAAGSV LKGLSHYPTD 180
HVIAPLIGSP LPSVFDIEDCI SSIPSYVPLN PSSPSCSYLS PGIGVYMPPP GSLNTALSAD 240
SSGLFGGNIL LGSELQAHLE DYQENGIGIF CTDSIQRVFN PPDQLQALGE TQKLAVAGAGS 300
SATLTPEISH LEDSNLKVKG LSVEQRKEKI HRYMKKRNER NFSKKIKYAC RKTLDASRP 360
VRGRFAKND EFGESHRQGS NHEEDDEEII VKEDDDMVDS SDIFAHISGV NSFKNYSIQ 420
SLI 423

```

```

SEQ ID NO: 6      moltype = DNA length = 1272
FEATURE
source           Location/Qualifiers
                 1..1272
                 mol_type = genomic DNA
                 organism = Glycine max

```

```

SEQUENCE: 6
atgttgacagg atgttatcca cccctcaaca ccggctgagc aactcccat tgatgagatt 60
tcaagcccga ttagtgctcg aattttcgaa ctttgcgagc ctgatttctt ccagacaca 120
ctgcaaaaatt cagatgttcc ttccagctca aattgttgcc atgaagagaa gtccctcatat 180
gccacaacca tatctccacc tttagattta gttagacaaca agatcaatat caataacaat 240
agcaacatag tcaactactac ctcatctagc actaccacaa ccagcaccac aaccaacaac 300
aacaacaaca acacaacgaa cagcaataac ctgtccatcc tctttgacac tcaagatgaa 360
attgacaatg acatctcagc ctccatagac ttctcatcat gtctgatttt agttgttcca 420
ccacttctct caatctcaac tcagcaggat cagtttgatt tctcttcagc tcagccacag 480
gtgcaactat cagcagcagc aggttcagtt ttgaagggcc tctctcacta ccctacagat 540
catgtgattg caccctctat tggatctccg ttaccatctg tttttgatga agattgcata 600
tcttccatcc cttcttatgt gccctcaca ccatcatcac cctcttgctc ttatctcagt 660
cctggcatag gagtgtacat gccctcctct gggtccctta aactgcctt atctgctgac 720
agttctggat tggttggtgg gaacattcta ctggggctcg aactgcaggc acatgaattg 780
gactatcagg gagaaaatgg tggaaatatt tgcaacagatt caattcagag ggtgtttaac 840
ccccagatc tcaggcactc tggtactgag actcagaaac ttgtagctgg ggttggaagt 900
ctgtccactt tgacaccaga aatctcacac ttggaggact ctaacttgaa agttggaaaa 960
ctctctgttg agcagaggaa ggaaaagatt catagataca tgaagaagag aaatgaaaga 1020
aatctcagca agaaaaatcaa gtatgcttgc cgcaaaactt tagcagatag ccggccccgg 1080
gttagaggaa ggtttgcaaa gaatgatgag tttggagaga gccatagaca aggaagtagc 1140
aatcatgaag aagatgatga agaaataatt gtgaaagaag atgatgatg ggttgattcc 1200
tcagatatct ttgcacatat cagtgaggatg aactctttca aatgcaacta ttccatccag 1260
tccttgattt ga 1272

```

```

SEQ ID NO: 7      moltype = DNA length = 19178
FEATURE
source           Location/Qualifiers
                 1..19178
                 mol_type = other DNA
                 organism = synthetic construct

```

```

SEQUENCE: 7
gtgccgaatt cggatccagc gatctggatt ttagtactgg attttggttt taggaattag 60
aaattttatt gatagaagta ttttacaat acaaatatcat actaagggtt tcttatatgc 120
tcaacacatg agcgaaaccc tataggaacc ctaattccct tatctgggaa ctactcacac 180
attattatgg agaaactcga gcttgctgat cgaactcagc tagagaagct cactttccaa 240
caaccttagt aatttctccc ttaacataat gaacaaattc ttcaagttag tcagctctag 300
aagcaagtct atcttcttct tgcctcaagt aagcttgctc agcttcaaga ataactgggt 360
gatattgagc tggaggtctt tccatagccc aatcagcagc aacatccttt ggagcaatct 420
ttccagtaac agcagaatac caaattctag aaagagtaga aacaacattt ctttcatctc 480
cagcccaatc tgggtggagaa ttccaaagag taagagtctt attaagagct tcaaaaagat 540
cttgttctgg aactggatca aaaagtctct cagcagctgg tccaacaaga gcaacagaa 600
gttctctagc cttagttaga agaatagcaa gatcaatct aatagtagct ggttcaaaaa 660
ttccagcaag aatatcattt ctttgccatt ctccaaattg aagttctctc ttagctggat 720

```

-continued

atctccatgg	aataatatca	tcatgaacaa	caatagtaac	ttcaacagct	ctaagaattt	780
cagatttctcc	tggagaagca	gaagtttcaa	gaagatcatt	aataagagct	cttctagtag	840
tttcatcaag	tctaacagta	acagtaacaa	gaagatcaat	atcagaatgt	ggcttaagtc	900
ctccatcaac	agcagatcca	taaagatgaa	cagcaagaag	agtaggttca	agatgtcttt	960
caataactcc	aacaacttca	gaaagttgag	tagaaacttc	agcaataaca	gcttctctca	1020
tgaagcagct	agcaacagaa	gcagaatttc	tgaagaacaa	gaacttttgc	atgaaaaatg	1080
aatcttctt	aagaacaagc	atagaattag	cagagttctt	aagcttctta	gatccgaaaa	1140
caagaaaaaga	agatgactta	ggaacttggt	gcttatggaa	attagagtta	ggattaagag	1200
tttgaattcc	ttgagccata	ttgttaattt	gagccattgt	atcgataatt	gtaaatgtaa	1260
ttgtaattgt	gtttgttgtt	tgttgttgtt	ggtaattgtt	gtaaaaatga	gctcttatac	1320
tcgagcgtgt	cctctccaaa	tgaatgaa	ttccttatat	agaggaaggg	tcttgcaag	1380
gatagtggga	ttgtgcgtca	tcctttacgt	cagtggagat	gtcacatcaa	tcacttgc	1440
ttgtagacgt	ggttggaacc	tcttctttt	ccacgatgct	cctcgtgggt	gggggtccat	1500
ctttgggacc	actgtccgca	gagagatctt	gaatgatagc	ctttccttta	tcgcaatgat	1560
ggcattttgt	ggagccacct	tccttttcta	ctgtccttcc	gatgaagtga	cagatagctg	1620
ggcaatggaa	tcagaggagg	tttccgaaa	ttatccttgg	tgaaaagtc	tcaatagccc	1680
tttgatcttc	tggactgta	tctttgacat	ttttggagta	gaccagagtg	tcgtgctcca	1740
ccatgttato	acatcaatcc	acttgccttg	tagacgtggg	tggaacctct	tctttttcca	1800
cgatgctct	cgtgggtggg	ggtccatctt	tgggacct	gtcggcagag	agatcttgaa	1860
tgatagcctt	tcctttatcg	caatgatggc	attttagtag	gccaccttcc	ttttctactg	1920
tcctttcgat	gaagtgcag	atagctgggc	aatggaatcc	gaggagggtt	cccgaaatta	1980
tcctttgttg	aaaagtctca	atagcccttt	gatcttctga	gactgtatct	ttgacatttt	2040
tggagttagac	cagagtgtcg	tgtccacca	tgttgacctc	cgaagaatt	caagcttagc	2100
gctgtgttgg	atcatctaaa	tgtgcattcc	gcaagccaat	taatgttcca	actgaagcag	2160
attgcttaca	taatttatat	gcataccaa	atatgcttca	cgagcataat	ttttaagttg	2220
ctacctcttg	gtaaggggca	gctaattgaa	actaggatcc	atgatctccg	tttgttcaaa	2280
aatctcttgg	tgtgttttca	aaatatcaat	ttagaatcta	atccaagagt	tctccctacc	2340
taataaataat	ctttaacctca	taataccaaa	atctaataat	tcgtgggatg	atcacatcc	2400
ttcacacaa	tcattccctgc	ttttctctcg	tgataagcat	gaagaataaa	aaacccacta	2460
atatacgcca	attcactgga	ttgagaacaa	aattcaaaaca	caatatgcca	gcatacagtt	2520
gataattaca	tcctccacct	acaaattgat	aacaacacgt	agtaacatta	atgggacaat	2580
agagttctac	acaaaccaa	cttattttca	aaggtcatta	cagaccatt	acacatcaca	2640
tttcttctta	aaatgtttgg	gaaaatccta	catcgaaat	cacacagaaa	tcagctccag	2700
caagctctac	tcctccacct	tccttctctc	tttggttgca	gatctctctc	cagttgggac	2760
agatcgatcc	tagtctcgta	cagtcacgtg	atggattggg	ggatcagagt	agcatccagc	2820
acctccttag	tcgaagtgtg	tctcttctta	tcgatagtag	tatcgaaagta	cttgaagaca	2880
gctggagctc	ccaggttagt	cagagtgaac	aggtggatga	tgttctcagc	ttgctctctg	2940
attggcttat	ctctgtgctt	gttgtaagcg	gacagcacct	tatccagggt	agcatcagcc	3000
aggatcactc	tcttgagaaa	ctcggagatt	tgtctgatga	tctcatccag	gtagtcttgc	3060
tgttgctcca	ccaacagttg	ttattgctcg	ttatctctcg	gggatccctt	cagcttctcg	3120
tagtgggaag	ccaggttagc	gaagttcacg	tacttgtagt	gcagagccag	ctcgtttccc	3180
ttttgcagct	ctccagcgga	agccagcatt	ctcttctctc	cggtctccag	ctcgaacagg	3240
gagtactttg	gcagcttgat	gatcagatcc	ttcttcacct	ccttgatcc	cttagcctcc	3300
aggaaatcga	ttgggttctt	ctcgaaggag	gatctctcca	tgatagtgat	tcacagcagc	3360
tctctcagcg	actcagctt	cttggaactt	ccctctccca	ccttagccac	caccagcagc	3420
gagtaagcca	cagttgggga	atcgaaatct	ccgtacttct	ttggatccca	atccttcttt	3480
ctagcgatca	gcttatcgga	gtttctcttt	ggcaggatgg	actccttggg	gaatcctcca	3540
gtttgcacct	cagcttcttt	cacgatgttc	acttgtggca	tggaagcagc	ctttctcaca	3600
gtagcgaaat	ctcttccctt	atccacacg	atctctccag	tctctccgtt	agtctcgatc	3660
agtggtcttt	ttctgatctc	tcctgttagc	agagtgatct	cagctctgaa	gaagttcatg	3720
atgttgagtg	agaagaagta	cttagcagta	gcctttccga	tctctgtctc	ggacttagcg	3780
atcatctttt	tcacatcgta	caccttgtaa	tctccgtaca	cgaactcgga	ctccagcttt	3840
gggtacttct	tgatcagagc	agttccacc	acagcgttca	ggtaagcatc	gtgagcgtgg	3900
tggtagttgt	tgatctctct	caccttgtag	aattggaaat	cctttctgaa	atcggacacc	3960
agcttggaact	tcagagtgtat	caccttcacc	tctctgatca	gcttatcggt	ctcatcgta	4020
ttagtgttca	ttctggaaat	caggatttga	gccacgtgct	tagtgatttg	tctagtctcc	4080
accagttgtc	tcttgatgaa	tcacgcttca	tcacgctcgg	acagtctctc	tctctcagcc	4140
ttagtcaagt	tatcgaaact	tctttgagtg	atcagcttag	cgttcagcag	ttgtctccag	4200
tagttcttca	tcttcttctc	caactcctcg	gatggcacgt	tatcggactt	tctctgttcc	4260
ttatcgatcc	tagtcagcac	cttggttatcg	atggaatcat	ccttcaggaa	ggattgtggc	4320
acgatgtgat	ccacatcgta	atcggacagt	ctggttgata	ccagctcttg	atccacgtac	4380
atatctcttc	cgttttgcag	gtagtacagg	tacagcttct	cgttttgcag	ttgagtgttc	4440
tcactcgggt	gctccttcag	gatttgggat	cccagctcct	tgattccctc	ctcgattctc	4500
ttcattctct	ctctggagtt	cttttgctcc	ttttgagtag	tttgggtctc	tctagccatc	4560
tcgatccaga	tgttctctgg	ccatcaccct	ccatcaccct	tcaccagctc	atccaccacc	4620
ttcacagttt	gcaggattcc	cttcttgata	gctggggatc	cagccaggtt	agcagtggtc	4680
tcgtgcaggg	aatctccttg	tcgggacact	tgagcctttt	ggatatcctc	cttgaagatc	4740
agggaaatcat	cgtggatcag	ttgcatgaag	tttctgttag	cgaatccatc	ggacttcagg	4800
aaatccagga	tagtctttcc	ggattgtctta	tctctgatcc	cgttgatcag	ctttctggac	4860
agtctacccc	atccagtgta	tcttcttctc	ttcagttgct	tcacacactt	atcatcgaa	4920
aggtgagcgt	aagtcttaag	tctctcctcg	atcatctctc	tatcctcgaa	cagagtcaga	4980
gtcagcacga	tatcctccag	gatatacctc	ttctcctcgt	tatccaggaa	atccttatcc	5040
ttgatgatct	tcagcagatc	gtggtaagtt	cccagggaag	cgttgaaatc	atcctccact	5100
ccggagatct	ccacggaaat	gaagcactcg	atcttcttga	agtaatcctc	cttcagttgc	5160
ttcacagtca	ccttctctgt	agttctgaa	agcagatcca	cgatagcctt	cttttgcctc	5220
ccggacagga	aagctggctt	tctcattccc	tcagtcacgt	acttcacctt	agtcagctcg	5280

-continued

ttgtacacag	tgaagtactc	gtacagcagg	gagtgccttg	gcagcacctt	ctcgtttggc	5340
agggttcctat	cgaagtttagt	cattctctcg	atgaaggatt	gagcggaagc	tcccttatcc	5400
accacctgca	catcaacaaa	ttttgggtcat	atattagaaa	agttataaat	taaaataatac	5460
acacttataa	actacagaaa	agcaattgct	atatactaca	ttcttttatt	ttgaaaaaaa	5520
tatttgaaat	attatattac	tactaattaa	tgataattat	tatatatata	tcaaaggtag	5580
aagcagaaa	ttacctctc	gaagttccat	ggagtगतag	tctctcgga	ctttctagtc	5640
atccaagcga	atctggagtt	tcctctagcc	agtggtccca	cgtagtatgg	gattctgaaa	5700
gtcaggatct	tctcgatctt	ctctctgtta	tccttcagga	atgggtagaa	atcctcttgt	5760
ctcctcagga	tagcgtgcag	ctctcccagg	tggattttgt	gtgggatgga	tccgttatcg	5820
aaagttcttt	gctttctcag	cagatcctct	ctgttcagct	tcaccagcag	ctcctcagtt	5880
ccatccatct	tctccaggat	tggcttgatg	aacttgtaga	actcctcttg	ggaagctcct	5940
ccatcgatgt	atccagcgta	tccgttcttg	gattgatcga	agaagatctc	cttgactctc	6000
ctgggcagtt	gttgtctcac	cagagccttc	agcagagtca	gatcttggtg	gtgctcatcg	6060
tatctctga	tcatggaaag	ggacagtgga	gccttagtga	tctcagtggt	cactctcagg	6120
atatcgga	gcaggatagc	atcggaacag	ttcttagcag	ccaggaacag	atcagcgat	6180
tgatctccga	tttgagccag	caggttatcc	agatcatcat	cgtaagtatc	cttgacagtt	6240
tgacgcttag	catcctcagc	cagatcgaa	ttggacttga	agtttgaggt	cagtcacagg	6300
gacagagcga	tcagggttcc	gaacagtcct	ttctctctct	ctcctggcag	ttgagcgatc	6360
aggttctcca	gttctctgga	cttggaacag	ctagcggaca	ggatagcctt	agcatccact	6420
ccggaagcgt	tgattgggtt	ctcctcgaa	agttgggtgt	aagtttgcac	cagttggatg	6480
aacagcttat	ccacatcgga	gttatctggg	ttcagatctc	cctcgatcag	gaagtgctct	6540
ctgaacttga	ctatgtgagc	cagagccagg	tagatcagtc	tcagatcagc	cttatcagtg	6600
gaatccacca	gctctctctt	caggtggtag	atagttgggt	actctctcgt	gtaagccacc	6660
tcacccacga	tggttccgaa	gattgggtgt	ctctcgtgct	tcttatcctc	ctccaccagg	6720
aaaggactct	ccagtcgtgt	gaagaaggaa	tcacccacct	tagccatctc	gttggaagaag	6780
atctcttgca	ggtagcagat	tcgttctctt	cttctagtg	atctctctct	agcagttctc	6840
ttcagtcctag	tagcctcagc	agtcctctcg	gaatcgaa	gcagagctcc	gatcagggtc	6900
ttcttgatgg	agtgtctatc	actgtttccc	agcacttga	acttcttga	tggcaccttg	6960
tactcatcag	tgatcacagc	ccatcccacg	gagttagtcc	cgatatccag	tccgatggag	7020
tactcttata	ccattctgtc	gagtcaacaa	tcacagataa	atcagatca	aaatccatta	7080
cgtaacttta	agtcaggatc	tgtaagcaatc	tcgtgaaaaa	aaacccctaat	ccacggcaat	7140
agaatcaaca	aacaaggcag	ttgtcaacag	atctgagcaa	tatctctcga	aaaacgatat	7200
agaatcaacg	aataaggctt	tttgtcaacg	gatctgaaca	atttcttgaa	aaaccacaga	7260
agtataaaaa	ataaggctgt	ttgttaacgg	atctgaagta	attcatgaaa	aagccctgat	7320
ctatgggtac	gtttaacgga	tctgaagtaa	tcaatgaaaa	aagccctaat	ctacgattaa	7380
tgaatcaaca	attaaggctt	gttgtaaaag	gatctgacat	gaaaaagctc	taattagagc	7440
aaatctacga	ttaaagaatc	atcaaattaa	ggacgttggt	gtaagataaa	taaaagctga	7500
atctgattat	gaaaaattga	gatcgatgaa	accaaaca	acacaatcaa	atcaacatcg	7560
tattgatcaa	agaacatata	caattcatgg	aaagaacgaa	gaagaaaaac	agaagaaatc	7620
gactaacctt	gaagggagag	gtagggaaat	acacagagag	aatggaatat	attggaattg	7680
aactgcgaag	gaagggaggag	gggtgtgggg	cgttttatag	cataaggtgt	ggaacccctc	7740
tcagtcata	ttgtgacgag	tggtcatgta	tggtgtcact	tggtgcctat	gacacgggta	7800
atatttggtg	ttggatgcgt	tttcggagag	tccttctagg	aaattgagaa	ttgcaatttg	7860
cgtggtagtt	ggcggtttcg	taactctgac	ggagtcacat	gcaccttcaa	gcttcgcttt	7920
atctgttacg	acgtcgttgt	tattattggg	ccaccccggt	gcaagtgagg	cgtccgaagt	7980
tgatacgggc	cccactatcc	attgcctgaa	gtcggtgct	tttacattat	taaaaatcta	8040
tggaaaaagct	gtcatgctag	atttttttta	ttatggacca	gaaaaagaata	attgaaaaagt	8100
tttcttatta	tggagcaaaa	aaatctttaa	taattatttt	ttaataaaaa	tcataattat	8160
tcactctaat	tctttcttat	ttaatctcac	cttgtaagat	atattttcag	tagatttttc	8220
aagtattttt	ttacataaca	ttacttgtat	acatttagtt	acagtttata	taaaattatt	8280
catttaaaaa	attatttagat	attttttaaa	aaataattca	aaatattttt	tttaattgaac	8340
tttctttttt	octaattttt	ctttcttatt	tcttattaga	tggttttact	accaatcggt	8400
tagggaagaa	ttttttctc	taatgaggag	atgtgtttaa	tgatgcaact	taaggacaga	8460
ttcaccaata	acataagact	aaaagatagc	tcattgacaa	aaacaataat	gatgttaaag	8520
aaaacgaatc	catttccatt	cggtataagcc	acaccattta	cgaacaagtt	gcctactaat	8580
aatatatggt	gtcagttttt	tttttttttg	ggctgaattg	gaggggaaat	tatccccaaa	8640
gtttgattaa	aaatcacgag	acaaagtaac	aaaactttta	ttccgctaaa	aaaattagtc	8700
aaagtcaatt	aaatctggta	ttagaattac	ttttccgcta	atgatataac	cacaataaca	8760
atgtattaat	attgaaaaaa	ataaagttaa	tagagtgtga	ctttttcttt	cattttttacg	8820
taaaattgaaa	aaaaaatcaa	gcctaataata	taaactttca	ttgatctca	cctaaatcct	8880
aaagtctctg	agcccaattta	aaaaactttac	tggaggataa	aggcccaaac	ttcttttttt	8940
ttttaggaac	agagcatgta	atgggtaatg	aggcagccta	agcccataaac	catgttagcc	9000
cgtatataatt	tttgtaatt	agatttccgg	ttcttatttt	tttttttagat	ttcagttttt	9060
gtttctcaaa	aaaataaaat	aacttttagt	gctgtattaa	ctttctgttaa	ataatttttag	9120
tctcaataaaa	gatttaattct	gaataataaaa	aataaacgag	gacaattttt	ttcacaactt	9180
ttaattcttt	atttattttt	tacacttaat	tttgggaaac	atggacctag	tgtcaaaaga	9240
ctcaaaactca	acattacaaa	aacttactac	atattcaagg	atcatggtct	gaactaatct	9300
ccacataaca	tataatcaca	tttttcaatt	aaaaagaaga	attaataatg	tattaaatga	9360
tgatcgacaa	gatcacatga	tttgagattt	ttttttatga	tgggaaaaat	atgatattgag	9420
attaggtaac	aatctttcca	tttttagatag	acttatgtct	ccactagaat	tcgagctcgg	9480
aggacgtcgt	tgtggttggt	gctttccctta	cattctgagc	ctctttccctt	ctaactccact	9540
catctgcac	ttcttctgtc	cttactaata	cctcattggt	tccaaattcc	ctccctttta	9600
gcaccagctc	gtttctgttc	ttccacagcc	tcccaagtat	ccaagggact	aaagcctcca	9660
cattcttcag	atcaggatat	tctgttttaa	gatgttgaa	tctatggagg	tttgatgaa	9720
ctgatgatct	aggacgggat	aagttccctt	cttcatacag	aacttattca	aagaatggtt	9780
tgtgtatcat	tctgtttaca	ttgttattaa	tgaaaaaata	ttattggtca	ttggactgaa	9840

-continued

cacgagtggt	aaatatggac	caggccccc	ataagatcca	ttgatatatg	aattaaataa	9900
caagaataaaa	tcgagtcacc	aaaccacttg	ccttttttaa	cgagacttgt	tcaccaactt	9960
gatacaaaaag	tcattatcct	atgc aaatca	ataatcatac	aaaaatatcc	aataacacta	10020
aaaaaattaa	aagaatagg	taatttcaca	atatgttata	cgataaagaa	gttacttttc	10080
caagaaattc	actgatttta	taagcccact	tcgattagat	aaatggcaaa	aaaaaacaaa	10140
aaggaaaaga	aataaagcac	gaagaattct	agaaaaacg	aaatcacgct	caatgcagtg	10200
ggacccacgg	ttcaattatt	gccaattttc	agctccaccg	tatatttaaa	aaataaaacg	10260
ataatgctaa	aaaaatataa	atcgtaacga	tcggtaaatc	tcaacggctg	gatcttatga	10320
cgaccgttaa	ggaaattgtg	gttgctcgac	gaagtccagt	aataaacggc	gtcaaaagtg	10380
ttgcagccgg	cacacacgag	tcgtgtttat	caactcaaa	cacaaatact	ttctctcaac	10440
ctaaaaataa	ggcaattgag	caaaaacaac	tttgcggtga	aacaacgctc	aatacacgtg	10500
tcattttatt	attagctatt	gcttcaccgc	cttagctttc	tcgtgaccta	gtcgctctcg	10560
tcttttcttc	tcttctcttc	ataaaaacaat	acccaaagag	ctcttctctc	tcacaattca	10620
gatttcaatt	tctcaaaatc	ttaaaaactt	tctctcaatt	ctctctaccg	tgatcaaggt	10680
aaatttctgt	gttctctatt	ctctcaaaat	cttctgattt	gttttctgtc	gatcccaatt	10740
tcgtatatgt	tctttgggtt	agattctgtt	aatcttagat	cgtagacgat	tttctgggtt	10800
tgatcggtag	atatcatctt	aattctcgat	taggggttca	tagatatcat	ccgatttgtt	10860
caaaataattt	gagttttgtc	gaataattac	tcttctgatt	gtgatttcta	tctagatctg	10920
gtgttagttt	ctagtttgtg	cgatcgaaat	tgtcgattaa	tctgagtttt	tctgattaac	10980
agatgcagat	cttaaatgaca	aagcaccagt	ggtctagtgg	tagaatagta	ccctgccacg	11040
ttacagaccc	gggttcgatt	cccggctggt	gcagctcagc	cggtgttgag	gggggttttag	11100
agctagaat	agcaagttta	ataaaggcta	gtccgttatc	aactgaaaa	agtggcaccg	11160
agtccggtga	acaaagcacc	agtgggtctag	tggtagaata	gtaccctgcc	acgggtacaga	11220
cccggttctg	attcccggtc	ggtgcaggta	acggagatcc	aataagggtt	tagagctaga	11280
aatagcaagt	taaaataaagg	ctagtcggtt	atcaacttga	aaaagtggca	ccgagtcggt	11340
gcaacaaaagc	accagtggtc	tagtggtaga	atagtagcct	gccacggtag	agacccgggt	11400
tcgattcccg	gctgggtcag	gagtgtagat	gcctctcccg	ttttagagct	agaaatagca	11460
agttaaaata	aggctagtcg	gtttcaact	tgaaaaagtg	gcaccgagtc	ggtgccaaca	11520
aagcaccagt	ggctctagtgg	tagaaatagta	ccctgccacg	gtacagaccc	gggttctgatt	11580
cccggtctgt	gcagctcttc	tagctagagt	cgatcgacaa	gctcgagttt	ctccataata	11640
atgtgtgagt	agttcccaga	taagggaatt	agggttctca	taggggttctg	ctcatgtgtt	11700
gagcatataa	gaaacccctta	gtatgtattt	gtatttgtaa	aatacttcta	tcaataaaat	11760
ttctaatctc	taaaacccaa	atccagtagt	aaaatccaga	tcgctttacg	aattcccatg	11820
gggaggtcaa	catggtggag	cacgacactc	tggtctactc	caaaaatgtc	aaagatcacg	11880
tctcagaaga	tcaaaaggct	attgagactt	ttcaacaaag	gataatttcg	ggaaacctcc	11940
tcggattcca	ttgcccagct	atctgtcaact	tcatcgaaag	gacagtagaa	aaggaaggtg	12000
gctctacaaa	atgctcatcat	tcgcgataaag	gaaaggctat	cattcaagat	ctctctgccg	12060
acagtggtcc	caaagatgga	ccccaccaca	cgaggagcat	cgtggaaaaa	gaagaggttc	12120
caaccacgtc	tacaaagcaa	gtggattgat	gtgataacat	gggtggagcac	gacactctgg	12180
tctactccaa	aaatgtcaca	gatacagctc	cagaagatca	aagggtctatt	gagacttttc	12240
aacaaaggat	aatctcggga	aacctcctcg	gattccattg	cccagctatc	tgtcacttca	12300
tcgaaaggac	agtagaagag	gaaggtggct	cctacaaatg	ccatcatatg	gataaaggaa	12360
aggctatcat	tcaagatctc	tctgccgaca	gtggtcccaa	agatggaccc	cccccacga	12420
ggagcatcgt	ggaaaaagaa	gagggtccaa	ccacgtctac	aaagcaagtg	gattgatgtg	12480
acatctccac	tgcgtaagag	gatgacgcac	aatcccacta	tccttcgcaa	gaccttctct	12540
ctatataagg	aagttctatt	catttggaga	ggacacgctc	gagtagaaga	gctcattttt	12600
acaacaatta	ccaacaacaa	caaacacaaa	acaacattac	aattacattt	acaattatcg	12660
atacaatggt	caagactaat	cttttctctt	tcttcatctt	ctcgtctctc	ctctcacttt	12720
ccagcgcgga	catggtgttc	aagggcggaag	aggtgattaa	agagttcatg	aggttcaaa	12780
tgagaatgga	gggcagtagt	aacggccatg	agttcgagat	cgaaggtgaa	ggcgaaggtg	12840
gaccatacga	gggcacccag	acggcgaaac	tgaaggtgac	gaagggcggc	ccactgccct	12900
tcgctcgga	catctagagt	ccacaattta	tgtacgggtc	caaggcatatc	ggttaagcacc	12960
ccgcccagac	ccctgattac	aaaaagctgt	cgttcccgga	gggggtttaa	tgggaaagag	13020
tgatgaattt	tgaggacgga	gggctcgtca	ctgtgactca	agactcctca	cttcaagacg	13080
gcacccta	ttacaaggtg	aagatgaggg	gaacaaattt	ccccctgatt	ggcctgttta	13140
tcgagaaaaa	aacaatggga	tgggagggat	ccaccgagag	actttaccgc	cgcgacggag	13200
tgctcaaaag	cgaaatccat	caggcactga	aactaaagga	cggcgggcac	tacctgggtg	13260
agttcaaaac	tatatatag	cccagaagac	ccgtccagtt	gcccggatat	tactatgtgg	13320
acacaaaact	ggatataacg	tcccacacag	aagattacac	catagtggaa	cagtagtagc	13380
gcagtgaggg	tcgtccacat	ctattcctcg	gacatggcac	gggttccacg	ggctcgggtt	13440
ccagcgccac	ggcagctcgc	gaggacaaca	acatggcggt	gatcaaaag	ttcatgctgt	13500
ttaaagtcgc	tattggagggc	agcatgaatg	ggcacgaatt	tgaatttgag	ggagaaggcg	13560
agggccgacc	atatgagggg	acacaaaccg	cgaactaaa	agtcaacaaa	ggtgggcccgc	13620
ttccttttgc	ctgggacatt	ctctctcttc	agttcatgta	tggtccaag	gcctacgtta	13680
aaatcccgcc	tgatattccc	gattataaga	aattgtcttt	cccgaggggg	tttaaatggg	13740
agcggttat	gaatttcgag	gatggtggac	tgggtactgt	tacacaagat	tcgagtcctg	13800
aggatggcac	attgatatac	aaagtcaaaa	tgaggggtac	gaattttccc	ccggacggcc	13860
cggtagtga	gaaaaaaact	atgggctggg	aggccagcac	ggagaggctg	taccocaggg	13920
acgggggttt	gaaaggggag	atacaccagg	cgtttaaact	gaaggacggg	gggcactatc	13980
tgggtggaatt	taagacgata	tatatggcga	aaaagcccg	gcagctacct	ggctactact	14040
acgtcgat	caactcgat	ataactcttc	ataacgagga	ctacactatc	gtggagcaat	14100
atgaaagatc	agaaggacgt	caccaccttt	ttctatacgg	aatggacgag	ctttacaaaa	14160
aggacgagtt	gtaagcttgt	cctgctttaa	tgagatatgc	gagaagccta	tgatcgcatg	14220
atatttgctt	tcaattctgt	tgtgcacgtt	gtaaaaaac	tgagcatgtg	tagctcagat	14280
ccttacccgc	gggttcgggt	cattctaagt	aatatatcac	ccgttactat	cgtattttta	14340
tgaataatat	tctccgttca	atttactgat	tgtaccctac	tacttatatg	tacaatatta	14400

-continued

aaatgaaaac	aatatattgt	gctgaatagg	tttatagcga	catctatgat	agagcgccac	14460
aataacaaaac	aattgctgtt	tattattaca	aatccaattt	taaaaaaagc	ggcagaaccg	14520
gtcaaaccta	aaagactgat	tacataaatc	ttattcaaat	ttcaaaagtg	ccccaggggc	14580
tagtatctac	gacacaccga	gcggcgaaact	aataacgctc	actgaaggga	actccggttc	14640
cccgccggcg	cgcattgggtg	agattccctg	aagttgagta	ttggccgtcc	gctctaccga	14700
aagttacggg	caccattcaa	cccggtccag	cacggcgggc	gggttaaccga	cttgctgccc	14760
cgagaattat	gcagcatttt	tttgtgtat	gtgggcccc	aatgaagtgc	aggtcaaacc	14820
ttgacagtga	cgacaaaatc	ttgggcgggt	ccaggggcga	ttttgcgaca	acatgtcgag	14880
gctcagcagg	accgctcaga	gaggatgcac	atgtgaccga	gggacacgaa	gtgatccgtt	14940
taaatatca	gtgtttgaca	ggatatattg	gcgggtaaac	ctaagagaaa	agagcgttta	15000
ttagaataat	cggatattta	aaagggcggtg	aaaagggtta	tccgttcgtc	catttgtatg	15060
tgccagccgc	ctttgcgacg	ctcaccgggc	tggttgccct	cgccgctggg	ctggcgggcg	15120
tctatggccc	tgcaaacgcg	ccgaaaacgc	cgtcgaagcc	gtgtgcgaga	caccgcggcc	15180
gcggcgctg	tgcataccct	gcggaaaact	tgccctcac	tgacagatga	ggggcgagcg	15240
ttgacacttg	agggggcgac	tcaccgcggc	cgccgttgac	agatgagggg	caggctcgat	15300
ttcgccggcg	gacgtggagc	tgccagcct	cgcaaatcgg	cgaaaacgcc	tgattttacg	15360
cgagttttcc	acagatgatg	tggaacaagc	tggggataag	tgccctgcgg	tattgacact	15420
tgagggggcg	gactactgac	agatgagggg	cgccatcctt	gacacttgag	gggcagagtg	15480
ctgacagatg	agggggcgac	ctattgacat	ttgaggggct	gtccacaggc	agaaaatcca	15540
gcatttgcaa	gggtttccgc	ccgtttttcg	gccaccgcta	acctgtcttt	taacctgctt	15600
ttaaaccaat	atttataaac	cttggtttta	accagggtcg	cgccctgtgc	gcgtgaccgc	15660
gcacgccgaa	gggggggtgc	cccccttctc	gaaccctccc	ggcccgctaa	cgccggccctc	15720
ccatcccccc	aggggctgcg	ccccctggcc	gcgaacggcc	tcaccccaaa	aatggcgagcg	15780
ctggccaatt	cccagtgctg	cggaacccct	atgtgtttat	ttttctaaat	acattcaaat	15840
atgtatccgc	tcattgagaca	ataaccctga	taaatgcttc	aataatattg	aaaaaggaag	15900
agtatggcta	aaatgagaat	atcacccgaa	ttgaaaaaac	tgatcgaaaa	ataccgctgc	15960
gtaaaagata	cgaaggaat	gtctcctgct	aaggtatata	agctgggtgg	agaaaatgaa	16020
aaacctatatt	taaaaatgac	ggacagcccg	tataaaaggga	ccacctatga	tgtggaacgg	16080
gaaaaggaca	tgatgctatg	gctgggaagg	aagctgcctg	ttccaaaggt	cctgcacttt	16140
gaacggcatg	atggctggag	caatctgctc	atgagtgagg	ccgatggcgt	cctttgctcg	16200
gaagagtatg	aagatgaaca	aagccctgaa	aagattatcg	agctgtatgc	ggagtgcact	16260
aggctcttcc	actccatcga	catatcggat	tgtccctata	cgaatagctt	agacagccgc	16320
ttagccgaat	tggtattactt	actgaataac	gatctggccc	atgtggattg	cgaaaactgg	16380
gaagaggaca	ctccatttaa	agatccgcgc	gagctgtatg	attttttaaa	gacggaaaag	16440
cccgaaagg	aacttgtctt	ttccccggcc	gacctgggag	acagcaacat	ctttgtgaaa	16500
gatggcaaa	taagtggctt	tattgatctt	gggagaagcg	gcagggcgga	caagtgggat	16560
gacattgctc	tctgcgtccg	gtcgtccagg	gaggatatcg	gggaagaaca	gtatgtcgag	16620
ctattttttg	acttactggg	gatcaagcct	gattggggaga	aaataaaata	ttatatattta	16680
ctggatgaat	tggttttagct	gtcagaccaa	gtttactcat	atataactta	gatttgattta	16740
aaacttcatt	tttaatttaa	agggatctag	gtgaagatcc	tttttgataa	tctcatgacc	16800
aaaatccctt	aacgtgagtt	ttcgttccac	tgagcgtcag	accccgtaga	aaagatcaaa	16860
ggatcttctt	gagatccctt	ttttctgcgc	gtaatctgct	gcttgcaaac	aaaaaaacca	16920
ccgtaccag	cggtgggttt	tttgcgggat	caagagctac	caactctttt	tccgaaggta	16980
actggcttca	gcagagcgca	gataccaaat	actgtccctc	tagtgtagcc	gtagttaggc	17040
caccacttca	agaactctgt	agcaccgcct	acatacctcg	ctctgctaat	cctgttacc	17100
gtggctgctg	ccagtggcga	taagtctgtg	cttaccgggt	tggaactcaag	acgatagtta	17160
ccggataaagg	gcgagcggtc	gggctgaacg	gggggttcgt	gcacacagcc	cagcttggag	17220
cgaaacgacct	acaccgaact	gagataacct	cagcgtgagc	tatgagaaag	cgccacgctt	17280
cccgaaaggga	gaaaggcgga	acggtatccg	gtaagcggca	gggtcggaac	aggagagcgc	17340
acgaggggagc	ttccaggggg	aaacgcctgg	tatctttata	gtcctgtcgg	gtttccggcc	17400
ctctgacttg	agcgtcgatt	tttgtgatgc	tcgtcagggg	ggcgagagcct	atggaaaaac	17460
gccagcaacg	cgcccttttt	agcgttccctg	gcagatcccta	gatgtggcgc	aacgatgccg	17520
ggcacaagca	ggagcgcacc	gaacttcttc	gcatacaagt	ttttggctct	caggccgagg	17580
cccacggcaa	gtattttgggc	aaggggtcgc	tggtattcgt	gcagggcgaag	attcggaata	17640
ccaagtacga	gaaggacggc	gcagcggctc	acgggaccga	cttcattgcc	gataaggtgg	17700
attatctgga	caccaaggca	ccaggcgggt	caaatacagg	ataagggcac	attgccccgg	17760
cgtgagtcgg	ggcaatcccg	caaggagggt	gaatgaatcg	gacgtttgac	cggaaggcat	17820
acaggcaaga	actgatcttg	cgggggtttt	ccgccgagga	tgccgaaacc	atcgcaagcc	17880
gcacgcgtcat	gcgtgcgccc	cgcgaaacct	tccagtcctg	cggtctgatg	gtccagcaag	17940
ctacggccaa	gatcgagcgc	gacagcgtgc	aactggctcc	ccctgccctg	cccgccccat	18000
cgcccgccgt	ggagcgttgc	cgtcgtcttg	aacaggaggc	ggcaggtttg	gcgaagtcga	18060
tgaccatcga	cacgcgagga	actatgacga	ccaagaagcg	aaaaaccgcc	ggcgaggacc	18120
tgggaaaaca	ggtcagcgag	gccaaagcag	ccgcgttgct	gaacacacac	aagcagcaga	18180
tcaaggaaat	gcagctttcc	ttgttcgata	ttgcgcctg	gccggacacg	atgcgagcga	18240
tgccaaacga	cacggcccg	tctgccctgt	tcaccacgcg	caacaagaaa	atccccgcg	18300
aggcgtcgca	aaacaaggtc	attttccacg	tcaacaaggga	cgtgaagatc	acctacacgc	18360
gcgtcgagct	gcgggcccag	gatgacgaac	tggtgtggca	gcaggtgttg	gagtacgcga	18420
agcgcacccc	tatcggcgag	ccgatcacct	tcaogttcta	cgagctttgc	caggacctgg	18480
gctggtcgat	caatggcccg	tattacacga	aggccgagga	atgcctgtcg	cgctacaggg	18540
cgacggcgat	gggcttcacg	tccgacccgc	ttgggcacct	ggaatcgggt	tcgctgctgc	18600
accgcttcg	cgtcctggag	cgtggcaaga	aaacgtcccg	ttgccaggtc	ctgatcgacg	18660
aggaaatcgt	cgtgctgttt	gctggcgacc	actacacgaa	attcatatgg	gagaagtacc	18720
gcaagctgtc	gccgacggcc	cgacggatgt	tcgactattt	cagctcgcac	cgggagccgt	18780
acccgctcaa	gctggaacc	ttccgcctca	tgtgcggatc	ggattccacc	cgctggaaga	18840
agtggcgca	cgaggtcgcc	gaagcctgcg	aagagtttgc	aggcagcgcc	ctggtggaa	18900
acgcctgggt	caatgatgac	ctgggtgcatt	gcaaacgcga	gggccttggt	gggtcagttc	18960

-continued

```

cggtggggg ttcagcagcc cctgctcgga tctgttgga cggacagtag tcatggtga 19020
tgggctgct gtatcgagtg gtgattttgt gccgagctgc cggtcgggga gctgttggt 19080
ggctggtggc aggatattt gtggtgtaaa caaatgacg cttagacaac ttaataaac 19140
attgcggacg tttttaatgt actggggttg aacactct 19178

```

```

SEQ ID NO: 8      moltype = DNA length = 36
FEATURE          Location/Qualifiers
source           1..36
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 8
atgttcgagg atgttatcca cccacaccgg ctgagc 36

```

```

SEQ ID NO: 9      moltype = DNA length = 32
FEATURE          Location/Qualifiers
source           1..32
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 9
atcatgtgat tgcacccctt atctccgtta cc 32

```

```

SEQ ID NO: 10     moltype = DNA length = 50
FEATURE          Location/Qualifiers
source           1..50
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 10
ctcagtcctg gcataggagt gtacatgcct tcctgggtcc cttaacactg 50

```

```

SEQ ID NO: 11     moltype = DNA length = 35
FEATURE          Location/Qualifiers
source           1..35
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 11
atgttcgagg atgttatcca cccacaccggc tgagc 35

```

```

SEQ ID NO: 12     moltype = DNA length = 32
FEATURE          Location/Qualifiers
source           1..32
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 12
atcatgtgat tgcacccctt atctccgtta cc 32

```

```

SEQ ID NO: 13     moltype = DNA length = 33
FEATURE          Location/Qualifiers
source           1..33
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 13
ctcagatggt acttctggt tcccttaaca ctg 33

```

```

SEQ ID NO: 14     moltype = DNA length = 39
FEATURE          Location/Qualifiers
source           1..39
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 14
atgttcgagg atgttatcca ccccaaacac cggctgagc 39

```

```

SEQ ID NO: 15     moltype = DNA length = 32
FEATURE          Location/Qualifiers
source           1..32
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 15
atcatgtgat tgcacccctt atctccgtta cc 32

```

```

SEQ ID NO: 16     moltype = DNA length = 47
FEATURE          Location/Qualifiers
source           1..47
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 16
ctcagtcctg gcataggagt gtacatgtcc tggttccctt aacactg 47

```

SEQ ID NO: 17	moltype = DNA length = 23		
FEATURE	Location/Qualifiers		
source	1..23		
	mol_type = other DNA		
	organism = synthetic construct		
SEQUENCE: 17			
ccaccctca acaccggtg agc			23
SEQ ID NO: 18	moltype = DNA length = 23		
FEATURE	Location/Qualifiers		
source	1..23		
	mol_type = other DNA		
	organism = synthetic construct		
SEQUENCE: 18			
ccccttattg gatctcgtt acc			23
SEQ ID NO: 19	moltype = DNA length = 23		
FEATURE	Location/Qualifiers		
source	1..23		
	mol_type = other DNA		
	organism = synthetic construct		
SEQUENCE: 19			
ggagtgtaca tgcctctcc tgg			23
SEQ ID NO: 20	moltype = DNA length = 27		
FEATURE	Location/Qualifiers		
source	1..27		
	mol_type = other DNA		
	organism = synthetic construct		
SEQUENCE: 20			
tatccatattg acgttcaga ttacgcc			27
SEQ ID NO: 21	moltype = DNA length = 25		
FEATURE	Location/Qualifiers		
source	1..25		
	mol_type = other DNA		
	organism = synthetic construct		
SEQUENCE: 21			
acctcagaat tttgcagtg gtgtg			25
SEQ ID NO: 22	moltype = DNA length = 20		
FEATURE	Location/Qualifiers		
source	1..20		
	mol_type = other DNA		
	organism = synthetic construct		
SEQUENCE: 22			
ctgctgttga tggaggactt			20
SEQ ID NO: 23	moltype = DNA length = 20		
FEATURE	Location/Qualifiers		
source	1..20		
	mol_type = other DNA		
	organism = synthetic construct		
SEQUENCE: 23			
ctcctggaga agcagaagtt			20
SEQ ID NO: 24	moltype = DNA length = 17		
FEATURE	Location/Qualifiers		
source	1..17		
	mol_type = other DNA		
	organism = synthetic construct		
SEQUENCE: 24			
gtatgcttgc cgcaaaa			17
SEQ ID NO: 25	moltype = AA length = 419		
FEATURE	Location/Qualifiers		
source	1..419		
	mol_type = protein		
	organism = Glycine max		
SEQUENCE: 25			
MLQDVFDQPL PKAEPQQQH PIWLQDEISS PLSARIFELC NTDFFPETLP NSEVTSSSNC			60
CYEENSSYAT TNISVTVDVD NKLNSNSNTV TTPSTTTTNT TNNNTTNSN LSIIFDSQEE			120
IDNDISASID FSLSPSFNVP PFLPATSQQE QDFESSVQH QVLAACSVVE GFSQYPNDV			180
APPFMGAPLP SVFEEDCISS VPSYMALNPS SPCTYLSPGM PPYMPHGPLT TALSTGSSGI			240
FGGNILLGSE LOAQELEYGQ ENGRMYCTDS IQRVFNPPDL QALGTESQOL VPGAGSSASL			300

-continued

PEISNLEDSS	FKVGKLSVEQ	RKEKINRYMK	KRNERNFSKK	IKYACRKTLA	DSRPRVRGRF	360
AKNDDFGETN	RTTSSNHEED	DEBEVVVKDE	DDMVDSSDIF	AHISGVNSFK	CNYSFQSWI	419

SEQ ID NO: 26 moltype = DNA length = 4401
 FEATURE Location/Qualifiers
 source 1..4401
 mol_type = genomic DNA
 organism = Glycine max

SEQUENCE: 26

gtcaacccaa	aatccaactc	aagacaagga	gagtgccatt	ggcagatggc	acccgagagc	60
caagagggtt	tgcagtgaa	gaatggccca	ccttatgtct	cgctttgacg	ccagcttgac	120
gccttttagg	agaaccctct	tgccctatat	atgaaacata	actaattctt	tgagctcttg	180
acatccctcat	actctccatc	atccctagcta	actactcatc	atcatattca	taactttccc	240
ccctctttct	ctctcttttt	ccctatggag	tgctcttgag	cacatgtttt	aagtcaccaa	300
cacacgtgtg	aatttcccac	attttgcact	cctatatagt	actctctttc	ttctagata	360
tagctttaat	tttgtgtaca	ttttttttgt	attaaagttt	ctgggggtgc	ttgattgcct	420
cgtgcaaaag	gaaacataca	tacatacacc	atgttgccagg	atgtttttga	tcagccactg	480
ccaccaaaag	ccgaaccaca	acagcaacat	cccattgtaa	tcacctcatc	ttgccagcag	540
tctcaatatt	catctattat	gttaatgcat	acatgcatgt	gttcttccta	ccttaatgac	600
atagtgtaat	cctaactcaa	tattaatttc	atgaaaactt	tacaagtgtt	ttaatgtgtt	660
ccatttttga	acgaatctct	ttctcatatg	tttgaatata	ttcaaatcta	aggctaagtt	720
gtttgtttac	agttgtctcg	tgatattggc	ccacttgtga	ttctcttctt	ttttcatgce	780
atgcatgcgt	gaatgatata	tatatatata	tatatatata	tatatattat	tggtttgtgt	840
ttatgtttgt	tgcaagtttg	atgtattatt	tgccagtgtc	aatagaatta	aagccattct	900
caaatctctt	ttccaggcttc	cctgtcaatt	tgcatggaag	aacgggttct	ttctctcaa	960
tttatatagt	tggtggagcat	aatgcttatt	gactagcaat	aagaaagaac	gtattagaga	1020
ttccgcaata	actattagac	tttgtccttt	tttatatatt	atcctatcct	aaccaagaca	1080
atggcattca	ccacaatgtc	attttctgag	ggcaacattg	taaactgcaa	agtgatatca	1140
attttctacat	gatttccagt	gtgatagcta	gccatttaaa	cgtagctttc	gattattgtg	1200
gaatatatat	ctttctgtgt	ttgtctattt	gccaaatttg	atattcttat	tcatttagag	1260
aggggagagg	agacttgagg	aacacttcac	atgtacagca	ttatgtgttc	ttttaagtgc	1320
cttttaattc	ctattaaatc	agcttccata	cactgaacaa	acacaacgta	gcatggcatg	1380
agattccttc	tttgtttcaa	acatacaagt	aacattgtca	cattaaggca	taaatgccca	1440
tttttttcaa	tttcttaaaa	aagccaaacc	tcttgttgaa	tttgttatct	gtacatgaca	1500
aaatataaac	acaactgact	agaggacaca	gccaaaccaa	tatgtatttg	aagttcctca	1560
tgatggggta	ttctacattg	atttatctcg	ttgattttcta	gacaaaggtg	ttgtgtatgt	1620
atgaacataa	ctgatgctta	taggcgtggt	acttactcac	tttagaatgc	tcactatttg	1680
atgaagacat	gcaataaacc	aaagttaga	ctgataataa	caagagataa	ttatcttttg	1740
aaagttaga	ctctgatcat	tataggcatg	gtcccaattt	ctattttgct	gatagaaacg	1800
agtgctgata	gatcacatgg	ccatcccata	agggaaaata	aatgggtgtg	aatcagtcct	1860
ctacctgttt	ataaagggtc	tatcaaaatt	caaaatgtta	ttatcaattt	gttttagtagt	1920
tattttttgt	agtcgatgat	cttacattgt	ttttatatca	gtggctgcag	gatgaaattt	1980
caagccctct	tagtgcctga	attttcgaac	tttgtaaac	tgattttttc	ccagagactt	2040
tgccaaatcc	tgaggttact	tccagctcaa	actgttggtt	tgaggagaa	tcctcatatg	2100
ccacaacaaa	tatatctgta	actgtagatg	tagacaataa	gttgaatagc	aatagcaaca	2160
cagtcaccac	cccaactagc	actaccacca	ccaacaccaa	caacaacaca	accaacagta	2220
gtaacctgtc	tattatcttt	gactcccaag	aagaaattga	caatgacatc	tctgcctcca	2280
tagactcttc	actatctcca	tctttcaatg	ttccccattt	tctccagccc	acatcccagc	2340
aagaacagtt	gtattttctt	ctctgtgcag	atcagggtcca	actagcagca	tggttcagttg	2400
tgaggggctt	ctctcagtat	cccaatgact	ctgttgccac	tccttttatg	ggggctccat	2460
taccttctgt	tttgtgaagag	gactgcata	cttctgtccc	ttctttatat	gctcttaaac	2520
cctcatctcc	ttgcacttat	ctcagtcctg	gcatgccacc	atcacatgct	catggccccc	2580
ttactactgc	cttatcaact	ggtagttctg	gaatatgttg	tgaggaaact	cttcttggtt	2640
ctgaacttca	ggcacaagaa	ttggaatacc	aaggagaaaa	tggaagaatg	tattgtacag	2700
attcaattca	cgcgggtggt	aacctccag	accttcaggt	atggatgatt	ttcaaacat	2760
tttgttaagt	catcattgct	ttttgtagtt	ttgttaacaa	taccatcaaa	atgaagtgtt	2820
cagttcaaat	gttcaatgat	ctcagtaatt	agagaaaaag	aagtcocattg	tccaaggatt	2880
ttttttattc	taatgatctg	aagctgttgt	tgtaagcttg	ctcttttcagg	ttctagtct	2940
gaaaggttgc	agcagtaagg	ttcttgacaa	taaagagtga	ctttacaaga	ggtgaaaagt	3000
ttgcaaaaag	taacggccca	caaatctcta	agaagtttat	atgccccata	gaatatatgc	3060
ccatatccat	caaacagta	gctagcatat	caaccctttt	tataatctca	ttctcacctg	3120
taaaaagtag	cataaaatct	gagctagcta	gtagcaggaa	ggaatcaca	ctttccacaa	3180
aaattgataa	caatcatcat	cagaaaagac	ccacttcatg	agtggttctt	tttgcatact	3240
gctttccgga	acctctaact	aaagctaatg	acagtttgtt	atgaccaatt	ttctttgtcg	3300
tagcaagaaa	gtaactgata	ttgtttgtga	tgtctttcct	aattcctagg	ctcttggcac	3360
tgagagtcag	caactagtac	caggggctgg	aagttctgcc	agtctaccag	aaatctcaaa	3420
cttagaggac	tcctccttca	aggttggaaa	actttctgtt	gagcaaaagga	aagaaaagat	3480
caatagatcc	atgaagaaaa	gaaatgaaa	aaacttcagc	aagaaaatta	aggtacactc	3540
aagcttcttc	ccatcttcaa	aacagccata	tcatgttaca	tatgcatatg	cttataaaca	3600
accacttttt	attgcatact	ttcaagttca	actaatgatg	aaaccaaata	tgcatgatgc	3660
ctgccgcgaaa	actctggcag	atagccgacc	tcgagtcaga	ggaagggttg	caaagaatga	3720
tgacttttgg	gaaaccaata	gaactacaag	tagcaatcat	gaagaagatg	atgaagaaga	3780
agtaagattt	ccagggctca	ttcatttatt	aaacacttaa	caggacattt	tttattcaac	3840
ttgcccgaat	taaaaaaaa	aatccctatc	acttgtacct	tgtcaattat	ttgctgatat	3900
agttttttta	tcctgcagggt	agttgtgaaa	gatgaagatg	atatgggtga	ttcctcagat	3960
atctttgtctc	atatcagttg	ggtgaactcc	ttcaaatgca	actattcctt	ccagtcctgg	4020

-continued

```

atttgactta atagtttgat cagtatacct acatttagct tgtgtagatt ttttgcattg 4080
ccatagacca cccaggtttt ctcatactcg gttttgatgg atgaggaaac tgaataatta 4140
gggtcctaact actatacaat aagtgaatga agaacaattta gaaagaaact gtatacattt 4200
tttgcagaca ctattttcta ctacttaaat gtattaagaa tcacatgttg tattaacttt 4260
gtctccatgt actagtgagg ctcattcatg tactctcttt tataatatag tgatgaggaa 4320
ctagtttttg ttacattcaa gtgcttaaag cttaaagccc aagggtttaa atatttttac 4380
tggttttgaa actttttttt t 4401

```

SEQ ID NO: 27 moltype = DNA length = 1260

FEATURE Location/Qualifiers
 source 1..1260
 mol_type = genomic DNA
 organism = Glycine max

SEQUENCE: 27

```

atgttgacagg atgtttttga tcagccactg ccaccaaaagg ccgaaccaca acagcaacat 60
cccatttggc tgcaggatga aatttcaagc cctcttagtg ctgaaatttt cgaactttgt 120
aacactgatt ttttccaga gactttgcc aattctgagg ttacttccag ctcaaaactgt 180
tgttatgagg agaactcctc atatgccaca acaaatatat ctgtaactgt agatgtagac 240
aataagttga atagcaatag caacacagtc accaccccaa ctageactac caccaccaac 300
accaacaaca acacaacca cagtagtaac ctgtctatta tctttgactc ccaagaagaa 360
attgacaatg acatctctgc ctccatagac ttctcactat ctccatcttt caatgttccc 420
ccatttctcc cagccacatc ccagcaagaa cagtttgatt tctcttctgt gcagcatcag 480
gtccaactag cagcatgttc agttgtggag ggcttctctc agtatcccaa tgactctgtt 540
gcactccttt ttatgggggc tcattacact tctgtttttg aagaggactg catatcttct 600
gtcccttctt atatggctct taacctctca tctccttgca cttatctcag tcttgccatg 660
ccaccataca tgcctcatgg ccccttact actgccttat caactggtag ttctggaata 720
tttggtggga acattcttct tggttctgaa ctccaggcac aagaattgga ataccaagga 780
gaaaatggaa gaatgtattg tacagattca attcagcggg tgtttaacc tccagacctt 840
caggctcttg gcactgagag tcagcaacta gtaccagggg ctggaagtgc tgcagtccta 900
ccagaaatct caaacttaga ggactcctcc ttcaagggtg gaaaacttcc tgttgagcaa 960
aggaagaaaa agatcaatag atacatgaag aaaagaaatg aaagaaactt cagcaagaaa 1020
attaagtatg cgtcccgcaa aactctggca gatagccgac ctcgagtcag aggaagggtt 1080
gcaaagaatg atgactttgg ggaacccaat agaactacaa gtagcaatca tgaagaagat 1140
gatgaagaag aagtagtgtg gaaagatgaa gatgatagg ttgattcctc agatatcttt 1200
gtcctatcca gtgggggtgaa ctcttcaaaa tgcaactatt ccttcacgtc ctggatttga 1260

```

SEQ ID NO: 28 moltype = AA length = 421

FEATURE Location/Qualifiers
 source 1..421
 mol_type = protein
 organism = Glycine max

SEQUENCE: 28

```

MIPISLLLIIE TSLIRSHSH RKGKQMDIEIS SPLSARIFEL CNTDFFPEAL QNSEVTSSSN 60
CCYEENSSYA TTNISVTVDV DNKLNSNSNT VTTPSTTTT NTNNNTTNS NLSIFDSDQE 120
EIDNDISASI DFSLSPSFNV PPFLPVSSQQ EQDFSSVQH QVQLAACSVV EGFTQYPTDA 180
VAPPFMGAPL PSVFEEDCIS SVPSYMLPLN SPCTYLSPL MPPYMPHGPL TTALSTDSSG 240
IFGGNILLGS ELQTELEYQ GENGRMYCTD SIQRFVFNPPD LQALGTESQQ LVPGAGSGST 300
LTPISNLIED SSFVGLSLV EQRKEKINRY MKKRNERNFS KIKYACRKT LADSRPRVRG 360
RPAKNDDFGE TNRTTSSNHE EDDEEEVVMK DEDDMVDSSD IFAHISGLNS PKNYSIQSW 420
I 421

```

SEQ ID NO: 29 moltype = DNA length = 4011

FEATURE Location/Qualifiers
 source 1..4011
 mol_type = genomic DNA
 organism = Glycine max

SEQUENCE: 29

```

acaccctcat actctccatc atcctagcta actcatcatc atattcataa ctcccccccc 60
tctctctctc tctccctccc tccttcccta tggagtggcc ttgagcacat gtttttagtc 120
accaacacac gtgtgaattt tcccacattt tgcacttcca tatagctctt ttctgggggtg 180
ccttgattgc ctctgtcaag gggatacata cacacacaca cacaccatgt tgcaggatgt 240
ttttgatcag ccactgccac caaaggccga accacaacag caacttcccc ttgtaataat 300
ctccagccag cggctctcatt catatcttcc atgttaatgc atgcatgtgt tctgcctacc 360
ttaattaatt agttaatttg tgcctgact catgacatgg tgtgatctta acccaaaatt 420
catgaaactt tacaaagtgt ttgtgttcca ttttggaaac tctttctcag gaatatattc 480
aagtctaaagg ctaagtgtgt tacaattgtc tctgtgatatt ggaccacttg tgattccatt 540
ttttcatgca tgcagtgaat atatatatgg ttgttttatt ttatgtttt ttgcaagtt 600
tgatatatta tttccagtg ctaatagaat taaagccatt ctcaaatctt tttccaggct 660
tctctggcaa tttgcaggga ggaacgggtt attttctctc aatttataga gttggggggg 720
ggaggggggg ggggcataat gcttattgac tagcaataag aaagaacgta tttagagattc 780
cgcaataatt attagtcttt gtcccttttt ttctctata ttatcctatc caatccaaga 840
caatgccatt taccaaatgat gtcattttct gagggcaaca ttgtaaactg caagtgatat 900
caatttctac atgattttca gtgtgatagc tagccattta aacatagctt tcgattgttg 960
tggaatatat atctttctgt ggttgccat ttgccaat ttgatattctt attcatttag 1020
agaggggaga ggagacttga ggaacacttc acatatatag cattatgtgt tcttttaagt 1080
gccttttaatt ttaattccta ttaaatcagt ttccatacac taaacaaaca caacgtagca 1140

```

-continued

tggcatgagg	ttctctcttt	gttacaaca	tacaagtaac	attgtcacgc	taaggcataa	1200
atgcccatct	ttttcaattt	cctaaaaaaa	ccaacccctc	tgtagaattt	atctgtacat	1260
gaccaaata	aagcacaa	gatagaggac	acaaccaacc	taatctgtat	ttggagtcc	1320
tcatgatggg	gtattgtaca	ttgatttatt	tcgttgattt	ctagacaaag	ggcttgtagt	1380
gatgaacata	actgatgctt	ataggcggtg	tactcactat	ttgatgaaga	cacgcaataa	1440
accaaagttt	agactgataa	taacaagaga	atattattat	cttttgaag	tttagactct	1500
gatcattata	ggcatgatcc	caatttctct	tttgcgtgata	gaaacgagtc	tgataagatc	1560
acacagccat	ccccgtaagg	gaaaaacaa	gggtgtagaat	cagtcttcta	cctgtttata	1620
aaggggtctat	caaaatgtta	tcaatatgtg	tctaatagtt	tttttggtag	tcatgatgct	1680
tacattgttt	ttcatatcac	tggtgcagg	atgaaatttc	aagccctctt	agtgtctgaa	1740
tcttcgaact	ttgtaaacct	gatttttttc	cagaggcttt	gcagaattct	gaggttactt	1800
ccagctcaaa	ctggtgttac	gaggagaact	cctcatatgc	cacaacaaat	atatctgtaa	1860
ctgtagatgt	agacaataag	ttgaatagca	atagcaacac	agtcaccacc	ccaactagca	1920
ctaccaccac	caacaccaac	aacaacacaa	ccaacagtag	taacctgtct	attttctttg	1980
actcccaaga	agagattgac	aatgacattt	ctgcctccat	agacttctca	ctctctccat	2040
ctttcaatgt	tccccattt	ctccagctct	catcccagca	agaacagttt	gatttctctt	2100
ctgtgcagca	tcaggctcaa	ctagcagcat	gttcgggtgt	ggagggtctc	actcagtatc	2160
ccactgacgc	tggtgcacct	ccttttatgg	gggtcccat	accttctggt	tttgaagagg	2220
actgcatact	ttcagtcctc	tcttatatgc	ctcttaaccc	ctcatctcct	tgcaattatc	2280
tcagtcctgg	catgccacca	tacatgcctc	atggccccct	tactactgcc	ttatcaactg	2340
atagttctgg	aattatttgg	gggaacattc	ttcttggttc	tgaacttcag	acacaagaat	2400
ttggaatacca	aggagaaatgt	attgtacaga	ttcaattcag	cgggtgttta		2460
accctccaga	ccttcaggta	tgaatgattt	tcaaaactat	tagttaagtt	ctccatcatt	2520
gcctttgtta	acaataccat	cctaatagaaa	agtgttcggg	tcgttcaata	tgctcagtaa	2580
cgagagaaaa	agaagtcctt	gtccaaggat	ttttttttt	tatttcttat	gatgctaagc	2640
tggtgtgtgt	agctgtctct	ttcaggtttc	tagtctgaaa	gggtgcagca	gtaaggttct	2700
tgactataaa	gagtggtctt	acaagagggtg	aaaaatctgc	aaaagataac	gggtccacaaa	2760
tttctaagaa	gctgatatgc	cccatagaat	atatgcccat	atccatccat	acagtatgta	2820
gcataatcaac	cccttttata	atctcattct	cacctgtaaa	aagtagcata	aaatctgagc	2880
tagctagttag	caggaaggaa	tcacaacatt	ccacaaaaat	tgataacgaa	caacatcatc	2940
atcagaaaaag	accacattca	tgacagtttg	tgatgatcaa	tttccctttg	cgtagcaaga	3000
aagtaactga	ttttgtttgc	aatttctttc	ctaggcactt	ggtagtgaga	gccagcaact	3060
agtaaccggg	gctgggtggt	ctggcactct	aacaccagaa	atctcaaaat	tagaggactc	3120
ctccttcaag	ggttggaacaa	ttctgttga	gcaaaggaaa	gaaaagatca	atagatacat	3180
gaagaaaaga	aatgaaaaga	acttcagcaa	gaaaatttaag	gtacagtcac	gtcctctccc	3240
atcttcaaaa	cagccacatc	atcaaacaaa	attgaaatct	tatcatgtta	catatgcata	3300
tgcaaggtat	ctcttgctta	taaacaaacca	cttggtattg	catactttca	agttcaacta	3360
atgatgcaac	caaatatgca	gtatgcctgc	cgcaaaactc	tggcagatag	ccgacctcga	3420
gttagaggaa	gggttgcaaa	gaatgatgac	tttgagagaa	ccaatagaac	tactagtagc	3480
aatcatgaag	aagatgttga	gaagaaagta	agatttccag	gggttcattca	tttattaac	3540
acttaacagg	acgcttttta	ttcaactatt	tggtgatata	gttttttaac	cctgcaggta	3600
gttatgaaag	atgaagatga	tatggttgat	tcctcagata	tccttgctca	tatcagtggtg	3660
ctgaactcct	tcaaatgcac	ctattccatc	cagtcctgga	tttgacttaa	tagtttgatc	3720
agtataccta	catttagcct	ctgtagattt	tttgcattgc	catataccac	cccagatttc	3780
tctatctcgg	ttttgtgga	tgaggaaact	gaataattgg	gggtcctaata	ctatacaata	3840
agtgaatgaa	aaatatatga	aaaaaaaact	gtatacatct	tttgagagaa	ctacctttct	3900
tactacttgt	atgtactaag	aatcacatgt	tgtatttaact	caaacgattt	catttgtctc	3960
catgtactag	tggagctcat	tcatgtactc	tcttttataa	tatagtgata	a	4011

SEQ ID NO: 30

FEATURE

source

moltype = DNA length = 1266

Location/Qualifiers

1..1266

mol_type = genomic DNA

organism = Glycine max

SEQUENCE: 30

atgatcccaa	tttctctttt	gctgatagaa	acgagttctga	taagatcaca	cagccatccc	60
cgtaagggaa	aacaaatgga	tgaaatttca	agccctctta	gtgctcgaat	cttcgaaact	120
tgtaaacactg	atttttttcc	agaggctttg	cagaattctg	aggttacttc	cagctcaaac	180
tggtgttacg	aggagaactc	ctcatatgcc	acaacaaata	tatctgtaac	tgtagatgta	240
gacaataagt	tgaatagcaa	tagcaacaca	gtcaccaccc	caactagcac	taccaccacc	300
aacaccaaca	acaacacaac	caacagtagt	aacctgtcta	ttttctttga	ctcccaagaa	360
gagattgaca	atgacatttc	tgctcccata	gacttctcac	tctctccatc	tttcaatggt	420
ccccatttc	tccagctctc	atcccagcaa	gaacagtttg	atttctcttc	tgtgcagcat	480
cagggtccac	tagcagcatg	ttcggttggtg	gagggcttca	ctcagtatcc	cactgaacgt	540
ggtgcacctc	cttttatggg	ggctccatta	ccttctgttt	ttgaagagga	ctgcatactc	600
tcagtcctct	cttatatgcc	tcttaacccc	tcatctcctt	gcacttatct	cagtcctggc	660
atgccaccat	acatgcctca	tggtccctct	actactgcct	tatcaactga	tagttctgga	720
atatttggtg	ggaacattct	tcttggttct	gaacttcaga	cacaagaatt	ggaataccaa	780
ggagaaaatg	gaagaatgta	ttgtacagat	tcaattcagc	gggtgtttta	ccctccagac	840
cttcaggcac	ttggtactga	gagccagcaa	ctagtaaccg	gggtgtgttg	ttctggcact	900
ctaacaccag	aaatctcaaa	cttagaggac	tctctcttca	aggttggaac	actttctggt	960
gagcaaaagg	aagaaaagat	caatagatac	atgaagaaaa	gaaatgaaag	aaacttcagc	1020
aagaaaatta	agtagcctg	ccgcaaaact	ctggcagata	gccgacctcg	agttagagga	1080
aggtttgcaa	agaatgatga	ctttggagaa	accaatagaa	ctactagtag	caatcatgaa	1140
gaagatgatg	aagaagaagt	agttatgaaa	gatgaagatg	atatggttga	ttcctcagat	1200
atctttgtctc	atatcagtg	gctgaactcc	ttcaaatgca	actattccat	ccagtcctgg	1260

-continued

atttga 1266

SEQ ID NO: 31 moltype = DNA length = 3129
 FEATURE Location/Qualifiers
 source 1..3129
 mol_type = genomic DNA
 organism = Arabidopsis thaliana

SEQUENCE: 31

tataaataaa	ttggttacga	taacttccat	aaccttcatt	agtttctcca	gaagaacact	60
gtaagaaatt	tctttctctt	ctctgtcttt	aaacagattt	ttattctggt	ctgtttgatt	120
aattattatc	tcttttgttc	cagtatttta	attcttttaga	ttttcttggt	cagacatcac	180
tagctccact	accacattca	agaagcctcc	ttcaataatc	tttccctttt	cctgggggaat	240
tttctctctg	gattcattcc	cggaaaatca	ttcacaaatt	ggagaccata	aagaacacat	300
aaacatgtcg	caggacgtaa	tttcgtctca	tgagcagctc	tccatgggtac	gtaataatcc	360
gtgcaatggt	tattgtttgt	cctcgaatct	gtttttatgc	ccttgaatgg	tttctcttct	420
ttgggtttcc	tcttttctat	catacattgc	ttaaatgatt	aaaagcatga	accttttttt	480
cctgaaaatt	ttgcatgtat	atatctatga	tttattcgca	aggaataaaa	ttattttctg	540
aaatctttgt	atttttaggt	taacatcggt	ttcttaattt	tcttagttat	tcgattttat	600
cttttgtcaa	tttcttctat	tctttttact	ttcttttctg	tttctggtta	ttctagatac	660
aattcagttc	atattccaaa	agtcctgaaac	gataaattagt	aaagcaaaact	ttttagtgtc	720
aactcttttg	tacgggtgtc	atacgtagtt	acgtaccttc	tcgctagcca	accacaaatc	780
gtcacaaatc	ttaatcagat	gaatagaac	agtttatcag	aattcataag	atgctacatc	840
tcttctttta	gtaattccag	taattaaaa	ttgaactttt	atttatactt	tcgaatagaa	900
tcacacgaat	ttttattaga	aatatataat	gagtgtatca	aaagcaatga	ttcttgcctc	960
taacctggcc	tttattgtgt	gttaaaaaat	cagatgtagt	tttaaatag	ggaacgtttt	1020
ggttctaaag	aaacgcaaga	atccaaaaac	tttctgttta	tggaaactct	cccgccttat	1080
cgtaaatctc	ttctctagat	tcctcttgta	tgaaaaccac	atagaaggaa	caagatctaa	1140
tcgtataaat	acatagttct	ctctaaacga	ctgtgttaag	ttaaaactaa	ccttaggttt	1200
attcaaaagt	attgtttctt	gtttcacaa	aaaatccac	caattagctt	tttatgcaag	1260
ccttcaacgt	atgtcaataa	acaacaatt	agatgcata	atttacttgt	tctattttct	1320
gctaatttga	tctttctctt	gctgtgttta	ttttgtagga	tgagatcaca	agtcctctca	1380
ccgcacagat	cttcgatttc	tgcgattcac	agttatttca	agaaaccttc	aatcaaaact	1440
ctgaagtaac	ctctgcttca	aacggttgtg	gctatgtcga	aaacaacaac	accaacaact	1500
ttccagacaa	atccaaacgt	gggttcgaatc	aagatcatga	agacaacaac	gacaacgctg	1560
atztatcaat	catattcgat	tcacaagatg	atttcgataa	cgatatcaca	gcttctatcg	1620
atttctcttc	atccatccag	ttcccagcct	ccgatcaact	ccaagagcag	tttgatttca	1680
ccggaattca	acttcaccag	cctccgaaca	ctctctactc	ttctctctct	ggtagacctc	1740
tgctctcgcc	tttatcggtt	ttcgaagaag	attgtctctt	ttctgtgcc	agttacaatc	1800
ttgggttccat	aaacctctct	tctctctctt	gctctttttt	gggtaataac	ggtttaccga	1860
cttcatgac	cgtaaccggg	aattatgatga	acaccgggtt	aggatctggt	ttttactccg	1920
ggaatatcca	ttctcggtct	gatttcaaac	catcacatga	tcaattgatg	gaaattcaag	1980
ctgataatgg	tggattgttc	tgccagatc	cgattaaacc	catctttaat	ccaggagatc	2040
atcatctcca	ggtatataat	ccaaacctta	tatataaata	atgaaaggta	ttgctggcaa	2100
acaaatttgt	taatgaaac	agattagtaa	accggtctga	tactaacagg	ttatgggggt	2160
tgttttctta	atagggactt	gacggtgtcg	agaaccagaa	ccacatgggt	gcacagccgg	2220
ttctcccgca	gttagggacg	gagataaacg	gtttagacga	cccaggttcc	aacaaagtgc	2280
gaaaactctc	ggcagagcag	agaaaagaga	agattcatag	gtatatgaag	aagaggaaacg	2340
agaggaaatt	cagcaagaata	atcaagggtc	ttgtgcttaa	ccatcataac	cggattaaac	2400
cagtgggtat	ccatataccg	atttgttcac	atgttttttt	tggttatatt	tggtgtagta	2460
tgcatgtagg	aagacgttgg	cagatagtcg	tccaagagtt	agagggaagat	ttgcaaaaga	2520
tgatgaattt	ggggaaccaa	atagacaagc	ctgttcaagc	catcatgaag	atgatgatga	2580
tgatgtaaga	ttaactctgt	ttgattacaa	ttagaaaaca	gatttagttt	tggttgatat	2640
aataaagcgt	ttctaaacgg	gggtgtgggt	tgcagggtggg	ggtaaaaggaa	gaagaacagt	2700
tagttgatcc	ttcggacata	ttttcacaca	taagtgggtg	gaactctttc	aaatgcaatt	2760
atccaatcca	gtcttggatt	tgagttggat	tacttacttg	gtcttatgat	ttacaaaaat	2820
aaaacaaaaa	tagtatgtgt	atagaggtct	gtaatttgta	tatttattag	tacatttttt	2880
ggccatgact	catagtttgt	cgaataagta	gaacataaga	aaagagagag	atgaataatg	2940
gggttaggca	ggtaatcatg	aataaaaattg	tagtgagtga	gaggggttgt	ttatgttaca	3000
tagaacatta	caaaaatttg	tagatatata	ttatatggta	tgataaattat	gatgtctatt	3060
aatgtatttg	atgggtgttg	tataaataat	aatcatttga	acattataaa	taagcagttg	3120
tgttcaagt						3129

SEQ ID NO: 32 moltype = DNA length = 1161
 FEATURE Location/Qualifiers
 source 1..1161
 mol_type = genomic DNA
 organism = Arabidopsis thaliana

SEQUENCE: 32

atgtcgcagg	acgtaatttc	gtctcatgag	cagctctcca	tggatgagat	cacaagtcct	60
ctcaccgcac	agatctctga	ttctcgcgat	tcacagttat	ttcaagaaac	cttcaatcaa	120
actctgaaag	ttaacctctgc	ttcaaacggg	tgtggctatg	tcgaaaacaa	caacaccaac	180
aaactttccag	acaaatccaa	cagtgggttcg	aatcaagatc	atgaagacaa	caacgacaa	240
gctgatttat	caatcatatt	cgattcacaa	gatgatttcg	ataacgatat	cacagcttct	300
atcgatttct	cttcatccat	ccagttccca	gcctccgcatc	aactccaaga	gcagtttgat	360
ttcacccgaa	ttcaacttca	ccagcctccg	aacactctct	actcttctct	ctctggtgac	420
cttctgcctc	cgcttttctc	ggttttcgaa	gaagattgtc	tttcttctgt	gccaaagtac	480

-continued

```

aatcttgggt ccataaacc ttcttcct tcttgcct tttgggtaa taccggtta 540
ccgacttaca tgacogtaac cgggaatat atgaacaccg gtttaggatc tggtttttac 600
tccgggaata tccatctcgg ttctgatttc aaaccatcac atgatcaatt gatggaaatt 660
caagctgata atggtgggatt gttctgtcca gatccgatta aaccatctt taatccagga 720
gatcatcatc tccagggact tgacgggtgc gagaaccaga accacatggt ggcacagccg 780
gttctcccg c agttaggagac ggagataacc ggttagacg acccgagttt caacaaagtc 840
ggaaaactct cggcagagca gagaaaagag aagattcata ggtatatgaa gaagaggaaac 900
gagagggaatt tcagcaagaa aatcaagtat gcatgtagga agacgttggc agatagtcgt 960
ccaagagtta gaggaagatt tgcaagaat gatgaatttg gtgaaccaa tagacaagcc 1020
tgttcaagcc atcatgaaga tgatgatgat gatgtggggg taaaggaaga agaacagtta 1080
gttgattctt cggacatatt ttcacacata agtgggtgtga actctttcaa atgcaattat 1140
ccaatccagt cttggatttg a 1161

```

```

SEQ ID NO: 33      moltype = AA  length = 386
FEATURE           Location/Qualifiers
source            1..386
                  mol_type = protein
                  organism = Arabidopsis thaliana

```

```

SEQUENCE: 33
MSQDVISSHE QLSMDEITSP LTAQIFDFCD SQLFQETFNQ TSEVTSASNG CGYVENNNTN 60
NFPDKSNSGS NQDHEDNNDN ADLSIIFDSQ DDFDNDITAS IDFSSSIQFP ASDQLQEQFD 120
FTGIQLHQPP NTLYSSSSGD LLPPPLSVFE EDCLSSVPSY NLGSINPSSP SCSFLGNTGL 180
PTYMTVTGNM MNTGLGSGFY SGNHLGSDF KPSHDQLMEI QADNGGLFCP DPIKPIFNPG 240
DHHLQGLDGV ENQNHMVAQP VLPQLGTEIT GLDDPSFNKV GKLSAEQRKE KIHRYMKKRN 300
ERNFSKKIKY ACRKTLADSR PRVRGRFAKN DEFGEFNRQA CSSHHEDDDD DVGKKEEQQL 360
VDSSDIFSHI SGVNSPKCNY PIQSWI

```

```

SEQ ID NO: 34      moltype = DNA  length = 20
FEATURE           Location/Qualifiers
source            1..20
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 34
ctggcagaac ttccagcccc 20

```

```

SEQ ID NO: 35      moltype = DNA  length = 20
FEATURE           Location/Qualifiers
source            1..20
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 35
ccaggactga gataagtgca 20

```

```

SEQ ID NO: 36      moltype = DNA  length = 20
FEATURE           Location/Qualifiers
source            1..20
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 36
ccaggactga gataagtgca 20

```

```

SEQ ID NO: 37      moltype = DNA  length = 21
FEATURE           Location/Qualifiers
source            1..21
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 37
atttatcccg aatggtaacg c 21

```

```

SEQ ID NO: 38      moltype = DNA  length = 21
FEATURE           Location/Qualifiers
source            1..21
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 38
taccggagat gtgtttgaag g 21

```

```

SEQ ID NO: 39      moltype = DNA  length = 21
FEATURE           Location/Qualifiers
source            1..21
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 39
aacgtgaaac ggatttgta g 21

```

```

SEQ ID NO: 40      moltype = DNA  length = 21

```


-continued

FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 40		
aaatttggtg aaaccaagac c		21
SEQ ID NO: 41	moltype = DNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 41		
at ttg ccga t ttc ggaac		19
SEQ ID NO: 42	moltype = DNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 42		
atttatcccg aatggtaacg c		21
SEQ ID NO: 43	moltype = DNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 43		
taccggagat gtgtttgaag g		21
SEQ ID NO: 44	moltype = DNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 44		
aacgtgaaac ggatttgtca g		21
SEQ ID NO: 45	moltype = DNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 45		
aaatttggtg aaaccaagac c		21
SEQ ID NO: 46	moltype = DNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 46		
at ttg ccga t ttc ggaac		19

1. A genetically modified legume (Fabaceae) plant having an improved agronomic trait, the plant comprising a nucleic acid sequence encoding a CCT motif-containing protein (CCT protein), wherein the nucleic acid sequence encoding the CCT protein comprises a nucleic acid modification and wherein the nucleic acid modification modifies the expression of the CCT protein in the plant thereby improving the agronomic trait of the plant, wherein the CCT protein is Seed Protein-Oil-Weight-Regulator 1 (POWR1) protein, POWR2 protein, or both, and wherein the improved agronomic trait is improved protein, improved oil accumulation in seeds and improved seed yield.

2-6. (canceled)

7. The genetically modified plant of claim 1, wherein the legume is common bean, cowpea, soybean, chickpea, pea, or Medicago.

8. The genetically modified plant of claim 1, wherein the legume is a soybean species (*Glycine max*, *hispida*).

9. (canceled)

10. The genetically modified plant of claim 8, wherein the CCT protein is a GmCCT67 (POWR1) protein and wherein the GmCCT67 protein is encoded by a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 2.

11. The genetically modified plant of claim 10, wherein the nucleic acid modification reduces the expression of the GmCCT67 protein in the plant thereby increasing oil content of seeds and reducing protein content of seeds.

12. (canceled)

13. The genetically modified plant of claim 10, wherein the nucleic acid modification increases the expression of the

GmCCT67 protein in the plant thereby reducing oil content of seeds and increasing protein content of seeds.

14-16. (canceled)

17. The genetically modified plant of claim **11**, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT67 protein comprises a transposable element (TE) insertion, and wherein the nucleic acid sequence comprising the TE insertion comprises at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 3.

18. (canceled)

19. The genetically modified plant of claim **13**, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT67 protein comprises an expression construct for expression of the GmCCT67 protein, wherein the expression construct comprises a nucleic acid sequence encoding the GmCCT67 protein operably linked to a ubiquitin promoter or a native promoter and wherein the expression construct comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 4.

20. (canceled)

21. The genetically modified plant of claim **8**, wherein the CCT protein is a GmCCT34 (POWR2) protein and wherein the GmCCT34 protein is encoded by a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 6.

22. The genetically modified plant of claim **21**, wherein the nucleic acid modification reduces the expression of GmCCT34 (POWR2) in the plant, thereby increasing oil content of seeds and reducing protein content of seeds.

23. (canceled)

24. The genetically modified plant of claim **21**, wherein the nucleic acid modification increases the expression of GmCCT34 (POWR2) in the plant, thereby reducing oil content of seeds and increasing protein content of seeds.

25-27. (canceled)

28. The genetically modified plant of claim **21**, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises an expression construct for expression of the GmCCT34 protein, wherein the expression construct comprises a nucleic acid sequence encoding the GmCCT34 protein operably linked to a promoter and wherein the expression construct for expression of GmCCT34 (POWR2) comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 7.

29. (canceled)

30. The genetically modified plant of claim **21**, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a genomic deletion of a 1.3-Mb region of the soybean genome (Chr10: 35253890-36584337) comprising a nucleic acid sequence encoding the GmCCT34 CCT protein.

31. (canceled)

32. The genetically modified plant of claim **21**, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 8 to 10 or any combination thereof.

33. The genetically modified plant of claim **21**, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 11 to 13 or any combination thereof.

34. The genetically modified plant of claim **21**, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 14 to 16 or any combination thereof.

35-50. (canceled)

51. The genetically modified plant of claim **1**, wherein the plant is a soybean species (*Glycine max*, *hispida*), wherein

a. the CCT protein is GmCCT67 (POWR1) and the nucleic acid modification in the nucleic acid sequence encoding the GmCCT67 protein comprises a nucleic acid sequence comprising at least about 75% or more, at least about 85% or more, at least about 95% or more, or 100% sequence identity with the nucleic acid sequence of SEQ ID NO: 3; and

b. the CCT protein is GmCCT34 (POWR2), wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a genomic deletion of a 1.3-Mb region of the soybean genome (Chr10: 35253890-36584337) comprising a nucleic acid sequence encoding the GmCCT34 CCT protein, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 8 to 10, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 11 to 13, wherein the nucleic acid modification in the nucleic acid sequence encoding the GmCCT34 protein comprises a nucleic acid sequence of SEQ ID NO: 14 to 16, or any combination thereof.

52-88. (canceled)

89. A method of improving an agronomic trait of a legume plant, the method comprising: introducing a nucleic acid modification into a nucleic acid sequence encoding a CCT motif-containing protein (CCT protein), wherein the nucleic acid modification modifies the expression of the CCT protein in the plant thereby improving the agronomic trait of the plant, wherein the CCT protein is Seed Protein-Oil-Weight-Regulator 1 (POWR1) protein, POWR2 protein, or both, and wherein the improved agronomic trait is improved protein and oil accumulation in seeds.

90. A kit for improving an agronomic trait of a plant, the kit comprising:

one or more genetically modified plant having an improved agronomic trait of claim **1**.

* * * * *